**LBA13** **General Session**

# A randomized phase 2 study of an individualized neoantigen-targeting immunotherapy in patients with newly diagnosed metastatic microsatellite stable colorectal cancer (MSS-CRC).

Joel R Hecht, Alexander I. Spira, Anthony V. Nguyen, Lyudmyla Derby Berim, Alexander Starodub, Meredith Pelster, David Gallinson, Anup Kasi, Matthew Reilley, Agustin Pimentel, Ki Young Chung, Fa-Chyi Lee, Brian Chauder, Kyounghwa Bae, Desiree Schneck, Andrew R. Allen, Karin Jooss, Andrew R. Ferguson, J. Thaddeus Beck; David Geffen School of Medicine at UCLA, Jonsson Comprehensive Cancer Center, Santa Monica, CA; Virginia Cancer Specialists Research Institute, Fairfax, VA; Comprehensive Cancer Centers Nevada US Oncology, Henderson, NV; Rutgers Cancer Institute of New Jersey, New Brunswick, NJ; The Christ Hospital Cancer Center, Cincinnati, OH; Nashville-Centennial Clinic, Tennessee Oncology, Nashville, TN; Summit Medical Group, Florham Park, NJ; University of Kansas Cancer Center, Westwood, KS; University of Virginia Comprehensive Cancer Center, Charlottesville, VA; Sylvester Comprehensive Cancer Center, University of Miami, Miami, FL; PRISMA Health Cancer Institute, Institute for Translational Oncology Research, Boiling Springs, SC; University of California, Irvine, Orange, CA; Gritstone bio, Inc, Emeryville, CA; Gritstone bio, Emeryville, CA; Highlands Oncology Group, Springdale, AR

**The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.**

**LBA14** **Oral Abstract Session**

# Prognostic and predictive role of circulating tumor DNA (ctDNA) in stage III colon cancer treated with celecoxib: Findings from CALGB (Alliance)/SWOG 80702.

Jonathan Andrew Nowak, Qian Shi, Tyler Twombly, Levi D. Pederson, Chao Ma, Juha P. Väyrynen, Melissa M. Zhao, Yasutoshi Takashima, Ardaman Shergill, Pankaj Kumar, Felix Couture, J. Phillip Kuebler, Smitha S. Krishnamurthi, Benjamin R. Tan, Eileen M. O'Reilly, Anthony F. Shields, Shuji Ogino, Alexey Aleshin, Jeffrey A. Meyerhardt; Brigham and Women's Hospital, Boston, MA; Alliance Statistics and Data Management Center and Mayo Clinic Rochester, Rochester, MN; Dana-Farber Cancer Institute, Boston, MA; University of Oulu and Oulu University Hospital, Oulu, Finland; University of Chicago, Chicago, IL; Heartland Cancer Research NCORP, Peoria, IL; CHU de Québec, l'Hôtel-Dieu de Québec, Quebec, QC, Canada; Columbus NCI Community Oncology Research Program, Columbus, OH; Cleveland Clinic, Cleveland, OH; Siteman Cancer Center, Washington University School of Medicine, St. Louis, MO; Memorial Sloan Kettering Cancer Center, New York, NY; Barbara Ann Karmanos Cancer Institute, Wayne State University, Detroit, MI; Natera, Inc., Austin, TX; Department of Medical Oncology, Dana-Farber Cancer Institute, Harvard Medical School, Harvard University, Boston, MA

**The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.**

# Circulating tumor DNA for detection of molecular residual disease (MRD) in patients (pts) with stage II/III colorectal cancer (CRC): Final analysis of the BESPOKE CRC sub-cohort.

Purvi K. Shah, Vasily N. Aushev, Joe Ensor, Stephanie A Sanchez, Christopher Gene Wang, Timothy Lewis Cannon, Lyudmyla Derby Berim, Trevor Feinstein, Axel Grothey, Joseph William McCollom, Sujith R. Kalmadi, Ahmed Zakari, Farshid Dayyani, Don Gravenor, Minetta C. Liu, Adham A Jurdi, Alexey Aleshin, Janelle Marie Meyer, Saima Sharif, Scott Kopetz; Virginia Cancer Institute, Richmond, VA; Natera, Inc., Austin, TX; Alabama Oncology, Birmingham, AL; Inova Schar Cancer Institute, Fairfax, VA; Rutgers Cancer Institute of New Jersey, New Brunswick, NJ; Piedmont Cancer Institue, Atlanta, GA; West Cancer Center and Research Institute, Germantown, TN; Parkview Cancer Institute, Fort Wayne, IN; Ironwood Cancer & Research Centers, Chandler, AZ; AdventHealth Cancer Institute, Orlando, FL; University of California, Irvine Medical Center, Irvine, CA; Baptist Cancer Center, Memphis, TN; Oregon Oncology Specialists, Salem, OR; University of Iowa, Iowa City, IA; The University of Texas MD Anderson Cancer Center, Houston, TX

**Background:** BESPOKE CRC, a multicenter, prospective, observational study investigated the clinical utility of ctDNA for detecting MRD-based early recurrence in pts with surgically resected CRC. The primary endpoint was to assess the impact of ctDNA testing on treatment decisions and asymptomatic recurrence rates. The secondary endpoint was to assess the MRD clearance rate, survival of MRD-negative pts, overall survival, and patient-reported outcomes. **Methods:** Complete clinical and laboratory data were available for 1001 pts with stages II–III CRC. Longitudinal ctDNA testing was performed prospectively using a clinically validated, personalized, tumor-informed 16-plex mPCR-NGS assay (Signatera, Natera, Inc.). Plasma time points (n=8,536) were collected during the MRD (2-12 weeks postoperatively) and surveillance (post-adjuvant chemotherapy [ACT] completion/12 weeks postoperatively for pts observed) windows. We evaluated the correlation between ctDNA status and disease-free survival (DFS) as part of exploratory analysis. **Results:** Following curative resection, 62.4% (625/1001) pts received ACT: 25.9% (115/443) stage II, and 91.3% (510/558) stage III.Among pts with ctDNA results available during the MRD window with a median follow-up of 23.15 (range: 2.89-33.45) months, ctDNA-positivity was observed in 8.1% (34/420) of stage II and 24.9% (126/505) of stage III pts. A significant association between ctDNA positivity and inferior DFS was observed in stage II (HR=10.4; $p<0.0001$), and stage III (HR=10.1, $p<0.0001$) pts. 18/24 month DFS estimates for stages II-III combined were: (MRD-negative: 93.0%/91.7%; MRD-positive: 44.4%/41.4%). Analysis of surveillance was performed separately for pts observed (N=368) vs. ACT-treated (N=597). During surveillance, in the observation cohort, 6.8% (22/323) of stage II, and 33.3% of stage III (15/45) pts tested ctDNA-positive correlating with significantly worse DFS (stage II: HR=34.9; $p<0.0001$, stage III: HR=34.1; p=0.0008). Likewise, in the ACT cohort, 10.9% (12/110) of stage II and 21.1% of stage III (103/487) pts turned ctDNA-positive and had significantly worse DFS (stage II: HR=131.4; $p<0.0001$, stage III: HR=54.6; $p<0.0001$). Analysis of primary and secondary endpoints will be presented. **Conclusions:** ctDNA positivity was highly prognostic of DFS within MRD and surveillance windows in a subset of stages II-III pts enrolled in the BESPOKE CRC study. Our results highlight the potential value of ctDNA-based MRD detection for treatment-decision making, and findings relevant to clinical utility will be reported in the conference presentation. Clinical trial information: NCT04264702. Research Sponsor: None.

# BREAKWATER: Analysis of first-line encorafenib + cetuximab + chemotherapy in BRAF V600E-mutant metastatic colorectal cancer.

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**Background:** Encorafenib + cetuximab (EC) is approved for previously treated BRAF V600E-mutant metastatic colorectal cancer (mCRC) based on the BEACON phase 3 study (NCT02928224). Historically, first-line (1L) treatment of BRAF V600E-mutant mCRC with chemotherapy (chemo) regimens has had limited efficacy.BREAKWATER (NCT04607421) is an open-label, global, randomized, phase 3 study evaluating 1L EC with or without chemo vs standard of care (SOC; chemo with or without bevacizumab). Reported here are the primary analysis of objective response rate by blinded independent central review (ORR by BICR; dual primary endpoint [EP]), the first interim analysis of overall survival (OS; key secondary EP), other secondary EPs, and safety for the EC+FOLFOX (oxaliplatin, leucovorin, and 5-FU) vs SOC arms. **Methods:** Eligible patients (pts) had untreated BRAF V600E-mutant mCRC, measurable disease (RECIST 1.1), and ECOG PS 0-1. Pts were randomized 1:1:1 to receive EC, EC+FOLFOX, or SOC; EC arm enrollment was closed after a protocol amendment. Dual primary EPs were ORR (assessed in the first 110 pts randomized to each of the EC+FOLFOX and SOC arms) and progression-free survival by BICR (EC+FOLFOX vs SOC); OS was a key secondary EP (EC+FOLFOX vs SOC), other secondary EPs included response duration and time to response (TTR). **Results:** Four hundred seventy-nine pts were randomized to the EC+FOLFOX and SOC arms (EC+FOLFOX: n=236; SOC: n=243). Baseline demographics and disease characteristics were similar across arms (median age: 61.0 years; male: 50.5%; ECOG PS 0: 54.3%). At data cutoff (Dec 22, 2023), the EC+FOLFOX arm demonstrated a clinically meaningful and statistically significant improvement in confirmed ORR vs the SOC arm, 60.9% vs 40.0%, odds ratio=2.443, one-sided *P*-value=0.0008, meeting this dual primary EP. The response observed with EC+FOLFOX was rapid and durable. OS data were immature but indicated a sustained survival benefit with EC+FOLFOX vs SOC arm. Serious treatment-emergent adverse events (EC+FOLFOX: n=231; SOC: n=228) occurred in 37.7% vs 34.6% of pts in the respective arms. The safety profile was consistent with that known for each agent. **Conclusions:** BREAKWATER demonstrated a substantially improved response rate that was rapid and durable with EC+FOLFOX in BRAF V600E-mutant mCRC with manageable toxicities and no new safety signals. Clinical trial information: NCT04607421. Research Sponsor: Pfizer, Inc.

| | EC+FOLFOX n=110 | SOC n=110 |
|---|---|---|
| **ORR by BICR % (95% CI)** | 60.9 (51.6, 69.5) | 40.0 (31.3, 49.3) |
| **Odds ratio (95% CI)** | 2.443 (1.348, 4.380) | |
| ***P*-value[a]** | 0.0008 | |
| | **n=67** | **n=44** |
| **Estimated median response duration by BICR (95% CI), mo** | 13.9 (8.5, NE) | 11.1 (6.7, 12.7) |
| **Pts with a response duration of ≥6 mo, n (%)** | 46 (68.7) | 15 (34.1) |
| **Pts with a response duration of ≥12 mo, n (%)** | 15 (22.4) | 5 (11.4) |
| **Median TTR by BICR (range), weeks** | 7.1 (5.7-53.7) | 7.3 (5.4-48.0) |
| | **n=236** | **n=243** |
| **OS (95% CI), mo** | NE (19.8, NE) | 14.6 (13.4, NE) |
| **Hazard ratio (95% CI)** | 0.47 (0.318, 0.691) | |

[a]One-sided $\alpha$=0.001.
NE, not estimable.

# Final analysis of modified (m)-FOLFOXIRI plus cetuximab versus bevacizumab for *RAS* wild-type and left-sided metastatic colorectal cancer: The DEEPER trial (JACCRO CC-13).

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**Background:** The DEEPER trial (NCT02515734), which evaluated m-FOLFOXIRI (irinotecan 150 mg/m$^2$, oxaliplatin 85 mg/m$^2$, 5-FU 2400 mg/m$^2$) plus cetuximab (cet) vs. bevacizumab (bev) as initial therapy in terms of depth of response (DpR) as the primary endpoint in *RAS* wild-type metastatic colorectal cancer (mCRC), has demonstrated a significantly better DpR in the cet arm (ASCO 2021). Moreover, favorable progression-free survival (PFS) was reported in the cet arm for patients (pts) with *RAS*/*BRAF* wild-type and left-sided tumors (ESMO 2023). Due to the small number of overall survival (OS) events, it was decided to extend the observation period, and the final survival analysis was performed at the cutoff date of August 2024. **Methods:** Survival analysis was pre-planned in the per-protocol set (PPS), which consisted of pts evaluable for the DpR by an external review board. The clinical outcomes were evaluated according to clinical factors including primary tumor sidedness, liver metastasis status, and *BRAF* status using a log-rank test. All statistical tests were two-sided, and *P* values ≤ 0.05 were considered significant. **Results:** 321 of 359 enrolled pts were defined as PPS (median age 65 years, 64% male, performance status [PS] 0/1: 91%/9%, left/right primary: 84%/16%). In *RAS* wild-type and left-sided tumors, median PFS and OS were 13.9 months vs. 12.1 months (HR 0.81, 95% CI 0.63-1.05) and 45.3 months vs. 41.9 months (HR 0.85, 95% CI 0.64-1.12) in the cet vs. bev arm, respectively. *BRAF* status was available in 234 (73%) of the 321 pts in the PPS. An exploratory analysis showed that PFS was significantly better in the cet arm compared to the bev arm (median 14.8 months vs. 11.9 months, HR 0.71, 95% CI 0.52-0.97) in 178 pts with *RAS*/*BRAF* wild-type and left-sided tumors. Additionally, OS was 50.2 months vs. 40.2 months (HR 0.74, 95% CI 0.53-1.05). Moreover, according to liver disease status, m-FOLFOXIRI plus cet was associated with longer PFS and OS in pts with *RAS*/*BRAF* wild-type and left-sided mCRC with extra-hepatic metastases (median 15.1 months vs. 11.4 months, HR 0.66, 95% CI 0.46-0.95 and 50.2 months vs. 38.6 months, HR 0.60, 95% CI 0.40-0.90), but not liver-limited disease (median 14.5 months vs. 15.5 months, HR 0.79, 95% CI 0.44-1.42 and 52.2 months vs. 49.9 months, HR 1.17, 95% CI 0.61-2.24). **Conclusions:** The final survival analysis of the DEEPER trial demonstrated favorable PFS and OS with m-FOLFOXIRI plus cet in pts with *RAS*/*BRAF* wild-type and left-sided mCRC. The m-FOLFOXIRI plus cet regimen may be a good option for initial therapy, offering longer survival times in pts with extra-hepatic metastases. Clinical trial information: jRCTs061180022. Research Sponsor: Merck Biopharma Co., Ltd.

# Performance of a blood-based test for colorectal cancer screening adjusted to the US census age and sex distribution.

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**Background:** Despite availability of various screening options, nearly 40% of eligible U.S. adults were not up to date with the colorectal cancer (CRC) screening. Blood-based testing offers a promising complementary approach and may enhance patient adherence among unscreened individuals. In the PREEMPT CRC study, we evaluated clinical performance of an investigational blood-based screening test for detecting molecular signals of advanced colorectal neoplasia (ACN) in an average-risk population. Freenome blood-based CRC screening test met all primary endpoints demonstrating 79.2% sensitivity for CRC, 91.5% specificity for ACN, 90.8% negative predictive value (NPV) for ACN and 15.5% positive predictive value (PPV) for ACN. Sensitivity for advanced precancerous lesions (APLs) was 12.5%. To evaluate how the test would perform in the standard U.S. population, an analysis of test performance weighted to the U.S. census sex and age distribution was prespecified and performed. **Methods:** Between May 2020 and April 2022, the study enrolled participants aged 45-85 at average risk for CRC. Enrolled participants had a blood drawn before the bowel preparation for the standard of care screening colonoscopy (CS). CS and applicable histopathology reports underwent central pathologist review. Blood samples were processed blind to clinical findings. Primary endpoints included sensitivity for CRC, specificity for ACN, NPV for ACN and PPV for ACN. A secondary endpoint was sensitivity for APLs. **Results:** A subset of 32,731 sequentially enrolled participants were included in the clinical validation cohort. Out of those, 27,010 (82.5%) had evaluable blood samples and CS. The median age of the evaluable participants was 57.0 years, and 55.8% were women versus 50.5% in the U.S. Census. Age distribution included 11.0% participants in the 45-49 age group versus 15.3% in the U.S. Census, 33.0% in the 50-54 age group versus 15.6%, and 32.3% in the 55-64 age group versus 32.3%, 20.8% in the 65-74 age group versus 24.3%, 3.0% participants aged 75 or older versus 12.5%. After performance was weighted to match U.S. Census sex and age distributions, sensitivity for CRC was 81.1% (95% CI 71.3%-88.1%), specificity for those without ACN was 90.4% (95% CI 90.0%-90.7%), NPV for those without ACN was 90.5% (95% CI 90.5%-90.7%), and PPV for ACN was 15.5% (95% CI 13.8%-16.2%). Sensitivity for APL was 13.7% (95% CI 12.4%-15.0%). **Conclusions:** PREEMPT CRC is the largest study evaluating a CRC screening blood-based test to date. The study successfully met all primary endpoints, demonstrating acceptable clinical performance of the investigational blood-based test. The U.S. census sex-age adjusted clinical performance estimates showed improvement in CRC and APL sensitivity. This blood-based test may provide a convenient and effective option for CRC screening in the average-risk U.S. population. Clinical trial information: NCT04369053. Research Sponsor: None.

# Single-cycle neoadjuvant pembrolizumab in patients with stage I-III MMR-deficient colon cancer: Final analysis of the RESET-C study.

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**Background:** Neoadjuvant treatment with immune checkpoint inhibitors has shown remarkable responses in patients with deficient mismatch repair (dMMR) colorectal cancer. However, the optimal choice and duration of treatment have yet to be established. **Methods:** RESET-C (NCT05662527) was an investigator-initiated, phase II, single-arm, multicenter study investigating the efficacy and safety of single-cycle neoadjuvant pembrolizumab in 85 patients with resectable stage I-III dMMR colon cancer. Additional inclusion criteria were ≥ 18 years of age, no indication for neoadjuvant therapy, and Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. After inclusion, patients received one cycle of pembrolizumab 4mg/kg (maximum 400mg) and underwent surgery within three to five weeks. A tumor-response evaluation was done before surgery, including blood samples, a computed tomography scan of the chest and abdomen, and a colonoscopy. The primary endpoint was the pathological complete response (pCR) rate according to Mandard tumor regression grading (Clopper-Pearson method). Secondary endpoints included surgical complications, safety of pembrolizumab, major pathological response (Mandard tumor regression grade 1 or 2), and disease-free survival. **Results:** Between February 2023 and March 2024, 85 patients were included. The median age was 74 years (IQR, 68-79), 72% were women, and 60% had clinical stage III disease. All patients received pembrolizumab, and 84 patients proceeded to surgery. One patient with stage I disease decided not to undergo surgery. A pCR rate of 44% (37/84; 95% CI, 33-55) and a major pathological response rate of 57% (48/84; 95% CI, 46-68) were found. A significantly higher pCR rate was seen in patients with stage I-II (20/33) versus stage III (17/51) disease (61% vs 33%; p=0.02). A total of 41 surgical complications were seen in 31 patients (37%; 95% CI, 27-48). Of these, eight complications were Clavien-Dindo grade 3a or above, including three patients with anastomotic leakages and two deaths within 30 days. The patients who died were aged 80 and 81, had ECOG performance status 1, and clinical stage IIIB and IIIC disease, respectively. Seven out of 85 patients (8%; 95% CI, 3-16) experienced grade 3 adverse events, three of which were treatment-related. No grade 4 or 5 adverse events were registered. Finally, data on the association between endoscopically evaluated clinical complete response and pCR, along with the 1-year disease-free survival rate expected in January 2025, will be reported. **Conclusions:** A single cycle of neoadjuvant pembrolizumab was efficacious and safe in patients with localized dMMR colon cancer. For most patients with clinical stage I-II disease, a single cycle sufficed to achieve pCR. Clinical trial information: NCT05662527. Research Sponsor: Danish Cancer Society; R343-A19803; Eva and Henry Frænkel's Memorial Fund; Vissing Foundation; 532937 AHO/LAX; Helge Peetz and Verner Peetz and wife Vilma Peetz's Scholarship; Merchant L.F. Foght's Fund; 22.340; Medical Doctor Sofus Carl Emil Friis and Wife Olga Doris Friis' Foundation; Axel Muusfeldt's Foundation; 2022-0016 and 2023-0486; NEYE Foundation; Danish Comprehensive Cancer Center; 16; Dagmar Marshalls Foundation; Frimodt Heineke Foundation; Manufacturer Einar Willumsen's Memorial Fund; Timber Merchant Vilhelm Bang's Fund.

# Tumour-free ctDNA detection as a decision tool to support organ preservation in node-negative rectal cancer undergoing neoadjuvant chemotherapy, excision, and observation in the phase II NEO trial (CCTG CO.28).

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**Background:** In the NEO trial (CCTG CO.28, NCT03259035) patients with node negative (N0) T1-T3 rectal cancer were treated with neoadjuvant CAPOX/FOLFOX and transanal excision surgery (TES) with the goal of organ preservation. Patients without documented response following neoadjuvant chemotherapy were recommended total mesorectal excision (TME). We present a post-hoc assessment of ctDNA detection in this early-stage population and correlate kinetics with response and outcomes. **Methods:** Fifty-eight patients enrolled in CO.28. Whole blood was collected in EDTA tubes and processed for future ctDNA analysis at up to six timepoints: pre-chemotherapy, after neoadjuvant chemotherapy (and pre-TES), yearly during surveillance for 3 years and upon progression. A total of 195 samples were analyzed with the Guardant Reveal™ assay: a tissue-free epigenomic assay leveraging >20,000 epigenomic regions for ctDNA detection. **Results:** Of 48 available pre-treatment samples, ctDNA was detected in 22 (46%). Sensitivity by T-stage was 14% (1/7) for T1, 53% (17/32) for T2, 44% (4/9) for T3 cancers. ctDNA detection was lower following neoadjuvant chemotherapy (4/46, 9%, $p<0.001$ vs pre-chemotherapy). Of 41 patients with paired pre- and post-chemo samples, 49% (20/41) had undetectable ctDNA at both timepoints, 44% (18/41) had reduced tumour fraction following chemotherapy, the majority of which completely cleared ctDNA (94%; 17/18) and 7% (3/41) had an increased tumour fraction following chemotherapy (67%, 2/3 changing from negative to positive), all three of whom failed to respond to neoadjuvant chemotherapy and had TME recommended (100%). In contrast, of those that cleared ctDNA following chemotherapy, 35% had TME recommended (6/17, $p=0.074$). Of five recurrences (3 distant, 2 local), all had negative or reduced ctDNA tumour fraction during neoadjuvant therapy before surgical intervention. Among patients with local relapse who were managed with TES, 1/2 (50%) had detectable ctDNA at the time of local progression. No samples were available for those with distant recurrences (3/5) at time of relapse. **Conclusions:** The Guardant Reveal tissue-free assay was able to detect ctDNA even in this extremely early-stage cohort and identified cancers with inadequate response to therapy and in whom TME was recommended. A tissue-free approach may support timeliness of ctDNA results that would support ctDNA as an additional decision tool with endoscopic and MRI assessments to increase physician and patient comfort with organ preservation. Clinical trial information: 03259035. Research Sponsor: Canadian Cancer Society.

# WW vs TME in patients with rectal cancer with a complete or near-complete response to TNT: Pooled analysis of CAO/ARO/AIO-12 and OPRA trials.

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**Background:** Patients with locally advanced rectal cancer (LARC) and a clinical complete response (cCR) or near-complete response (nCR) to total neoadjuvant therapy (TNT) can be offered watch-and-wait (WW) with the intention of achieving organ preservation. However, one third of patients on WW develop tumor regrowth and require total mesorectal excision (TME). Assessing the safety of WW in patients with a cCR or nCR is challenging due to the difficulty of finding an adequate control group. Here we compare the survival outcomes of patients with LARC treated with TNT and either mandatory TME or selective WW stratified by tumor response. **Methods:** This is a pooled analysis of two multicenter, phase II trials (CAO/ARO/AIO-12 [CAO] and OPRA) that randomized patients with stage II/III rectal cancer to either induction or consolidation TNT. All patients in the CAO trial underwent TME within 6 weeks of TNT. Pathologic tumor regression grade (TRG) was categorized as complete (TRG 4), intermediate (TRG 2,3) or poor (TRG 0,1). Patients in the OPRA trial were restaged by endoscopy and MRI 8±4 weeks after end of TNT. Clinical response was graded as cCR, nCR, or incomplete clinical response (iCR). Patients with a cCR or nCR were offered WW; patients with an iCR were recommended for TME. Survival curves were estimated using the Kaplan-Meier method and the log-rank test. **Results:** The study included 628 patients (*n*=304 CAO; *n*=324 OPRA). Median follow-up was 3.6 (IQR 1.13) and 5.1 (IQR 2.2) years. Patients in the CAO trial were more likely to have cT3/4 and cN positive disease while patients in the OPRA trial had tumors closer to the anal verge. Compliance with TNT and rates of grade 3+ adverse events were similar between studies. A total of 144 (47%) patients in the OPRA trial achieved long term organ preservation. We found no differences in survival between trials. In addition, we found no differences in DFS between studies for patients with an excellent (TRG4 87% [95% CI, 79-97%], cCR 87% [95% CI, 81-93%]) or intermediate (TRG2-3 69% [95% CI, 62-76%], nCR 71% [95% CI, 68-80%]) response. **Conclusions:** This pooled analysis demonstrated that almost half of LARC patients treated with TNT can achieve organ preservation and that a selective WW strategy yields similar survival outcomes compared to mandatory TME. In addition, we found no differences in survival based on paired clinical and pathologic tumor response grades following TNT. This data provides further evidence supporting the safety of WW for patients with LARC and a cCR or nCR to TNT. Research Sponsor: None.

**Three-year survival outcomes in the CAO/ARO/AIO-12 and OPRA trials.**

| | CAO/ARO/AIO-12 (95% CI) | OPRA (95% CI) | *P* |
|---|---|---|---|
| **3y DFS** | 73 (71–81)% | 76 (68–78)% | 0.3 |
| **3y DRFS** | 82 (78–87)% | 82 (78–87)% | 0.7 |
| **3y LRFS** | 95 (92–98)% | 95 (92–97)% | 0.4 |
| **3y OS** | 92 (89–95)% | 94 (89–95)% | 0.5 |

DFS: disease-free survival, DRFS: distant recurrence-free survival, LRFS: local recurrence-free survival, OS: overall survival.

**LBA22** **Poster Session**

# A randomized, double-blind, phase III study comparing trifluridine/tipiracil (FTD/TPI) versus placebo in patients with molecular residual disease following curative resection of colorectal cancer (CRC): The ALTAIR study.

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**The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.**

# Preliminary results from a randomized, open-label, phase 2 study of botensilimab (BOT) with or without balstilimab (BAL) in refractory microsatellite stable metastatic colorectal cancer with no liver metastases (MSS mCRC NLM).

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**Background:** BOT is an Fc-enhanced, multifunctional anti-CTLA−4 antibody designed to improve Fc gamma receptor-mediated effector functions and extend the reach of I-O to tumor types such as MSS mCRC. Here we present preliminary data from a randomized, open-label, phase 2 study in patients (pts) with MSS mCRC NLM treated with BOT ± BAL (anti-PD−1; NCT05608044). The study aimed to inform dose and contribution of components based on the primary endpoint of objective response rate (ORR) by RECIST 1.1 per investigator, and safety, and was not powered for statistical comparisons between arms. **Methods:** A total of 234 pts (intent-to-treat [ITT]) were randomized to BOT (up to 4 doses) 75 or 150 mg every 6 wks (Q6W), BOT 75 or 150 mg Q6W plus BAL 240 mg Q2W (up to 2 years), or standard of care (SOC; regorafenib or trifluridine/tipiracil). **Results:** Median age was 58 yrs (range 23—90), 50% male, 39% rectal, 44% 3L+, 43% ECOG 1, 58% KRAS mutant, 4% NRAS mutant, 83% prior bev, all MSS and/or pMMR by local testing. Key characteristics were well balanced with some exceptions including median time from diagnosis of metastatic disease to study entry (30 mos across arms; 45 mos SOC) and presence of peritoneal metastases (34% across arms; 42% 75 mg BOT / 240 mg BAL; 27% SOC). As of July 29, 2024, median follow-up was 9.8 mos. Key efficacy and safety data are shown (Table). Image based endpoints by blinded independent review, as well as overall survival will be reported in the future. Grade ≥3 treatment-related adverse events (TRAEs) were highest with SOC followed by BOT + BAL combination, and then BOT monotherapy, with dose dependency. Treatment-related immune-mediated diarrhea/colitis (imDC) was manageable and highest with 150 mg BOT / 240 mg BAL. No new safety signals and no treatment-related deaths occurred. **Conclusions:** The study met the objectives of informing dose and contribution of components. Overall ORR was higher with BOT + BAL vs BOT monotherapy. ORR was highest with 75 mg BOT / 240 mg BAL with less toxicity as compared to 150 mg BOT / 240 mg BAL. Consistent with published data, there were no objective responses in SOC whereas most responses seen with BOT + BAL were ongoing, similar to the durable responses observed in the ph1 study. These responses are differentiated from previous I-O-only combinations and SOC, supporting further investigation of 75 mg BOT / 240 mg BAL vs SOC in a planned global ph3 trial. Clinical trial information: NCT05608044. Research Sponsor: Agenus Inc.

| | 75 mg BOT/240 mg BAL | 150 mg BOT/240 mg BAL | 75 mg BOT | 150 mg BOT | SOC |
|---|---|---|---|---|---|
| **ITT (Randomized)** | *n=62* | *n=61* | *38* | *n=40* | *n=33* |
| **ORR, %** | 19% | 8% | 0% | 8% | 0% |
| **(95% CI)** | (10—31) | (3—18) | (0—10) | (2—20) | (0—11) |
| **Safety (Treated)** | *n=62* | *n=60* | *n=37* | *n=39* | *n=21* |
| **Any Grade TRAEs, n (%)** | 54 (87) | 59 (98) | 28 (76) | 31 (79) | 19 (90) |
| **Grade ≥3 TRAEs, n (%)** | 22 (35) | 25 (42) | 8 (22) | 9 (23) | 12 (57) |
| **Any Grade Treatment-related IMDC, n (%)** | 19 (31) | 28 (47) | 13 (35) | 12 (31) | 0 (0) |

# Disparities in stage of diagnosis of colon cancer: Differences by Hispanic background, race, and ethnicity.

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**Background:** Given noted heterogeneity in Hispanic adults, cancer outcomes reported in aggregate may not detect disparities within specific Hispanic subgroups. The objective was to assess whether any potential differences in colon cancer (CC) stage at diagnosis among patients with Hispanic background (HBackground) could be explained by differences in socioeconomic determinants of health (SDOH) including income, education and insurance status. **Methods:** The National Cancer Database was queried for patients diagnosed with CC from 2004 to 2021. Two logistic regression models generated adjusted odds ratios of late-stage (III-IV) vs early-stage (I-II) CC at diagnosis, disaggregated either by HBackground or R&E, compared to non-Hispanic White (NHW) patients. Model 1 controlled for patient/facility characteristics; Model 2 controlled for additional SDOH. **Results:** Of the 442,742 CC patients identified, 6% were Hispanics. Compared to NH adults, Hispanic patients were more likely (6% higher odds, $p<0.0001$) to present with late-stage CC. When Hispanic patients were disaggregated by HBackground, some were more likely (Mexicans 16%; Cubans 20% and South/Central Americans 12%) than NHW adults to present with late-stage CC while others (Puerto Ricans and Dominicans) were not. Disaggregation by R&E indicated that only Hispanic-White patients were more likely (12%) to present with late-stage disease than NHW patients. Controlling for SDOH, R&E differences in stage at diagnosis persisted but HBackground differences became insignificant (Table). **Conclusions:** Disaggregation of Hispanic adults by HBackground identifies specific subgroups at increased risk for late-stage CC at diagnosis. Understanding modifiable SDOH may identify opportunities for improving early detection of CC and outcomes in specific Hispanic subgroups. Research Sponsor: None.

**Odds of presenting with late-stage colon cancer of "Hispanic" patients disaggregated by race and ethnicity or Hispanic background compared to non-Hispanic Whites.**

| Disaggregation based on: | Model 1: Adjusted for age, facility type, year of diagnosis, and Charlson-Deyo comorbidity aOR (95% CI) | p-value | Model 2: Adjusted for age, facility type, year of diagnosis, Charlson-Deyo comorbidity, neighborhood income, neighborhood education, and insurance status aOR (95% CI) | p-value |
|---|---|---|---|---|
| **Non-Hispanic White (n=418,046)** | (ref) | | (ref) | |
| **Hispanic White (n=24,058)** | 1.12 (1.09, 1.15) | <.0001 | 1.04 (1.01, 1.07) | 0.01 |
| **Hispanic Background** | | | | |
| **Mexican (n=3,612)** | 1.16 (1.08, 1.24) | <.0001 | 1.05 (0.97, 1.13) | 0.25 |
| **South/Central American (n= 1,734)** | 1.12 (1.02, 1.24) | 0.02 | 1.01 (0.91, 1.12) | 0.83 |
| **Cuban (n=937)** | 1.20 (1.05, 1.36) | 0.001 | 1.11 (0.97, 1.27) | 0.13 |
| **Puerto Rican (n=1,280)** | 1.07 (0.95, 1.19) | 0.28 | 1.01 (0.90, 1.14) | 0.88 |
| **Dominican (n=465)** | 1.14 (0.94, 1.39) | 0.18 | 1.02 (0.83, 1.25) | 0.85 |

aOR, Adjusted odds ratio; CI, confidence Interval.

# Association of social deprivation and colon cancer mortality: A population-based analysis.

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**Background:** Colon cancer mortality is largely influenced by social factors, with higher rates seen in deprived areas. The Social Deprivation Index (SDI) measures factors like poverty, education, and unemployment, which have been linked to poor health outcomes. This study examines the relationship between county level social deprivation and colon cancer mortality in Texas. **Methods:** We conducted a population-based cohort study of colon cancer deaths in Texas. We queried the National Cancer Institute Surveillance, Epidemiology and End Results Program for county level colon cancer deaths in Texas over the most recent available year, 2021. We calculated age-adjusted mortality rates (AAMRs) per 100,000 population using the direct standardization method based on the age group weights from the 2000 standard US population. Confidence intervals for AAMR were derived by estimating the standard error as the AAMR divided by square root of number of mortalities. The SDI by county was linked to mortality rate using Federal Information Processing Standards (FIPS) codes. The significance of the difference between two AAMR was evaluated by determining if the intersection of 95% confidence intervals was empty. The association between SDI and AAMR was tested using both unweighted and weighted Spearman correlation with county population as weight. Results are reported and Spearman correlation and 95% confidence interval ($\rho$ [95% CI]). Sensitivity analysis was performed by considering only the individual components of the SDI. Subgroup analyses included age group, sex, urban counties, rural counties, and race/ethnicity groups. **Results:** A total of 3,800 colon cancer mortalities were identified in 2023. The statewide AAMR was 12.5 (95% CI 12.1 to 12.9). Out of 254 counties, 29 (11.4%) had an SDI $\leq$ 25, 61 (24.0%) had 26 $\leq$ SDI $\leq$ 50, 87 (34.3%) had SDI 51 $\leq$ SDI $\leq$ 75, and 77 (30.3%) had 76 $\leq$ SDI $\leq$ 100. Total population in counties with SDI $\geq$ 76 was 15,106,726 (51.1%) with AAMR 12.959 (12.379 to 13.538). In contrast, the population in counties with SDI $\leq$ 25 was 4,771,473 (16.1%) with significantly lower mortality (AAMR 9.752 [95% CI 8.879 to 10.625]). The population weighted Spearman correlation between county level SDI and AAMR was positive ($\rho$ = 0.339 [95% CI 0.209 – 0.514]). The unweighted Spearman correlation between county level SDI and AAMR was also positive ($\rho$ = 0.178 [95% CI 0.089 – 0.238]). Households with no vehicle, poverty, and unemployment were most correlated with AAMR while renter occupied housing had no association. The correlation of county level SDI and AAMR were similar for rural counties, urban counties, males, females, Whites, Hispanics and Blacks. **Conclusions:** This study reveals a significant positive correlation between higher SDI and increased colon cancer mortality at the county level in Texas. The findings highlight the need for targeted public health interventions in deprived areas to reduce mortality and address disparities. Research Sponsor: None.

# Incidence trends of early-onset colorectal cancer (EOCRC) in the United States, by age, sex, race and ethnicity, and urbanicity (2000-2021).

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**Background:** Research is needed to understand the extent to which the incidence of early onset colorectal cancer (EOCRC)is rising across population sub-groups, including by age, sex, race/ethnicity, and urbanicity, to inform hypotheses regarding its causes and strategies for screening and prevention. **Methods:** We used data on colorectal cancer cases diagnosed among adults 20-49 years (yr) between 2000-2021 in the delay-adjusted Surveillance, Epidemiology, and End Results (SEER) database (April 2024). We calculated delay- and age-adjusted annual incidence rates (IR) per 100,000, overall and by age, sex, race/ethnicity, and urbanicity. Urbanicity was defined using USDA Rural-Urban Continuum Codes for county; 1-3 were classified as urban and 4-9, 98, and 99 were classified as non-urban. We used NCI's Joinpoint regression models (Joinpoint version 5.2.0.0) to estimate annual percent change (APC) and 95% confidence intervals (CI). We also calculated IRs for the most recent 5-year period (2017-2021) and used incidence rate ratios (IRR) to compare across groups; IRs and IRRs were calculated using SEER*Stat version 8.4.3. **Results:** Analyses included 155,500 cases of EOCRC. Most cases (72%) occurred among adults 40-49 yr; 22% occurred among those 30–39 yr and 6% in people 20-29 yr. Among adults 40-49 yr, the incidence of EOCRC increased 4.4% / yr on average between 2018-2021 (95%CI: 2.5, 5.6). Among adults 30-39 yr, incidence of EOCRC increased 3.0% / yr, on average, starting in 2010 (95%CI: 2.4, 4.5). For adults ages 20-29 yr, there were large increases in incidence from 2000-2016 (2000-2013, APC: 3.5; 95%CI: 2.4, 4.1; 2013-2016, APC: 11.2; 95%CI: 6.8, 13.5) but no change from 2016-2021. For the most recent 5-yr period (2017-2021), females had lower IR compared to males (IRR: 0.90; 95%CI: 0.88, 0.91), but trends over time were similar by sex. The incidence of EOCRC increased across races and ethnicities between 2000-2021, albeit at different rates. In the most recent period (2017-2021), the IR per 100,000 was 22.0 for Non-Hispanic American Indian and Alaskan Native (AIAN), 15.0 for Non-Hispanic White, 14.9 for Non-Hispanic Black, 12.4 for Hispanic or Latino, and 11.2 for Non-Hispanic Asian American or Pacific Islander adults. People in non-urban areas had a higher incidence of EOCRC compared to those in urban areas (IRR: 1.18; 95%CI: 1.14, 1.22), with similar trends over time. **Conclusions:** The incidence of EOCRC is highest among adults 40-49 yr, males, Non-Hispanic AIAN adults, and in non-urban areas. Trend analyses generally showed similar increases in incidence (3-5%/yr) in recent periods across sex, race/ethnicity, and urbanicity. Research Sponsor: National Cancer Institute; Centers for Disease Control and Prevention; California Department of Public Health.

27 Poster Session

# Comparative analysis of demographics and outcomes in hospitalized patients with young- versus average-onset colorectal cancer in New York state (2017-2022).

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**Background:** Recent studies have shown an increasing incidence of colorectal cancer (CRC) among young patients. This study investigates clinical outcomes and healthcare utilization among young (<50) versus average onset (≥50) CRC patients admitted to hospitals in New York State (NYS). **Methods:** We performed a retrospective analysis using the Statewide Planning and Research Cooperative System (SPARCS) database from 2017 to 2022. Patients were divided into two groups: young-onset colorectal cancer patients (YOCRC, <50 years) and average onset colorectal cancer patients (AOCRC, ≥50 years). The study population was further stratified by demographic and clinical characteristics. All associations were compared using the Kruskal-Wallis test, along with multivariate linear and logistic regression in RStudio version 4.4.1, at a significance level of ≤0.05. Clinical characteristics, including severity of illness and risk of mortality, were defined using the All Patient Refined (APR) grading system. **Results:** A total of 9,904 patients were identified (948 YOCRC and 8,956 AOCRC) with a primary admitting diagnosis of CRC from 2017 to 2022. In comparing the two age groups, AOCRC patients had a higher proportion of White, and female patients, whereas the YOCRC group included a higher proportion of Black and male patients ($p<0.001$). The median length of stay was greater for AOCRC patients (5 days) compared to YOCRC patients (4 days), $p<0.001$. Both age groups showed an upward trend in total cost of stay in the recent years. AOCRC patients were more likely to have increased severity of illness graded as major-extreme (60% vs 48%), while a higher proportion of YOCRC patients were classified as minor-moderate disease (51% vs 40%), $p<0.001$. Similarly, AOCRC patients experienced a higher risk of mortality compared to YOCRC patients ($p<0.001$). Regarding procedures, AOCRC patients needed more transfusions (13%) than YOCRC patients (9.3%), $p<0.001$. YOCRC patients were also more likely to be discharged home (71% vs 46%), while AOCRC patients often required home health services or skilled nursing ($p<0.001$). Notably, while the risk of mortality was higher for AOCRC patients, it has been decreasing overall from 2017 to 2022 [OR 0.92 (95% CI: 0.84-0.99), $p=0.039$]. **Conclusions:** AOCRC patients experienced greater severity of disease, higher mortality risk, and required more assistive care at discharge compared to YOCRC patients. The total cost of hospital stays has been rising for both groups. Policies focused on earlier outpatient interventions may help reduce the burden of inpatient care and rising costs. Research Sponsor: None.

**28** **Poster Session**

# Genetic differences in colorectal cancer across race and ethnicity.

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**Background:** Colorectal cancer (CRC) is the fourth most common cause of cancer death and cancer-related mortality is expected to increase exponentially. Ethnic minorities, in particular African Americans, have increased mortality from CRC. Although multifactorial, differences in somatic gene mutations could contribute to these racial discrepancies. Mutations in the critical oncogene KRAS are associated with worse prognosis in CRC. Recent literature suggests that in CRC patients, KRAS mutations are more frequently found in African Americans compared to Caucasians. This finding led to our hypothesis that molecular alterations across multiple races/ethnicities could explain differences in patient outcomes. This project will also inform a larger analysis using artificial intelligence learning models to predict race from tumor-specific genetic variants, as a tool to identify genetic drivers of race-specific patient outcome. **Methods:** We examined the molecular alterations in 304 patients diagnosed with CRC who had genetic testing between June 2021 and April 2023 at NYP Queens, NYP Brooklyn, and NYP Cornell. Race and ethnicity, genetic mutations, mortality, tumor location, and age, stage, and the presence of metastasis at diagnosis was collected. Patients self-identified as Non-Hispanic Black (NHB), White (NHW), Asian (NHA), American Indian (NHAI), or Hispanic, Other, or Declined. Median and interquartile range were used to summarize continuous variables, and frequency and proportion were used to summarize categorical variables. The significance of difference across ethnicity groups was tested using Kruskal-Wallis rank sum test for continuous variables and Fisher's exact test for categorical variables. Mutational correlates of race using a multiple-instance learning artificial intelligence approach (i.e. Anaya et al., Nature Biomedical Engineering, 2023), will be presented. This method will predict race from patients' tumor-specific genetic variants, allowing us to identify specific variants and patterns predictive of race, and potential drivers of patient outcome. **Results:** Amongst the 304 CRC patients, we identified 116 NHW, 51 NHA, 40 NHB, 35 Hispanic and 1 NHAI, for a total population of 243. BRAF mutations were almost absent in NHB (2.5%) and NHA (3.9%), compared to NHW (13.8%; p=0.02). KRAS mutations were also more prevalent in underserved populations; 65.7% Hispanic, 57.5% NHB, compared with 44.8% NHW (p=0.14). We further identified substantial differences in KRAS mutations when dividing the Asian population into East Asian (62.9%) and South Asian (37.5%). **Conclusions:** We identified significant differences in key molecular drivers between ethnicities. Our findings suggest that genomic and tumor specific differences across ethnicities could in part explain differences in patient survival across these groups. Mutational correlates of race and survival using natural language processing will be presented.
Research Sponsor: None.

29 Poster Session

# Adherence to the 2018 World Cancer Research Fund/American Institute for Cancer Research (WCRF/AICR) recommendations and overall mortality among adults with colorectal cancer in a multiethnic cohort study.

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**Background:** High adherence to AICR recommendations on diet, adiposity, and physical activity is associated with improved survival among adults with colorectal cancer (CRC). However, over 80% of participants in existing studies self-reported as non-Hispanic White (NHW). We studied the association between adherence to AICR recommendations and overall mortality among adults with CRC in a multiethnic cohort. **Methods:** This prospective analysis used data from the Multiethnic Cohort (MEC) study. The MEC is a racially/ ethnically diverse cohort of over 200,000 participants residing in Hawaii and Los Angeles. Lifestyle behaviors in the MEC were assessed by questionnaire at enrolment/Q1 (1993-1996) and periodically over follow-up. We included those with incident CRC before completing Q3 (2003-2008). Follow-up was from Q3 survey completion to December 2019. The main exposure was the standardized AICR score [0-7 points] at Q3. The primary outcome was overall mortality; CRC-specific mortality was secondary. We used multivariable Cox proportional hazards regression to estimate adjusted hazard ratios (HR) and 95% confidence intervals (CIs). **Results:** There were 1127 adults diagnosed with CRC prior to Q3. Of these, 1079 (95.7%) had non-missing data on the AICR score and comprised our study population. Median (IQR) age at diagnosis was 69 (62-75); median (IQR) follow-up was 12.2 (0.1-16.4) years. There were 489 (45.3%) women. By self-reported race/ethnicity, there were: 153 (14.2%) Black/African Americans, 453 (42.0%) Japanese Americans, 79 (7.3%) Latinos, 165 (15.3%) Native Hawaiians, and 229 (21.2%) NHWs. Only 48 (4.4%) adults had high adherence (AICR≥5 points). Those in the highest category of AICR scores (AICR≥4.5) had statistically significant lower risk of overall mortality relative to those in the lowest category (AICR≤2.5) (HR: 0.64, 95% CI, 0.47-0.89). A similar (but non-statistically significant) association was observed with CRC-specific mortality [HR (high vs low): 0.70, 95% CI, 0.31-1.60]. Those who increased their score between Q1&Q3 had lower overall mortality (HR: 0.87, 95% CI,0.79 - 0.96). Similar trends were observed in stratified analysis by race/ethnicity; however, statistical power was limited. **Conclusions:** High adherence to AICR recommendations was associated with lower risk of overall mortality in this multiethnic population of adults with CRC. Efforts to increase adherence to AICR recommendations in this population are needed. Research Sponsor: None.

# Demographic predictors of multi-target stool DNA test completion in an underserved primary care population: A retrospective analysis.

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**Background:** Colorectal cancer screening remains crucial for early detection, with multi-target stool DNA(Exact Sciences, Madison, WI, USA) serving as a non-invasive option. However, adherence to test completion may vary across different demographic groups. This study aims to assess the association between patient demographics and the completion of Cologuard testing in an underserved primary care population in order to further explore the disparities that exist. **Methods:** We conducted a retrospective analysis of 12071 eligible patients, aged 45-75, across Hartford Healthcare Primary Care Clinics across Connecticut, USA, between November 2022 and April 2024 for whom an order for this test was placed. Demographic data including gender, age, English proficiency, and insurance status(public/private vs no insurance) were analyzed. The outcome of interest was whether patients had results available within 120 days of placing the order. Logistic regression was used to examine the relationship between these variables and the likelihood of Cologuard test completion. **Results:** 11704(96.9%) of all patients were insured. Among insured patients, 6945 (59.9%) had results available, while 4759 (41.1%) did not have results at the 120 day mark. Among uninsured patients, 46 (12.5%) had results, while 321 (87.5%) did not. Insured patients were more likely to have test results available compared to uninsured patients (OR = 10.184, 95% CI: 7.460-13.901, $p < 0.01$). 10977(90.9%) patients were English proficient. They were more likely to have results compared to Non-English proficient patients (OR = 3.474, 95% CI: 3.038-3.972, $p < 0.01$). Gender was not a significant predictor of test completion (OR = 0.964, 95%CI: 0.895-1.039, $p = 0.341$), and older age demonstrated a small, non-significant effect (OR = 1.004, $p = 0.094$), with a test for non-linearity confirming a linear relationship between age and test completion ($p = 0.162$). **Conclusions:** Insured status and English proficiency are significant predictors of Cologuard test completion in this primary care cohort and are more likely complete testing. These findings underscore the need for targeted interventions to increase colorectal cancer screening adherence, particularly among this population. Research Sponsor: None.

# Comprehensive analysis of Native Hawaiians and other Pacific Islanders with early-onset colorectal cancer.

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**Background:** Rates of early-onset colorectal cancers (EOCRC) are increasing in Hawaii across all racial groups. Although previous studies have shown that Native Hawaiians have a higher mortality rate compared to other racial groups, the adjustment for sociodemographic factors have been limited. Our objective is to perform a comprehensive analysis of outcomes among patients with EOCRC in a racially diverse population accounting for tumor factors and patient sociodemographics. **Methods:** Patients diagnosed with EOCRC between 2000 and 2022 in Hawaii. Overall survival of Asians, Whites, and Native Hawaiian or Other Pacific Islanders (NHOPI) were calculated using the Kaplan-Meier method. Cox proportional hazard regression models assessed survival predictors adjusting for clinical and sociodemographic factors. **Results:** We included 379 patients in the final analysis. NHOPI had higher prevalence of underinsured status and higher histopathology grade compared to Whites and Asians. In unadjusted Cox regression, NHOPI race, underinsured status, grade, and stage were prognostic for survival. After adjusting for confounders, underinsured status, grade, and stage remained prognostic, but race was not significantly associated with survival. **Conclusions:** This study concludes that while NHOPI patients with EOCRC demonstrated poorer survival outcomes compared to other racial groups, this disparity was largely explained by the large percentage of underinsured NHOPI patients. Addressing treatment disparities in underinsured or uninsured is essential to improve survival outcomes. Research Sponsor: None.

**Univariate and multivariate analysis of survival.**

| | Univariate | | Multivariate | |
|---|---|---|---|---|
| | HR (95% CI) | p-value | HR (95% CI) | p-value |
| **Insurance** | | | | |
| **Private Insurance** | - | - | - | - |
| **Underinsured & Uninsured** | 1.865 (1.331-2.612) | <0.001 | 1.843 (1.285-2.643) | <0.001 |
| **Race** | | | | |
| **White** | - | - | - | - |
| **Asian** | 1.179 (0.746-1.865) | 0.48 | 0.936 (0.585-1.496) | 0.781 |
| **NHOPI** | 2.005 (1.231-3.265) | 0.005 | 1.051 (0.714-1.547) | 0.8 |
| **Age at diagnosis** | 0.996 (0.967-1.025) | 0.771 | 1.002 (0.971-1.034) | 0.901 |
| **Histopathology grade** | | | | |
| **Grade 1&2** | - | - | - | - |
| **Grade 3&4** | 2.785 (1.958-3.963) | <0.001 | 2.278 (1.571-3.303) | <0.001 |
| **Stage** | | | | |
| **Stage 1** | - | - | - | - |
| **Stage 2** | 1.104 (0.49-2.487) | 0.81 | 1.028 (0.451-2.342) | 0.948 |
| **Stage 3** | 3.255 (1.666-6.359) | <0.001 | 2.873 (1.447-5.706) | 0.003 |
| **Stage 4** | 14.851 (7.499-29.41) | <0.001 | 12.778 (6.283-25.987) | <0.001 |
| **Unknown** | 3.738 (1.506-9.275) | 0.004 | 3.008 (1.189-7.608) | 0.02 |
| **Sex** | | | | |
| **Male** | - | - | - | - |
| **Female** | 0.812 (0.587-1.123) | 0.22 | 0.852 (0.609-1.194) | 0.352 |
| **Microsatellite status** | | | | |
| **Stable** | - | - | | |
| **Unstable** | 0.481 (0.148-1.562) | 0.204 | | |
| **Not done/Unknown** | 1.08 (0.737-1.582) | 0.739 | | |
| **Time to Treatment** | | | | |
| **Less than 1 month** | - | - | - | - |
| **More than 1 month** | 1.221 (0.863-1.727) | 0.26 | 1.123 (0.784-1.609) | 0.527 |
| **Location of Tumors** | | | | |
| **Right** | - | - | - | - |
| **Left** | 1.239 (0.76-2.02) | 0.391 | 1.466 (0.886-2.425) | 0.137 |
| **Rectal** | 1.391 (0.86-2.249) | 0.178 | 1.211 (0.738-1.986) | 0.449 |

NHOPI: Native Hawaiians and Other Pacific Islanders; HR: Hazard Ratio; CI: Confidence Interval.

# Comprehensive analysis of socioeconomic and clinical factors contributing to treatment delay in early-onset colorectal cancer.

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**Background:** The prevalence of early-onset colorectal cancers (EOCRC) is rising globally. Disparities in social determinants of health can delay access to treatment and impacting outcomes. This study aims to analyze factors influencing time from diagnosis to treatment among EOCRC patients in a racially diverse population, considering tumor factors and socio-demographics. **Methods:** We analyzed EOCRC data from patients diagnosed between 2000 and 2022 at Queen's Medical Center in Honolulu, Hawaii. Time from diagnosis to treatment for Asians, Whites, and Native Hawaiian or Other Pacific Islanders (NHOPI) was assessed using the Kaplan-Meier Method. Cox proportional hazards regression models were used to identify predictors, adjusting for clinical and pathological factors. **Results:** A total of 379 patients were analyzed. NHOPI were more like to be underinsured compared to Whites and Asians. In the unadjusted Cox model, underinsured/uninsured status, older age, unknown stage, left sided, and rectal locations were associated with delayed treatment. After adjustment, underinsured/uninsured status, older age, and rectal location remained significant predictors of delayed treatment. **Conclusions:** EOCRC patients with underinsured/uninsured status, age, and rectal tumor location are significantly associated with delayed in cancer treatment. While age was statistically significant, its clinical impact appears minimal. The need for a multidisplinary approach and additional preoperative staging in rectal cancer may explain treatment delays even after adjustment. Improving treatment accessibility and reducing delays, especially for underinsured EOCRC patients, is important. Research Sponsor: None.

**Univariate and multivariate analysis of time from diagnosis to treatment.**

| | Univariate | | Multivariate | |
|---|---|---|---|---|
| | HR (95% CI) | p-value | HR (95% CI) | p-value |
| **Insurance** | - | - | - | - |
| **Private insurance** | - | - | - | - |
| **Underinsured/Uninsured** | 0.692 (0.545-0.878) | 0.003 | 0.706 (0.55-0.906) | 0.006 |
| **Race** | | | | |
| **White** | - | - | - | - |
| **Asian** | 0.912 (0.696-1.195) | 0.503 | 0.817 (0.619-1.077) | 0.151 |
| **NHOPI** | 0.749 (0.548-1.025) | 0.071 | 0.779 (0.562-1.08) | 0.134 |
| **Age** | 0.978 (0.96-0.997) | 0.021 | 0.968 (0.949-0.987) | 0.001 |
| **Histopathology grade** | | | | |
| **Grade 1&2** | - | - | - | - |
| **Grade 3&4** | 1.046 (0.798-1.371) | 0.746 | 1.083 (0.813-1.441) | 0.587 |
| **Stage** | | | | |
| **1** | - | - | - | - |
| **2** | 1.022 (0.718-1.456) | 0.902 | 1.071 (0.744-1.539) | 0.713 |
| **3** | 1.227 (0.896-1.682) | 0.203 | 1.387 (0.997-1.931) | 0.052 |
| **4** | 0.964 (0.673-1.381) | 0.842 | 1.039 (0.716-1.507) | 0.842 |
| **Unknown** | 0.631 (0.406-0.981) | 0.041 | 0.748 (0.473-1.182) | 0.213 |
| **Sex** | | | | |
| **Male** | - | - | - | - |
| **Female** | 1.029 (0.836-1.267) | 0.785 | 1.072 (0.868-1.325) | 0.518 |
| **Microsatellite status** | | | | |
| **Stable** | - | - | | |
| **Unstable** | 1.074 (0.616-1.875) | 0.8 | | |
| **Not done/Unknown** | 1.259 (1.004-1.577) | 0.046 | | |
| **Location of Tumors** | | | | |
| **Right** | - | - | - | - |
| **Left** | 0.631 (0.468-0.851) | 0.003 | 0.735 (0.535-1.011) | 0.058 |
| **Rectal** | 0.43 (0.317-0.582) | <0.001 | 0.448 (0.327-0.613) | <0.001 |

NHOPI: Native Hawaiians and Other Pacific Islanders. HR: Hazard Ratio CI: Confidence Interval.

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# Disparities in incidence and mortality of colorectal cancer in Appalachian Kentucky.

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**Background:** Since the late 1990s, both the incidence and mortality of invasive colorectal cancer (CRC) have decreased across the United States, including in Kentucky. Despite this, CRC incidence and mortality are above national average in Kentucky. Over the last two decades, a disparity in CRC incidence and mortality has emerged between Appalachian Kentucky (AK), a distinct eastern region, and Non-Appalachian Kentucky (NAK). This disparity is attributed to differences in social and structural determinants of health (SSDH) between these populations. **Methods:** Age-adjusted CRC incidence and mortality rates (2016-2020) for both Kentucky and the U.S. were obtained from the National Cancer Institute's public database. Data from the Kentucky Cancer Registry were used to compare CRC incidence and mortality rates in AK and NAK counties from 1994-2021. Incidence rate ratio analyses across discrete time periods demonstrated statistical differences between these regions. **Results:** Compared to national data (2016-2020), Kentucky's CRC age-adjusted incidence and mortality rates were significantly higher: 45.9 vs. 36.5/100,000, 25.8%, for incidence, and 16.2 vs. 13.1/100,000, 23.7%, for mortality (both $p<0.0001$). From 1995-2003 incidence and 1994-2003 mortality of CRC, were higher in NAK, these rates eventually decreased and equalized between NAK and AK. By 2010-2015 both incidence and mortality rates were higher in AK, 55.8 vs 48/100,000, 16.3%, ($p=<0.0001$) and 20.1 vs 15.8/100,000, 27.2%, ($p=<0.0001$) respectively. The most recent data (2016-2021) shows that these significant disparities persist, with AK's incidence and mortality 51.8 vs 44.1/100,000, 17.5% ($p=<0.0001$) and mortality 19.0 vs 15.1/100,000, 25.8% ($p=<0.0001$) higher, respectively. **Conclusions:** While CRC rates have decreased nationwide and in Kentucky, a pronounced gap in both incidence and mortality has developed between Appalachian and Non-Appalachian Kentucky. Appalachian counties—which are more rural and impoverished—face greater barriers to accessing preventive healthcare, contributing to this disparity. Additionally, AK has higher rates of obesity, poverty, smoking, and lower educational attainment and screening rates . Recent public health efforts, such as promoting stool-based screening and increasing preventive care discussions, aim to reduce these disparities. However, delays in CRC screenings during the COVID-19 pandemic— resulting in approximately 1.7 million missed colonoscopies nationally— may have skewed the most recent data, necessitating further analysis in the coming years. Research Sponsor: None.

# Differences in disease characteristics in young vs older adults with colorectal carcinoma.

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**Background:** Colorectal carcinoma (CRC) exhibits distinct disease characteristics between younger (≤50 years) and older (>50 years) adults, which may significantly influence treatment decisions and outcomes. This study aims to investigate differences in disease characteristics of colorectal carcinoma among patients aged ≤50 years compared to those aged >50 years, utilizing data from the Surveillance, Epidemiology, and End Results (SEER) database. **Methods:** A retrospective analysis was performed using the SEER database covering the years 2000 to 2020. A total of 150,000 diagnosed CRC cases were included. Key variables analyzed comprised stage at diagnosis, tumor grade, histological subtype, and 5-year survival rates. Rates of CRC diagnoses in various age brackets were calculated for three time periods: 2000-2010, 2010-2020, and overall from 2000-2020. **Results:** Within the total cohort, 12% (18,000 cases) were diagnosed in young adults, while 88% (132,000 cases) were in older adults. An analysis of age-specific rates showed: 2000-2010: Young adults (≤50 years) represented 10% of CRC cases, with 16% presenting at advanced stages (Stage III or IV). 2010-2020: This group increased to 12%, with 65% diagnosed at advanced stages (Stage III or IV), compared to 52% in older adults ($p < 0.001$). In terms of tumor characteristics, poorly differentiated tumors were more prevalent in young adults (28% vs. 18%, $p < 0.001$). The prevalence of microsatellite instability-high (MSI-H) tumors was significantly greater in young adults at 18%, compared to 12% in older adults ($p < 0.05$). The 5-year overall survival rate for young adults was 68%, notably higher than the 48% for older adults ($p < 0.001$). **Conclusions:** This study underscores significant differences in disease characteristics between young and older adults with colorectal carcinoma. Young patients are more likely to present with advanced disease but have better survival outcomes. These findings indicate the necessity for an age specific approach to screening and treatment strategies. Research Sponsor: None.

**Differences in stage at diagnosis, tumor grade, presence of MSI-H, and survival outcomes between younger and older adults diagnosed with colorectal carcinoma.**

| Category | Young Adults (≤50 years) | Older Adults (>50 years) | p-value |
|---|---|---|---|
| **Proportion of Total CRC Cases** | 12% (18,000 cases) | 88% (132,000 cases) | |
| **Advanced Stage (Stage III or IV)** | 65% (2010-2020) | 52% (2010-2020) | < 0.001 |
| **Poorly Differentiated Tumors** | 28% | 18% | < 0.001 |
| **Microsatellite Instability-High (MSI-H)** | 18% | 12% | < 0.05 |
| **5-Year Overall Survival Rate** | 68% | 48% | < 0.001 |

35 Poster Session

# Disparities and barriers in germline testing among patients with early-onset gastrointestinal cancers.

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**Background:** Early-onset gastrointestinal cancers (EOGIC) are strongly associated with hereditary predisposition, making germline testing essential for treatment and familial risk management. Disparities in germline testing persist. This study investigates disparities and barriers to germline testing in EOGIC. **Methods:** This IRB-approved retrospective analysis included 864 patients aged 18-49 with GI cancers at the University of Chicago (2018-2023). Data collected included race, gender, age, insurance, family cancer history, and subtype. Outcomes were physician referrals for germline testing and completion. Chi-square tests and logistic regression models assessed predictors of recommendations and completion. **Results:** Of 864 patients, 473 (54.7%) received germline testing recommendations, and 323 (37.4%) completed testing. Completion rates: 30.2% for no insurance, 29.1% for Medicare/Medicaid, and 40.3% for private insurance (p=0.032). Patients with family history of cancer were more likely to be recommended for testing (62.9% vs. 39.5%, $p<0.001$) and complete it (44.0% vs. 23.7%, $p<0.001$). In our cohort, no significant racial disparities were observed (p=0.712 and p=0.358). Females were more likely to receive recommendations (59.1%) and complete testing (41.2%) than males (50.9% referral, 34.0% completion, p=0.016 and p=0.029). Patients under 30 had the highest rates of testing recommendation (66.1%) and completion (46.4%), followed by 30-39 (61.7%, 44.4%) and 40-49 (51.2%, 34.0%), p=0.006 and p=0.009. By cancer subtype, colorectal cancer (CRC) patients had the highest referral (66.1%) and completion (45.9%) rates, while anal cancer had the lowest (13.2% referral, 5.3% completion, OR 0.08, $p<0.0001$). Gastric cancer (52.3% referral, 29.2% completion, OR 0.49, p=0.005) and esophageal cancer (36.9% referral, 20.0% completion, OR 0.31, $p<0.0001$) also had lower testing rates. Liver cancer (35.4% referral, 27.1% completion, OR 0.38, $p<0.0001$) and cholangiocarcinoma (45.5% referral, 22.7% completion, OR 0.28, p=0.016) showed similarly low rates. **Conclusions:** Disparities in germline testing were seen among uninsured patients, males, and those with public insurance. CRC had the highest testing rates, while anal, esophageal, liver, and cholangiocarcinoma had lower rates, likely due to lower suspicion of hereditary predisposition. Targeted interventions are needed to improve testing uptake and completion, especially among uninsured and underrepresented populations. Research Sponsor: Bucksbaum Pilot Grant.

**36** **Poster Session**

# Temporal trends in rural vs urban colon cancer mortality in Texas: A SEER database analysis from 2012 to 2021.

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**Background:** Colorectal cancer is a major concern for the U.S. healthcare system, being the fourth most common cancer by incidence in the country with 152,810 cases in 2024. Measuring the burden this disease places on patients and health resources may serve to optimize our approach to the detection, management and resource allocation for combating this ailment. **Methods:** Our study is a population-based study cohort of colon cancer deaths in Texas. We queried the National Cancer Institute Surveillance, Epidemiology and End Results Program (SEER) for colon cancer deaths in Texas over the most recent available decade, 2012 through 2021. The National Center for Health Statistics Urban-Rural Classification Scheme was used to classify counties according to the 2013 US Census classification: urban (≥ 50,000) and rural (< 50,000). We calculated age-adjusted mortality rates (AAMRs) per 100,000 population using the direct standardization method based on the age group weights from the 2000 standard US population. Confidence intervals for AAMR were derived by estimating the standard error as the AAMR divided by square root of number of mortalities. Differences between AAMR were tested by comparing the 95% confidence intervals (95% CI) for the individual rates. The annual percent change (APC) in AAMR was tested using negative binomial regression. Subgroup analyses included age group, sex, and race/ethnicity. **Results:** Over the study period, 10.8% of the study population lived in rural areas and 89.2% lived in urban areas. A total of 33,591 colon-cancer deaths were reported during the study period with 5,423 (16.1%) in rural counties and 28,168 (83.9%) in urban counties. At the beginning of the study period the rural AAMR (13.9 [12.7 to 15.1]) was significantly higher than urban AAMR 12.1 [95% CI 11.7 - 12.6]). The rural AAMR rose significantly over the study period (APC 1.1% [95% CI 0.3% - 1.9%]; p = 0.0050) to AAMR 14.9 (95% CI 13.7 to 16.1). In contrast, the urban AAMR did not change over study period (APC -0.3% [95% CI -0.7 to 0.1]; p = 0.1444) and was equal to the initial AAMR 12.1 (95% CI 11.7 to 12.5). On subgroup analyses, males, people ages 25-64 years, Hispanics, Non-Hispanic Blacks, Non-Hispanic Asians, and Non-Hispanic Native Americans in rural areas each had significant and increasing trends in AAMR. **Conclusions:** Our analysis revealed an increase in AAMR in rural counties of Texas, reflecting a concerning trend with regard to colorectal cancer in this region. This may be due to higher poverty rates, lower level of education, more distance from healthcare facilities and difficulties with transportation faced by rural counties. Furthermore, rural areas also report higher rates of obesity and poor dietary habits. Rural areas also face a shortage of Primary Care Physicians (PCP's), with metropolitan counties retaining 1.7 times as many PCP's as rural counties, and 82.5% of counties without a PCP being rural. Research Sponsor: None.

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# Disparities in organ preservation rate for high-risk stage I rectal cancer.

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**Background:** Guidelines for the treatment of high-risk stage I rectal cancer is a curative oncologic resection, while recommendations for stage II/III rectal cancer include neoadjuvant therapy with the option for organ preservation. The paradoxical nature of treating less advanced stage cancer with more morbid surgery calls into question the use of organ preservation strategies (OP) for less advanced disease. While sphincter preservation is appealing, benefits from OP require commitment to preoperative therapy, which may require more significant time and financial commitments than a single-stage curative treatment. This study aims to evaluate disparities in social determinants of health regarding receipt of OP compared to major oncologic resection for high-risk stage I rectal cancer. **Methods:** The National Cancer Database was queried to analyze all patients with high-risk stage 1 rectal cancer between 2004-2021. High-risk stage 1 disease was defined as either T2 or T1 tumors with lymphovascular invasion, or grade 3 or 4 differentiation. Covariates in patients who underwent OP, defined as chemoradiation therapy without oncologic resection or local excision, were compared to those who underwent major oncologic resection. Additional variables included age, sex, race, great circle distance, insurance status, education, facility type, income, rurality, and comorbidity index. Univariate and multivariate analyses were utilized to compare association of covariates to OP. **Results:** Of the 63,762 patients included, 9,261 (14.5%) underwent OP. Increased rates of OP were seen with increasing age (OR 1.029, $p<0.0001$), female sex (OR 1.147, $p<0.0001$), higher education level (OR 1.179, $p=0.0034$), rurality (OR 1.273, $p=0.0003$) and lower income (OR 0.757 for income >74K, $p<0.0001$). Interestingly, OP was more frequently utilized at community cancer centers in comparison to comprehensive cancer centers (OR 0.78, $p<0.0001$) and academic centers (OR 0.842, $p=0.0021$). Race, insurance status, distance to treatment, and comorbidity index were not statistically significant predictors of OP. Of note, those who underwent OP had a considerable delay in time to first treatment, with median time to treatment of 36 days in comparison to 13 days for oncologic resection. **Conclusions:** Socioeconomic and demographic variables are associated with receipt of OP. While plausible that a treatment strategy that requires frequent office visits, travel expenses, and time off from work may preclude OP, lower income and rurality had increased odds of OP. Future aims should be directed to identifying root causes of disparities in treatment and their clinical consequences.
Research Sponsor: None.

# Colorectal cancer care disparities in an urban diverse population: A tale of two hospitals in the nation's capital.

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**Background:** Colorectal cancer (CRC) disparities by race and ethnicity are well-documented nationally. In the District of Columbia (DC), Black Americans have an 18% higher incidence and 46% higher mortality rate compared to overall rates among all races/ethnicities. These disparities often stem from long-standing inequities in the social determinants of health, which affect access to cancer prevention, early detection, and treatment. MedStar Georgetown serves a large, urban, diverse population at 2 hospitals: the MedStar Georgetown University Hospital (MGUH) and the MedStar Washington Hospital Center (MWHC). MWHC serves a higher proportion of minoritized patients (pts) with multiple social determinant barriers to care, including those without private health insurance. We hypothesized that disparities would be found in CRC pt care and outcomes when comparing MWHC with MGUH. **Methods:** We conducted a retrospective study of all CRC pts treated during the 2021-2023 period at MGUH and MWHC using the COTA real-world database. Pt demographics, insurance status, and selected clinical data, including but not limited to treatment site, disease stage, tumor molecular testing, and pt survival, were abstracted for analysis. Descriptive, bivariate, and multivariate analyses focused on gender, race/ethnicity, insurance status, receipt of tumor testing, and the interval between diagnosis and therapy initiation and explored survival by these factors in addition to the site of care (MGUH versus MWHC). **Results:** Of the 379 CRC pts receiving care at the Lombardi Comprehensive Cancer Center during the study period, 53% were treated at MGUH and 47% at MWHC. CRC pts at MWHC were more likely than CRC pts at MGUH to be Black Americans (65% vs. 28%, $p<0.001$) and have 3 or more comorbidities (16% vs. 8%, $p=0.22$) but were less likely to have early onset CRC ($p<0.01$) and be on private insurance (36% vs. 23%, $p=0.01$). Pts treated at advanced stages (stages 3-4) were similar between the two hospitals (64%). In spite of these differences, we did not find any differences between MWHC and MGUH in molecular tumor testing (83% vs. 86%, $p=0.41$), median time (days) to treatment initiation (28 vs. 26 days, $p=0.18$), or short-term survival (90% vs. 91%, $p=0.95$). In analyses comparing care/treatment-related factors with race/ethnicity and insurance status across the 2 hospitals, we did not find any statistically significant differences. **Conclusions:** Contrary to our expectations of finding significant cancer care disparities, we did not observe any differences in molecular tumor testing rates, time to initiation of therapy, or short-term survival by treatment site or race/ethnicity status. Our results suggest that integrated, specialized, guideline-concordant care in comprehensive cancer centers has the potential to eliminate CRC outcome disparities among pts diagnosed at these centers. Research Sponsor: None.

# Survival associated with metastatic early-onset colorectal cancer among Black patients: A real-world account.

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**Background:** Early-onset colorectal cancer (EOCRC), defined as colorectal cancer (CRC) diagnosed under the age of 50, is a growing health concern. The current study explored the racial differences in EOCRC survival in the United States (US). **Methods:** This retrospective cohort study was conducted using the TriNetX database, a platform that provides access to deidentified medical records on more than 130 million patients across 94 healthcare organizations in the US. The study cohort included patients with metastatic CRC (mCRC), aged 20 to 50, diagnosed between January 1, 2002, and January 1, 2020, and received frontline chemotherapy with oxaliplatin and 5- fluorouracil or capecitabine. The cohort was divided into Black or African American cohort and White cohort. The primary endpoint, 5-year overall survival (OS), was compared between these cohorts after performing 1:1 propensity score matching (PSM) for baseline demographics and comorbid conditions (Table). Kaplan-Meier curves, log-rank chi-square statistics, and hazard ratios (HR) were used to compare the outcome. **Results:** The entire cohort consisted of 2,923 patients with a mean age of 44 (±6) years; 48.58% were female. The majority were white (56.38%), followed by 15.56% of unknown race, Black/African American (9.16%), Hispanic (9.34%), and Asian (8.96%). After propensity score matching (PSM), there were 262 patients in both the Black and White groups. In the matched population, the Kaplan-Meier survival curves showed that Black patients had significantly lower 5-year OS rates (24.84%) compared to White patients (33.76%) [Hazard Ratio (HR): 1.39, 95% CI (1.06, 1.85), $p<0.05$]. There is no significant difference between Black and White population over age of 50 [ HR: 1.06, 95% CI (0.966-1.179), P=0.2]. **Conclusions:** Survival among Black patients with EOCRC appears to be worse than White patients despite adjustment for confounding variables, supporting further research focusing onto this patient population. Research Sponsor: None.

**Characteristics of the cohorts before and after matching.**

| | Black cohort (n = 265) and White cohort (n = 1,632) characteristics before propensity score matching | | | Black cohort (n = 262) and White cohort (n = 262) characteristics after propensity score matching | | |
|---|---|---|---|---|---|---|
| | Black patients, n (%) | White patients, n (%) | Sth diff. | Black Patients, n (%) | White patients, n (%) | Sth diff. |
| **Age (Mean ± SD)** | 36.2 ± 6.18 | 36.3 ± 6.01 | 0.022 | 36.1 ± 6.17 | 36.2 ± 6.43 | 0.013 |
| **Female** | 144 (54.5%) | 774 (47.4%) | 0.142 | 143 (54.5%) | 143 (54.5%) | <0.0001 |
| **Tobacco use** | 15 (5.6%) | 96 (5.8%) | 0.0087 | 15 (5.7%) | 11 (4.2%) | 0.0703 |
| **Hypertensive diseases** | 67 (25.3%) | 214 (13.12%) | 0.3147 | 65 (24.8%) | 64 (24.4%) | 0.0089 |
| **Ischemic heart diseases** | 10 (3.7%) | 40 (2.4%) | 0.0769 | 10 (3.8%) | 10 (3.8%) | < 0.0001 |
| **Type 2 diabetes mellitus** | 24 (9%) | 89 (5.4%) | 0.1403 | 22 (8.3%) | 19 (7.2%) | 0.0426 |
| **Diseases of arteries, arterioles and capillaries** | 12 (4.5%) | 47 (2.8%) | 0.0881 | 11 (4.1%) | 10 (3.8%) | 0.0195 |

# Impact of insurance status on mortality following surgical treatment of colorectal cancers.

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**Background:** Healthcare disparities in colorectal cancer, driven by insurance status and socioeconomic factors, lead to delayed diagnoses and poorer surgical outcomes. We sought to examine the impact of insurance status on presentation, treatment, and in-hospital mortality in CRC patients undergoing colectomy using the Nationwide Inpatient Sample (NIS) database. **Methods:** We included patients aged 18-65 years diagnosed with colon cancer and undergoing colectomy, as identified by ICD-9-CM codes, who had private insurance, Medicaid, or no insurance from January 1, 2005, through December 31, 2014. The primary variable was insurance status; patient characteristics such as age, sex, race, income, and tumor type were analyzed. We used the Elixhauser comorbidity index to control for confounding and performed a subset analysis on patients without major comorbidities. The primary outcome was in-hospital postoperative death. Associations between this outcome and insurance status were analyzed using the Cox proportional hazard model, both in the full cohort and in a subset of patients without comorbidities, with models stratified by hospitals to account for clustering effects from variations in access to care. **Results:** The study cohort included 301,304 patients, of whom 238,158 (79.0%) were privately insured, 40,417 (13.4%) on Medicaid, and 22,729 (7.6%) were uninsured. Most patients were White (71.6%), followed by African American (12.6%), Hispanic (8.4%), Asian/Pacific Islander (3.8%), and Native American (0.5%). A total of 55.4% of cases took place in teaching hospitals. In the unadjusted analysis, the mortality rate for privately insured patients was 0.7% (95% CI, 0.6%-0.7%) compared with 2.1% for uninsured patients (95% CI, 1.7%-2.5%) and 1.5% for Medicaid recipients (95% CI, 1.2%-1.8%; p=0.001). After adjusting for patient characteristics and stratifying by hospital in patients with no comorbidity, uninsured patients still had a higher risk of experiencing in-hospital death (HR, 1.60; 95% CI, 1.24-2.07) compared with privately insured patients, while no significant disparity was found in Medicaid recipients (HR, 0.95; 95% CI, 0.75-1.22) (Table). **Conclusions:** Uninsured patients undergoing colectomy for colon cancer experienced the highest in-hospital mortality, a disparity not fully explained by overall health differences. These findings underscore the critical role of insurance coverage in improving surgical outcomes and highlight the need for policy interventions to reduce mortality disparities. Research Sponsor: None.

**In-hospital mortality after surgery for colorectal cancer.**

| | Full sample | Patients With No Comorbidity | Patients With No Comorbidity in Urban Teaching Hospitals |
|---|---|---|---|
| **Covariate** | | **Hazard Ratio (95% CI)** | |
| **Insurance** | | | |
| **Private** | Reference | Reference | Reference |
| **Medicaid** | 0.95 (0.75-1.22) | 1.38 (0.91-2.11) | 1.66 (0.91-3.02) |
| **Uninsured** | 1.60 (1.24-2.07) | 2.12 (1.42-3.15) | 2.28 (1.25-4.16) |

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# Regional variation in trajectories of colon cancer mortality, 2012-2021.

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**Background:** Colorectal cancer (CRC) is the fourth most common cancer by incidence in the United States in 2024, and the second most common cause of cancer mortality in Texas. Rates of advanced CRC at diagnosis and cancer related mortality are higher in rural areas. Here we explore the variation in mortality trends across the eleven public health regions (PHR) of Texas from 2012 to 2021. **Methods:** We conducted a population-based cohort study of colon cancer mortality in Texas over the period 2012 through 2021. We queried the National Cancer Institute Surveillance, Epidemiology and End Results Program (SEER) for county level colon cancer deaths in Texas over the most recent available decade. Counties were aggregated into the eleven public health regions defined by the Texas Department of State Health Services. We calculated age-adjusted mortality rates (AAMRs) per 100,000 population using the direct standardization method based on the age group weights from the 2000 standard US population. Confidence intervals for AAMR were derived by estimating the standard error as the AAMR divided by square root of number of mortalities. The annual percent changes (APC) in AAMR were tested using negative binomial regression. Subgroup analyses included age group, sex, and race/ethnicity. **Results:** A total of 33,591 colon cancer mortalities were included in the study. Total colon cancer mortalities rose from 2,973 in 2012 to 3,800 in 2021. The statewide AAMR was initially 12.4 (95% CI 12.0 to 12.9), and did not change over the study period (APC -0.1% [95% CI -0.5% to 0.2%]; p = 0.5155) ending at (AAMR 12.5 [95% CI 12.1 to 12.9]). Divergent trajectories for AAMR across public health regions were observed. The largest decline in AAMR (APC -3.9% [95% CI -6.3 to -1.6]; p = 0.0013) occurred in region 10, where the AAMR fell from 14.2% [95% CI 8.6 to 19.7]) to AAMR 11.0 [95% CI 6.8 to 15.1]. Region 6 also saw a significant decrease (APC -1.1% [95% CI -1.8% to -0.4%]; p = 0.0029) falling from AAMR 12.2 [95% CI 11.3 to 13.2] to AAMR 11.2 [95% CI 10.4 to 11.9]. The largest significant increasing trend (APC +2.3% [95% CI 0.6% to 4.1%]; p = 0.0089) for AAMR was observed for region 9 rising from AAMR 12.4 (95% CI 9.6 to 15.3) to AAMR 15.4 (95% CI 12.3 to 18.4). **Conclusions:** Our analysis showed a significant and increasing trend in CRC related mortality in PHR 9. This region, colloquially termed West Texas, consists of 30 counties of which over 86% are designated as rural. Rural areas struggle with higher poverty, lower degree of higher education attainment, less access to healthcare and physicians, and a higher frequency of uninsured status. West Texas in particular also has a higher rate of smoking and obesity. However, PHR 10, which shares many similarities with PHR 9 such as rurality (83.3%) and similar socioeconomic and geographic challenges, witnessed the greatest decline in AAMR of the eleven PHR's in the same time frame, suggesting the presence of other factors which may be contributing to this increase. Research Sponsor: None.

# Trends in early onset colorectal cancer–related mortality among adults in the United States from 1999-2020.

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**Background:** Colorectal cancer (CRC) has witnessed a decrease in incidence and mortality in the United States (US) for the past few decades likely due to increased screening and improvements in treatment . Despite the overall burden of CRC has declined and survival has been made for older adults, there is a simultaneous and alarming increase in CRCs diagnosed in individuals younger than 50 years of age. These Early Onset CRC (EOCRC) which now account for roughly 10% of new CRC cases in the US. The purpose of this study was to assess the trends and regional differences in EOCRC-related mortality among adults in the United States. **Methods:** Death certificates from the CDC WONDER (Centers for Disease Control and Prevention Wide-Ranging OnLine Data for Epidemiologic Research) database were examined from 1999 to 2020 for -related mortality in adults <45 years of age. Age-adjusted mortality rates (AAMRs) per 10,000 persons and annual percent change (APC) were calculated and stratified by year, sex, race/ethnicity, and geographic region. **Results:** Between 1999 and 2020, 32,850 EOCRC-related deaths occurred among adults 25-44 years. The AAMR increased from 2.1 in 1999 to 2.6 in 2020. Men had consistently higher AAMR than women from 1999 (AAMR men: 2.0 vs women: 1.6) to 2020 (AAMR men: 2.6 vs women: 2.0). Non Hispanic (NH) African Americans had the highest overall AAMR (3.1), followed NH White (2.2), NH Asians (1.6) and Hispanics (AAMR 1.5) Caucasians have had a significant increase in mortality over the past 2 decades ( APC 1.4 (95%Cl 1.1-.7)) whereas the Hispanics have shown an APC of 4 (95% Cl 2.3-5.8 from 2012-2020). AAMR also varied substantially by region (overall AAMR: Midwest 2.1; South: 2.4; West: 1.9; Northeast: 2.0). States in the top 90th percentile of EOCRC-related AAMR were Mississippi, Arkansas, Oklahoma, South Carolina, Alabama West Virginia, Louisiana ,Kentucky and Tennessee which had approximately double the AAMRs compared with states that fell into the lower 10th percentile. **Conclusions:** EOCRC-related mortality in U.S. adults has been steadily increasing over the past two decades . The highest AAMRs were observed among African American adults and men, and among patients living in the Southern , Midwestern regions and those living in the nonmetropolitan United States. Targeted strategies are needed to prevent and treat EOCRC among older adults to curb increasing levels of EOCRC-related mortality. Research Sponsor: None.

# Association between visceral adipose tissue radiodensity and overall survival among older adults with colorectal carcinoma.

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**Background:** Central obesity is a known risk factor for colorectal carcinoma (CRC); however, the association between visceral adipose tissue (VAT) and survival outcomes remains unclear. This study aims to investigate the relationship between VAT and overall survival (OS) among older adults with CRC. **Methods:** We analyzed data from the Cancer and Aging Resilience Evaluation registry, including adults aged ≥ 60 years with newly diagnosed CRC who were seen at the University of Alabama at Birmingham for initial consultation between October 2017 and April 2024. Using single-slice computed tomography staging scans, we calculated visceral adipose tissue radiodensity (VATD; in Hounsfield units, HU) at the third lumbar vertebra (L3) using validated methods. A lower VATD (more negative score) indicates higher visceral adipose tissue. Participants were followed until death or the study cutoff date of July 10, 2024. Kaplan-Meier survival analysis was performed to estimate OS for those with low VAT (VATD ≥ median VATD) versus high VAT (VATD < median VATD). Additionally, Cox proportional hazards models were utilized to evaluate the association between VATD and OS, adjusting for age at diagnosis, sex, race/ethnicity, skeletal muscle index (SMI), skeletal muscle density (SMD), and cancer stage. **Results:** A total of 190 patients were included, with a median age at diagnosis of 68 years (IQR: 64-75), 56% male, 72% non-Hispanic White, and 40% diagnosed with Stage IV disease. The median SMI and SMD were 42.06 (IQR: 33.81 - 49.52) $cm^2/m^2$ and 38.68 HU (IQR: 31.38 - 44.89) respectively. The median VATD was -86.14 HU (IQR: -93.78 – -76.39). Over a median follow-up of 60 months, 96 patients (51%) died. Median OS was 27 months for patients with low VAT and 62 months for those with high VAT (log-rank test, $p < .001$). After adjusting for confounders, VATD remained independently associated with OS (adjusted Hazard Ratio: 1.41 per IQR decrease; 95% CI: 1.06 – 1.86; $p \leq .05$), indicating that each IQR decrease in VATD scores (implying an increase in VAT) is associated with a 41% increased risk of all-cause mortality. **Conclusions:** Our findings demonstrate that increased visceral adipose tissue, indicated by more negative VATD scores, is significantly associated with poorer survival outcomes among older adults with CRC. These results underscore the importance of dietary and weight loss interventions to improve outcomes in this population. Research Sponsor: None.

# Multi-site blinded validation of a deep learning approach for clinical-grade MSI/dMMR detection in colorectal cancer from H&E-stained pathology images.

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**Background:** Testing for microsatellite instability (MSI) or mismatch repair deficiency (dMMR) is part of the diagnosis and clinical management of patients with colorectal cancer (CRC). Healthcare services recommend MSI or dMMR testing for all CRC patients to guide therapeutic choices and assist in identifying Lynch Syndrome. However, in clinical practice, high costs and the demand for timely test results, combined with the rising prevalence of CRC and a shrinking pathology workforce, present a barrier to universal adoption. This highlights the need for rapid and affordable alternatives. PANProfiler CRC (PPC) is a deep learning-based solution for detecting MSI/dMMR in CRC tumors that only requires whole slide images (WSIs) of haematoxylin and eosin (H&E)-stained tissue to provide test results. Using only WSIs, PPC offers an efficient alternative to standard testing. This study evaluates PPC's performance in a multi-site blinded setting. **Methods:** Blinded validation was performed using 3246 WSIs of H&E-stained CRC specimens. PPC provided outputs as "Stable", "Unstable", or "Indeterminate", with "Unstable" indicating dMMR or MSI-High, and "Stable" indicating proficient mismatch repair or non-MSI-High. "Indeterminate" was returned when PPC did not have a definitive result. PPC was evaluated by comparison to standard MSI/dMMR tests. Validation data spanned three cohorts from two sites (Table). St James's University Hospital (SJUH), Leeds, UK, supplied Cohorts 1 and 2; Cohort 3 was sourced from Wales Cancer Biobank (WCB), UK. Blinded analysis was performed at SJUH. **Results:** Results are given (Table). PPC demonstrated an overall percent agreement of 93.91%, a positive percent agreement of 92.17%, and a negative percent agreement of 94.15%, returning a definitive result for 88.05% of WSIs. **Conclusions:** This real-world, multi-site, blinded validation study demonstrates PPC's remarkable performance, comparable to standard tests for detecting MSI/dMMR in CRC, with high test replacement rates. In the clinical setting, PPC could significantly accelerate testing and enable timely delivery of stratified treatment plans. This accurate and cost-effective diagnostic solution promises to revolutionize MSI/dMMR testing in CRC. Research Sponsor: None.

**Blinded validation results of PPC with confidence intervals (CI) at 95%.**

| Site | Cohort | Sample Size (Unstable; Stable) | Overall Percent Agreement % (CI) | Positive Percent Agreement % (CI) | Negative Percent Agreement % (CI) | Test Replacement Rate % |
|---|---|---|---|---|---|---|
| **SJUH** | 1 | 488 (78; 410) | 92.79 (89.86-95.08) | 90.16 (79.81-96.30) | 93.24 (90.11-95.62) | 85.25 |
| **SJUH** | 2 | 2704 (318; 2386) | 94.31 (93.31-95.21) | 92.31 (88.48-95.18) | 94.57 (93.52-95.50) | 88.42 |
| **WCB** | 3 | 54 (11; 43) | 84.31 (71.41-92.98) | 100.00 (71.51-100.00) | 80.00 (64.35-90.95) | 94.44 |
| | All | 3246 (407; 2839) | 93.91 (92.97-94.76) | 92.17 (88.82-94.78) | 94.15 (93.16-95.04) | 88.05 |

# Utility of computed tomography (CT) –based staging of mismatch repair deficient (dMMR) colon cancer.

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**Background:** dMMR tumors are characterized by rapid primary tumor growth with reduced likelihood for nodal and distant metastases compared to MMR proficient (pMMR) tumors. Neoadjuvant immune checkpoint inhibitors (ICIs) have demonstrated promising efficacy for localized colon cancer with high pathologic complete response rates and low relapse rates. This raises the question if surgical treatment could be avoided with accurate radiographic staging while avoiding overtreatment of early stage tumors. This study assesses the utility of radiographic staging by CT for dMMR tumors compared to pathologic staging. **Methods:** Patients with stage I-III colon cancer treated with upfront surgical resection were retrospectively reviewed from a single institution from 2012 to 2023 in two cohorts: 1) dMMR colon cancer and 2) pMMR colon cancer limited to similar sample size as matched controls. Clinical stage was determined based on CT scans prior to surgical resection by 2 independent radiologists blinded to pathologic stage, and the results were correlated to pathologic stage. Clinical characteristics were extracted from the electronic medical record. Statistical analysis was performed using SPSS. **Results:** We identified 80 patients with dMMR colon cancer and 66 with pMMR tumors. CT clinical tumor staging was discordant from pathologic staging for 68% of dMMR tumors and 61% for pMMR tumors. For T4 dMMR tumors, CT had a 74% sensitivity and 44% specificity. T4 tumors were understaged by radiology to less than T4 for 56% of cases. For dMMR nodal staging, CT had 89% sensitivity and 50% specificity (Table 1). For pathologic node positive tumors, 14% were understaged by radiologist CT read to node negative. For pathologic node negative tumors, 50% of tumors were overstaged to node positive by radiologist CT read. There was poor inter-rater reliability for both T stage and N stage for dMMR and pMMR tumors. The weakest agreement between radiologists was for T stage (dMMR kappa = 0.246, mMMR kappa = 0.145). **Conclusions:** Radiologic clinical staging of dMMR and pMMR colon cancer does not correlate well with pathologic staging. There were high rates of understaging by CT evaluation for patients who may be candidates for neoadjuvant treatment (T4, N+). Furthermore, there was poor inter-rater reliability between radiologists, indicating that CT staging alone may not be sufficient. Additional diagnostic modalities for nodal detection may be necessary to accurately clinically stage patients before neoadjuvant ICIs in patients with locally advanced disease.
Research Sponsor: None.

**Comparing utility of pMMR versus dMMR tumors for T4 and node positive disease.**

| | pMMR (n = 66) | dMMR (n = 80) |
|---|---|---|
| **T4 versus T3 or less** | | |
| **Sensitivity (%)** | 92 | 74 |
| **Specificity (%)** | 51 | 47 |
| **PPV (%)** | 80 | 77 |
| **NPV (%)** | 82 | 63 |
| **Node positive versus node negative** | | |
| **Sensitivity (%)** | 60 | 50 |
| **Specificity (%)** | 75 | 89 |
| **PPV (%)** | 81 | 89 |
| **NPV (%)** | 53 | 50 |

Abbreviations: PPV = positive predictive value; NPV = negative predictive value.

# A protein-based blood panel test for the early detection of colorectal cancer and advanced neoplasia.

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**Background:** Early detection of colorectal cancer (CRC) is key to survival. CRC screening compliance is still low because of either the invasive procedure (colonoscopy) or the so-called "ick factor" (stool-based tests). Stool-based tests also lack ability to detect advanced adenoma (CRC precursor). A blood-based test could improve adherence to screening guidelines, but no one has so far been recommended due to a lack of sufficient sensitivity and specificity. Here we report the potential clinical utility of a blood-based biomarker panel providing high sensitivity (SE) and specificity (SP) for CRC and advanced adenomas (AA) as a screening triage option addressing positive patients to colonoscopy. **Methods:** A total of 241 retrospective serum samples (179 CRC, 22 adenoma, 40 healthy controls) were obtained from the Cooperative Human Tissue Network (CHTN) biobank and tested with a proprietary sandwich ELISA that detects dual CRC specific biomarkers (BF7 and CC3). A standard curve was generated and biomarkers concentration in the samples were calculated using a 4-parameter logistic method. The median biomarker concentration values in each patient group were compared to determine whether differences were statistically significant ($p< 0.05$). Statistical analysis of two-group and multi-group comparisons was carried out using the non-parametric Mann-Whitney U test and the Kruskal-Wallis test, respectively. Receiver operating characteristic (ROC) curves were calculated to evaluate the diagnostic performance, by determining sensitivity (SE), specificity (SP), area under the curve (AUC), and confidence interval (CI). Statistical analysis, ROC curves and scatter plots were performed with GraphPad Prism 7.0. **Results:** Both biomarker's median concentration difference between CRC versus healthy samples groups were found to be statistically significant ($p<0.05$). Both biomarkers were also elevated in AA as compared to healthy samples, but difference in median concentration was only statistically significant for CC3 ($p<0.0001$). BF7 serum levels in cancer patients were significantly associated with TMN stage ($p=0.019$), N stage ($p=0.039$) and M stage ($p=0.0006$). No significant associations were found between this biomarker levels and other clinical characteristics of the CRC patients including age. Our test discriminated CRC (all stages) from healthy individuals with a SE of 80% and a SP of 90%. Sensitivity for early (n=95) and late-stage CRC (n=80) was 79% and 84%, respectively. Our test was also able to detect 81% of serum from advanced adenoma patients who are at increased risk of CRC. **Conclusions:** This study illustrates the potential clinical utility of our blood-based assay for the early detection of CRC which has a better sensitivity for advanced adenoma than tests in current clinical use. A large prospective cohort study is planned to further validate the clinical performance of this test. Research Sponsor: None.

# The role of artificial intelligence in colorectal cancer and polyp detection: A systematic review.

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**Background:** Colorectal cancer (CRC) remains a leading cause of cancer-related mortality worldwide. Early detection and accurate diagnosis are essential for improving survival rates and optimizing therapeutic strategies. Recently, artificial intelligence (AI) technologies, such as machine learning algorithms, convolutional neural networks (CNNs), and computer-assisted diagnostic (CAD) systems, have significantly enhanced traditional diagnostic tools like colonoscopy and histopathology. This systematic review evaluates the current role of AI in CRC management, focusing on diagnostic improvements, predictive modeling, and outcome prediction. **Methods:** A comprehensive literature review was conducted using PubMed, Google Scholar, and MEDLINE using MeSH term to identify studies utilizing AI in CRC diagnosis and management, with a focus on diagnostic imaging, histopathological analysis, and predictive modeling. Studies were evaluated for the accuracy of AI-driven diagnostic systems and the predictive performance of AI models in clinical outcomes. **Results:** Out of 147 studies identified, 37 met the eligibility criteria for this review. AI tools such as CNNs, CAD systems, and EndoBRAIN showed high accuracy in CRC detection and polyp classification. Kudo et al. (2020) reported 98% CRC detection accuracy using EndoBRAIN. Blanes-Vidal et al. (2019) achieved 96.4% accuracy in polyp detection via DCNN. Additionally, Wang et al. (2019, 2020) demonstrated improved Adenoma Detection Rates (ADR) of 34.1% and 29.1%, respectively, using AI over traditional methods. For outcome prediction, AI models, including CNNs, predicted patient prognosis with reasonable accuracy, as highlighted by Skrede et al. (2020) with 76% accuracy in prognosis prediction. **Conclusions:** The systematic review and analysis of AI-assisted diagnostic modalities in colorectal cancer and polyp detection reveal promising results, with most models demonstrating high sensitivity, specificity, and accuracy. CNN-based models and CAD systems, in particular, show strong potential for improving detection rates and clinical outcomes in CRC screening. The studies included in this review demonstrate that AI can enhance diagnostic precision, particularly in identifying polyps and predicting clinical outcomes, with high validation rates. However, more external validation and long-term clinical studies are needed to fully establish the robustness of these AI tools in routine clinical practice. Research Sponsor: None.

# Molecular characteristics and prognostic impact of *GNAS* mutations in colorectal cancer: An international collaborative study between United States and Japan.

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**Background:** *GNAS* (Guanine nucleotide-binding protein, alpha stimulating) encodes the alpha subunit of the stimulatory G protein, and gain of function mutations in this gene have been previously identified, especially those located at the R201 codon. Although *GNAS*$^{R201C/H}$ has been associated with mucinous histology and poor response to chemotherapy in appendiceal cancer, the impact of *GNAS* mutations in colorectal cancer (CRC) remains unclear. **Methods:** We conducted a comparative, integrative analysis of *GNAS* mutations in CRC using two cohorts: MD Anderson Cancer Center (MDA) across all stages (N=5,249) and the GALAXY study involving only resectable disease (UMIN000039205, N=2,599). We assessed the correlation between *GNAS* pathogenic mutations and microsatellite stability, CRC sidedness, co-mutations, and survival outcomes. The Chi-squared test was used for categorical variables, and the log-rank test for overall survival (OS) and disease-free survival (DFS). **Results:** In the MDA and GALAXY cohorts, *GNAS* mutations were identified in 107 (2.0%) and 61 (2.4%) patients, respectively. Prevalence of *GNAS* mutations was higher in right sided tumors (MDA: 3.8% vs. 1.3%, GALAXY: 6.2% vs 2.8%, $p<0.01$ for both cohorts) and microsatellite instability-high (MSI-H) tumors (MDA: 6.5% vs. 1.8%, GALAXY: 16.5% vs. 1.4%, $p<0.01$ for both cohorts). There was not an association of *GNAS* mutations with stage in either cohort. *GNAS* was significantly enriched for co-mutation with *KRAS* in the MDA cohort (OR=3.26, $P<0.01$), similar to prior observations in appendix cancer, but was not significantly associated with *KRAS* mutations in the GALAXY cohort (OR=1.35, P=0.3). *GNAS* mutations were mutually exclusive with *TP53* in both cohorts (MDA: OR=0.58, P=0.026; GALAXY: OR=0.21, $P<0.01$). With regards to survival, in the MDA cohort, *GNAS* mutant tumors had a trend towards worse OS in the overall population (20 mo vs. 33 mo, HR=1.27, 95% CI=0.61-1, P=0.085). The survival impact of *GNAS* mutations was greatest in metastatic patients (Stage IV: 19 mo vs. 38 mo, HR=1.49, 95% CI=0.47-0.94, P=0.021 vs. Stage I-III: HR=1.35, 95% CI=0.46-1.2, P=0.22). The GALAXY cohort showed no significant association between *GNAS* mutations and DFS (HR=1.05, 95% CI=0.6-1.8, P=0.9). In both cohorts, *GNAS* mutations had a worse prognosis in left-sided CRC (MDA: HR for OS = 1.45, P = 0.091; GALAXY: HR for DFS=2.15, P=0.09) with no significant effect observed in right-sided CRC (MDA: HR for OS=0.91, P=0.68; GALAXY: HR for DFS=0.65, P=0.3). **Conclusions:** Our findings highlighted the intricate role of *GNAS* mutations in CRC, demonstrating its higher prevalence in MSI-H and right-sided colon, significant association with *KRAS* mutations, and divergent prognostic impacts based on tumor sidedness. Further research is needed to elucidate the mechanisms behind the worsened survival in left-sided *GNAS*-mutated CRC. Research Sponsor: Conquer Cancer Career Development Award; (2022CDA-7604125121); SCRUM-Japan Funds.

# Landscape of metastatic colorectal cancer (CRC) using comprehensive circulating tumor DNA (ctDNA) next-generation sequencing (NGS) in India: Expanding beyond RAS and RAF.

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**Background:** Traditional testing in mCRC primarily focuses on RAS and RAF mutations. However, a broader range of emerging biomarkers, collectively termed the "Pressing panel" (PP), identifies patients with RAS/RAF wild-type (wt) tumors who may particularly benefit from anti-EGFR therapy. Comprehensive liquid biopsies provide a non-invasive method for identifying traditional and emerging biomarkers. Limited genomic data, including the prevalence of PP alterations (alts), is available for mCRC patients in India tested by ctDNA NGS. **Methods:** We reviewed the results of Guardant360 (Guardant Health, Inc.) ordered for patients with mCRC as part of routine clinical practice in India through July 2024. NGS analyzed plasma samples at a CLIA-certified, CAP-accredited central laboratory. Guardant360 analyzes ctDNA for mutations (mts), insertions and deletions, amplifications (amps), and fusions in up to 83 genes, including PP genes, i.e., KRAS, NRAS, BRAF, ERBB2, PIK3CA, PTEN, AKT1, ALK, ROS1, NTRK, RET, and MSI status. The Indian cohort's prevalence of PP mts was compared to that from other countries in Asia and Middle East (AME). Timing of testing relative to line of therapy was unknown. **Results:** Among 145 samples from India, ctDNA was detected in 92% (n=134). Median age was 59 years, with women comprising 40%. The median turnaround time from blood collection to result was 7 days. The mean mts count per sample was 5, and the median variant allele frequency was 1.5%. Frequent mts were in APC (53%), KRAS (56%), and SMAD4 (13%), while amps of EGFR (16%) and FGFR1 (9%) were common. Clinically relevant fusions were observed in 3% of patients. KRAS, NRAS, and/or BRAF V600E alterations were found in 63% of both India and AME samples. PP alts were identified in 24% and 27% of Indian and AME patients, respectively. Most frequent PP alts included PIK3CA (15%), PTEN (3%), and MET amp (2%). The prevalence of most of the PP alts was similar between the Indian and AME cohorts (Table), with minor differences observed in PIK3CA (15% vs 18%) and MSI-High (3% vs 1%). In patients with RAS/RAF wild-type lacking EGFR-amp, 9.5% (India) and 18% (AME) harbored at least one PP alt. **Conclusions:** This study highlights the role of liquid CGP for characterizing the mutational landscape and identifying PP mts for patients from India with advanced CRC. Research Sponsor: None.

**Pressing panel alterations.**

| PP Alteration | INDIA % (n=134) | AME % (n=720) |
|---|---|---|
| ERBB2 Amp | 1% | 2% |
| MET Amp | 2% | 3% |
| ALK Fusion | 0% | 0% |
| ROS1 Fusion | 0% | 0% |
| NTRK1 Fusion | 1% | 0% |
| RET Fusion | 0% | 1% |
| ERBB2 Mut | 1% | 2% |
| PIK3CA Mut | 15% | 18% |
| PTEN Mut | 3% | 3% |
| AKT1 Mut | 0% | 1% |
| MSI-High | 3% | 1% |

# Clinical validation of plasma circulating-tumor DNA assay using highly sensitive Safe-SeqS technology for detecting *RAS* and *BRAF* V600E in metastatic colorectal cancer.

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**Background:** We developed a circulating-tumor DNA (ctDNA) test assay for *RAS/BRAF* mutations using next-generation sequencing-based Safe-SeqS technology. The clinical performance of detecting *RAS* and *BRAF* mutations were evaluated versus approved liquid biopsy and tissue testing, respectively. **Methods:** Metastatic colorectal cancer (mCRC) patients (pts) were enrolled. The concordance of the *RAS* mutational status in ctDNA between Safe-SeqS assay and OncoBEAM RAS CRC kit, which is approved as a companion diagnostic in Japan, was evaluated using archival plasma samples. The primary endpoints were positive percent agreement (PPA) and negative percent agreement (NPA). The concordance of the *BRAF* V600E mutational status between plasma-Safe-SeqS assay and tissue-based testing using MEBGEN RASKET-B kit, which is approved as a companion diagnostic in Japan, was prospectively evaluated. The primary endpoints were sensitivity and specificity of Safe-SeqS assay compared to tissue-based testing. Multivariate analysis using LASSO regression model was performed to evaluate the discordance of *BRAF* V600E mutational status between Safe-SeqS assay and tissue-based testing. **Results:** In 125 eligible plasma samples, the PPA and NPA for *RAS* mutational status in ctDNA were 87.1% (95% confidence interval [CI]: 76.1–94.3) and 95.2% (95% CI: 86.7–99.0), respectively. On the other hand, in 103 eligible pts, the sensitivity and specificity of Safe-SeqS assay for *BRAF* V600E mutational status were 69.2% (95% CI: 48.2–85.7) and 98.7% (95% CI: 93.0–100.0), respectively. Multivariate analysis revealed the baseline longest diameter and the number of lesions in pts with peritoneal and/or lung metastasis were factors associated with discordance. The sensitivity for *BRAF* V600E mutations increased to 81.0% (95% CI: 58.1–94.6, fisher's exact test $p$=0.02) when the 5 pts with lower tumor burden (peritoneal metastasis with the longest diameter <20 mm, and lung metastasis with the longest diameter <20 mm and <10 lesions, according to the previous report [1] ) were excluded. **Conclusions:** The clinical validity of the ctDNA assay for detecting *RAS/BRAF* V600E mutations using Safe-SeqS technology was confirmed in pts with mCRC. Careful attention should be paid for mCRC pts with only peritoneal and/or lung metastases having fewer metastases or smaller diameter lesions. 1. Kagawa Y, et al. Clin Cancer Res 2021. Research Sponsor: None.

# Communication of cancer biomarker testing results: What patients need.

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**Background:** As colorectal cancer (CRC) biomarker testing becomes more widespread and important for treatment decisions, understanding testing results can improve patient engagement, shared decision-making, and facilitate clinical trial enrollment. Our experience in the online 12,000+ member CRC community COLONTOWN (CT), suggests that communication of biomarker results to patients is not ideal. To learn more, we surveyed CT members on their biomarker testing experience. **Methods:** An online survey posted in CT in September 2024 had questions on testing received (what was tested and how), test results experience (communication before testing and after results, patient understanding and resulting impact), suggested provider statements on biomarker results (statements patients prefer) and disease and respondent details. **Results:** Of 83 respondents, 70% had testing that included at least 3 common CRC biomarkers; the following are results from this group. Until receipt of results, 30% were unaware that biomarker testing was being done, and 56% first reviewed results without a clinician. Only 17% of respondents felt that nothing was lacking in the clinician communication of biomarker testing results. Upon receipt of biomarker test results, respondents' feelings improved (30%,18%), did not change (63%, 55%) or worsened (7%, 27%) about treatment options and prognosis, respectively. Test results made 51% feel more confident in their care, and reduced stress for 25%, but increased stress for 39%, with 1/3 citing sub-par communication of results as at least partly driving the stress increase. The value of clear communication was evident, as the most appealing sample provider statements on results with no actionable biomarkers included confirmation of proposed treatment plan and clarity on non-actionable biomarkers. Not surprisingly, 58% learned the most about their biomarker results from COLONTOWN. **Conclusions:** Our survey highlights the gaps in communication of biomarker testing results and provides insights on what may improve them. The perceived value of biomarker test results for patients seem to go beyond actionability as traditionally defined. Context-specific, nuanced communication of "what the results really mean" and a discussion of its impact on patient experience needs to be part of clinician communication of test results. We plan to develop resources to help, and we implore clinicians and testing companies to consider our findings and make changes in how they share test results with patients. Research Sponsor: None.

# A method for classifying colorectal cancer and gastric/esophageal cancer using blood-based testing.

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**Background:** Identifying the cancer signal of origin is of great value in the clinical use of blood-based cancer testing, especially in those tests that have potential to assess for more than one cancer type. Thus, an approach that can accurately identify cancer signal of origin in order to guide subsequent diagnostic workup is impactful to the adoption of this new technology. Here, we report feasibility data using the DNA methylation signature of plasma cell-free DNA to differentiate between colorectal cancer (CRC) and gastric/esophageal cancer (GEC) types. **Methods:** We developed an algorithm to distinguish between CRC and GEC based on the DNA methylation signature of plasma cell-free DNA molecules from more than 3,000 differentially methylated regions. First, a regression model (a multi-cancer classifier) was trained to identify the samples with sufficient tumor derived methylation signal using 6,822 cancer samples of 24 cancer types and the samples from 4,423 cancer-free individuals. A second regression model (a CRC/GEC classifier) was trained to distinguish CRC from GEC cases on the samples with positive calls from the multi-cancer classifier. The second model was trained on methylation signals from advanced stage CRC samples, advanced stage GEC samples (including esophageal, gastroesophageal, and gastric cancers), and samples from cancer-free individuals. **Results:** The multi-cancer classifier identified 92% (2,954/3,204) of CRC and GEC samples as positive at 90% target specificity, including 93% (2,086/2,253) of CRC samples and 91% (868/951) of gastric/esophageal cancer samples. The performance of the CRC/GEC classifier is evaluated through 10-fold cross validation on these 3,204 cancer samples, where in each fold, an additional 6,298 samples from cancer-free individuals were also used in training. Among the 2,954 CRC and GEC samples detected by the multi-cancer classifier, 2,698 (91%) were accurately classified by the CRC/GEC classifier. Among the detected CRC samples, 91% (1,907/2,086) were correctly classified as CRC. Similarly, 91% (791/868) of the detected GEC samples were correctly classified. The precision of the CRC/GE prediction was also assessed and 91% (2,698/2,954) of the detected cancer samples were correctly classified. Of the 1,984 detected cancer samples predicted as CRC, 96% (1,907/1,984) were true CRC samples. Of the 970 detected cancer samples classified as GEC, 82% (791/970) were correct. **Conclusions:** This assay with high sensitivity and accurate CRC/GEC classification can effectively guide subsequent diagnostic workups following a positive result, maximizing potential clinical utility. Ongoing work will continue further refinements. Research Sponsor: None.

# Use of molecular residual disease (MRD) testing with measurement of circulating tumor DNA (ctDNA) post-operatively in stage II colon cancer (SIICC) in the community-based setting.

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**Background:** 80% of adult oncology patients (pts) are treated in the community-based setting. Data on the implementation of emerging technologies such as ctDNA testing is scant in this population. Use of adjuvant chemotherapy in SIICC pts historically was based on clinical/pathological risk factors however new data presented in mid 2022 utilizes the presence or absence of ctDNA 6-8 weeks post operatively to help risk stratify pts. The absence of ctDNA predicts a high chance of disease-free survival. Use of ctDNA was incorporated into the NCCN Colon Cancer guidelines as of 2023. In a private practice setting, we retrospectively looked at the use of ctDNA assay for adjuvant chemotherapy decision-making in SIICC pts. **Methods:** We conducted a retrospective study to quantify ctDNA testing frequency in SIICC pts in our community based oncology practice. The historical cohort was comprised of pts diagnosed prior to the NCCN guideline inclusion of ctDNA; the current cohort included pts diagnosed after the NCCN guideline change. For all pts, we reviewed clinicopathologic risk factors deemed to be high risk (positive margin, $<$ 12 LN removed, tumor budding, lymphovascular invasion, poorly differentiated features; perforation or obstruction) and used this for stratification. Low-risk pts had no high risk features. Medical record data (office notes, pathology, molecular testing) was evaluated to determine ctDNA results including timing and use of adjuvant chemotherapy. Primary endpoint was the percent (%) of current pts who were low-risk with ctDNA testing sent compared to historical control. Secondary endpoint was the % of ctDNA testing sent within 8 weeks of curative surgery in both cohorts. **Results:** In a private practice setting, we identified low-risk SIICC pts spanning a 36-month period and categorized them into two groups: the historic cohort (A) pts were diagnosed between July 2021-Dec 2022, and the active cohort (B) pts were diagnosed between Jan 2023-July 2024. 26 pts were identified in cohort A and 36 pts were identified in cohort B to have SIICC. Out of 26 pts in cohort A, we identified 13 low risk SIICC pts; for cohort B, out of 36 pts, we identified 25 low-risk SIICC pts. The percentage of pts in each group that had ctDNA testing sent was 79.6% (10/13 pts) in cohort A and 92% (23/25 pts) in cohort B. In cohort A, 80% (8/10 pts) and in cohort B, 90.9% (20/22 pts) had ctDNA sent within 8 weeks of surgery. **Conclusions:** Incorporation of ctDNA testing in low-risk SIICC pts in a community-based setting occurred quickly. As ctDNA testing was implemented into practice, a majority of pts had the testing sent in the appropriate time frame. Future best practices, especially in the community, should incorporate tumor specific clinician oversight with use of updated pathways to ensure any recent changes to standard of care are followed. Research Sponsor: None.

# Shifting the paradigm: Early identification of colorectal cancer with C the Signs clinical decision support.

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**Background:** The incidence of colorectal cancer is increasing rapidly, particularly amongst younger people. Despite a screening program being widely available, most patients are diagnosed with cancer through symptomatic presentation. Leveraging electronic medical records (EMRs) through clinical decision support platforms offers potential for identifying high-risk individuals before clinical suspicion develops. This study aims to explore the challenges associated with early colorectal cancer diagnosis and evaluate the effectiveness of the cancer case finding feature of a cancer management platform called C the Signs. **Methods:** A retrospective analysis was conducted using data from the Mayo Data Platform, comprising records of 894,275 patients, including 7,348 diagnosed with colorectal cancer. C the Signs was employed to identify patients at risk of colorectal cancer based on EMR data. Sensitivity and specificity analyses were performed to assess the platform's performance in identifying high-risk patients. Additionally, the study evaluated the temporal discrepancy in colorectal cancer diagnoses between those flagged by C the Signs and those detected by primary care physicians. **Results:** The C the Signs screening platform had a sensitivity of 93.8% and a specificity of 19.7% in identifying patients at risk of colorectal cancer. Notably, 29.4% of patients with colorectal cancer were identified as being at risk up to 5 years earlier by C the Signs compared to diagnoses made by primary care physicians, highlighting the platform's potential for early detection. **Conclusions:** Early identification of high-risk individuals holds promise for improving patient outcomes and reducing the burden of colorectal cancer. Whilst the specificity of the C the Signs platform for colorectal cancer is relatively low, the sensitivity is comparable to the most favorable rates reported for colonoscopy. In addition, if C the Signs was used in conjunction with a relatively cheap screening test such as Faecal Immunochemical Testing, this would significantly drive up the specificity whilst maintaining the sensitivity. Further studies are needed to assess its efficacy in conjunction with tests available in primary care. Research Sponsor: None.

# Long-term follow-up of the randomized trial on open versus laparoscopic surgery for primary tumor resection in patients with non-curable stage IV colon cancer (JCOG1107).

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**Background:** The Japan Clinical Oncology Group (JCOG) 1107 (UMIN-CTR: UMIN000009715) was a non-inferiority phase III trial to compare laparoscopic surgery (LAP) with open surgery (OP) for symptomatic, non-curable stage IV colon cancer, using progression-free survival (PFS) as the primary endpoint based on the intention-to-treat principle. The planned primary analysis confirmed the non-inferiority of the LAP over the OP. Additionally, the present study aimed to compare the LAP and OP using long-term 3-year follow-up data from the JCOG1107. **Methods:** Eligibility criteria included pathologically proven adenocarcinoma or adenosquamous carcinoma; primary tumor located at the cecum, ascending, transverse, descending, sigmoid, and rectosigmoid colon; primary tumor with bowel stenosis and/or bleeding; and having at least one to three non-curable factors. The sample size was 194 with a power of 70%, one-sided alpha of 5%, and non-inferiority margin of the hazard ratio (HR) of 1.38. **Results:** In total, 195 patients were randomized (OP 95, LAP 100) between January 2013 and January 2021. OP was performed in 92 and LAP in 98 patients. Postoperative chemotherapy was administered in 82 OP and 86 LAP patients. As the data cut-off date of Feb 2024, 93 (98%) and 100 (100%) PFS event occurred in OP and LAP. The 3-year PFS were 5.3% (95% CI, 2.0-11.0) and 3.0% (0.8-7.8) for OP and LAP arms, respectively (HR: 1.028, 95% confidence interval [CI]: 0.772–1.370; P for non-inferiority = 0.0222). Regarding OS, 87 (92%) and 90 (90%) death occurred in OP and LAP. The 3-year OS were 31.5% (22.5-41.0) and 28.5% (20.0-37.6) for OP and LAP arms, respectively (HR: 1.048; 95% CI: 0.780–1.410). **Conclusions:** Long-term follow-up data supported the non-inferiority of LAP over OP for primary tumor resection in patients with non-curable stage IV colon cancer. Our study findings indicate that LAP is acceptable as a standard treatment for symptomatic, non-curable stage IV colon cancer. Clinical trial information: UMIN000009715.
Research Sponsor: Ministry of Health, Labour and Welfare, Japan.

# Survival benefits of asymptomatic primary tumor resection after bevacizumab plus FOLFIRI as first-line therapy for patients with metastatic colorectal cancer with synchronous unresectable metastasis.

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**Background:** Metastatic colorectal cancer (mCRC) poses a clinical challenge and requires a combination of systemic therapy and conversion surgery. Although first-line chemotherapy and target therapy are considered the standard treatments for mCRC, the role of primary tumor resection (PTR) in asymptomatic synchronous mCRC with unresectable metastatic lesion after initial therapy remains relatively underexplored. **Methods:** A retrospective observational study was conducted from January 2015 to January 2021, involving 74 patients with synchronous mCRC who received bevacizumab plus FOLFIRI as first-line systemic therapy. All 74 patients had unresectable metastatic lesions confirmed through multidisciplinary team (MDT) discussion. Patients' characteristics, PTR data, and radiotherapy (RT) and overall survival (OS) outcomes were analyzed. The patients were categorized into a "PTR" group and a "No PTR" group and then further stratified into IVA, IVB, and IVC subgroups based on the initial mCRC stage. Additionally, four subgroups - namely "PTR(+)/RT(+)", "PTR(+)/RT(-)", "PTR(-)/RT(+)", and "PTR(-)/RT(-)" were formed to assess the combined effects of PTR and RT. **Results:** The median OS for all the patients was 23.8 months (20.5 - 27.1 months). The "PTR" group exhibited a significantly higher median OS of 25.9 months (21.3 - 30.5 months) compared with 21.4 months (15.8 - 27.1 months) in the "No PTR" group ($P$ = 0.048). Subgroup analyses revealed a trend of improved survival with PTR in patients with stage IVA and IVB; however, the results were not statistically significant ($P$ = 0.116 and 0.493, respectively). A subgroup analysis of PTR and RT combinations revealed no significant difference in median OS rates. **Conclusions:** For asymptomatic mCRC with synchronous unresectable distant metastasis. PTR following first-line therapy with bevacizumab plus FOLFIRI can provide a survival benefit, particularly in stage IVA/IVB patients compared with stage IVC patients. Additionally, RT for primary tumor did not provide an additional OS benefit in mCRC with unresectable metastasis. A prospective randomized trial with a larger sample size is essential to further elucidate the role of PTR in this context. Research Sponsor: None.

# Effectiveness of anti-epidermal growth factor receptor agents in elderly patients (age ≥ 70 years) with *RAS* wild-type metastatic colorectal cancer: Nationwide real-world data.

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**Background:** This study investigated the efficacy and safety of anti-epidermal growth factor receptor (EGFR) agents plus chemotherapy for elderly patients (aged > or = 70 years) with *RAS* wild-type metastatic colorectal cancer (mCRC) from nationwide data in Taiwan. **Methods:** Total 756 patients with *RAS* wild-type mCRC who received first-line therapy with anti-EGFR agents (Cetuximab) plus chemotherapy at nationwide multiple institutions from November 2016 to January 2021. The study explored the prognostic impact between two age groups (≥70 vs <70). Progression-free survival (PFS) and cancer-specific survival (CSS) were compared. Additionally, severe adverse events (SAEs) were also compared between two age groups. **Results:** The median PFS and CSS were comparable between the two age groups (14.0 vs 14.0 months, $p$ = 0.098; 32.0 vs 35.0 months, $p$ = 0.226, respectively). Moreover, subgroup analysis revealed similar PFS and CSS between two age groups in patients with synchronous or metachronous mCRC and left-sided or right-sided tumors (all $p > 0.05$). Regarding hematologic SAEs, a significant higher incidence of grade ≥ 3 anemia was found in elderly patient group (13.3% vs 5.0%, $p$ = 0.003). However, no significant difference was observed in the incidence of non-hematologic SAEs (all $p > 0.05$). **Conclusions:** According to this nationwide real-world data, for patients carrying *RAS* wild-type mCRC treated with cetuximab plus chemotherapy, elderly patients could have comparable benefit as younger ones. Except for grade ≥ 3 anemia, other hematologic SAEs and non-hematologic SAEs revealed no significant difference between two age groups. Research Sponsor: None.

# Real-world evidence on genomic profiles and survival outcomes in patients with HER2-positive metastatic colorectal cancer.

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**Background:** Patients (pts) with HER2-positive (HER2+) metastatic colorectal cancer (mCRC) represent a small, but increasingly important subgroup due to emerging HER2-targeted therapies. Although HER2 positivity is associated with resistance to anti-EGFR therapy (EGFRi), it is unclear if EGFRi or bevacizumab (BEV) is preferable as first-line (1L) therapy. Here, we offer novel evidence on the genomic profiles of pts with HER2+ mCRC and examine differences in survival between key treatment groups. **Methods:** This study used the US-based deidentified Flatiron Health-Foundation Medicine mCRC clinico-genomic database. The deidentified data originated from approximately 280 US cancer clinics (~800 sites of care). Included pts were 18 years or older, had evidence of stage IV or recurrent mCRC diagnosed between 01 Jan 2012 and 31 Mar 2022, and underwent tissue-based Comprehensive Genomic Profiling testing. HER2+ was defined as an *ERBB2* copy number of ≥5 for diploid tumors. A doublet regimen was defined as fluoropyrimidines plus oxaliplatin or irinotecan. Data cut-off was 31 Mar 2024. Kaplan-Meier methods were used to examine real-world overall survival (rwOS) and progression-free survival (rwPFS) overall, by 1L treatment class and across key subgroups defined by tumor sidedness and genomic profile, indexed to the start of 1L with pts censored at their last EHR-recorded activity. **Results:** Among a cohort of 5545 pts, HER2+ was detected in 171 (3.1%) pts. Median age (years) was 57 for HER2+ vs 61 for HER2-; 69.3% of HER2+ pts were White, compared to 72.1% in HER2-. Among pts with HER2+, 31 (18.1%) had *RAS* mutation (mt), *BRAF* V600E mt, or microsatellite instability-high. *ERBB2* copy number was significantly lower in pts with *RAS* mt or *BRAF* V600E mt compared to those without these mutations ($p < 0.01$, $p = 0.045$, respectively). Among pts who received 1L treatment (n=4748), rwPFS significantly differed between HER2+ (n=144) and HER2-negative (HER2-; n=4604) groups (7.6 vs 8.7 months, unadjusted HR (uHR) 1.20, p=0.04). This difference was consistent in pts with left-sided *RAS/BRAF* V600E wild-type microsatellite stable (MSS) mCRC and treated with 1L doublet plus EGFRi (8.7 vs 12.5 months, uHR 2.18, p=0.02; n=10 HER2+,n=117 HER2-) or 1L doublet plus BEV (8.9 vs 10.5 months, uHR 1.65, p=0.04; n=20 HER2+, n=387 HER2-). No significant rwPFS difference was observed between EGFRi and BEV in the HER2+ subgroup. rwOS showed no significant difference between HER2+ and HER2- groups (overall: 26.1 vs 24.5 months, uHR 0.94, p=0.52; n=144 HER2+, n=4604 HER2-). **Conclusions:** Although about 80% of pts with HER2-positive mCRC are *RAS/BRAF* V600E wild-type and eligible for EGFRi, no survival benefit was observed in EGFRi relative to BEV as first-line combination. While the impact of potential confounders needs to be examined in further analyses, these findings highlight the need for specific treatments for pts with HER2+ mCRC. Research Sponsor: None.

# Understanding the unmet needs in colorectal cancer: Insights from patient and survivor experiences.

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**Background:** Colorectal cancer (CRC) is the fourth most commonly diagnosed cancer and the second leading cause of cancer-related deaths worldwide among men and women combined. Despite improved survival rates, there remains a gap in understanding the specific unmet needs of CRC patients and survivors. These needs include emotional support, information on treatment options, coping strategies, quality of life (QoL), and overall healthcare experience. The stress of a diagnosis is often compounded by the emotional and physical demands of treatment, which extend into survivorship when the disease is not active. **Methods:** The Colorectal Cancer Alliance launched a survey for colorectal cancer patients, survivors, and caregivers. This IRB-approved cross-sectional study, comprising more than 153 questions, aimed to inform on patient profiles, diagnosis experience, resources, QoL, access to care, and treatment experience. Participation in the survey was solicited from the Alliance network of social media, email, and web-based communities, underscoring the collective effort to understand and address the needs of CRC patients and survivors. **Results:** The survey included 283 participants, primarily aged 46-55. Key findings showed that 74% of participants had difficulty finding someone who understood their experience, and 41% reported reduced support from others after treatment ended. Additionally, 54% struggled with fatigue, and 51% coped with ongoing stress. CRC significantly impacted careers (64%), dating lives (51%), desire to have children (43%), relationships with spouses or partners (58%), sex lives (80%), and participation in social activities (65%). Although many felt informed before treatment, 46% reported unmet needs for information on complementary or alternative therapies. **Conclusions:** These results highlight the profound psychosocial and physical challenges faced by CRC patients and survivors, emphasizing the need for healthcare providers to develop tailored treatment and survivorship plans and comprehensive support services. Insights from this survey will guide the Colorectal Cancer Alliance (Alliance) in creating targeted actions, care programs, and support initiatives to address these needs. The Alliance aims to enhance patient care and improve CRC patients' and survivors' QoL, potentially reshaping the future of CRC management and survivorship care. Research Sponsor: None.

# Clinical outcomes of hypofractionated radiotherapy (5x5 Gy) with a simultaneous integrated boost (5x6 Gy) in locally advanced rectal cancer.

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**Background:** Hypofractionated RT (5x5 Gy) with or without subsequent chemotherapy is a standard of care used at the National Institute of Oncology, Warsaw, Poland, as a preoperative treatment for locally advanced rectal cancer. To maximize the chance of achieving a complete response (CR), when surgery is not planned or when there are clinically involved lymph nodes located outside the standard TME field, RT can be augmented with a simultaneous integrated boost (SIB) delivering 5x6 Gy. This report aims to evaluate the clinical outcomes of hypofractionated RT with a SIB in rectal cancer. **Methods:** From Dec 2018 to Sep 2022, 78 consecutive pts with a median age of 67 years (range 36–89 years), received hypofractionated RT utilizing the SIB-VMAT technique. The radiation dose fractionation was 5x5 Gy to the rectum, mesorectum and elective lymph nodes and a SIB to the GTV for a total dose of 30 Gy. **Results:** 30 pts (38.5%) received a SIB for the rectal tumor, of which 15 pts (19.2%) were deemed unfit for surgery, 10 pts (12.8%) refused surgical treatment, 3 pts (3.8%) had an oligometastatic disease, and 2 pts (2.6%) had an advanced primary tumor. 48 pts (61.5%) received a SIB to the clinically involved lymph nodes located outside the standard TME field. All 78 pts completed RT as planned without interruptions or dose modifications and with an acceptable toxicity profile. There was a 26.9% (n = 21) incidence of G2 toxicities (diarrhea, proctitis, skin toxicity, cystitis or lumbosacral plexus neuropathy) and a 3.8% (n = 3) incidence of G3 toxicities (diarrhea or proctitis). After a median follow-up of 917 days (range 33–1927 days), an assessment of response was conducted in 68 cases. In 94.1% (n = 64) there was no progression at the site irradiated to a dose of 30 Gy. After completion of RT, mr-TRG was evaluated in 28 pts. In 46.4% (n = 13) mr-TRG 1-2 (cCR) was achieved. Of the 24 pts qualified for the watch and wait strategy, 17 had follow-up. Of this group, 64.7% (n = 11) remained progression-free, and 11 pts had an adequate evaluation for cCR (imaging and endoscopy), resulting in a 63.6% (n = 7) rate of cCR. 43 pts underwent surgery, of which 97.7% (n = 42) had R0 resection. In 41 cases pathological TRG was assessed according to AJCC or Ryan, of which 41.4% (n = 17) achieved TRG 0-1 (pCR). In 40 cases the operated pts had no initial metastatic spread, among them 67.5% (n = 27) remain disease-free. 67 pts reported disease-related symptoms before RT. After treatment, 89.6% (n = 60) of those pts achieved symptomatic improvement in either rectal bleeding, bowel frequency, or pain control. **Conclusions:** Hypofractionated RT with SIB-VMAT delivering a dose of 30 Gy/25 Gy/5 fractions provides good local control and is a safe and viable option for both young and elderly patients who will not undergo surgery and for patients with involved lymph nodes located outside the standard TME field to maximize the chance of achieving CR. Research Sponsor: None.

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# Investigating the real-world outcomes of single- versus multi-agent preoperative chemoradiotherapy for locally advanced rectal cancer: NCDB analysis from 2004-2020.

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**Background:** Fluorouracil-based chemoradiotherapy is the standard initial treatment for locally advanced rectal cancer, demonstrating improved pathological response and reduced local recurrences compared to radiation alone. Fluorouracil-based chemo-radiation therapy has no impact on distant recurrences. Oxaliplatin was tested in addition to fluorouracil-based chemo-radiotherapy in clinical trials to improve pathological complete response (pCR), distant recurrence-free survival, and overall survival. The trial results are conflicting. In this study, we aim to compare single-agent (SA) versus multi-agent (MA) chemotherapy using real-world data from the National Cancer Database (NCDB). **Methods:** The NCDB was used to identify 39,667 patients with rectal cancer who received concurrent chemoradiotherapy between 2004 and 2020. We compared patients who received SA chemotherapy with those who received MA concurrent chemotherapy. The primary outcomes of interest were overall survival and pCR. The groups were compared using univariate analysis and Cox proportional hazard models to adjust for potential confounding factors. **Results:** Of 39,667 patients with rectal cancer, 24,452 received single-agent (SA) chemotherapy and 14,215 received multi-agent (MA) chemotherapy. Patients in the SA group were of advanced age, had a greater number of comorbidities (as calculated by the Charleson-Dayo score), were treated within community facilities, had a higher income status, were covered by government insurance, were situated on the east coast, and had lower educational attainment compared to those in the MA group ($p<0.05$). SA group had higher T4 and N1 disease, however MA group had more N2 disease. Upon survival analysis, patient in MA group had higher survival (41.2 vs 38.9 months, $p=0.005$), however pCR was significantly higher in SA group (10.4% vs 6.4%, $p<0.001$). **Conclusions:** MA chemoradiotherapy for rectal cancer improves survival, with the SA regimen showing enhanced complete pathological response. Differences in baseline characteristics may explain the outcomes. While MA chemotherapy offers modest overall survival benefits, potential toxicity needs to be considered and confirmed in future studies. Research Sponsor: None.

**Comparison between single agent vs multi-agent chemo-radiotherapy group.**

| Outcome | Single-agent chemotherapy | Multiagent chemo-therapy | Chi Squared p-value |
|---|---|---|---|
| No readmission within 30 days of surgical discharge | 22,727 (89.3 %) | 12,921 (90.9 %) | <0.001 |
| 30-day mortality after surgery | 360 (1.5 %) | 59 (0.5 %) | <0.001 |
| 90-day mortality after surgery | 716 (3.0 %) | 158 (1.3%) | <0.001 |
| Median duration of surgical admission | 6.0 | 5.0 | <0.001 |
| Median survival-vital = deceased (months) | 38.9 | 41.2 | 0.005 |
| Median survival-vital = alive (months) | 66.9 | 52.0 | <0.001 |
| Pathological complete response (pCR) | 2,644 (10.4%) | 904 (6.4%) | <0.001 |

# Real-world (RW) treatment patterns and clinical outcomes of patients treated with sequential regorafenib and trifluridine/tipiracil ± bevacizumab (BEV) for metastatic colorectal cancer (mCRC) in the United States (US) community oncology setting: The SEQRT2 study.

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**Background:** Both Regorafenib (R) and Trifluridine/Tipiracil (T) ± BEV are widely used treatments for mCRC in later lines. This RW study evaluated optimal sequencing and associated outcomes among patients who initiated R and T in sequential lines of therapy (LOT) between LOT1-LOT6 within community oncology settings in the US. **Methods:** A retrospective study utilizing electronic medical record data from The US Oncology Network examined adult patients with mCRC who initiated R and T ± BEV sequentially between 09/01/2015 and 11/30/2022 (index date), with follow-up until 05/31/2023. Treatment patterns, overall survival (OS) and time to next treatment (TTNT) after sequence were assessed overall, R first (R-T) and T first (T-R). Multivariable Cox regression adjusting for age, sex, ECOG, tumor stage, and prior medication, assessed the association of factors with OS and TTNT. **Results:** In total, 308 mCRC patients initiating R and T (R-T, 156 pts [140 R-T and 16 R-T+BEV]; T-R, 152 pts [136 T-R and 16 T+BEV-R]) met study criteria. Baseline demographic (mean age 63 yrs, males 55%) and clinical characteristics (70% colon cancer, 58% ECOG 0-1) were similar between the cohorts. Most patients initiated index therapies at the third line of therapy (LOT3) (49% R-T, 45% T-R). The median (IQR) sequence duration was 7.4 (5.6), 6.7 (4.4), and 7.0 (4.7) months, respectively in patients with R-T, T-R, and overall. The median (95% CI) OS was 12.8 (11.2, 14.1) months among R-T and 10.2 (8.8, 11.9) months among T-R patients. Compared to R-T, the adjusted HR for the T-R sequence was 1.2 (95% CI 0.9–1.6; p=0.2). The median (95% CI) TTNT was 9.3 (8.4, 10.3) and 8.6 (7.8, 9.4) months, for R-T and T-R, respectively; the adjusted HR was 1.1 (95% CI 0.8–1.4; p=0.62). **Conclusions:** Patients initiating R first appeared to have a longer duration of treatment and survival, although not statistically significant. The one-year median duration of survival in this real-world setting confirms the importance of access to these treatments in patients with mCRC and can inform physicians regarding the choice of medications while sequencing therapies in mCRC. Research Sponsor: Bayer Pharmaceuticals.

**Clinical outcomes among R-T and T-R mCRC patients.**

| Variable | Level | R-T (± BEV) [N[1] = 156] | T (± BEV) -R [N[1] = 152] | Adjusted[3,4] HR[5] (95% CI) [R-T Reference] | Effect |
|---|---|---|---|---|---|
| **Overall Survival** | Months – median (95% CI) | 12.8 (11.2, 14.1) | 10.2 (8.8, 11.9) | 1.20 (0.91, 1.58) | 0.20 |
| **TTNT[2]** | Months – median (95% CI) | 9.3 (8.4, 10.3) | 8.6 (7.8, 9.4) | 1.07 (0.83, 1.37) | 0.62 |

[1]Entire R-T and T-R cohorts are presented since 16 patients in each cohort initiated T+BEV.
[2]Time To Next Treatment.
[3]Overall Survival adjusting for age, gender, stage at diagnosis, ECOG, prior anti-VEGF treatment.
[4]Time To Next Treatment adjusting for age, gender, stage at diagnosis, and line of therapy.
[5]Hazard Ratio.

# Effects of the delay of surgery in localized rectal cancer.

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**Background:** The treatment landscape for localized rectal cancer has evolved significantly in recent years, with long-course neoadjuvant chemoradiation (CRT) and total neoadjuvant therapy (TNT) emerging as the primary options. However, the optimal timing of surgery remains an important clinical question, particularly in resource-limited settings where surgical delays can occur. Data from previous studies have shown conflicting results. This retrospective study aimed to assess the impact of delaying surgery beyond the longer interval used in previous studies, focusing on its effects on Pathologic Complete Response (pCR), DFS, and OS. **Methods:** We conducted a retrospective cohort study using electronic medical records from three centers in Brazil between the years 2013-2024. The study included 236 patients with localized rectal cancer treated with neoadjuvant CRT and/or chemotherapy. Patients were divided into two pre-specified groups based on the interval between the end of neoadjuvant therapy and surgery: ≤11 weeks and >11 weeks. The primary endpoint was DFS. Secondary endpoints were pCR rates, and OS. Descriptive statistics were used for population analysis, while the Kaplan-Meier method estimated DFS and OS, with comparisons made using the log-rank test. A Cox proportional-hazards model was used to estimate hazard ratios and 95% confidence intervals. **Results:** The median age of the study population was 62 years, with males comprising 56.4% of the sample. The majority of the sample consisted of stage III patients, making up 72.2%, followed by stage II patients at 25.8%, and stage I patients at 1.9%. Most patients (77.8%) had surgery more than 11 weeks after completing neoadjuvant therapy, with a median time to surgery of 16 weeks. The pCR rates did not significantly differ between the two groups (29% for ≤11 weeks vs. 21.5% for >11 weeks; p = 0.271). After a median follow-up of 30 months, the median DFS was not reached, and there was no significant difference between the groups (HR: 0.86; 95% CI: 0.48-1.5; p = 0.63). Similarly, the median OS was not reached, and no difference was observed between the groups (HR: 0.85; 95% CI: 0.47-1.54; p = 0.88). **Conclusions:** Delaying surgery beyond 11 weeks did not result in greater tumor downstaging in patients with localized rectal cancer. Furthermore, DFS and OS were not significantly impacted by extended intervals between neoadjuvant therapy and surgery. Research Sponsor: None.

# A real-world study of treatment patterns among patients (pts) with metastatic colorectal cancer (mCRC) receiving third-line (3L) or fourth-line (4L) treatment in the US.

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**Background:** There are limited real-world data describing 3L and 4L treatments for pts with mCRC. However, the landscape is evolving with new options becoming available, including fruquintinib, a highly selective oral inhibitor of all 3 VEGF receptors, approved by the US FDA in November 2023. This study aimed to understand pt characteristics and real-world treatment patterns in 3L and 4L to characterize the standard of care, adherence to guidelines, and identify unmet needs among pts with mCRC in the US, prior to recent changes in the landscape. **Methods:** A retrospective study of adults with mCRC who had received ≥3L of therapy between 1/1/2018 and 3/31/2023 in the US-based Flatiron electronic health records database. In two cohorts, defined by initiating 3L and 4L systemic treatment, pt demographics, clinical characteristics, and treatment by line of therapy were summarized. **Results:** Overall, 5401 pts were included in the 3L cohort and 2868 pts in the 4L cohort: median age at baseline was 63 and 62 years (yrs), 56% and 56% were male, 43% and 43% had a *KRAS* variant, and 46% and 49% had liver metastasis, respectively. 96% of pts in the 3L cohort received standard chemotherapy (CT) +/- targeted therapy as 1L treatment (55% CT + anti-vascular endothelial growth factor [VEGF] therapy; 34% CT only; 7% CT + anti-epidermal growth factor receptor [EGFR] therapy). In 2L, 87% of pts had standard CT-containing regimens (53% CT + anti-VEGF therapy; 19% CT alone; 15% CT + anti-EGFR therapy). A similar treatment history was observed for the 4L cohort. Commonly used 3L regimens in the 3L cohort were trifluridine/tipiracil (TAS-102; 11%), regorafenib (rego; 10%), and standard CT-containing regimens (54%; including: 12% FOLFIRI + bevacizumab [bev], 7% FOLFOX + bev, 4% FOLFIRI, 3% FOLFOX, and 15% standard CT + anti-EGFR therapy). Approximately 47% (n=2513) of pts did not receive subsequent treatments after 3L. In the 4L cohort, use of TAS-102 (16%) and rego (15%) increased slightly in 4L, accompanied with a moderate decrease in CT-containing regimens (42%; including 6% FOLFOX + bev, 5% FOLFIRI + bev, 3% FOLFOX, 3% FOLFIRI, and 12% standard CT + anti-EGFR therapy). Approximately 4% of pts in the 3L and 4L cohorts received TAS-102 + bev in 3L and 4L, respectively. Pts receiving standard CT-containing regimens were slightly younger (<65 yrs: 3L 57%, 4L 58%) and had better Eastern Cooperative Oncology Group performance status (ECOG PS; 0-1: 3L 77%, 4L 77%) vs pts receiving TAS-102 (<65 yrs: 3L 54%, 4L 58%; ECOG PS 0-1: 3L 70%, 4L 68%) or rego (<65 yrs: 3L 50%, 4L 53%; ECOG PS 0-1: 3L 68%, 4L 70%). **Conclusions:** Characterization of treatment patterns in US pts from 1/2018 to 3/2023 showed that standard CT +/- targeted therapy was used frequently in 3L and 4L, demonstrating an unmet need for additional therapies. Approval of novel treatments, such as fruquintinib, may help to address this need. Research Sponsor: Takeda Development Center Americas, Inc. (TDCA), Lexington, MA, USA.

# Personalized frailty risk assessment in long-term survivors of colorectal cancer.

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**Background:** Frailty, a pathologic form of aging, is associated with reduced quality of life and increased risk of death. Its incidence increases with age. Frailty is a concern for cancer survivors, especially since this population is surviving longer and increasing in size. There is a need to predict which patients are at risk of frailty to tailor preventative measures, particularly in early-stage colon and rectal cancer survivors, the largest group in gastrointestinal cancer survivors. **Methods:** This was a retrospective cohort study of individuals aged 66 and older in the SEER-Medicare linked database, diagnosed with stage I-III colon or rectal cancer between 2003-2012. Patients included in the study received definitive surgical treatment and survived for at least 5 years after diagnosis. Frailty was assessed using administrative claims codes with the Kim frailty index. Patients already frail at year 5 were excluded from the analysis. An increase in frailty score, indicating the onset of frailty or worsening to moderate or severe frailty, occurring within 5-10 years following cancer diagnosis was the primary outcome. Predictors of frailty were identified using restricted mean survival time (RMST) regression, with results less than 1 indicating a shorter time to frailty, and significant factors were used to develop a clinical prediction model and stratify patients into risk tertiles. **Results:** At 10 years or end of available follow-up, 58% of the patients had developed new onset or worsening frailty. There were no significant differences in RMST by patient race, sex, cancer stage and grade. Receipt of systemic therapy 4-5 years after diagnosis, having an ostomy present in years 4-5, advancing age, comorbidities, and living in an area with a greater proportion of residents below the federal poverty line were all associated with shorter time to frailty; the largest effects were seen with advancing age and comorbidities (Table). Our clinical prediction model incorporated age>85 and having multiple comorbidities for both cohorts and ostomy in years 4-5 for the colon cohort. **Conclusions:** Our population level analysis provided a clinical prediction model for frailty 5-10 years after cancer diagnosis in colon and rectal cancer survivors and may help inform long-term survivorship management. Research Sponsor: American Cancer Society; RSG-21-039-01-CPPB.

**RMST for selected variables.**

| Selected Variables | RMST for Colon Cohort (95% CI) | RMST for Rectal Cohort (95% CI) |
|---|---|---|
| **Age 85-89 (ref: age 71-74)** | 0.76 (0.73, 0.79) | 0.79 (0.74, 0.84) |
| **Age 90+ (ref: age 71-74)** | 0.67 (0.64, 0.70) | 0.74 (0.66, 0.82) |
| **20-100% area-level poverty (ref: 0-<5%)** | 0.95 (0.92, 0.98) | 0.96 (0.91, 1.02) |
| **Receiving systemic therapy in years 4-5** | 0.89 (0.85, 0.92) | Not selected |
| **Ostomy in years 4-5** | 0.80 (0.69, 0.92) | 0.97 (0.92, 1.02) |
| **Elixhauser Comorbidity Index (ECI) 1-2 (reference: ECI 0)** | 0.77 (0.75, 0.79) | 0.79 (0.76, 0.82) |
| **ECI 3 (ref: ECI 0)** | 0.60 (0.57, 0.64) | 0.67 (0.59, 0.76) |

# Predictors of late recurrence of colorectal cancer (CRC) among older long-term survivors using claims-based evidence.

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**Background:** There are established features that predict early recurrence of CRC in the first 5 years from diagnosis, but predictors of late recurrence 5-10 years after diagnosis are less well understood. Long-term CRC survivors require multidisciplinary follow-up accounting for both cancer and non-cancer related conditions. Identifying predictors of cancer recurrence after 5 years of disease-free survival may help inform survivorship care. **Methods:** This was a retrospective cohort study of patients aged ≥66 in the SEER-Medicare database with non-metastatic CRC diagnosed between 2003 and 2011. As SEER does not collect recurrence data, we used a validated algorithm to identify treated recurrence in Medicare claims. Patients with early recurrence or a second primary malignancy in years 0-5 were excluded. Demographic and cancer-specific clinical data were collected at time of cancer diagnosis. Comorbidities were identified between 4 -5 years after cancer diagnosis and recurrence was modeled starting at year 5 using restricted mean survival time (RMST) regression. Estimates from RMST models were converted to RMST ratios (ratio <1 indicates shorter time to event). The Fine-Gray model was used to estimate cumulative incidence of CRC recurrence or mortality between 5 and 10 years after cancer diagnosis. **Results:** 20,255 patients with non-metastatic CRC were included: 16,158 with colon cancer (33.4% stage I, 41.7% stage II, 25.0% stage III), and 4,097 with rectal cancer (43.1% stage I, 30.1% stage II, 26.7% stage III). Mean age at cancer diagnosis was 77 years. In those who remained disease free five years after initial diagnosis, late recurrence developed in 4.8% (973/20,255) of the combined CRC cohort, 4.2% (679/16,158) of those with colon cancer, and 7.2% (294/4,097) of those with rectal cancer. In colon cancer, stage III disease was associated with a shorter time to recurrence (RMST ratio = 0.97, CI = 0.96-0.98). In both colon (RMST ratio = 0.98, 95% CI 0.96-0.99) and rectal cancer (RMST ratio = 0.97, 95% CI 0.94-0.99) having ≥3 comorbidities was associated with a shorter RMST, consistent across disease stages. For colon cancer, the cumulative incidence of CRC-specific mortality vs. all-cause mortality in years 5-10 from diagnosis was 3% (95% CI 2.8-3.0) vs. 35% (95% CI 34.0-36.0), respectively. For rectal cancer, during the same period, the corresponding rates were 6% (95% CI 5.4-7.0) vs. 30% (95% CI 28.4-31.0). **Conclusions:** Late CRC recurrence is rare in the SEER-Medicare population. Primary risk factors are advanced stage at diagnosis and multiple co-morbidities. Patients who survive 5 years from their diagnosis have a higher risk of all-cause mortality compared to CRC-specific mortality. Interpretation of data is limited by the validated algorithm used to detect recurrence, as it may underestimate recurrence in patients who did not undergo treatment or workup for recurrent disease. Research Sponsor: American Cancer Society.

# Survival outcomes for right- versus left-sided colon cancer and rectal cancer in patients receiving regorafenib and/or trifluridine/tipiracil for refractory metastatic colorectal cancer: Findings from the multicenter retrospective ReTrITa study.

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**Background:** Several studies show that patients (pts) with colon cancer, particularly pts with right colon cancer (RCC), have lower survival rates than those with rectal cancer. This argues that colorectal cancer should be divided into three clinical entities: RCC, left colon cancer (LCC) and rectal cancer (RC). In refractory metastatic colorectal cancer (mCRC), regorafenib (R) and trifluridine/piracil (T) were found to improve survival. This real-life subgroup survival analysis focused on treatment with R and T, sequential or not, according to the above three primary tumor sites. **Methods:** Clinical data of pts diagnosed with mCRC and treated with R and/or T between 2012 and 2023 were retrospectively collected at 17 Italian cancer centers. The purpose of this analysis was to compare the overall survival (OS) and progression-free survival (PFS) of pts who received R and/or T as third line and beyond, according to their primary tumor site. **Results:** The entire ReTrITA study enrolled 1156 patients. 261 (22.5%) of them received the T/R sequence (T/R), 155 (13.4%) the reverse sequence (R/T), 427 (37%) T, and 313 (27.1%) R. For the purpose of this subgroup analysis we identified 381 RCC pts (32.9%), 531 LCC pts (45.9%) and 244 RC pts (21.2%). When comparing the groups treated with the sequential treatment, we observed a statistically significant OS improvement for pts receiving R prior to T: 17,4 vs. 12,2 months (mos) in RCC pts (HR = 0,67; 95% CI = 0,46-0,99; P = 0,0461); 16,1 vs. 15,1 mos in LCC pts (HR = 0,71; 95% CI = 0,51-1,00; P = 0,0559) and 16,6 vs. 11,4 mos in RC pts (HR = 0,57; 95% CI = 0,36-0,89; P = 0,0153). R/T sequence was also found to provide a considerable PFS benefit: 11,8 vs. 7,8 mos in RCC pts (HR = 0,53; 95% CI = 0,37-0,76; P = 0,0006); 11,7 vs. 8,9 mos in LCC pts (HR = 0,64; 95% CI = 0,47-0,88; P = 0,0071) and 10,7 vs. 8,2 mos in RC pts (HR = 0,63; 95% CI = 0,42-0,96; P = 0,0342). The non-sequential administration of R and T had no statistically significant impact on survival outcomes in the three groups. **Conclusions:** Our real-world subanalysis suggests that the R/T sequence significantly increases survival for pts with primary colon and rectal cancer, even after third-line treatment. Examining the pts groups treated with R and T as monotherapy based on the primary tumour site, however, did not allow us to make any significant conclusions. Treatment decisions should, however, constantly include the patient's characteristics, such as sex, ECOG PS, and the extent of metastatic disease. However, in order to verify our findings, prospective studies are required. Research Sponsor: None.

# Vascular endothelial growth factor receptor (VEGFR) -tyrosine kinase inhibitors combined with programmed death receptor-1 (PD-1) inhibitors as third- or later-line treatment in patients with microsatellite-stable (MSS) metastatic colorectal cancer (mCRC): A retrospective study.

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**Background:** Microsatellite-stable (MSS) metastatic colorectal cancer (mCRC) shows a low response to programmed death receptor-1 (PD-1) inhibitors. Recent studies suggest that combining vascular endothelial growth factor receptor (VEGFR)-tyrosine kinase inhibitors (TKIs) with PD-1 inhibitors may enhance the anti-cancer activity for mCRC patients. Multiple antiangiogenic drugs have been approved by the FDA and NMPA for the treatment of later-line mCRC. Nevertheless, the effects of various anti-angiogenic drugs when used in combination with immunotherapy for mCRC treatment is not yet fully established. **Methods:** We conducted a non-interventional, retrospective study. Data were compiled from patients with mCRC at Shandong Cancer Hospital. Eligible patients must have received a combination of anti-angiogenic drugs and PD-1 inhibitors following at least two lines of standard treatment. The treatment period spanned from August 2019 to April 2024. The primary endpoint was overall survival (OS), and the secondary endpoint was progression-free survival (PFS). **Results:** At the cut-off date (April 29th, 2024), 79 eligible patients were available for analysis. Of these, 49 patients received fruquintinib in combination with PD-1 inhibitors (F group), 30 patients received investigator-selected other anti-angiogenic agents in combination with PD-1 inhibitors (NF group). The baseline characteristics of the two groups were well balanced. The median overall survival (OS) was significantly longer in the F group compared to the NF group (hazard ratio [HR]: 0.36, 95% CI 0.14-0.96, P=0.042). The median OS was not reached (95% CI 21.1-NA) in the F group, whereas it was 19.1 months (95% CI 12.6-NA) in the NF group. Subgroup analyses of OS were generally consistent with benefit in the whole population across nearly all subgroups. In contrast to NF group, individuals in F group demonstrated a significant enhancement in overall survival (OS) when patients had received prior bevacizumab (HR: 0.26, 95% CI 0.07-0.75, P=0.015), had no prior treatment with cetuximab (HR: 0.23, 95% CI 0.26-0.97, P=0.045), or had experienced surgical resection at baseline (HR: 0.34, 95% CI 0.12-0.95, P=0.041). The median PFS between two groups showed no statistically significant difference (HR: 0.65, 95% CI 0.38-1.13, P=0.127), F group 6.1 months (95% CI 4.6-10.3) versus NF group 4.4 months (95% CI, 3.0-7.5). **Conclusions:** Fruquintinib demonstrated improved OS over other anti-angiogenic agents when combined with immunotherapy for third or later-line MSS mCRC treatment. Research Sponsor: None.

# A multi-institutional retrospective study evaluating clinical outcomes and prognostic factors in patients with metastatic appendiceal adenocarcinoma.

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**Background:** Appendiceal adenocarcinoma (AA), a rare cancer, has been reported to have high incidences of mucinous type histology, peritoneal metastasis, and relatively good prognosis. However, its chemotherapeutic efficacy and prognostic factors is not well known. **Methods:** We collected patients with metastatic AA treated with chemotherapy between February 2005 and March 2023 in two high volume cancer centers in Japan. We retrospectively analyzed the clinical outcomes of the chemotherapy of patients with metastatic AA and the clinical factors associated with treatment efficacy and prognosis by multivariate analysis. **Results:** A total of 66 patients with a median age of 54 years, were enrolled. 40 patients (60.6%) received an oxaliplatin-based regimen in combination with bevacizumab. The response rate was 22.0% and the disease control rate was 87.8%. With median follow-up period was 15.5 months [1.9-103.1], median progression-free survival (PFS) was 10.9 months [95% CI: 7.2-15.2] and median overall survival (OS) was 23.0 months [95% CI: 19.6-35.8]. 77.4% of patients received subsequent chemotherapy. In multivariate analysis, pretreatment carcinoembryonic antigen (CEA) and/or carbohydrate antigen 19-9 (CA19-9) were associated with significantly shorter PFS (HR: 2.93, $P$ = 0.0049). Presence of primary tumor (HR:2.50, $P$ = 0.015), ECOG performance status (PS) ≥1 (HR: 2.67, $P$ = 0.0091), moderate to massive ascites (HR: 3.06, $P$ = 0.0043) and high tumor marker levels (HR: 2.46, $P$ = 0.026) were associated with significantly shorter OS. Classifying the patients by number of these prognostic factors for OS into three groups (0, 1-2, and 3-4), the group with prognostic factors (1-2 or 3-4) had a significantly shorter prognosis compared to the group without any these factors (0 vs 1-2: HR: 4.70, $P$ = 0.038, 0 vs 3-4: HR: 17.6, $P < 0.01$). In addition, the group with 3-4 prognostic factors had relatively lower rate of subsequent chemotherapy ratio than the group without these factors (66.7% vs 100%, $P$ = 0.25). **Conclusions:** We confirmed our treatment outcomes were consistent with previous studies and elevation of tumor markers presence of primary tumor, ECOG PS ≥1, moderate to massive ascites were significantly associated with survival in patients with metastatic AA. Research Sponsor: None.

# Real world characteristics and outcomes of patients with BRAFV600E-mutant metastatic colorectal cancer in Australia: The COALA project.

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**Background:** BRAFV600E-mutant metastatic colorectal cancer (mCRC) represents a unique molecular subset with poor prognosis and less responsiveness to chemotherapy. The BEACON CRC study demonstrated that encorafenib plus cetuximab (EC) significantly improved overall survival (OS) compared to FOLFIRI plus cetuximab in BRAF-mutant mCRC as 2$^{nd}$ or 3$^{rd}$ line therapy. This combination was reimbursed in Australia from January 2022. **Methods:** We analysed data from the Australian TRACC mCRC clinical registry, encompassing patients (pts) with BRAF-mutant mCRC diagnosed in Australia from 2009 to 2024 at 31 hospitals. We assessed baseline demographics, uptake of EC following reimbursement and efficacy analysis of progression-free survival (PFS) and OS for pts that received EC and those who did not. Pts receiving EC on a clinical trial were excluded. **Results:** Among 2,800 pts with known BRAF mutation status, 365 (13%) were BRAF mutated. The BRAF mutant cohort had a median age of 63 years, with a significantly higher mutation rate in the 20-39 age group than other age groups (age 20-39, 25.5%; age 40-59, 10%; age 60+, 13.2%; $P<0.001$); 54.5% were female, 58% right-sided tumours, 64% with synchronous metastatic disease and 21.7% deficient mismatch repair (dMMR). Overall, 314 (86%) pts with BRAF mutant tumours and 2,079 (85%) BRAF wild-type (WT) tumours received 1$^{st}$ line treatment, and 156 (43%) BRAF mutant and 1,144 (47%) BRAF WT pts received 2$^{nd}$ line therapy. The median OS from diagnosis was 17.3 months for BRAF mutant pts versus 34.7 months for BRAF WT pts ($p < 0.0001$). All dMMR BRAF mutant pts received 1$^{st}$ line pembrolizumab since its reimbursement in August 2021. Among 41 BRAF mutant pts progressing on 1$^{st}$ line therapy since January 2022,(32 pMMR, 9 dMMR), 33 (80%) were subsequently treated with EC, and 4 (10%) did not receive further treatment. OS from the start of 2$^{nd}$ line therapy was 9.6 months (95% CI: 7.2-12.5) for pts who received EC and 7.1 months (95% CI: 6.4-8.6) for those who received other systemic treatment. Median PFS for pts receiving EC was 4.6 months (95% CI: 3.4-5.8) and 3.3 months (95% CI: 2.5-5.3) for other 2$^{nd}$ line treatments. **Conclusions:** mCRC pts younger than 40 have a high rate of BRAF mutation, where 1 in 4 tumours harbour the mutation. BRAF mutant mCRC has a poorer outcome than BRAF WT despite a similar proportion receiving 1$^{st}$ and 2$^{nd}$ line treatment. The majority of eligible BRAF mutant mCRC pts received EC following reimbursement in Australia, with PFS and OS outcomes consistent with the BEACON CRC trial. Research Sponsor: None.

71 Poster Session

# Aflibercept (AFL) vs. ramucirumab (RAM) -based therapy in patients with advanced colorectal cancer: Efficacy comparison using real-world data evidence from TriNetX platform.

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**Background:** FOLFIRI in combination with Aflibercept (AFL) or Ramucirumab (RAM) prolongs overall survival compared with FOLFIRI alone in patients with advanced colorectal cancer (aCCR) as second-line therapy, however, there are no comparative studies between them. **Methods:** Utilizing the TriNetX Global Collaborative Network, a platform that operates globally based on anonymized and aggregated clinical data, a sample of aCCR patients from 128 healthcare organizations (HCOs) who met the initial criteria was selected. 5-year overall survival (OS) was analyzed between these cohorts using a Kaplan-Meier analysis. Hazard Ratio (HR) and its 95% confidence interval (95%CI) were calculated to evaluate the difference between cohorts. Propensity Score Matching (PSM) was used to balance the cohorts based on age, gender, and race mitigating possible cofounding variables. All statistical analyses were conducted utilizing the TriNetX Analytics function in the online research platform. **Results:** A total of 1046 patients met the study criteria and were included in our study. 523 received AFL and 523r received RAM-based therapy. Patients treated with AFL showed a significant better OS compared to patients treated with RAM, both previous to PSM and before PSM (post-PMS median OS 430 days vs 317 days, HR 0.797, 95%CI 0.683-0.930). **Conclusions:** In this study, based on Real World data, Aflibercept showed superior results to Ramucirumab in advanced colorectal cancer patients treatment. To date this is the largest study comparison between this two therapeutic options. Research Sponsor: None.

# Patient-reported outcomes in physical, cognitive, and sexual health in early-onset gastrointestinal cancers (EOGIC).

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**Background:** Incidence of early-onset (ages 18-49) gastrointestinal cancers is alarmingly increasing. To assess unmet needs in health-related quality of life (HRQoL), we conducted this pilot study using PRO measures in patients with EOGIC. **Methods:** Patients ages 18-49 with any GI cancer seen at our center during a clinic visit were invited to participate in this cross-sectional study. Validated PRO tools PROMIS 29+2, GP5 from FACT-G, AYA Sexual Health PRO Battery and Adult Neuro-QOL Cognitive Function short form were administered. Participants' PROMIS 29+2 responses were scored to generate a T score, and two-tailed t-tests were used to compare T scores with the average score from a large general US population. The electronic medical record was reviewed to collect demographic, diagnosis and treatment information. **Results:** 39 (85% response rate) patients were enrolled from 03-08/24. Table 1 describes demographics, diagnoses, and treatment. In the PROMIS 29+2, EOGIC patients reported significantly lower physical function and higher anxiety, depression, fatigue and pain interference compared to the general US population: t-score (T-test p-value) 40.4 (p 0.0003), 57.7 (p 0.005), 55.7 (p 0.017), 55.1 (p 0.033) and 58.5 (p 0.0001), respectively. 56% reported being at least somewhat "bothered by side effects of treatment," and this measure has a known correlation with clinician reported adverse events. Regarding sexual health (SH) in prior 30 days, 79% of participants felt less whole or "damaged" due to their disease, 64% were sexually active, 56% had some distress due to change in SH, and 50% were worried about their romantic relationship(s). Patients identified discussing safe sex with low counts, body image, contraception and addressing sexual problems as areas of unmet need during their care. They reported not receiving counseling from their care teams: only 15% discussed safe sex with low counts, 28% contraception, 13% STI prevention, and 18% discussed effect of cancer on body image. No significant change in cognitive functioning over the preceding 7 days was reported in this study. **Conclusions:** PROs are an important tool in clinical and survivorship care of young patients with GI cancers. We discovered that decreased physical function, increased anxiety, depression, fatigue and pain significantly effect HRQoL patients with EOGIC. We also found high unmet need to address sexual health in patients with EOGIC. Our ongoing work aims to better understand and address these needs for our patients. Research Sponsor: Bucksbaum Institute for Clinical Excellence, University of Chicago.

**Demographic, diagnostic, and therapeutic characteristics of participants.**

| Age | At study participation | 24-49 | median 43 |
|---|---|---|---|
| **Gender** | Female | 26 | 66.7% |
| **Primary cancer** | Colorectal | 19 | 48.7% |
| | Pancreatic | 5 | 12.8% |
| | Biliary tract cancers | 5 | 12.8% |
| | Esophageal/Gastric | 3 | 7.7% |
| | Other (including anal, NET, GIST, appendiceal) | 7 | 17.9% |
| **Stage** | Metastatic | 22 | 56.4% |
| **Treatment** | On any active treatment | 24 | 61.5% |
| | On chemotherapy | 20 | 51.2% |

73 Poster Session

# Real-world survival comparison of second-line treatment strategies for patients with RAS/RAF wild-type, right-sided metastatic colorectal cancer.

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**Background:** For patients with RAS/RAF wild type metastatic colorectal cancer (mCRC), systemic therapy usually includes fluoropyrimidine-based chemo + anti-VEGF or anti-EGFR targeted therapy. The PARADIGM trial established tumor sidedness as an important predictive biomarker for patients with RAS/RAF wild-type disease; in right-sided tumors, there was no survival difference between first-line chemo + anti-EGFR vs. chemo + anti-VEGF. However, there is no comparative data to guide second-line treatment decisions in this population. We aim to compare the effectiveness of second-line chemo + anti-EGFR versus chemo + anti-VEGF therapy for patients with RAS/RAF wild-type, right-sided mCRC who received first-line chemo + anti-VEGF. **Methods:** We used the nationwide Flatiron Health electronic health record-derived database, comprising de-identified patient-level structured and unstructured data obtained from ~280 cancer clinics and curated via technology-enabled abstraction. Patients ≥18 years old with RAS/RAF wild-type, right-sided mCRC who received first-line therapy consisting of chemotherapy ((FOLFIRI or FOLFOX or CAPEOX) + anti-VEGF and initiated second-line chemotherapy with either anti-EGFR or anti-VEGF targeted therapy between January 2013-May 2024 were included. Multiple imputation with chained equations imputed missing values for sidedness, RAS/RAF status, and propensity score covariates (age, gender, year of diagnosis, synchronous/metachronous disease, MMR/MSI status, ECOG score, CEA level, and first-line therapy duration). Cox proportional hazards modeling with stabilized inverse probability of treatment weighting (IPTW) assessed the association of anti-EGFR vs anti-VEGF treatment with overall survival. **Results:** 4,444 patients received appropriate first-line treatment and received chemo + anti-EGFR or chemo + anti-VEGF in the second line. Across 25 imputations, an average of 444 patients met inclusion criteria: 175 patients received chemo + anti-EGFR and 269 patients received chemo + anti-VEGF. Following IPTW, baseline characteristics were balanced between treatment groups. Patients who received chemotherapy + anti-EGFR had a 24% increased hazard of death when compared with patients who received chemotherapy + anti-VEGF, though this was not statistically significant (HR 1.24, 95% CI 0.96 – 1.61, p=0.097). **Conclusions:** Among patients with RAS/RAF wild-type, right-sided mCRC who received first-line chemotherapy + anti-VEGF, there is some evidence to support continuing anti-VEGF therapy vs. switching to anti-EGFR therapy in the second line, though the result was not statistically significant. Future studies should explore predictive biomarkers for EGFR vs VEGF-directed treatment for this population to determine the patients most likely to benefit from anti-EGFR therapy during their disease course. Research Sponsor: None.

# Physical activity and sedentary behavior patterns after early-onset colorectal cancer diagnosis.

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**Background:** Sedentary lifestyle and physical inactivity are established risk factors for colorectal cancer diagnosed among adults younger than age 50 years (early-onset CRC). However, patterns of physical activity (PA) and sedentary behavior (SB) in patients after an early-onset CRC diagnosis are less understood. **Methods:** The Reproductive Health After Cancer Diagnosis and Treatment (REACT) Study is a patient-partnered, cross-sectional study in cooperation with 23 community partners and patient advocates that recruited adults with a first primary cancer (diagnosis age: 18-49 years) over an 8-week period (October to December 2021). Patient-reported sociodemographic, cancer and cancer impact, PA and SB questions were prospectively collected via questionnaire. Frequency, duration and intensity of leisure-time, work-related and transportation-related PA were summed as minutes per week (min/wk). Time spent on SB was calculated by hours spent sitting on a typical day. Diagnosis age- and year-adjusted Spearman correlation coefficients were calculated between PA and SB by cancer treatment status. **Results:** A total of 131 patients were diagnosed with a first primary early-onset CRC between 2016 and 2021 (median [IQR] age, 40.0 [35-45] years; 82.4% female) in REACT. One hundred thirteen patients (86.3%) were concerned about their physical fitness or getting enough exercise after cancer diagnosis—of whom 62.8% (n=71 of 113) were somewhat or very concerned about their physical fitness/exercise. Among 80 patients with early-onset CRC and complete PA/SB data (86.3% female), 28.8% were not adherent to PA guidelines for aerobic activity (≥150 min/wk) and 56.3% reported long sedentary time (>6 hr/day). The proportion of patients across PA/SB groups also statistically significantly differed by treatment status among those diagnosed between 2019 and 2021 (p=0.049), but not between 2016 and 2018 (p=0.25). While PA was not significantly correlated with SB for early-onset CRC patients who were not actively undergoing treatment (r=-0.16, $p$=0.32), there was a statistically significant moderate inverse correlation between PA with SB among early-onset CRC patients under active cancer treatment (r=-0.58, $p$=0.004). **Conclusions:** One in every 2 early-onset CRC patients reported long sedentary time and one in every 4 early-onset CRC patients in this study did not meet the Physical Activity Guidelines for Americans. With over 80% of patients expressing concern about their physical activity and exercise after an early-onset CRC diagnosis, and differences in PA/SB patterns observed by treatment status, these findings warrant prospective, longitudinal studies of objectively-measured PA and SB among young patients in order to support tailored strategies (e.g., exercise prescriptions) that improve patient health and outcomes across the CRC continuum. Research Sponsor: American Cancer Society; IRG-19-139-59; National Cancer Institute/U.S. National Institutes of Health; P50 CA236733; National Institutes of Health/Eunice Kennedy Shriver National Institute of Child Health and Human Development; K12 HD043483.

# Patterns of metastatic spread and survival outcomes for stage IV left-sided colon vs. rectal cancers: A real-world analysis.

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**Background:** A growing body of literature has reported differences in biology, metastatic potential, and outcomes among patients with colorectal cancer based on the sidedness of their primary tumor (right vs. left). Left-sided primary tumors have comprised both colon and rectal tumors, despite differences in venous and lymphatic drainage patterns and treatment paradigms. The present work sought to investigate patterns of metastases and survival outcomes among patients with left sided colon primary tumors and rectal primary tumors. We hypothesized that these two disease sites metastasize differently and exhibit different survival patterns based on metastatic site. **Methods:** The National Cancer Database (NCDB) was queried for all patients with a diagnosis of metastatic left-sided colon (splenic flexure, descending colon, rectosigmoid) or rectal adenocarcinoma with known primary site and known metastatic site(s) between 2010 and 2021 using parameters outlined in the NCDB Data Dictionary. Patients with left-sided colon primaries (LP) were compared to patients with rectal primaries (RP) along clinicodemographic parameters. Incidence of metastases to the liver, lungs, bone, brain, and non-regional lymph nodes were also compared. Multivariable survival analyses were performed to analyze differences in overall survival (OS). **Results:** Among 84,958 patients, 51,931 (61.1%) had LP while 33,027 (38.9%) had RP. Patients with RP were less likely to have liver metastases than those with LP (79.4% vs. 90.3%), but more likely to have lung (40.7% vs. 29.2%), bone (8.3% vs. 6.0%), and distant lymph node metastases (11.3% vs. 8.6%) (all $p<0.001$). Compared to patients with LPs, patients with RP were less likely to undergo surgery to the primary tumor (25.1% vs. 46.3%) or any metastatic site (15.3% vs. 18.8%), more likely to receive chemotherapy for metastatic disease (80.8% vs. 75.2%), and more likely to receive radiation to the primary site (17.7% vs. 4.3%) (all $p<0.001$). Additionally, patients with RP had longer median OS than those with LP (27.3 months vs. 26.4 months, $p<0.001$). On multivariable analysis controlling for age, gender, tumor grade, RAS mutation status, MSI status, CEA level, clinical T and N category, and treatment details, patients with RP with liver metastases (HR 0.91, 95% CI 0.89-0.92), lung metastases (HR 0.85, 95% CI 0.82-0.87), and bone metastases (HR 0.89, 95% CI 0.84-0.95) (all $p<0.001$) demonstrated improved OS compared to those with LP. **Conclusions:** Patterns of metastases, treatment regimens, and, in certain instances, survival outcomes, differ among patients with stage IV left-sided colon cancer and rectal cancer. Future work evaluating the implications of sidedness in colorectal cancer as well as future staging efforts should consider assessing left-sided colon cancer separately from rectal cancer, as these demonstrate distinct clinical behavior and outcomes. Research Sponsor: None.

# Treatment sequences in patients with metastatic colorectal cancer in Japan: Real-world evidence of 1st- to 5th-line treatments.

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**Background:** The development of later-line drugs for metastatic colorectal cancer (mCRC) has expanded treatment options for systemic chemotherapy in patients with mCRC. However, few studies have investigated patients who received later-line treatment in a real-world setting, and available information on treatment sequences and transition rates from early- to later-line treatments is limited. **Methods:** This was a retrospective study using hospital administrative data collected by MDV (Medical Data Vision, Tokyo, Japan) from April 2008 through February 2023 in Japan. Treatment regimens were derived from standard treatments recommended in the Japanese guidelines. The study population was determined using an algorithm from a similar study previously conducted in Japan. Our analysis focused on patients who started CRC surgery or 1st-line treatment after January 2017 to reflect recent late-line real-world practice. Transition rates from 1st- to 5th-line treatment were calculated, and treatment sequences were summarized using a Sankey diagram. Logistic regression was performed to identify the factors associated with the transition from 2nd- to 3rd-line treatment. **Results:** A total of 722,005 patients with a CRC diagnosis record were identified in the MDV database. Of these, 26,456 patients were included in the study population. The median age was 69 years and 61.4% of patients were male. The rates of transition to subsequent lines from 1st to 5th line ranged from 65.7% to 70.4% (1st to 2nd 68.3%; 2nd to 3rd 70.4%; 3rd to 4th 69.9%; 4th to 5th 65.7%). Among 8528 patients who received 2nd-line treatment, ~70% of patients (6006) continued 3rd-line treatment, and 30% of patients (2522) received best supportive care. Comparing these 2 patient groups, younger age (<65 years, OR: 1.50, 95%CI: 1.35-1.67; $p<0.001$) and longer treatment durations of 1st (≥180 days, OR: 1.26, 95%CI: 1.14-1.39; $p<0.001$) and 2nd (≥120 days, OR: 1.67, 95%CI: 1.52-1.84; $p<0.001$) lines were significant factors for 3rd-line treatment continuation. Prior therapy with oxaliplatin or irinotecan plus molecular targeted drugs was also associated with a higher likelihood of proceeding to 3rd-line treatment (OR: 1.42, 95%CI: 1.28-1.58; $p<0.001$). **Conclusions:** In this study, ~70% of patients were able to transition to subsequent lines at the time of each line transition. However, 30% of patients were unable to continue treatment in real-world practice. To increase treatment continuation rates and maintain patient quality of life, new treatment options and further research are necessary to meet patients' treatment needs, especially in the later-line treatment setting where unfavorable effects from previous treatments are accumulated. Research Sponsor: Takeda Pharmaceutical Company Limited.

# Bevacizumab plus mFOLFOX6/FOLFIRI sequential bevacizumab plus capecitabine maintenance in RAS-mutated advanced colorectal cancer: A real-world study of first-line/second-line conversion therapy.

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**Background:** Currently, the standard first-line/second-line treatment for advanced colorectal cancer with RAS mutations is chemotherapy combined with bevacizumab. Patients with colorectal cancer confirmed maintenance therapy can effectively improve progression-free survival, improve patient quality of life. Therefore, this study aims to evaluate the efficacy and safety of bevacizumab combined with mFOLFOX6/FOLFIRI sequential bevacizumab combined with capecitabine as first-line/second-line conversion therapy for RAS-mutated advanced colorectal cancer. **Methods:** This is an open-label, single-center study in previously untreated patients with RAS mutant mCRC. Eligible patients received first-line chemotherapy of bevacizumab (5mg/m$^2$) combined with mFOLFOX6/FOLFIRI for 12 cycles randomly, during which the efficacy was assessed every 4 cycles, then followed by bevacizumab(7.5mg/m$^2$) in combination with capecitabine maintenance therapy which assessed every 2 cycles until disease progression, and then switched to the second-line therapy FOLFIRI/mFOLFOX6 combined with bevacizumab which was assessed every 4 cycles until disease progression or intolerable toxicity. Imaging studies were performed to assess treatment efficacy according to RECIST1.1 criteria, and the primary endpoint was progression-free survival (PFS), including first-line PFS and second-line PFS after progression, and the secondary endpoints included disease response rate (ORR), disease control rate (DCR), overall survival (OS), and safety. **Results:** This study recruited 78 subjects from January 2021 to May 2024. As of September 10, 2024, a total of 70 subjects were analyzable for the first-line treatment, of which 6 subjects met the surgical criteria after treatment and achieved NED status; 45 patients reached the first-line PFS endpoint, with a median PFS (mPFS) of 249 days (95% CI: 204.4-293.6), an ORR of 38.6%, and a DCR of 95.7% . Twenty patients entered the first-line maintenance therapy, with a mPFS of 358 days (95% CI: 53.8-662.2). For the second-line treatment, a total of 43 subjects were analyzable, of which 35 patients reached the second-line PFS endpoint, with a second-line mPFS of 171 days (95% CI: 89.8-252.1). 41 subjects underwent at least one efficacy evaluation, with ORR of 14.6% and DCR of 75.6%. The most common grade 3 treatment-related adverse events were neutropenia, leukopenia, anemia, hypertension, diarrhea, and thrombocytopenia. There were no treatment-related deaths. No new adverse events occurred during the entire combination treatment period. **Conclusions:** This real world study has demonstrated good tolerability and encouraging clinical efficacy, especially with potential implications for treatment planning and management strategies for advanced colorectal cancer with RAS mutations. Clinical trial information: ChiCTR2100042553. Research Sponsor: None.

# Comparative efficacy of TAS-102 and chemotherapy rechallenge in the third-line setting and beyond in patients with advanced colorectal cancer: A retrospective study.

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**Background:** The optimal approach to third line therapy for patients with metastatic colorectal cancer remains a subject of debate. Options include Trifluridine/Tipiracil (TAS-102), TAS-102 with bevacizumab, regorafenib, chemotherapy rechallenge with or without an anti-EGFR agent, or fruquintinib. This study aims to evaluate the efficacy of standard of care TAS-102, TAS-102 with bevacizumab, or regorafenib versus chemotherapy rechallenge in patients with metastatic colorectal cancer in the third line setting and beyond. **Methods:** We conducted a retrospective analysis of patients with metastatic colorectal cancer who received either TAS-102 with or without bevacizumab, regorafenib or chemotherapy rechallenge with or without an anti-EGFR agent between 2018 and 2023, at the Ministry of National Guard for Health Affairs in Riyadh, Saudi Arabia. Patients received such agents in the third line setting and beyond. **Results:** A total of 75 patients were included in this study, with 39 patients receiving TAS-102 with or without bevacizumab, or regorafenib, and 36 receiving chemotherapy rechallenge. The median patient age was 58 years (range 28-81), with the majority being males at 55%. Among the chemotherapy rechallenge group, 31% were treated with cetuximab and 6% had panitumumab rechallenge, some of which were guided by cell free DNA liquid biopsy testing. In the other group, 66% received regorafenib, 26% TAS-102 alone, and 8% TAS-102 with bevacizumab. At the time of analysis, 56% of patients had passed away. Median overall survival for patients who received TAS-102 with our without bevacizumab, and regorafenib in the third line setting was found to be 13.1 months. Whereas for patients who received chemotherapy rechallenge, their median overall survival was 19.5 months. **Conclusions:** Patients with metastatic colorectal cancer who received chemotherapy rechallenge in the third line setting survived longer compared with patients who received TAS-102 with or without bevacizumab, or regorafenib.
Research Sponsor: None.

# Real-world clinical outcomes of patients (Pts) with metastatic colorectal cancer (mCRC) who received trifluridine-tipiracil (FTD-TPI) monotherapy or FTD-TPI + bevacizumab (FTD-TPI+bev) combination therapy.

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**Background:** FTD-TPI+bev has shown survival benefit when compared to FTD-TPI alone in pts with mCRC and was approved in the U.S. in 2023. However, there is a lack of real-world data on outcomes for pts who received FTD-TPI+bev as compared to pts who received FTD-TPI. **Methods:** This retrospective study used data abstracted from electronic medical records and claims in the ConcertAI RWD360 dataset. Adult pts with a diagnosis of mCRC and exposure to FTD-TPI were included. Pts were categorized as having received FTD-TPI or FTD-TPI+bev based on first exposure to FTD-TPI (index date). Kaplan-Meier analyses were used to describe the overall survival (rwOS), real-world time to discontinuation (rwTTD), and time to next treatment or death (rwTTNTD) from the index date. Multivariate Cox regression analyses were used to control for patient characteristics, including demographics, ECOG performance status (PS), comorbidities, sites of metastatic disease, time from mCRC to index date, lab results, and receipt of prior regorafenib. **Results:** This study included 3,680 pts. The FTD-TPI cohort included 3,151 pts (median age 62 years; male 56.3%; White 63.3%; any comorbidities 40.6%; ECOG PS 0-1 34.8%; 2+ metastatic sites 25.4%; median time from mCRC diagnosis to index date 699 days). The FTD-TPI+bev cohort included 529 pts (median age 60 years; male 53.9%; White 68.1%; any comorbidities 34.4%; ECOG PS 0-1 51.4%; 2+ metastatic sites 25.1%; median time from mCRC diagnosis to index date 609 days). Prior to index date, 18.5% and 6.9% of pts had been exposed to regorafenib in the FTD-TPI and FTD-TPI+bev cohorts, respectively. FTD-TPI+bev significantly increased rwOS (median 9.4 [95% CI 8.0-10.1] vs 6.4 [95% CI 6.1-6.6] months; $p<0.0001$) and significantly decreased the risk of death (HR=0.68; $p<0.0001$) in the adjusted model. Other significant factors for rwOS included 2+ metastatic sites (vs.1 site, HR=1.16; p=0.002) and no receipt of regorafenib prior to index date of FTD-TPI (HR=0.80; $p<0.0001$). FTD-TPI+bev also significantly prolonged rwTTD as compared to FTD-TPI (median 3.5 [95% CI: 3.3-3.8] vs 2.4 [95% CI: 2.3-2.6] months; $p<0.0001$) and decreased risk of discontinuation (HR=0.65; $p<0.0001$) after adjusting for patient demographic and clinical characteristics. Additionally, the FTD-TPI+bev cohort had a significantly longer rwTTNTD as compared to the FTD-TPI cohort (median 5.0 [95% CI: 4.6-5.5] vs 3.9 [95%CI: 3.8-4.1] months; $p<0.0001$) in the adjusted model (HR=0.73; $p<0.0001$). **Conclusions:** This is the largest study to compare the clinical outcomes of pts receiving FTD-TPI and FTD-TPI+bev in the real-world setting. FTD-TPI+bev improved clinical outcomes including rwOS, rwTTD, and rwTTNTD, as compared to FTD-TPI in pts with mCRC. These findings align with the results of the SUNLIGHT trial. Research Sponsor: Taiho Oncology, Inc.

# Treatment experiences and decision-making among patients with metastatic colorectal cancer: Results of an online US patient survey.

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**Background:** Developing treatment plans for patients (pts) with metastatic colorectal cancer (mCRC) is complex. Using an online survey, we aimed to understand the treatment experience of pts with mCRC, including pt involvement in treatment decision-making. **Methods:** Adults (≥18 yrs) with mCRC took part in an online survey conducted in the US by The Harris Poll between Jan 17 and Feb 7, 2024. Of 344 respondents, 90 were recruited from patient advocacy groups (PAGs). The 60-question survey explored pts' experience of mCRC. We report results from questions focused on treatment. Raw data were not weighted and are only representative of the participants. All surveys are subject to sources of error. **Results:** Of 256/344 pts who received mCRC treatment, chemotherapy and surgery were the most common therapies(Table). Among pts who received therapy, 67% reported their ability to take treatment was impacted by non-medical factors such as finances or mental health. Overall, 58% of pts have had to decide between affording treatment or meeting basic needs, e.g. safe housing, food, transportation, especially Black and Hispanic pts (Table). Of all 344 pts, 87% reported reliance on their healthcare provider (HCP) when making treatment decisions, primarily their oncology surgeon (48%) or oncologist (45%), despite 268 (78%) pts carrying out independent research on new treatments or clinical trials. Although 266 pts (77%) were involved in treatment decision making, 231 pts (67%) wished for more power in the process. Overall, 64 pts (19%) wished for more support from PAGs during treatment, while 59 pts (17%) relied on PAGs to help guide treatment decisions. The most common factors affecting pt treatment decisions related to costs or insurance coverage (52%), impacts on physical health and quality of life (each 39%), fear of being a burden on family (35%), and side effects (32%). **Conclusions:** Pts with mCRC rely on their HCPs when making treatment decisions and wish for more power in the process. Personal finance and insurance coverage play dominant roles in treatment for mCRC. PAGs are an important resource for treatment information that are underutilized. Research Sponsor: Takeda Development Center Americas, Inc. (TDCA), Lexington, MA, USA.

**Treatment experience of pts with mCRC in an online US survey.**

| n (%) | All pts N=344 | Black pts n=87[c] | Hispanic pts n=75[c] | White pts n=158 |
|---|---|---|---|---|
| **Received treatment** | 256 (74) | 52 (60) | 60 (80) | 129 (82) |
| **Chemotherapy[a]** | 169 (66) | 23 (44) | 33 (55) | 104 (81) |
| **Surgery[a]** | 147 (57) | 23 (44) | 32 (53) | 86 (67) |
| **Radiotherapy[a]** | 101 (39) | 25 (48) | 15 (25) | 52 (40) |
| **Immunotherapy[a]** | 72 (28) | 16 (31) | 20 (33) | 30 (23) |
| **Targeted therapy[a]** | 61 (24) | 12 (23) | 14 (23) | 30 (23) |
| **Other[a]** | 13 (5) | 0 | 0 | 12 (9) |
| **Stopped treatment due to overall burden[a,b]** | 92 (36) | 28 (54) | 33 (55) | 27 (21) |
| **Had to decide between treatment or basic needs** | 198 (58) | 60 (69) | 52 (69) | 69 (44) |

[a]Among pts who received treatment.
[b]Financial impact, quality of life, etc.
[c]Base size <100 pts; interpret results directionally.

# Real-world clinical outcomes of trifluridine-tipiracil monotherapy (FTD-TPI) and FTD-TPI + bevacizumab combination therapy (FTD-TPI+bev) in 639 black patients (pts) with metastatic colorectal cancer (mCRC).

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**Background:** In the phase 3 SUNLIGHT trial, FTD-TPI+bev improved overall survival (OS) as compared to FTD-TPI in treatment-refractory mCRC. Out of 492 pts in the study, only 7 (1.4%) pts self-identified as Black; therefore, the potential benefit of FTD-TPI+bev in this pt population was not well-examined. We present the largest analysis to date of real-world clinical outcomes of FTD-TPI and FTD-TPI+bev in Black pts with mCRC. **Methods:** We retrospectively identified Black adult pts with a documented exposure to FTD-TPI and diagnosis of mCRC at or prior to the initiation of FTD-TPI using United States-based electronic medical records and claims in the ConcertAI RWD360 dataset. Pts were categorized as having received FTD-TPI or FTD-TPI+bev. Kaplan-Meier analyses were used to compare the real-world overall survival (rwOS) and time to discontinuation (rwTTD) from the date of first exposure to FTD-TPI (index). Multivariate Cox regression analyses were used to control for pt characteristics including demographics, ECOG performance status (PS), comorbidities, metastatic sites, time from mCRC diagnosis to index, lab results, and prior receipt of regorafenib. **Results:** Out of 639 Black pts included in our study, 551 received FTD-TPI and 88 received FTD-TPI+bev. Index years were 2021-2023 for 26% and 92% pts receiving FTD-TPI and FTD-TPI+bev, respectively. Pts receiving FTD-TPI had median age 61 years; male 51.2%; any comorbidities 43.9%; ECOG PS 0-1 36.7%; median time from mCRC diagnosis to index 632 days. Pts receiving FTD-TPI+bev had median age 59 years; male 50.0%; any comorbidities 42.0%; ECOG PS 0-1 60.3 %; median time from mCRC diagnosis to index 579.5 days. The rwOS was significantly improved with FTD-TPI+bev as compared with FTD-TPI (10.8 vs 6.2 months, respectively; p=0.0001), similar to the SUNLIGHT study (Table). In the adjusted model, pts who received FTD-TPI+bev had a significantly lower risk of death as compared to pts who received FTD-TPI (hazard ratio [HR] =0.45; p<0.0001). The rwTTD was also significantly prolonged with FTD-TPI+bev as compared with FTD-TPI (3.8 vs 2.4 months, respectively; p<0.0001). In the adjusted model, pts who received FTD-TPI+bev had about half the risk of discontinuation as compared with pts who received FTD-TPI (HR=0.51; p<0.0001). **Conclusions:** This is the first study to compare the clinical outcomes with FTD-TPI vs. FTD-TPI+bev in Black pts with mCRC in the real-world setting. Our study confirmed OS improvement with FTD-TPI+bev as compared to FTD-TPI in Black pts with mCRC, as seen in the overall population of the SUNLIGHT trial. Research Sponsor: Taiho Oncology, Inc.

| | Black n | Current Study rwOS months (95% CI) | Black n | SUNLIGHT OS months (95% CI) |
|---|---|---|---|---|
| **FTD-TPI** | 551 | 6.2 (5.7-6.9) | 3 | 7.5 (6.3-8.6) |
| **FTD-TPI+bev** | 88 | 10.8 (7.8-16.0) | 4 | 10.8 (9.4-11.8) |
| | | HR 0.45 (95% CI 0.33-0.63; p<0.0001) | | HR 0.61 (95% CI 0.49-0.77; p<0.001) |

# Clinical-pathological characteristics, follow-up, and survival of patients with early-onset vs late-onset colorectal cancer: Institutional experience in Argentina.

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**Background:** The incidence of colorectal cancer (CRC) in young adults aged 50 or younger (Early Onset - EO) is increasing worldwide, while in patients over 50 years old (Late Onset - LO), it is declining. The objective is to compare the clinicopathological characteristics, time to diagnosis, and survival between Early Onset (EO) and Late Onset (LO) patients diagnosed with colorectal cancer (CRC). **Methods:** The sample was described using measures of central tendency and dispersion for continuous numerical variables and percentages for categorical variables. Clinical characteristics between Early Onset (EO) and Late Onset (LO) groups were compared using the Student's t-test or Chi-square test. Kaplan-Meier curves and the Log-Rank Test were utilized to analyze mortality time. For multivariable analysis, a Cox regression model was applied. A p-value of less than 0.05 was considered statistically significant. The analysis was performed using RStudio. **Results:** A total of 460 patients were evaluated, with 343 diagnosed with colon adenocarcinoma and 117 with rectal cancer. Patients were grouped into two categories: the LO group with 408 (88.6%) patients and a mean age of 69.3 (8.60 SD) years, and the EO group with 52 (11.3%) patients and a mean age of 41.9 (6.23 SD) years. The EO group presented a higher percentage of patients with a PS of 0 compared to the LO group. In the EO group, there was a predominance of left-sided colon cancer (40.4% vs. 40.0%) and rectal cancer, especially in the middle and lower regions (30% vs. 18.8%), with disease diagnosed at stage III (26.9% vs. 26.5%) and IV (5.8% vs. 4.4%) compared to the LO group. Molecular testing for KRAS, NRAS, and BRAF mutations was more frequent in the EO group (19.2%, 3.8%, and 3.8%, respectively) compared to the LO group, as well as mismatch repair (MMR) status at 9.6% in the EO group vs. 6.9% in the LO group. The median time to diagnosis was 186 days in the LO group and 341 days in the EO group (p=0.21). Follow-up was 858 days in the LO group and 725 days in the EO group (p=0.18); while mortality was 200 (49.0%) in the LO group and 24 (46.2%) in the EO group, with a median survival of 1193 days for the LO group and 1414 days for the EO group, respectively (p=0.7). PS and stage were the most influential prognostic factors for mortality. **Conclusions:** In the comparison, the EO group showed a higher incidence of left-sided colon and rectal cancers, more advanced stages, better PS, and a greater number of molecular alterations, with a longer time to diagnosis and shorter follow-up. Although these differences were not statistically significant, they highlight the need to optimize prevention strategies and genetic counseling. Additional studies with a larger number of EO patients in real-life contexts are needed to generate more evidence that can improve clinical decision-making. Research Sponsor: None.

# Annual incidence of colorectal cancer among patients with average risk in the United States between 2017 and 2023.

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**Background:** Colorectal cancer (CRC) is the second most common cause of cancer-related mortality and third most common cancer in the United States. In 2021, 141,902 new cases of CRC were reported. With recent increases in early onset of CRC incidence, a shift in patterns amongst various demographics has been observed. Together with an estimated 60% increase in burden of CRC globally, it is important to continuously assess the nationwide incidence rates and monitor the changing patterns. In this study, we examined the annual incidence rates of CRC in average risk adults using a large national claims database and report patterns of incidence based on various demographic characteristics. **Methods:** CRC diagnoses were retrospectively identified from a large national claims database, which covers over 165 million lives and is representative of the U.S. population, for each calendar year during the study period from 2017 to 2023. Individuals aged 45 to 75 years, with an average risk of CRC before each index calendar year were included in the study. Individuals were required to be continuously enrolled in the same health plan for at least 1 year prior to and 1 year following the index year. Annual CRC incidence rates per 100,000 people were calculated for each year from 2017 to 2023 within various subgroups based on baseline demographics. **Results:** Overall, annual CRC incidences remained consistent during the study period with 186.86 (95% CI, 184.87-188.86) cases in 2017 and 183.40 (95% CI, 181.32-185.5) cases in 2023 per 100,000. Among younger adults aged 45-49 years, the annual CRC incidence increased by 66% from 2017 (92.04 cases/100,000) to 2023 (153.12 cases/100,000). Conversely, annual CRC incidence decreased by 25% between 2017 (328.52 cases/100,000) and 2023 (244.93/100,000) in older adults aged 65-75 years. Over the study period, CRC incidence in men was higher than women with an overall rate ratio (RR) of 1.206 (95% CI, 1.122-1.297). African American adults had higher CRC incidence than White Americans with an overall rate ratio (RR) of 1.216 (95% CI, 1.152-1.283). **Conclusions:** Based on this large study of national claims data, CRC incidence rates remained relatively stable in the total population among average-risk patients from 2017 to 2023. The continuing increase in recommended CRC screening utilization showed a decreased incidence rate in the older population, while observing increased incidences among younger patients. Research Sponsor: Exact Sciences Corporation.

# Long-term outcomes of total neoadjuvant treatment (TNT) for locally advanced rectal cancer: Maria Sklodowska Curie National Cancer Research Institute (MSCNCRI) cohort from Polish-II study.

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**Background:** The data on long-term outcomes of TNT for locally advanced rectal cancer are still limited. The Polish-II study randomized 515 cT4/fixed cT3 rectal cancer to pre-operative chemoradiation (CRT) with 5fu/oxaliplatin or 5x5 Gy radiotherapy with consolidating chemotherapy (TNT). Despite the early separation of survival curves with benefit for TNT, the outcomes at the longer observation were similar reaching 49% in both arms surviving at 8 years. Here, we present the long-term oncological and functional outcomes of patients recruited to the study in MSCNCRI along with the quality of life in survivors at >10 years of observation. **Methods:** Of 180 patients treated in the study in MSCNCRI 88 and 92 pts were randomized to TNT and CRT arms, respectively. The long-term follow-up was performed according to the standard protocol, FACT-C and EORTC QLQ-C30 scales were used to assess the functional outcomes. **Results:** Median survival in the TNT group and CRT was 115 and 61 months (p=0.191). At the cut-off in March 2023, 53.4% and 61.1% of pts died in the TNT and CRT group (p = 0.375). In the TNT group, 73.9% of deaths were attributed to cancer corresponding to 81.8% in CRT (p = 0.725). Late relapses diagnosed over 5 years from the end of the active treatment were identified in 5 of 94 patients in the follow-up (1 local recurrence in the TNT group, 4 patients with lung metastasis – 3 in TNT with one having also CNS relapse - and 1 in CRT). All late relapses in the TNT group could be salvaged with local therapies. Regarding long-term quality of life, no differences were observed in FACT-C and EORTC QLQ C30 in patients treated with either arm: QLQ C30 – TNT: 50.68 ± 12.74 in the range 34.00- 79.00; CRT: 48.61 ± 11.39 in the range 34.00-70.00 (p = 0.507); FACT-C - TNT: 113.00 in the range 42.00-136.00; CRT: 118.00 in the range 74.00-136.00; p=0.252). No difference in physical, family, and emotional status was observed. At the ultimate observation, stoma was present in 40.5% and 40.6% of TNT and CRT groups, respectively (p>0.999). **Conclusions:** In this randomized cohort, after more than 10 years of observation TNT led to numerically longer OS than CRT, although this was not statistically significant. The functional outcomes of the long-term survivors are similar in both cohorts. Late relapses are rare but could be salvaged in most cases. This study further supports the use of TNT in patients with locally advanced rectal cancer. Clinical trial information: NCT00833131. Research Sponsor: None.

# Management of patients with locally advanced rectal cancer in the total neo-adjuvant therapy (TNT) era: Experience from Argentina.

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**Background:** In the last years a paradigm shift has changed the way we manage non metastatic rectal cancer. The varied options within TNT have gradually been adapted in our country. This study aims to depict the strategies implemented. **Methods:** A prospective and retrospective observational study was performed from 2020 to 2024 including stage II-III rectal cancer patients treated at three centers in Buenos Aires, Argentina. Data was extracted from medical records. **Results:** 68 patients were included in the analysis. Median age was 59 years (IQR 48–71). 72% were male. 88% (60) had stage III disease, and 12% (8) stage II. Geriatric assessment was performed in 7 of 19 patients aged over 70 (5 frail). All patients were staged with high resolution MRI and discussed in multidisciplinary tumor board. Mismatch repair was assessed in 72%(49) of patients, only one had MLH1 deficiency. 79%(54) patients received TNT: 22%(12) short course vs 78%(42) long course radiotherapy; 24%(13) induction vs 76%(41) consolidation chemotherapy. Preferred chemotherapy regimen was CAPOX both in induction and consolidation chemotherapy (69.7% and 63.4%, respectively) Despite small numbers non significant differences in risk factors were found between TNT vs non TNT groups (Table). We registered 13 Grade 3-4 toxicity events: 3 in induction chemotherapy, 7 in consolidation and 3 in radiotherapy. Clinical complete response was achieved in 25.9% of TNT (11 consolidation, 3 induction) and 7.1%(1) of non-TNT (p=0.25). Watch and Wait was employed in 9 patients in TNT and 1 in non TNT group. 40 patients underwent surgery in the TNT and 11 in the non-TNT group; complete pathological response was reached in 17.5% (7) and 9% (1), respectively. Overall relapse rate was 19.14% (13): 2.94% (2) local and 16.2% (11) systemic. (TNT: 1.85%(1) local vs 18.5%(10) metastatic; nonTNT: 7.14%(1) both local and metastatic) p=0.4. At a median follow-up of 20 months (IQR 15–32 months), 2-year progression-free survival (PFS) rate was 87% (95% CI: 38–98%) in the non-TNT group and 78% (95% CI: 61–88%) in the TNT group. **Conclusions:** This study highlights the feasibility of worldwide adoption of TNT. We describe the preferred regimens and results of TNT in three high volume centers in Argentina. Long course radiotherapy and consolidation treatment with CAPOX are the predominant strategies. Non operative management is routinely offered to complete responders. Our efficacy results are comparable with published literature. Comprehensive geriatric assessment is recommended but still partially applied. In line with recent findings MMR testing has been widely adopted as an initial workup parameter. Research Sponsor: None.

| | TNT %(n=54) | non-TNT %(n=14) | p |
|---|---|---|---|
| **T4 disease** | 26 (14) | 21 (3) | 0.99 |
| **N2** | 37 (20) | 29 (4) | 0.75 |
| **EMVI positivity** | 49 (26) | 18 (3) | 0.16 |
| **Circumferential resection margin <1 mm** | 29.6 (16) | 28.57 (4) | 0.23 |
| **Low-lying tumors** | 33 (18) | 43 (6) | 0.54 |

# The efficacy of anti-EGFR therapy across different lines of treatment in *RAS* wild-type left-sided metastatic colorectal cancer.

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**Background:** Anti-epidermal growth factor receptor monoclonal antibodies (anti-EGFR mAbs) are recommended as a first-line treatment in *RAS wild-type* (*RASwt*) left-sided metastatic colorectal cancer (mCRC). However, the efficacy of these therapies across different treatment lines remains unclear. This study aims to evaluate the efficacy of anti-EGFR mAb across first, second, third, and later-line treatments. **Methods:** We conducted a retrospective analysis of patients diagnosed with *RASwt* mCRC who received anti-EGFR mAb in all treatment lines at Siriraj Hospital between 2008 and 2023. The impact of anti-EGFR mAb treatment line on progression-free survival (PFS) and overall survival (OS) was determined using the Kaplan-Meier method and compared with the log-rank test. **Results:** Of the 350 patients with *RASwt* mCRC receiving anti-EGFR mAb, 33% (115 of 350) were treated in the first-line, while 16%, 45% and 6% were treated in the second-, third- and later-line settings, respectively. Eighty-six percent of patients had left-sided tumors, and anti-vascular growth factors were used in 34% of patients. The median follow-up time was 30.4 months (mo). Among patients with left-sided tumors, those receiving first-line anti-EGFR mAb treatment exhibited a significantly longer mPFS (11.9 mo, 95% CI 7.4-16.3, $p<0.001$). The mPFS in the second-, third- and later-line settings were 6.1 mo (95% CI 4.6-7.5), 4.9 mo (95% CI 3.8-6.0), and 5.1 mo (95% CI 2.3-7.9), respectively. There were no significant differences in terms of OS across treatment lines ($p=0.27$). **Conclusions:** Although first-line treatment with anti-EGFR mAb had the longest mPFS in left-sided mCRC, OS did not significantly differ among treatment lines. Therefore, anti-EGFR mAb can be incorporated as a part of therapy at any line of treatment. Research Sponsor: None.

87 Poster Session

# Digesting danger: A two-decade study exploring the association between food processing and colon cancer.

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**Background:** Dietary patterns play a significant role in cancer prevention and progression, yet the specific association of food processing with colon cancer remains largely under-investigated. The NOVA classification system categorizes foods by their level of processing, providing a unique way to examine this relationship. Colon cancer remains a leading global health concern, and emerging evidence points to diet as potential contributors. This study explores associations between NOVA-classified foods and colon cancer, offering new insights into the role of food processing in cancer development. **Methods:** Survey data from the NHANES database (from 1999 to 2018) was probed for demographic and clinically relevant data, and day one food journals were converted into NOVA classifications. Participants aged 20 and older were included for analysis, including those with and without colon cancer (n=339 and 47615, respectively). Independent two-sample t-tests were conducted to compare the mean proportion of daily energy consumption of various NOVA food categories between the two groups, and a Benjamini-Hochberg correction was applied to adjust for multiple comparisons at $p<0.05$. Sensitivity analyses were performed using total grams and percent total caloric intake as inputs. Subgroup analyses were conducted via a logistic regression model including BMI, age, ethnicity, gender, education, country of birth, and military service. **Results:** Individuals with colon cancer had higher intake of ham and salted meats (NOVA 3, adjusted p=0.028) compared to those without cancer. Conversely, individuals without cancer consumed more legumes (NOVA 1, adjusted p=0.028), oils (NOVA 2, adjusted p=0.023), and "other processed foods" that did not fall into a specific processed food category (NOVA 3, adjusted p=0.038). Subgroups analyses revealed a greater association with colon cancer for age greater than 50 and non-hispanic White ethnicity. **Conclusions:** To our knowledge, this is the only study at this scale that evaluates colon cancer prevalence in the context of dietary behaviors stratified by level of industrial processing. The significant associations between ham and salted meats consumption, age greater than 50, and non-Hispanic White ethnicity with increased cancer prevalence highlight the impact of dietary and demographic patterns on colorectal carcinogenesis. Additional studies are needed to confirm these hypotheses in the prospective setting. Research Sponsor: None.

**88** **Poster Session**

# Multi-target stool-DNA test for colorectal cancer screening and effects on health and economic outcomes compared to blood-based tests: Influence of adherence.

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**Background:** There are an estimated 60 million individuals in the US who are not up-to-date with average-risk colorectal cancer (CRC) screening, as screening rates and adherence have remained stubbornly below the national target of 80%. Efforts are ongoing to improve CRC screening performance and participation, including the development of new blood-based tests. Despite high expectations for these tests, performance remains lower than other guideline-recommended strategies, particularly with respect to advanced precancerous lesion (APL) sensitivity. We conducted a simulation study of estimated clinical and economical outcomes for CRC screening with a blood-based test (at perfect adherence) compared to the multi-target stool-DNA (mt-sDNA) test (at published adherence rates). **Methods:** Utilizing CRC-AIM, a calibrated and validated microsimulation model, CRC screening outcomes were calculated for 1 million average-risk individuals screened between ages 45-75 years with triennial blood-based versus mt-sDNA testing. CRC and APL sensitivity and specificity inputs were derived from two large clinical validation studies for these screening modalities: ECLIPSE (NCT04136002) and DeeP-C (NCT01397747), respectively. To demonstrate the maximum benefits and burdens of blood-based screening, adherence to initial screening and follow-up colonoscopy after a positive blood test was modeled at 100%, while real-world adherence estimates of 65.6% were used for the mt-sDNA test. Outcomes of interest included life years-gained (LYG) and CRC cases missed by the blood-based test compared to the stool-based test, additional treatment costs imposed by theblood-based test, and additional CRC-related deaths per 1 million screened. **Results:** Compared to triennial screening with mt-sDNA at real-world adherence, blood-based screening at perfect adherence resulted in a higher number of incident (n=18,464) and fatal (n=6,483) CRC cases that would been avoided with the mt-sDNA test. Over a lifetime, screening with blood-based tests required 1,276,310 more tests (21% more) and 52,942 additional follow-up colonoscopies (7% more) compared to mt-sDNA screening. This increased testing burden did not generate additional benefits and instead reduced life-years gained by 67,645 per 1 million screened, resulting in an additional $1.6 billion in treatment costs compared to the mt-sDNA strategy. **Conclusions:** Data from this CRC-AIM modeling study show that even with perfect adherence, blood-based screening yields inferior clinical and economic benefits compared to mt-sDNA screening, due to suboptimal APL detection with the former test. Research Sponsor: Exact Sciences Corp.

# Comparison of screening outcomes associated with advanced precancerous lesion versus colorectal cancer sensitivity for a blood-based test: A simulation study.

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**Background:** With estimates of over 150,000 incident and 50,000 fatal colorectal cancer (CRC) cases in 2024, efforts are ongoing to improve the national screening performance and engagement in the US. The Centers for Medicare & Medicaid Services (CMS) have established criteria for covering new blood-based CRC screening tests if minimum CRC sensitivity and specificity thresholds are met. However, the current blood-based tests are challenged by their relative inability to detect advanced precancerous lesions (APLs). At present, the importance of APL versus CRC sensitivity on screening outcomes remains undetermined. To address this knowledge gap, we conducted a simulation study to compare the screening effectiveness and outcomes of a hypothetical blood-based test across a range of sensitivities for CRC and APL detection. **Methods:** Outcomes were simulated using the CRC-AIM microsimulation model, which has been previously calibrated and validated. Predicted outcomes were calculated for average-risk individuals screened triennially with a blood-based test between ages 45-75 years, assuming perfect adherence. Base-case screening sensitivity and specificity inputs were based on CMS minimum thresholds (74% CRC sensitivity, 90% specificity) with assumed 10% APL sensitivity. Additional scenarios considered an increase in either APL or CRC sensitivity. Outcomes of interest included life years-gained (LYG), CRC incidence, and mortality reductions. **Results:** Compared to the CMS minimum performance (Table), increasing APL sensitivity by 1% (from 10% to 11%) resulted in a similar increase in LYG as compared to increasing CRC sensitivity by 11% (from 74% to 85%). Additionally, increasing APL sensitivity by 1% resulted in greater reductions in CRC cases and deaths (3% decrease) as compared to increasing CRC sensitivity by 11% (0% and 1% reductions, respectively). **Conclusions:** Data from this study show that increased APL sensitivity is a key determinant of screening-related outcomes, such as LYG, derived from a blood-based strategy, and that similar results would require much larger increases in CRC sensitivity. With the current CMS minimum performance threshold, blood-based tests that are designed to have high APL sensitivity can outperform tests with higher CRC sensitivity and little to no APL sensitivity. Research Sponsor: Exact Sciences Corp.

**Comparison of CRC screening health outcomes with CMS proposed minimum threshold blood-test versus hypothetical tests with either increased APL and CRC sensitivity.**

| APL Sensitivity | CRC Sensitivity | CRC Specificity | Follow-up Colonoscopies | CRC Cases (% reduction) | CRC Deaths (% reduction) | Life-Years Gained (LYG) (% increase) |
|---|---|---|---|---|---|---|
| **10%** | 74% | 90% | 809 | 50 (ref) | 18 (ref) | 220 (ref) |
| **11%** | 74% | 90% | 813 | 48 (-3%) | 17 (-3%) | 226 (3%) |
| **10%** | 85% | 90% | 809 | 50 (0%) | 18 (-1%) | 227 (3%) |

APL: Advanced precancerous lesions; CRC: Colorectal cancer.

# Birth cohort effects in appendiceal adenocarcinoma incidence across the United States.

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**Background:** Appendiceal adenocarcinoma incidence rates are increasing across all age groups, sexes and histological subtypes in the United States. Birth cohort patterns of appendiceal adenocarcinomas can provide us with new, etiologic clues into these rising rates but have not been examined for this rare malignant tumor type. We estimated appendiceal adenocarcinoma incidence rates across birth cohorts in the United States. **Methods:** Using the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) Program data from eight population-based cancer registries, we identified a total of 4,858 persons aged 20+ years when diagnosed with a pathologically-confirmed primary appendiceal adenocarcinoma (non-mucinous, mucinous, goblet cell, signet ring cell carcinoma) between 1975 and 2019. Birth cohorts (1900-1904 to 1990-1994) were created using five-year age groups and time periods. The ratio of age-specific incidence rates was estimated in each birth cohort relative to the 1945-1949 birth cohort and reported as incidence rate ratios (IRRs) with 95% confidence intervals (CI). **Results:** Compared to persons born in 1945-1949, appendiceal adenocarcinoma incidence rates more than tripled among persons born in 1980-1984 (IRR 3.41, 95%CI 2.54-4.56) and quadrupled among persons born in 1985-1989 (IRR 4.62, 95%CI 3.12-6.82). For all tumor histological types, age-specific appendiceal adenocarcinoma incidence rates increased across successive birth cohorts after 1945-1949—although to varying degrees reported across histological types (e.g., 1980-1984: mucinous adenocarcinoma: IRR 2.71, 95% CI 1.73-4.26; non-mucinous adenocarcinoma: IRR 2.60, 95% CI 1.46-4.63; goblet cell adenocarcinoma: IRR 4.95, 95% CI 2.77-8.85). **Conclusions:** There are strong, yet distinct birth cohort effects for appendiceal adenocarcinomas across histological subtypes that remain unexplained. These patterns signal that recent exposure changes may be influencing this increasing disease risk for Generation X and Millennials now entering mid-adulthood. With the trend toward more non-operative management of appendicitis, another significant consideration for providers is to keep occult appendiceal tumors in the differential diagnosis of patients who present in this way. The current absence of appendiceal cancer prevention and screening modalities emphasizes the importance of efforts to support earlier detection in a rare malignancy where clinical trials have been historically very limited. Moreover, as similar trends have been reported in other gastrointestinal cancers, this is also suggestive of potential shared exposures contributing to this rising cancer burden across generations. Research Sponsor: Appendix Cancer Pseudomyxoma Peritonei (ACPMP) Research Foundation; National Cancer Institute/U.S. National Institutes of Health; P50 CA236733; National Institutes of Health/Eunice Kennedy Shriver National Institute of Child Health and Human Development; K12 HD043483.

# Impact of COVID-19 on colorectal cancer screening using Medicare Centers for Medicare and Medicaid Services (CMS) claims data.

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**Background:** When COVID-19 pandemic started in 2020 in the US, the Centers for Medicare and Medicaid Services (CMS) recommended that all non-urgent procedures, including screening colonoscopies, be delayed until pandemic stabilization. This was done to prevent the spread of the virus via colonoscopy since the presence of the virus in fecal samples has been reported in infected patients. Little is known of the impact of COVID-19 on the number of colorectal cancer (CRC) screening performed in the US. **Methods:** We utilized the Centers for Medicare & Medicaid Services Database which provides Medicare claims data for 2013-2022. We searched "Medicare Physician & Other Practitioners - by Geography and Service" using CPT code for colonoscopy(G0121, G0105), fecal occult blood test (FOBT; G0328, 82270), stool-based gene analysis (81528), sigmoidoscopy (G0104), and barium enema (G0120) and collected the number of CRC screening claims at national level and calculated the year-over-year (YOY) change. **Results:** The number of CRC screening claims in the US from 2013 to 2022 is listed (Table). The total number of CRC screening claims increased an average of 0.5% YOY from 2014 to 2019 then decreased 23.4% in 2020. From 2014 to 2019, the average YOY change in colonoscopy, FOBT, stool-based gene analysis, sigmoidoscopy, and barium enema claims were -0.5%, -6.0%, 58.7%, 8.1%, and -6.0%, respectively whereas in 2020, the rates were -27.5%, -25.0%, -14.4%, -25.3%, and -1.7%, respectively. The total number of CRC screening claims in year 2021 and 2022 increased from 2020, however, was still lower than 2019. **Conclusions:** CRC screening claims decreased in the US in 2020 likely due to COVID-19 and not yet returned to pre-pandemic levels. Research Sponsor: None.

**Number of CRC screening claims in the US from 2013 to 2022.**

| Screening test | 2013 | 2014 | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|---|---|---|---|---|---|---|---|---|---|---|
| **Colonoscopy** | 759,883 | 736,400 | 743,873 | 765,974 | 767,063 | 755,806 | 736,247 | 533,972 | 685,439 | 707,672 |
| **FOBT** | 1,307,054 | 1,263,602 | 1,161,214 | 1,130,130 | 1,085,777 | 996,016 | 897,993 | 673,614 | 646,550 | 556,950 |
| **Stool-Based Gene analysis** | | | | 123,748 | 231,765 | 335,459 | 482,761 | 413,224 | 497,748 | 536,348 |
| **Sigmoidoscopy** | 2,666 | 2,316 | 3,254 | 3,390 | 3,835 | 3,784 | 3,975 | 2,968 | 3,893 | 3,680 |
| **Barium enema** | 107 | 93 | 80 | 116 | 128 | 97 | 58 | 57 | 44 | 65 |
| **Total CRC screening** | 2,069,710 | 2,002,411 | 1,908,421 | 2,023,358 | 2,088,568 | 2,091,162 | 2,121,034 | 1,623,835 | 1,833,674 | 1,804,715 |

Data are expressed in numbers (N).
CRC; colorectal cancer; FOBT; fecal occult blood test.

# Gender imbalance in colon cancer screening rates in Texas: A population-based cohort study.

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**Background:** Males have a higher risk of acquiring Colorectal Cancer (CRC) than females. In Texas, males are diagnosed with CRC more than females, regardless of race or ethnicity. Texas CRC screening compliance rates are lower than the national average of 72.2%. It is important to understand the gender trends in screening to apply more specific strategies. **Methods:** We used the Texas Outpatient Public Use Data File (TOPUDF), a deidentified and publicly available statewide dataset, to conduct a population-based cohort study of outpatient colon cancer screenings of residents age 45 -74 years old at hospitals and ambulatory surgery centers during 2021. Colon cancer screenings were identified using Current Procedural Terminology (CPT) and Healthcare Common Procedure Coding System (HCPCS) codes. We queried the National Cancer Institute Surveillance, Epidemiology and End Results Program (SEER) for age grouped populations in 2021 stratified by gender, race, age group, and county of residence. We calculated age-adjusted outpatient colon cancer screening rates per 100,000 population (AASRs) using the direct standardization method based on the age group weights from the 2000 standard US population. Confidence intervals for AASR were derived by estimating the standard error as the AASR divided by square root of number of colon cancer screenings. The statistical significance of the difference between female and male AASR were tested by comparing the intersection of 95% confidence intervals (95% CI) for the individual rates. Subgroup analyses included age group, rural *vs* urban residence, and race/ethnicity. **Results:** A total 514,601 outpatient colon cancer screenings were included in the study of which 275, 189 (53.5%) were for female residents. During 2021, 51.1% of state residents age 45-74 years old were female. The overall statewide AASR in 2021 was 6071.5 (95% CI 6054.6 – 6088.4)) with the female AASR (6282.1 [95% CI 6258.1 – 6306.0]) significantly higher than the male AASR (5847.7 [95% CI 5823.9 – 5871.5]). For residents aged ≥ 45 years the statewide AASR was 5580.8 (95% CI 5565.5 – 5596) with the female AASR (5735.2 [95% CI 5715.9 – 5754.4]) significantly higher than the male AASR (5429.8 [95% CI 5410.1 – 5449.6]). Female residents had higher AASR amongst White, Black, and Hispanic race/ethnicity groups with the largest difference observed for Blacks; female AASR 5843.9 (5777.7 - 5910.1) *vs* male AASR 4733.5 (4668.5 - 4798.5). The difference between male and female outpatient screening rates was significant among both rural and urban residents. **Conclusions:** Despite a clear relationship between male gender and CRC, females have a higher rate of colon cancer screening. The Black population has the largest gender difference among race/ethnicity groups. Research Sponsor: None.

# Cost-effectiveness of blood-based colorectal cancer screening: A simulation model incorporating real-world longitudinal adherence.

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**Background:** Colorectal cancer (CRC) is a predominant cause of cancer-related mortality worldwide. Regular screening improves outcomes. However, despite there being many U.S. Preventive Services Task Force (USPSTF) recommended CRC screening options, many people elect not to use any of them. Most cost-effectiveness models assume perfect (100%) adherence initially and over time (longitudinally), which is implausible as approximately one-third of eligible individuals are not up to date with CRC screening. Our study evaluates the cost-effectiveness of Shield, an FDA-approved blood-based CRC test, while incorporating real-world adherence. **Methods:** The CAN-SCREEN (Colorectal cANcer SCReening Economics and adherENce) model is a validated, discrete-event simulation model designed to evaluate the clinical and economic outcomes of CRC screening strategies under real-world adherence scenarios. We simulated individual lifetime outcomes for a cohort of 10,000 people beginning screening at age 45, running 4,000 trials per cohort. We conducted a cost-effectiveness analysis using the CAN-SCREEN model, and compared the Shield blood-based test administered every three years to no screening. Costs were calculated from a healthcare perspective and adjusted to 2023 US dollars. Shield was considered cost-effective if the incremental cost-effectiveness ratio (ICER) was below $100,000 per quality-adjusted life-year (QALY) gained. The analysis also assessed the maximum price Shield could maintain while remaining cost-effective. **Results:** Compared to no screening, Shield increased the number of QALYs by 154 per 1,000 individuals and raised costs by $7.5 million per 1,000 individuals at its commercial test cost. With an incremental cost of $48,662 per QALY gained, Shield was found to be cost-effective relative to no screening at $100,000 cost per QALY willingness-to-pay threshold. Furthermore, the unit cost of Shield can be as high as $3,241 and $4,942 to be considered cost-effective at $100,000 and $150,000 per QALY gained, respectively. **Conclusions:** This study used the CAN-SCREEN model to compare Shield, an FDA-approved blood-based CRC screening test, against no screening, finding that Shield is cost-effective. With about one-third of the U.S. population not up to date on CRC screening, Shield's noninvasive approach offers a promising, cost-effective solution to increase adherence and reduce CRC mortality. Research Sponsor: None.

**Economic outcomes of Shield against no screening using CAN-SCREEN model.**

| Intervention | Test Unit Cost | QALYs/ Person | Cost ($)/Person | Cost ($)/QALY Gained vs No screening |
|---|---|---|---|---|
| **No Intervention** | - | 15.269 | $6,312.18 | - |
| **Shield** | $1,495 | 15.423 | $13,786.72 | $48,662 |
| **Shield** | $3,241 | 15.423 | $21,672.34 | *$100,000.00* |
| **Shield** | $4,942 | 15.423 | $29,352.42 | *$150,000.00* |

# Influence of preoperative Lynch syndrome diagnosis on surgery in patients with colorectal cancer.

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**Background:** Lynch Syndrome (LS) can guide surgery for colorectal cancer (CRC), particularly for *MLH1/MSH2* carriers, who may benefit from extended procedures: total colectomy (TC) or total proctocolectomy (TP). We investigated timing of germline genetic testing (GGT) and surgical approach in patients (pts) with LS and CRC. **Methods:** Integrated GGT (Labcorp, formerly Invitae Corp.) and insurance claims (Komodo Healthcare Map) data for adult pts with non-metastatic colon (CC) or rectal (RC) cancer and CRC surgery from 2015-23, ≥ 6 months of claims pre-CRC diagnosis, and GGT for *EPCAM, MLH1, MSH2, MSH6,* and *PMS2*. $X^2$ and t tests and multivariable logistic regression compared GGT results and surgical approach. **Results:** Of 1616 CRC pts (1553 CC, 63 RC), 15% were LS positive (15% CC, 11% RC). Compared to pts with negative GGT, pts with LS were more likely male, Black, younger at diagnosis and GGT, to have family history of GI/any cancer and to have GGT pre-CRC surgery. Of 129 LS pts with CC and *MLH1/MSH2/EPCAM*, 41 (32%) had GGT pre-surgery and 20/41 (49%) had TC/TP (Table). 5/6 (83%) of the corresponding RC pts had GGT pre-surgery and 3/5 (60%) had TP. 88 CC *MLH1/MSH2/EPCAM* pts had GGT post-surgery and 17/88 (19%) had TC/TP. 1 *MLH1/MSH2/EPCAM* RC patient had GGT post-surgery and 0 had TP. More CC pts had TC/TP than RC pts ($p<0.001$). Compared to pts with negative GGT, pts with LS and CRC were more likely to have TC/TP (21% vs. 6%, $p<0.001$), particularly with GGT pre-surgery (43% vs. 13%, $p<0.001$). RC LS pts with GGT pre-surgery had shorter mean months from GGT to surgery than pts with negative GGT (6 vs 14, $p=0.026$). CC pts with *MLH1* or *MSH2/EPCAM* and GGT pre-surgery had higher odds of undergoing TC/TP (odds ratio (OR): 6, confidence interval (CI): 3-10; OR: 7, CI: 4-14; OR: 4, CI: 2-6) than pts with negative or post-surgery GGT. **Conclusions:** GGT performed pre-surgery for a new diagnosis of CRC was more likely to result in extended procedures, especially in *MLH1/MSH2/EPCAM* carriers. More data in RC is needed to better understand the influence of GGT on surgical approach. Research Sponsor: None.

**CC surgery type and timing in relation to GGT.**

| | GGT pre-surgery (N=295) | | | | GGT post-surgery (N=1258) | | | |
|---|---|---|---|---|---|---|---|---|
| **Pts w/ CC surgery** | **Total** | **Partial Colectomy** (N=245) N (% total) | **TC/TP** (N=50) N (% total) | **P-value: partial vs TC/TP^** | **Total** | **Partial Colectomy** (N=1203) N (% total) | **TC/TP** (N=55) N (% total) | **P-value: partial vs TC/TP^** |
| ***MLH1* (N=75)** | 24 (32) | 13 (54) | 11 (46) | **0.001** | 51 (68) | 42 (82) | 9 (18) | **<0.001** |
| ***MSH2/ EPCAM* (N=54)** | 17 (32) | 8 (47) | 9 (53) | **0.006** | 37 (69) | 29 (78) | 8 (22) | **<0.001** |
| ***MSH6* (N=49)** | 13 (27) | 11 (85) | 2 (15) | 1 | 36 (74) | 35 (97) | 1 (3) | 1 |
| ***PMS2* (N=51)** | 6 (12) | 3 (50) | 3 (50) | 0.252 | 45 (88) | 43 (96) | 2 (4) | 1 |
| **LS positive (N=229)** | 60 (26) | 35 (58) | 25 (42) | **<0.001** | 169 (74) | 149 (88) | 20 (12) | **<0.001** |
| **LS VUS (N=61)** | 6 (10) | 6 (100) | 0 (0) | 1 | 55 (90) | 55 (100) | 0 (0) | 0.526 |
| **LS Negative (N=1263)** | 229 (18) | 204 (89) | 25 (11) | **<0.001** | 1034 (82) | 995 (96) | 39 (4) | **0.006** |

Note: Excludes pts with RC due to small N.
^Adjusted for multiple testing.

# Comparison of risk factors, treatment strategy, and outcomes of early- vs. older-onset colorectal cancer within the prospective ColoCare study.

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**Background:** The incidence of colorectal cancer among young adults aged 20-49 has continually increased in recent decades. Early-onset colorectal cancer (EOCRC) patients are more likely to be diagnosed at a more advanced stage of disease, have hereditary cancer syndromes, and have different molecular pathological features of disease than later-onset patients. Herein we compare the risk factors, treatment approaches, and responses to treatment of early-onset versus later-onset colorectal cancer within. **Methods:** The ColoCare Study is a prospective cohort enrolling individuals diagnosed with primary colorectal cancer at six different cancer centers in the United States and one in Germany. Serial questionnaires assess demographic characteristics and symptoms experienced prior to diagnosis, while clinical characteristics (including staging, pathology, treatments given) are abstracted from medical records. Restricting to patients with available age and stage, patients diagnosed <50 years old were labeled as "early-onset" and ≥50yo as "later-onset." Associations between patient age and characteristics of interest were tested using Fisher's Exact Test for count data. **Results:** Of the3,173 patients included in this analysis, 78% were later-onset (mean: 65 years) and 22% were early-onset (mean: 42 years). Demographic data was comparable between the two groups, with 43% being female. Though similar rates of ever alcohol use were seen between the two groups, early-onset patients reported consuming fewer weekly alcoholic beverages than later-onset patients (mean (standard deviation) of 5.5 (6.6) vs. 8.4 (9.3)). Early-onset patients were less likely to have ever smoked tobacco (38% vs 51%, $p<0.001$) and had lower rates of obesity (27% vs 31%) or overweight (34% vs 40%) than later-onset patients ($p<0.001$). For stage II/III disease, early-onset patients were more likely to have received chemotherapy and/or radiation treatment compared to later-onset patients (87% vs 70%, $p<0.001$), though both groups had similar rates of surgery. Despite this, younger patients did not appear to experience better outcomes, with 21% of younger patients experiencing a recurrence or new metastasis during follow-up versus 20% of later-onset patients. **Conclusions:** The risingincidence of colorectal cancer in early-onset patients does not appear to be explained by traditional risk factors such as alcohol use, tobacco use, or high BMI. Early-onset patients are more likely to receive chemotherapy or radiation, but observed outcomes are notably poorer compared to later-onset patients. Further statistical analysis on specifics of treatments by stage is ongoing. Research Sponsor: None.

# dMMR/MSI tumour occurrence during follow-up of patients with Lynch syndrome treated with immune checkpoint inhibitors for a metastatic digestive cancer between 2015 and 2024: A retrospective monocentric cohort.

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**Background:** Immune check point inhibitor(s) (ICIs) transformed the prognosis of patients with metastatic dMMR/MSI cancers especially for patients with Lynch syndrome (LS). The aim of our study was to describe a population of LS carrier patients treated with ICI(s) for metastatic digestive cancer, specifically assessing the risk of developing metachronous cancer. **Methods:** We collected data of 93 patients with LS, treated by ICI(s) for a metastatic digestive tumour. These patients were selected from the "immunoMSI" patient cohort, which included all patients presenting with a MSI and/or dMMR digestive cancer treated with ICI(s) between February 2015 and April 2024 in the medical oncology department of Saint-Antoine Hospital (Paris, France). Our primary outcome was the cumulative risk of metachronous MSI/dMMR cancer; secondary outcomes included cumulative risk and time to de novo colorectal polyps, as well as PFS and OS under ICI(s). We performed a cox model to assess the association of predefined variables with OS and PFS under ICI(s). **Results:** Patients most frequently carried a germline pathogenic variant on the *MSH2* (or *EPCAM*) gene (47%) or on the *MLH1* gene (30%). The majority presented with colorectal cancer (CRC) as their primary cancer (83%), followed by small intestine tumours (6%), stomach and pancreatic cancers (4%) and bile duct cancer (1%). Median age at cancer index was 62 years-old (26-77). A total of 12 out of 93 patients presented metachronous cancers during or after treatment with ICI(s). Ten of these metachronous cancers were dMMR or within the LS spectrum (urothelial n=4, CRC n=2, bilio-pancreatic n=1 and 3 skin cancers including 2 kerato-acanthomas). All patients who developed a metachronous cancer had a complete or partial response under ICI(s) for the index cancer. During a median follow-up of 47.8 months, 44 patients had colonoscopies and a total of 17 patients (39,5 %) developed pre neoplastic colorectal polyps. In multivariate analysis, PS $\geq$ 2 and age $\geq$ 70 years were associated with poorer OS with HRs of 6.3 (95% CI [1.5-25.9], p=0.01) and 5.5 (95% CI [1.6-18.9], p=0.006) respectively. **Conclusions:** Our results show that ICI(s) for patients with LS treated for an MSI/dMMR metastatic digestive cancer can still develop metachronous cancers and pre neoplastic polyps, which underlines the importance of maintaining dedicated follow-up after ICI(s) treatment in this population. Research Sponsor: None.

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# Effectiveness of the Colorectal Cancer Alliance's digital screening quiz and navigation model for promoting CRC screening.

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**Background:** Evaluating the effectiveness of a Colorectal Cancer screening navigation model delivered in a digital platform offered by a patient advocacy organization. Designed as a digital screening quiz tool and live navigation to promote CRC screening, the Colorectal Cancer Alliance's Screening Quiz is a risk stratification tool that connects respondents to live certified cancer care navigators delivering evidence-based interventions to increase CRC screening, education and patient and family support. The framework utilizes small and mass media, one-on-one education and patient navigation services to help address and reduce barriers to care. Research on these critical interventions shows that on-time screening for CRC leads to better outcomes. **Methods:** The Screening Quiz has been promoted through social sharing, community outreach, and media, including influencer campaigns to increase outreach to at-risk communities. Questions assess personal and family history, hereditary syndromes, and demographics to provide a CRC screening recommendation based on current USPSTF guidelines. Respondents overdue for screening are triaged to the Alliance's patient navigation team. This interaction includes phone calls and emails for symptomatic individuals/high risk for CRC, telephone and email for average-risk uninsured, and emails for average-risk insured. At-risk individuals are connected to resources, such as financial assistance, free at-home screening tests, and colonoscopy scheduling services. **Results:** July 2022 to July 2024, the Screening Quiz provided over 50,000 screening recommendations. The median age of respondents was 46; 32% of respondents were between the ages of 45 to 64. Of the respondents, 29% were male and 71% were female. Most were White (79.2%), but nearly half of those who opted into navigation identified as BIPOC, including 16.7% Black/AA, 6.6% Asian, and 2.1% AI/AN. Additionally, 8.4% of respondents identified as ethnically Hispanic/Latino with another race. Most respondents were symptomatic (66.6%) but 6.5% were asymptomatic high-risk. These two groups represented 88% of 1,869 respondents triaged to patient navigation. **Conclusions:** The digital Screening Quiz, combined with live navigation, represents a promising approach for increasing CRC screening rates and addressing care gaps compared to traditional methods. The Alliance's ability to engage at-risk communities through its national network of volunteers and partners, combined with evidence-based interventions, positions it well to continue closing screening gaps. Going forward, the Alliance aims to leverage its relationships with healthcare providers and community organizations further to expand the reach of its CRC screening efforts. Research Sponsor: None.

# Trends in utilization of colorectal cancer screening modalities among patients with average risk in the United States from 2017 to 2023.

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**Background:** Colorectal cancer (CRC) is the second leading cause of cancer mortality in the US. The national CRC screening rate (58%) is still below 80%, which is the goal set by the National Colorectal Cancer Roundtable. National guidelines recommend multiple strategies for CRC screening including colonoscopy, sigmoidoscopy, CT colonography, fecal occult blood test (FOBT), fecal immunochemical tests (FIT), or multi-target stool DNA (mt-sDNA) test. To provide updated insights regarding proportional utilization of different CRC screening strategies, we analyzed modality-specific CRC screening rates across demographic subgroups between 2017 and 2023. **Methods:** A retrospective review was conducted using a large national claims database that covers over 165 million lives and is representative of the US population for the study period from 2017 to 2023. Individuals were included if they were aged 45-75 years, at average CRC risk in the baseline period of 1 year before the index date and continuously enrolled in a health plan for at least 1 year after the index date. The index date was defined as date of the first claim for a CRC screening test. CRC screening uptake by modality and newly screened individuals each calendar year were examined among the entire eligible population and across demographic subgroups. **Results:** Overall rates of CRC screening were relatively stable across the study years (approximately 20%) with the lowest annual proportion of screen eligible patients in 2020 (16.6%) and highest in 2023 (24.4%). For CRC-screened individuals, colonoscopy was the most commonly used screening modality, increasing from 1,943,733 (53%) in 2017 to 2,319,695 (58.7%) in 2023. The mt-sDNA test use increased significantly from 87,769 (2.4%) in 2017 to 807,227 (20.4%) in 2023; while FIT/FOBT use declined from 1,611,730 (44%) in 2017 to 805,150 (20.4%) in 2023. Utilization of colonoscopy and mt-sDNA tests increased notably among the 45–49 age group starting in 2021, while FIT/FOBT use simultaneously declined in this age group. Observed trends were similar to the above findings across additional subgroups defined by gender, race/ethnicity, payer type, and geographic region. **Conclusions:** Data from this large study of national claims data provides updated results for overall and proportional utilization of guideline-endorsed CRC screening strategies. While colonoscopy usage remained stable , the use of the non-invasive mt-sDNA test has increased significantly as FIT/FOBT usage declined over time, reflecting growing interest in this noninvasive, home-based option. Research Sponsor: Exact Sciences Corporation.

# Multi-target stool DNA test adherence in average-risk African Americans from 2017-2024.

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**Background:** Colorectal cancer (CRC) stands as the second most common cause of cancer-related mortality in the United States, while being the third most common cancer in the country. African Americans (AA) are disproportionately affected with CRC (incidence and mortality), yet the disease can be prevented through screening and early detection. Data reveals that only 59% of AA between ages 45-75 were up to date with CRC screening as of 2021. Broad availability of the navigation-supported, at home multi-target stool DNA (mt-sDNA) test may help mitigate barriers to screening adherence in this population. We examined multi-target stool-DNA (mt-sDNA) test adherence rates in a nationally representative AA population. **Methods:** Using a national claims database of over 165 million individuals linked with Exact Sciences Laboratories data, mt-sDNA orders for AA patients were retrospectively identified from 2016 to 2024. Average CRC risk individuals aged ≥45 years, and new to mt-sDNA, were included in this analysis. Primary outcome of interest was adherence, defined as mt-sDNA kit return rate within 365-days from shipment. Time to test return was examined as the secondary outcome. Demographics were age, sex, ordering provider, residential geography, payor, and outreach preference. Covariates on adherence were examined using logistic regression. **Results:** Among 434,951 mt-sDNA orders from 2017-2024, the overall adherence rate was 62%, and the average time to test return was 27.6 days. The cohort included higher percentages of women (61.7%), ages 50-64 years (57.8%), located in Southern US (51.4%), and metro areas (90.9%). Most patients were also commercially insured (56.1%) and received their mt-sDNA order from their primary care provider (65.4%). Digital SMS was the most preferred outreach option (47%). Patients ages ≥76 years demonstrated the highest adherence (70.3%), with all age categories near or above 60% (45-49 yrs [59.4%]; 50-64 yrs [60.8%], 65-75 yrs [64.3%]). Across payors, patients with Medicare had the highest adherence rates (66.5%). Across outreach, highest adherence was among those who preferred 'Digital SMS + Email' outreach (64.3%). Men were quicker to return their test than women (26.8 days vs 28 days, $p<0.001$). Those ages ≥76 years and with tests ordered by GIs had the shortest time to test return (21.3 and 22.2 days, respectively). Adjusted analyses revealed multiple factors were associated with increased odds of adherence, including older age (≥76 years; OR=1.54; $p<0.001$), 'Digital SMS + Email' outreach (OR=1.25; $p<0.001$), rural residence (OR=1.23; $p<0.001$), and males (OR=1.05; $p<0.001$). **Conclusions:** Mt-sDNA adherence was higher for AA patients in this large, national cohort as compared to previously reported overall CRC screening adherence data for this population group. Findings suggest mt-sDNA utilization, with its included patient navigation, may help reduce screening disparities among AA patients. Research Sponsor: Exact Sciences Corporation.

# Adherence to the multi-target stool DNA test in the US Hispanic population from 2016-2024.

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**Background:** With colorectal cancer (CRC) as the second most common cause of cancer-related mortality in the US, adherence to CRC screening is a vital component to alleviating the cancer burden. Statistics show that only 52% of Hispanic or Latino individuals ≥45 years were up-to-date with CRC screening in 2021. A guideline-recommended, at-home CRC screening option, such as the multi-target stool DNA test (mt-sDNA), could be an efficient modality for screening average-risk individuals. Using a large national claims database, we examined CRC screening adherence with mt-sDNA in the US Hispanic population. **Methods:** Data was sourced from a database of over 165 million individuals linked with Exact Sciences Laboratories data. Hispanic persons who received mt-sDNA orders and resided in the US were retrospectively identified from 2016 to 2024. The analyses included persons who were ≥45 years of age, new to mt-sDNA and average risk for CRC during the 12-month pre-index period. Primary outcome of interest was adherence, which was defined as the mt-sDNA kit return rate within 365-days from shipment. All patients received a mailed letter along with digital outreach through the accompanying patient navigation program, according to their communication preference. The modes of digital outreach included: SMS, email, or both. Baseline demographics collected included age, sex, ordering provider, residential geography, payor, and outreach preference. Adjusted analyses were used to examine effects of covariates on adherence. **Results:** A total of 447,968 mt-sDNA orders were shipped during the study period and the overall adherence rate was 62.5%. Average time to test return was 28.5 days. Women comprised the majority of the cohort (59.9%), resided mostly in the Southern US (46.9%), aged 50-64 (60.9%), resided in a metropolitan area (91.2%), and received their mt-sDNA order from their primary care provider (62.7%). Across evaluated age groups, adherence was above 60% for all: ages 45-49 [60.5%]; 50-64 [62.6%], 65-75 [62.6%]), and ≥ 76 years [66.8]. By payor, Medicare-insured patients demonstrated the highest adherence (65.5%). Digital SMS was the preferred contact method (51.2%), and those opting for SMS + Email had the highest adherence rate (64.2%). Of provider specialties, individuals having their tests ordered by GIs produced the shortest time to adherence (23.2 days). Regression analyses revealed that older age (≥ 76 years), living in a rural area, and receiving fully or partially digital outreach were associated with increased mt-sDNA adherence. **Conclusions:** This large, national analysis reveals high mt-sDNA adherence amongst Hispanic individuals in the US, despite reports of low national CRC screening rates in this population. These data support the important role of mt-sDNA in facilitating screening engagement, while emphasizing the importance of preference-based navigation in facilitating screening test completion. Research Sponsor: Exact Sciences Corporation.

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# Adherence to multi-target stool DNA test in the US Asian population from 2017-2024.

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**Background:** In 2024, incidence and mortality cases related to colorectal cancer (CRC) in the US are estimated to reach 152,810 and 53,010, respectively. CRC screening can help identify malignant lesions at earlier stages; however, in 2021, only 48% of Asian American persons between the ages of 45-75 were up-to-date. Multi-target stool DNA test (mt-sDNA), a highly sensitive and guideline recommended CRC screening modality, may improve screening adherence in average-risk individuals within this population. Herein, we examined CRC screening adherence with mt-sDNA in Asian Americans using national claims data. **Methods:** A large claims database of over 165 million individuals linked with Exact Sciences Laboratories data, was used to source mt-sDNA orders for Asian Americans from 2017 to 2024. Eligible individuals were aged ≥45 years, new to mt-sDNA and average risk for CRC during the baseline period of 12 months. Primary outcome was adherence to mt-sDNA, defined as return of mt-sDNA kit within 365-days from shipment. The secondary outcome was time to kit return in days. All individuals received a standard mailed letter along with additional digital outreach through the accompanying mt-sDNA patient navigation program, according to their desired mode of communication. Baseline demographic measures included age, sex, ordering provider, residential geography, payor, and outreach preference. Association of adherence with patient characteristics was evaluated using multivariable regression. **Results:** Of the 182,259 eligible mt-sDNA orders shipped during the study period, overall adherence was 68.9%. Average time to test return was 27.1 days. The cohort was comprised mostly of women (56.7%), aged 50-64 years (58.42%), primary care provider (74.7%), and preferred 'Digital SMS' outreach (50.9%). Average adherence for all age groups, 45-49 years, 50-64 years, 65-75 years, and 76+ years, exceeded 65%, at 67.2%, 68.8%, 69.7%, and 72%, respectively. Medicare recipients demonstrated the highest adherence by payor type (71%, $p < 0.0001$), and persons who opted for 'Digital SMS + Email' communication were most adherent (70.9%; $p < 0.0001$). Of all provider specialties, individuals having their tests ordered by gastroenterologists produced the shortest time to adherence (21.8 days). Multivariable regression revealed that age, digital outreach (compared to none), micropolitan residence, and ordering provider were independent predictors of increased mt-sDNA adherence. **Conclusions:** The results of this large cohort and nationally representative analysis demonstrated high mt-sDNA adherence amongst Asian American persons, despite a low percentage of this population being current with CRC screening overall. The use of mt-sDNA, in concordance with its included patient navigation, is a viable modality to encourage test completion through screening engagement – which has the potential to reduce screening barriers in this population. Research Sponsor: Exact Sciences Corporation.

# Multi-target stool DNA test adherence among average-risk 45- to 49-year-old patients from 2017-2023.

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**Background:** Colorectal cancer (CRC) is the third most common cancer in the United States and the second most common cause of cancer-related mortality. In 2021, the US Preventive Services Task Force (USPSTF) lowered the age of initial CRC screening from 50 to 45 years of age for average risk individuals. At that time, only 20% of adults aged 45-49 years were up-to-date with CRC screening. With its at-home use, accessibility, and patient navigation, the multi-target stool DNA test (mt-sDNA) could help support screening uptake in this population. Here, we analyzed national claims data to investigate mt-sDNA adherence rates among average-risk adults aged 45-50 years. **Methods:** Using a national claims database of over 165 million individuals linked with Exact Sciences Laboratories data, mt-sDNA orders for AA patients were retrospectively identified from 2017 to 2023. Average CRC risk individuals aged ≥45 years, and new to mt-sDNA, were included in this analysis. Primary outcome of interest was adherence, defined as mt-sDNA kit return rate within 365-days from shipment. Test return time was examined as the secondary outcome. Demographics were age, sex, ordering provider, residential geography, payor, and outreach preference. Covariates on adherence were examined using logistic regression. **Results:** A total of 607,388 mt-sDNA orders were prescribed during the study period, with an overall 365-day return rate of 65.4%. Average time to test return was 30.4 days. Women outnumbered men in the cohort (58.6%), and most mt-sDNA orders were prescribed by primary care providers (59.4%). The majority of individuals preferred 'Digital SMS' communication (62%), and nearly 90% of the cohort was Commercially insured (87.6%). Among those in the cohort participants' self-reported race/ethnicity was: White (37.3%), Hispanic or Latino (10.1%), Black or African American (8.8%), unknown (8.8%), Asian or pacific islander (4.6%), and other (3.7%). Adherences for the four identified races/ethnicities were all near, or above 60%. Men demonstrated higher adherence than women (66.7% vs 64.5%; $p<0.0001$). Time to test return was shortest in men (28.9 days) and African American persons (28.5 days). Tests ordered by gastroenterologists yielded the quickest adherence times amongst providers (26 days). After controlling for race, gender, payor class, setting (urban/rural classification), and preferred language, digital outreach and provider type were significantly associated with increased mt-sDNA adherence. **Conclusions:** This large, national analysis reports a substantially higher rate of adherence among individuals ages 45-49 with the mt-sDNA test than the current nationally reported rates. At-home testing, along with standard and preference-based navigation may have contributed to high mt-sDNA completion rates. More research is warranted to explore screening behaviors and the benefits of screening within this younger population. Research Sponsor: Exact Sciences Corporation.

**103** **Poster Session**

# Prevalence of extracolonic malignancies in patients with familial adenomatous polyposis.

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**Background:** Familial adenomatous polyposis (FAP) is an autosomal-dominant inherited colorectal cancer syndrome resulting from mutations in the adenomatous polyposis coli (*APC*) gene. Patients with FAP have nearly 100% lifetime risk of colorectal cancer, typically managed with prophylactic colectomies. As patient survival has improved, the incidence of extracolonic tumors, including thyroid malignancies and desmoid tumors, has increased. However, limited research exists on the prevalence of extracolonic malignancies in FAP patients. **Methods:** We prospectively enrolled 219 patients with FAP in the Hereditary Gastrointestinal Cancers Cohort at the MD Anderson Cancer Center between December 2015 and July 2024. Inclusion criteria included either a clinical or molecular diagnosis of FAP. Data on demographics, cancer history (including cancers diagnosed both prior to and during the study period), BMI, tobacco use, medications, endoscopy results, thyroid ultrasound findings, and abdominal/pelvic CT results were collected. Screening for thyroid and desmoid tumors was provider-dependent, with additional screenings as deemed appropriate. Descriptive and inferential statistics were used to identify associations between demographic and clinical factors and the prevalence of extracolonic malignancies. **Results:** Among 219 patients, 20.5% developed desmoid tumors, 18.7% developed colorectal cancer, and 6.8% developed well-differentiated thyroid cancer (6.4% papillary thyroid cancer, 0.4% follicular thyroid cancer). Prophylactic colorectal surgery was performed in 151 patients (68.9%). Per chart review, benign thyroid disease and hypothyroidism were present in 21% and 11.4% of patients, respectively. Female sex was significantly associated with hypothyroidism ($p=0.007$) and thyroid cancer ($p=0.009$). BMI $>$ 30 was also linked to increased thyroid cancer risk ($p=0.046$). Desmoid tumors were found in 20.5% of patients, most commonly in the abdominal wall. Female sex, BMI $>$ 30, and history of colorectal surgery were significantly associated with desmoid tumor development ($p=0.004$, 0.0064, 0.0038). **Conclusions:** FAP patients exhibit a higher prevalence of benign thyroid disease, thyroid cancer, and desmoid tumors compared to the general population. Given these findings, adherence to existing surveillance guidelines, such as those outlined by the National Comprehensive Cancer Network (NCCN), is essential. Our findings further support the importance of these established surveillance strategies in the management of FAP patients. Research Sponsor: None.

# Refining colorectal cancer screening strategies using polygenic risk scores and classical risk factors: A proof-of-concept study in the UK Biobank cohort.

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**Background:** While the rising incidence of CRC in younger populations has led to a push for lowering the screening age to 40, current clinical guidelines lack sufficient consideration of individual risk variations, raising concerns about the potential harms of over-diagnosis and false positives. Our proof-of-concept study evaluates the utility of risk stratification for multi-target stool DNA (mt-sDNA) testing by integrating polygenic risk scores and traditional clinical risk factors, addressing gaps in current clinical CRC screening guidelines. **Methods:** Our study utilized the UK Biobank prospective cohort, consisting of 311,544 unrelated individuals of White British ancestry, aged 40-69, excluding those with any prior malignant cancer diagnosis. We developed a sex-specific Cox proportional hazards model to estimate the risk of developing CRC, with the primary outcome being the first incidence of CRC. Three models were assessed: a polygenic risk score (PRS)-only model, a classical risk factors (RF)-only model, and a combined (PRS + RF) model. The traditional risk factors considered were BMI, smoking status, pack years of smoking, and family history of bowel cancer in first-degree non-adoptive relatives. Model performance was evaluated using the area under the curve (AUC), hazard ratios (HR), and the positive and negative predictive values of mt-sDNA across various risk profiles. **Results:** The combined model for early CRC screening using mt-sDNA demonstrated superior risk stratification (Table) and improved PPV compared to the risk factors-only model, with enhanced performance even among younger individuals at lower baseline risk. If risk stratification was not used, the average PPV for the test in the general population (ages 40-80) would be 2.13% for males and 1.44% for females. In contrast, individuals in the top 1% of risk would have significantly higher PPVs: 5.62% for males and 3.51% for females. If we limit testing to only the high-risk group, per 100,000 individuals, approximately 3,490 additional cases for males and 2,070 additional cases for females would be detected compared to the average-risk population without stratification. **Conclusions:** The findings suggest that risk stratification may enhance the effectiveness of screening by identifying high-risk individuals who would derive the greatest benefit. In contrast, broad application of the test across the general population may not achieve an optimal risk-benefit ratio, given the relatively low PPV in average-risk individuals. Research Sponsor: None.

**Model performance of the PRS-only, RF-only, and PRS + RF (Combined) models for females and males, respectively.**

| | Hazard Ratio (95% CI) | C-index (SE) | 5-Year AUC |
|---|---|---|---|
| **Female** | | | |
| **PRS-Only** | 1.50 (1.37 - 1.64) | 0.62 (0.01) | 0.61 |
| **RF + PRS** | 1.49 (1.36 - 1.63) | 0.62 (0.01) | 0.61 |
| **RF Only** | 1.09 (0.99 - 1.20) | 0.53 (0.02) | 0.51 |
| **Male** | | | |
| **PRS-Only** | 1.49 (1.38 - 1.61) | 0.63 (0.01) | 0.69 |
| **RF + PRS** | 1.49 (1.38 - 1.61) | 0.64 (0.01) | 0.71 |
| **RF Only** | 1.23 (1.15 - 1.31) | 0.57 (0.01) | 0.57 |

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# The TAB-EOAO study: Trends in use and benefit of adjuvant chemotherapy between early onset and average onset locally advanced colon cancer.

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**Background:** Despite the declining incidence rates (IR) in average onset colon cancer (AO-CC), IR for early onset colon cancer (EO-CC), defined as patients (pts) younger than 50-years-old at diagnosis, has been increasing. Treatment for high risk locally advanced colon cancer (CC) is surgical resection followed by adjuvant chemotherapy (AC). Data indicates a more permissive chemotherapy use in EO-CC. There is limited data about treatment and survival trends in EO-CC. Thus, we aimed to assess the use and benefit of AC between in EO vs AO CC in locally advanced disease. **Methods:** CC diagnosed between 2000-2020 were identified in the Surveillance, Epidemiology, and End Results Program (SEER) Database. Eligible pts were adults (≥18), Stage II or III at diagnosis. Stage II disease was further stratified as low risk (pT1-3) and high risk (pT4). Logistic regression methods were used to assess Odds Ratio (OR) of AC between EO-CC and AO-CC; Cox models were used to compare overall survival (OS) and Cancer-specific survival (CSS) between both groups during 2000-2010 and 2011-2020. **Results:** Our study included 136,676 pts (AO-CC= 121,606; EO-CC =15,070). The median age for EO-CC was 44 years versus 71 years for AO-CC population. The use of AC increased over time for both the EO-CC (63% vs 66%) and AO-CC (33% vs 40%). Moreover, pts with AO-CC received less AC than EO-CC in the entire study time (see table). In pts with AO-CC, AC improved CSS (HR=0.69 for 2000-2010, HR=0.50 2011-2020). However, among those with EO-CC, AC improved CSS in 2011-2020 (HR=0.76) but did not improve CSS in 2000-2010 (HR=1.02). **Conclusions:** Over the past 20 years EO-CC has been managed aggressively, especially for stage II low risk patients where current guidelines do not support the use of AC. In this real world data set we show that for EO-CC AC did not improve CSS in the decade 2000- 2010 but did improve CSS in the 2011-2020 period. Further analysis may clarify stage specific benefits of AC. There clearly remains a need for better risk stratification, including the use of ctDNA, to identify the subpopulations of EO-CC patients that will best benefit from AC. Research Sponsor: None.

**Multivariable logistic regression model of chemotherapy by timeframe with interaction term: Age* stage.**

| | 2000-2010 | | 2011-2020 | |
|---|---|---|---|---|
| **AO-CC vs EO-CC** | **OR** | **95% CI** | **OR** | **95% CI** |
| **Stage II low risk** | 0.23 | (0.21, 0.26) | 0.28 | (0.24, 0.32) |
| **Stage II high risk** | 0.28 | (0.22, 0.36) | 0.28 | (0.21, 0.36) |
| **Stage III** | 0.36 | (0.32, 0.4) | 0.30 | (0.26, 0.34) |

**106** **Poster Session**

# Frequency of surveillance CT scan ordering in patients with stage II/III colorectal cancer.

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**Background:** Patients with stage II/III colorectal cancer (CRC) undergo surveillance CT scans to detect early asymptomatic oligometastatic spread, which can be treated with curative intent. However, randomized clinical trials show no difference in 5-year mortality in those that undergo low intensity surveillance (annual) compared a higher intensity strategy (every 6 months). In Ontario, the Program in Evidence-Based Care (PEBC), Ontario Health (Cancer Care Ontario) guidelines recommend surveillance CT scan for patients with stage II/III CRC every 12 months. We aimed to determine the frequency at which clinicians in our Ontario-based hospital system ordered surveillance CT scans for patients with stage II/III CRC. **Methods:** This single centre retrospective cohort study included all patients that underwent surgical resection for CRC from April 1, 2018 – March 31, 2019 at our academic tertiary care centre in Ontario, Canada. We excluded patients that were followed at an outside hospital, had stage IV disease, had disease recurrence, had a concurrent neoplasm, or did not have CRC. The primary outcome was the proportion of surveillance CT scans completed 10+ months after the previous scan. We also described sociodemographic and medical characteristics of included patients. Follow-up was until August 2023. **Results:** We included 200 patients. The median age was 62 years (IQR: 52-70), 48% (N=96) were female, the rectum was the most commonly involved disease site (43%, N=86), 38% (N=76) had stage III disease, and 54% (N=107) received adjuvant chemotherapy. A total of 1286 scans were ordered during the study period. Of these, 67% (N=860) were ordered as surveillance. Of these, 18% (n=155) were completed 10 months or longer from the previous scan. The median time interval between surveillance scans was 6.6 months (IQR: 5.9-7.9). Approximately 20% (n=137 of 630) of surveillance scans ordered by general surgeons were completed 10 months or longer from the previous scan, whereas only 9% (n=18 of 205) of scans ordered by medical oncologists were completed 10 months or longer from the previous scan. Less than 1% of surveillance scans (n=3) identified new metastatic disease. **Conclusions:** The minority of surveillance CT scans ordered in patients with stage II/III CRC in our hospital were timed in concordance with Ontario-based guidelines. Future studies should seek to standardize ordering habits such that they align with best practices. Research Sponsor: Sunnybrook AFP Association through the Innovation Fund of the Alternative Funding Plan from the Academic Health Sciences Centres of Ontario.

**107** **Poster Session**

# Care delivery and biomarker testing for people with colorectal cancer receiving 4R care sequence plans.

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**Background:** The 4R Oncology model has shown improvements in patient knowledge of care and timing/sequence of key care events (1,2) but has not been studied in colorectal cancer specific. 4R is Right Info/Care/Patient/Time. The 4R Care Sequence is a patient-facing care plan describing the course and timing/sequence of care. Here we report the impact of using 4R for patients with colorectal cancer. **Methods:** We tested the impact of 4R care sequence plans on patient reported biomarker testing and care delivery referrals and receipt of services at a colorectal cancer program via patient surveys. 4R Plans were provided to patients from Sept '23 – Aug '24 (4R cohort). Patient-reported data were analyzed to compare the 4R cohort (N = 19) to a control cohort of patients who received care pre-4R, Sept '22 – Aug '23 (N = 60). Used Fisher's exact test. **Results:** Response rates: 4R cohort 79% (19/24), Control cohort 87% (41/47). Comparing the 4R cohort to the control cohort respectively, patients were male (47%, 54%), White (89%, 76%), average age (57, 52), and reported income less than $30,000 (21%, 24%). The 4R cohort significantly improved care delivery in five metrics, and directionally improved in care delivery metrics compared to the control cohort (table). One biomarker testing metric significantly improved and other biomarker testing metrics directionally improved in the 4R cohort vs. the control cohort (Table). **Conclusions:** The use of 4R Care Sequence plans for patients with colorectal cancer facilitated improvements in care delivery and patient 's awareness of biomarker testing. While there were directional improvements in all metrics, there is opportunity and room for additional improvements. The team will continue to work on efforts to improve these metrics and optimize the use of 4R Care Sequence plans for patients with colorectal cancer. 1. Trosman 2021. 2. Liu 2023. Research Sponsor: Pfizer and Association of Cancer Care Centers.

**Referral and receipt of services, biomarker testing.**

| | 4R chort N=19 | Control cohort N=41 | Pvalue |
|---|---|---|---|
| **Did your provider recommend you see a nutritionist/dietitian?** | 89% | 20% | **0.0001** |
| ***If Yes, did you see a nutritionist/dietitian ?*** | *84%* | *20%* | ***0.001*** |
| **Did your provider recommend you see your primary care physician about staying as healthy as possible during care treatment?** | 95% | 29% | **0.0001** |
| ***If Yes, did you see your primary care physician?*** | *95%* | *67%* | ***0.04*** |
| **Did your provider recommend you get support for transportation, food, housing, insurance, or other needs?** | 42% | 12% | **0.01** |
| ***If Yes, did you get support for these needs?*** | *88%* | *67%* | *0.4* |
| **Did you provider recommend you get help for emotional distress or worry?** | 53% | 46% | 0.4 |
| ***If Yes, did you get help for emotional distress or worry?*** | *70%* | *47%* | *0.2* |
| **Did your care providers talk to you about biomarker testing after your cancer diagnosis?** | 53% | 37% | 0.2 |
| ***If yes, how clear was it to you why biomarker testing was or was not recommended for you?*** | *90%* | *73%* | *0.3* |
| **Did you get biomarker testing?** | 63% | 34% | **0.03** |
| **Did your care providers discuss results of biomarker tests with you?** | 92% | 87% | 0.6 |

**108** **Poster Session**

# Real-world genomic testing performance in colorectal cancer (CRC): The MultiTEAM Systems Framework Precision Oncology Reflex Testing (TEAMSPORT).

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**Background:** Genomic testing plays a pivotal role in personalizing treatment for CRC patients. Despite recommendations from NCCN, adoption remains uneven. Research indicates that only 64.7% of patients with metastatic CRC (mCRC) underwent molecular testing nationwide. TEAMSPORT seeks to address this gap by evaluating the feasibility of reflex testing and examining how pathology and oncology teams integrate genomic testing into routine practice. **Methods:** This quasi-experimental study tracks testing over time. Eligible patients were 18 years or older, with a histological diagnosis of adenocarcinoma, who received treatment at the University of Kansas Health System (TUKHS). Exclusion criteria included patients discharged to hospice after diagnosis, those whose last contact with TUKHS occurred within 30 days of diagnosis, and patients who passed away within 30 days of diagnosis. Testing rates were calculated as the number of tests performed divided by the number of testing opportunities, where "testing opportunities" refer to NCCN-recommended tests based on cancer stage. An intervention to increase adherence to NCCN-recommended testing began in December 2022. **Results:** From January 2022 to December 2023, 564 colorectal cancer (CRC) cases were eligible for genomic testing, resulting in 1,525 testing opportunities. In localized CRC, testing rates steadily improved, from 92% in January-June 2022 to 98% by July-December 2023 for patients diagnosed outside of TUKHS. At TUKHS, the rates remained high, reaching 100% in the first two periods and staying over 90% in subsequent periods. Patients with mCRC saw similar trends, particularly in the control arm where rates went from 61% to 71%. Specific molecular NCCN-recommended tests were assessed. At TUKHS, testing rates were high for microsatellite instability (MSI - 98%) which is caused by deficiency of the DNA mismatch repair (dMMR) system. Lower testing rates were observed for other markers like *KRAS* (70%), *BRAF* (67%) and *HER2* (48%). Testing rates were assessed based on primary insurance. At TUKHS, the testing rates for Medicaid patients reached 94%, private insurance patients at 78%, and Tricare/VA at 100%. Outside of TUKHS, Medicaid patients had lower testing rates at 21%, private insurance patients at 74%, and CMS patients at 66%. Time to test by processing lab will be presented at the meeting. **Conclusions:** Despite the high testing rates for MSI/dMMR, the overall genomic testing in CRC remains imperfect, with lower rates for *BRAF* and *HER2*. This discrepancy may be related to the fact that targeted therapies for *BRAF* and *HER2* are not yet part of first-line treatment for CRC. Causes for the discrepancy in testing rates by primary insurance have yet to be determined. Future research should address these gaps to enhance precision oncology in CRC management. Research Sponsor: National Cancer Institute; R01CA274560.

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# Disparities in prophylactic tetracycline use among patients with metastatic colorectal cancer undergoing treatment with EGFR inhibitors.

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**Background:** Epidermal growth factor receptor (EGFR) inhibitors have been shown to improve survival in metastatic colorectal cancer and are considered a standard of care option for patients with RAS/RAF wild-type disease. However, anti-EGFR agents are associated with an acneiform rash that can lead to decreased quality of life and early discontinuation of treatment. Prophylactic doxycycline has been shown in a randomized clinical trial to decrease any skin toxicity ≥ grade 2 by 50%. In this study we sought to determine the rate of prophylactic tetracycline use in metastatic colorectal cancer patients receiving anti-EGFR therapy in the United States. **Methods:** The Flatiron Health Electronic Health Record-derived, de-identified database was used to analyze the records of patients who were at least 18 years old, diagnosed with metastatic colorectal cancer, and received treatment with an anti-EGFR agent (cetuximab or panitumumab). Patients were stratified by receipt of prophylactic tetracycline versus not. This was defined by an order for a tetracycline 28 days before to 1 day after initiation of an anti-EGFR agent. Demographic and clinical characteristics were compared via multivariable logistic regression. The change in percentage of patients who received a prophylactic tetracycline over time was assessed and was compared to the rate of anti-emetic prescription which served as a control to ensure drug administration data was accurately captured. **Results:** The records of 6,713 patients who met inclusion criteria and received an anti-EGFR agent between 2013 and 2023 were included in our analysis. A logistic regression model assessing the association between patient characteristics and the receipt of a prophylactic tetracycline suggested that older age, race (black/African American or Other), presence of RAS mutation, and higher line of therapy were all significantly associated with not receiving prophylactic tetracyclines. Other patient characteristics including gender, ECOG status, mismatch repair status, and RAF mutation status did not appear to impact the likelihood of receiving a prophylactic tetracycline. The rate of prophylactic tetracycline use increased from 10.9% in 2013 to 22.3% in 2023 while the rate of prophylactic anti-emetic remained stable at around 82%. **Conclusions:** This real-world analysis suggests the use of preemptive treatment for skin toxicity in patients with metastatic colorectal cancer who are treated with an anti-EGFR in US is suboptimal. Disparities in cancer care are also evidenced by lower use of tetracyclines in elderly and African Americans. Randomized controlled trials support the use of prophylactic tetracycline in these patients; thus, further research into the reason for this practice gap is necessary. Educational efforts should be undertaken to improve tetracycline prescribing rates in this population. Research Sponsor: University of Utah Study Design and Biostatistics Center, with funding in part from the National Center for Research Resources and the National Center for Advancing Translational Sciences, National Institutes of Health; UM1TR004409 (formerly UL1TR002538).

# Increasing RAS mutation result availability in newly diagnosed patients with metastatic colorectal cancer by their first outpatient medical oncology clinic visit at a large safety net hospital.

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**Background:** Rat sarcoma virus mutation testing (RASmt) is critical in guiding systemic therapy selection for patients with metastatic colorectal cancer (mCRC) in the first-line setting. Missing results at the initial medical oncology visit can negatively impact patient outcomes by delaying treatment or leading to inappropriate treatment selection. This quality improvement (QI) project aims to increase the percentage of patients with newly diagnosed mCRC with RASmt results available by their first outpatient medical oncology appointment at a large safety net hospital to 20% over an 8-month period. **Methods:** A multidisciplinary team of key stakeholders including clinicians, a pathologist, and administrators was assembled to determine factors associated with failure to obtain RASmt. A baseline analysis of all patients referred to the gastrointestinal (GI) oncology clinic from June 2022 to April 2023, with manual review of electronic medical records by investigators, showed that only 6% (3/46) of newly diagnosed patients with mCRC had RASmt results available by their first GI oncology appointment. Two Plan-Do-Study-Act (PDSA) cycles were conducted to assess interventions that were implemented based on process mapping and reasons for unavailable test results. To increase the percentage of patients with RASmt results available, starting December 2023, patients initially diagnosed with mCRC as inpatients who were being followed by the oncology consult service had RASmt ordered on the tissue biopsy prior to discharge (PDSA1). Starting April 2024, the medical directorfor outpatient GI oncology clinics ordered RASmt for all patients with mCRC referred to the clinic at the point of referral (PDSA2). New patient referrals were reviewed through July 3, 2024. **Results:** At baseline, the most common reason for unavailable RASmt results was the lack of test orders, with 70% (32/46) of new patients with mCRC not having RASmt ordered prior to their first appointment. Once RASmt was ordered, the average time for results was 13 days. After the first intervention, 4% (1/23) of patients with mCRC had RASmt results available at their first appointment. After the second intervention, 42% (5/12) of patients had RASmt results available. Additionally, median time from diagnosis of mCRC to first appointment decreased from 23 days at baseline to 17 days during PDSA1 to 12 days during PDSA2. **Conclusions:** By ordering RASmt at the point of referral, the percentage of patients who had RASmt available at their first appointment markedly increased from 6% to 42%. This QI initiative additionally demonstrates the importance of multidisciplinary teams to enhance patient care efficiency at a large safety net hospital. Further efforts are ongoing to increase RASmt availability beyond 42% with a focus on reducing the turnaround time of the test.
Research Sponsor: None.

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# Impact of unplanned readmission to the same facility within 30 days following robot-assisted and laparoscopic hemicolectomy for early-onset non-metastatic colon cancer.

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**Background:** Colon cancer in younger populations has risen in recent years, prompting changes in screening guidelines. Hemicolectomy remains integral for standard treatment for non-metastatic early-onset colon cancer, and many institutions have adopted robot-assisted approaches for this procedure. However, high costs associated with this technology along with mixed data regarding survival benefits and post-surgical complications warrant further investigations on the specific advantages of robot-assisted surgery. This study aims to examine the impact of unplanned readmission to the same facility within 30 days following surgery for patients receiving robot-assisted and laparoscopic hemicolectomy on long-term survival for early-onset non-metastatic colon cancer to see how current paradigms of colon cancer treatment extend to early-onset patients. **Methods:** The National Cancer Database was used to identify patients under the age of 50 receiving a minimally invasive hemicolectomy for primary non-metastatic colon cancer diagnosed between 2016-2020. Patients were stratified based on surgical approach and whether they underwent an unplanned readmission to the same facility within 30 days following surgery, and survival differences were calculated log-rank test and Cox regression analysis that included co-variates related to tumor characteristics, patient demographics, and facility characteristics. **Results:** 5,076 patients were identified, with 577 receiving a robot-assisted approach and 4,499 receiving a laparoscopic approach. Multivariate analysis found no significant difference in long-term survival between patients receiving a laparoscopic versus robot-assisted approach (HR: 0.893, 95% CI: 0.687-1.161, p=0.398). Additionally, multivariate analysis found that unplanned readmission within 30 days did not have a significant impact on long-term survival outcomes for patients receiving a robot-assisted approach (HR: 0.856, 95% CI: 0.257-2.848, p=0.800), but it did find worsened survival outcomes for the laparoscopic approach (HR: 1.535, 95% CI: 1.070-2.203, p=0.020). **Conclusions:** Patients receiving laparoscopic hemicolectomy for early-onset non-metastatic colon cancer appear to have worsened overall long-term survival outcomes when undergoing an unplanned readmission to the same facility within 30 days of surgery when compared to patients receiving a robot-assisted surgical approach. However, it also appears that overall, both surgical approaches have similar overall long-term survival. Further research on how robot-assisted surgery impacts survival and post-surgical complications in patients with early-onset non-metastatic colon cancer is warranted to examine how this advancing technology could improve cancer care. Research Sponsor: None.

# Nationwide analysis of Medicare expenditure and utilization of EGFR inhibitors from 2018-2022.

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**Background:** EGFR inhibitors are used extensively for treatment of metastatic colorectal cancer ranging from first line drugs in combination with chemotherapy to monotherapy for refractory disease. As we increasingly move towards targeted treatments for cancer, financial implications of these drugs must be taken into consideration. In this study we aim to analyze changes in Medicare spending for Cetuximab and Panitumumab to better understand the financial impact of these therapies. **Methods:** We obtained data from the publicly available Medicare Part D database from Centers for Medicare and Medicaid Services. Data on total spending, total claims, total beneficiaries, average spending per claim, and average spending per beneficiary was extracted from the database. We adjusted the data for 2018 for 2022 using a standard annual inflation rate. Percentage change was calculated using the adjusted data. **Results:** For Cetuximab, we saw a sharp increase in total spending by 38.77%, from $4,912,903.87 in 2018 to $6,811,781.05 in 2022. The number of claims rose by 31.96%, while the total number of beneficiaries saw a modest increase of 2.44%. Average spending per claim increased by 16.04%, and average spending per beneficiary grew by 49.44%. Panitumumab showed a similar rise in total spending by 43.23%, from $1,594,347.66 in 2018 to $2,283,142.83 in 2022. Claims increased by 25.42%, and beneficiaries grew by 47.83%. Average spending per claim increased by 25.97%, and average spending per beneficiary saw a substantial rise of 65.75%. **Conclusions:** Our results indicate a significant increase in the financial burden related to EGFR inhibitors. As we move away from chemotherapy and shift our focus to targeted therapies, it is imperative to explore generic alternatives and direct policy changes to ensure equitable access across various racial and socio-economic groups. These findings highlight the need for further research into healthcare policy reforms to ease the burden on our health system. Research Sponsor: None.

| Drug | Year | Total Spending ($) | Total Claims | Total Beneficiaries | Avg Spending/ Claim ($) | Avg Spending/ Beneficiary ($) |
|---|---|---|---|---|---|---|
| **Cetuximab** | 2018 | 4,912,903.87 | 551 | 123 | 8,072.80 | 36,163.51 |
| | 2022 | 6,811,781.05 | 727 | 126 | 9,369.71 | 54,061.75 |
| | % Change | 0.3877 | 0.3196 | 0.0244 | 0.1604 | 0.4944 |
| **Panitumumab** | 2018 | 1,594,347.66 | 5,955 | 184 | 242.72 | 30,714.91 |
| | 2022 | 2,283,142.83 | 7,470 | 272 | 305.81 | 50,736.51 |
| | % Change | 0.4323 | 0.2542 | 0.4783 | 0.2597 | 0.6575 |

# Integrating pharmacogenetic testing in the management of gastrointestinal cancers.

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**Background:** Gastrointestinal cancers are among the most prevalent malignancies, often treated with chemotherapeutic agents whose efficacy and toxicity can be influenced by genetic variations. Pharmacogenetic testing for key metabolic genes, including DPYD, UGT1A1, and G6PD, can guide personalized treatment approaches. **Methods:** We analyzed samples from 3,519 individuals for genotyping using semiconductor-based Next-Generation Sequencing (NGS) technology. High-quality genomic DNA was extracted and subjected to target enrichment via high multiplex PCR amplification using an NGS panel targeting variants of key metabolic genes, including DPYD, G6PD, and UGT1A1. **Results:** For the DPYD gene, 96% of patients were normal metabolizers. However, 4% intermediate and 0.17% individuals had poor metabolizer status. Poor metabolizers possess homozygous non-functional alleles, leading to complete dihydropyrimidine dehydrogenase (DPD) deficiency and increased toxicity risk with fluoropyrimidine drugs like 5-fluorouracil (5-FU), Capecitabine, and Tegafur. Patients with certain homozygous or compound heterozygous variants in the DPYD gene are at increased risk for acute early-onset toxicity and serious, including fatal, adverse reactions due to fluorouracil. 5-FU is not recommended for use in patients known to have certain homozygous or compound heterozygous DPYD variants that result in complete absence of DPD activity. Intermediate metabolizers exhibit approximately 50% reduced DPD activity, necessitating a 50% reduction in starting doses followed by careful titration based on toxicity. UGT1A1 analysis revealed 46% normal, 42% intermediate, and 12% poor metabolizers, affecting Irinotecan metabolism and toxicity risks. Poor metabolizers show decreased UGT1A1 activity, resulting in reduced clearance of Irinotecan and increased risk of dose-limiting toxicities. These patients are also at higher risk for neutropenia, diarrhea, and asthenia, requiring starting dose adjustments. Additionally, they may experience elevated hyperbilirubinemia with treatments like Regorafenib. Recognizing G6PD deficiency in GI cancers can prevent serious complications and ensure safer treatment options. G6PD testing indicated that 98% were homozygous wildtype, 2% heterozygous, and 0.31% homozygous/hemizygous deficient. Patients with homozygous/hemizygous deficient genotype may be at increased risk for drug-induced hemolysis, particularly when treated with Dabrafenib. **Conclusions:** These findings highlight the therapeutic significance of incorporating genetic testing into clinical practice to tailor treatments, enhance drug efficacy, and minimize side effects in GI cancer management. By integrating pharmacogenetic insights, the safety and effectiveness of therapies can be significantly improved, ultimately leading to better patient care in GI oncology. Research Sponsor: None.

# Genetic variants and clinical factors affecting the response to 5-fluorouracil–based treatment in Chilean patients with advanced colorectal cancer.

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**Background:** Colorectal cancer (CRC) ranks as the second most common cancer in Chile, affecting both men and women. Late diagnosis results in approximately 25% of patients presenting with metastatic disease, with a five-year survival rate of about 14%. Standard treatment includes tumor resection followed by adjuvant chemotherapy, commonly involving 5-fluorouracil (5-FU) combined with drugs like oxaliplatin or irinotecan. However, patient responses to 5-FU vary greatly due to genetic polymorphisms in enzymes like thymidylate synthase (TYMS) and dihydropyrimidine dehydrogenase (DPD), which affect the drug's pharmacokinetics and pharmacodynamics. Polymorphisms in genes related to oxaliplatin elimination and efficacy, such as glutathione-S-transferases (GSTs) and DNA repair enzymes, also play a role. However, the relationship between genetic variants and treatment outcomes remains unclear, often varying by population and cancer stage. Although predictive models for treatment response and risk exist for various diseases, no comprehensive model combining genetic and clinical variables to predict chemotherapy safety in Chilean CRC patients has been developed. **Objective:** This study aimed to identify relevant genetic variants in the genes *TYMS*, *TYMP*, *DPYD*, *GSTP1*, *MTHFR*,*ERCC2*, *ABCB1*, *ABCC2*, *ABCC4*, and *ABCG2*, which, along with clinical variables, could help create a predictive model for the safety of 5-FU-based chemotherapy in advanced CRC patients. **Methods:** A retrospective nested case-control study was conducted on 82 advanced CRC patients. Sixteen genetic variants were analyzed to assess their influence on adverse reactions (ADRs) and their severity, using logistic regression to identify potential associations. Multivariate models were developed to predict chemotherapy safety. **Results:** Among the 16 variants analyzed in 82 patients, several key findings emerged: The G allele of *GSTP1* (rs1695) was found to be protective against neuropathy (OR = 0.147; p = 0.012) but increased the risk of mucositis (OR = 2.27; p = 0.036). The C allele of *DPYD* (rs1801265) was associated with an elevated risk of neuropathy (OR = 4.50; p = 0.05). The *TYMS* deletion (rs151264360) provided protection against cutaneous (OR = 0.029; $p < 0.0001$) and hematological ADRs (OR = 0.098; p = 0.005). Additionally, *TYMS* deletion was protective against severe toxic ADRs (OR = 0.098; p = 0.005). Two multivariate models were created to predict anemia (p = 0.027) and pain (p = 0.01) development. **Conclusions:** This study represents an initial step toward developing predictive models for ADRs related to 5-FU, such as neuropathy, mucositis, and hematological and skin toxicities. In the future, these findings may contribute to pharmacogenetic-based dose adjustment models aimed at reducing ADRs in Chilean CRC patients undergoing 5-FU-based treatment. Research Sponsor: FONDECYT Regular Project N°1211948.

# A protocol for pretreatment testing for antibodies to galactose-alpha-1,3-galactose to mitigate the risk of cetuximab hypersensitivity reactions.

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**Background:** Cetuximab has been shown to improve survival in patients with KRAS wildtype metastatic colorectal cancer. However, high rates of hypersensitivity reactions (HSRs) limit its use, with HSR rates up to 10-20%. A major driver of cetuximab HSR is from pre-formed antibody response to galactose-1,3-alpha-galactose (alpha-gal). Evidence from retrospective studies supports alpha-gal pre-screening in this setting. We are the first to report on the impact of prospective alpha-gal antibody screening on cetuximab HSR. **Methods:** Records were reviewed across three medical oncology centres that have adopted alpha-gal antibody screening measures. Data for patients with metastatic colorectal cancer treated with cetuximab were retrieved. All centres assessed alpha-gal antibody status using the ImmunoCAP ELISA assay. Due to lack of a shared screening approach across study sites, a pre-determined protocol was retrospectively applied to all cases. This protocol allowed cetuximab administration if alpha-gal levels were ≤0.1 kUA/L, but prohibited it if alpha-gal levels were >0.1 kUA/L in favour of panitumumab administration. Patients were allocated to either the Protocol Applied or Protocol Not Applied cohorts based on protocol requirements being met. The primary outcome between these patient groups was the incidence of cetuximab HSRs. **Results:** Of 254 assessable patients, 39 underwent the pre-treatment screening protocol. Of the Protocol Applied group, 3% (n=1/38) experienced a cetuximab HSR compared to 16% (n=35/215) in the Protocol Not Applied group (Odds ratio (OR) 7.16; 95% CI 1.13–299.61, $p$=0.02). Patients with alpha-gal antibody titres >0.1 kUA/L were more likely to experience a cetuximab HSR (OR 69.71; 95% CI 5.18–4296.81, $p$=0.0001). **Conclusions:** Pre-treatment screening for alpha-gal antibodies significantly reduces the incidence of cetuximab HSRs. A testing threshold of 0.1 kUA/L is effective in identifying patients at risk. Implementing this protocol can improve the safety of cetuximab therapy in high-risk populations. Research Sponsor: None.

# Sex differences in toxicities and survival outcomes among patients with stage III colorectal cancer receiving adjuvant fluoropyrimidine monotherapy: A pooled analysis of four randomized controlled trials (JCOG2310A).

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**Background:** Fluoropyrimidine agents are the key component of adjuvant chemotherapy for pathological stage III colorectal cancer (CRC). Western studies have shown that female sex is associated with favorable survival outcomes after surgery, yet a risk factor for fluoropyrimidine-related adverse events (AEs) during adjuvant chemotherapy. However, it remains unclear whether there are sex differences in the treatment outcomes in this setting among the Japanese population. **Methods:** Patients enrolled in four randomized controlled trials (JCOG0205, JCOG0404, JCOG0910, and JCOG1006) who received adjuvant fluoropyrimidine monotherapy for pathological stage III CRC were analyzed. Incidences of AEs and survival outcomes (relapse-free survival [RFS], and overall survival [OS]) were compared between males and females. Multivariable logistic regression analyses were performed to evaluate associations between sex and the development of AEs. Multivariable Cox regression analyses were performed using Cox proportional hazard models to evaluate associations between sex and survival outcomes. **Results:** A total of 3170 patients (male, n = 1654; female, n = 1516) were included in the analysis. Compared with male patients, female patients were more likely to require dose reduction (male, 20.4%; female: 28.8%, $p < 0.0001$) and had low completion proportions of adjuvant chemotherapy (male, 81.7%; female, 77.0%, p = 0.0011). Female patients experienced higher incidences of any-grade hematological and non-hematological AEs (Table). Female sex was associated with higher incidences of grade 3/4 hematological AEs (odds ratio [OR]: 1.702 [95% CI 1.272–2.278], p = 0.0003) and non-hematological AEs (OR: 1.666 [95% CI 1.373–2.020], $p < 0.0001$). Female sex was identified as a favorable prognostic factor for RFS (hazard ratio [HR]: 0.729 [0.627–0.848], $p < 0.0001$) and OS (HR: 0.796 [0.653–0.970], p = 0.0238). **Conclusions:** Sex differences had implications for the development of AEs and survival outcomes of Japanese patients with pathological stage III CRC who received adjuvant fluoropyrimidine monotherapy. Research Sponsor: National Cancer Center; 2023-J-03.

**Incidences of adverse events.**

| Adverse event | Any grade | | | Grade 3/4 | | |
|---|---|---|---|---|---|---|
| | Male % (95% CI) | Female % (95% CI) | p-value | Male % (95% CI) | Female % (95% CI) | p-value |
| Any hematological toxicity | 63.7 (61.3–66.0) | 83.9 (82.0–85.8) | <0.0001 | 5.0 (4.0–6.1) | 8.9 (7.5–10.4) | <0.0001 |
| Any non-hematological toxicity | 93.3 (92.0–94.5) | 96.1 (95.0–97.0) | 0.0006 | 13.3 (11.7–15.1) | 20.8 (18.8–22.9) | <0.0001 |

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# Meta-analysis of phase II/III randomized controlled trials (RCTs) to evaluate the incidence of hand-foot skin reaction (HFSR) or palmar-plantar erythrodysesthesia (PPE) syndrome in patients with gastrointestinal (GI) cancers treated with fruquintinib.

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**Background:** Fruquintinib is a selective inhibitor of vascular endothelial growth factor receptors 1, 2, and 3 blocks the angiogenesis associated with tumor proliferation. Recently, the US FDA approved fruquintinib as a third-line treatment for metastatic colorectal cancer (mCRC). As with any other novel cancer therapies, adverse events are always a concern and all patients should be monitored in order to early detect and appropriately treat any adverse events. This meta-analysis aims to assess the risk of HFSR or PPE syndrome in patients with GI cancers treated with fruquintinib. **Methods:** We conducted a comprehensive literature search using MEDLINE, EMBASE, and COCHRANE databases from inception through August 12th, 2024. Phase II/III RCTs utilizing fruquintinib in GI cancers mentioning HFSR or PPE as an adverse effect were included. Mantel-Haenszel method was used to calculate the estimated pooled risk ratio (RR) with 95% confidence interval (CI). Heterogeneity was assessed with Cochran's Q-statistic. Random effects model was applied. **Results:** A total of 1872 patients, of whom 1131 (60%) received fruquintinib, from 3 phase III RCTs (FRESCO, FRESCO-2, FRUTIGA) and 1 phase II RCT were eligible for evaluation of HFSR or PPE incidence. FRESCO, FRESCO-2, and phase II trials involved patients with mCRC, they compared fruquintinib vs placebo. FRUTIGA trial involved patients with gastric or gastroesophageal junction adenocarcinoma, it compared fruquintinib + paclitaxel vs placebo + paclitaxel. The incidence of any-grade and high-grade HFSR or PPE was higher in the fruquintinib group compared to the placebo group, 27.93% vs 3.64% (RR, 7.64; 95% CI: 4.08-14.33; $P<0.00001$), and 8.5% vs 0% (RR, 26.0; 95% CI: 6.42-105.29; $P<0.00001$), respectively. In the mCRC subgroup analysis, the higher incidence of any-grade and high-grade HFSR or PPE in the fruquintinib arm was also statistically significant, 29.70% vs 2.55% (RR, 10.27; 95% CI: 5.59-18.88; $P<0.00001$), and 8.45% vs 0% (RR, 19.34; 95% CI: 3.84-97.36; $P=0.0003$), respectively. **Conclusions:** This meta-analysis revealed increased incidence of any-grade and high-grade HFSR or PPE in patients with GI cancers including mCRC treated with fruquintinib. In the FRESCO trial, this association was also shown to confer survival benefit in Chinese patients with mCRC. Prompt identification and providing proper supportive care is crucial in managing this adverse event and ultimately, preserving the patients' quality of life. Research Sponsor: None.

# Meta-analysis of randomized controlled trials (RCTs) to evaluate the incidence of hemorrhage and venous thromboembolism (VTE) events in patients with gastrointestinal (GI) cancers treated with fruquintinib.

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**Background:** Fruquintinib is a novel cancer therapy that was recently approved by the US FDA as a third-line treatment for metastatic colorectal cancer (mCRC). It works by inhibiting the vascular endothelial growth factor receptors, that leads to blocking of angiogenesis which is essential for the tumor proliferation. Close monitoring is always crucial in order to promptly detect and manage any adverse events. This study aims to assess the risk of hemorrhagic and VTE events in patients with GI cancers treated with fruquintinib. **Methods:** A comprehensive literature search was conducted using MEDLINE, EMBASE, and COCHRANE databases from inception through August 12th, 2024. Phase II/III RCTs utilizing fruquintinib in GI cancers reporting hemorrhage or VTE events as an adverse event were included. Mantel-Haenszel method was used to calculate the estimated pooled risk ratio (RR) with 95% confidence interval (CI). Heterogeneity was assessed with Cochran's Q-statistic. Random effects model was applied. **Results:** A total of 1872 patients were included. Data were pooled from three phase III RCTs (FRESCO, FRESCO-2, FRUTIGA) and one phase II RCT, all reporting VTE events including deep vein thrombosis and pulmonary embolism and hemorrhage. FRESCO, FRESCO-2, and the phase II trials involved patients with mCRC, comparing fruquintinib to placebo. FRUTIGA trial incorporated patients with gastric or gastroesophageal junction adenocarcinoma, comparing fruquintinib + paclitaxel to placebo + paclitaxel. High-grade hemorrhage incidence was higher in the fruquintinib arm compared to the control arm (29.00% vs 21.18%, RR 1.62; 95% CI: 1.26-2.08, P=0.0001). In the mCRC subgroup, high-grade hemorrhage rate was also higher and statistically significant in the fruquintinib arm 20.87% vs 11.22% (RR 1.84; 95% CI: 1.25-2.70; P=0.002). Incidence of any grade or high grade VTE events was neither statistically significant in GI cancer population or mCRC subgroup. Incidence of any grade VTE events in GI cancer group and mCRC subgroup was 2.74% vs 2.15% (RR, 1.23; 95% CI: 0.53-2.88; P=0.63) and 2.68% vs 1.53% (RR, 0.95; 95% CI: 0.09-9.67; P=0.97), respectively. Rate of high grade VTE events in GI cancer cohort was 1.67% vs 0.67% (RR, 1.96; 95% CI: 0.52-7.36; P=0.32), while in mCRC subgroup was 1.79% vs 0.76% (RR, 1.08; 95% CI: 0.06-19.98; P=0.96). **Conclusions:** This meta-analysis demonstrates that patients with GI cancers, including mCRC, treated with fruquintinib have higher incidence of high-grade hemorrhage. These findings underscore the importance of individualized risk assessment, especially for patients predisposed to hemorrhagic complications. No increased risk of VTE events was noted in patients treated with fruquintinib compared to placebo. Research Sponsor: None.

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# Meta-analysis of phase II/III randomized controlled trials (RCTs) to determine the incidence of hypertension and proteinuria in patients with gastrointestinal (GI) cancers treated with fruquintinib.

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**Background:** Fruquintinib is a selective inhibitor of vascular endothelial growth factor receptors which impedes angiogenesis associated with tumor growth. On November 8th, 2023, the US FDA approved fruquintinib for use as a third-line treatment for metastatic colorectal cancer (mCRC). With the introduction of novel cancer therapies, monitoring for adverse events is critical. This meta-analysis aims to assess the risk of hypertension (HTN) and proteinuria in patients with GI cancers treated with fruquintinib. **Methods:** We conducted a comprehensive search of MEDLINE, EMBASE, and COCHRANE databases from inception through August 12th, 2024, identifying Phase II/III RCTs that employed fruquintinib in GI cancers and reported HTN and proteinuria as adverse effects. The Mantel-Haenszel method was used to calculate pooled risk ratios (RR) with 95% confidence intervals (CI). Heterogeneity was assessed using Cochran's Q-statistic, and a random effects model was applied. **Results:** Our analysis included 1872 patients, 1131 (60%) of whom received fruquintinib, from three Phase III RCTs (FRESCO, FRESCO-2, FRUTIGA) and one Phase II RCT. FRESCO, FRESCO-2, and the phase II trials compared fruquintinib to placebo in patients with mCRC. FRUTIGA trial involved patients with gastric or gastroesophageal junction adenocarcinoma, comparing fruquintinib + paclitaxel to placebo + paclitaxel. The incidence of any-grade HTN was higher in the fruquintinib group, 34.92% vs 7.55% (RR 3.96; 95% CI: 3.05-5.13; $P<0.00001$); the rate of high-grade HTN was 13.96% vs 1.21% (RR 9.01; 95% CI: 4.67-17.40; $P<0.00001$). In the mCRC subgroup, incidence of any-grade and high-grade HTN was 43.02% vs 10.45% (RR 3.97; 95% CI: 2.95-5.33; $P<0.00001$), and 17.28% vs. 1.27% (RR 12.06; 95% CI: 5.19-28.01; $P<0.00001$), respectively. Incidence of any-grade proteinuria was also found to be statistically significant and higher in the fruquintinib arm in both GI cancer cohort and mCRC subgroup: 27.3% vs 16.19% (RR 1.89; 95% CI: 1.30-2.74; P=0.0008), and 25.22% vs 11.73% (RR 2.29; 95% CI: 1.17-4.51; P=0.02). The rate of high-grade proteinuria was not statistically significant in both cohorts. In GI cancer group, 1.85% vs 0.53% (RR, 2.49; 95% CI: 0.86-7.16; P=0.09), and mCRC subgroup 2.17% vs 0.51% (RR, 2.87; 95% CI: 0.74-11.12; P=0.13). **Conclusions:** This meta-analysis revealed a higher incidence of any-grade and high-grade HTN with fruquintinib in GI cancer patients, with significant association in the mCRC subgroup. Fruquintinib was also noted to be associated with increased risk of any-grade proteinuria, however, incidence of high-grade proteinuria was not different in patients treated with fruquintinib compared to placebo. Prompt identification and early management of these adverse events are crucial. Research Sponsor: None.

# Quality of life following complete clinical response to chemoradiotherapy for rectal cancer: Patient-reported outcomes.

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**Background:** Complete clinical response (cCR) after neoadjuvant therapy (chemoradiotherapy/total neoadjuvant treatment) for rectal cancer may permit an organ-preserving approach with "watch and wait" (W&W) surveillance. This confers comparable survival, while avoiding the sequelae of pelvic surgery. Data on quality of life (QOL) for this cohort are lacking. **Methods:** Patient-reported outcomes were collated for patients on W&W following cCR in a single hospital network. Validated questionnaires evaluating surgical, radiation and chemotherapy-related QOL were completed, including Low Anterior Resection Syndrome (LARS) score, Functional Assessment of Chemotherapy: Colorectal (FACT-C), EORTC QLQ-CIPN20, EQ-5D-5L and a linear analogue and ordinal scale ranking self-perception of health. **Results:** From 2016-2023, 76 patients were enrolled on W&W. Median follow-up is 37 months (IQR20-63). 12 patients (15.8%) had local regrowth, of whom 11 (92%) had salvage surgery. Five patients (6.5%) developed metastases at median 8 months (IQR3-12) from W&W entry. Of 57 patients (75%) with sustained cCR on surveillance, 44 (77%) responded to structured questionnaires. Two with stomas were excluded from LARS. 74% of respondents had no LARS, 9% minor LARS and 17% major LARS. 43% reported faecal urgency and 33% faecal incontinence. 41% experienced functional impairment due to chronic chemotherapy-related side effects (Table). Median EORTC CIPN20 score was 23 (IQR20-28; possible range 19-76, high score = poor QOL). Median sensory score was 11 (IQR9-15.5, PR 9-36), median motor score 8 (IQR8-11, PR 8-32) and median autonomic score 4 (IQR3-6) for males (PR 3-12) and 2 (IQR2-2.5) for females (PR 2-8). 54% of men reported erectile dysfunction, of whom 62% experienced severe dysfunction (33% of all males). Median FACT-C score was 123.5 (IQR108-131; PR 0-136, high score = better QOL). Median overall health score was 80% (IQR70-95). 36% of patients reported difficulty with mobility, 27% with usual activities and 11% with self-care. 30% declared ongoing physical pain/discomfort, while 23% live with anxiety/depression related to their diagnosis. Overall, 86% of reported problems were mild/moderate and 14% severe/very severe. **Conclusions:** This study highlights self-reported QOL in patients on W&W pathway following neoadjuvant treatment of rectal cancer. It provides important insight to side-effects of treatment escalation in hope of organ preservation, and presents vital information for the counselling of future patients. Research Sponsor: None.

| | Patients (n=41) |
|---|---|
| **Upper limb paraesthesia** | 12 (29%) |
| **Lower limb paraesthesia** | 17 (41%) |
| ***Difficulty with:*** | |
| **- Walking/standing (foot drop or impaired sensation)** | 10 (24%) |
| **- Stairs/sit-to-stand (leg weakness)** | 11 (27%) |
| **- Hot/cold differentiation** | 4 (10%) |
| **- Holding pen/manipulating small objects/opening jar/bottle** | 12 (29%) |
| **Hearing impairment** | 13 (32%) |
| **Blurred vision** | 4 (10%) |

# Safety and tolerability of different total neoadjuvant therapy strategies in patients with locally advanced rectal cancer.

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**Background:** Total neoadjuvant therapy (TNT) leads to improved local control and organ preservation for locally advanced rectal cancer (LARC). The chemotherapy part of TNT may play a critical role in achieving local control and in improving OS. Few studies have focused on the toxicity associated with the chemotherapy of TNT. Furthermore, the differences in safety and tolerability between radiotherapy-first and chemotherapy-first strategies remain unclear. **Methods:** LARC patients treated with TNT between June 2020 and July 2023 at a single cancer institute were retrospectively enrolled. TNT consisted of induction chemotherapy followed by long-course chemoradiotherapy (INCT-LC-CRT) or long-course chemoradiotherapy/short-course radiotherapy followed by consolidation chemotherapy (LC-CRT-CNCT/SC-RT-CNCT). Concurrent chemotherapy in LC-CRT was capecitabine (825 mg/m$^2$ oral twice daily). INCT/CNCT consisted of six cycles of CAPOX (capecitabine 1000 mg/m$^2$ oral twice daily on days 1–14 and oxaliplatin 130 mg/m$^2$ intravenously on day 1 every 21 days). INCT allowed the addition of bevacizumab (7.5 mg/kg intravenously on day 1). We compared adverse events (AEs) and treatment exposure between INCT-LC-CRT, LC-CRT-CNCT and SC-RT-CNCT. **Results:** A total of 93 patients were included. Patients with INCT-LC-CRT and LC-CRT-CNCT/SC-RT-CNCT were 28 (30.1%) and 45 (48.4%)/20 (21.5%), respectively. Patients who were treated with INCT-LC-CRT (53.6%) had more N2 stage than those with LC-CRT-CNCT/SC-RT-CNCT (21.6%). Grade 3–4 AEs during chemotherapy were significantly more frequent in LC-CRT-CNCT/ SC-RT-CNCT group than in INCT-LC-CRT group (50.8% vs. 21.4%; $p = 0.02$). Neutropenia was the most common Grade 3–4 AEs during chemotherapy in INCT-LC-CRT group (7.4%) and LC-CRT-CNCT/SC-RT-CNCT group (27.7%). Grade 3–4 AEs during chemotherapy were not significantly different between the LC-CRT-CNCT and SC-RT-CNCT groups (36.9% vs. 40.0%; $p = 0.42$). All patients underwent the full radiation dose. The completion rate of six cycles of chemotherapy was no significant difference between INCT-LC-CRT group and LC-CRT-CNCT/SC-RT-CNCT group (92.9% vs. 84.6%; $p = 0.37$). However, LC-CRT-CNCT/SC-RT-CNCT group required significantly more dose modification during chemotherapy than INCT-LC-CRT group (75.4% vs. 46.4%; $p < 0.01$). Dose modification during chemotherapy between LC-CRT-CNCT and SC-RT-CNCT group was no significant difference (modification rate 77.8% vs. 70.0%; $p = 0.54$). Chemotherapy discontinuations due to AEs in INCT-LC-CRT group and LC-CRT-CNCT/SC-RT-CNCT group were 1(3.6%) and 6 (9.2%), respectively. **Conclusions:** Grade 3–4 AEs during chemotherapy were more frequent and required dose modification of chemotherapy in a radiotherapy-first strategy. Safety and tolerability of chemotherapy part of TNT may differ between radiotherapy-first and chemotherapy-first strategies. Research Sponsor: None.

# Tumor lysis syndrome in colorectal cancer: A systematic review of clinical characteristics.

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**Background:** Tumor lysis syndrome (TLS) is an oncologic emergency resulting from the release of potassium, phosphorus, and nucleic acids from tumor cells. While rare in colorectal cancer (CRC), TLS is linked with high mortality, and literature is limited. This study aims to summarize the clinical characteristics of patients with TLS in CRC. **Methods:** We conducted a systematic review of TLS in CRC through a comprehensive literature search in PubMed and manual searches from January 1st, 1993, up to September 13th, 2024. Keywords included "Tumor Lysis Syndrome," "Colorectal Cancer," and "Colon Cancer." 32 studies were found, identifying 11 spontaneous and 22 therapy-related TLS cases. Quantitative and qualitative data were extracted for descriptive analysis. **Results:** A total of 33 patients (16 males and 15 females) had a mean age of 56.5 ± 17.8 years at TLS diagnosis. The mean ages for males and females were 58.3 ± 7.1 and 55.8 ± 13.7 years, respectively (p=0.59). The primary histology was adenocarcinoma (92%) followed by neuroendocrine (8%) (data unavailable for 9 patients). Among 15 patients with grading information, Grade 2 tumors were most common (53%), followed by Grade 3 (40%) and Grade 1 (7%). Staging data was available for 32 patients. Most patients had Stage IV disease (94%), with frequent metastases to the liver (93%), lungs (30%), and bones (10%). Hyperuricemia is present in all patients (1 patient without data). Acute kidney injury (AKI) was present in all patients (4 patients without data). Hyperphosphatemia was found in 82% of patients (5 patients without data). Hypocalcemia was noted in 52% (10 patients without data), and hyperkalemia was found in 74% (6 patients without data). Out of the 33 patients, 11 experienced spontaneous TLS (33%) and 22 had therapy-related TLS (67%). Duration of TLS onset following therapy was available for 17 patients, including one outlier at 30 days on Capecitabine. The mean duration for therapy-related TLS, excluding the outlier, was 3.8 ± 2.5 days. Among the 22 patients on therapy, 59% received 5-Fluorouracil (5-FU)-based treatment. 31% of those receiving 5-FU had combination therapy with Bevacizumab. Death occurred in 59% of patients, with data missing for 4 patients. **Conclusions:** TLS in colorectal cancer (CRC) is associated with a mortality rate of up to 59% and shows no gender preference. Identifying predictive factors is crucial for TLS prophylaxis. Patients with TLS in CRC often present with Stage IV disease, liver metastasis, hyperuricemia, and acute kidney injury (AKI). Research Sponsor: None.

# First-in-human, phase 1a, dose escalation study of BGB-B167, a CEA x 4-1BB bispecific antibody, as monotherapy or combined with tislelizumab (anti-PD-1), in patients with selected advanced or metastatic solid tumors.

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**Background:** BGB-B167 is potentially a first-in-class IgG-based bispecific antibody that targets CEA, a tumor-associated antigen overexpressed in many cancers, and 4-1BB, a key T-cell co-stimulatory receptor expressed on activated CD4+/CD8+ T lymphocytes. Preclinical data suggest BGB-B167 binds to these target proteins with high specificity/affinity, inducing T-cell activation and antitumor activity, with potential for mitigating strong AEs, eg CRS and hepatoxicity. We present results of a first-in-human, phase 1a, open-label, multicenter trial of BGB-B167 given as monotherapy (Part A [A]) or with tislelizumab (TIS) (Part B [B]) (NCT05494762). **Methods:** Eligibility: ≥18 y with unresectable/metastatic CRC, CEA+ GC or CEA+ NSCLC previously treated with standard systemic therapy or for whom treatment is unavailable; pts with CRC and known MSI-H/dMMR status must have received prior ICI, if available. Primary objectives: safety/tolerability; MTD and RP2D for BGB-B167 ± TIS; key secondary objective: antitumor activity (RECIST v1.1). **Results:** As of July 7, 2024, 54 pts were enrolled (31 in A, 23 in B) with BGB-B167 dose ranges of 5–1200 mg IV QW assessed in A and 50–600 mg IV QW in B. In A and B, respectively, 29/31 (93.5%) and 18/23 (78.3%) had CRC, 2/31 (6.5%) and 3/23 (13%) had GC, 0/31 (0%) and 2/23 (8.7%) had NSCLC; 19/31 (61.3%) and 18/23 (78.3%) had liver metastases; 20/31 (64.5%) and 14/23 (60.9%) received ≥3 lines of prior therapy. TEAEs occurred in all pts (Table). Most common treatment-related TEAEs were nausea (4/31; 12.9%) and fatigue (3/31; 9.7%) in A and pruritus (5/23; 21.7%) and diarrhea (4/23; 17.4%) in B; there were no treatment-related TEAEs indicative of hepatotoxicity or CRS. There was a single DLT (gr 3 SJS) in a pt who received 600 mg BGB-B167 + TIS (resolved after treatment discontinuation). MTD was not reached. 3 pts (1 with CRC [A]; 1 with CRC and 1 with GC [B]) had confirmed PRs. DoR was ≥5.6 mo for the pt in A and ≥5.6 mo and ≥6.9 mo for the pts in B; all remained on treatment with ongoing responses as of the data cutoff date. DCR (95% CI) was 33.3% (17.3–52.8) in A and 40.9% (20.7–63.6) in B. Serum exposure of BGB-B167 increased dose dependently from 150 to 1200 mg. **Conclusions:** Overall, BGB-B167 ± TIS was well tolerated and has demonstrated limited antitumor activity in pts with advanced/metastatic CEA+ solid tumors. Clinical trial information: NCT05494762. Research Sponsor: BeiGene, Ltd.

**Safety.**

| | Part A BGB-B167 monotherapy (N=31) | Part B BGB-B167 + TIS (N=23) |
|---|---|---|
| **Any TEAE** | 31 (100.0) | 23 (100.0) |
| **Any treatment-related TEAE** | 17 (54.8) | 15 (65.2) |
| **Gr ≥3** | 4 (12.9) | 1 (4.3) |
| **Serious** | 0 (0) | 2 (8.7) |
| **Leading to death** | 0 (0) | 0 (0) |
| **Leading to treatment discontinuation** | 0 (0) | 1 (4.3) |
| **Any imAE** | 2 (6.5) | 5 (21.7) |
| **IRRs** | 4 (12.9) | 6 (26.1) |

Patients with multiple adverse events (AEs) are counted once. All AEs are listed as n (%).
imAE, immune-mediated AE; IRR, infusion-related reaction; TEAE, treatment-emergent AE.

# Phase II study of FOLFIRI with low-dose irinotecan plus ramucirumab as second-line treatment in Japanese patients with metastatic colorectal cancer (RINDO study).

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**Background:** Phase III trials (ML18147, VELOUR, and RAISE trials) of second-line combination therapy with molecular-targeted agents after first-line treatment with bevacizumab (BEV) for metastatic colorectal cancer (mCRC) demonstrated significant improvements in overall survival (OS). In the RAISE trial (irinotecan (IRI) dose: 180 mg/m$^2$), the relative dose intensity (RDI) of IRI was lower (63.8%) and the incidence rates of adverse events leading to discontinuation of cytotoxic agents was higher (48.6%) in the Japanese population compared to all patients. Based on these results, we conducted a prospective trial to evaluate the efficacy and safety of fluorouracil, levofolinate, and IRI (150 mg/m$^2$, standard dose in Japan) (FOLFIRI) plus ramucirumab (RAM) as second-line treatment for mCRC in Japanese patients. **Methods:** On day 1 of each 2-week cycle, patients with unresectable mCRC who were refractory to oxaliplatin and fluoropyrimidine in combination with BEV or anti-epidermal growth factor receptor (EGFR) antibodies as first-line treatment received 8 mg/kg RAM, followed by the FOLFIRI regimen with low-dose IRI (150 mg/m$^2$). The primary endpoint was progression-free survival (PFS), and secondary endpoints were OS, treatment compliance, and safety. We hypothesized an RFS threshold of 4.3 months and expected RFS of 5.7 months based on data from a Japanese subgroup of the RAISE trial. The protocol treatment was considered to be effective if the lower limit of the 95% confidence interval (CI) exceeded the 4.3-month threshold. **Results:** A total of 62 patients were enrolled from 15 institutions between January 2018 and August 2021. The intent-to-treat and safety populations included 61 and 58 patients, respectively. The cutoff date for the primary analysis was December 2023. Median PFS and OS were 5.9 months (95% CI, 4.8-6.9 months) and 17.0 months (95%CI, 12.0-21.0 months), respectively. Median PFS was 5.7 months (95% CI, 4.4-6.8 months) in patients treated with first-line chemotherapy with BEV and 7.4 months (95% CI, 4.6-11.0 months) in those treated with first-line chemotherapy with anti-EGFR antibodies (hazard ratio [HR], 1.17; 95% CI, 0.64-2.12; p = 0.60), and median OS was 19.8 months (95% CI, 10.4-22.4 months) and 17.5 months (95% CI, 11.5-26.1 months), respectively (HR, 0.96; 95% CI, 0.52-1.78; p = 0.91). The objective response rate and disease control rate were 8.2% and 74%, respectively. Median RDI of IRI, 5-fluorouracil, and RAM were 73.8% (range, 40.3-102.4%), 58.5% (range, 22.8-102.4%), and 80.8% (range, 36.1-102.4%), respectively. The observed Grade ≥3 adverse events included neutropenia (40%), anemia (1.7%), diarrhea (8.6%), fatigue (6.9%), decreased appetite (10%), hypertension (6.9%), and proteinuria (3.4%). **Conclusions:** FOLFIRI with low-dose IRI plus RAM is a feasible second-line treatment in Japanese patients with mCRC. Clinical trial information: jRCTs041180074. Research Sponsor: None.

**LBA125** **Rapid Oral Abstract Session**

# Low-dose aspirin to reduce recurrence rate in colorectal cancer patients with PI3K pathway alterations: 3-year results from a randomized placebo-controlled trial.

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**The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.**

# Retrospective analysis of HAIP therapy in metastatic colorectal cancer: Efficacy, safety, and prognostic factors.

Matthew Sawyer, Nilesh Verma, Amy Little Jones, Anna Moscowitz, Timothy J. Brown, Radhika Kainthla, Salwan Al Matar, Syed Mohammad Ali Kazmi; UT Southwestern Medical Center, Dallas, TX; University of Texas Southwestern Medical Center, Dallas, TX; UT Southwestern Medical School, Dallas, TX

**Background:** HAIP (hepatic arterial infusion pump) therapy is a promising intervention for metastatic colorectal cancer (mCRC) with liver metastases, addressing significant challenges in management and prognosis. It offers targeted treatment to maximize efficacy while minimizing systemic toxicity. We present a retrospective analysis of patients with metastatic colorectal adenocarcinoma treated with HAIP therapy at an academic center and safety net hospital. **Methods:** Patients with mCRC and hepatic metastasis who received at least one HAIP floxuridine (FUDR) cycle from 2013 to 2023 were included. Baseline data, including demographics, tumor characteristics, and treatment history, were collected pre-HAIP. The main outcome was overall survival (OS) post-pump placement, with secondary outcomes being real-world progression-free survival (rwPFS) and HAIP-related complications. **Results:** In total, 88 patients were evaluated for inclusion, of whom 63 met inclusion criteria and were included for analysis. The median age of included patients was 46 years (interquartile range (IQR) 41-57) and 54% were male. Prior to HAIP, 38 (60%) of patients received one line of systemic therapy, while 18 (29%) had two. The median OS from HAIP placement was 28.0 months (95% CI 21.9-NR) and median rwPFS was 11.2 months (95% CI 7.1-16.1). Baseline bilirubin, KRAS mutant status, and CEA were associated with OS in the multivariate cox model. The most common complication from HAIP placement was hematoma (n=4, 6.3%), followed by pump infection (n=2, 3.2%). Grade 3 toxicities from FUDR infusion were infrequent, involving alkaline phosphatase (n=6), AST or ALT elevation (n=3), bilirubin increase (n=2), and platelet count decrease (n=1). **Conclusions:** HAIP treatment is associated with durable responses and few complications, making it well-tolerated by patients. For individuals with colorectal cancer and hepatic metastases, HAIP chemotherapy using FUDR presents a viable and effective treatment option, potentially improving overall outcomes and quality of life. Further prospective and randomized work may establish the role for HAIP in mCRC with liver metastases. Research Sponsor: None.

127 Poster Session

# Zanzalintinib (XL092) alone or in combination with atezolizumab in patients (pts) with refractory metastatic colorectal cancer (mCRC): Results from an expansion cohort of the phase 1 STELLAR-001 study.

Elisa Fontana, Marie Robert, Alejandro Falcon Gonzalez, Damien Pouessel, Iosune Baraibar, Alexander Stein, Haeseong Park, Matthew Reilley, Benoit You, Rasha Cosman, Zhong Wang, Taylor Jew, Yijia Wang, Anwaar Saeed; Sarah Cannon Research Institute, London, United Kingdom; Institut de Cancérologie de l'Ouest & ARPEGO Network, Saint-Herblain, France; Hospital Universitario Virgen del Rocío, Seville, Spain; Institut Universitaire du Cancer de Toulouse - Oncopole, Oncopole Claudius Regaud, Toulouse, France; Vall d'Hebron, Institute of Oncology, Barcelona, Spain; Facharztzentrum Eppendorf, University Cancer Center, Hamburg, Germany; Dana-Farber Cancer Institute, Boston, MA; University of Virginia Cancer Center, Charlottesville, VA; Hospices Civils de Lyon, CITOHL, EPSILYON, Claude Bernard University Lyon 1, Lyon, France; Kinghorn Cancer Centre, St. Vincent's Hospital, University of New South Wales, Sydney, Australia; Exelixis, Inc., Alameda, CA; University of Pittsburgh Medical Center (UPMC) and UPMC Hillman Cancer Center, Pittsburgh, PA

**Background:** VEGFR tyrosine kinase inhibitors (TKIs) in combination with immune checkpoint inhibitors (ICIs) have demonstrated clinical activity in pts with previously treated mCRC, particularly in pts without liver metastases (LM). Zanzalintinib (zanza) is a novel, oral, multi-targeted TKI with activity against VEGFR, MET, and TAM kinases. Here, we present results from the randomized expansion cohort of pts with previously treated non-MSI-H/dMMR mCRC receiving zanza ± atezolizumab (atezo) in the phase 1 STELLAR-001 study (NCT03845166). **Methods:** Adults (aged ≥18 y) with locally advanced or metastatic CRC that was *RAS*-wildtype, who were refractory/intolerant to prior standard of care (5-FU–based treatment ± targeted therapies) and had an ECOG PS of 0/1, were enrolled. Pts with MSI-high or dMMR CRC, or who had received prior treatment with a PD-L1/PD-1 targeting ICI, regorafenib, and/or TAS-102, were excluded. Pts were randomized 1:1 to receive zanza+atezo (100 mg QD + 1200 mg Q3W) or single-agent zanza (100 mg QD). Objectives were to evaluate median (m)OS, investigator-assessed ORR, and mPFS per RECIST 1.1, and safety. **Results:** As of May 6, 2024, 107 pts were enrolled (zanza+atezo, n=54; zanza, n=53). In the zanza+atezo/zanza groups, median age was 61/60 years, 39%/51% had an ECOG PS of 1, pts had a median 2.5/3.0 (range, 1–5) prior lines of therapy, and 31%/32% did not have LM. With a median follow-up of 15 months across both arms, the mOS was 14.3 and 11.1 months for zanza+atezo and zanza, respectively (HR 0.75, 95% CI 0.45–1.26), confirmed ORR was 7.4% and 1.9% (all partial responses), mPFS was 4.0 and 3.0 months (HR 0.68, 0.44–1.04), and DCR was 59% and 55%. In pts without LM, mOS was not reached and 12.5 months (HR 0.46, 0.15–1.36) with 6-month survival rates of 88% and 65%, confirmed ORR was 18.0% and 5.9%, mPFS was 8.2 and 3.3 months (HR 0.40, 0.16–1.0), and DCR was 76% and 59%. The most common treatment-related adverse events (TRAEs) were diarrhea (zanza+atezo, 52%; zanza, 49%), nausea (54%, 36%), and decreased appetite (41%, 36%). Grade 3/4 TRAEs occurred in 48% of pts with zanza+atezo and 40% with zanza (Grade 4 in 3.7% and 0); a Grade 5 TRAE occurred in 1 pt (2%) in each group. 89% and 94% of pts had discontinued study treatment. Zanza was discontinued due to TRAEs in 11% and 8% of pts; atezo was discontinued due to TRAEs in 9% (zanza+atezo). Biomarker analysis is ongoing and results will be presented. **Conclusions:** Clinical activity was observed with zanza both as a single agent and in combination with atezo in this cohort of heavily pretreated pts with mCRC. Greater activity was seen with zanza+atezo versus zanza alone, particularly in pts without LM. Both treatments were generally well tolerated. Evaluation of zanza+atezo versus regorafenib is ongoing in the phase 3 STELLAR-303 study in non-MSI-H/dMMR mCRC (NCT05425940). Clinical trial information: NCT03845166. Research Sponsor: Exelixis, Inc.

# Phase Ib/II study of fruquintinib plus 5-fluorouracil/leucovorin after progression on fruquintinib monotherapy in metastatic colorectal cancer.

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**Background:** Fruquintinib is a standard third-line treatment in metastatic colorectal cancer (mCRC). However, no standard therapeutic regimen is available for mCRC patients who progressed on third-line therapy. Preclinical experiments supported a rationale for combining antiangiogenic therapy and 5-fluorouracil (5FU) in refractory mCRC with evidence for synergy. This open-label phase Ib/II study was designed to determine the safety and efficacy of fruquintinib plus 5-fluorouracil/leucovorin for mCRC patients who progressed on fruquintinib monotherapy. **Methods:** The enrolled patients would receive fruquintinib at a fixed dose of 4mg once daily (three weeks on, one week off) and 5FU at two dose levels (1800mg/m$^2$ and 2400mg/m$^2$) CIV 46 hours on day one and day 15, with leucovorin 400mg/m$^2$ iv 2 hours before 5FU. Treatments were repeated every 28 days. The primary endpoint was determining the recommended phase II dose (R2PD) of 5FU in phase Ib and the six-month OS rate in phase II. Secondary endpoints for phase II included objective response rate (ORR), disease control rate (DCR), progression-free survival (PFS), and safety. Planned enrollment was 39 patients. **Results:** No dose-limiting toxicities occurred in the phase Ib dose-escalation study, and the R2PD of 5FU was 2400mg/m$^2$. Between April 2020 and June 2024, 24 patients were enrolled in the phase II stage. The median age was 57.5 years old (from 35 to 68), and 58.33% were female. Left-sided colorectal cancer patients accounted for 75%. Sixteen (66.7%) patients harbored KRAS mutation, and one (4%) had BRAF V600E mutation. Regarding the number of lines of systemic therapy, 83.3% received this study regimen as fourth-line therapy and 16.7% as fifth-line or later therapy. As of June 30, 2024, efficacy was assessed in all 24 patients with a DCR of 62.5% (15 stable disease, nine progression disease). The median follow-up time was 16.5 months (95% CI:12.553-20.447). The primary endpoint six-month OS rate was 91.3%, the median PFS was 4.4 months (95% CI: 1.243-7.557), and the median OS was 17.3 months (95% CI: 12.892-21.708). The most common TRAEs were nausea (25%), hand-foot syndrome (16.7%), abnormal liver function (8.3%), and proteinuria (8.3%). One patient experienced grade 3 mucositis. No treatment-related death occurred. **Conclusions:** Fruquintinib plus 5-fluorouracil/leucovorin regimen preliminarily demonstrated promising efficacy with a tolerable safety profile for mCRC patients despite progression on prior fruquintinib monotherapy. Further investigation into this possible later-line regimen in mCRC is warranted. Clinical trial information: ChiCTR2000032640. Research Sponsor: None.

# Tetravalent death receptor 5 (DR5) agonist ozekibart (INBRX-109) + FOLFIRI in 2L+ colorectal adenocarcinoma (CRC): Preliminary results from a phase 1 study.

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**Background:** DR5, a proapoptotic receptor, is selectively expressed on tumor vs normal cells, making it an attractive target. DR5 binds to its ligand, TRAIL, resulting in DR5 multimerization and apoptosis; greater DR5 clustering enhances downstream effects. Ozekibart is a next-generation, recombinant, humanized, tetravalent DR5 agonist that is based on a single-domain antibody platform and precisely engineered to balance agonism and safety. Ozekibart previously demonstrated an acceptable safety profile and clinical benefit in chondrosarcoma (1), prompting a phase 2, randomized trial (ChonDRAgon; NCT04950075). An ongoing phase 1 study (NCT03715933) is evaluating ozekibart in various solid tumors, including 2L+ CRC, for which effective therapies are lacking. We present preliminary results of ozekibart + FOLFIRI in 2L+ CRC (data cutoff: Aug 9, 2024). **Methods:** Parts 1 (single-agent dose escalation) and 2 (single-agent dose expansion; recommended phase 2 dose [RP2D] ozekibart 3 mg/kg IV Q3W) of this 3-part, open-label trial are complete. Part 3 (combination dose expansion with chemo) is ongoing in pts with CRC, Ewing sarcoma, or SDH-deficient GIST. Pts with locally advanced or metastatic, unresectable CRC who received oxaliplatin-based chemo were eligible for part 3. Pts with chronic liver disease were ineligible. Pts in the part 3 CRC cohorts (N=13; enrollment complete) received ozekibart 1 mg/kg (n=5) or 3 mg/kg (n=8) IV Q28D (1 cycle) + FOLFIRI Q14D (FOLFIRI could be stopped after ≥6 cycles); in 4/5 pts, ozekibart dose was escalated to 3 mg/kg (RP2D). The primary endpoint is safety; clinical response is an exploratory endpoint. **Results:** In total, 13 pts with CRC (male, 62%) received ozekibart + FOLFIRI. Median age was 60. All pts had metastatic disease; median number of prior lines of therapy was 2 (range, 1-6). At data cutoff, 2 pts remained on treatment. Ozekibart-related adverse events (AEs) were reported in 85% of pts (grade ≥3, 31%) and led to ozekibart interruption in 3 pts and discontinuation in 1 pt. One combo-related AE (neutropenic sepsis) led to death. The most common ozekibart-related AEs were nausea (n=5) and diarrhea, fatigue, and increased alanine aminotransferase (each n=4). Four pts (31%) had responses (all partial). Disease control rate (response + stable disease) was 77% (10/13) and durable (>180 days) in 46% (6/13) of pts. Most pts had tumor shrinkage. Median PFS was 7.85 mo. **Conclusions:** Ozekibart + FOLFIRI showed encouraging preliminary efficacy, including PRs and durable disease control, as 2L+ therapy in pts with CRC. Given these promising data, a new expansion cohort (C4d) is open to validate these findings in a more homogeneous population. Eligible pts will have had 2 or 3 prior lines of systemic therapy (irinotecan allowed but not in the immediately prior line) and must have archival or fresh biopsy tissue. 1. Subbiah. *Clin Cancer Res.* 2023. Clinical trial information: NCT03715933. Research Sponsor: Inhibrx Biosciences, Inc.

**130** **Poster Session**

# Genetic mutations and cytotoxic chemotherapy response for advanced solid tumors: A detailed analysis using the C-CAT database in Japan.

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**Background:** Chemotherapy regimens are selected based on information about the primary organ or its histological type and are administered based on the results of previous clinical trials. Even now, as cancer genomic research has progressed and precision medicine, which optimizes therapeutic interventions based on tumor genome profiling using next-generation sequencing(NGS), has become standard treatment, cytotoxic anticancer drugs still remain key drugs. We examined the real-world database of genomic and clinical information established by the national central datacenter, the Center for Cancer Genomics and Advanced Therapy (C-CAT) to systematically analyze the effects of mutations across multiple cancers and therapies. **Methods:** This study delineated the association between genetic mutations and the therapeutic efficacy of cytotoxic chemotherapy in advanced solid tumors in a real-world Japanese clinical context. We incorporated data from 15,474 patients enrolled in the C-CAT database between June 2019 and June 2022. We analyzed the overall response and time to next treatment (TNT) for genetic alterations and five categories of cytotoxic anticancer drugs in gastrointestinal cancers. **Results:** In the C-CAT data, gastrointestinal cancers were frequently registered, with colorectal cancer being the most common, followed by pancreatic cancer, esophageal/gastric cancer, and biliary tract cancer. For platinum-based drugs, in colorectal cancer, the presence of gene X was associated with not only a higher response rate but also a longer TNT (hazard ratio 0.82, $p<0.001$). In the case of colorectal cancer and pancreatic cancer with KRAS gene mutations, the ORR was low, and the TNT was also short in the case of pancreatic cancer (hazard ratio 1.33, $p<0.001$). Interestingly, when BRCA2 genes are mutant, TNT tended to be longer in many organs, suggesting that platinum-based drugs are effective across organs in carcinomas with BRCA2 mutations. Additionally, gene X showed differences in efficacy not only for platinum but also for antimetabolites and topoisomerase inhibitors. Moreover, KRAS mutation was observed in almost all pancreatic cancers and was a treatment-resistant mechanism for all platinum drugs and antimetabolites. **Conclusions:** This study demonstrated that the effects of cytotoxic anticancer drugs differ depending on genetic mutations. When determining chemotherapy regimens, considering not only the organ but also genetic mutations may enable more efficient drug selection. Research Sponsor: None.

# Adagrasib (Ada) + cetuximab (Cetux) for *KRAS*$^{G12C}$-mutated metastatic colorectal cancer (mCRC): Longer follow-up analysis from KRYSTAL-1.

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**Background:** *KRAS*$^{G12C}$ mutations occur in 3%–4% of CRC cases and are associated with poor prognosis. In the phase 1/2 KRYSTAL-1 study (NCT03785249), at a median follow-up of 11.9 months (mo), Ada (irreversible inhibitor of *KRAS*$^{G12C}$) in combination with Cetux (anti-EGFR antibody) demonstrated promising clinical activity (objective response rate [ORR] of 34% per blinded independent central review [BICR] and 43% per investigator [INV]) and was well tolerated in patients (pts) with previously treated *KRAS*$^{G12C}$-mutated mCRC. Based on these findings, Ada + Cetux was granted accelerated approval in the United States for these pts. Here, we present longer-term follow-up analyses from this study. **Methods:** Adults with previously treated *KRAS*$^{G12C}$-mutated mCRC and ECOG performance status of 0 or 1 were treated with Ada (600 mg BID) in combination with Cetux (400 mg/m$^2$ followed by 250 mg/m$^2$ QW or 500 mg/m$^2$ Q2W) until disease progression, unacceptable toxicity, withdrawal of consent, or death, in separate phase 1 and phase 2 cohorts. Primary endpoints were safety (phase 1) and ORR per BICR (phase 2). Secondary endpoints were duration of response (DOR), progression-free survival (PFS), overall survival (OS), and safety (phase 2). **Results:** A total of 94 pts received Ada + Cetux. The median number of prior lines of systemic therapy was 3 (range, 1–9); 23 pts (24%) had received ≥ 4 prior lines of systemic therapy. The most frequent sites of metastases at baseline were the lung (71%) and liver (64%). At a median follow-up of 20.4 mo, ORR per INV was 43% (95% CI 32–53) and all were partial responses; median DOR was 5.9 (95% CI 5.5–7.6) mo. Disease control rate per INV was 86% (95% CI 78–92). Median PFS per INV was 6.9 (95% CI 5.9–7.4) mo; 6- and 12-mo PFS rates were 61% and 19%, respectively. Median OS was 16.0 (95% CI 13.3–18.8) mo; 6- and 12-mo OS rates were 88% and 66%, respectively. Any-grade treatment-related adverse events (TRAEs) were reported in 100% of pts, 28% of which were grade 3/4. TRAEs led to discontinuation in 10% of pts. Efficacy analyses by BICR, subgroup analyses, and additional safety results will be presented. **Conclusions:** In heavily pretreated pts with *KRAS*$^{G12C}$-mutated mCRC, Ada + Cetux continued to demonstrate clinically meaningful activity and tolerable safety with longer follow-up. These updated results are consistent with those from the primary analysis and constitute the longest duration of follow-up for dual *KRAS*$^{G12C}$/EGFR blockade in this setting. Efficacy of second-line Ada + Cetux vs chemotherapy in *KRAS*$^{G12C}$-mutated mCRC is being investigated in the phase 3 KRYSTAL-10 study (NCT04793958). Clinical trial information: NCT03785249. Research Sponsor: Mirati Therapeutics, a Bristol Myers Squibb company.

# Adverse events (AEs) –derived biomarker in predicting clinical outcomes in patients with colorectal cancer (CRC) treated with immunotherapy (IC).

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**Background:** There are insufficient studies evaluating the role of AEs in CRC patients treated with IC. This study aims to transform AEs from traditional toxicity evaluation tools to informative biomarkers for assessing treatment efficacy by utilizing modern analytical strategies. **Methods:** We analyzed a single arm study of 51 mismatch repair proficient, refractory CRC patients (NCT03712943) treated with regorafenib and nivolumab. Data for analysis included CTCAE version 5 AE data, RECIST treatment response, progression-free survival (PFS), and overall survival (OS). Analytic approach leveraged multiple AE parameters to develop a set of innovative AE biomarkers. One AE biomarker of interest was treatment related low-grade early AEs (TR-LG-AE) which was defined as frequency of all TR-LG-AEs occurring between the 1st day and day 30 after the 1st treatment. **Results:** TR-LG-AE was associated with treatment response ($p<0.001$ by ANOVA trend test). Patients with higher AE events tended to be a responder, compared to PD patients with a lower frequency of AE events. The TR-LG-AE also correlated with improved PFS and OS with hazard ratio of 0.83 ($p=0.01$) and 0.89 ($p=0.03$), respectively. Accordingly, patients were grouped into high versus low TR-LG-AE based on optimal cutoff of 6 events for further analysis. The dichotomized TR-LG-AE (high vs low) showed significant association with survival outcomes (median PFS: 15.3 vs 3.7 months; $p<0.001$; median OS: 21.9 vs 9 months; $p=0.02$). Further interaction analysis showed patients with high TR-LG-AE had improved PFS regardless of occurrence of high-grade AE (HG-AE) ($p=0.01$), suggesting its independence of HG-AE. Individual AE analysis identified patients experienced with early low-grade rash maculo-papular, lipase increased, or hypertension had improved PFS ($p<0.001$). In biomarker analysis, one key mutation biomarker, RAS, did not show significant association with survival outcomes (PFS: $p=0.66$; OS: $p=0.81$). In contrast, integration with TR-LG-AE led to remarkable improvement. Specifically, the TR-LG-AE was able to help improve RAS classification by identifying patients with better treatment benefit. Specifically, patients with RAS mutation and high TR-LG-AE experiences had a median PFS of 15.3 months compared to others (median PFS of 3.7 months; $p=0.007$), and a median OS of 21.9 months compared to others (median OS of 9.4 months; $p=0.04$). **Conclusions:** The study demonstrated utility of AE in predicting clinical outcomes in CRC, similar to our previous findings in lung cancer. Moreover, it discovered that integration of AE-derived biomarkers with molecular biomarkers could produce a more robust classifier than when considered individually. This analysis is the first of its kind to identify subgroups of CRC patients that benefit from IC treatment. Research Sponsor: National Cancer Institute; 1R21CA286417; Division of Community Health Promotion/Public Health Research/Biomedical Research Unit; MOAAX.

# Total neoadjuvant therapy with FOLFOX plus bevacizumab and short-course radiotherapy for Ras mutant, high-risk, locally advanced rectal cancer: The TRAINER trial.

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**Background:** Total neoadjuvant therapy (TNT) intensified by the combination of targeted therapy or immunotherapy is warranted to achieve satisfying local and systemic control for high-risk locally advanced rectal cancer (LARC). Our study aimed to explore the efficacy and safety of a FOLFOX-based TNT regimen plus bevacizumab and short-course radiotherapy (SCRT) in Ras-mutant high-risk LARC patients. **Methods:** In this single-arm, single-center, phase II trial, LARC patients with at least one risk factor (cT3c-4, mesorectal fascia positive, cN2, lateral lymph node involvement, and extramural venous invasion positive) were recruited and only Ras-mutant patients were included. Four cycles of FOLFOX plus bevacizumab were arranged as induction chemotherapy followed by SCRT (25Gy/5Fx) and two cycles of FOLFOX as consolidation chemotherapy. Surgery was performed 8 weeks after completion of radiotherapy. After surgery, high-risk stage III patients (T4 or N2) were recommended to receive an extra 6 cycles of FOLFOX as adjuvant chemotherapy, while low-risk stage III patients were scheduled for regular follow-up. Patients with clinical complete response(cCR) were allowed to choose the "Watch & Wait" (W&W) strategy instead of surgery. The primary endpoint was the complete response (CR) rate, defined as the total rate of cCR plus pathological complete response (pCR). Secondary endpoints included adverse events of TNT, postoperative complications, 2-year disease-free survival, and overall survival. **Results:** Between April 2, 2021, and March 12, 2024, 47 of 51 enrolled patients were eligible for treatment. All patients completed the prescribed TNT cycles, and six (12.8%) of them had dose reduction. The CR was achieved in 18 (38.3%) patients, including 14 (29.8%) pCR. No grade IV-V toxicity was observed and the incidence of grade III toxicities was 34.0%. Four patients (8.5%) achieved cCR and chose the W&W strategy. The other 43 patients underwent curative surgery and anal preservation was achieved in 39 (90.7%) patients. Seven patients (16.3%) experienced postoperative complications, including four (9.3%) cases of anastomotic leakage. **Conclusions:** The intensified TNT regimen combined with bevacizumab achieved a satisfying CR rate with acceptable toxicities and complications for Ras-mutant high-risk LARC patients. Long-term survival data of this cohort are needed to validate the efficacy of this regimen further. Clinical trial information: NCT04923620. Research Sponsor: None.

# Nirogacestat in patients with desmoid tumors and mutations of adenomatous polyposis coli (*APC*): Findings from the DeFi trial.

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**Background:** Nirogacestat (niro) is an oral, targeted gamma secretase inhibitor FDA-approved for adults with progressing desmoid tumors (DT) who require systemic treatment. In the phase 3 DeFi trial (NCT03785964), niro demonstrated significant improvement vs placebo (pbo) in progression-free survival (PFS; HR 0.29 [95% CI: 0.15–0.55], $P<.001$) and objective response rate (ORR; 41% vs 8%, $P<.001$). Most DT are sporadic tumors characterized by somatic mutations in the *CTNNB1* gene. About 10%–20% of DT are associated with *APC* mutations, the majority being germline that may cause familial adenomatous polyposis (FAP), an inheritable trait linked to an increased risk of colorectal cancer and may also confer more aggressive DT behavior. **Methods:** In DeFi, adult patients (pts) were randomized to oral niro (150 mg) or pbo twice daily. In pts with evaluable blood and tumor samples, descriptive post hoc analyses assessed effects of niro in pts with germline and/or somatic *APC* mutations, including pts with co-occurring somatic mutations of *APC* and *CTNNB1*. **Results:** Of the 142 pts in DeFi, 29 pts had *APC* mutations (niro=13, pbo=16; 22 somatic, 21 germline, and 14 somatic and germline), including 3 pts (niro=2, pbo=1) with co-occurring somatic mutations of *APC* and *CTNNB1*. Of these 29 pts, 19 (66%) were female, 16 (55%) were aged ≤30 y, 22 (76%) had a family history of FAP, and 22 (76%) were refractory to prior therapy (median of 3 prior lines of therapy). PFS was improved with niro vs pbo (HR 0.21 [95% CI: 0.05–1.00], $P=.016$). Confirmed ORR was 38% (5/13) for niro vs 13% (2/16) for pbo; median time to response with niro was 8.31 months. All 3 pts with co-occurring somatic mutations of *APC* and *CTNNB1* were female, aged 18–56 y, had DT in the upper extremity or abdominal wall, and had no family history of FAP. The 2 pts on niro had received prior systemic therapy and/or surgery and the 1 pt on pbo had no prior therapy. Of the 2 pts on niro, both achieved a partial response (median time to response: 9.9 months). The 1 pt randomized to pbo experienced disease progression in 2.6 months. In pts with *APC* mutations, diarrhea was the most frequently reported adverse event. In niro-treated pts, increased rates of skin events (maculopapular rash, 62%; dermatitis acneiform, 38%) and stomatitis (46%) were reported in pts with *APC* mutations compared with those in the overall DeFi population (32%, 22%, and 29%, respectively). **Conclusions:** Improvement in PFS and ORR was observed with niro vs pbo in pts with DT harboring *APC* mutations. Efficacy and safety of niro in pts with *APC* mutations were generally consistent with findings for the overall DeFi population. Although analyses were limited due to small sample size, these results suggest that niro can provide clinically meaningful benefit to pts with progressing DT and *APC* mutations, including those with co-occurring somatic mutations of *APC* and *CTNNB1*. Clinical trial information: NCT03785964. Research Sponsor: SpringWorks Therapeutics, Inc.

# Phase Ib study of gevokizumab (GEVO) in combination with standard-of-care (SoC) anticancer therapies in patients (pts) with metastatic colorectal cancer (mCRC), metastatic gastroesophageal cancer (mGEC), and metastatic renal cell cancer (mRCC).

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**Background:** GEVO, a humanized monoclonal antibody, binds to interleukin-1β (IL-1β) and inhibits its activity. We report results from a phase Ib study of GEVO + SoC anticancer therapies in pts with mCRC/mGEC/mRCC. **Methods:** This open-label study enrolled pts aged ≥18 years in cohorts A (first line mCRC, [A]), B (second line [2L] mCRC, [B]), C (2L mGEC, [C]) and D (2L/third line mRCC, [D]). The primary objectives were to determine the pharmacodynamically-active dose (PAD) of GEVO monotherapy in A/B, the safety and tolerability, and the recommended dose for expansion (RDE) of GEVO + SoC in A/B/C/D, and the efficacy of GEVO + SoC at RDE in A/B/C measured by the progression-free survival (PFS) rate. The proof of concept (PoC) criteria for PFS rates were ≥42% at 15 months (m) (lower 80% CI limit ≥29%) in A, ≥48% at ~9 m (lower 80% CI limit ≥37%) in B, ≥51% at 6 m (lower 60% CI limit ≥38%) in C. The relationship between baseline levels of high-sensitivity C-reactive protein (hs-CRP), interleukin-6 (IL-6) and IL-1β, and on-treatment decrease of circulating tumor DNA (ctDNA) levels with clinical response was explored. **Results:** As of March 1, 2023, 71 (A), 62 (B), 26 (C) and 7 pts (D) were treated. The PAD was established as 120 mg GEVO intravenous (IV) every 4 weeks (Q4W) and the RDE as 120 mg GEVO IV Q4W + SoC. There were 2 dose-limiting toxicities: grade 4 decreased neutrophil count (1 pt in C) and grade 3 hyponatremia (1 pt in D). The most frequent grade ≥3 treatment-related adverse events (≥15% of all pts) were neutropenia (A/B: 21.1%/22.6%), decreased neutrophil count (A/B/C: 18.3%/17.7%/15.4%) and hyponatremia (D: 28.6%). The PFS rates were 29.0% (80% CI: 20.4, 38.2), 39.4% (80% CI: 29.8, 48.9), 20.0% (60% CI: 13.3, 27.6) in A/B/C, respectively. In B, median PFS and overall survival (OS) were significantly longer in pts with baseline hs-CRP <10 mg/L (low [L]) than ≥10 mg/L (high [H]), IL-6 <9.58 pg/mL (L) than ≥9.58 pg/mL (H), and IL-1β <0.12 pg/mL (L) than ≥0.12 pg/mL (H). In A + B, a median on-treatment decrease of ctDNA fraction by >72% (H) was associated with significantly longer OS. **Conclusions:** The safety and tolerability of GEVO combinations was acceptable. The primary PFS rates did not meet the PoC criteria. Baseline levels of hs-CRP, IL-6 and IL-1β in pts with 2L mCRC, and on-treatment reduction of ctDNA in pts with 1L + 2L mCRC may have prognostic value. Clinical trial information: NCT03798626. Research Sponsor: Novartis Pharmaceuticals Corporation.

| Variables | PFS HR (95% CI) | p-value | OS HR (95% CI) | p-value |
|---|---|---|---|---|
| **A** | | | | |
| **hs-CRP L / H** | 1.74 (0.81, 3.73) | 0.155 | 1.99 (0.66, 5.98) | 0.221 |
| **IL-6 L / H** | 1.38 (0.70, 2.72) | 0.352 | 1.73 (0.71, 4.20) | 0.225 |
| **IL-1β L / H** | 1.50 (0.74, 3.01) | 0.259 | 0.98 (0.38, 2.52) | 0.973 |
| **B** | | | | |
| **hs-CRP L / H** | 2.56 (1.27, 5.18) | 0.009 | 3.04 (1.40, 6.60) | 0.005 |
| **IL-6 L / H** | 2.92 (1.33, 6.40) | 0.007 | 3.25 (1.43, 7.36) | 0.005 |
| **IL-1β L / H** | 2.41 (1.13, 5.13) | 0.022 | 3.60 (1.54, 8.44) | 0.003 |
| **A + B** | | | | |
| **ctDNA decrease L / H** | 0.65 (0.39, 1.10) | 0.108 | 0.45 (0.24, 0.83) | 0.011 |

# Risk of residual tumor following removal of a malignant colorectal polyp with high-risk features.

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**Background:** National guidelines recommend surgery when pathology analysis of a colorectal polyp reveals high-risk features: positive margin/piecemeal resection (PM/PR), lymphovascular/perineural invasion (LVI/PNI), poor differentiation (PD), or tumor budding (TB). However, some patients choose to avoid surgery, with its morbidity and quality-of-life implications, despite the risk of residual disease. **Methods:** This single-center cohort study determined the rates of luminal or lymph node disease and distant metastasis in patients who underwent surgery and patients who underwent observation following removal of a malignant colorectal polyp with high-risk features between 2015 and 2022. **Results:** Of 336 patients who underwent a polypectomy for a malignant polyp in the colon (n = 226) or rectum (n = 110), 312 had a PM/PR, 96 had LVI/PNI, 47 had PD, and 45 had TB; 208 patients (62%) underwent surgery, and 128 (38%) underwent observation. Patients with a malignant polyp in the rectum were more likely to undergo observation than patients with a malignant polyp in the colon (79 [72%] vs. 49 [22%] patients; $P < 0.01$). In the surgery group, 19 patients (9%) had residual luminal disease, 39 patients (19%) had lymph node disease, and 5 patients (2%) had distant metastasis. In the observation group, disease was identified in 14 patients (11%): regrowth in the lumen, 7 patients (9%); regrowth in lymph nodes, 5 patients (6%); distant metastasis, 1 patient (1%); regrowth with distant metastasis, 1 patient (1%). The 12 cases of regrowth were salvaged: resection, 4 patients; local excision, 2 patients; chemoradiotherapy, 6 patients. **Conclusions:** The risk of residual disease following removal of a malignant colorectal polyp with high-risk features is considerable. Patients concerned about the morbidity of surgery and postsurgery quality of life should be counseled about this risk, as well as the availability of effective salvage options. Research Sponsor: None.

| High-risk features (HRF) | Surgery (n = 208) | Observation (n = 128) | Luminal/LN disease at surgery (n = 51; 25%) | Luminal/LN regrowth during observation (n = 12; 9.4%) | Distant metastasis (n = 7; 2.1%) |
|---|---|---|---|---|---|
| **PM/PR (n = 312; 93%)** | 193 | 119 | 49 (25%) | 12 (10%) | 7 (2.2%) |
| **LVI/PNI (n = 96; 29%)** | 56 | 40 | 16 (29%) | 3 (7.5%) | 3 (3.1%) |
| **PD (n = 47; 14%)** | 33 | 14 | 11 (33%) | 0 | 1 (2.1%) |
| **TB (n = 45; 13%)** | 20 | 25 | 5 (25%) | 1 (4.0%) | 1 (2.2%) |
| **1 HRF (n = 215; 64%)** | 141 | 74 | 31 (22%) | 8 (11%) | 4 (1.9%) |
| **2 HRF (n = 78; 23%)** | 39 | 39 | 11 (28%) | 4 (10%) | 1 (1.3%) |
| **3 HRF (n = 35; 10%)** | 25 | 10 | 8 (32%) | 0 | 2 (5.7%) |
| **4 HRF (n = 6; 1.8%)** | 2 | 4 | 1 (50%) | 0 | 0 |

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# Pattern and predictors of D3 lymph node station positivity in patients undergoing curative surgery for adenocarcinoma of colon.

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**Background:** The prognosis and surgical management of colon cancer is intricately related to its lymphatic drainage. The oncological outcome of apical node positivity has been evaluated in different studies with varying results. There is no published regional data on the rate of apical node positivity in colon cancer. An insight into the probability of apical node metastases and the risk factors for the same may help us in prognosticating colon cancer patients better. **Methods:** This was a prospective observational study conducted at the Regional Cancer Centre Thiruvananthapuram from September 2022 to June 2024. Our primary objective was to determine the prevalence of apical lymph node positivity in patients undergoing surgery for colon cancer with D3 lymphadenectomy. The secondary objectives were to find out the factors associated with the risk of apical node metastasis, the incidence of skip metastases to apical nodes, the adequacy of lymph node harvest and the postoperative complications. Patients undergoing primary surgery for adenocarcinoma of the colon were included. Patients with metastatic disease, synchronous colon or rectal cancers, past history of colonic surgery were excluded. All patients underwent D3 lymphadenectomy along with colectomy depending upon the tumour location. Lymphatic tissue corresponding to the D3 station was retrieved and sent separately after grossing the fresh specimen on the side table in the operating room. **Results:** A total of 139 patients were included in the study. Fifty two (37.4%) patients had stage III disease, 38.9% had stage II disease and the rest had stage I colon cancer. The mean nodal harvest was 17.86 and mean apical node harvest was 2.04. Node harvest was adequate in 89.21% patients. The apical node positivity rate was 7.19% (10 patients). Skip metastasis to the apical nodes was found in 2 patients. Lymphovascular invasion, number of positive lymph nodes and young age (≤45 years) had a significant correlation with apical node positivity on multivariate analysis. Clavien Dindo grade 3 and above complications were seen in 5.03%. **Conclusions:** Our study showed an apical node positivity rate of 7.19%, with a skip metastasis rate of 1.4%. This is the first Indian data on apical node positivity rate in colon cancer and its predictors. The prognostic value of apical nodes and the occurrence of skip metastasis to the apical nodes are compelling reasons to advocate D3 dissection, even with relatively low prevalence of apical node positivity. Research Sponsor: None.

# What is the optimal treatment strategy of salvage line treatment after progression on anti-epidermal growth factor receptor monoclonal antibody in patients with tissue *RAS/BRAF* wild-type metastatic colorectal cancer?

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**Background:** Trifluridine/tipiracil (FTD/TPI) plus bevacizumab (BEV) and anti-epidermal growth factor receptor (EGFR) monoclonal antibody (mAb) re-challenge are treatment options for tissue *RAS/BRAF* wild-type (WT) metastatic colorectal cancer (mCRC) in salvage line, although the optimal treatment sequence of salvage line remains unclear. **Methods:** *RAS/BRAF* WT mCRC patients refractory to anti-EGFR mAb were retrospectively enrolled at a single cancer institute from November 2017 to April 2024. We compared responses, progression-free survival (PFS), and overall survival (OS) between Group A (anti-EGFR mAb re-challenge given first) and B (FTD/TPI plus BEV given first). We compared responses between Group C (treated with anti-EGFR mAb re-challenge at any line) and D (treated with FTD/TPI plus BEV at any line). We explored the clinical factors related to the treatment efficacy using multivariate analysis. **Results:** A total of 89 patients (median age, 63 years) were included. The cohort included 74 patients with a primary lesion in the left-sided colon, while 15 patients had a primary lesion in the right-sided colon. The response rate (RR) was 3.4%. The disease control rate (DCR) was 52.8%. The median PFS (mPFS) and OS (mOS) were 3.5 months (95% confidence interval [CI], 2.8-4.2) and 12.8 months (95% CI, 8.8-15.7) respectively. Two of 18 patients (11.1%) in Group A and 1 of 71 patients (1.4%) in Group B achieved partial response (PR). The RR of Group C was significantly higher than Group D (9.6% vs 1.2%; p = 0.031). Both PFS and OS showed no significant difference between Group A and B (mPFS: 3.3 vs 3.9 months; hazard ratio [HR], 1.08; 95% CI, 0.63-1.84; p = 0.79) (mOS: 12.8 vs 15.0 months; HR, 0.90; 95% CI, 0.49-1.65; p = 0.72). Multivariate analysis identified that primary tumor resection was associated with PFS (HR, 0.44; 95% CI, 0.25–0.80; p = 0.0068) and primary tumor resection (HR, 0.34; 95% CI, 0.18–0.65; p = 0.0010) and serum CA19-9 level (HR, 1.75; 95% CI, 1.02-3.00; p = 0.043) were associated with OS. **Conclusions:** No significant survival difference was observed between patients administered anti-EGFR mAb re-challenge first and those administered FTD/TPI plus BEV first, whereas the RR of anti-EGFR mAb re-challenge is significantly higher than that of FTD/TPI plus BEV. Research Sponsor: None.

# Neoadjuvant chemoradiotherapy with ipilimumab and nivolumab in rectal cancer (CHINOREC): A prospective randomized, open-label, multicenter, phase II clinical trial.

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**Background:** Immune checkpoint inhibitors (ICIs) seem only effective in a primed tumor immune microenvironment (TIME). Neoadjuvant radiotherapy (RT) has been shown to induce an immunogenic cell death (ICD) and thereby restores the susceptibility to ICI. This study evaluates the safety of neoadjuvant chemoradiotherapy (CRT) with concomitant ipilimumab (IPI) and nivolumab (NIVO) in curative rectal cancer (RC) patients. **Methods:** The CHINOREC study (ClinicalTrials.gov identifier NCT04124601) is a prospective, randomized (ratio 50:30), open-label, multicenter, phase II investigator-initiated trial (IIT). Patients with RC received neoadjuvant CRT (50 Gy in 2 Gy fractions with concurrent capecitabine 1650 mg/m$^2$/d) alone or in combination with IPI (1 mg/kg IV at day 7), following 3 cycles of NIVO (3 mg/kg IV Q2W, starting on day 14). Surgical resection was performed 10-12 weeks post CRT. The primary endpoint was surgical safety and feasibility (Clavien-Dindo Classification) of neoadjuvant CRT with sequential IPI and NIVO following surgical resection. Secondary outcome was complete response rate (clinical and pathological). **Results:** Between June 2020 and November 2023, 145 patients were screened and 80 randomly assigned to CRT (n=30) or CRT+IPI/NIVO (n=50). The primary endpoint (any surgical complication) did not differ between the 2 groups (78% vs. 77%). The reoperation rate (≥Grade IIIb) was comparable low in both groups (8% vs. 7%). No new safety signals were identified. The rate of major pathological response (MPR) and complete response (CR) was also comparable high in both arms (38% vs. 37% and 30% vs. 22%, respectively). **Conclusions:** Neoadjuvant IPI/NIVO can be safely applied concomitant with CRT in patients with RC, as it does not increase the rate of reoperation or surgical complications. The validity of the encouraging CR rate needs to be elucidated over time. Further translational analyses will elucidate mechanistic insight regarding improved fraction, dosing and timing of CRT and ICI. Clinical trial information: NCT04124601. Research Sponsor: None.

# Survival benefits of adjuvant chemotherapy after surgical resection of para-aortic lymph node metastasis from colorectal cancer: A multicenter retrospective study.

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**Background:** Surgical removal of metastasized paraaortic lymph nodes (PALNs) can prolong the survival of certain patients with colorectal cancer (CRC). However, the role of postoperative chemotherapy in such patients remains unknown. **Methods:** This multi-center retrospective study examined97 patients with PALN metastasis from CRC who underwent surgical resection at 36 centers in Japan between 2010 and 2015. Based on adjuvant chemotherapy (AC) after the lymphadenectomy, the patients were classified into Non-AC and AC groups (27 and 70 patients, respectively). After the exclusion of patients receiving irinotecan, the latter group was further categorized into 5-fluorouracil (5-FU) and oxaliplatin (L-OHP) subgroups (14 and 52 patients, respectively) according to the use of L-OHP. Background characteristics and postoperative survival were compared among the groups. **Results:** Marked differences were not seen in background characteristics, except for neoadjuvant treatment, between the Non-AC and AC groups. The AC group exhibited better recurrence-free survival (RFS, $p = 0.009$) and overall survival (OS, $p = 0.040$) than the Non-AC group. However, RFS and OS of the 5-FU group did not differ from those of the L-OHP group ($p = 0.73$ and $p = 0.92$, respectively). **Conclusions:** AC contributed to the improved prognosis of patients after the removal of PALN metastasis from CRC, but L-OHP did not offer additional survival benefits. Prospective studies comparing Non-AC with 5-FU- and L-OHP-based AC are needed to confirm these findings. Research Sponsor: None.

# AGITG SPAR: A randomized, placebo-controlled, phase II trial of simvastatin in addition to neoadjuvant chemotherapy and radiation for rectal cancer.

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**Background:** Retrospective clinical studies and preclinical studies demonstrated that statin use during preoperative (chemo)radiation (pCRT) for rectal cancer is associated with improved survival, response, and toxicity. Tumor regression following pCRT has strong prognostic significance and can be assessed using MRI-based tumor regression grading (mrTRG), including with non-operative management. SPAR was designed to prospectively evaluate the benefits of adding simvastatin (SIM) to pCRT on tumor regression and gastrointestinal (GI) adverse events (AE). **Methods:** SPAR is a double-blind randomized phase 2 trial investigating SIM/placebo (PBO) in addition to long-course fluoropyrimidine-based pCRT for rectal adenocarcinoma. Stratification included trial site, clinical T stage (<4 vs 4), clinical N stage (<2 vs 2), either mesorectal fascia involvement (MRFI) or extramural venous invasion (EMVI) on MRI (yes vs no), and total neoadjuvant therapy (TNT): induction vs consolidation chemotherapy vs none. Study treatment was SIM 40mg/PBO daily for 90 days, starting 1 week prior to pCRT; recent statin use was excluded. Pelvic MRI was repeated 6-8 weeks after pCRT to determine mrTRG. An amendment in January 2022 allowed TNT with either induction or consolidation chemotherapy; the timing of post-pCRT MRI remained unchanged. The design was amended to open-label due to PBO supply issues. Primary objective: rate of centrally assessed grade 1-2 mrTRG. Secondary objectives include centrally assessed favorable pathologic TRG (pathTRG), safety and cancer outcomes. Analysis was by intention-to-treat. **Results:** Between April 2018 - November 2023, 135 of 222 planned participants from 17 sites in Australia and New Zealand were randomized (68 SIM; 67 no SIM). Recruitment was hampered by the COVID-19 pandemic and adoption of TNT as standard care before the protocol amendment. Participant characteristics: median age 59 years; 85 (63%) males; 118 (87%) T2-3 disease, 80 (59%) N0-1 disease; 55 (41%) MRFI or EMVI; 25 (19%) had TNT. Rates of grades 1-2 mrTRG with SIM vs no SIM were 38.5% and 29.7% ($X^2$ = 1.09, p = 0.30), respectively. There was no significant difference in rates of > grade 2 GI and non-GI AE. Four serious AE were recorded, none related to study treatment. Median follow-up was 3.2 years. 3-year local recurrence rates (LRR) were low: 1 (2.2%) and 3 (5.5%) with SIM vs no SIM, respectively (HR: 0.29, 95% CI: 0.03 to 2.67, p = 0.26). 3-year disease-free survival (DFS) with SIM vs no SIM was 84% and 72%, respectively (HR 0.48; 95% CI: 0.21–1.08, p = 0.07). **Conclusions:** The rates of favorable mrTRG, 3-year LRR and DFS were numerically better with the addition of SIM to pCRT, though the differences were not statistically significant. SIM was well tolerated. Interpretation is limited by reduced sample size and statistical power. PathTRG and other key outcomes will be presented at the meeting. Clinical trial information: ACTRN12617001087347. Research Sponsor: Health Research Council of New Zealand; 18/233; Cancer Society of New Zealand; 15.17; Cancer Australia; 1125958; Cancer Council New South Wales; RG 17-07.

# Fruquintinib plus best supportive care for patients with metastatic colorectal cancer: Characterization of patients who had an overall survival of ≥10 months in the FRESCO-2 study.

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**Background:** Fruquintinib (F) is a highly selective, oral tyrosine kinase inhibitor of all 3 vascular endothelial growth factor receptors (VEGFRs -1, -2, -3) that is approved by the US FDA and in the EU, UK, and Japan for previously treated metastatic colorectal cancer (mCRC), regardless of biomarker status. FRESCO-2 (NCT04322539) met its primary endpoint demonstrating significantly improved overall survival (OS) with F + best supportive care (BSC) vs placebo (P) + BSC. This subgroup analysis assessed landmark survival of all FRESCO-2 patients (pts), and baseline (BL) characteristics and safety data of pts treated with F who had OS ≥10 months (mos) compared with the intent-to-treat (ITT) population (pop). **Methods:** Pts were randomized 2:1 to receive F 5 mg or matching P, by mouth, once daily, 3 weeks on, 1 week off, + BSC. Pts had received prior chemotherapy, anti-VEGF therapy and, if *RAS* wild type, anti-EGFR therapy; and had prior exposure to trifluridine/tipiracil (TAS-102) and/or regorafenib. The hazard ratio (HR) between the two treatment arms was calculated from a stratified Cox proportional hazard model with treatment group as the only covariate in the model. **Results:** In the ITT pop, OS was improved with F vs P (HR 0.66); OS rates (95% confidence intervals [CIs]) at 6 and 9 mos were 60.4% (55.9–64.9) vs 41.5% (35.0–48.0) and 41.1% (36.4–45.8) vs 28.2% (22.1–34.3), respectively. Progression-free survival (PFS) rates (95% CIs) at 6 and 9 mos were 23.8% (19.7–28.0) vs 1.1% (0–2.6) and 11.3% (8.1–14.6) vs 0.5% (0–1.6), respectively. Of 461 pts who received F, 113 (24.5%) had OS ≥10 mos (range 10.0–18.9) with a median duration of F treatment of 6.3 mos (range 0.7–19.1). Among pts receiving F, a higher proportion (≥10% difference) of pts with OS ≥10 mos had no liver metastases (mets) and an Eastern Cooperative Oncology Group performance status (ECOG PS) of 0 at BL compared with the ITT pop (Table). In pts with OS ≥10 mos, 61.9% experienced a grade ≥3 treatment-emergent adverse event compared with 62.7% in the ITT pop. **Conclusions:** Longer-term OS and PFS rates at 6 and 9 mos were higher with F vs P in the FRESCO-2 ITT pop. The data show a higher proportion of pts with OS ≥10 mos had an absence of liver mets and ECOG PS 0 at BL than the ITT pop. The safety profile of F in pts with OS ≥10 mos was consistent with the ITT pop. Clinical trial information: NCT04322539. Research Sponsor: HUTCHMED.

**BL characteristics of pts treated with F with OS ≥10 mos vs ITT pop.**

| BL characteristic | Pts with OS ≥10 mos (n=113) | ITT pop (n=461) |
|---|---|---|
| ECOG PS (0 / 1), % | 54.0 / 46.0 | 42.5 / 57.5 |
| Median time since 1st CRC diagnosis, mos | 52.0 | 47.2 |
| Primary tumor location at 1st diagnosis (colon / rectum / both), % | 51.3 / 35.4 / 13.3 | 60.5 / 31.0 / 8.5 |
| Primary site at 1st diagnosis (colon left / colon right), % | 42.5 / 15.9 | 41.6 / 21.0 |
| Liver mets, % | 58.4 | 73.5 |
| Median duration of mCRC, mos | 42.7 | 37.9 |
| Prior lines of therapy for mCRC (≤3 / >3), % | 26.5 / 73.5 | 27.1 / 72.9 |
| Prior TAS-102, % | 46.9 | 52.1 |
| Prior regorafenib, % | 12.4 | 8.7 |

**LBA143** **Oral Abstract Session**

# First results of nivolumab (NIVO) plus ipilimumab (IPI) vs NIVO monotherapy for microsatellite instability-high/mismatch repair-deficient (MSI-H/dMMR) metastatic colorectal cancer (mCRC) from CheckMate 8HW.

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**The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.**

# Neoadjuvant therapy in T3 and T4 colon cancer: Is there a promising future?

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**Background:** Neoadjuvant chemotherapy (NAC) is increasingly being considered over standard upfront surgery followed by adjuvant chemotherapy for locally advanced colon cancer. Studies have been underpowered to look at the impact of NAC on overall survival (OS) in individual clinical T stages. **Methods:** A nationwide retrospective analysis using the National Cancer Database to evaluate patients with clinical stage T3 and T4 colon adenocarcinomas from 2010 to 2020. Cox proportional hazards model was used to evaluate the impact of NAC approach compared to upfront surgery on OS. Clinical and pathological T stages were compared among those who underwent upfront surgery to evaluate accuracy of clinical T staging. **Results:** Total of 60,557 patients were identified - 2,313 patients undergoing NAC and 58,245 undergoing upfront surgery. In the past decade, use of NAC increased from 1.73% to 12.13%. In our multivariable survival analysis combining clinical T3 and T4 patients, we observed a worse OS in the NAC group (HR 1.16, 95% CI 1.07-1.26). We found no OS benefit for patients undergoing NAC approach for clinical stage T3 (HR 1.13, 95% CI 0.99-1.28) but found a significant OS benefit for patients undergoing NAC approach for clinical stage T4 (HR 0.85, 95% CI 0.77-0.95). We observed that clinical T4 stage was accurately staged in 90.48% with overstaging in 9.52%; clinical T3 stage was accurately staged in 95.24% with overstaging in 2.18% of patients. **Conclusions:** Our analysis demonstrated OS benefit of the NAC approach in T4 colon adenocarcinoma but not in T3. The study highlights the importance of evaluating survival benefit of NAC separately for T3 and T4 colon adenocarcinoma in future studies, and the challenges of accurate clinical staging of colon cancer for NAC planning. Research Sponsor: None.

# Updated analysis of a phase 2 study of fruquintinib plus trifluridine/tipiracil (TAS-102) as third-line treatment in patients with metastatic colorectal adenocarcinoma.

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**Background:** Fruquintinib and TAS-102 are two standard therapies for patients (pts) with previous-treated mCRC worldwide. The study aimed to evaluate the efficacy and safety of fruquintinib plus TAS-102 as third-line treatment for mCRC. We previously reported that fruquintinib plus TAS-102 was well tolerated with encouraging clinical activity in pretreated mCRC. Here, we reported the updated results at the data cutoff of Sep 3, 2024. **Methods:** In this open-label, single-arm, multi-center, phase 2 trial (NCT05004831), pts with mCRC who had failed at least two prior standard treatment regimens were enrolled. Eligible pts received fruquintinib (4mg, qd, d1-21) and TAS-102 (35mg/m$^2$, bid, d1-5, 8-12) orally every 4 weeks until disease progression or unacceptable toxicity. Primary endpoint was PFS (per RECIST v1.1). Secondary endpoints were OS, ORR, DCR and safety (per NCI-CTCAE v5.0). **Results:** From Mar 2022 to Aug 2023, 50 eligible pts were enrolled. Median age was 60 years (range 39-76) and 58.0% were male. 82% pts were of left-sided colon and rectal cancer, 42.0% pts were *RAS* mutant, 58.0% had liver metastasis and 18.0% had peritoneal metastasis. Prior treatments and proportion were standard chemotherapy 100%, anti-VEGF 88.0% and anti-EGFR 26.0%. As of Sep 3, 2024, ORR was 10.9% (95%Cl: 3.6-23.6) and DCR was 73.9% (95%Cl: 58.9-85.7). The median PFS was 6.33 months (m) (95% CI: 4.20-8.62). The 6-m, 9-m and 12-m PFS rates were 53.0% (95% CI: 40.2-70.0), 28.3% (95% CI: 17.4-45.9) and 23.1% (95% CI: 13.2-40.5), respectively. At a median follow-up of 17.6m, median OS was 18.4m (95% CI: 12.0-NA). The 6-m, 9-m and 12-m OS rates were 87.0% (95% CI: 77.8-97.3), 66.9% (95% CI: 54.0-82.9) and 64.3% (95% CI: 51.1-80.8), respectively. Median PFS was comparable in liver metastasis (LM) and non-LM pts (6.33m [95%CI: 4.13-8.62] vs. 6.46m [95%CI: 3.74-NA], $P$=0.54). Similar results were observed in peritoneal metastasis (PM) and non-PM pts (6.07m [95%CI: 3.74-NA] vs. 6.33m [95%CI: 4.20-8.27], $P$=0.95). None of *RAS* gene mutation, prior treatment lines ≥ 3, age ≥ 65 years, metastatic sites ≥ 2, peritoneal metastasis and right-sidedness were identified as factors significantly associated with PFS or OS (by univariable Cox proportional-hazards model). The most common treatment-related adverse events (TRAEs) of any grade and grade ≥ 3 were mainly hematological toxicities. TRAEs (any grade, grade ≥ 3) in ≥ 50% of pts including neutrophil count decreased (80.0%, 54.0%), white blood cell count decreased (70.0%, 26.0%), anemia (58.0%, 20.0%) and proteinuria (50.0%, 4%). No treatment-related deaths were observed. **Conclusions:** The updated analysis demonstrated encouraging survival benefits of fruquintinib plus TAS-102 as third-line treatment in patients with mCRC with acceptable toxicities. This regimen could be an alternative therapeutic approach for these patients. Clinical trial information: NCT05004831. Research Sponsor: None.

# Phase I/II study of cabozantinib alone or in combination with panitumumab in patients with *MET*-amplified metastatic colorectal cancer.

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**Background:** Despite the introduction of anti-EGFR therapies, resistance remains a significant challenge in metastatic colorectal cancer (mCRC). MET amplification is a recognized resistance mechanism. Cabozantinib, a multi-kinase inhibitor targeting MET, has shown promise in overcoming resistance to anti-EGFR therapy. This study aimed to evaluate the efficacy and safety of cabozantinib alone (Cabo) or in combination with panitumumab (Cabo+Pmab) in patients with MET-amplified mCRC. Patients and **Methods:** The study population included patients with RAS and BRAF wild-type mCRC who had progressed on anti-EGFR therapy and subsequently confirmed MET amplification by circulating tumor DNA. The phase I part aimed to establish the recommended cabozantinib dose in combination with panitumumab by assessing dose-limiting toxicity (DLT) within the first four weeks. The primary endpoint of phase II part was objective response rate (ORR) in patients receiving the study treatment as third-line therapy or later. In phase II part, patients were randomized to Cabo or Cabo+Pmab group. Based on a threshold ORR of 1.6% response rate and an expected efficacy of 30%, the planned sample size was 14 patients with one-sided alpha of 1.25% and power of 80% in each arm, Cabo and Cabo+Pmab. A tumor response of at least 3 out of 14 patients would warrant further investigation of these treatments. **Results:** A total of 35 patients were enrolled (6 in phase I and 29 in phase II, 28 for the primary analysis; 13 in Cabo and 15 in Cabo+Pmab). In phase I, no DLTs were observed. This allowed researchers to set the maximum tolerated dose and recommend a dosage of 60 mg/day of cabozantinib in combination with panitumumab for phase II. Of 28 pts for the primary analysis, ORR was 7.7% (1/13; 95%CI 0.2-36%) in Cabo group and 0% (0/15; 95%CI 0-21.8%) in Cabo+Pmab group, median progression-free survival was 3.1 months in both groups, and cumulative proportion of pts with grade ≥3 treatment-related adverse events was 38.5% (including hypertension [23.1%]) in Cabo and 60% (including hypomagnesaemia [20%]) in Cabo+Pmab group. **Conclusions:** The combination of cabozantinib 60 mg plus panitumumab had manageable safety profile. However, this treatment did not demonstrate significant clinical benefit in patients with MET-amplified mCRC. Further investigation is warranted to explore alternative therapeutic strategies for this patient population. Clinical trial information: jRCT1080224637. Research Sponsor: Takeda Pharmaceutical Co., Ltd.

# A multicenter, prospective phase II trial of second-line aflibercept plus FOLFIRI in patients with metastatic colorectal cancer refractory to anti-EGFR agents (HGCSG1801): Updated analyses.

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**Background:** The HGCSG1801 trial evaluated the treatment outcomes of aflibercept (AFL) plus FOLFIRI for patients(pts) with metastatic colorectal cancer (mCRC) refractory to an oxaliplatin-based regimen combined with an anti-EGFR agent. Here we report results of the updated efficacy, safety, and a biomarker analysis. **Methods:** This was a prospective open-label phase II trial. AFL (4 mg/kg iv) followed by FOLFIRI (irinotecan 180 mg/m$^2$, leucovorin 200 mg/m$^2$ iv, bolus 5-fluorouracil [5-FU] 400 mg/m$^2$, and infusional 5-FU 2400 mg/m$^2$/46 h) was given every 2 weeks until progression or unacceptable toxicities. The primary endpoint was 6-month progression-free survival (PFS) rate, and the secondary endpoints included overall survival (OS), PFS, overall response rate (ORR), disease control rate (DCR), and adverse events. Angiogenic factors were analyzed in plasma sample using a Luminex multiplex assay using the Cox proportional hazards model. This study was sponsored by the Non-Profit Organization Hokkaido Gastrointestinal Cancer Study Group and supported by a grant from Sanofi. **Results:** Forty-three pts were enrolled between November 2019 and October 2022. The data cut-off date was April 30, 2024. The 6-month PFS rate was 59.0% (90% confidence interval [CI], 45.8–72.1%). Median PFS and OS were 7.3 months (95% CI, 5.5–11.0 months) and 18.8 months (95% CI, 12.9–26.6 months), respectively. The ORR was 20.9% (95% CI, 10.0–36.0%) and DCR was 88.4% (95% CI, 74.9–96.1%). No deaths and no new safety signals with a causal relation to the study treatment were observed. A biomarker analysis using pretreatment plasma samples showed a trend toward better PFS in pts with low VEGF-A (hazard ratio [HR] 0.40 [95% CI, 0.18-0.91]), TSP-2 (HR 0.40 [95% CI, 0.18-0.88]) or IL-8 levels (HR 0.43 [95% CI, 0.20-0.93]). There was also a trend toward better OS in pts with low levels of TSP-2 (HR 0.30 [95% CI, 0.11-0.81]) or TIMP-1 (HR 0.20 [95% CI, 0.06-0.69]). **Conclusions:** These updated data further support the activity and manageable safety profile of AFL plus FOLFIRI for pts with mCRC who failed an oxaliplatin-based regimen combined with an anti-EGFR agent. It was suggested the association between some angiogenic factors in plasma samples and the activity of AFL plus FOLFIRI in mCRC. Clinical trial information: jRCTs011190006. Research Sponsor: Non-Profit Organization Hokkaido Gastrointestinal Cancer Study Group; Sanofi; SGZ- 2017-11875.

# Phase II study of organ preservation (OP) in node-negative (NN) low rectal cancer (RC): Updated clinical and patient reported outcomes (PROs).

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**Background:** Total mesorectal excision (TME) is a highly effective treatment for rectal cancer (RC) but is associated with significant morbidity and mortality. Organ preservation (OP) approaches for locally advanced RC have been successful but not well studied for node-negative (NN) low RC. The objective of this investigator-initiated trial (IIT) is to determine the feasibility of performing successful local excision (LE) after neoadjuvant chemotherapy (CTX) in NN low RC. **Methods:** With IRB approval, patients (pts) with clinical stage T1-3, N0 low rectal adenocarcinomas (<6 cm from anal verge) were included in this single arm phase II IIT. Pts received 6 cycles of FOLFOX (5-FU bolus 400 mg/m2 and infusion 2400 mg/m2, Leucovorin 400 mg/m2, Oxaliplatin 85 mg/m2). Those with evidence of a response underwent LE (transanal excision) 6-12 weeks later. To target occult nodal metastases and reduce in-bowel recurrences, LE was followed by chemoradiotherapy ([CXRT] capecitabine 825 mg/m2 and long course RT to 54 Gy). The primary endpoint was the proportion of pts with successful LE after neoadjuvant CTX (NCT03548961). **Results:** Nineteen pts with low RC were enrolled; nine were female, and the mean age at diagnosis was 65 years. T stage was as follows: T1- 1(5%), T2 - 11(58%), T3 - 7(36%). Eighteen (95%) pts completed at least 5 cycles of neoadjuvant Ctx and 16/19 (84%) underwent LE. Negative margins were achieved in 79% (15/19), meeting our primary endpoint ($p < 0.001$). Ten (53%) were downstaged, and complete pathological response was achieved in 5/16 (31%) of pts who underwent LE. Four (21%) pts (all T2 stage) did not meet the primary endpoint (3 with inadequate response to CTX and 1 with LE and positive margins). Of these, three were salvaged (2 with CXRT; 1 with TME), and one died of infection (unrelated complication). All 16 who underwent LE completed CXRT. Of the 15 pts with successful OP at a median follow-up of 25.5 months, no local recurrences have been reported and 1 pt with lung nodules below the threshold for metastases at study entry ultimately developed clear lung metastases. Median DFS was not reached. Patient reported outcomes (PRO) assessment revealed a slight decline in physical health scores after neoadjuvant CTX but improved to baseline at follow-up. No decline was seen in mental health scores. Symptom specific QOL (sexual and bowel function) and long-term survival data will be presented. **Conclusions:** Neoadjuvant CTX and LE allows for OP in NN low RC and results in a margin negative LE in over three fourths of pts with durable long term local control and preserved QOL. Ongoing trials are investigating an OP approach in NN RC (NCT03259035). Our approach adds to the mounting evidence and will provide early data of the benefit of adjuvant chemoradiation to target occult nodes in this setting. Clinical trial information: NCT03548961. Research Sponsor: Colon Cancer Coalition and Institutional Funds.

# Optimal cut-off value for lymph node harvesting in colon cancer based on tumor stage and sidedness: A retrospective analysis of a Japanese nationwide cohort.

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**Background:** The presence of fewer than 12 harvested lymph nodes (LN) is currently considered a risk factor in stage II colon cancer (CC). However, this cut-off value does not take into consideration the biological differences associated with tumor sidedness. Moreover, the applicability of this cut-off value to stage III CC is unknown. **Methods:** Patients with pathological stage II/III CC who underwent curative surgical resection were assessed using data from the Japanese Society for Cancer of the Colon and Rectum database (2001–2015). The optimal cut-off value of harvested LN was evaluated by analyzing all cut-off values (i.e., 3–40), taking into account the point estimate of the hazard ratio (HR) and lower limit of the 95% confidence interval (CI), according to the tumor stage (II/III) and sidedness (right/left). A Cox proportional hazards model was used to investigate the impact of the number of harvested LN on overall survival, adjusting for age, sex, T stage, histology, size of tumor, sidedness, adjuvant chemotherapy and extent of colon resection. Multivariate analysis was also conducted. **Results:** From the data, 4249 (left/right; 3248/1001) and 3549 (left/right; 2798/751) patients with stage II and III CC, respectively, were identified. The median number of harvested LN was identical between stage II (18, interquartile range [IQR] (12–15)) and III (18, IQR (13–25)). However, this number was higher in right-sided CC (21, IQR (14–31)) versus left-sided CC (17, IQR (12–24)). In stage II, the conventional cut-off value (i.e., 12 LN) was identified as a prognostic factor in the multivariate analysis (HR 1.45; 95% CI: 1.20–1.75; $p<0.001$). Nevertheless, the prognostic impact of the number of harvested LN was stronger when the optimal cut-off value was set to 7 (HR 1.88; 95% CI: 1.44–2.44; $p<0.001$). In stage III, the optimal cut-off value was 12 (HR 1.88; 95% CI: 1.44–2.44; $p<0.001$). When evaluated based on tumor sidedness, the optimal cut-off value differed, especially for stage III. Data are shown in the table. Similar results were obtained following the exclusion of the extent of bowel resection as variable in Multivariate analysis. **Conclusions:** While the presence of 12 harvested LNs remains a valid prognostic factor, the optimal cut-off value can be further evaluated based on tumor stage and sidedness. Research Sponsor: None.

| | Stage II | | | Stage III | | |
|---|---|---|---|---|---|---|
| | All | Left | Right | All | Left | Right |
| **Cut-off value** | 12 | 12 | 12 | 12 | 12 | 12 |
| **N (>cut-off/<cut-off)** | 3209/1040 | 2371/877 | 838/163 | 2814/735 | 2181/617 | 633/118 |
| **HR (>cut-off/<cut-off) (95% CI)** | 1.45 (1.20-1.75) | 1.49 (1.20-1.84) | 1.31 (0.85-2.02) | 1.30 (1.09-1.55) | 1.27 (1.05-1.54) | 1.43 (0.97-2.13) |
| **P value** | <0.001 | <0.001 | 0.216 | <0.003 | 0.015 | 0.073 |
| **Cut-off value** | 7 | 7 | 10 | | 20 | 13 |
| **N (>cut-off/<cut off)** | 3930/319 | 2977/271 | 889/112 | | 1200/1598 | 609/142 |
| **HR (>cut-off/<cut-off) (95% CI)** | 1.88 (1.44-2.44) | 2.07 (1.56-2.74) | 1.39 (0.84-2.27) | | 1.35 (1.13-1.60) | 1.59 (1.11-2.28) |
| **P value** | <0.001 | <0.001 | 0.198 | | <0.001 | 0.011 |

# Short-course radiotherapy-based total neoadjuvant therapy combined with tislelizumab in the treatment of locally advanced rectal cancer (Neo-STAR): Early outcomes of a prospective, randomized phase II trial.

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**Background:** It has been found that radiotherapy possess synergistic anticancer effect with immune checkpoint inhibitors (ICIs). Therefore, this phase II randomized clinical trial (RCT) (ClinicalTrials.gov identifier: NCT05086627) aimed to evaluate the efficacy and safety of SCRT followed by CAPOX and tislelizumab in patients with LARC. **Methods:** Patients initially diagnosed rectal adenocarcinoma with cT1-2N+M0 or cT3-4NanyM0 were recruited, randomly assigned to receive SCRT (25Gy/5f), followed by four cycles of CAPOX combined with tislelizumab or CAPOX alone, respectively. After total mesorectal excision (TME), two cycles of postoperative adjuvant chemotherapy were given according to the patient's preferences. The pCR rate was set as the primary endpoint, and the major pathological response (MPR) (tumor regression grade (TRG) 0+1), 3-year progression free survival (PFS), 3-year overall survival (OS) and treatment safety were set as the secondary endpoints. **Results:**Between September 2021 and March 2024, 118 patients were randomly assigned and 111 patients started allocated treatment, with 53 in the trial group and 58 in the control group. Ninety-six patients (86.5%) completed SCRT and 4 cycles of CAPOX chemotherapy with/without tislelizumab, including 49 patients (92.5%, 49/53) in the trial group and 47 patients (81.0%, 47/58) in the control group. Eighty-two patients (85.4%) had surgical resection, including 42 patients (85.7%, 42/49) in the trial group and 40 patients (85.1%, 40/47) in the control group. Among 82 patients who underwent surgery, 54.8% (23/42) patients in the trial group achieved pCR compared with 37.5% (15/40) in the control group. During the neoadjuvant treatment period, the incidence of grade 3-4 AEs were 9.4% in the trial group and 10.3% in the control group. **Conclusions:** SCRT-based TNT combined with tislelizumab followed by TME exhibited a favorable pCR rate in LARC. Clinical trial information: NCT05086627. Research Sponsor: Hebei Province Science and Technology Department; 22377733D.

**Surgical and postoperative pathological outcomes.**

| Characteristic | Experimental group (n=42) | Control group (n=40) |
|---|---|---|
| **R0 resection** | | |
| **Yes** | 42 (100%) | 40 (100%) |
| **No** | 0 (0%) | 0 (0%) |
| **pCR** | | |
| **Yes** | 23 (54.8%) | 15 (37.5%) |
| **No** | 19 (45.2%) | 25 (62.5%) |
| **TRG score** | | |
| **0** | 23 (54.8%) | 15 (37.5%) |
| **1** | 3 (7.1%) | 2 (5%) |
| **2** | 14 (33.3%) | 15 (37.5%) |
| **3** | 2 (4.8%) | 8 (20%) |
| **Pathological T stage** | | |
| **T0** | 24 (57.1%) | 15 (37.5%) |
| **Tis** | 1 (2.4%) | 0 (0%) |
| **T1** | 0 (0%) | 1 (2.5%) |
| **T2** | 9 (21.4%) | 6 (15%) |
| **T3** | 6 (14.3%) | 10 (25%) |
| **T4** | 2 (4.8%) | 8 (20%) |
| **Pathological N stage** | | |
| **N0** | 39 (92.9%) | 36 (90%) |
| **N1** | 3 (7.5%) | 3 (7.5%) |
| **N2** | 0 (0%) | 1 (2.5%) |
| **Type of surgery** | | |
| **Anterior resection** | 36 (85.7%) | 24 (60%) |
| **Abdominoperineal resection** | 6 (14.3%) | 13 (32.5%) |
| **Hartmann procedure** | 0 (0%) | 3 (7.5%) |
| **Anus preservation** | | |
| **Yes** | 36 (85.7%) | 27 (67.5%) |
| **No** | 6 (14.3%) | 13 (32.5%) |

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# Fruquintinib combined with FOLFIRI/mFOLFOX6 as second-line treatment in patients with RAS-mutant metastatic colorectal cancer (mCRC): A multicenter, open-label, phase II study.

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**Background:** Anti-angiogenic drugs combined with chemotherapy could bring good clinical benefit to patients (pts) with mCRC, but the efficacy in pts with RAS mutant mCRC was limited. Fruquintinib is a highly selective oral inhibitor of vascular endothelial growth factor receptors (VEGFRs -1, -2, and -3) and was approved for previously treated mCRC. Our study was to explore the efficacy and safety of fruquintinib combined with FOLFIRI/mFOLFOX6 in the second line treatment of RAS-mutant mCRC (NCT05634590). **Methods:** Eligible pts with histologically confirmed RAS-mutant mCRC who had received first-line standard chemotherapy regimens were included in the study, and pts received Fru (4 mg, p.o. qd, 3 weeks on/1 week off, q4w) plus FOLFIRI/mFOLFOX6 (d1-15, q4w) until disease progression, unacceptable toxicities, or withdrew consent. The primary endpoint was progression-free survival (PFS). Secondary endpoints included objective response rate (ORR), disease control rate (DCR), overall survival (OS)and safety. **Results:** As of Aug 30, 2024, 25 eligible pts were enrolled and received at least one time treatment. Baseline characteristics included median age (66.0 [range: 35-73]), female/male (56.0% / 44.0%), ECOG PS 1(68.0%), left-sided and right-sided colon (72.0% / 28.0%), liver metastasis (64.0%), prior anti-VEGF therapies (76.0%). 14 pts had received at least one tumor assessment. Five pts achieved partial response (PR), 9 pts achieved stable disease (SD), and the ORR was 35.7% (5/14) (95% confidence interval [CI]: 12.8-64.9), the DCR was 100.0% (14/14) (95% CI: 76.8-100.0). Compared to liver metastases, the ORR was higher without liver metastases (42.9% [95%CI 9.9-81.6]). The median PFS was 6.41 months (95% CI: 6.37 - NA), and median OS was not reached yet. In the subgroup analysis, Median PFS was also longer without liver metastases pts than with liver metastases (8.84 mo vs 6.41 mo; HR=1.230; 95%CI 0.2462-6.148; p=0.7775). Additionally, Grade ≥3 treatment-emergent adverse events (TEAEs) included platelet count decreased (18.2%), hypertension (9.1%) and neutrophil count decreased (9.1%). **Conclusions:** Fruquintinib combined with FOLFIRI/mFOLFOX6 as second-line in patients with RAS-mutant mCRC achieved a promising clinical benefit, with a manageable safety profile. This trial is ongoing and more updated clinical data will be presented in the future. Clinical trial information: NCT05634590. Research Sponsor: None.

# A retrospective observational study to evaluate the efficacy of trifluridine/tipiracil ± bevacizumab in metastatic colorectal cancer with MSI-high/deficient MMR.

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**Background:** MSI-high (MSI-H)/deficient MMR (dMMR) metastatic colorectal cancer (mCRC) have reported to be resistant to various cytotoxic agents including fluorouracil. On the other hand, preclinical study showed that trifluridine was effective to fluorouracil-refractory dMMR CRC cell lines. Previous studies on trifluridine/tipiracil (FTD/TPI) ± bevacizumab (BEV) for mCRC as later line treatment showed an objective response rate (ORR) of 1.1-5.6% and disease control rate (DCR) of 44-76.6%. However, there are no reports on the efficacy of FTD/TPI± bevacizumab in MSI-H/dMMR mCRC patients. **Methods:** We retrospectively evaluated the efficacy and safety of FTD/TPI ± BEV as second- or later-line treatment which patients with MSI-H/dMMR mCRC received between June 2012 and January 2023 at the 10 institutions. The primary endpoint was investigator-assessed objective response rate (ORR). The secondary endpoints were progression-free survival (PFS), overall survival (OS), disease control rate (DCR), and adverse event rates. We also compared the FTD/TPI + BEV (FTB group) with FTD/TPI (FT group). **Results:** A total of 18 patients (FTB/FT, 9/9) were included. Patient characteristics were as follows (FTB/FT): median age, 70 (70/70) years; ECOG PS of 0/1, 7 (5/2)/11(4/7); primary tumor location of right side/left side, 12(6/6)/6(3/3); *BRAF* V600E mutant/wild-type, 6(3/3)/11(6/5); number of previous treatment lines of 1/≥2, 2(1/1)/16(8/8); number of metastatic sites of 1-2/≥3, 8(4/4)/10(5/5); prior use of anti-PD-1 therapy, 10(3/7). Efficacies in the whole population were as follows: ORR, 16.7% (95%CI, 3.6-41.4); DCR, 72.2%; median (m) PFS, 5.5 months; mOS, 11.8 months. Efficacies in each group (FTB vs. FT) were as follows: ORR (22.2% vs. 11.1%) and DCR (88.9% vs. 55.6%) were favored in FTB group than in FT group, while the median PFS were similar between two groups (5.6 months vs. 5.2 months). OS in FTB group was longer than in FT group (median, 18.9 months vs. 7.1 months; hazard ratio, 0.22; p value=0.03). Of 11 (FTB/FT, 7/4) patients who received subsequent treatment, 6 in FTB group and 1 in FT group received anti-PD-1 therapy. The most common grade 3 or more adverse events in each group (FTB vs. FT) were neutropenia (77.8% vs.77.8%), anemia (22.2% vs. 33.3%) and febrile neutropenia (0% vs. 22.2%). **Conclusions:** FTD/TPI ± BEV showed a promising efficacy and favorable safety. Although this was retrospective study with a small sample size, FTD/TPI ± BEV therapy may have better efficacies for MSI-H/dMMR mCRC than those for MSS/proficient MMR mCRC in previous prospective studies. Next generation sequencing as biomarker analysis using pretreated tissue samples is ongoing. Research Sponsor: TAIHO PHARMACEUTICAL.

**153** **Poster Session**

# Cetuximab every second week in combination with daily encorafenib in patients with BRAF V600E mutated metastatic colorectal cancer: Interim analysis from NEW BEACON.

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**Background:** Weekly administered cetuximab in combination with the BRAF inhibitor encorafenib has improved survival in pre-treated patients (pts) with BRAF V600E mutated (BRAF-mut) metastatic colorectal cancer (mCRC), with a median progression free survival (PFS) of 4.3 months in the BEACON study. It is now standard-of-care in the second line treatment. However, a regimen with cetuximab administered every second week may demonstrate comparable efficacy and is more convenient for the pts. Thus, we initiated the NEW BEACON study investigating cetuximab given every second week together with daily encorafenib. Details on the rationale and aims of the study have previously been published (doi: 10.1186/s12885-022-10420-x.). **Methods:** NEW BEACON is an open-label, single-arm, multicenter, phase II study including pts with pre-treated BRAFmut mCRC. The primary end point is 2 months PFS rate. We report here the result of a pre-specified interim analysis after treatment of the first 19 included pts. We collected fresh tumor tissues for comprehensive molecular profiling in order to explore features associated with response and resistance. Baseline genomic variants will be reported. **Results:** From February 2021 to February 2024 20 pts were included in the study. 1 patient never started study treatment due to rapid disease progression. At the data cut-off date 19 pts had initiated study treatment. The 2 months PFS rate was 89% (17/19 pts). We had tumor tissue for molecular profiling available from 12 pts. Pathogenic variants in TP53 were the most frequently detected co-mutations at baseline in 83% of pts (10/12 pts). Available preliminary safety, efficacy, and survival data and more extensive genomic data will be presented at the meeting. **Conclusions:** The preliminary PFS data from pts with BRAFmut mCRC treated with cetuximab every second week together with daily encorafenib is promising demonstrating a 2 months PFS rate of 89%. This is similar to what was reported from the BEACON study (2 months PFS rate of 84%). Loss-of-function mutations in tumor suppressor genes were frequently detected as co-mutations at baseline. Clinical trial information: EudraCT: 2020-003283-10. Research Sponsor: Kræftens Bekæmpelse (The Danish Cancer Society); Pierre Fabre.

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# Phase II study of trifluridine/tipiracil (TAS-102) plus irinotecan in combination with bevacizumab as a second-line therapy for patients with metastatic colorectal cancer.

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**Background:** Metastatic colorectal cancer (mCRC) patients have poor prognoses with limited response to second-line chemotherapy. Therefore, it is urgent to refine and design an optimal second-line treatment regimen to improve the response rate and prolong the survival of patients with mCRC. This study assessed the safety and efficacy of the trifluridine/tipiracil (TAS-102), irinotecan, and bevacizumab regimen in second-line settings for mCRC patients. **Methods:** This was a multicenter, single-arm, phase II trial. The mCRC patients who are refractory to standard first-line treatment, including fluoropyrimidine and oxaliplatin, are eligible. Based on the recommended phase II dose established in phase I, enrolled patients received TAS-102 (30 mg/m2 twice daily on days 1-5) and irinotecan (150 mg/m2 on day 1) combined with bevacizumab (5 mg/kg on day 1) every two weeks. The primary endpoint was the objective response rate (ORR). **Results:** Between October 1, 2023, and August 31, 2024, 60 patients were enrolled, and all of them were evaluated for efficacy. As of August 31, 2024, efficacy was assessed in all 60 participants with an ORR of 18.3% (2 had complete response, CR; 9 had partial response, PR; 39 had stable disease, SD). Among the 39 patients with SD, 29 patients experienced tumor shrinkage. The disease control rate (DCR) was 83.3%. The median follow-up was 8.6 months (95% confidence interval [CI], 6.743-10.457). The six-month progression-free survival (PFS) was 59.6%. The median PFS was 6.9 months (95% CI 5.016-8.784), whereas the median overall survival (OS) was not reached. The 1-year OS rate was 92%. At the time of analysis, four patients have died, and 56 patients are still alive. All patients were evaluable for safety assessment. The most common treatment-related adverse events (TRAE) were nausea (100%), neutropenia (86.7%), and anemia (83.3%). The most frequent grade 3/4 TRAE were neutropenia (48.3%), febrile neutropenia (8.3%), nausea (3.3%), and anemia (3.3%). No treatment-related death occurred. **Conclusions:** The regimen of irinotecan, TAS-102, plus bevacizumab preliminarily demonstrated promising efficacy with tolerable toxicity for mCRC patients as second-line treatment. This regimen warrants further exploration in refractory mCRC patients. Clinical trial information: NCT06202001. Research Sponsor: None.

# Long-course chemoradiation vs short-course radiation in total neoadjuvant therapy for rectal cancer: Propensity score case-matched analysis.

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**Background:** Short-course radiotherapy (Short-TNT) is not widely used in the United States as part of total neoadjuvant therapy (TNT) for rectal cancer, due to concerns for lower tumor regression, increased local recurrence, and poorer quality of surgical specimens compared to long-course chemoradiotherapy (Long-TNT). However, there have been few studies that directly compared outcomes of Long-TNT versus Short-TNT. This study aimed to evaluate outcomes of Short-TNT versus Long-TNT. **Methods:** Consecutive patients with stage II-III rectal cancer who underwent TNT at an NCI-designated comprehensive cancer center from 2019 to 2022 were identified. Outcomes after Short-TNT and Long-TNT were compared using 1:1 propensity score matching. **Results:** Among 318 patients, 170 (85 Short-TNT, 85 Long-TNT) were eligible for the matched analysis (Table). The median follow-up was 31 months. The two groups were similar in tumor distance from anal verge (5cm vs. 6cm), clinical stage (stage III: 90.6% vs. 89.4%), radiologic high-risk features, and treatment sequence (induction chemotherapy: 30.5% vs. 28.2%). The Long-TNT was associate with a higher proportion of non-operative management (38.8% vs. 24.7%, p=0.03). A local regrowth rate with long-TNT was not lower than short-TNT (39.4% vs. 28.6%, p=0.42). There were no differences in rates of clinical complete plus near-complete response (52.9% vs. 52.9%, p=1.00), overall complete response (pathologic plus sustained clinical complete response; 32.2% vs. 29.9%, p=0.74), and surgical outcomes including retrieved lymph nodes number (19 vs. 22, p=0.27), sphincter preservation (46.2% vs. 62.5%, p=0.10), distal margin (3cm vs. 3.5cm, p=0.77), CRM+ (83.8% vs. 3.1%, p=0.42) and postoperative complications (overall: 34.6% vs. 25.0%, p=0.14; ≥Grade3: 5.8% vs. 9.4%, p=0.60). **Conclusions:** Short-TNT had a comparable overall complete response rate with equivalent surgical outcomes to Long-TNT. Short-TNT should be revisited in the US as a viable alternative TNT strategy both for nonoperative management and for treatment of locally advanced rectal cancer. Research Sponsor: None.

**Propensity score case-matched analysis.**

| Outcomes | Total (n=170) | Long-TNT (n=85) | Short-TNT (n=85) | p value |
|---|---|---|---|---|
| Clinical complete/near complete response, number (%) | 90 (52.9) | 45 (52.9) | 45 (52.9) | 1.00 |
| Non-operative management, number (%) | 54 (31.8) | 33 (38.8) | 21 (24.7) | 0.03 |
| Overall complete response, number (%)* | 54 (31.0) | 26 (30.6) | 28 (32.9) | 0.74 |
| **Surgical outcomes** | **Surgery (n=116)** | **Surgery after Long-TNT (n=52)** | **Surgery after Short-TNT (n=64)** | |
| Sphincter-preserving surgery, number (%) | 64 (55.2) | 24 (46.2) | 40 (62.5) | 0.10 |
| Circumferential resection margin +, number (%) | 4 (3.4) | 2 (3.8) | 2 (3.1) | 0.42 |
| Distal margin, cm (range) | 3.4 (0.1-12) | 3 (0.2-6.5) | 3.5 (0.1-12) | 0.77 |
| Major complications (≥Grade3), number (%) | 9 (7.8) | 3 (5.8) | 6 (9.4) | 0.60 |

*Pathologic + sustained clinical complete response.

# Prognosis of patients with liver single-organ metastases: Updated survival results of BBCAPX-II.

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**Background:** Although immunotherapy has changed the treatment strategy for many cancers with great success, patients with microsatellite stable (MSS) and RAS-mutant metastatic colorectal cancer (mCRC) especially with liver metastases have a low response rate to immunotherapy. Sintilimab plus bevacizumab and CapeOX (BBCAPX) has proved its efficacy and safety in above unresectable mCRC patients (1). Here, we report subgroup analysis of updated survival results of this single arm, open-label, phase 2 trial. **Methods:** Eligible patients were histologically confirmed unresectable metastatic colorectal adenocarcinoma by multidisciplinary team, and had RAS gene mutation and confirmed MSS status. All patients received treatment with sintilimab plus bevacizumab, oxaliplatin and capecitabine of each 21-day cycle. ORR, DCR, PFS, OS, and subgroup analyses based on metastasis site were performed. **Results:** From April 2021 to December 2021, 25 patients were enrolled. The ORR was 84% and the DCR was 100%. Six (24%) patients including 5 patients with liver single organ metastases underwent surgical treatment and unexpectedly achieved no evidence of disease status. At the data cut-off day (September 13, 2024), patients with liver single organ metastases presented better prognosis. Median PFS was 25.3 months (95% CI, 4.8-NA) in the patients with liver single organ metastases and 11.5 months (95% CI, 4.83-24.3) in the patients with other metastasis. Median OS was not reached in the patients with liver single organ metastases and 35.5 months (95% CI, 9.36-NA) in the patients with other metastasis. The OS rate at 24 months was 72% (95% CI, 56.4-91.9) in FAS. Median OS was not reached in FAS and 35.5 months (95% CI, 11.4-NA) in PPS. 32% patients had at least one grade 3 or 4 TRAEs. No grade 5 adverse events occurred during the study. **Conclusions:** This study provides a highly promising regimen for patients with RAS-mutant, MSS, unresectable mCRC, especially those with liver single organ metastases. Furthermore, we are launching a phase III, randomized, open-label, multicentric clinical trial (NCT05171660) to further analyze the effects, safety, and prognostic biomarkers of this regimen. 1. Xuefeng Fang et al. ASCO 2022. Clinical trial information: NCT04194359. Research Sponsor: National Natural Science Foundation of China; 82373415; National Natural Science Foundation of China; 82403806.

**PFS and OS results of BBCAPX-II study.**

| | Median PFS | 6m PFS rate (%) | 12m PFS rate (%) | Median OS | 12m OS rate (%) | 24m OS rate (%) |
|---|---|---|---|---|---|---|
| **Full analysis set (FAS), n = 25** | 17.9 (95% CI, 8.84-27) | 84 (70.8-99.7) | 56 (39.6-79.3) | NA | 76 (61-94.7) | 72 (56.4-91.9) |
| **Per-protocol set (PPS), n = 19** | 9.79 (95% CI, 6.44-27.9) | 78.9 (62.6-99.6) | 42.1 (24.9-71.3) | 35.5 (95% CI, 11.4-NA) | 68.4 (50.4-92.9) | 63.2 (44.8-89) |
| **Classified by metastatic organs (FAS, n = 25)** | | | | | | |
| **Liver single organ metastases, n = 10** | 25.3 (95% CI, 4.8-NA) | 90 (73.2-100) | 70 (46.7-100) | NA (11.4-NA) | 90 (73.2-100) | 80 (58.7-100) |
| **Other metastases, n = 15** | 11.5 (95% CI, 4.83-24.3) | 80 (62.1-100) | 46.7 (27.2-80.2) | 35.5 (9.36-NA) | 66.7 (46.6-95.3) | 66.7 (46.6-95.3) |

# Preoperative botensilimab (BOT) with or without balstilimab (BAL) for patients with resectable locally advanced pMMR or dMMR colon cancer: Results from the UNICORN trial by GONO.

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**Background:** Standard treatment for patients (pts) with localized colon cancer (CC) is surgery and adjuvant chemotherapy where indicated. Neoadjuvant strategies allow for earlier exposure to systemic therapy, potentially leading to tumor downstaging, micro-metastases eradication, improved survival, and may pave the way for non-operative management (NOM). BOT, a novel Fc-enhanced multifunctional anti-CTLA-4 antibody plus BAL, an anti-PD-1 antibody, have demonstrated activity in proficient mismatch repair (pMMR) CC pts with durable responses in metastatic CC and unprecedented rates of pathological complete response (pCR) in the localized setting. **Methods:** UNICORN is a window-of-opportunity, multicohort, umbrella platform phase II trial enrolling non-metastatic, radiologically staged rT3-4 N0-2, resectable CC pts to be treated with a short course preoperative targeted treatment according to a prespecified molecular profile assessed by immunohistochemistry and next-generation sequencing on tumor biopsy. Pts with pMMR or deficient (dMMR) tumors received intravenous (IV) BOT at 1 mg/kg on day 1 (cohorts 4 or 6) or IV BOT 1 mg/kg on day 1 and BAL 3 mg/kg on days 1 and 15 (cohorts 5 or 7, enrolled after 4 and 6) and underwent radical surgery on day 35 ± 5. Pathological response (pR), major response (pMR) and pCR rates were defined as ≤ 50%, ≤10% and 0% residual viable tumor. The primary endpoint was centrally assessed pMR rate in each cohort. According to a Fleming 1-stage design, choosing $\beta$=80% and $\alpha$=5%, 14 pts were enrolled in each cohort and the treatment was judged promising if ≥5/14 pMRs were observed. **Results:** All 56 enrolled pts completed preoperative treatment and underwent surgery. Timely surgery was performed in most pts (98%), except 1 who underwent surgery with a delay < 4 weeks due to treatment-related hyperthyroidism. Among pts with pMMR disease, activity was limited for BOT monotherapy (pMR 0%, pR 43%). With BOT + BAL, pCR was 29%, pMR 36% and pR 71%. Among those with dMMR status, BOT led to pCR, pMR and pR in 29%, 36% and 64% pts, respectively. Notably, BOT + BAL led to pCR and pMR in 93% and 100% pts. Adverse events (AEs) of any grade occurred in 28/56 pts (50%), and 22 (39%) were deemed immune-related. Serious AEs occurred in 9 pts (16%) and were treatment-related in 3 pts (5%). **Conclusions:** The primary endpoint was met in all cohorts except for BOT monotherapy in pts with pMMR status. Despite limited sample size, the activity of 1 neoadjuvant cycle with BOT + BAL favorably compares to ipilimumab/nivolumab in both pMMR and dMMR subgroups. pCR rate was remarkable in pMMR and the highest ever reported in dMMR pts, paving the way to NOM studies irrespective of MMR status. Clinical trial information: EU-CT 2022-501308-90-00. Research Sponsor: Fondazione GONO; AGENUS.

| Molecular status | n° | Treatment | pCR, n (%) | pMR, n (%) | pR, n (%) |
|---|---|---|---|---|---|
| **pMMR** | 14 | BOT | 0 (0) | 0 (0) | 6 (43) |
| | 14 | BOT/BAL | 4 (29) | 5 (36) | 10 (71) |
| **dMMR** | 14 | BOT | 4 (29) | 5 (36) | 9 (64) |
| | 14 | BOT/BAL | 13 (93) | 14 (100) | 14 (100) |

# The crossover of the bevacizumab and panitumumab survival curves in the PARADIGM study: A clue to time-related effects of bevacizumab on the risk of tumor progression.

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**Background:** The PARADIGM study in CRC patients demonstrated a cross over of the bevacizumab survival curve below the panitumumab survival curve at 28 months, followed by a steady decrease of the bevacizumab survival curve below the panitumumab survival curve up to 11% by 5 years. Similar differences were seen in the FIRE3, CALGB-80405 and PEAK trials starting at 20 months, suggesting an unsuspected time-related effect of bevacizumab on patient's survival starting at about 2 years after its 1st administration. **Methods:** Review of time related effects of bevacizumab in 1st and 2nd line trials conducted in CRC, breast, ovary and renal cancer showed a significant PFS but no OS survival benefit, with a detrimental effect on survival in the adjuvant setting (AVANT, NSABP-08 studies). Preclinical data on tumor biology, intratumor-heterogeneity, clonal evolution and the cross talk between bevacizumab and the diverse tumor microenvironments, was correlated to the clinical data, in an attempt to explain the time-related effects of the biological. **Results:** In the NSABP-08 trial, bevacizumab reduced relapse for 6 months, effect diminished and lost at 12 months. At 24 months after bevacizumab's 1st administration, the relapse of the bevacizumab treated population increased steadily over the control. Identical kinetics were shown in AVANT study and metastatic ovarian patients suggesting that these time-related effects of bevacizumab are universal. The biologic explanation of the clinical data is that short-term treatment with bevacizumab (plus chemotherapy) induces initial tumor stasis and shrinkage, followed by selection and emergence of more aggressive and resistant tumor clones, characterized by an accelerated growth, invasiveness and metastasis shifting the overall tumor growth kinetics to a steeper angle. Bevacizumab appears to be effective for the time of its administration, followed by a slow rebound after its discontinuation. The initial survival benefit will be lost at about 20-24 months after its first administration, as clearly demonstrated in the PARADIGM, FIRE-3, CALGB 80405 and the PEAK studies. **Conclusions:** As bevacizumab benefit persist for about 2 years after its 1$^{st}$ administration, it explains the "survival benefit" obtained in studies with with median survival of 20 months but not in studies with survivals >30 mo. (PARADIGM, FIRE-3, CALGB 80405 and the PEAK). This phenomenon also explains the crossover of the survival curves in PARADIGM study and the benefits of bevacizumab obtained preferentially in patients with short survival (rt. sided tumors), high tumor burden, and 2$^{nd}$ and 3$^{rd}$ line CRC. Research Sponsor: None.

# Outcomes of simultaneous, adaptive, MRI-guided liver SBRT in patients with heavily pretreated colorectal cancer and four or more liver metastases.

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**Background:** About one-third of colorectal cancer (CRC) patients (pts) develop metastases, most commonly in the liver (CRLM). Ablative treatment of multiple liver lesions is difficult and often not done with conventional CT-based radiation therapy, but feasible with MRI-guided adaptive radiation. This study reviews our experience treating pts with high burden CRLM using MRI-guided stereotactic body radiation therapy (MRgRT). **Methods:** This study identified pts with 4 or more CRLM treated with MRgRT between 7/2019 and 4/2024, all of whom had progressive disease during or after standard-of-care therapy. Ablative radiation was delivered to all sites of disease with no concurrent systemic therapy. All with Child-Pugh score 5-6 (class A) liver function. Local control was defined as no progression in any treated lesion on a per-patient basis. Descriptive statistics summarized the data. A Cox proportional hazards model was used to identify predictors of survival, while the Kaplan-Meier method was used to estimate survival functions. Statistical analysis was performed using SPSS. **Results:** Seventeen pts with 110 lesions were included, with a median follow-up of 10.7 months (range, 1.7–36.9). The majority (n=15) received systemic therapy in the year before MRgRT. The median number of prior lines of systemic therapy was 2 (range, 1–5). Eight pts (47.1%) had nodal and/or lung progression in addition to CRLM, and 4 pts (23.5%) had at least one prior course of MRgRT. A median of 6 liver lesions (range, 4–11) were prescribed a median dose of 60 Gy (range, 60–80), each delivered in 5 fractions. Median total gross tumor volume per pt was 23.8 $cm^3$ (range, 0.97–60.1). Seventy-four fractions were available for radiation delivery analysis, of which 55 (74.3%) were adapted. At least one target was modified in 51 (92.7%) of the adapted fractions. Acute toxicity occurred in 7 pts, all with self-limited fatigue and nausea. Progression in Child-Pugh classification from A to B occurred in three pts 3-6 months post-MRgRT. Late toxicity occurred in 1 patient with mild, possibly radiation-related peripheral biliary dilation. Two pts developed malignant biliary strictures. Per-patient local control was 50.8% and 34.8% at 1 and 2 years, respectively, while liver control was 28.7% at 2 years. Six-month systemic therapy-free survival was 31.9%. Two-year overall survival (OS) was 43.9%, with a median of 17.1 months (range, 8.6–25.7). The size of the largest tumor correlated with OS (HR 1.23, p=0.014). **Conclusions:** In this small series of CRC pts with 4+ CRLM, MRgRT is well-tolerated and shows promise as astrategy to consolidate disease and delay the immediate need for systemic therapy changes. Patient selection is critical to identify those most likely to benefit, though the overall impact on survival and disease trajectory remains uncertain. Research Sponsor: None.

# Risk of peritoneal recurrence following laparoscopic versus open surgery for stage II or III colorectal cancer (JCOG2310A): An integrated analysis of three phase III randomized controlled trials.

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**Background:** Peritoneal recurrence in colorectal cancer (CRC) has long been a concern. Although studies have examined the higher risk of peritoneal recurrence following laparoscopic (LAP) compared with open (OP) surgery, the issue remains unresolved, largely due to the heterogeneity of existing data and the limited number of observed peritoneal recurrence events, preventing any definitive conclusions. This study aimed to evaluate whether LAP for pStage II/III CRC presents a potential risk factor for peritoneal recurrence compared with OP. **Methods:** Data from three phase III trials (JCOG0404, JCOG0910, JCOG1006) were extracted from an ancillary research database (JCOG2310A). From this cohort, patients with pStage II or III CRC who underwent curative surgery were included, and the incidence of peritoneal recurrence was compared between LAP and OP. A multivariate analysis was conducted to adjust covariates between the two groups. The primary outcome was the cumulative incidence of peritoneal recurrence, and the secondary outcome was recurrence in specific high-risk subgroups identified through a univariate analysis. **Results:** From 3061 patients (OP: 1734; LAP: 1327), pT4 (25% vs. 17%), pStage III (65% vs. 83%), and right-sided tumor (63% vs. 70%) were different between OP and LAP, respectively. With a median follow-up for peritoneal recurrence of 6.2 years, the 5-year cumulative incidence of peritoneal recurrence was 2.8% in the OP group and 2.9% in the LAP group (hazard ratio (HR): 0.940, 95% confidence interval (CI): 0.621-1.425 in the univariate analysis, and HR: 1.086, 95% CI: 0.683-1.727 in the multivariate analysis. The univariate analysis identified pT4, pN2, pStage III, right-sided tumor, poorly differentiated (por/sig/muc), and ECOG-PS 1 as risk factors for peritoneal recurrence. No significant difference was observed in the subgroups (see table). An exploratory propensity score matching analysis with 1066 patients in each group also showed no difference between the groups (the 5-year cumulative incidence of peritoneal recurrence was 2.8% in the OP group and 3.0% in the LAP group [HR: 1.002, 95% CI: 0.625-1.607]). **Conclusions:** This study revealed that LAP was not a risk factor regarding peritoneal recurrence following curative surgery for Stage II/III CRC.
Research Sponsor: None.

**Subgroup analysis of cumulative incidence of peritoneal recurrence.**

| Outcome Measures | Subgroups | | | | | |
|---|---|---|---|---|---|---|
| | pT4 | pN2 | pStage III | Right-sided | Poorly differentiated | ECOG PS 1 |
| N (OP vs. LAP) | 428 vs. 229 | 274 vs. 184 | 1133 vs. 1105 | 635 vs. 403 | 122 vs. 66 | 56 vs. 25 |
| Event (OP vs. LAP) | 31 vs. 23 | 13 vs. 10 | 45 vs. 32 | 28 vs. 18 | 10 vs. 2 | 7 vs. 2 |
| 5-year peritoneal recurrence rate (OP vs. LAP)(%) | 6.8 vs. 10.1 | 4.8 vs. 5.4 | 3.6 vs. 2.9 | 4.1 vs. 4.5 | 7.5 vs. 3.0 | 9.1 vs. 8.0 |
| HR (95% CI) | 1.403 (0.819-2.402) | 1.144 (0.503-2.603) | 0.725 (0.461-1.139) | 1.014 (0.562-1.829) | 0.372 (0.081-1.699) | 0.609 (0.130-2.862) |

# Nelmastobart (hSTC810) combined with capecitabine therapy in metastatic colorectal cancer with resistance or intolerance to oxaliplatin and irinotecan-based chemotherapy: A phase 1b clinical trial.

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**Background:** Treatment options for patients with metastatic colorectal cancer (mCRC) who experienced disease progression after receiving oxaliplatin and irinotecan-based chemotherapy are limited. Nelmastobart (hSTC810), a humanized anti-BTN1A1 (butyrophilin subfamily 1 member A1) monoclonal antibody of the IgG4 isotype, showed acceptable safety profile and the disease control rate was 39.3% in the first-in-human study, suggesting its potential efficacy. This phase Ib study was designed to evaluate the safety and explore synergistic anti-tumor activity of nelmastobart in combination capecitabine. The study aims to determine the maximal tolerated dose (MTD) and recommended phase ll dose (RP2D) of this combination therapy in patients with mCRC who are refractory to standard therapy. **Methods:** Dose escalation phases were conducted in patients with mCRC who were administered orally twice daily capecitabine (850mg, 1000mg, 1250mg orally, days 1-14) and nelmastobart (400mg, 800mg intravenously every 3 weeks). Standard "3 + 3" dose escalation was used to define the MTD. In this phase 1b, 12 participants enrolled between February 5 2024 and June 11 2024 were 18 years of age or older with mCRC previously treated oxaliplatin and irinotecan in the metastatic setting. **Results:** The first 3 patients at the dose level 0 (nelmastobart 400mg, capecitabine 1,000 mg/m$^2$ twice daily) and level 1 (nelmastobart increased to 800 mg, capecitabine 1,000mg/m$^2$ twice daily) had no dose-limiting toxicities (DLTs). In dose level 2 (nelmastobart 400mg, capecitabine 1,250mg/m$^2$ twice daily), 2 of 3 had DLTs (grade 3 hand-foot syndrome), which were attributed to capecitabine. When dose level 2 was expanded to 3 additional patients, 2 more patients experienced DLTs (grade 3 mucositis and hand-foot syndrome). Consequently, the combination of nelmastobart 800 mg once every three weeks and capecitabine 1,000 mg/m$^2$ twice daily (2 weeks on 1 week off) was established as the MTD/RP2D in patients with mCRC. The most common AEs were hand-foot syndrome (n=4), nausea (n=3) and stomatitis (n=2) which were related with capecitabine. Four patients experienced grade 1 fatigue, which was considered to be related to nelmastobart. Two patients (16.7%) out of 12 showed partial response and 8 patients have maintained stable disease for more than 4 months among 9 patients who had a median follow-up of more than 4 months (as of August 30). **Conclusions:** The combination of hSTC810 and capecitabine is well tolerated at the MTD, without unexpected AEs and shows promising antitumor activity in patients with heavily treated mCRC. The phase 2 study is currently in progress. Clinical trial information: NCT0599054. Research Sponsor: STCube.

# Immunotherapy outcomes in dMMR/MSI-H and/or TMB-H metastatic colorectal cancer (CRC) with peritoneal metastases (PM).

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**Background:** CRC with PM has a poor prognosis and limited treatment options. Immunotherapy has shown promise in improving outcomes for patients with microsatellite instability-high (MSI-H), mismatch repair-deficient (dMMR), and high tumor mutational burden (TMB-H) tumors. However the peritoneal niche has a unique microenvironment, which can influence the effectiveness of immune check point inhibitors (ICIs). We assessed the impact of ICI on progression-free survival (PFS) and overall survival (OS) in CRC patients (pts) and PM, considering key clinical and molecular characteristics. **Methods:** A retrospective review aiming to identify CRC pts with MSI-H/dMMR and PM treated with immunotherapy between 2015 and 2023 was performed. Data collected included pt demographics, tumor characteristics, and treatment details. Responses were evaluated using RECIST v1.1 criteria. **Results:** 37 pts were included, 9 (24%) had Lynch syndrome. The median age of diagnosis was 61 years and most pts were females (65%). Most pts had MSI-H/dMMR tumors, while one pt was treated based on ultra-hypermutated phenotype (TMB 202) and POLE mutation. Among those with available TMB data, the median TMB was 52 mutations/Mb (0-202). Common mutations were *PMS2* (51%), *MLH1* (38%), *MSH6* (16.2%), and *MSH2* (16.2%). Right-sided tumors were in 54%, mucinous histology in 62%, and ascites in 54%. Pts received a median of 11 cycles of ICI (2-52). Pembrolizumab was the most frequently used ICI (78%), while Nivolumab was used either alone (11%) or in combination with Ipilimumab (11%). Median PFS in our cohort was 7.8 months and median OS was 29.9 months. The disease control rate (DCR) was 81%, with an objective response rate (ORR) of 51.3% (9 complete responses, 12 partial responses). At the time of data collection, 30% of pts were off systemic therapy and 5-FU-based chemotherapy combined with a biologic agent was the most common subsequent treatment in 32% who had PD. Univariate analysis showed no significant differences in PFS or OS based on tumor sidedness, isolated PM, liver metastases, mucinous vs. non mucinous histology or *BRAF*/*RAS* status (WT vs mutant). Malignant ascites was associated with worse OS(HR 0.3; p= 0.03). **Conclusions:** Treatment with ICIs in pts with MSI-H/dMMR and/or TMB-H CRC and PM improved disease control in CRC pts with PM, however, as compared to historical controls for patients with MSI-H/d-MMR disease treated with ICI, this group showed less favorable outcomes. Malignant ascites was associated with poorer overall survival. Further research of the immune response and tumor microenvironment in the peritoneal cavity is warranted. Research Sponsor: None.

# Encorafenib and cetuximab with/without binimetinib therapies for patients with *BRAF*-mutated metastatic colorectal cancer with prognostic factors: The BEETS trial (JACCRO CC-18).

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**Background:** BRAF inhibitor combination therapies, triplet (encorafenib, binimetinib and cetuximab) and doublet (encorafenib and cetuximab), have been approved for *BRAF*-mutated metastatic colorectal cancer (mCRC) in Japan. The treatment with triplet is recommended for patients (pts) with clinical prognostic factors according to the sub-group analysis of the BEACON CRC trial. However, it is unclear whether the treatment strategy is useful. We therefore conducted a study on the use of combination therapy with triplet or doublet based on observational study that accumulated real world data. **Methods:** The BEETS trial was a multicenter observational study to prospectively evaluate the efficacy and safety of BRAF inhibitor combination therapy as a 2nd- or 3rd-line treatment in pts with *BRAF*-mutated mCRC in clinical practice [UMIN000045530]. Two hundred pts were assigned to either triplet or doublet therapy based on the physician's choice. The primary endpoint was overall survival (OS). The secondary endpoints included overall response rate (ORR), disease control rate, progression-free survival (PFS), and safety. The exploratory analysis was performed to evaluate the efficacy of triplet and doublet therapies in all pts and pts with clinical prognostic factors (PS1, 3 or more metastatic sites, elevated CRP value, or primary tumor present) using inverse probability weighting (IPW) analysis based on propensity scores to account for potential cofounding factors. **Results:** Among 203 pts enrolled from Oct 2021 to Nov 2023, 195 pts were evaluable (median age 67 years old, 52% male, ECOG PS 0/1/2 55%/39%/6%, no. of prior regimens 1/2 85%/15%, right-sided primary 61%, MSI-high 7%, triplet regimen 57%). Median OS was 13.2 months (95%CI 11.4-17.4) with 1y-survival rate of 53%, and median PFS was 4.9 months. Comparing the triplet and doublet using IPW, the median OS and PFS were comparable between the two treatments (triplet vs. doublet; 16.0 months vs. 13.6 months [HR 0.88, 95%CI 0.53-1.48] and 5.7 months vs. 4.8 months [HR 0.87, 95%CI 0.59-1.28]). In 139 pts with any prognostic factors, median OS and 1y-survival rate were 13.2 months vs. 9.6 months [HR 0.80, 95%CI 0.45-1.43] and 54% vs. 42%, median PFS and ORR were 5.3 months vs. 4.2 months [HR 0.75, 95%CI 0.48-1.19] and 34% vs. 42%, in the triplet vs. doublet therapy, respectively. **Conclusions:** Our study demonstrated that BRAF inhibitor doublet therapy is a reasonable treatment for *BRAF*-mutated mCRC. However, triplet therapy may be a clinically useful for pts with clinical prognostic factors. Clinical trial information: UMIN000045530. Research Sponsor: Ono Pharmaceutical Co. ltd., Bristol-Myers Squibb.

# Efficacy and safety of fruquintinib alternating with bevacizumab plus capecitabine as maintenance therapy after first-line treatment in metastatic colorectal cancer (mCRC): A multicenter, open-label, phase II study.

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**Background:** Maintenance therapy with bevacizumab (Bev) plus capecitabine (Cap) is widely recommended to unresectable mCRC patients (pts). Fruquintinib (Fru) is a highly selective TKI that inhibits vascular endothelial growth factor receptor (VEGFR)-1,2,3. This study is to compare the therapeutic potential of alternating treatment with fruquintinib and bevacizumab plus capecitabine as maintenance therapy for mCRC (NCT05659290). **Methods:** Eligible mCRC pts aged 18-75 years with stable disease or better after induction treatment with chemotherapy in combination with Bev, ECOG PS 0-2, adequate bone marrow, liver, and renal function were enrolled. Forty patients were included (20 in phase IIa, 40 in phase IIb). In phase IIa, pts were orally administered with Fru (5 mg, qd, d1-14, q3w) alternating with Bev (7.5 mg/kg, iv.gtt, d1, q3w) plus Cap (850 mg/m$^2$, orally, twice daily, d1-14, q3w). In phase IIb, pts were randomly assigned (1:1) to either maintenance treatment with Fru alternating with Bev plus Cap or Bev plus Cap. The primary endpoint was progression-free survival (PFS). Secondary endpoints included objective response rate (ORR), disease control rate (DCR) and safety. **Results:** At cutoff date of Sep 5, 2024, In phase IIa, 20 pts (14 males and 6 females) were enrolled with the median age was 59.0 years (range 27-75), ECOG PS 1 (95.0%), liver metastasis (55.0%), left-sided colon and rectal primary (70.0%). 11 pts had received at least one tumor assessment. the DCR was 100.0% (11/11) and the mPFS was immature, but 4 pts showed the PFS of ≥ 8 months (8.3, 8.6, 9.2, 13.4 m, respectively). The most common treatment-emergent adverse events (TEAEs) were proteinuria (60.0%), hypoalbuminemia (40.0%), hypertension (35.0%); The most common grade ≥ 3 adverse events were hypertension (10.0%), proteinuria (5.0%), and platelet count decreased (5.0%). After fully considering about patient's tolerance and safety, the phase IIb of Fru was adjusted from 5mg to 3mg, these results provided further evidence that 3mg can ensure the safety and tolerance. **Conclusions:** Fruquintinib alternating with bevacizumab plus capecitabine as maintenance therapy after first-line treatment in mCRC showed preliminary anti-tumor activity and manageable toxicity. The phase IIb is ongoing and warrants further exploration in mCRC. Clinical trial information: NCT05659290. Research Sponsor: None.

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# Updated survival results of ZL-IRIAN: Irinotecan plus anlotinib or in combination with penpulimab as second-line treatment of metastatic colorectal cancer.

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**Background:** The irinotecan based chemotherapy in combination with bevacizumab had become the standard second-line treatment for metastatic colorectal cancer (mCRC), but the efficacy was limited. Anlotinib, an oral anti-angiogenic tyrosine kinase inhibitor, had initially shown promising clinical effects when combined with chemotherapy in the first-line treatment of mCRC. Whether anlotinib could bring clinical benefit in the second line treatment warrants further exploration. The PD-1 antibody might have partial effects when combined with anti-angiogenic agents with or without chemotherapy in first- and third-line treatment for proficient mismatch repair (pMMR)/microsatellite stable (MSS) mCRC. Our study was to explore the rationality of anlotinib plus chemotherapy or further combined with a novel anti-PD-1 IgG1 antibody—penpulimab in the second line treatment of mCRC. **Methods:** This study was an open label, non-randomized, multi-cohorts, phase II prospective clinical trial that included pts with pMMR/MSS mCRC who were eligible for the second-line systemic treatment. Two cohorts were established as follows: Cohort A consisted of anlotinib(10mg daily, d1-10, q2w) + irinotecan (180mg/m d6, q2w) (n = 23); Cohort B consisted of anlotinib (8mg daily, d1-10, q2w)+irinotecan (180mg/m d6, q2w) + penpulimab (200mg, d6, q2w) (n = 23). The primary endpoint was ORR. Secondary endpoints included progress free survival (PFS), overall survival (OS), duration of response (DoR), disease control rate (DCR) and safety. The ORR was reported at 2024 ASCO, and PFS and OS have been updated in this report. **Results:** At the cutoff date on August 2024, a total of 23 Pts were enrolled in A cohort, and 23 pts were enrolled B cohort but 21 pts received at least one dose of treatment. The median PFS was 7.87 months (95% CI 5.52-10.22) in A cohort, compared to 7.4 months (95% CI 3.17-11.63) in B cohort, p=0.141. The 6-month, 12-month and 18-month PFS rates in A cohort and B cohort were 69.3% vs 80.7%, 27.5% vs 40.3% and 5.5% vs 22.4%, respectively. The median OS of A cohort was 21.73 months (95% CI 9.43-34.03), compared to 19.17 months (95% CI 11.44-26.90 months) in B cohort, p=0.987. The 12-month and 24-month OS rates in A cohort and B cohort were 64.7% vs 73.1% and 20.2% vs 54.8%, respectively. Safety profile indicated that 65.2% (15/23) and 66.7% (14/21) experienced grade≥3 adverse events in A cohort and B cohort, respectively. Common grade ≥ 3 adverse reactions (≥ 20%) mainly include hypertension (4.35% vs 14.29%), diarrhea (13.04% vs 9.52%), neutrophilopenia (39.13% vs 33.33%), and leukopenia (26.09% vs 9.52%). **Conclusions:** There was no statistically significant difference in overall efficacy between the two cohorts, but the long-term PFS and OS rates were higher in group B cohort, indicating that immunotherapy may bring long-term survival benefits to certain patients. Clinical trial information: NCT05229003. Research Sponsor: None.

# Dual versus single immune checkpoint blockade for patients with metastatic colorectal cancer harboring microsatellite instability: The Saint-Antoine immunoMSI score.

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**Background:** Anti-PD1 monotherapy (mono) and anti-PD1 plus anti-CTLA4 therapies (combo) are validated for metastatic colorectal cancer (mCRC) harboring microsatellite instability and/or mismatch repair deficiency (MSI/dMMR). We aimed at assessing the efficacy of combined anti-CTLA4 and anti-PD1 blockade versus anti-PD1 monotherapy and identifying which subgroups benefit more from the combo. **Methods:** All patients (pts) with MSI/dMMR mCRC treated by anti-PD1 ± anti-CTLA4 included in the prospective monocenter immunoMSI cohort were analyzed. The choice between monotherapy or combination therapy varied according to open studies and regulatory approvals, without taking pt characteristics into account. Datalock date was April 23, 2024. Main endpoints were progression-free survival (PFS) per iRECIST and overall survival (OS) estimated with the Kaplan-Meier method. Univariate Cox proportional hazards models were used to estimate hazard ratios with CI. The interactions between studied subgroups and the treatment group were modelized by adding in a single Cox model the studied subgroups, the treatment group parameters, and their interaction term. **Results:** Among 210 pts, 97 were treated combo and 113 with mono. Baseline characteristics were similar in both groups. 25% and 17% were in first line, respectively. Median follow-up was 4.4 years: 4.1 years for combo and 5.4 years for mono. Combo was associated with better PFS (HR=0.48; 95%CI 0.31-0.76) and OS (HR=0.49; 0.29-0.84). Best observed response rates were: 78% of objective responses, 17% of stable diseases and 5% disease progressions with combo versus 60%, 25% and 15% with mono. Subgroup analyses showed significant differential effect of the treatment group for gender (PFS HR=0.21 vs 0.87 for females and males, P-interaction: 0.0067), liver metastases (PFS HR=0.30 vs 0.79 for pts without liver mets and those with liver mets, P-interaction: 0.0479) and sidedness (PFS HR=0.27 vs 0.64 for left-sided and right-sided mCRC, P-interaction=0.0922), but not *RAS*/*RAF* mutations. We computed a predictive score based on gender (female: 2 points), no liver met (1 point) and left side (1 point). PFS HR was 0.1 (0.03-0.33) for the group ≥3 points (N=67), 0.66 (0.36-1.23) for the group with 1-2 points (N=113) and 1.54 (0.50-4.70) for the group with 0 point (N=30). **Conclusions:** Combined CTLA4 and PD1 blockade is associated with improved PFS and OS compared to anti-PD1 alone in MSI/dMMR mCRC. The subgroup deriving the highest benefit of the combo seems to be female pts with left sided tumor and no liver met. Data from the randomization between nivolumab versus nivolumab + ipilumumab in the CheckMate-8HW phase III study are expected. Research Sponsor: None.

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# Seven-year efficacy and safety in a randomized phase III trial investigating duration of adjuvant oxaliplatin-based therapy (3 vs. 6 months) for patients with high-risk stage II colon cancer: ACHIEVE-2 trial.

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**Background:** ACHIEVE-2 was conducted as one of four phase III trials for patients with high-risk stage II colon cancer (CC) investigating the duration of adjuvant (adj) oxaliplatin-based therapy in a prospective pooled analysis, IDEA collaboration. The results of the 7-year follow-up are presented here. **Methods:** From Feb 2014 to Jan 2017, 525 Asian patients with high-risk stage II CC (T4, inadequate nodal harvest, poorly differentiated, obstruction, perforation, or vascular invasion) were randomly assigned to 3- or 6-month mFOLFOX6/CAPOX treatment after curative surgery. The cutoff date for the data was Jan 2024. **Results:** Of the 525 randomized patients, 11 were not treated. Among the 514 participants (255 in the 3-month arm; 259 in the 6-montharm), 432 (84%) received CAPOX and 184 (36%) presented with T4 as a high-risk factor for recurrence. The baseline characteristics of the patients in each arm were well balanced between the treatment arms and two therapy regimens. The 7-year disease-free survival (DFS) rates were 83.4% and 81.8% in the 3- and 6-month arms, respectively (hazard ratio [HR]= 1.00 [95% CI, 0.66-1.53]). The 7-year overall survival (OS) rates were 88.4% and 88.8% in the 3- and 6-month arms, respectively (HR= 1.16 [95% CI, 0.70-1.93]). With CAPOX, the HR for DFS and OS of the 3-month arm compared with the 6-month arm were 0.97 (95% CI, 0.61-1.52) and 1.09 (95% CI, 0.64-1.89), respectively. Multivariate analysis of the six high-risk factors for recurrence showed that T4 and inadequate nodal harvest were independent risk factors for both DFS and OS. The rates of any grade of peripheral sensory neuropathy (PSN) lasting up to 96 months in the 3- vs. 6-month arms were 7.9% vs. 25.0% (*P=0.0189*). With CAPOX, the incidence of PSN lasting up to 96 months in the 3- vs. 6-month arms were 9.4% vs. 27.5%. **Conclusions:** The incidence of long-lasting PSN was significantly lower at 3 months than at 6 months of therapy. Three months of CAPOX therapy may be appropriate for high-risk stage II CC, as shorter treatment did not worsen the outcome, even after a long follow-up. Clinical trial information: UMIN000013036. Research Sponsor: Japanese Foundation for Multidisciplinary Treatment of Cancer under the contract with Yakult Honsha.

# Updated efficacy and subgroup analysis of first-line serplulimab plus bevacizumab and XELOX versus placebo plus bevacizumab and XELOX in metastatic colorectal cancer: A phase 2/3 study.

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**Background:** This is a randomized, double-blind, multicenter phase 2/3 study comparing the efficacy and safety of serplulimab (a novel anti-PD-1 antibody) plus bevacizumab and XELOX chemotherapy vs. placebo plus bevacizumab and XELOX as first-line treatment for metastatic colorectal cancer (mCRC). Our previous presentation at the 2024 ASCO Meeting showed the progression-free survival results. Here we present the updated efficacy and safety findings together with subgroup analysis results after a median follow-up of 31.0 months. **Methods:** A total of 114 patients with mCRC and no prior systemic therapy were randomized 1:1 (serplulimab arm, n = 57; placebo arm, n = 57) to receive intravenous (IV) serplulimab (300 mg) plus bevacizumab (7.5 mg/kg) and XELOX (IV oxaliplatin [130 mg/m$^2$] and oral capecitabine [1000 mg/m$^2$]) (group A) or placebo plus bevacizumab and XELOX (group B) once every 3 weeks. Stratification factors were PD-L1 expression level, ECOG PS score, and primary tumor site. The primary endpoint was independent radiological review committee (IRRC)-assessed PFS per RECIST 1.1. Secondary endpoints included other efficacy endpoints, safety, pharmacokinetics, biomarker explorations, and quality-of-life assessments. **Results:** In the phase 2 part, by the data cutoff of June 30, 2024, sustained improvements in PFS (16.6 vs. 10.7 months, stratified HR 0.66, 95% CI 0.37–1.19) and DOR (17.7 vs. 11.3 months, stratified HR 0.45, 95% CI 0.20–0.98) were observed for patients in group A compared to group B in the modified intent-to-treat population (n = 112; two patients in group A did not receive any intended study treatment). 32/55 (58.2%) patients in group A and 35/57 (61.4%) in group B have died. Median overall survival (OS) was 25.6 months in group A and 21.2 months in group B (stratified HR 0.86, 95% CI 0.53–1.42). A trend of PFS and OS benefits was similarly observed for the patients with a microsatellite stable (MSS) status (PFS: 16.8 vs. 10.1 months, stratified HR 0.65, 95% CI 0.33–1.29; OS: 23.5 vs. 20.2 months, stratified HR 0.79, 95% CI 0.45–1.38). 39 (70.9%) patients in group A and 34 (59.6%) in group B had grade ≥3 treatment-related adverse events. Serplulimab/placebo-related serious TRAEs occurred in 13 (23.6%) patients in group A and 14 (24.6%) patients in group B. **Conclusions:** With a follow-up duration of 31.0 months, improved survival benefits demonstrated with the addition of serplulimab were maintained in the first-line treatment of mCRC patients including those with MSS status alongside a manageable safety profile. The phase 3 part of this study is currently ongoing to further evaluate serplulimab plus bevacizumab and XELOX as a first-line treatment option in mCRC. Clinical trial information: NCT04547166. Research Sponsor: None.

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# Overall survival with fruquintinib versus placebo after adjusting for subsequent anticancer therapy in patients with refractory metastatic colorectal cancer in the FRESCO-2 study.

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**Background:** Fruquintinib is a highly selective, oral inhibitor of all 3 vascular endothelial growth factor receptors (VEGFRs -1, -2, and -3) that is approved by the US FDA and in the EU, UK, and Japan for previously treated metastatic colorectal cancer (mCRC), regardless of biomarker status. The phase 3 FRESCO-2 study (NCT04322539) met its primary endpoint demonstrating significantly improved overall survival with fruquintinib + best supportive care (BSC) vs placebo + BSC. As patients may have received subsequent anticancer therapy after progression, understanding the impact on overall survival is important. This analysis examined overall survival data from FRESCO-2 after adjusting for subsequent anticancer therapy. **Methods:** In FRESCO-2, patients were randomized 2:1 to receive fruquintinib (5 mg) or matching placebo by mouth, once daily, 3 weeks on, 1 week off, + BSC. Eligible patients had received prior chemotherapy, anti-VEGF therapy, anti-EGFR therapies (if indicated), and TAS-102 and/or regorafenib. We assessed the impact of subsequent anticancer therapy on overall survival by excluding patients who received subsequent therapies, censoring these patients, and determining the causal hazard ratio using inverse probability of censoring weights (IPCW) and marginal structural models (MSM) to adjust for the potential bias introduced by subsequent treatments. **Results:** In FRESCO-2, a lower proportion of patients in the fruquintinib vs placebo arm received subsequent anticancer therapy (29.4% vs 34.3%). Among these patients, the most common subsequent anticancer therapies in the fruquintinib vs placebo arms were fluorouracil (7.7% vs 9.6%) and regorafenib (7.5% vs 7.8%). The overall survival benefit seen with fruquintinib vs placebo in the intent-to-treat population (hazard ratio [HR]: 0.66; 95% confidence intervals [CI]: 0.55–0.80) was improved when patients who had received subsequent anticancer therapy were excluded (HR: 0.45; 95% CI: 0.36–0.57) or censored (HR: 0.63; 95% CI: 0.50–0.79). In addition, the overall survival benefit seen with fruquintinib vs placebo was improved after adjusting for subsequent anticancer therapy using IPCW (HR: 0.43; 95% CI: 0.33–0.55) and MSM (HR: 0.48; 95% CI: 0.38–0.60). Patients receiving fruquintinib had similar rates of grade ≥3 adverse events, whether or not they received subsequent anticancer therapy (51.5% and 67.4%, respectively). **Conclusions:** Consistent with the primary analysis of the FRESCO-2 intent-to-treat population, fruquintinib improved overall survival vs placebo after adjusting for the impact of subsequent anticancer therapy. Furthermore, these analyses are robust with consistent results reported using IPCW and MSM. These findings support fruquintinib as a new treatment option for patients with previously treated mCRC. Clinical trial information: NCT04322539. Research Sponsor: HUTCHMED.

# Survival analysis using neoadjuvant rectal score in patients with locally advanced rectal cancer who underwent surgery after neoadjuvant chemoradiotherapy: An ambispective study.

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**Background:** Colorectal cancer, the third most common malignancy and a major cause of cancer deaths, is treated using a standard protocol of neoadjuvant chemoradiotherapy (NACTRT) or short-course radiotherapy (SCRT) followed by total mesorectal excision (TME) for locally advanced rectal cancers (LARC). Long-term follow-up is essential to determine study endpoints, necessitating surrogate markers for short-term outcomes. This study aims to stratify patients into risk groups based on their Neoadjuvant Rectal Cancer (NAR) score and evaluate its predictive value for oncological outcomes, including overall survival, disease-free survival (DFS), locoregional relapse-free interval (LRFI), and distant metastasis-free interval (DMFI), while also correlating clinicopathological characteristics with these outcomes. **Methods:** This ambispective observational study at the Regional Cancer Centre, Thiruvananthapuram, involved 459 patients with biopsy-proven, locally advanced rectal adenocarcinoma (T3/T4 and/or node positive) who underwent long-course neoadjuvant chemoradiotherapy (NACTRT) followed by curative surgery (Anterior Resection or Abdomino-Perineal Resection) between 2010 and 2015, with a minimum follow-up of three years. Exclusion criteria included previous malignancies, synchronous colorectal cancers, oligometastatic disease, and loss to follow-up. The study aimed to determine the predictive value of the Neoadjuvant Rectal Cancer (NAR) score for oncological outcomes. Clinicopathological details were collected from case records, imaging, and operative notes, and postoperative histopathological records were used to calculate the NAR score. Patients were categorized into three groups based on their NAR scores and followed prospectively for recurrence or death. Data were analyzed using frequency, percentages, mean, standard deviation, Chi-square test, Kaplan-Meier method, log-rank test, and Cox regression analysis, with a significance level set at $p < 0.05$. **Results:** 459 patients were included in the study, with 41.4% females, and 44% being less than 60 years. Most of the patients had lesion 5-10 cm from anal verge (46%). 77% of patients had T3 disease prior to the start of NATCTRT. 51.9% had underwent APR, and 80.4% had open surgery. 11.1% had PCR attained. 27% had disease recurrence of which most were distant metastasis. NAR was a significant predictor of overall survival, disease free survival, distant metastasis free interval and locoregional failure free interval. Other significant predictors of overall survival were pre-NACTRT T stage, pN stage, PLNR, CRM status and PCR, but only NAR was found to be significant on multivariate analysis. **Conclusions:** NAR score is a significant predictor of OS, DFS,LRFI and DMFI and can be used as a short term surrogate for survival data in anticipated trials. Research Sponsor: None.

# A cross-sectional analysis from a real-world cohort of patients with microsatellite instability colorectal cancer (MSI-CRC) with localized disease from the Spanish RETUD registry.

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**Background:** Microsatellite instability (MSI) localized colorectal cancer (CRC) constitutes a subgroup of patients (pts) with different clinical and molecular characteristics and prognosis compared to microsatellite stable (MSS) disease. Here, we present our real world data regarding the management and clinical outcomes of a cohort of MSI-h localized CRC patients included in the Spanish Group of Treatment of Digestive Tumors (TTD) Registry (RETUD). **Methods:** RETUD is a national, multicenter registry for gastrointestinal tumors from the Spanish TTD Group. In this cross-sectional analysis we evaluated a real-world cohort of MSI-h localized CRC pts diagnosed from 1st January 2017 to 29th April 2024. Baseline characteristics and neo/adjuvant treatments are descriptively presented. Disease-free survival (DFS) and overall survival (OS) analyzed by Kaplan-Meier method are presented along with 95% confidence intervals (CI). **Results:** Five hundred and sixty-one (561) evaluable pts out of 679 MSI-CRC included in RETUD had a localized disease as initial diagnosis. Pts had a median age of 73.1 years and were predominantly Caucasian (97.7%) and female (52.6%). Fifty (12.4%) pts had Lynch Syndrome. Main tumor baseline characteristics are described (Table). Local-locoregional therapeutic procedures were: primary tumor resection in 531 (99.1%) pts and radiotherapy in 18 (3.2%) pts. A total of 198 (35.3%) pts received systemic therapy, mostly chemotherapy (n=165 83.3%) but 18 (9.1%) were treated with pembrolizumab due to unresectable disease. At database cut-off, 123 (21.9%) patients died, mostly due to not related intercurrent illness, and 96 (17.1%) pts developed metastasis. After a median follow up period of 28.5 months, the median (95% CI) OS was not reached and DFS was 78.2 (55.4-NA) months. The impact of adjuvant systemic therapy (chemotherapy vs. non-chemotherapy) for both stage II and stage III pts is under evaluation. **Conclusions:** Our work provides valuable real-world data from a cohort of MSI-h localized CRC pts treated under clinical practice conditions in Spain, reiterating the good disease prognosis exhibited by this subset of pts in contrast to MSS pts. Research Sponsor: Bayer; MDS Pharma Services; Roche.

| Main tumor characteristics. | |
|---|---|
| **Stage at initial diagnosis, n (%)** | |
| **I/II n (%)** | 55 (9.8) / 258 (46.0) |
| **III, n (%)** | 248 (44.2) |
| **Location of primary tumor[a], n (%)** | |
| **right colon /left colon /rectum** | 447 (78.4) / 82 (14.4) / 41 (7.2) |
| **Type histological[b], n (%)** | |
| **Intestinal** | 271 (48.9) |
| **Mucinous (colloid) adenocarcinoma (>50% mucinous)** | 115 (20.8) |
| **Signet ring cell carcinoma (>50% signet ring)** | 14 (2.5) |
| **Medullary carcinoma** | 18 (3.2) |

[a]Pts with more than 1 primary tumor; the percentage may be over 100%. [b]Unknown histology in 132 (23.8%) pts and missing data in 7 pts.

# Exploratory study of TAS-102 combined with intermittent administration of fruquintinib in the treatment of third-line metastatic colorectal cancer.

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**Background:** Fruquintinib and TAS-102 are two globally recognized standard therapies for mCRC patients who have previously undergone standard chemotherapy. A phase II exploratory study suggested that the combination of TAS-102 and fruquintinib could prolong progression-free survival (PFS) and overall survival (OS) in these patients. However, the benefits are sometimes limited due to tolerability issues associated with continuous dosing. This study aimed to investigate the efficacy and safety of intermittent administration of TAS-102 combined with fruquintinib as a third-line treatment for mCRC patients. **Methods:** This open-label, single-arm, single-center, exploratory clinical trial enrolled mCRC patients who had failed at least two standard treatment regimens. Eligible patients received oral fruquintinib (3 mg, once daily, on days 1-5 and 8-12) and TAS-102 (35 mg/m$^2$, twice daily, on days 1-5) every two weeks until disease progression or unacceptable toxicity occurred. The primary endpoint was objective response rate (ORR). Secondary endpoints included OS, PFS , DCR, and safety. **Results:** From August 2023 to September 2024, 26 eligible patients were enrolled. The median age of the patients was 58 years (range: 19-77), with 61.5% being female. RAS mutations were present in 57.7% of patients, 57.7% had left-sided colon or rectal cancer, 57.7% had liver metastases, and 76.9% had multiple metastases. As of September 10, 2024, 22 patients had undergone at least one tumor assessment, and 12 patients were still on treatment. Among them, 18.2% (4/22) achieved partial response , and 50% (11/22) had stable disease as the best response. Median PFS (mPFS) was 135 days (95% CI: 60.52-209.48). mPFS in RAS-mutant and RAS wild-type patients was 161 days (95% CI: 63.37-258.63) and 135 days (95% CI: 60.20-209.80), respectively, with ORRs of 25% and 12.5%. mPFS in patients with liver metastases (LM) and non-LM was 135 days (95% CI: 45.42-224.58) and 161 days (95% CI: 87.84-234.16), respectively, with ORRs of 14.3% and 25%. mPFS in patients with multiple metastases was 135 days (95% CI: 71.86-198.14), with an ORR of 17.6%. Median OS was not yet mature. Treatment-related adverse events (TRAEs) were generally manageable and tolerable. The most common hematological TRAEs (any grade, grades 3-4) included leukopenia (76.9%, 11.5%), neutropenia (73.1%, 11.5%), and anemia (53.8%, 7.7%). Non-hematological TRAEs were mostly grades 1-2, including anorexia (46.2%) and fatigue (19.2%). One case of grade 3 hypertension was reported. No treatment-related deaths were observed. **Conclusions:** These preliminary results suggest that intermittent administration of TAS-102 in combination with fruquinitinib as a third-line treatment for patients with mCRC reduces hematological toxicity and is well tolerated, with encouraging clinical results. Clinical studies are ongoing. Clinical trial information: ChiCTR2300078241.
Research Sponsor: None.

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# A phase 1 dose-escalation study of GCC19CART: A novel CAR T-cell therapy for metastatic colorectal cancer in the United States.

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**Background:** GCC19CART, the first clinical candidate from the CoupledCAR solid tumor platform, pairs solid tumor CAR T-cells with CD19-targeting CAR T-cells to enhance proliferation, activation, and persistence against solid tumors. GCC19CART targets guanylate cyclase-C (GCC), a known mucosal marker. Building on the promising results from a clinical trial in China, a phase 1 study was initiated in the US for patients with metastatic colorectal cancer (mCRC) refractory to standard chemotherapy. As of August 14, 2024, a total of 9 subjects have been treated. **Methods:** Subjects were initially screened for GCC expression, however screening was discontinued after >95% of subjects with mCRC demonstrated GCC expression by immunohistochemistry (95/98). Eligible subjects underwent leukapheresis, lymphodepleting chemotherapy (fludarabine 30mg/m$^2$ and cyclophosphamide 300mg/m$^2$ on day-3), and then infusion of a single dose GCC19CART. Objective response was assessed by RECIST v1.1. **Results:** Four subjects have been enrolled at dose level 1 (1x10$^6$ cells/kg) and 5 subjects at dose level 2 (2x10$^6$ cells/kg). The most common adverse events were cytokine release syndrome (CRS) in 9/9 subjects (Grade 1: 6/9 [66.7%] and Grade 2: 3/9 [33.3%]) and diarrhea in 8/9 subjects (Grade 1: 3/9 [33.3%], Grade 2: 3/9 [33.3%], Grade 3: 2/9 [22.2%]). Immune effector cell-associated neurotoxicity syndrome (ICANS) occurred in 2/9 subjects (Grade 2: 1/9 [11.1%] or Grade 3: 1/9 [11.1%]) and resolved rapidly with corticosteroid and dasatinib treatment. Three out of 9 subjects had a mild tremor (Grade 1) that resolved with corticosteroids. One subject experienced a dose-limiting toxicity (DLT) at dose level 2, presenting with Grade 3 diarrhea, Grade 4 enterocolitis and Grade 5 sepsis, and died from fungal sepsis 48 days post-infusion. At dose level 1, the overall response rate (ORR) was 50.0% (2/4). Two subjects demonstrated a partial response (PR) with one confirmed PR, while 1 additional subject had partial metabolic response on PET/CT with stable disease (SD) and 1 patient had progressive disease. At dose level 2, the ORR was 80% (4/5), including 1 subject with a pathological complete response, and 3 with a confirmed partial response, of which 1 had a complete metabolic response on PETCT. **Conclusions:** Preliminary results indicate that GCC19CART exhibits significant clinical activity in patients with refractory mCRC. Adverse events were related to GCC19CART mechanism and in line with what has been observed with other CAR T-cell products. Strategies to further optimize the management of treatment-related diarrhea are currently being implemented. The trial is ongoing and updated data will be presented. Clinical trial information: NCT05319314. Research Sponsor: None.

# A phase Ib study of immunotherapy with ex-vivo pre-activated and expanded CBNK cells in combination with cetuximab, in patients with colorectal cancer (CRC) with minimal residual disease (MRD): Final report.

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**Background:** Colorectal cancer (CRC) is the second most common type of cancer. Even following successful curative treatment, patients often relapse, likely because of undetected micro-metastatic foci. Tumor-informed ctDNA testing has helped to identify those patients before they developed radiographic evidence of disease, introducing the concept of minimal residual disease (MRD) in solid tumors. No standard-of-care treatment options are available for MRD patients. Therefore, we have designed a phase IB trial of the combination of *in vitro* pre-activated and pre-expanded cord blood natural killer (pre-A+E CBNK) cells with cetuximab. We hypothesized that we could achieve CAR-like efficacy without genetic engineering, by pre-activating NK cells with inflammatory cytokines and by using antibodies to redirect NK cell specificity and antibody-dependent cellular cytotoxicity. The highly immunosuppressive CRC tumor microenvironment is understudied, and novel NK-cell therapy could be highly effective in MRD clearance. **Methods:** 15 patients with CRC-MRD were enrolled in this trial and completed treatment. All the patients received immunodepleting treatment prior to an infusion of pre-A+E CBNK cells, up to a dose of 1x10^8/kg. Patients were monitored inpatient for 24 hours after treatment for infusion-related reaction, and were then seen for toxicity follow-up in the outpatient clinic twice/week for 4 weeks (DLT window). Samples for correlative analysis were collected at different times. **Results:** 8/15 patients enrolled were female and 7 were male, age range 26-69. They all had no measurable radiographic evidence of disease and a positive ctDNA Signatera test at baseline. 11/15 were previously treated for metastatic disease; 13 patients had a history of metastases (including 7 liver, 2 lymph nodal, 2 peritoneal). All of them received 5-FU-based chemotherapy. 7/15 were RAS/RAF WT, 4/10 were KRAS mut, 1/10 had BRAF mutation, 1/10 had HER2 amplification. They were all pMMR. No DLT was observed, only 1 patient had G2 CRS requiring tocilizumab. 9/15 patients treated so far had ctDNA clearance at some point. One patient had negative ctDNA at 3 and 6 months, 4/15 were negative at d28 and 7/15 at d14. 4/15 patients have not met the 90-day time point yet. 10/12 patients analyzed up to the primary endpoint had decreased ctDNA over baseline CBNK cells in blood peaked 3 hours post-infusion and persisted at 3 days. Phenotype analysis of CBNK pre-post, donor NK and other immune cells is ongoing. **Conclusions:** Cetuximab in combination with pre-A+E CBNK is safe and showed promising results for the treatment of MRD-CRC. Clinical trial information: NCT050468.
Research Sponsor: MD Anderson Moonshot Program, K12 Paul Calabrese and ASCO CDA Conquer.

# OP-TNT: Phase II study of total neoadjuvant chemoradiotherapy followed by local resection for early distal rectal carcinoma.

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**Background:** Surgery is standard of care for pMMR early distal rectal cancer (RC), which can be partially exempted by totally neoadjuvant chemoradiotherapy (TNT) followed by watch and wait. But long-term data showed that 1/3 patients had local growth( mostly in the rectal wall)in two years. This study investigates the efficacy and safety of TNT followed by local excision( LE) for early distal rectal carcinoma (cT1-3N0M0) in order to preserve organs. **Methods:** In this single-arm, phase 2 prospective trial, patients were eligible if they were aged 18 years or older, with a performance status of 0-1,newly diagnosed, biopsy proven pMMR/MSS RC which located ≤5cm from the anal verge and staged as cT1-3N0M0. Patients were assigned to receive TNT: 6 courses of CapOx(capecitabine 1000 g/m2 PO BID D1-14, oxaliplatin 130 mg/m2 IV QD D1,Q3W), combined with concurrent long-course radiotherapy (LCRT)(45 Gy/25 Fx to the pelvis with an SIB of 50 Gy/25 Fx to the tumor bed). Evaluation was performed 12 weeks after completion of LCRT. Patients who achieved a clinical complete response (cCR) or near clinical complete response (ncCR)were assigned to LE, whereas those with >ypT1, positive margin, or neurovascular invasion after LE were recommended for total mesorectal excision (TME). The primary endpoint is the 3-year organ-preserving rate. Secondary endpoints include the pathological complete response (pCR)(encompassing response after LE and TME), adverse effects rate, 3y-disease-free survival (DFS)and 3y-overall survival (OS)rate. **Results:** From October 2022 to August 2024, 52 patients had completed therapy. Median age at evaluation and follow-up duration were 58.3 years and 11 months. Of the patients 78.8% (41/52) were defined as MRI stage T3, and 76.9%(40/52) located ≤3cm from the anal verge. Of 52 evaluable patients ,82.7% (43/52) achieved cCR/ncCR and had LE (6 via endoscopic resection and 37 through trans-anal minimally invasive surgery). Among these patients, 25.6% (11/43) were found to have residual tumors after LE, of which 6 were radical-operated while 5 refused this intervention. Nine patients who did not achieve cCR or ncCR received TME resection, and 6 of them were confirmed as pCR. The total pCR rate was 73.1% (38/52). Of all patients, 3 underwent abdominoperineal resection, and sphincter-preserving rate was 94.2%(49/52). No grade 3 or 4 adverse events were reported, and there were no treatment-related deaths. The main adverse event after local resection was the 1-3 months anterior resection syndrome. **Conclusions:** The results of interim analysis showed that tumor residue was found in patients with cCR/ncCR after TNT, similar to the regrowth rate previously reported. LE strategy may become prevention. Long-term outcomes are being followed up. Clinical trial information: NCT05563922. Research Sponsor: None.

# A cross-sectional analysis from a real-world cohort of patients with microsatellite instability colorectal cancer (MSI-CRC) with metastatic disease from the Spanish RETUD registry.

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**Background:** Survival of microsatellite instability (MSI) metastatic colorectal cancer (mCRC) patients (pts) has remarkably increased with immune checkpoint inhibitors (ICI). Here, we present our real-world data regarding the management and clinical outcomes of a cohort of 214 MSI-mCRC patients included in the Spanish Group of Treatment of Digestive Tumors TTD Registry (RETUD). **Methods:** RETUD is a national, multicenter registry for gastrointestinal tumors from the Spanish TTD Group. In this cross-sectional analysis we evaluated a real-world cohort of MSI-mCRC pts diagnosed from 1st January 2017 to 29th April 2024. Baseline characteristics, treatments and treatment response are descriptively presented. Tumoral response was evaluated according to RECIST 1.1 criteria. First line progression-free survival (PFS) and overall survival (OS) analyzed by Kaplan-Meier method are presented along with 95% confidence intervals (CI). **Results:** Among 679 MSI-CRC pts included in RETUD, a total of 214 mCRC pts were evaluable. Pts' age at diagnosis was 69.6 (26-96) years, and they were predominantly Caucasian (97.7%) and female (53.7%). Eastern Cooperative Oncology Group (ECOG) performance status was 0-1 for 168 (78.5%) pts. Seventeen (10.7%) pts presented Lynch syndrome. Main tumor biological characteristics are described at Table 1 and the molecular profile (when available) was: KRAS mutation (24.3%), NRAS mutation (3.5%) and $BRAF^{v600E}$ mutation (53.0%). Surgical resections: 158 (73.8%) pts for primary tumor and 42 (19.6%) pts for metastasis. First line systemic treatment was administered to 187 (97.1%) pts: 92 (49.2%) pembrolizumab, 83 (44.4%) chemotherapy (CT), 4 (2.1%) other ICIs, 8 (4.3%) not reported. With a median follow up period of 17.4 months, the median (95% CI) OS and first-line PFS of the total mCRC population were 32.6 (22-72.4) and 11.1 (8.2-17.9) months (m), respectively. In immunotherapy (IT) pts mOS was not achieved (22.0-NA) and PFS was 26.5 m (12.9-NA) while the mOS was 28.9 m (16.3-69.3) and PFS 7.5 (5.4-9.3) m in CT pts. The overall response rate (ORR) was 54.2% in IT pts vs. 37.3% in CT pts. **Conclusions:** The introduction of IT has changed the evolution of MSI-H mCRC management, offering a beneficial impact in the OS and PFS survival in contrast to other conventional therapies in a real-world context. Research Sponsor: Bayer; MDS Pharma Services; Roche.

| Tumor characteristics at initial diagnosis of CRC. | |
|---|---|
| **Stage at initial diagnosis of CRC** | |
| **I/II, n (%)** | 4 (1.9) / 26 (12.1) |
| **III/IV, n (%)** | 66 (30.8) / 118 (55.1) |
| **Location of primary tumor[a], n (%)** | |
| **right colon /left colon /rectum** | 160 (74.8) /37 (17.3) /19 (8.9) |
| **Main metastatic location[b], n (%)** | |
| **Liver** | 84 (39.3) |
| **Peritoneal** | 76 (35.5) |
| **Lymph** | 71 (33.2) |
| **Lungs** | 41 (19.2) |

[a]Pts with more than 1 primary tumor; the percentage may be over 100%. Data missing for 4 (1.9%) pts. [b]Pts with more than 1 metastatic site; the percentage may be over 100%.

**179** **Poster Session**

# Neoadjuvant immunotherapy for locally advanced/metastatic mismatch repair deficient colorectal cancer: A two-year institutional experience.

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**Background:** MMRd is a tumour agnostic biomarker predictive of response to immune checkpoint inhibitors (ICIs). Given the poor response to 5-FU based chemotherapy, ICIs are now being explored in early MSI-H CRC. The outcomes on this strategy are limited. **Methods:** We retrospectively evaluated early outcomes for patients receiving neoadjuvant ICIs for unresectable locally advanced/oligometastatic MMRd colorectal cancer (CRC), where intention of treatment was downstaging to permit curative resection. Following discussion at regional MDT, suitable patients were administered a PD-1 inhibitor 6-weekly until disease progression or improvement rendering suitability for surgical resection (to max of 2 years). **Results:** From October 2022-September 2024, ten patients with MMRd CRC were commenced on neoadjuvant ICIs. These comprised six right, one left sided colonic and three rectal tumours. Median age at diagnosis was 59 (IQR54–68). Four patients had family history of CRC. One patient had clinical stage II, seven stage III and two patients oligometastatic stage IV disease. All were patients with initially unresectable disease, with potential for curative resection pending response to ICI. Histologically, two tumours were poorly differentiated, two mucinous phenotype and six moderately differentiated adenocarcinomas. BRAF mutation was identified in three patients. None had KRAS mutation. Three rectal cancers received radiotherapy in addition to immunotherapy. Five required a defunctioning stoma either prior to or on ICI. Median follow-up was 12.5 months (IQR7-17). Objective response rate by RECIST 1.1 was 100%, with significant downstaging in 9 of 10 patients. The remaining patient is early in treatment course with no reimaging. To date, complete clinical response (cCR) has been observed in three patients (33%). All 10 patients are alive. Grade 3-4 treatment-related adverse events occurred in two patients (20%) who developed treatment-related strictures (one also had tumour perforation). This was successfully managed with defunctioning ileostomy and antibiotics, with subsequent cCR. **Conclusions:** We report impressive early outcomes of immunotherapy for locally advanced/metastatic MMRd CRC, with excellent downstaging and high predicted cCR. This highlights the importance of screening all CRC for MSI-H/MMRd. In the setting of initially unresectable CRC, an ICI-first approach can render patients eligible for surgery, however deep responses can result in local effects including strictures and/or perforation. Research Sponsor: None.

# A phase 1 trial of folinic acid, fluorouracil, oxaliplatin, bevacizumab, botensilimab, balstilimab (FOLFOX-3B) in microsatellite stable metastatic colorectal cancer.

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**Background:** Botensilimab (BOT) is a novel Fc fragment engineered CTLA4 inhibitor designed to boost innate and adaptive immune response. The combination of BOT and the PD-1 antibody balstilimab (BAL) has been associated with durable and deep responses in microsatellite stable (MSS) metastatic colorectal cancer (mCRC). Given the potential synergistic activity between checkpoint inhibitors and oxaliplatin and VEGF(R) inhibitors, we performed a phase 1 clinical trial of folinic acid, fluorouracil, oxaliplatin, bevacizumab, botensilimab, and balstilimab (FOLFOX-3B). **Methods:** We enrolled ECOG performance status 0-1 patients (pts) with measurable MSS mCRC, up to 2 prior lines of therapy, and without progression following prior oxaliplatin. FOLFOX was given at a fixed dose, every 2 weeks, across the study: folinic acid at 400 mg/m$^2$ x 2 hours (hr), oxaliplatin at 85 mg/m$^2$ x 2 hr, and fluorouracil at 2400 mg/m2 x 46 hrs. BOT was administered at 2 dose levels (DL) of 25 mg and 75 mg IV Q6 weeks x 2. BAL was administered at a fixed dose of 240 mg IV Q2 weeks. The study followed a 3 x 3 escalation design with up to 9 patients per DL. A dose level was deemed safe if $<$ 1 out of 6 pts or $<$ 2 out of 9 pts had a DLT (expansion to 9pts only if 2 DLTs encountered in the 1$^{st}$ 6 pts). The DLT period consisted of 6 weeks. Pts should have received all intended BOT/BAL and FOLFOX doses in the 1$^{st}$ 6 weeks to be deemed DLT-evaluable. **Results:** 14 pts were enrolled on study: 9/14 with liver metastatic disease, 5/14 *RAS*-mutated, all TMB low. 7 were treated on DL1 and 7 on DL2. 6/7 pts at DL1 had received 1-2 prior lines of therapy, 4 of whom included oxaliplatin. 6/7 of pts at DL2 had 1-2 prior systemic therapy, all of whom received prior oxaliplatin. No DLT were noted at DL1 or DL2. 1 pt at each DL was not DLT-evaluable as they were not able to receive all intended therapy due to treatment interruptions. Grade (G)2 and above immune related toxicities included: one pt at DL1 with transient G3 AST/ALT elevation and transient G2 colitis (resolved with steroids and infliximab), one pt at DL1 with G2 hypothyroidism and one patient at DL2 with G2 hyperthyroidism. At DL1, 4/7 had a partial response (PR) - including 2/4 with prior oxaliplatin exposure. At DL2, 6/7 pts had a PR – including 5/6 with prior oxaliplatin exposure. 3 patients proceeded to surgical resection or ablation and are currently followed with observation only. In the combined cohort of DL1 and DL2, the median progression free survival (PFS) is 8 months and remains unreached for DL2. Overall survival outcome has not matured at the time of this analysis. **Conclusions:** FOLFOX-3B with BOT at 75 mg IV Q6 weeks x 2 in combination with BAL 240 mg Q2 weeks is well-tolerated and is associated with promising clinical activity with 5/6 PR and prolonged PFS. The study was amended to explore additional DL that explore higher doses of BOT (150 mg) or longer duration of BOT 75mg (4 doses). Clinical trial information: NCT05627635. Research Sponsor: Agenus.

**181** **Poster Session**

# The role of stereotactic radiotherapy in brain metastases from gastrointestinal cancer.

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**Background:** The incidence of brain metastases from Gastrointestinal primary tumors (GI-BMs) is approximately of 6%. Appropriate management of these patients, who often presents at advanced disease stage, is poorly defined. Stereotactic Radiotherapy (SRT) could be proposed to improve symptoms and extend disease control. The aim of this study is to assess the impact of SRT in this population. **Methods:** Data from consecutive patients treated for GI-BMs with SRT from March 2015 to March 2024 were retrospectively collected. Dose was expressed as Equivalent Dose in 2Gy Fractions (EQD2). lntra-Cranial Control (IC) and Overall Survival (OS) were calculated from date of SRT to event (intracranial relapse and death respectively) or last follow-up. **Results:** Fifty-four patients (median age 68 years, range 46-84) accounting for 98 GI-BMs were included. Primary tumors were colorectal and gastro-oesophageal adenocarcinoma in 37 (68.5%) and 17 (31.5%) patients respectively. Eighteen (33%) patients presented with a single GI-BM. Extra-cranial disease was present in 41 (76%) patients, including unresected/relapsing primary tumor in 9 (17%) cases. Detection of BMs occurred at diagnosis in 8 (15%) patients, as first site of relapse in 27 (50%) or as progression of known metastatic disease in 19 (35%) patients. Before SRT, surgery and/or WBRT were performed in 10 (19%) and 5 (9%) patients respectively. For each SRT course a median of 1 (range 1-5) GI-BMs was treated. Dose regimens consisted of 12-30 Gy in 1-5 fractions, to a median EQD2 of 50 $Gy_{10}$ (range 22-68). A second SRT course for out-of-field intracranial progression was performed in 7 patients (13%). First- and further chemotherapy lines were administered in 39 (72%) and 15 (28%) patients respectively. IC rate was 88% at 6 months and 70% at 1 year. At univariate analysis (UVA) no variables were correlated with IC. Median OS was 11 months (IC 95% 9-16), 6-months and 1-year OS rate was respectively 76% and 48%. Only systemic treatment beyond first line was correlated with OS at UVA (4 versus 15 months, p=0.02). Two (4%) patients developed mild symptomatic radionecrosis. **Conclusions:** Onset of GI-BMs can occur at any stage of disease. Despite poor overall outcome, selected patients may benefit from SRT to improve IC, particularly in association with first-line systemic treatment. Multiple SRT courses can be delivered in case of further intracranial recurrence. Symptomatic radionecrosis may occur in less than 5% of cases. Research Sponsor: None.

# Phase I/II trial of encorafenib, cetuximab, and nivolumab in microsatellite stable *BRAF*$^{V600E}$ metastatic colorectal cancer following progression on prior BRAF+EGFR targeted therapies.

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**Background:** A *BRAF*$^{V600E}$ mutation, present in approximately 8-10% of all colorectal cancers (CRC), is a poor prognostic biomarker for patients with metastatic CRC. The BRAF$^{V600E}$ inhibitor encorafenib (E) and the anti-EGFR antibody cetuximab (C) are approved for refractory, *BRAF*$^{V600E}$ mCRC based upon on overall response rate (ORR) of 20% and median progression-free survival (PFS) of 4.2 months. A single-arm trial adding the anti-PD1 antibody nivolumab (N) to E+C in the same (BRAF-inhibitor naïve) population showed promising efficacy, with an ORR of 50% and median PFS of 7.4 months. We evaluated the efficacy of E+C+N in patients with microsatellite stable (MSS), *BRAF*$^{V600E}$ mCRC who had progressed on prior E+C. **Methods:** Patients with MSS, *BRAF*$^{V600E}$ mCRC with documented prior progression on E+C were eligible for treatment with E (300 mg PO daily), C (500 mg/m$^2$ q14 days), and N (480 mg IV q28days) were included in the study. Primary endpoint was best overall response by RECIST 1.1. Secondary endpoints were PFS, overall survival (OS), and toxicity (by CTCAE v5). Circulating tumor DNA assessment of molecular alterations in the MAPK pathway in association with response were performed by NGS. Descriptive statistics were used to summarize outcomes. **Results:** Among the 12 participants, none achieved a radiographic response, and 5 experienced stable disease. Disease control rate was 41.7% (95% CI, 13.8-69.6). Median PFS was 3.7 months (95% CI, 1.7-5.7), and median OS was 8.3 months (NE-NE). Four patients (16.7%) had concomitant MAPK-activating alterations at baseline, and *APC*, *TP53*, and *PIK3CA* mutations were present in 25.0%, 66.7%, and 33.3% of patients, respectively. The presence of MAPK-activating mutation was not associated with disease progression (odds ratio 1.3; 95% CI 0.11-16; p=0.82). The most common grade 1-2 treatment-related adverse events (AEs) were anemia (N=5), arthralgia (N=5), rash (N=4), headache (N=4), and fatigue (N=3); 1 of the 12 patients experienced a grade 3 small bowel obstruction. **Conclusions:** In this pilot study, addition of N to E+C did not appear to overcome resistance to prior BRAF + EGFR targeted therapies in patients with MSS, *BRAF*$^{V600E}$ mCRC. While the addition of immunotherapy has demonstrated promising signal in early-phase clinical trials for this population, the benefit appears unlikely when offered sequentially after progression on BRAF combination therapies. Clinical trial information: NCT04017650. Research Sponsor: None.

# Safety, tolerability, and preliminary efficacy of tinodasertib as a monotherapy or in combination with pembrolizumab or irinotecan in metastatic colorectal cancer: Interim results from a phase II open-label, dose-finding, run-in and cohort expansion study.

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**Background:** MNK inhibition has been shown to downregulate phosphorylation at serine 209 of eIF4E, a potent regulatory point of CAP-mediated translation of 5' polyadenylated mRNA associated with growth factor and pro-oncogenic signals in cells. Elevated levels of eIF4E phosphorylation are observed in a broad range of tumors. MNK1/2 knock out mice are healthy and viable, as cellular house-keeping mRNA translation occurs readily using the IRES mechanism, whereas mRNA with more complex and longer 5' untranslated ends such as those involved in growth factor and pro-oncogenic signaling, are MNK activated. Cells from MNK knockout animals become relatively resistant to subsequent oncogenic transformation. In a program of SAD and MAD Phase I studies conducted in normal healthy volunteers, Tinodasertib was found to be safe and well tolerated with no dose-limiting toxicity (DLT). The objectives of this ongoing Phase 2 study are to evaluate safety, and preliminary efficacy of Tinodasertib as monotherapy and in combination with either pembrolizumab or irinotecan in patients with advanced colorectal cancer (CRC). **Methods:** Patients had to have advanced CRC and previously received ≥2 lines of therapy. The dose escalation phase of the study was open to all patients with CRC. As of 23 July 2024, 22 patients were dosed in a modified 3X3 dose escalation design with Tinodasertib from 20 to 80 mg on alternate days. Out of 22 patients, 12 received monotherapy, 4 were treated with Tinodasertib and Irinotecan and 6 were treated with Tinodasertib and pembrolizumab. Majority (19) had MSS CRC and 3 had unknown status. Nine patients had KRAS-mut CRC. **Results:** No DLTs were observed and MTD was not reached. Grade 3 treatment-related adverse events (TRAEs) were observed in 2 (9%) patients and were related to irinotecan while no Grade 3 AEs were attributed to Tinodasertib by either the investigator or the sponsor. Most common TRAEs were related to gastrointestinal system organ class. There were no Grade 4-5 TRAEs. Of the 22 patients, 18 were evaluable for RECIST 1.1 response. No patient had an objective response to treatment, 12 had stable disease with disease control rate of 67% and a progression free survival of 2.99 months. Patients remained on therapy for up to 28 weeks. Overall survival (OS) at 52 weeks was 52% (CI 29 to 93). **Conclusions:** Tinodasertib either as monotherapy or combined with irinotecan or pembrolizumab, was well tolerated with no DLTs at the dose levels evaluated. Prolonged median time to progression and OS compared to historical controls were observed even during the dose escalation phase for the study population as a whole and for each of the monotherapy and combination arms. Enrolment in the dose escalation phase continues. Clinical trial information: NCT05462236. Research Sponsor: AUM Biosciences Pte. Ltd.

# Rechallenge with epidermal growth factor receptor inhibitors for metastatic colorectal cancer: A systematic review and meta-analysis.

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**Background:** Clonal evolution is a mechanism of treatment resistance against epidermal growth factor receptor inhibitors (EGFRi) in metastatic colorectal cancer (mCRC). Upon EGFRi discontinuation, EGFRi-resistant subclones decay with time, which allows for EGFRi rechallenge. We conducted a meta-analysis to investigate the efficacy of EGFRi rechallenge for patients with mCRC. **Methods:** Clinical trials and cohort studies investigating the efficacy of EGFRi rechallenge were extracted from PubMed and Embase from inception to January 31, 2024. Hazard ratios (HR) of overall survival (OS) and progression-free survival (PFS) were aggregated using random-effects models. **Results:** Among 179 studies initially retrieved from the databases, 23 studies (665 patients) were included in the final analysis. There were 11 single-arm trials, 5 randomized controlled trials (including 3 post-hoc analyses), and 7 observational studies. At least 444 patients (66.8%) had left-sided mCRC. In 19 studies, EGFRi rechallenge was given only for patients previously progressing on 5-fluoropyrimadine, oxaliplatin and irinotecan. RAS/RAF mutation status was assessed with circulating tumor DNA (ctDNA) or tissue-based sequencing right before EGFRi rechallenge in 13 studies. Ten clinical trials only enrolled patients with RAS/RAF-wild type mCRC for EGFRi rechallenge. The median EGFRi-free interval before EGFRi rechallenge was ≧4 months in 13 studies (≥10 months in 9 studies) and between 2 to 4 months in 3 studies. Thirteen studies reported an objective response rate (ORR) up to 70%. Eight out of the 13 studies (61.5%) reported an ORR >10% (7 studies with ORR >20%), which is numerically higher than that achieved by FDA-approved third-line agents for mCRC (ORR <10%). EGFRi rechallenge was associated with significantly longer pooled OS (HR 0.62; 95%CI, 0.42-0.92; $I^2$=51.9%) and pooled PFS (HR 0.55; 95%CI, 0.38-0.79; $I^2$=5.5%) than non-EGFRi systemic therapy. Patients without detectable RAS/RAF mutations by ctDNA right before EGFRi rechallenge had a significantly longer OS (HR, 0.39; 95%CI, 0.23-0.66; I2=32%) and PFS (HR, 0.41; 95%CI, 0.29-0.59; I2=0%) than those with detectable RAS/RAF mutations by ctDNA. Acneiform rash was the most commonly reported non-hematologic adverse event (up to 38.9%). No treatment-related deaths were reported. **Conclusions:** EGFRi rechallenge is associated with significantly longer OS and PFS than conventional systemic therapy and a clinically meaningful ORR in heavily treated patients with mCRC, particularly for patients with RAS/RAF-wild type mCRC tested right before EGFRi rechallenge. Research Sponsor: None.

185 Poster Session

# Neoadjuvant PD-1 blockade in mismatch repair-deficient or microsatellite instability-high, locally advanced and resectable colorectal cancer: A retrospective real-world analysis.

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**Background:** Neoadjuvant PD-1 blockade therapy has shown encouraging efficacy in patients with mismatch repair- deficient (dMMR)/microsatellite instability-high (MSI-H) locally advanced colorectal cancer. However, data from the real-world setting are scarce. **Methods:** We retrospectively collected clinical data of patients with dMMR/MSI-H locally advanced colorectal cancer diagnosed between 2022-2024 that received neoadjuvant anti-PD-1 monotherapy at the First Affiliated Hospital of Zhengzhou University. Pathological complete response, clinical complete response, R0 resection rate, and safety were analyzed. All statistical analyses were conducted using SPSS Statistics, version 22.0. **Results:** A total of 45 patients were enrolled. As of the data cutoff (Sep 15, 2024), the median follow-up was 14.0 months (range: 4.0-31.4). The median age was 53 years (range: 29.0–78.0) and 25 (55.6%) of 45 patients were male. 32 (71.1%) patients had clinical stage III disease and 39 (86.7%) patients had colon cancer. A higher proportion of patients (53.3%) had clinical T4 stage. 37 patients underwent the surgery of whom 29(78.9%) had a pathological complete response. The R0 resection was 100%. The median treatment duration was 4 cycles (range: 2-10). Among the 18 patients who received equal to or less than 4 cycles, the pCR rate was 77.8%, while for the remaining 17 patients who had more than 4 cycles of treatment, the pCR rate was 84.2%. Eight other patients had a clinical complete response and chose the watch and wait strategy. Treatment-related adverse events ≥3-4grade were observed in 3 patients. All patients survived without any disease recurrence. **Conclusions:** In this retrospective analysis, anti-PD-1 monotherapy is effective and tolerable for patients with dMMR/MSI-H locally advanced colorectal cancer. This study shows that pCR rate increases is associated with a longer treatment cycles probably. Further analysis of larger real-world datasets with longer follow-up will be essential to validate these results. Research Sponsor: None.

# Immunotherapy for MSI-H colorectal cancer with isolated peritoneal metastases.

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**Background:** NCCN guidelines recommend immunotherapy for MSI-H colon cancer (CC) with isolated peritoneal metastases (iPM); however, limited data support its benefit. This study aimed to evaluate the impact of immunotherapy in this population. **Methods:** Patients diagnosed with MSI-H CC with iPM from 2016-2019 were selected from the NCDB. Patients who received immunotherapy were compared to those who did not. Overall survival was assessed using Kaplan-Meier analysis and multivariate Cox proportional hazard regression. The trend of immunotherapy utilization from 2016-2020 was analyzed using the Cochran-Armitage test. **Results:** We analyzed a cohort of 444 patients, with 161 (36%) receiving immunotherapy. Patients in the immunotherapy group were, on average, younger (63 vs. 71 years; $P<0.001$), with balanced gender distribution between groups. Among those who received immunotherapy, 70% were treated as a first-line agent, and 30% after chemotherapy. Immunotherapy was more commonly administered in the adjuvant setting (66%) compared to neoadjuvant (7%) or without surgery (27%). Immunotherapy was associated with significantly longer median survival (33.6 vs. 14.9 months; $P<0.001$). In multivariate analysis, immunotherapy was associated with improved overall survival, whether given as a first-line agent (HR 0.54, 95% CI 0.39-0.74; $P<0.001$) or second-line after chemotherapy (HR 0.59, 95% CI 0.37-0.94; P=0.03). Additionally, primary tumor resection (HR 0.44, 95% CI 0.32-0.60; $P<0.001$), but not metastatectomy (HR 0.96, 95% CI 0.71-1.29; P=0.8), was associated with prolonged survival. Patients who received both immunotherapy and primary tumor resection (n=118) had a median survival of 43.8 months. From 2016 to 2020, the utilization of immunotherapy in this cohort increased from 29% to 37% ($\chi^2=0.23$, P=0.63). **Conclusions:** Utilization of immunotherapy remains low in MSI-H CRC with iPM and did not change significantly during the study period. Immunotherapy is associated with improved overall survival and should be considered for appropriate candidates. The combination of primary tumor resection and immunotherapy yielded the best overall survival. Research Sponsor: None.

# Optimal duration of neoadjuvant chemotherapy prior to CRS±HIPEC for colorectal cancer: An assessment of survival and postoperative outcomes.

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**Background:** Cytoreductive surgery (CRS) with or without heated intraperitoneal chemotherapy (HIPEC) represents a viable treatment option for select patients with colorectal cancer peritoneal metastases. Given high recurrence rates after surgery, it is common practice to administer neoadjuvant chemotherapy (NAC). The optimal duration, however, when considering oncologic value and postoperative outcomes, is not known. **Methods:** A single institution database (2009-2024) of colorectal cancer patients that underwent CRS±HIPEC was reviewed. Patients with colorectal cancer undergoing CRS±HIPEC for curative intent (completeness of cytoreduction 0 or 1) with known NAC duration were included. Analysis was stratified by NAC duration of 0-3 months or greater than 3 months. Co-primary outcomes were recurrence free survival (RFS) and overall survival (OS). Secondary outcomes included clinically significant complications, defined as Grade 3 or greater adverse events based on Common Terminology Criteria for Adverse Events Version 5.0. **Results:** From 2009-2024, 108 patients underwent CRS±HIPEC for colorectal cancer. Final analysis included 84 patients that underwent curative intent surgery and had a known duration of NAC ( > 3 months NAC: 55 patients, 0-3 months NAC: 29 patients). Median peritoneal cancer index (PCI) score was similar between the two cohorts ( > 3 months: median PCI 11 versus (vs) 0-3 months: median PCI 12; p = 0.42). Recurrence-free survival was 9 months for the entire cohort and not significantly different based on duration of NAC ( > 3 months: 8 months vs 0-3 months: 15 months, p = 0.14). Overall survival was also similar between both cohorts ( > 3 months: 26 months vs 0-3 months: 37 months; p = 0.11). On univariate analysis, > 3 months of NAC was associated with an increased rate of complications of any severity ( > 3 months: 83.6% vs 0-3 months: 55.2%; p = 0.01), clinically significant complications ( > 3 months: 54.5% vs 0-3 months: 31.0%; p = 0.07), and increased median length of stay ( > 3 months: 10 days vs 0-3 months: 8 days; p = 0.01). When controlling for other perioperative variables on multivariable analysis, > 3 months of NAC trended towards an increased risk of clinically significant complications (HR 1.76, 95% CI 0.59-5.30; p = 0.32). **Conclusions:** Even with a similar disease burden, as measured by PCI, an extended duration of neoadjuvant chemotherapy prior to CRS±HIPEC does not appear to be associated with any improved recurrence-free or overall survival. Conversely, postoperative outcomes are worse with higher complication rates and increased length of stay. Thus, despite high recurrence rates after CRS±HIPEC for this disease, it still appears that in well selected patients, an extended duration of neoadjuvant chemotherapy may not be the best treatment strategy. Research Sponsor: None.

**189** **Poster Session**

# TAS102 in combination with oxaliplatin (TASOX) for refractory metastatic colorectal cancer.

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**Background:** TAS102 (trifluridine/tipiracil hydrochloride) is approved for use alone or with bevacizumab in late line metastatic colorectal cancer (mCRC), with overall survival benefit in phase 3 trials (RECOURSE and SUNLIGHT). Oxaliplatin is frequently reintroduced after progression, resolution of limiting neuropathy, or disease recurrence post adjuvant therapy. In preclinical studies, oxaliplatin combined with TAS102 has shown synergism. We hypothesized TASOX may be an option for patients who have progressed/recurred after FOLFOX. **Methods:** We assessed safety and efficacy of TAS102 with oxaliplatin, enrolling patients with metastatic CRC progressed on ≥2 lines of therapy including 5FU, oxaliplatin and irinotecan. Patients with recurrence during or within 6 months of adjuvant chemotherapy were allowed. Exclusion: Prior TAS102 exposure, functional impairing peripheral neuropathy, ≥Grade 3 hypersensitivity to oxaliplatin, or uncontrollable grade 1-2 hypersensitivity to oxaliplatin. Treatment continued until disease progression or unacceptable toxicity. Patients received oxaliplatin 85 mg/m2 and TAS-102 35 mg/m2 bid days 1-5 every two weeks, and bevacizumab per MD choice. The primary endpoint was overall response rate (ORR) by RECIST (by independent radiologists). Secondary endpoints included disease control rate (DCR), safety and tolerability. **Results:** 54 patients enrolled; 53 received treatment on study and 48 had ≥1 disease assessment (median age 59, 58 % male, tumor RAS mutated 69%). Median time on study was 4 months (8 cycles; range 1-37 cycles). ORR was 6% (n=3), with average change in target lesions -45% (range -35 to -67%). 71% (n=34) had SD defined on study, with average change in target lesions -3% (range -29% to +20%). In those treated with bevacizumab (59%; n=28 ), 79% (n=22) had disease control vs. 75% (n=15). The most common reason for study discontinuation was progressive disease (23%, n=11). 68% (n=36) had a treatment-related adverse event (TRAE) ≥Grade 3 at any point during therapy. The most common G3 TRAEs were anemia 19% (n=10) and leukopenia 28% (n=15) at any time. Two (4%) developed grade 3 neuropathy. **Conclusions:** TASOX is effective as a 3rd line therapy for patients with mCRC. In our study, overall disease control rate was 77% with up to 19 months on study. Anemia and leukopenia are common but manageable. Bevacizumab did not seem to have as great an effect in our trial as observed in SUNLIGHT. Future studies would warrant TASOX in combination to bevacizumab in randomized trials in candidates for oxaliplatin retreatment. Clinical trial information: NCT02848079. Research Sponsor: None.

# QL1203 vs placebo plus mFOLFOX6 as first-line therapy in RAS wild-type, metastatic colorectal cancer (mCRC): Interim analysis (IA) of a multicenter, randomized, double-blinded, parallel, phase 3 trial.

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**Background:** Panitumumab plus FOLFOX or FOLFIRI has been approved as the first-line therapy for RAS wild-type mCRC. QL1203 is a panitumumab biosimilar showing similarity to the originator panitumumab (Vectibix, Amgen) in preclinical studies and phase 1 clinical pharmacokinetic study. Here we present results from interim analysis of a phase 3 trial (NCT04233151) investigating QL1203 vs placebo plus mFOLFOX6 as first-line therapy in Chinese patients (pts) with RAS wild-type mCRC. **Methods:** Pts with treatment-naïve, RAS and BRAF wild-type, mCRC unsuitable for radical resection and local therapy were enrolled and randomly assigned (2:1) to receive QL1203 (6 mg/kg) plus mFOLFOX6 once every 2 weeks (Q2W) or placebo plus mFOLFOX6 Q2W, stratified by primary tumor site, liver metastases, and previous neoadjuvant/adjuvant therapy. The primary endpoint was PFS assessed by a Blinded Independ Review Committee (BICR) per RECIST v1.1. Key secondary endpoints include PFS by investigator (INV), OS, ORR, DoR, and safety. This planned IA was conducted at 338 (81.25%) of 416 PFS events occurred. **Results:** From Jan 8, 2020 to Jun 12, 2023, 641 pts were randomized (QL1203/placebo, n=426/215). 551 (86.0%) pts had left-sided tumors (QL1203/placebo, 85.9%/86.0%). 433 (67.6%) pts had liver metastases (QL1203/placebo, 68.1%/66.5%). As of data cutoff of this IA (Mar 22, 2024), median follow-up was 23.8 months. Median PFS by BIRC was 11.20 months for QL1203 and 8.34 months for placebo (hazard ratio [HR], 0.61 [97.42% confidence interval [CI], 0.47-0.79]; stratified one-sided log-rank $p<0.0001$). Other efficacy results are presented (Table). Incidence of grade≥ 3 adverse events (AEs) related to study drug was 59.9% for QL1203 and 32.9% for placebo. The most common grade≥3 AE related to QL1203 was neutrophil count decreased (20.3%). **Conclusions:** QL1203 demonstrated significantly improved PFS versus placebo and higher ORR. Clinical trial information: NCT04233151. Research Sponsor: None.

| | QL1203 (n=426) | | Placebo (n=215) |
|---|---|---|---|
| **PFS by BIRC** | | | |
| **mPFS, months (95% CI)** | 11.20 (9.89-13.11) | | 8.34 (7.26-8.54) |
| **HR (97.42% CI)** | | 0.61 (0.47-0.79) | |
| **PFS by INV** | | | |
| **mPFS, months (95% CI)** | 10.91 (9.69-11.30) | | 8.41 (7.39-9.07) |
| **HR (95% CI)** | | 0.74 (0.60-0.91) | |
| **mOS, months (95% CI)** | 27.66 (24.74-32.79) | | 24.54 (20.17-27.79) |
| **HR (95% CI)** | | 0.82 (0.64-1.06) | |
| **ORR by BIRC, n (%)** | 68.31% | | 47.91% |
| **95% CI** | 63.66-72.70 | | 41.07-54.81 |
| **ORR by INV, n (%)** | 64.08% | | 49.30% |
| **95% CI** | 59.33-68.65 | | 42.44-56.19 |

# Efficacy and safety of glecirasib (JAB-21822) monotherapy and in combination with cetuximab in patients with KRAS G12C-mutated advanced colorectal cancer.

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**Background:** Glecirasib (JAB-21822), a highly selective, covalent and oral KRAS G12C inhibitor, has demonstrated a favorable safety profile and promising efficacy result as a monotherapy in NSCLC. We previously reported preliminary data of glecirasib monotherapy (NCT05009329) and in combination with cetuximab (NCT05194995) in KRAS G12C-mutated advanced Colorectal Cancer (CRC); here, we present the updated efficacy results. **Methods:** Patients (pts) with advanced CRC harboring KRAS G12C mutation were enrolled and treated with either single agent glecirasib (800mg QD) or glecirasib (800mg QD) combined with cetuximab. The primary endpoint was overall response rate (ORR). The secondary endpoints included duration of response (DoR), disease control rate (DCR), progression-free survival (PFS), overall survival (OS) and safety. **Results:** As of June 30, 2024, 44 pts received glecirasib (median follow up was 21.9 months). The median age was 55.5 years with female (38.6%) /male (61.4%), ECOG PS 0 (25.0%) /1 (75.0.%), and 79.5% having received ≥ 2 prior lines of therapy. The confirmed ORR and DCR were 22.7% (10/44) and 86.4% (38/44), respectively. Median DoR was 4.4months (95%CI: 4.2, 9.7), median PFS was 5.6 months (95%CI: 4.1, 7.0), and median OS was 16.0 months (95%CI: 8.8, 26.3). Any grade treatment-related adverse events (TRAEs) occurred in 41 (93.2%) pts, the most common ones (>20%) being anemia, blood bilirubin increased, hypertriglyceridemia and white blood cell count decreased. Grade 3-4 TRAEs occurred in 9 (20.5%) pts. No grade 5 TRAEs occurred. No patient discontinued treatment due to TRAEs. Among the47 pts treated with combination regimen (median follow up was 18.7 months), the median age was 54.8 years with female (53.2%)/ male (46.8%), ECOG PS 0 (44.7%) /1 (55.3%), and 74.5% having received ≥ 2 prior lines of therapy. The confirmed ORR and DCR were 50% (23/46) and 87.0% (40/46), respectively. Median DoR was 5.1 months (95%CI: 4.1, 6.9), median PFS was 6.9 months (95%CI: 5.4-6.9), and median OS was 19.3 months (95%CI: 13.1, NE). Any grade TRAEs occurred in 46 (97.9%) pts, the most common ones (>20%) being rash, skin fissures, blood bilirubin increased, ALT increased, AST increased, anemia and proteinuria. Grade 3-4 TRAEs occurred in 9 (19.1%) pts. No grade 5 TRAEs occurred. 2(4.3%) pts discontinued treatment due to TRAEs. **Conclusions:** Glecirasib in combination with cetuximab demonstrated better efficacy compared with glecirasib monotherapy in advanced KRAS G12C mutated advanced CRC, while maintaining a favorable safety profile. Clinical trial information: NCT05009329, NCT05194995. Research Sponsor: None.

# Short-course radiotherapy (SCRT) followed by fruquintinib plus adebrelimab and CAPOX in the total neoadjuvant therapy of locally advanced rectal cancer (LARC): A multicenter, single-arm, open-label, phase II study (UNION TNT).

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**Background:** We have demonstrated that neoadjuvant SCRT followed by PD-1 inhibitor plus CAPOX for LARC patients (pts) showed a higher pCR rate and a well-tolerated safety profile in phase III UNION study. Fruquintinib is a potent and highly selective VEGFR inhibitor, which had been approved for previously treated mCRC pts. This study aimed to investigate the efficacy and safety of SCRT followed by fruquintinib plus adebrelimab and CAPOX as a total neoadjuvant therapy for high-risk LARC pts. **Methods:** In this prospective, open-label, multi-centers, single-arm phase 2 study (NCT06234007), pts diagnosed with newly diagnosed and high risk LARC were recruited. Pts received SCRT (25Gy/5f) followed by six cycles of fruquintinib (4mg, qd, po, q3w,for the initial 6 pts in the safety run-in period, dosage would be adjusted based on dose-limited toxicities in the 1st cycle) combined with adebrelimab (1200mg, iv, d1, q3w) and CAPOX (q3w). Primary endpoint was CR rate (including clinical CR & pathologic CR). Secondary endpoints included 3-year EFS rate, OS, R0 resection rate and safety. **Results:** As of 30 Aug 2024, 45 pts were enrolled (median age 59 years [range: 24-75], 26 males, 23 clinical T4 stage and 20 clinical N2 stage, 36 with EMVI positive, 31 with MRF positive, 22 tumor located ≤5cm from the anal verge). Up to 20 Sep 2024, all pts completed SCRT, 73.33% (33/45), 57.78% (26/45) and 31.11% (14/45) pts completed two, four and six cycles of combination therapy, respectively. At data cut-off, of 33 pts included in the efficacy analysis, 28 pts underwent MRI examination and mrTRG results were available, the remaining 5 pts had early surgery due to AEs/serious AEs. 35.7% (10/28) and 46.43% (13/28) pts were classified as mrTRG1 and mrTRG2, respectively. Of 19 pts who underwent surgery, R0 rate was achieved in all 19 pts (100%) and pCR was achieved in 12 pts (63.16%). Moreover, 1 pt achieved cCR during neoadjuvant therapy, thus the treatment strategy provided a CR rate of 65% (1 cCR plus 12 pCR divided by 20). Grade 3/4 adverse effects (AEs) occurred in 16 pts (47.06%), and most commonly were lymphocyte count decreased (n=6, 18.18%), diarrhea (n=5, 15.15%) and AST increased (n=2, 6.06%). 4 serious AEs reported with intestinal perforation and all achieved pCR after surgery therapy. No treatment related death was reported. **Conclusions:** SCRT followed by fruquintinib combined with adebrelimab and CAPOX as a total neoadjuvant therapy showed promising CR rate and favorable pCR for high-risk LARC pts, and the safety profile was generally manageable. This treatment strategy might be a potential option as neoadjuvant therapy for high-risk LARC pts. Updated data will be presented in the future. Clinical trial information: NCT06234007. Research Sponsor: None.

# Update of phase 1b/2 study of muzastotug (ADG126, an anti-CTLA-4 SAFEbody) in combination with pembrolizumab in advanced/metastatic MSS CRC.

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**Background:** ADG126 is an anti-CTLA-4 IgG1 masked antibody with cleavable masking peptides that is preferentially activated in the tumor microenvironment, which in turn binds to a unique epitope to block CTLA-4 function, prime T cells and deplete Tregs. We previously reported that repeated dosing of the IO doublet therapy ADG126 plus pembrolizumab (Pembro) was well-tolerated with promising clinical efficacy in 3L MSS CRC patients (pts) free of liver metastasis (NCT05405595)[1,2]. Here we update clinical safety and efficacy of the study as result of continued dose optimization for the IO doublet therapy. **Methods:** This is a Phase 1b/2, open-label, multicenter dose escalation (DE) and expansion (EXP) study. Primary endpoints were safety and tolerability. Secondary endpoints were PK, ADA, ORR, DCR, DOR, PFS and OS. **Results:** A total of 72 pts were dosed with ADG126/Pembro (Table). 27.8% pts had ≥ 3 prior therapies and 6.9% pts had prior IO therapies. There was no Grade 4/5 TRAE, and MTD was not reached. Grade 3 TRAEs were 5.9% (1/17), 16.2% (6/37), 33% (4/12) for 10 mg/kg Q6W, 10 mg/kg Q3W and 20 mg/kg loading dose (LD)[&] cohorts, respectively. Repeated dosing of ADG126 at 20 mg/kg (Q3W)/Pembro was not well-tolerated. In MSS CRC EXP cohorts, ADG126 of 10 mg/kg Q3W/Pembro treatment resulted in 4 confirmed PRs (n=24 EE), 17% ORR (95% CI: 5-37%), 75% DCR, 4.7-mon mPFS and OS rate of 88% (6 mons) and 75% (9 mons). Subgroup analysis revealed that in pts without liver and peritoneal metastasis (NLPM), ADG126 of 10 mg/kg Q6W/Pembro (10 EE) showed 70% SD and ~ 100% OS (6 and 9 mons); ADG126 of 10 mg/kg Q3W/Pembro (17 EE) showed 24% ORR (4 PR (95% CI: 7-50%)), 88% DCR, 47% CBR, 8.5-mon mPFS and OS rate of 89% (6 mons) and 83% (9 mons). Detailed safety and efficacy results, including those from ADG126 20 mg/kg LD cohort, will be reported. **Conclusions:** Extensive effort on ADG126 dose optimization, aided by PK-informed dosing strategy, suggested that ADG126 10 mg/kg Q6W and Q3W are appropriate dose/regimen for the IO doublet; this is supported by the well-balanced safety and efficacy (ORR, PFS and OS) profile observed in MSS CRC. ADG126 20 mg/kg Q3W appears to have reached safety ceiling for the IO doublet. The totality of the data further verifies that ADG126 possesses potential best-in-class therapeutic index, and provides basis for continued development of the IO doublet in MSS CRC and for additional exploration of the IO doublet in combination with SOCs/other anti-cancer agents in broader patient populations. Clinical trial information: NCT05405595. Research Sponsor: None.

**Patient number for safety and efficacy evaluation.**

| | ADG126 Dose/Pembro* | | | | |
|---|---|---|---|---|---|
| **Evaluated for** | **10 mg/kg Q6W** | **10 mg/kg Q3W** | **20 mg/kg Q3W** | **20 mg/kg LD[&]** | **Overall** |
| **Safety (all pts of DE & EXP)** | 17 | 37 | 6 | 12 | 72 |
| **Efficacy (MSS CRC NLM, EXP)** | 10 | 24 | / | *5* | 39 |

*200mg, Q3W, IV; [&]20 mg/kg LD: 20 mg/kg x1 cycle followed by 10 mg/kg Q3W. NLM: liver metastasis-free; DE: Dose Escalation; EXP: Dose Expansion; EE: efficacy evaluable. Data cut date: Aug 30, 2024.

# Efficacy and safety of anlotinib as a later-line treatment for advanced colorectal cancer.

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**Background:** Colorectal cancer (CRC) is increasingly prevalent in China, with rising incidence and mortality rates. Often, early-stage CRC presents with no symptoms, leading to most patients being diagnosed at an advanced stage, where prognosis is poor. First- and second-line treatments for advanced CRC typically include chemotherapy combined with cetuximab or bevacizumab. However, third-line therapies such as regorafenib, furosemide, and TAS102 have shown limited efficacy. Anlotinib, a multi-targeted tyrosine kinase inhibitor (TKI), has demonstrated improved progression-free survival (PFS) in refractory metastatic CRC (mCRC) in a phase III trial (NCT02332499), showing both efficacy and manageable toxicity. This report provides updated results from extended follow-up. **Methods:** This study enrolled 73 patients with advanced colorectal cancer who had undergone at least two prior standard treatments and were treated with oral anlotinib hydrochloride (8-12 mg daily, days 1-14 of a 21-day cycle) between November 2021 and March 2024 at Tangdu Hospital. Patients were treated until disease progression or intolerable adverse events. Efficacy was assessed using RECIST version 1.1, and adverse reactions were evaluated using CTCAE version 5.0. The primary endpoints were PFS, disease control rate (DCR), and adverse events (AEs), while secondary endpoints included overall survival (OS). **Results:** Of the 73 patients included, 38 had an ECOG performance status (PS) of 0-1, and 35 had a PS of 2. Clinical staging identified 18 patients with stage IIIB/IIIC and 55 with stage IV disease. Patients received anlotinib doses of 8 mg (n=11), 10 mg (n=44), or 12 mg (n=18), with 48 patients also receiving chemotherapy and 25 receiving anlotinib alone. The median PFS was 4.0 months (95% CI: 3.4-4.6), and the median OS was 7.0 months (95% CI: 5.7-8.3). Subgroup analysis indicated that patients under 60 years of age, those without liver metastases, and those with ECOG PS 0-1 derived greater benefit from anlotinib. The most common treatment-related adverse event was hypertension (19.2%). Other notable adverse events included elevated AST (11.0%), proteinuria (6.8%), and jaundice (6.8%). Most adverse reactions were grade 1-2, although 1 patient experienced grade ≥3 gastrointestinal bleeding. **Conclusions:** Anlotinib showed promising efficacy and a favorable safety profile as a later-line treatment for advanced colorectal cancer. These findings suggest that anlotinib may be a viable clinical option for advanced CRC, though larger-scale clinical trials are necessary to confirm these results and further define its role in treatment. Research Sponsor: None.

# Liposomal irinotecan + 5-fluorouracil + leucovorin + bevacizumab as second-line therapy in metastatic colorectal cancer (IRIS): A multi-center, single-arm, prospective, phase II study.

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**Background:** In patients with advanced colorectal cancer (CRC), the recommended second-line treatment following first-line therapy with oxaliplatin-based regimens is irinotecan-based therapy. Liposomal irinotecan, a novel formulation of traditional irinotecan, has demonstrated potential to enhance efficacy while minimizing toxicity. This study aims to investigate the efficacy and safety of liposomal irinotecan in combination with 5-FU/LV and bevacizumab as a second-line treatment option for metastatic CRC. **Methods:** This is a multi-center, single-arm, prospective, phase II study. Patients with metastatic CRC who received oxaliplatin-based chemotherapy as first-line treatment were enrolled to receive liposomal irinotecan + 5-fluorouracil + leucovorin + bevacizumab regimen until disease progression and/or unacceptable toxicity. The primary endpoint is objective response rate (ORR), secondary endpoints include disease control rate (DCR), progression-free survival (PFS), overall survival (OS), and safety. **Results:** A total of 50 patients were enrolled from Jan 2024 to Jul 2024 across 6 sites in China. The median age was 56.5 years (range: 30.0-76.0), with 58.0% male and 36.0% ECOG 0. Among the patients, 38.0% had left-sided colon cancer, 24.0% had right-sided colon cancer, and 38.0% had rectal cancer. As of Sept 14, 2024, 39 patients had at least one tumor assessment and 19 patients were still on treatment. ORR and DCR were 20.5% (8/39, 95% CI: 9.3%-36.5%) and 84.6% (33/39, 95% CI: 69.5%-94.1%), respectively. Eleven patients had disease progression and 5 patients died, and the median PFS and OS were not reached. The relative dose intensity of liposomal irinotecan and 5-fluorouracil were 92.8% (range: 23.6%-109.6%) and 89.5% (range: 13.3%-116.8%), respectively. During treatment, 43 (86.0%) patients had at least one adverse event (AE) and 24 (48.0%) had grade 3-4 AE. Most common (≥ 10%) grade 3-4 AEs were neutropenia (20.0%), leukocytopenia (16.0%) and diarrhea (10.0%). No unexpected toxicities observed in this study. **Conclusions:** These preliminary results reveled that liposomal irinotecan + 5-fluorouracil + leucovorin + bevacizumab after oxaliplatin-based treatment for metastatic CRC was well tolerated with encouraging clinical activity, which warrants further follow up. Clinical trial information: NCT06184698. Research Sponsor: None.

# Amivantamab with or without chemotherapy in right-sided metastatic colorectal cancer: Updated results from OrigAMI-1, an open-label, phase 1b/2 study.

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**Background:** Amivantamab (ami) is an EGFR-MET bispecific antibody with immune cell-directing activity and is FDA approved in *EGFR*-mutated advanced non-small cell lung cancer. High MET expression is observed in ~68% of patients (pts) with metastatic colorectal cancer (mCRC). Additionally, MET amplification occurs in up to 23% of EGFR-resistant mCRC and is implicated in driving resistance to EGFR-targeting antibodies (EGFRi). Compared to left (L)-sided disease, right (R)-sided disease is less responsive to EGFRi and associated with poorer outcomes. We present longer follow-up data among pts with R-sided mCRC. **Methods:** OrigAMI-1 (NCT05379595) is assessing ami as monotherapy and combined with chemotherapy in mCRC. All pts were wild-type for *KRAS*, *NRAS*, *BRAF*, and *EGFR* ectodomain by central ctDNA testing, without *ERBB2*/*HER2* amplification. One ami monotherapy cohort (Cohort C) enrolled only pts with R-sided disease (all pts must have 2-3 prior lines; prior EGFRi allowed). The ami plus chemotherapy cohorts (Cohorts D [with FOLFOX] and E [with FOLFIRI) enrolled both L- and R-sided disease (1 prior line max; prior EGFRi use was exclusionary). Primary tumor locations of cecum, ascending colon, hepatic flexure, and transverse colon were considered R-sided. Response was assessed by the investigator per RECIST v1.1. **Results:** As of 26-Aug-2024, 23 pts with R-sided mCRC received ami monotherapy (median follow-up of 8.1 months [mo]). Median number of prior lines was 2, and 43% had prior EGFRi. There were 7 pts with R-sided disease who received ami plus FOLFOX or FOLFIRI (median follow-up of 6.5 mo); all 7 had received 1 prior line. Among pts receiving ami monotherapy, objective response rate (ORR) was 22% (5/23) and disease control rate (DCR) was 78% (18/23), with 1 achieving a complete response. Median duration of response (DoR) is 7.4 mo; response and treatment are ongoing for 3 of the 5 responders. In pts receiving ami plus FOLFOX or FOLFIRI, ORR was 43% (3/7) and DCR was 86% (6/7). Median DoR is 5.8 mo, with all 3 responders on treatment, of which 2 are ongoing response. Additional details are in the Table. Biomarker data will be presented at the meeting. Safety profile among R-sided disease was consistent with prior reports, with the most common ami-related grade 3+ AEs being rash and hypoalbuminemia (2 pts each). **Conclusions:** Ami monotherapy or combined with FOLFOX or FOLFIRI demonstrated durable antitumor activity in R-sided mCRC. Clinical trial information: NCT05379595. Research Sponsor: J&J Innovative Medicine.

**Efficacy.**

| | Ami monotherapy (n=23) | Ami + FOLFOX or FOLFIRI (n=7) |
|---|---|---|
| **Median follow-up, mo (range)** | 8.1 (0.6–17.5) | 6.5 (3.2–8.8) |
| **Median number prior lines** | 2 | 1 |
| **Prior EGFRi, n (%)** | 10 (43) | 0 |
| **ORR, % (95% CI)** | 22 (8–44) | 43 (10–82) |
| **Median DoR, mo (95% CI)** | 7.4 (3.7–NE) | 5.8 (NE–NE)[a] |
| **DCR, % (95% CI)** | 78 (56–93) | 86 (42–100) |
| **Median progression-free survival, mo (95% CI)** | 3.7 (3.4–5.5) | 7.4 (1.8–NE) |

[a]All 3 responders remain on treatment; 2 of 3 responders are ongoing response.

# Preliminary safety and efficacy of upfront FOLFOXIRI and bevacizumab with cadonilimab in patients with proficient mismatch repair/microsatellite stable (pMMR/MSS) metastatic colorectal cancer: A multicenter phase II study (SYLT-026).

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**Background:** The AtezoTRIBE phase II trial reported that adding Atezolizumab to upfront FOLFOXIRI plus Bevacizumab significantly improved overall survival (OS) in patients with metastatic colorectal cancer (mCRC). However, the improvement in OS was modest within the pMMR/MSS subgroup (HR, 0.80 [80% CI, 0.63–1.02]; P = 0.117) [1]. Cadonilimab, a humanized bispecific antibody targeting both PD-1 and CTLA-4, may offer enhanced therapeutic benefits. Therefore, this study aims to investigate whether the addition of Cadonilimab to upfront FOLFOXIRI plus Bevacizumab improves efficacy while maintaining safety in patients with pMMR/MSS mCRC. **Methods:** This multicenter, single-arm phase II study enrolled treatment-naive patients with pMMR/MSS mCRC, aged 18–75, who had at least one measurable lesion according to RECIST 1.1, an ECOG performance status (PS) of 0–2, and adequate organ function. Patients with dMMR/MSI-H tumors or prior immunotherapy were excluded. Eligible participants received FOLFOXIRI (irinotecan 165 mg/m$^2$, oxaliplatin 85 mg/m$^2$, leucovorin 200 mg/m$^2$, and fluorouracil 2400 mg/m$^2$ as a 48-hour infusion), along with Bevacizumab (5 mg/kg), and Cadonilimab (6 mg/kg). Treatment was administered for up to twelve 14-day cycles, followed by maintenance therapy with fluoropyrimidine and Bevacizumab plus Cadonilimab for up to 52 weeks, or until disease progression, unacceptable toxicity, or withdrawal of consent. Patients were equally enrolled, with 10 having liver metastases and 10 without. The primary endpoint was objective response rate (ORR) per RECIST 1.1. **Results:** As of August 30, 2024, 20 patients were enrolled, with a median age of 65.5 years (range: 33–74). Of these, 40% had an ECOG PS of 1 or 2, 55% had three or more metastatic organs, and 7 patients (35%) had RAS or BRAF mutations. The median follow-up was 5.0 months. Fifteen patients underwent at least two imaging assessments, revealing a confirmed ORR of 100% (15/15) and a disease control rate (DCR) of 100%. The median progression-free survival (PFS) and OS data were not yet mature. All 20 patients experienced treatment-emergent adverse events, most of which were grade 1 or 2. One patient (5%) discontinued treatment due to immune-mediated colitis. The most common grade 3/4 adverse events included neutropenia (55%), leukopenia (15%), infusion reactions (5%), dermatitis (5%), colitis (5%), anemia (5%), elevated ALT (5%), and diarrhea (5%). No new safety signals were identified. **Conclusions:** In conclusion, Cadonilimab in combination with FOLFOXIRI and Bevacizumab demonstrated promising efficacy and a manageable safety profile as a first-line treatment for patients with pMMR/MSS mCRC. Further investigation in larger trials is warranted. 1. Antoniotti, et al, J Clin Oncol, 2024. Clinical trial information: NCT05839470. Research Sponsor: None.

# Avutometinib plus cetuximab after chemotherapy in patients with *KRAS*-mutated metastatic colorectal cancer: Results of phase 1 dose escalation.

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**Background:** Anti-EGFR antibodies like cetuximab are ineffective in *KRAS* mutated (mt) colorectal cancer (CRC) due to constitutive activation of downstream pathways. Avutometinib is a first in class, oral, novel, dual RAF/MEK inhibitor. In patient-derived xenograft models of *KRAS*mt CRC, the combination of an anti-EGFR antibody with avutometinib conferred stronger tumor growth inhibition than either agent alone. We present results from the dose escalation phase (1b) of an ongoing phase 1b/2 trial evaluating the combination avutometinib plus cetuximab in *KRAS*mt metastatic (m) CRC. **Methods:** This is a single-center, open-label, phase 1b/2 study evaluating the safety and combined tolerability of avutometinib and cetuximab. Patients received avutometinib 2.4mg (at DL0) or 3.2mg (at DL1) orally twice a week for 3 weeks out of a 4-week cycle + cetuximab 500mg/m$^2$ intravenously q2week. Patients with disease progression on, or intolerance to 5-FU, oxaliplatin, and irinotecan, and measurable disease per RECIST v1.1 were enrolled. The primary endpoint for phase 1b was to establish the maximum tolerated dose (MTD) and hence determine the recommended phase 2 dose (RP2D). Dose limiting toxicity (DLT) period was 28 days. A 3+3 dose escalation design was used with dose level (DLs) 0 (2.4mg avutometinib) as the starting level and one dose escalation to DL+1 (3.2mg avutometinib). **Results:** We report safety results of phase 1b. 10 patients, 7 at DL0 and 3 at DL+1, were enrolled. Median age was 52.5 yrs (range 41 – 75). Any grade (G) treatment related adverse events (TRAEs) were seen in all 10 patients. Most common TRAEs were acneiform rash (100%), chills (30%), diarrhea (30%), and fatigue (30%). G3 TRAEs at DL0 were seen in 4/7 (57.1%) patients and at DL+1 in 2/3 (66.7%) patients. No G4 TRAEs were seen. G3 TRAEs seen at DL0 are noted in the table. One patient from the initial three enrolled was not evaluable due to underdosing. Another patient was later found to not meet eligibility criteria, but they did clear the DLT period. Hence, 6/7 patients at DL0 were evaluated for AE assessment, of which 1 had a DLT. 3 additional patients were enrolled in DL+1. 2/3 patients in DL+1 had grade 3 toxicities (both DLTs) of rash. Hence DL0 was established as the MTD/RP2D. **Conclusions:** This is the first trial evaluating the combination avutometinib plus cetuximab in *KRAS*mt mCRC. RP2D for avutometinib was established at 2.4mg orally twice a week for 3 weeks out of a 4-week cycle when used in combination with q2week 500mg/m$^2$ cetuximab for *KRAS*mt mCRC. Enrollment continues currently for the dose expansion phase for efficacy assessment. Clinical trial information: NCT05200442. Research Sponsor: Verastem Oncology (funding and drug), Eli Lilly and Company (drug only).

**G3 TRAEs at DL0 attributed to both drugs unless specified.**

| G3 TRAE | # of patients |
|---|---|
| **Acneiform rash** | 3/7 (42.9%) |
| **Hand-Foot syndrome** | 1/7 (14.3%) |
| **Extremity edema** | 1/7 (14.3%) |
| **Cellulitis** | 1/7 (14.3%) |
| **Sepsis** | 1/7 (14.3%) |
| **Anemia** | 1/7 (14.3%) |
| **Creatine phosphokinase elevation** | 1/7 (14.3% – avutometinib only) |

**201** **Poster Session**

# Neoadjuvant chemotherapy with FOLFOXIRI for high-risk relapsed locally advanced colon cancer: A single-arm phase II trial.

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**Background:** To evaluate the safety and efficacy of neoadjuvant chemotherapy (NAC) with three drug-intensity chemotherapy (oxaliplatin, irinotecan, and fluorouracil [FOLFOXIRI]) in patients with high-risk factors for recurrence and metastasis of locally advanced colon cancer (hr-LACC). **Methods:** A single-arm, open-label, phase II trial was conducted at West China Hospital of Sichuan University from July 2020 to Oct 2023. Patients with T4a/b or (and) N2/fused lymph nodes) and no distant metastases were considered as the hr-LACC patients. Eligible enrolled patients received four cycles of NAC with FOLFOXIRI followed by surgery and adjuvant chemotherapy with capecitabine and oxaliplatin (CAPOX) for two to five cycles or single-agent capecitabine for five cycles or observation. The primary endpoint was the rate of TRG grade 0-2 according to the criteria of NCCN guideline. **Results:** Of 72 patients enrolled (median age 53 [18-74] years,65 were T4a/b [90.2%], 66 were N2 [91.6%]), 58 patients (80.6%) completed four cycles of FOLFOXIRI. Three patients (4.2%) achieved clinical complete response. Twenty-eight patients (39.4%) obtained clinical partial response. The most common adverse event (AE) was nausea (86.1%, 62/72); grade 3-4 AEs occurred in 37 of 72 patients (51.4%).R0 resection was achieved in 64 of 64 patients, and the TRG (0-2) rate was 87.5% (56/64). The pathologic complete response rate was 7.8% (5/64). The proportions of pT0-2 and pN0 were 21.9% and 68.8%, respectively. Only three (4.7%) patients had prolonged hospital stay due to perioperative complications. No patient died within 30 days and 90 days after surgery. The 2-year tumor recurrence rate was 17.2%. **Conclusions:** For patients with hr-LACC, NAC with FOLFOXIRI regimen was safe and effective. Longer follow-up and larger clinical studies are needed to validate this innovative regimen. Clinical trial information: NCT05018182. Research Sponsor: Department of Science and Technology of Sichuan Province; 2024NSFSC1902; the Nation-Sponsored Postdoctor Researcher Program; GZC20231836.

# Immune checkpoint inhibitors in digestive tumors with mismatch repair deficiency or microsatellite instability (dMMR/MSI-H): Effectiveness, safety, and prognostic factors in real-world practice.

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**Background:** ICIs have shown durable responses in dMMR/MSI-H digestive tumors in phase II-III clinical trials. The aim of this study is to evaluate the efficacy, safety and prognostic factors in the context of real-world practice. **Methods:** We conducted a retrospective multicenter observational study of patients with dMMR/MSI-H digestive tumors treated with ICIs in routine clinical practice across seven university hospitals in northwest Spain. We analyzed clinicopathological characteristics, treatment response data, efficacy, and adverse events. **Results:** A total of 122 patients treated with ICIs between November 2015 and February 2024 were included. The median age was 70.4 years (range 29-89), and 52.4% were male. The most frequent origin was colorectal (54.9%; 67.2% located in the right colon and 37.3% BRAF V600mt), followed by gastroesophageal adenocarcinoma (34.4%). 16.4% had >3 metastatic sites (including 30.3% liver metastases and 44.3% peritoneal metastases). Baseline performance status (ECOG PS) was 0/1/2 in 21.7%/58.7%/19.8%, respectively. 39.3% had received one prior line of treatment. 92.6% received pembrolizumab monotherapy. The objective response rate (ORR) was 77.7% (including 31.3% complete response, 46.4% partial response, 13.4% stable disease, and 8.9% progressive disease), and the disease control rate (DCR) was 90.1%. Median overall survival (OS) was 53.2 months (95% CI: 42.6 – 63.7 months), and progression-free survival (PFS) was 46 months (95% CI: 30.6 – 61.4 months). The most common grade 3-4 treatment-related adverse events included nephritis (4.0%), hepatitis (3.3%), asthenia (2.5%), and pneumonitis (1.6%). 81.1% of patients discontinued treatment (including 24.6% due to progression, 21.3% after completing 2 years of treatment, and 14.8% due to toxicity). Potential prognostic factors identified in univariate analysis included ECOG PS, primary tumor resection, number of metastatic sites, and best treatment response. In multivariate analysis, a significant association was confirmed between ECOG PS, the presence of liver metastases, and best response to treatment with ICIs and overall survival. **Conclusions:** Our series confirms the efficacy and safety of ICIs in dMMR/MSI-H digestive tumors in routine clinical practice. The identified prognostic factors may be useful for identifying patients who could benefit the most from ICI treatment. Research Sponsor: None.

**203** **Poster Session**

# Real-world outcomes of targeted therapies and local interventions in BRAF V600E–mutated metastatic colorectal cancer: A single-center retrospective study.

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**Background:** Metastatic colorectal cancer (mCRC) patients with BRAF V600E mutations, especially those with microsatellite stable (MSS) tumors, have poor prognoses. Chemotherapy has limited efficacy in this subgroup, especially for second-line or subsequent-line treatments. Current research is increasingly focusing on the integration of targeted therapies and local interventions, but real-world evidence remains scarce, particularly from Asian cohorts. This study leverages real-world data to evaluate the impact of novel therapeutic combinations, including triplet and doublet targeted therapies, alongside local treatments in this high-risk patient population. **Methods:** This retrospective study includes MSS-type mCRC patients with BRAF V600E mutations treated at the Sixth Affiliated Hospital of Sun Yat-sen University between April 2014 and May 2024. Patients receiving targeted therapies were stratified into three groups: Triplet group (BRAF, EGFR, and MEK inhibitors, including Dabrafenib, Cetuximab and Trametinib), Doublet group (BRAF and EGFR inhibitors, including Dabrafenib and Cetuximab or Encorafenib and Cetuximab), and Doublet target-chemotherapy group (BRAF and EGFR inhibitors combined with chemotherapy, including Vemurafenib, Irinotecan, and Cetuximab). Primary endpoint was progression-free survival (PFS). Secondary endpoints included overall survival (OS) and objective response rate (ORR). **Results:** Among 239 enrolled patients, 58 received targeted therapy and 68 underwent conventional chemotherapy as second-line treatment. 63.5% (80/126) of these patients underwent local interventions, including primary tumor resection, metastasectomy, CRS±HIPEC, and liver ablation. Targeted therapy demonstrated significantly improved mPFS compared to conventional chemotherapy (5.5 vs. 4.2 months, HR = 0.44; 95% CI: 0.29-0.68, $p < 0.001$). OS did not differ significantly between the two treatment arms (12.7 vs. 12.1 months, HR = 0.80; 95% CI: 0.50-1.25, $p = 0.330$). In the triplet group, ORR, mPFS, and mOS were 21%, 5.8 months and 18.0 months, respectively, showing a trend towards enhanced efficacy compared to the other targeted therapy groups, albeit without statistical significance. Patients who received targeted therapy (n=35) or chemotherapy (n=45) in combination with local interventions demonstrated significantly prolonged OS compared to those without local therapy (mOS: 22.3 vs. 9.6 months; HR = 0.19; 95% CI: 0.09-0.42; $p < 0.001$ and 14.3 vs. 10.9 months; HR = 0.53; 95% CI: 0.29-0.98; $p = 0.042$, respectively). **Conclusions:** Optimal treatment for BRAF V600E mCRC remains under investigation. This study suggests that combining targeted therapies with selective local interventions may significantly improve patient survival in this challenging subgroup, warranting further prospective evaluation. Research Sponsor: None.

# Efficacy and safety of trifluridine/tipiracil and bevacizumab (FTD/TPI+BEV) in patients with metastatic colorectal cancer (mCRC) in real-world practice.

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**Background:** In Sunlight trial, the combination of FTD/TPI + BEV showed a significant improvement in Overall Survival (OS) and Progression-Free Survival (PFS), along with better Objective Response Rate (ORR) and Disease Control Rate (DCR), compared to FTD/TPI alone, in patients with mCRC progressing after two chemotherapy regimens. The BeTAS trial aimed to evaluate the effectiveness and safety of the FTD/TPI + BEV combination in real-world practice. **Methods:** This was a retrospective, observational, multicenter study of mCRC patients treated with FTD/TPI+BEV in routine clinical practice across 11 Spanish university hospitals. It included patients with mCRC who were refractory or intolerant to standard therapies. **Results:** A total of 208 patients treated with FTD/TPI+BEV between July 2019 and December 2023 were included. The median age was 69 years (range 33 to 88 years), with 59.8% being male. 15.3% had an ECOG performance status of 2, 55.9% had RAS mutations, 30.3% had three or more metastatic sites, and 81.2% had liver metastases. Additionally, 19.8% had a time from diagnosis of first metastasis of less than 18 months, and 55.3% were categorized in Tabernero's Unfavorable Prognostic Subgroup. Most patients (81.3%) received FTD/TPI+BEV as third-line therapy, with 90.5% having previously received anti-VEGF therapy. The median number of treatment cycles was 4 (range 1 to 31 cycles). The ORR was 4.6%, and the DCR was 50%. The median PFS was 4.9 months (95% CI, 4.0-5.7 months), and the median OS was 11.1 months (95% CI, 9.5-12.8 months). The most common grade 3-4 toxicities included neutropenia (41.8%), asthenia (7.2%), and hepatic toxicity (5.5%). No treatment-related deaths were reported. **Conclusions:** Our series confirms the efficacy and safety of FTD/TPI + BEV in routine clinical practice, even in a population with poor prognosis and an extensive history of previous treatments. Research Sponsor: None.

# The importance of treatment handling and compliance on overall response rate in a phase III study of metastatic colorectal cancer: Post-hoc per protocol analyses of the AGENT trial.

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**Background:** To study the compliance in handling the investigational medicinal product (IMP) and its impact on outcomes in patients with metastatic colorectal cancer participating in the AGENT trial (NCT03750786). **Methods:** In a randomized, multicenter, multinational, phase III study, patients were randomized to receive either arfolitixorin (n=245) or leucovorin (n=245) with 5-fluorouracil (5-FU), oxaliplatin and bevacizumab as a first-line treatment. Logistic regression was applied for overall response rate (ORR) and Cox regression for all-cause death, and all-cause death or progression, adjusted for randomization strata. **Results:** Missing or incorrect IMP handling was reported for 28 (11%) patients in the arfolitixorin and 23 (9%) in the leucovorin arm, <80% compliance for duration of bolus 5-FU 2-4min for 101 (42%) and 97 (41%), respectively. Considering the time between the bolus 5-FU and arfolitixorin, non-compliance, defined by <80% doses given within 25-35min PP window, was reported for 92 (38%) in arfolitixorin patients. Adherence to the timing between the first and second arfolitixorin dose (30-60min) was high. In total, 91 (37%) patients in the arfolitixorin and 134 (55%) in the leucovorin arm fulfilled PP definitions (cPP population). The bolus 5-FU dose was equally likely to be reduced in both the arfolitixorin and leucovorin arms across regions (ranging from 22% to 39%), except for Japan where 55% of patients in the arfolitixorin arm had their bolus 5-FU dose reduced, compared to 28% in the leucovorin arm. In the cPP population, the ORR was achieved by 54 (59%) in the arfolitixorin and 69 (51%) in the leucovorin arm, adjusted odds ratio (aOR) 1.40 (95% CI 0.81-2.45), p=0.23. Excluding Japan, the ORR was 41 (62%) in the arfolitixorin and 49 (46%) in the leucovorin arm, aOR 2.11 (95% CI 1.09-4.06), p=0.026. Corresponding aOR for Europe was 1.50 (95% CI 0.71-3.15), p=0.29, and for North America aOR 7.95 (95% CI 1.45-43.62), p=0.017. In Japan, the ORR had better outcome in the leucovorin arm (71.4%) compared to the arfolitixorin arm (52.0%), aOR 0.42 (95% CI 0.13-1.31), p=0.14. However, the relationship was numerically opposite for patients without bolus 5-FU dose reduced. All-cause death during the complete follow-up was reported for 37 (41%) in the arfolitixorin and 61 (46%) in the leucovorin arm. **Conclusions:** In the AGENT trial, 46% of the included patients met per protocol compliance for treatment handling. In this cPP population, there was a non-significant, but clinically relevant, increase in the overall response rate in the arfolitixorin compared to the leucovorin arm. Notably, in all regions excluding Japan, the overall response rate was higher for arfolitixorin, with statistical significance. Adherence to the treatment protocol is crucial for the outcomes of patients with metastatic colorectal cancer. Clinical trial information: NCT03750786. Research Sponsor: Isofol Medical.

# Phase 1/2 study of XTX101, a tumor-activated, Fc-enhanced anti-CTLA-4 monoclonal antibody, in combination with atezolizumab in patients with advanced solid tumors and in MSS CRC.

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**Background:** XTX101 is an investigational tumor-activated, Fc-enhanced, high affinity binding aCTLA-4 designed to block CTLA-4 and deplete regulatory T cells upon activation in the tumor by proteases. XTX101 also contains mutations designed to enhance Fcγ receptor binding. Fc enhancement augments FcγR co-engagement on antigen presenting cells and has been linked with efficacy of aCTLA-4 combinations in patients (pts) with "cold tumors", including microsatellite stable (MSS) colorectal cancer (CRC). Parts 1A/1B of the study evaluated XTX101 monotherapy in pts with advanced solid tumors. XTX101 was generally well tolerated with limited peripheral XTX101 activation (~13%) compared to evidence of XTX101 activation in the tumor microenvironment (73-96%, in n=2 on-treatment tumor biopsies). A durable partial response (PR) and resolution of hepatic metastases was observed in a pt with PD-L1 negative NSCLC at the monotherapy recommended phase 2 dose (RP2D) of 150 mg once every six weeks (Q6W) [1]. **Methods:** Part 1C evaluated the safety and feasibility of XTX101 in combination with atezolizumab in pts with advanced solid tumors using 3+3 dose escalation to establish the RP2D. Phase 2 is enrolling pts with MSS CRC with at least 1 prior regimen for metastatic CRC. Initial safety and anti-tumor activity data from Phase 2 in ~20 pts will be presented. **Results:** As of July 15, 2024, 15 pts were enrolled in Part 1C: MSS CRC (n=12), esophageal cancer, NSCLC, and ampullary carcinoma (n=1 each). Median age 69 and 3 prior lines of therapy (1-12). Across all XTX101 dose levels (75 mg to 150 mg), treatment-related adverse events (TRAEs) of any grade with >10% incidence included infusion reactions (n=4), fatigue and diarrhea (n=2 each), no G4 or G5 TRAEs were reported. In Part 1C, 2 pts experienced a dose limiting toxicity: 1 pt had G3 colitis, and 1 pt had G3 liver enzyme elevation (both were at the XTX101 150 mg dose level and resolved with immunosuppression). Two unconfirmed PRs were reported: in a pt with MSS CRC, including resolution of a liver target lesion, and in a pt with ampullary carcinoma (accompanied by a Ca 19-9 decrease from 700 to 41). In addition, a decrease in intra-tumoral Tregs was observed in paired tumor biopsies from a pt. The combination RP2D was established as XTX101 100 mg Q6W and atezolizumab 1200 mg once every three weeks (Q3W), and enrollment in the Phase 2 part of the study for pts with MSS CRC is ongoing at the RP2D. **Conclusions:** In Part 1C dose escalation, the combination of XTX101, a tumor-activated, Fc-enhanced aCTLA-4, and atezolizumab was generally well tolerated in pts with advanced solid tumors and demonstrated initial evidence of anti-tumor activity in cold tumors. These data support the initiation of an ongoing Phase 2 study evaluating the combination in pts with MSS CRC with and without liver metastases. 1. Davar ESMO IO 2023. Clinical trial information: NCT04896697. Research Sponsor: Xilio Development, Inc.

# Neoadjuvant botensilimab (BOT) plus balstilimab (BAL) in resectable mismatch repair proficient (pMMR) and deficient (dMMR) colorectal cancer (CRC): NEST clinical trial update.

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**Background:** Checkpoint inhibitors (CPI) have been transformative for localized dMMR CRC (microsatellite instability-high [MSI-H]), but not for localized pMMR CRC (microsatellite stable [MSS]). The combination of BOT, a novel Fc-enhanced multifunctional CTLA-4 antibody, and BAL, an anti-PD-1 antibody, has demonstrated significant activity in metastatic CRC. We examined this novel dual CPI therapy in patients with MSI-H and MSS localized CRC. **Methods:** This is a single arm trial of neoadjuvant BOT 75 mg/m$^2$ (day 1) and BAL 240 mg/m$^2$ every 2 weeks x2 (NEST1) or x4 (NEST2) in patients with localized CRC who were candidates for surgery. The primary endpoint was major pathologic response (MPR), defined as ≥90% pathologic tumor regression. The 95% confidence interval (CI) was computed based on exact method for binomial distribution. **Results:** NEST1 (n=12: 9 MSS / 3 MSI-H) accrued from 3/21/23 to 9/25/23 and NEST2 (n=14: 13 MSS / 1 MSI-H; 2 MSS patients had synchronous CRC) accrued from 2/15/24 to 5/6/24; 2 patients in NEST1 were not evaluable (1 with rectal cancer not resected [ypT0N0 on local excision] and 1 patient with occult lung metastasis). In total, 7 MSS CRC tumors were evaluable from NEST1 and 15 from NEST2; and 3 MSI-H tumors were evaluable from NEST1 and 1 from NEST2, leaving a final evaluable study population of 24 patients (20 MSS / 4 MSI-H) and 26 tumors (22 MSS / 4 MSI-H). The Table provides pathologic response rates across NEST1, NEST2, and MSI-H. With a median follow-up of 13.1 months (NEST1) and 4.8 months (NEST2), no patients have recurred and ctDNA testing remains negative. No grade 4 toxicities were observed and the only treatment-related grade 3 toxicity was diarrhea/colitis in 15% and fever in 4%. Treatment-related grade 2 diarrhea/colitis occurred in 8%, grade 2 fever in 15%, and grade 2 fatigue in 4% of patients. Diarrhea/colitis resolved in all cases with steroids ± anti-TNFα therapy. All patients underwent laparoscopic resection, with 1 conversion to open. No surgeries were delayed due to treatment-related adverse events. Median length of stay was 2 days (range 1-14). Translational data including pathologic evaluation of tumor infiltrate and response will be presented. **Conclusions:** Neoadjuvant BOT/BAL is safe, with no delays to surgery, and effective. We observed high MPR rates in both MSS and MSI-H CRC with no recurrences to date. The MPR and pCR rate improved with extended time to surgery. Clinical trial information: NCT05571293. Research Sponsor: Agenus, Inc.

**Pathologic response to neoadjuvant BOT/BAL in localized CRC.**

| Pathologic Response | NEST 1 n=7 MSS tumors | NEST 2 n=15 MSS tumors | MSI-H (3 NEST1, 1 NEST2) |
|---|---|---|---|
| **100% (CR%, 95%CI)** | 1 (14%, 0.4-58%) | 6* (40%, 16-68%) | 3** (75%, 19-99%) |
| **≥ 90% (MPR%, 95%CI)** | 2 (29%, 4-71%) | 7 (47%, 21-73%) | 4^ (100%, 40-100%) |
| **≥ 50%** | 4 (57%) | 9 (60%) | 4 (100%) |
| **Median days to Surgery (range)** | 29 (21-37) | 57 (45-104) | 46 (34-78) |

*2 with carcinoma *in situ*, no tumor bed. ^MPR was in NEST1. **1 was rectal.

# Outcomes in management of locally advanced rectal cancer with total neoadjuvant therapy in an underserved population.

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**Background:** The use of total neoadjuvant therapy is now standard of care and is favored when treating locally advanced rectal cancer but its treatment is seldom studied in an underserved population. **Methods:** A retrospective study was conducted on patients with Stage II and III locally advanced colorectal cancer treated with either total neoadjuvant therapy (TNT) followed by resection or conventional therapy (CT) with chemoradiation followed by surgery, with or without adjuvant treatment afterwards. Patients completed treatment between 2015 and 2023 at the Hartford Hospital Helen and Harry Gray Cancer Center at Hartford, Connecticut. Data collected included demographics, type of treatment, tumor location (high vs mid vs low) and pathological response to treatment spread out among the two groups. Propensity score matching was used to minimize baseline differences between the two groups, adjusted for age, gender, ethnicity and tumor location and Fisher's exact test was performed to then check for significance. **Results:** Out of the 133 patients analyzed, 90 underwent TNT, and 43 received CT. The average age at diagnosis was 57.5 years (SD: 11.15). In the TNT group, 79 patients (87.7%) achieved either a partial or complete response (PCR), while 11 patients (12.3%) had No response (NR). Tumor locations in the TNT group were distributed as follows: 13.4% (n=12) had high tumors, 34.4% (n=31) had mid tumors, and 52.2% (n=47) had low tumors. In the CT group, 31 patients (72.9%) achieved PCR, while 12 patients (27.9%) had NR. Tumor locations in the CT group were distributed as follows: 9.3% (n=4) had high tumors, 51.2% (n=22) had mid tumors, and 39.5% (n=17) had low tumors. After propensity score matching and adjusting for demographic covariates, the TNT group demonstrated a higher proportion of responders compared to the CT group (p=0.048). Furthermore, 40% (n=36) of patients in the TNT group achieved a complete response, compared to 32% (n=14) in the CT group. Partial response rates were 39.5% (n=17) in the TNT group and 47.8% (n=43) in the CT group. **Conclusions:** TNT demonstrated superior pathological response rates compared to CT, which supports its growing role in the management of locally advanced rectal cancer even in this underserved population.
Research Sponsor: None.

# Efficacy of plogosertib (CYC149462), a novel orally bioavailable PLK1 inhibitor, on patient-derived models of colorectal cancer.

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**Background:** Colorectal cancer (CRC) is the third most common cancer in adults in the United States. Given the relative rapid cell division in cancer cells compared to most normal cells, drugs targeting mitosis and other cell cycle checkpoints are of interest in CRC treatment. One multi-tasking protein in cell division is polo-like kinase 1 (plk1). Plk1 is a serine threonine kinase that is highly expressed during mitosis and involved in a variety of functions, including spindle assembly, chromosome segregation, G2/M transition, and cytokinesis. Encouragingly, its expression is also significantly higher in CRC compared to normal mucosa, which makes it an attractive cancer-specific target in CRC. In this study, we sought to evaluate the efficacy of plogosertib (CYC149462), a novel plk1 inhibitor, in a series of preclinical models of advanced CRC. **Methods:** Fourteen CRC patient-derived organoids (PDOs) were generated from patients who underwent biopsy or resection for their primary or metastatic CRC under an IRB approved protocol at Duke University. Diagnosis of CRC is confirmed in PDOs using H&E and IHC. Subsequently, PDOs were treated with plogosertib in a range of concentrations from 256pM to 100μM. Drug screens were performed for standard of care agents 5FU, oxaliplatin, and SN38. Viability was assessed with Cell-Titer Glo (CTG) 72 hours after treatment. Using a matched PDX, mice were treated with vehicle or 40 mg/kg daily of plogosertib via oral gavage for 2 weeks, 5 days per week. Cell cycle analysis was performed by flow cytometry, and localization of DNA during mitosis was done by fluorescence staining. **Results:** CRC PDOs were more sensitive to plogosertib (CYC149462) than 5FU, and oxaliplatin, with significantly lower $IC_{50}$ values (518.86±377.47nM for plogosertib vs. 38.87±45.63μM for 5FU, and 37.78±39.61μM for oxaliplatin respectively (ANOVA, $P < 0.05$). Subsequently, we validated our findings *in vivo* using a matched PDX in which plogosertib treatment led to significant tumor growth inhibition compared to vehicle (t-test, $p < 0.05$) without serious adverse effects. Furthermore, we showed that plogosertib induces dysfunction in alignment of chromosomes and subsequent G2/M cell cycle arrest ($p < 0.05$). **Conclusions:** Together, our study shows that pharmacological inhibition of Plk1 using plogosertib represents a promising therapeutic approach for advanced CRC.
Research Sponsor: None.

**LBA210** **Poster Session**

# Short-course radiation(SCRT) followed by 6 cycles of cadonilimab plus mFOLFOX6 as neoadjuvant therapy for patients with locally advanced rectal cancer (LARC): A multicenter, single arm, phase II trial (NeoCaCRT).

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**The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.**

**211** **Poster Session**

# Sensitizing K-Ras G12D mutant colorectal patient-derived organoids via ferroptosis pathway.

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**Background:** Colorectal cancer remains the third most fatal form of cancer for males and females combined, with K-Ras mutations present in approximately 40% of all cases. These mutations drive cancer progression and confer resistance to conventional therapies, underscoring the need for novel treatment strategies. Ferroptosis, a regulated non-apoptotic form of cell death characterized as iron-dependent lipid peroxidation, has emerged as a target for cancer therapy. A study published in 2023 found that K-Ras mutant pancreatic ductal adenocarcinoma and non-small cell lung cancer cells lack ferroptosis induced lipid peroxidation and concurrently upregulated ferroptosis suppressing protein (FSP-1). It is unknown if K-Ras mutant colorectal tumors also evade ferroptosis-induced cell death as a method of cancer progression and therapy resistance. This study aims to measure the effects of inhibiting K-Ras G12D, the most common K-Ras mutation found in CRC, on ferroptosis through a patient derived colorectal cancer organoid (PDO) model. We hypothesize that blocking K-Ras G12D using a MRTX-1133, a K-Ras G12D specific inhibitor, can induce ferroptosis and synergistically work with ferroptosis inducers. **Methods:** As a pilot study, two patient derived organoid lines were established. Surveillance screening and whole exome sequencing was performed to verify K-Ras mutation. CRC9T has wild-type and CRC30T has mutant K-Ras. Additionally, Columbia Pathology Core confirmed tissues used to generate organoids were moderately differentiated colorectal adenocarcinoma. CTG-3D assays were performed to measure IC50 using ferroptosis inducers RSL3, iFSP, and K-Ras inhibitor MRTX-1133. Organoids were cultured in standard gut media for 72 hours, treated, and collected after 72 hours. After treatment, organoids were imaged and stained with BODIPY-C11, a lipid peroxidation fluorescence dye. Flow cytometry was performed, and results were analyzed using FlowJo to show ferroptosis-induced lipid peroxidation in stained organoids. **Results:** Inhibiting K-Ras G12D and inducing ferroptosis showed a synergistic effect on ferroptosis-induced lipid peroxidation. CRC9T was more sensitive (IC50 RLS3: 2.96 uM, iFSP 3.46 uM) to ferroptosis inducers than CRC30T (IC50: RSL3: 5.68 uM, iFSP: 8.6 uM). Blocking K-Ras in was only efficacious in CRC30T (IC50: MRTX1133: 9.73 uM). Cells from the mutant line showed greatest lipid peroxidation when combining ferroptosis inducers with K-Ras G12D inhibitors in flow cytometry analysis. **Conclusions:** These findings demonstrate a synergistic effect when blocking K-Ras G12D and inducing ferroptosis and align with previous studies that showed a connection between ferroptosis suppression and oncogenic KRAS expression. We plan to validate these findings in other patient derived organoid model systems and to assess the effects of pan-KRAS inhibition and ferroptosis inducer on non-K-Ras G12D PDOs. Research Sponsor: Herbert Irving Comprehensive Cancer Center Support Grant; P30CA013696 (JG, DD).

# Oncomine panel testing in the clinical management of colorectal cancer: An Irish experience.

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**Background:** Early onset colorectal cancer (EOCRC) is defined as colorectal cancer (CRC) in those under the age of 50. It has been hypothesized that the molecular signatures of EOCRC could differ from those of late onset CRC (LOCRC). Next Generation Sequencing (NGS) using the Oncomine Precision Assay is a useful tool to guide treatment decisions based on the presence or absence of oncogene drivers. While not all of these mutations currently influence the clinical management of CRC patients, ongoing research on the function of these genes and the development of new drugs could offer insights into patient prognosis, treatment choices and prediction of therapy resistance. **Methods:** Of >1500 CRC identified between 2010 and 2024, we investigated the clinical and genetic characteristics of 38 EOCRC and 41 LOCRC cases. All cases were microsatellite stable and staged I-III. NGS was performed with the Ion Torrent Genexus integrated platform using the Oncomine 45-gene Precision Assay panel. DNA hotspot mutations and copy number variations (CNVs) were assessed, followed by the classification of clinically relevant mutations using the genetic variant OncoKB database. **Results:** There were 38 EOCRC and 41 LOCRC cases. The age range was 27 to 49 and 50 to 91, with median ages of 44 and 72, in the EOCRC and LOCRC groups, respectively. Based on the stratification of NGS mutation levels as per the OncoKB classification, all patients in both groups (100%) had level 1 actionable mutations. NRAS wild type (WT) status was present in 100% of patients in both groups. However, differences were observed in KRAS status: 84% (32/38) of EOCRC were KRAS WT, while only 65% (27/41) of LOCRC were KRAS WT. While 3 % (n=1) of the EOCRC cohort had an ERBB2 amplification, none were detected in the LOCRC group. BRAF V600E mutations were present in 11% (4/38) of EOCRC and 12% (5/41) of LOCRC. Level 4 mutations were detected in 18% (n=7) of EOCRC, all of which were PIK3CA mutations, while these were seen in 24% (n=10) of the LOCRC group, including 9 PIK3CA mutations and 1 FGFR1 mutation. Regarding concurrent mutations, 2 patients in the EOCRC group had two mutations each: one with concurrent KRAS and PIK3CA mutations, and another with concurrent JAK2 and KRAS mutations. The LOCRC group had a higher incidence of concurrent mutations, with 8 patients affected: 6 had concurrent KRAS and PIK3CA mutations, and 2 had concurrent BRAF and PIK3CA mutations. **Conclusions:** Our study demonstrates that EOCRC have higher KRAS WT status and fewer concurrent mutations compared to LOCRC, who exhibit more frequent KRAS non-G12C oncogenic mutations. It also underscores the importance of routine NGS mutation panel analysis in CRC management. Additionally, identifying specific mutations in patients that relapse can guide targeted therapies, potentially improving outcomes and offering personalized treatment options. Research Sponsor: None.

# Genomic profiling of matched tumor samples in patients with rectal cancer undergoing total neoadjuvant therapy.

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**Background:** Total neoadjuvant therapy (TNT) is increasingly used for locally advanced rectal cancer. However, which patients benefit most from TNT remains to be elucidated. In this study we sequenced a TNT cohort with long-term follow-up to identify genomic correlates of TNT efficacy. **Methods:** Patients with locally advanced rectal cancer who underwent TNT followed by surgery (Oct 2016-June 2020) were identified at our institution. MSS patients with paired tissues (normal, biopsy, and surgery) with successful DNA extraction and sequencing were analyzed. Retrospective chart review collected patient demographics and clinical outcomes. Genomic analyses were performed with the R maftools package. Descriptive statistics were calculated with Fisher's exact, Wilcoxon rank sum, and Wilcoxon rank sum exact tests. Univariate survival analyses (UVA) were calculated with Cox proportional hazards regressions. All statistical analyses were performed using R (v4.3.2). **Results:** 53 patients met inclusion criteria and 32 patients with paired tissues (64 samples) were obtained and successfully sequenced. All patients received chemotherapy, most commonly FOLFOX (81.3%) as part of TNT and a 5-FU based chemoradiation therapy (RT) regimen. Almost all patients received 50.4 Gy (93.8%). Median followup was 57 months (IQR: 47-68). There were 24 pathological responders (R) [complete/partial] and 8 nonresponders (NR) [stable disease/progression]. R vs NR were not statistically different across demographic or treatment characteristics. Local recurrences were infrequent (9.4% 3/32). No covariates were prognostic on UVA survival analyses. However, NR was associated with worse distant metastases free survival (DMFS) (HR = 4.11, CI 1.01-16.8, p = 0.048) on UVA. 5-year DMFS was 83% for R and 50% for NR, and the median DMFS was not reached for R, 41 mo for NR. Tumor mutation burden (TMB) did not change with treatment (pre = 4.5 vs post = 6.2 mutations/Mb; r, 1.7-10.3, p=0.59) and was not different on biopsy between R and NR (4.1 vs 5.9, p=0.09). Pre-treatment, the most mutated gene was *APC* (81%), [majority nonsense/frameshift (88%)], followed by *TP53* (69%), [missense (59%)]. When comparing frequent mutations ($n \geq 5$), *PIEZO1* was more mutated in NR (63%, 5/8) than R (4% 1/24 patients) (p = 0.023 [multiple testing corrected]). Putative radio-resistant *KRAS-TP53* co-mutations were uncommon with similar representation in both groups (R 12.5%, 3/24, NR 12.5%, 1/8). **Conclusions:** Pathologic response positively correlated with DMFS in this cohort with long-term followup. *PIEZO1* mutations were negatively prognostic for pathologic NR to TNT. This uncovers a potential new role in chemoRT resistance in addition to a previously reported association with metastatic spread. Further evaluation of *PIEZO1* as a prognostic biomarker may help personalize appropriate treatment for patients with locally advanced rectal cancer. Research Sponsor: None.

# Plasma shallow whole-genome sequencing profiling in patients with *RAS*$^{WT}$ metastatic colorectal cancer (mCRC) undergoing first-line anti-EGFR therapy.

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**Background:** Acquired resistance (AR) to first-line (1L) chemotherapy (CT) + anti-EGFR inhibitors is a major challenge in the management of *RAS*$^{WT}$ mCRC patients (pts). While AR in later lines has been explained with point mutations in the *EGFR* pathway, its real mechanisms in 1L remain poorly understood. Tumor molecular biology can be dynamically studied through circulating tumor DNA (ctDNA) analysis, a real-time, non-invasive approach to detect AR. In this context, we conducted the multicenter prospective PLATFORM-B study, which enrolled 100 *RAS*$^{WT}$ mCRC pts receiving 1L CT + cetuximab across 15 Spanish Medical Oncology Departments, alongside 30 *RAS*$^{MUT}$ pts (control) treated with 1L CT + bevacizumab. Peripheral blood samples have been collected at baseline (BL), week 8, and disease progression (PD). Next-generation sequencing analysis (Ion S5 system) of ctDNA at PD revealed AR mutations in only 10% of pts, suggesting that alterations in the EGFR pathway may only partially explain AR in 1L. Therefore, we conducted shallow whole-genome sequencing (shWGS) to further characterize genomic profiles possibly related with novel mechanisms of AR. **Methods:** Baseline and PD samples with sufficient remaining plasma were considered for this analysis. shWGS was performed with a coverage ranging 0.5X to 2.0X and analyzed using ichorCNA pipeline to evaluate various genomic characteristics such as copy number alterations (CNA), tumor fraction (TF), subclonal fraction (SF), and ploidy. Molecular results were correlated to clinical-pathological features and outcomes. **Results:** Overall, 35 mCRC pts (27 *RAS*$^{WT}$ and 8 *RAS*$^{MUT}$) were included in the study. At BL ($N$ = 26), median TF (mTF) was 0.19, median ploidy 2, median SC fraction 0.5, median SC genome fraction 0.21, and median subclonal CNA fraction 0.41. At PD ($N$ = 33), mTF was 0.08, median ploidy 2, median SC fraction 0.5, median SC genome fraction 0.17, and median subclonal CNA fraction 0.38. Of all the features analyzed, only the TF showed potential prognostic value. At BL, pts with TF levels above the median had poorer overall survival (OS) compared to those with lower levels, approaching statistical significance ($P$ = .055). At PD, higher TF levels were statistically significantly associated with worse OS ($P$ = .02). No meaningful differences between the *RAS*$^{WT}$ and *RAS*$^{MUT}$ groups were observed. Specific variations in CNA, TF, SF, or ploidy under treatment are being correlated to clinical data to seek for predictive value. **Conclusions:** This preliminary analysis of the PLATFORM-B using shWGS showed the prognostic value of TF. Further analyses are undergoing to better understand the molecular landscape of acquired resistance to 1L anti-EGFR-based therapy in *RAS*$^{WT}$ mCRC pts.
Research Sponsor: None.

# Risk factors for relapse based on the molecular profiling of Japanese patients with stage II colorectal cancer.

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**Background:** The association between the molecular profiles and prognosis of Stage II colorectal cancer (CRC) remains unclear. This study aimed to evaluate the long-term outcomes of Stage II CRC based on the molecular profiling and to examine the risk factors for relapse. **Methods:** We retrospectively enrolled Japanese patients with pStage II CRC without pre- and/or postoperative adjuvant therapy and whose surgically resected specimens were evaluated using gene expression and whole exome analyses from January 2014 to December 2018. We evaluated the 5-year overall survival (OS) and relapse-free survival (RFS), and examined the risk factors for RFS. **Results:** We evaluated 322 patients with pStage II CRC, including 126 (39.1%) with right colon cancer, 133 (41.3%) with left colon cancer and 63 (19.6%) with rectal cancer. There were 87 patients (27.0%) with pT4, 175 (54.3%) with venous invasion, 120 (37.3%) with lymphatic invasion, and 68 (21.1%) with perineural invasion. Whole exome analyses revealed that there were 48 patients (14.9%) with MSI-High, and the presence of mutations in key genes for colorectal cancer development was as follows: *APC*, 245 (76.1%); *TP53*, 208 (64.6%); *KRAS*, 134 (6.5%); *PIK3CA*, 64 (19.9%); *FBXW7*, 27 (8.4%); and *BRAF*, 21 (6.5%). The 5-year OS and RFS rates were 98.0% and 90.7%, respectively. According to the consensus molecular subtype (CMS) classification based on gene expression, 76 patients (23.6%) had CMS4 and a significantly lower RFS than the other patients (5-year RFS: 83.8% vs. 92.9%, p=0.017). The independent risk factors for RFS were perineural invasion (Hazard ratio [HR] 5.316, $p<0.001$) and CMS4 (HR 2.399, p=0.020). **Conclusions:** Molecular profiling of Stage II CRC to assess the risk factors for relapse may contribute to the indication and drug selection for postoperative adjuvant chemotherapy and lead to personalized treatment to improve prognosis. Research Sponsor: Shizuoka Prefecture.

# Patient body fat and AI ensemble technique and the influence on false-positive rate in AI second observer for colorectal cancer detection.

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**Background:** Colorectal cancer (CRC) is the second leading cause of cancer-related deaths. We previously found that delayed diagnosis due to lack of radiological identification results in significantly worse outcome for patients. We had developed a rudimentary, AI second observer which demonstrated potential for detecting CRC on routine CT abdomen/pelvis (CTAP). However, the AI algorithm detected many false positives. In this study, we analyzed the data using TCIA as test cases and evaluated whether patient peritoneal fat content influenced the false positive rate. This could serve as a guide for future training of AI second observer to minimize false positive detection. **Methods:** 2D U-Net convolutional neural network (CNN) containing 31 million trainable parameters was trained with 58 CRC CT images from Banner MD Anderson (AZ) and MD Anderson Cancer Center (TX) (51 used for training and 7 for validation) and 59 normal CT scans from Banner MD Anderson Cancer Center. 18 of the 25 CRC cases from public domain data (The Cancer Genome Atlas) were used to evaluate the performance of the models (5 had no identifiable cancer and 2 were rejected for having no contrast). The CRC was segmented using ITK-SNAP open-source software (v. 3.8). To apply the deep ensemble approach, five CNN models were trained independently with random initialization using the same U-Net architect and the same training data. Given a testing CT scan, each of the five trained CNN models was applied to produce tumor segmentation for the testing CT scan. The tumor segmentation results produced by the trained CNN models were then fused using a simple majority voting rule (up to 2 voters) to produce consensus tumor segmentation results. The segmentation was analyzed for the number and location of false positives per case. The peritoneal fat content was classified at the level of aortic bifurcation by the distance of fat between adjacent small bowel loops ($\leq$ or $>$ 1 cm). Chi-square test was performed testing fat volume and number of voters as the intervention. **Results:** Our results showed that the higher volume of peritoneal fat ($>$ 1 cm, N=6) decreases the rate of false positive compared with low volume ($\leq$ 1 cm, N=12, p=0.013). When comparing between having one voter and two voter ensemble using low fat volume data, two voter ensemble also decreased the number of false positives but not statistically significant (p=0.286). **Conclusions:** Our results show that AI-based second observer generates more false positives when patients have lower peritoneal fat volume; this implies that future training may require higher percentage of cases with low peritoneal fat to improve second observer precision. Our analysis also showed that increasing the number of voter in the ensemble also decrease the number of false positives per case. Research Sponsor: None.

**217** **Poster Session**

# Development of new therapeutic strategies to enhance the effectiveness of irinotecan in colorectal cancer treatment.

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**Background:** Chemotherapy plays an important role in the treatment of colorectal cancer. Irinotecan, a topoisomerase I inhibitor, is used in therapies such as FOLFIRI and FOLFOXIRI. However, many patients become resistant to irinotecan, and the detailed mechanisms of resistance are not fully understood. The aim is to develop new therapeutic strategies by elucidating the mechanisms of irinotecan resistance. **Methods:** We used in vitro approach to demonstrate the mechanism of topoI degradation. Immunohistochemistry staining was used to explore the biomarker for irinotecan, Finally, in vivo approach was used to develop a new strategy for irinotecan treatment. **Results:** Mechanism of topoisomerase I (topoI) degradation: We investigated the molecular mechanism of topoI degradation in detail. After irinotecan administration, double-strand breaks occur in the DNA, activating DNA-PK. This activated DNA-PK phosphorylates the S10 residue of topoI, and BRCA1-BARD1 ubiquitinates the phosphorylated topoI. Eventually, topoI is degraded by the proteasome. Colorectal cancer cell lines where topoI degradation occurred after irinotecan administration showed irinotecan resistance. Development of a biomarker for irinotecan sensitivity: We created a monoclonal antibody that labels the phosphorylated S10 residue of topoI (topoI-pS10). We examined topoI-pS10 expression in colorectal cancer cases with a history of irinotecan treatment. The results showed a sensitivity of 87.5%, accuracy of 70%, positive predictive value of 70.7%, and negative predictive value of 87%, indicating that topoI-pS10 expression could serve as a biomarker for irinotecan sensitivity.Development of new therapeutic strategies: The degradation of topoI, which causes irinotecan resistance, is carried out via the ubiquitin-proteasome pathway. It is expected that combining irinotecan with a proteasome inhibitor, which blocks this pathway, will enhance irinotecan's effectiveness. When irinotecan was combined with a proteasome inhibitor in irinotecan-resistant colorectal cancer cell lines, cell death occurred at a higher rate compared to when they were not combined. Similar results were observed in experiments using mice. **Conclusions:** Based on the mechanism of topoI degradation, we developed a biomarker for irinotecan sensitivity and a new therapeutic strategy. This study suggests the potential for further improvements in irinotecan therapy. Research Sponsor: None.

# Diverse stromal phenotypes at spatial resolution in colorectal cancer peritoneal metastasis.

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**Background:** Peritoneal metastases (PM) in colorectal cancer (CRC) portend a poor prognosis. We sought to elucidate spatially resolved molecular features facilitating transcoelomic dissemination and progression. **Methods:** 50 PT, 116 PM and 37 primary normal samples from 96 patients were retrieved and profiled with digital spatial profiling (GeoMx DSP, Nanostring Technologies). Through unsupervised clustering, we characterized the microenvironment of tumor-stroma compartments and studied the roles of stromal phenotypes in promulgating tumorigenesis. These findings were orthogonally validated through spatial proteomics (multiplex IHC [mIHC] - COMET, Lunaphore Technologies) of 10 paired PT-PM samples retrieved from 6 patients. **Results:** A median of 37 region of interests (ROIs) (IQR: 32-38) were selected from a total of 9 tissue microarray (TMA) slides. A total of 269 ROIs (181 tumor, 88 stroma) from PM samples, 45 ROIs (34 tumor, 11 stroma) from PT samples and 28 (all stroma) ROIs from primary normal samples were retrieved and profiled. Through consensus clustering of stroma ROIs, we identify a fibro-collagenous and immune infiltrated stromal phenotype (stromal cluster [SC] 2) characterized by increased cancer associated fibroblasts (CAFs), memory B cells, M2 macrophages and T-cell exhaustion. Patients with SC2 stroma had poorer survival (p=0.036). SC2 stroma was also observed to support adjacent tumor compartments with enriched oncogenic pathways such as TGF-beta, TNF-alpha, hypoxia and JAK-STAT. Through discriminatory gene expression profiles, we developed a 20-gene composite bidirectional signature of SC2. The prognostic significance of the SC2-signature was externally validated through several cohorts including TCGA and MSI-H only cohorts. Next, by inspecting mIHC retrieved immune cell type densities across ROIs, we confirmed a similar phenomenon in which two distinct stromal clusters were found – one being fibroblast infiltrated but T-cell depleted (SC2-like, n=67 ROI) and another which is T-cell infiltrated but fibroblast depleted (SC1-like, n=104 ROI). Diverging spatial distributions of T-cells and fibroblasts between SC1 and SC2-like stroma was appreciated as well. Between both PT and PM, we find close clustering of T cells in SC1-like stromal compartments and fibroblasts in SC2-like stromal compartments. **Conclusions:** We describe SC2, a pro-tumorigenic stromal phenotype characterized by increased CAFs, T cell exhaustion and is associated with poor prognosis in CRC PM. Research Sponsor: National University Health System; NUHSRO/2024/008/RO5+6/Seed-Sep23/01; National University Hospital; JRA/Sep23/002; Yong Loo Lin School of Medicine, National University of Singapore; National Medical Research Council; MOH-CIRG21jun-0005; National Medical Research Council; MOH-CIRG24jan-0044; National Medical Research Council; MOH-CSAINV22jul-0005; National Medical Research Council; MOH-OFLCG18May-0003 and MOH-OFLCG23-may0031; National Medical Research Council; NMRC/CIRG23Jul-0035.

# Longitudinal ctDNA measurements for treatment response monitoring in patients with metastatic colorectal cancer undergoing systemic therapy: The ORCA trial.

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**Background:** Precise monitoring of treatment response may support and improve treatment decision-making in patients with metastatic colorectal cancer (mCRC). The PLCRC-ORCA study aimed to explore the clinical applicability of personalized ctDNA testing for treatment response monitoring in patients with mCRC. **Methods:** The prospective, observational ORCA study is a substudy of the Prospective Dutch ColoRectal Cancer cohort (PLCRC). After informed consent, blood samples were collected before initiating first line of treatment (baseline) and subsequently every 8-9 weeks during routine clinical care. ctDNA levels were analyzed using a clinically validated, personalized, tumor-informed 16-plex PCR NGS assay (Signatera™, Natera, Inc). **Results:** The majority of patients (45/49) had blood samples collected at both baseline and the first on-treatment time point. At baseline, 43/45 (95.6%) patients were ctDNA-positive with a median concentration of 63.5 mean tumor molecules per mL (MTM/mL; range 0.08 to 16,200). In a multivariate linear regression model including metastatic site at baseline and resection status of the primary tumor, ctDNA levels at baseline were lower in patients with metastatic disease limited to the lung ($n = 4$; $p < 0.001$) or peritoneum ($n = 6$; $p = 0.002$) compared to patients with liver-limited metastases. Moreover, primary tumor resection was independently associated with lower levels of ctDNA at baseline ($p = 0.05$). At the first measurement on-treatment, 13/45 patients (28.9%) were ctDNA-negative and 32/45 (71.1%) patients were ctDNA-positive with ctDNA levels ranging from 0.04 to 318 MTM/mL. Reduction in ctDNA levels relative to baseline was observed in 42/45 (85.7%) patients while two patients exhibited increased ctDNA levels. One patient had undetectable ctDNA levels at baseline and on treatment. Among the 30 patients with detectable but reduced ctDNA levels on treatment, the median ctDNA levels decreased 55-fold (95% CI: 26–600). **Conclusions:** The levels of ctDNA in patients with mCRC are influenced by the metastatic site and are lowest among patients with metastases limited to the lung and peritoneum. Our data suggest that clinical applications of ctDNA testing for treatment response monitoring should take the baseline ctDNA levels into account when interpreting ctDNA dynamics. Correlation of ctDNA status with radiological treatment response is underway and will be reported during the conference. Research Sponsor: None.

# The use of liquid biopsy comprehensive genomic profiling to identify the molecular landscape of patients with metastatic colorectal cancer who are candidates for anti-EGFR rechallange therapy: Findings from the CAVE-2 GOIM trial.

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**Background:** In the era of precision medicine, biomarker guided therapies are necessary for patients with chemorefractory metastatic colorectal cancer (mCRC). Circulating tumor DNA (ctDNA) comprehensive genomic profiling (CGP) may contribute to the development of more effective treatments. **Methods:** CAVE-2 GOIM trial (ClinicalTrials.gov ID: NCT05291156) is the first randomized phase II study to assess the efficacy of avelumab plus cetuximab compared to cetuximab alone as rechallenge strategy in pre-treated plasma ctDNA *RAS, BRAF* WT mCRC patients. Before being enrolled in the trial, patients were selected by baseline plasma ctDNA analysis by Foundation One Liquid CDx (F1L CDx) 324 genes to identify ctDNA *RAS/BRAF* WT cases. **Results:** As of July 31th 2024, 229 (out of approximately 310) patients have been screened. A total of 1562 pathogenic variants (PV) were detected, with *TP53* (86.0%), *APC* (84.2%), *DNMT3A* (30.5%) representing the most common altered genes. In these anti-EGFR pretreated patients different acquired mechanisms of resistance were observed. In particular, 72/229 (31.4%) patients had *RAS* or $BRAF^{V600E}$ PV (102 *KRAS*, 21 *NRAS* and 8 $BRAF^{V600E}$ mutations, respectively). Additionally, in these 72 patients, multiple $RAS/BRAF^{V600E}$ gene alterations (median of 1 PV, range 1-7) were observed in 41.6% cases. *MAPK*, *RTK*, and *PI3K/AKT/mTOR* PV were detected in 43.2%, 24.5% and 24.5% patients, respectively. Of note, the variant allele frequency of resistant PV was lower than other gene alterations, suggesting a high tumor heterogeneity as a consequence of the development and/or expansion of EGFR-inhibitor resistant cancer cell sub-clones. All patients were assessed for tumor mutational burden (TMB). Median TMB was 6 (IQR 4-10) mut/Mb, with 29.6% cases showing a TMB ≥10 mut/Mb. Finally, a total of 258 actionable genomic alterations were detected in 52,4% of patients according to the ESMO Scale for Clinical Actionability of Molecular Targets (ESCAT). Eighty-one tier I alterations were detected in 32.3% patients, which included TMB high (29,7%), $BRAF^{V600E}$ (3,5%), $KRAS^{G12C}$ (6,5%), and *RET* fusions (0,4%). **Conclusions:** These findings support liquid biopsy-based GCP for individualized therapeutic choices in chemorefractory mCRC. Up-dated final analyses of the entire screened patient population of the CAVE-2 GOIM trial will be presented. Clinical trial information: NCT05291156. Research Sponsor: CAVE-2 GOIM clinical trial is partially supported by a research grant from Merck KGaA Italy.

# Can overall survival (OS) benefit be predicted from improvements in progression-free survival (PFS) for previously untreated metastatic colorectal cancer (mCRC)?

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**Background:** With improvements in treatment of mCRC, OS maintains its gold standard efficacy measure but takes longer to mature than intermediate endpoints. This study analyzed the association between treatment effects on PFS and OS using aggregate-level data from RCTs in previously untreated mCRC patients. **Methods:** A systematic literature review identified RCTs in previously untreated mCRC patients published from 2010–2021 reporting hazard ratios on PFS ($HR_{PFS}$) and OS ($HR_{OS}$). All treatments in comparison to chemotherapy alone or chemotherapy with anti-VEGF (bevacizumab) or anti-EGFR (cetuximab) targeted therapy were considered. Correlation between $HR_{PFS}$ and $HR_{OS}$ was evaluated using bivariate random-effects meta-analysis (BRMA) and weighted linear regression (WLR). Predictive performance of the surrogacy equations from WLR was assessed using leave-one-out cross-validation (LOOCV). Surrogate threshold effects (STE), defined as the minimum PFS benefit that would translate into a statistically significant OS benefit with 95% probability, were also derived to gauge the practical utility of the models. Primary analysis consisted of all included trials. Sensitivity analyses omitted trials that (I) had anti-EGFR medications, (II) had anti-VEGF medications, (III) violated proportional hazards assumptions, and (IV) permitted treatment crossover. **Results:** In the primary analysis 47 trials were included. The estimated correlation between PFS and OS was 0.67 (95% CI: 0.48–0.80) using BRMA and 0.70 (95% CI: 0.48–0.84) using WLR. The surrogacy equation derived from WLR was $\log(HR_{OS}) = -0.03 + 0.56 \log(HR_{PFS})$ with a statistically insignificant intercept and significant slope. Estimated STEs corresponding to sample sizes of 200 and 300 patients were 0.55 and 0.62, respectively. Observed $HR_{OS}$'s were within their 95% prediction intervals predicted from $HR_{PFS}$ for 93.6% of studies in LOOCV. Sensitivity analyses produced moderate correlations with >90% coverage in LOOCV (Table). **Conclusions:** Moderate correlations were found between $HR_{PFS}$ and $HR_{OS}$ using both modeling approaches, highlighting the stability of the findings. Cross-validations of surrogacy equations indicated promising predictive value of PFS benefit for OS benefit in previously untreated mCRC. Research Sponsor: Bristol Myers Squibb.

| Analysis Set | # of Studies | Correlation (95% CI) | | STE | | LOOCV Coverage Rate |
|---|---|---|---|---|---|---|
| | | BRMA | WLR | N = 200 | N = 300 | |
| **Sensitivity Analysis I** | 34 | 0.75 (0.57, 0.86) | 0.78 (0.55, 0.90) | 0.55 | 0.61 | 94.1% |
| **Sensitivity Analysis II** | 14 | 0.44 (-0.08, 0.77) | 0.62 (-0.05, 0.90) | 0.44 | 0.52 | 92.9% |
| **Sensitivity Analysis III** | 36 | 0.61 (0.36, 0.77) | 0.59 (0.26, 0.80) | 0.46 | 0.55 | 94.4% |
| **Sensitivity Analysis IV** | 43 | 0.68 (0.48, 0.81) | 0.67 (0.41, 0.83) | 0.52 | 0.60 | 93.0% |

# Alterations in gut microbiome in age- and diet-induced colorectal cancer (CRC) progression in a preclinical model.

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**Background:** CRC is the third leading cause of cancer related deaths worldwide with age and diet among the strongest risk factors. Emerging evidence suggests gut microbiome plays important role in CRC and is highly impacted by the exposome primarily the food intake. Since diet and age plays an important role in gut microbiome diversity, the present study aimed to investigate the alterations in the gut microbiome in young versus old mice harboring CRC allografts and fed with different diets. **Methods:** Two cohorts of C57BL/6 including young (n = 9, age = 6 weeks) and old (n = 9, age = 20-24 months) mice were implanted with $1x10^6$ MSI (microsatellite instable) CRC MC38 tumor cells. Mice in both cohorts were randomized into three different diet groups: normal diet (ND; standard chow), high-fat (HF) and calorie-restricted (CR: 30% reduction in total calories). Mice were housed individually under standard laboratory conditions. Mice fecal samples were collected and subjected to DNA isolation (ZymoBIOMICS-96 MagBead DNA Kit), followed by metagenomic shotgun and whole genome sequencing (ZymoBIOMICSe and Illumina NovaSeq, respectively). $P \leq 0.05$ were considered as statistically significant. **Results:** Fecal microbiome analysis showed that old microbiome has higher alpha diversity compared to the young mice. No statistically significant differences in were found in microbiome diversities between the diet groups. Pairwise permanova results showed statistically significant differences between microbiomes of old vs. young groups (*P*: 0.001), CR vs. HF groups (*P*: 0.012) and HF vs. ND groups (*P*: 0.021). ANCOM-BC analysis determined the differentiating features and microbial functional pathways in the young mice compared to the old mice group and in different diet groups with relative abundance differences of larger than $\log_{10}2$. *Bacteroides thetaiotaomicron* (*P*: 1.43E-39) and *Parabacteroides goldsteinii* (*P*: 2.20E-23) were enriched in young and old groups, respectively. *Akkermansia muciniphila* (*P*: 0.008) was significantly enriched in ND compared to CR group. *Lactococcus lactis* (*P*: 9.94E-160) and *Lachnospiraceae bacterium A4* (*P*: 4.95E-10) were significantly enriched in HF compared to ND diet groups and vice-versa, respectively. Among the functional pathways, CMP-legionaminate biosynthesis I (*P*: 2.38E-13) and L-arginine biosynthesis IV (archaebacteria) (*P*: 4.77E-11) were the most significantly enriched in young and old groups, respectively. **Conclusions:** Our findings highlight the differential diversity and microbial features of gut microbiome in age- and diet-induced CRC progression. Collectively, alteration in diet and age of the host lead to changes in gut microbiota which has a direct impact on activation of specific signaling pathways, metabolism and local and systemic immune responses, which may in turn affect chemosensitivity and outcome in CRC Research Sponsor: National Cancer Institute; P30CA014089; Ming Hsieh research fund; National Cancer Institute; U2CCA252971; Gloria Borges WunderGlo Foundation; Dhont Family Foundation; Gene Gregg Pancreas Research Fund; San Pedro Peninsula Cancer Guild; Daniel Butler Research Fund; V foundation for cancer research; Victoria and Philip Wilson Research Fund; Fong research project.

# Impact of expression of metabolic genes on patient (pts) outcomes in metastatic colorectal cancer (mCRC): Data from CALGB/SWOG 80405 (Alliance).

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**Background:** Metabolic pathways are reprogrammed during CRC development, leading to increased glycolysis, glutaminolysis, and fatty acid synthesis. Understanding the impact of metabolic reprogramming on treatment outcomes is paramount. Hence, we investigated whether the tumor gene expression of 4 selected major metabolic genes (*PKM*, *SLC2A1*, *SLC16A1*, and *CAV1*) could affect treatment response in pts enrolled in the CALGB/SWOG 80405 trial. **Methods:** 433 mCRC pts treated with either bevacizumab (bev, n = 226) or cetuximab (cet, n = 207) in combination with first-line chemotherapy were analyzed. RNA was isolated from FFPE tumor samples and sequenced on the HiSeq 2500 (Illumina). Overall survival (OS) and progression-free survival (PFS) were compared between groups of pts categorized by tertiles of gene expression (high [H], medium [M] and low [L]). Logrank *P*-values provide a non-parametric unadjusted assessment of differences. Likelihood ratio tests (LRT) were computed from multivariate Cox proportional hazards models, adjusting for age, sex, ECOG, tumor side, number of metastatic sites, *KRAS*, CMS, and treatment. **Results:** *PKM* gene expression was associated with cet treatment outcomes with *PKM*-low tumors showing significantly longer PFS and OS (median PFS: 14.3 vs 9.8 vs 8.1 months, L vs M vs H, $P < 0.0001$, LRT $P = 0.011$; median OS: 45.9 vs 31.2 vs 20.9 months, $P < 0.0001$, LRT $P = 0.12$). Low *SLC2A1* was also associated with longer OS in cet-treated pts (32.4 vs 35.8 vs 25.2 months, $P = 0.02$, LRT $P < 0.0001$). Similar results were observed in bev-treated pts where low *SLC2A1* expressing tumors had longer OS (37.4 vs 26.1 vs 29 months, $P = 0.037$, LRT $P = 0.069$) and a non-significant trend for longer PFS (13.1 vs 11 vs 9.5 months, $P = 0.076$). *CAV1* low gene expression was associated with longer OS in cet-treated pts (37.4 vs 34 vs 21.5 months, $P = 0.0048$, LRT $P = 0.013$), while no significant associations were found in bev-treated (interaction LRT $P = 0.0009$). No significant results were observed for *SLC16A1*. **Conclusions:** The intratumoral expression of metabolic genes was prognostic and predictive in mCRC pts treated with first-line therapy. GLUT1 (encoded by *SLC2A1*) and PKM2 (encoded by *PKM*) promote aerobic glycolysis of cancer cells, known as Warburg effect, supporting rapid cell proliferation and survival. CAV1 has been shown to be involved in mitochondrial bioenergetics and fatty acid metabolism and to play either a tumor suppressor or oncogenic role depending on the cancer type and stage. Lower expression of *SLC2A1* and *PKM* was associated with improved survival and increased benefit from cet treatment in our cohort. Low *CAV1* expression was also associated with increase survival suggesting a cancer-promoting role in mCRC. Our results suggest that targeting cancer cell metabolism through novel inhibitors of the aforementioned pathways may be a promising therapeutic strategy. Research Sponsor: U.S. National Institutes of Health; P30CA014089, U10CA180821, U10CA180882, U10CA180820 (ECOG-ACRIN); U10CA180888 (SWOG); Pfizer, Genentech; https://acknowledgments.alliancefound.org.

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# Impact of education on clinical confidence utilizing HER2-targeted therapies in colorectal cancer.

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**Background:** Novel targeted therapies entered the treatment landscape in patients with HER2-positive metastatic colorectal cancer (mCRC). To ensure that these therapies are used effectively, clinicians must assess tumor HER2 status and recognize how HER2-directed treatment modalities fit into the current paradigm. To further enable clinicians to integrate HER2-targeted therapies, an educational program was designed in 2023. **Methods:** A one-hour online, video-based program was designed for clinicians and their teams and hosted on MedLive from February 2023 for one year. Knowledge and competence questions were administered pre-, and immediate post-activity. Additional questions assessed attitudes, barriers, and intended practice changes related to HER-positive mCRC. **Results:** Over 4,000 participants engaged in the educational activity, 67% of whom were physicians, and 63% noting their specialty as oncology. Approximately 68% identified as treaters seeing 14 patients with CRC per week with 78% of those patients have HER2-positive CRC. All pre-post questions showed significant improvements in knowledge and competence related to HER2 amplification testing methodologies, trial eligibility criteria, identification of patients who would benefit from anti-HER2 targeted regimens, and adverse events associated with anti-HER2 agents. Following the activity, 51% were more likely to incorporate testing for HER2 expression level or amplification prior to the selection of targeted therapy for their patients. The greatest challenges faced when managing patients with HER2-positive mCRC were affordability of therapy for patients (50%), adherence to treatment schedules (25%) and patient anxiety about treatment efficacy (24%). The most common barriers to patient enrollment in clinical trials identified by clinicians were lack of trials at their institutions (35%), lack of trials in their geographic region (21%), and patient lack of interest (20%). Qualitative insights on intended practice changes were shared by learners post-activity. **Conclusions:** The outcomes of this educational activity demonstrate the effectiveness of education in improving knowledge and confidence from testing to adverse event management to better integrate new regimens into practice. Encouraging practice changes were also seen from the write-in responses from learners. Additional needs were also identified to further solidify HER2 testing as a prerequisite for anti-HER2 targeted therapies, selecting eligible patients, and timely management of adverse events. Research Sponsor: Seagen Inc.

# Enhancing a low-cost, minimally invasive screening test for dihydropyrimidine dehydrogenase mutations linked to 5-fluorouracil sensitivity by integrating computational mutation predictions.

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**Background:** 5-Fluorouracil (5-FU), a common chemotherapy for solid tumors, is metabolized primarily by dihydropyrimidine dehydrogenase (DPD), encoded by the *DPYD* gene. With > 200 known variants, individuals with nonfunctional *DPYD* alleles exhibit impaired 5-FU metabolism and are at risk of severe toxicity. Approximately 35% of the population has partial DPD deficiency, and 0.2% completely lack enzymatic activity. Despite this, *DPYD* genotyping is not standard practice. This study evaluates the use of non-invasive saliva samples for *DPYD* genotyping and combines AI-based molecular prediction modeling to search for novel variants within *DPYD* coding region. Saliva samples were collected from fifty-six healthy individuals and from gastrointestinal (GI) cancer patients including a family cohort of nineteen individuals and three unrelated patients undergoing chemotherapy. **Methods:** gDNA was sequenced for nine *DPYD* variants reported to reduced or abrogate DPD activity: c.1905+1G > A, c.1679T > G, c.2846A > T, c.1129-5923C > G, c.1236G > A, c.299_302delTCAT, c.703C > T, c.2983G > T, and c.557A > G. cBioPortal was utilized to search the TCGA database for studies involving colon cancer patients with novel DPYD mutations exhibiting a total FIS score of at least 2 or identified through 3D modeling of DPD using PyMol based on their proximity to or the presence of polar contacts within the active site. Putative pathogenic mutations were analyzed using AlphaMissense and ChimeraX to assign a RSMD score, assessing their potential negative impact on DPD function. **Results:** Computational studies identified three mutations in DPYD from TCGA colon cancer patients with an unknown impact on DPD and a FIS greater than 4 (c.198G > C/T, c.2161G > A, c.2185G > A), suggesting potential disruption of its function. Four mutations (c.424T > G, c.427T > G, c.2460G > C, c.2909C > A) were identified through protein modeling using PyMol, but only two had a FIS greater than 2. Of these seven identified mutations, four (c.198G > C/T, c.2161G > A, c.2185G > A, c.2909C > A) also scored highly using AlphaMissense and ChimeraX. *In silico* prediction models from VarSome listed c.2185G > A as Pathogenic Moderate and c.2909C > A as Pathogenic Very Strong. Sequencing of saliva samples indicated mutations c.1129-5923C > G and c.1236G > A were present in four and three volunteers, respectively, and c.1905+1G > A in one case. Mutations identified in GI cancer patients were c.1129-5923C > G and c.1236G > A. **Conclusions:** By combining computational modeling with the analysis of naturally occurring mutations in colon cancer patients, we have successfully identified potentially pathogenic mutations in the *DPYD* gene. This information can enhance existing clinical diagnostic tests, providing a more comprehensive assessment of DPYD mutations, including novel variants. Research Sponsor: Fralin Biomedical Research Institute; Virginia Commonwealth Clinical Research Network.

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# Spatial transcriptomics and lethality-associated stromal remodeling of colorectal cancer.

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**Background:** Colorectal cancer (CRC) is surgically resectable and eminently screenable yet remains a lethal entity with high affinity to metastasise synchronously and metachronously. It has been proposed that cellular and extracellular components of the microenvironment contribute to metastatic potential. Dissociative profiling techniques such as bulk transcriptomic and single-cell RNA sequencing have contributed insight but limited therapeutic progress. Spatial transcriptomic (ST) assessment allows molecular profiling of tissue while preserving tissue architecture. Here we employ ST approaches to interrogate tumor compartments of primary resectable and matched synchronous CRC demonstrating stromal signatures with distinct collagen expression associated with outcome. **Methods:** 25 patients with primary resectable CRC and 4 patients with matched primary CRC and liver metastasis (CRLM) underwent single-cell spatial transcriptomics using the Nanostring CosMx Spatial Molecular Imager (SMI, 1000plex gene panel) (Discovery cohort). 71681 epithelial and 88806 stromal cells with intact spatial resolution were analysed. The spatial single-cell signatures were reconstituted in 89 patients using the Nanostring GeoMx Digital Spatial Profiler (Validation cohort). 3 GeoMx compartments were analysed: Epithelial (PanCK+); Stroma (aSMA+); Immune (PanCK-aSMA-). **Results:** CosMx demonstrated 2 distinct collagen signatures: COL1A1,COL1A2,COL3A1 associated with normal fibroblasts; COL9A2 associated with cancer-related stroma and lethal subtypes of epithelial cell. In the GeoMx validation cohort, 42 patients expressed the COL1A1 signature in the aSMA compartment and 47 did not (5 year recurrence free survival: 0.88 vs 0.49, $p < 0.005$) with different morphological patterns of each distinct aSMA group. In patients with favourable prognosis, chemokine high epithelial subtypes expressed CXCL5 and recruited COL9A2-, IL6+ neutrophils to the surrounding microenvironment. In contrast, epithelial cells in patients with poor prognosis expressed CXCL8 and recruited COL9A2+ neutrophils which were frequently in contact with SPP1+ macrophages. **Conclusions:** We have used ST approaches to interrogate cellular compartments of CRC demonstrating stromal subtypes that impact outcome. These insights could be used in the clinical setting to quantify COL9A2 expression in aSMA+ cells to prognosticate patients. Targeting of tumor collagen has been proposed to augment existing anti-cancer therapies and this work has demonstrated COL9A2 as a potential target in need of further investigation. Research Sponsor: None.

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# Unbiased transcriptomic analysis of primary early-onset vs late-onset colorectal cancer.

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**Background:** Colorectal cancer is the 3rd most diagnosed and 2nd leading cause of cancer death. Despite the decline in incidence late-onset colorectal cancer (LOCRC), there is an increase in early onset of colorectal cancer (EORC). Clinicopathologic examinations from single centers revealed left-sided and rectal enrichment, poorly differentiated, and higher stage at presentation. Genomic studies examining the potential germline mutations/variants for EOCRC demonstrated mixed results. Transcriptomic analysis of the EOCRC compared to LOCRC remains limited. We hypothesized that EOCRC present with enrichment in aggressive molecular signatures consisting of CMS4 and iCMS3 signatures. To understand the mechanism behind EOCRC, we performed an unbiased RNA-Seq analysis in age groups of 32-82 in 95 patients. **Methods:** We utilized RNA-Seq data from the Orien Network, a collaborative between 19 NCI/NCCN cancer centers. We identified 95 patients diagnosed with colorectal cancer ranging in age from 32-82 and undergone RNA-sequencing of their primary tumors. Clinicopathologic assessment were analogously performed. We collected unnormalized raw-count RNA-Seq data from these patients. To account for differences in sequencing depth across samples, raw counts were scaled using the Reads Per Million (RPM) formula. The scaled reads in RPM were then utilized to assess the Consensus Molecular Subtypes (CMS) classification through two distinct approaches: the R package CMScaller and a deconvolution analysis using the R package SpatialDecon with the iCMS gene signature. Differences in age at the sample collection across patients with different CMS classifications were assessed by performing two-tailed unpaired T-tests. Additionally, differential expression analysis between EOCRC and LOCRC patients, defined by the upper and lower quartiles of the patients' ages at the sample collection, was conducted using the LIMMA R package. **Results:** Clinicopathologic assessment of the primary tumors revealed left-sided tumors with poorly differentiated phenotype. Differential gene expression analysis between EOCRC and LOCRC demonstrated no significant differences. Stratification of transcriptomic data by decades also did not reveal any significant differences. Furthermore, there were no differences in CMS or iCMS classifications between different age groups. **Conclusions:** The RNA-Seq analysis comparing EOCRC and LOCRC did not identify any differential gene expression patterns. There was no significant differences were observed in CMS or iCMS classifications. These findings suggest that EOCRC and LOCRC share similar tumor biology based on molecular phenotypes. This implies that the initiating events for the disease may occur earlier in life while triggering a same disease process. As such, the findings underscores the importance of considering preventive measures, including earlier colonoscopy screenings. Research Sponsor: None.

**229** **Poster Session**

# Phase II study of cabozantinib and nivolumab in refractory metastatic microsatellite stable (MSS) colorectal cancer (CRC).

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**Background:** Current data suggest that combining tyrosine kinase inhibitors with immune checkpoint inhibitors may be a promising treatment strategy in patients with metastatic MSS CRC. This prospective, open-label, single-arm, phase II study (NCT04963283) evaluated the efficacy of cabozantinib in combination with nivolumab in patients with metastatic refractory MSS CRC. **Methods:** Patients with metastatic/unresectable MSS CRC who were refractory to chemotherapy in the 3$^{rd}$ line setting and beyond were eligible. Patients were treated with cabozantinib 40 mg orally daily and nivolumab 480 mg IV every 28 days. Tumor assessments were obtained every 8 weeks for the first 16 weeks, and then every 12 weeks thereafter. Response was assessed via RECIST v1.1. The primary endpoint was 16-week disease control rate (16-wk DCR); secondary endpoints were objective response rate (ORR), progression free survival (PFS), overall survival (OS), safety and tolerability, and correlative studies (to be presented at a future date). **Results:** Between June 2021 and January 2024, 49 pts were enrolled; 63% were male; median age 55 [50-64]. Median prior treatment regimens were 4 [2-14]. Most patients (84%) had not received prior trifluridine/tipiracil; only 1 patient had received prior regorafenib. The 16-wk DCR was 40% (19/47) (95% CI 0.264 – 0.557). The 16-wk ORR was 8.5% (4/47) (95% CI 0.024 – 0.204). mPFS was 3.4 months (95% CI 1.8-5.0) and mOS was 10.9 months (95% CI 4.7-13.8). Thirty nine percent of patients experienced a serious adverse event (SAE), with 4% Gr 1, 37% Gr 2, 35% Gr 3 and 4% Gr 4; there were no Gr 5 treatment related adverse events. The most common AEs were diarrhea (53%), fatigue (43%), hypothyroidism (31%), weight loss (29%), nausea (29%), and hypertension (27%). Two patients remain on study at time of abstract submission. **Conclusions:** The combination of cabozantinib and nivolumabwas well tolerated and provided modest benefit in a population of patients with metastatic, chemotherapy refractory, MSS CRC. Clinical trial information: NCT04963283. Research Sponsor: None.

# Preclinical analysis and clinical validation to identify biomarkers for trifluridine/tipiracil (FTD/TPI) efficacy with or without bevacizumab in patients with metastatic colorectal cancer.

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**Background:** We previously identified candidate biomarkers for FTD/TPI efficacy through preclinical and translational studies in patients with metastatic colorectal cancer (mCRC). Recent studies reported that FTD/TPI plus bevacizumab (BEV) improved survival compared to FTD/TPI alone. This study evaluated the clinical significance of these biomarkers in patients with refractory mCRC receiving either FTD/TPI plus BEV or FTD/TPI alone. **Methods:** Candidate biomarkers were selected based on our transcriptomic and cell biological analyses. We validated these biomarkers in mCRC patients treated with FTD/TPI plus BEV (cohort A) or FTD/TPI alone (cohort B). Blood samples were collected at baseline (BL), before the second cycle (2nd), and progressive disease (PD), with serum biomarker levels measured using ELISA. Change patterns were classified as 'increased' or 'decreased' from BL. To address the unbalanced distribution of baseline characteristics between cohorts, propensity score matching (PSM) was performed. **Results:** A total of 112 patients were included, with 34 in each cohort after PSM. Cohort A showed significantly longer overall survival (OS) (14.4 vs 6.2 months, HR 0.53, 95%CI: 0.30-0.93) and higher disease control (DC) (58.8 vs 34.4%, P=0.047) compared to cohort B. In univariate analysis, increased IL-8 or AREG at 2nd and increased IFI16 at PD were associated with poorer clinical outcomes. Grade ≥3 neutropenia (NEU) was linked to better survival in both cohorts. Multivariable analysis identified decreased AREG and Grade ≥3 NEU as better markers for progression-free survival and OS. OS was significantly longer in cohort A compared to B (12.8 vs 5.8 months, HR 0.43, 95%CI: 0.21-0.89) when IL-8 increased at 2nd. Cohort A also had longer OS compared to cohort B when EREG (12.8 vs 6.2 months, HR0.27, 95%CI: 0.11-0,65) or AREG (14.2 vs 9.7 months, HR 0.40, 95%CI: 0.17-0.95) decreased at 2nd. There was no significant interaction between the regimen and biomarkers (IL-8, EREG, and AREG) for OS. **Conclusions:** Our data suggest that AREG may serve as a prognostic marker for FTD/TPI efficacy. Changes in IL-8, EREG and AREG potentially contribute to the extended survival benefit of combining BEV with FTD/TPI. Research Sponsor: JSPS KAKENHI.

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# Proto-oncogene HRAS transcript level and overall survival in stage II and III colorectal cancer.

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**Background:** Mutational landscape is prognostic in colorectal cancer (CRC). Driver mutations in the RAS proto-oncogenes such as KRAS and NRAS portend poor survival outcomes, whereas HRAS mutations are extremely rare and their prognostic value remains uncertain. The prognostic implication of RAS gene activity at the transcript level is also obscure. Here, we retrospectively review stage II and III CRC tumor RNA-Seq data and demonstrate that high transcript levels of HRAS are associated with superior overall survival (OS). **Methods:** Tumor RNA-Seq of 734 stage II and III CRC cases and their associated clinical data were retrieved from the ORIEN AVATAR database (Aster Insights). Transcript level was determined by batch-corrected TPM (transcripts per million) of each gene, and cohort median was used as a reference to determine "high" versus "low" in subgroup analyses. The p-value and hazard ratio (HR) derived from Kaplan-Meier survival analysis are log-rank. **Results:** The 4-year HR for death in patients with "very high" HRAS transcript level (top 5%) was 0.15 (95% CI 0.07-0.34, p=0.03) compared to the rest, and this superior OS was independent from TNM stage or peri-operative 5-FU treatment history. In subgroup analyses, "high" HRAS expression was associated with superior OS only in patients with concomitant "low" KRAS expression in the absence of pathologic KRAS mutations in codons 12, 13 and 61 – the 5-year HR in "low" KRAS group was 0.42 (95% CI 0.18-0.96, p=0.03) compared to "high" KRAS group. NRAS transcript level did not influence OS. When further sub-grouped based on primary disease location, this superior OS was present only in the right-sided primary disease group with 5-year HR of "low" KRAS group at 0.17 (95% CI 0.03-0.99, p=0.06) compared "high" KRAS group. **Conclusions:** Contrary to the notion that RAS family genes are proto-oncogenic, we demonstrate that high HRAS transcript levels are associated with superior OS in stage II and III CRC. The high HRAS-associated OS benefit was most pronounced in patients with right-sided primary expressing low KRAS transcript levels in the absence of pathologic KRAS mutations. We speculate that when expressed at high levels, HRAS may counteract proto-oncogene KRAS but not oncogene KRAS at the transcript level. The potential of HRAS as a prognostic biomarker should be explored further. Research Sponsor: None.

# Performance of a blood-based screening test for the early detection of colorectal cancer.

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**Background:** While the clinical benefit of highly sensitive screening tools for colorectal cancer (CRC) has been established, adherence rates remain low. The convenience and familiarity of blood-based testing may reduce disease burden through greater access and compliance to regular and recommended asymptomatic screening. We report on preliminary findings from a minimally invasive blood-based early cancer detection (ECD) screening test for CRC. **Methods:** Plasma samples from healthy individuals and CRC patients sourced from biobanks were included. A targeted panel composed of differentially methylated genomic regions was developed. Using an independent cohort of plasma samples, a machine-learning model was trained to classify patient samples into cancer-free and CRC based on observed differential methylation patterns in circulating cell-free DNA (cfDNA). The assay performance was calculated based on concordance with clinical diagnosis (CRC, cancer-free). **Results:** The test set included 440 individuals: 127 with a confirmed CRC diagnosis (47% stage I/II, mean age 65±12 years, male 46%) and 313 without cancer (mean age 56±10 years, male 41%). Overall sensitivity was 95% (95% CI: 88-100%), and specificity was 91% (95% CI: 84-98%). Among patients with stage I disease, sensitivity was 92% overall, and 88% in screen-detected individuals. Sensitivity was 100% (n=14/14) in microsatellite-high tumors and 95% (n=107/113) in microsatellite-stable tumors. Multivariate analysis demonstrated associations between methylation signal and tumor size, cancer stage, and sex. Compared with a tumor-informed, variant-based circulating tumor DNA (ctDNA) test (Signatera), we observed a PPA (positive percent agreement) of 100%. Moreover, the differentially methylated allele fraction (DMAF) across target regions strongly correlated with variant allele frequencies (VAFs) from the tumor-informed test (Spearman's correlation coefficient, 0.85). **Conclusions:** Using epigenetic markers specific for CRC, we developed a sensitive and specific blood-based test to detect cfDNA from early-stage and asymptomatic disease. The methylation signal strongly correlated with ctDNA VAFs from a clinically validated tumor-informed test and increased with cancer stage and disease burden. This test could serve as a means to increase patient adherence rates for CRC screening, thereby improving patient outcomes. Research Sponsor: None.

# PK/PD analyses of circulating tumor DNA (ctDNA) in patients with metastatic colorectal cancer (mCRC) treated with telisotuzumab adizutecan (ABBV-400).

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**Background:** Telisotuzumab adizutecan (ABBV-400) is an antibody drug conjugate (ADC) consisting of a c-Met targeting antibody telisotuzumab conjugated to a potent topoisomerase 1 inhibitor payload adizutecan. The ongoing first in human phase 1 trial (Study M21-404; NCT05029882) shows that ABBV-400 has encouraging radiographic response and ctDNA molecular response (MR) as monotherapy in patients with advanced solid tumors, including mCRC. Herein, correlation between ABBV-400 exposures and % change in ctDNA from baseline and PK/PD analyses using MR are presented for mCRC patients. Impact of c-MET expression and *MET* amplification (*MET*-amp) status on exposures was also evaluated. **Methods:** Analyses utilized preliminary data in mCRC patients (N=122) from study M21-404 where subjects were treated with ABBV-400 (1.6 – 6.0 mg/kg Q3W). Serial PK samples were collected in cycles 1 (C1) and 3 (C3) and sparse samples across other cycles. C1 exposure metrics ($C_{max}$, $C_{avg}$, and $C_{trough}$) for both ABBV-400 conjugate and unconjugated payload were obtained by non-compartmental PK analysis. Baseline and C3 day 1 plasma samples were collected and analyzed using the Guardant INFINITY assay for biomarker analysis. ctDNA (circulating tumor fraction [cTF]) was estimated based on 1) variant allele frequency of somatic mutations in a 74 gene panel and 2) methylation signals across targeted regions of the Guardant INFINITY methylation panel. MR was defined as a 50% decrease in cTF from baseline. PK/PD correlations with % change in cTF at C3 and MR at C3 were evaluated. c-Met protein expression was assessed by IHC (VENTANA MET SP44 RxDx Assay) and *MET*-amp status was based on available local testing results and retrospective ctDNA testing. **Results:** Higher exposures of ABBV-400 conjugate and payload (primarily $C_{avg}$ for both analytes and $C_{max}$ for conjugate) correlated with greater decrease in % change in cTF at C3 from baseline, as assessed by the 74 gene panel (N =66) or methylation panel (N=74). PK/PD analysis showed that higher exposure was strongly correlated with higher probability of MR (n=66-74, nominal $p < 0.05$). ABBV-400 conjugate and payload exposures were comparable across c-Met expression H-score (range 9 – 295), cut-offs (≥90% with 1+ intensity, ≥50% with 2+, ≥10% with 3+, ≥25% with 3+, and ≥50% with 3+), and *MET*-amp status. **Conclusions:** Exposure-response analyses have previously shown correlation between ABBV-400 conjugate exposures and probability of efficacy (objective response rate) as well as safety in mCRC subjects. The current PK/PD analyses demonstrate that higher ABBV-400 conjugate and payload exposures are also correlated with liquid biopsy data, i.e., ctDNA and MR based on ctDNA. c-MET expression and MET-amp did not have impact on ABBV-400 exposures. Clinical trial information: NCT05029882. Research Sponsor: AbbVie.

# A novel immune-based score and its immune contexture exploration in colorectal cancer.

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**Background:** The majority of currently available biomarkers in colorectal cancer (CRC) perform poorly with respect to prognostication and prediction of response. Immunoscore (IS) has been proposed as a signature of host immune cancer response reflecting a favorable prognosis. Next, through AtezoTRIBE, 'Immunoscore immune checkpoint' (IS-IC) was introduced, indicating its potential predictive value for immune checkpoint blockade (ICB) in pMMR metastatic CRC. IS-IC is computed through quantitative and spatial variables related to density and proximity of CD8 cells and programmed cell death-ligand 1 (PD-L1) cells in the tumor core (TC). There are little confirmatory studies. In this work we test the application of IS-IC through multiplex immunofluorescence (mIF) on our own cohort of patients. Next we explore immune contexture differences through subgroup analysis. **Methods:** 15 prospectively collected, treatment naïve, primary CRC samples (9 dMMR/MSI-H, 6 pMMR/MSS) where analyzed with mIF. Cell density, proximity between cells, cell clustering and spatial distribution of selected mIF biomarkers (CD3, CD20, CD11c, CD163, CD15, CD4, CD8, Foxp3, CD56, Granzyme B, PD-L1, programmed cell death protein 1 (PD-1), pan-cytokeratin (CK) and Ki67) were analyzed both at the invasive margin (IM) and TC using digital pathology. All slides subjected to digital image analysis met predefined quality control criteria. **Results:** Serial mIF showed PD-L1 was enriched in stromal cells ($CK^-$) within IM ($p$=0.0148) and confirmed its co-localization with macrophages ($CD163^+$) ($p$=0.0092). Hence, both TC and IM were included to quantify the densities of CD8, PD-L1 and the proximity between them, as a suggested novel IS-$IC_{TC+IM}$. During exploration of the immune context we found that both IS-IC and IS-$IC_{TC+IM}$ high groups exhibited significantly increased infiltration of PD-$1^+$ cells ($p$=0.0074 and $p$=0.0058, respectively) and $CD8^+$PD-$1^+$ cells ($p$=0.0101 and $p$=0.0062, respectively), compared to the corresponding low groups. Whereas only the IS-$IC_{TC+IM}$ high group showed increased natural killer (NK) cells ($CD56^+$) ($p$=0.0192), found mainly at IM ($p$=0.0167) in contrast to TC ($p$=0.0824). The novel IS-$IC_{TC+IM}$ scoring system captures the immunosuppressive and immune responsive interaction among PD-$1^+$ cells, PD-$L1^+$ cells, NK cells and $CD8^+$ cells in TC and IM from the cell numbers and cell-cell proximity. A validation cohort is underway. **Conclusions:** In this study we confirm the ease of replicating IS-IC and the presence of ICB sensitive stroma in the high score group. This study further explored the immune context in both TC and IM, emphasizing the importance of regional differences. Additionally it was found that NK cells were only increased at IM. Altogether we suggests adding dual quantification in TC and IM as a novel IS-$IC_{TC+IM}$, which may be a more sufficient reflection of the tumor micro environment. Research Sponsor: None.

# A comprehensive molecular and clinical study of patients with young-onset CRC.

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**Background:** Young-onset colorectal cancer (YO-CRC), defined as colorectal cancer diagnosed in individuals under 50 years of age, has emerged as a distinct clinical entity, often presenting at advanced stages. Despite the increasing incidence, molecular and clinical underpinnings of YO-CRC remain underexplored. This study aims to characterize the clinical and molecular features of YO-CRC and to evaluate their impact on OS. **Methods:** We retrospectively reviewed 110 patients diagnosed with YO-CRCfrom our institution's molecular database. All patients underwent next-generation sequencing. Demographic, clinical, and molecular data, including age, gender, race, tumor location, cancer stage, and mutation status (KRAS, NRAS, BRAF, POLE, ERBB-2/HER2, microsatellite status), were collected by reviewing electronic medical records. For OS analysis, we focused on patients diagnosed with de novo Stage 4. Cox proportional hazards regression and Kaplan-Meier survival analysis were utilized to assess the association of these factors with OS, with statistical significance determined by a p-value threshold of $<0.05$. **Results:** Among 110 patients, N=44 (40%) presented with local disease (stage 1-3), while N=66 (60%) patients presented with de novo metastatic disease at the time of diagnosis. The median age at diagnosis was 44.5 years. The cohort consisted of 64% males and 36% females, with 84% of patients identified as White. Most tumors were left-sided (77%), including the distal colon/sigmoid (44%) and rectum (33%). KRAS and BRAF mutations were present 36% and 5.5%, respectively. ERBB-2/HER2 amplification and microsatellite instability were observed in 4.5% and 6.4% respectively. Tumor mutation burden (TMB) was $<10$ in 57% of patients, with 14% having TMB $>20$. CNV analysis revealed that 14% of patients had copy gains, 12% had concurrent gains/losses, and 31% had copy losses. Among the 66 Stage 4 patients, 44% had died by the time of analysis, with a median overall survival (OS) of 43.6 months (95% CI, 28.7 - not reached). KRAS mutation was found to be significantly associated with worse survival outcomes. Cox regression analysis reveals the prognostic significance of KRAS status, with a hazard ratio (HR) of 3.52 (95% CI: 1.59-7.76, $P = 0.002 < 0.05$), indicating a significantly higher risk of death for KRAS-mutant YO-CRC patients. **Conclusions:** KRAS mutations are a key prognostic factor in YO-CRC, highlighting the need for therapeutic need to improve outcomes in this high-risk group. Research Sponsor: None.

# Role of cfDNA and DNA integrity index (DII) in colon cancer in response to surgery.

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**Background:** Colorectal cancer diagnosis requires new biomarkers for improved detection and monitoring. cfDNA has shown potential in detecting minimal residual disease, evaluating treatment response, and providing prognostic information. Recent studies highlight cfDNA as a promising molecular marker for early disease detection and progression monitoring. This study aims to evaluate the role of cfDNA and DII in assessing surgical response in CRC patients. The primary objective is to assess cfDNA levels pre- and post-curative resection. Secondary objectives include identifying factors influencing baseline cfDNA levels, evaluating factors affecting DII, and comparing cfDNA and CEA levels in response to surgery. Statistical analyses included descriptive statistics, ROC curve, Student T tests and Fisher's exact test. **Methods:** The study was designed as a prospective investigation conducted over a period from September 2022 to April 2024 at the Regional Cancer Centre. Forty pathologically proven non metastatic colonic carcinoma planned for primary surgery were included and patients with sepsis and chronic diseases were excluded from the study. Blood samples (10 ml) were collected pre-surgery and at four weeks post-surgery. Plasma cfDNA was extracted, quantified, and analyzed via quantitative real-time PCR using human β-actin as a reference. The DII was calculated as the ratio of 394 bp to 99 bp amplicons of β-actin. **Results:** The mean age was 62.15 years (range 39-85), with 55% over 60 years. High-risk features included poor differentiation (4.2%), pre-operative CEA >5 ng/ml (17.5%), LVI (20.8%), and PNI (6.3%). Surgical approaches included 10 laparoscopic, 29 open, and 1 converted from laparoscopic to open due to excessive bleeding. Overall stage grouping showed 18 patients in Stage III, and 11 each in Stage I and II. Median pre- and postoperative cfDNA levels were 24.2 ng/μl and 29.6 ng/μl (p=0.83), and median DNA integrity index levels were 1.1 and 1.0 (p=0.95). Preoperative cfDNA levels averaged 24.2 ng/μl and postoperative levels 29.6 ng/μl (p = 0.83); DII levels were 1.1 preoperatively and 1.0 postoperatively (p = 0.95). Postoperative cfDNA levels decreased in 21 patients and increased in 19, while DII decreased in 20 and increased in 20. Significant associations were noted for cfDNA changes with surgery type and high-risk features. ROC analysis revealed cfDNA cut-off at 275 ng/μl, DII at 1.5, and CEA at 4 μg/L. Sensitivities for cfDNA and DII were 66.67% and 100%, respectively, with combined sensitivity reaching 100%. **Conclusions:** Persistent high cfDNA levels post-surgery suggest potential minimal residual disease despite radical resection. The residual disease is a harbinger of systemic failure. Monitoring cfDNA and DII can identify patients at risk for recurrence, guiding intensified adjuvant therapy. Further studies with extended follow-up are recommended to validate these findings. Clinical trial information: CTRI/2023/10/058468. Research Sponsor: None.

# Clinical relevance of tumor fraction assessment from circulating tumor DNA in metastatic colorectal cancer.

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**Background:** Both tissue (TBx) and liquid (LBx) biopsies are indicated for detecting actionable alterations in metastatic colorectal cancer (mCRC). LBx is easier to obtain and often has faster turnaround time, but its informative results depend on the ctDNA tumor fraction (TF) content. We aimed to (1) evaluate the concordance between LBx and TBx in CRC; and (2) determine baseline lab and clinical factors associated with TF. **Methods:** CRC patients (pts) with both tissue and liquid Foundation Medicine comprehensive genomic profiling within an interval of up to 90 days between samples collection were analyzed. Positive percent agreement (PPA) and negative predictive value (NPV) for RAS (KRAS/NRAS) and BRAFV600E detection were calculated with tissue as reference. Clinical data was obtained by the US-wide de-identified Flatiron Health-Foundation Medicine real-world clinicogenomic CRC database, from ~280 cancer clinics (~800 sites of care) between 1/2011-12/2023. mCRC pts with LBx collected up to 60 days prior to therapy (Tx) start were analyzed for association between TF and baseline factors using logistic regression with TF≥1% as the binary outcome. Baseline factors include age, ECOG, race/ethnicity, line of therapy, practice type, metachronous vs. synchronous disease, tumor side, albumin, alkaline phosphatase (AP), serum creatinine, hemoglobin, lactate dehydrogenase (LDH), neutrophil-to-lymphocyte ratio, opioid and steroid pre-Tx. Foundation Medicine's ctDNA TF is a composite algorithm prioritizing aneuploidy at higher levels to avoid germline signal and prioritizing variant allele frequency of canonical alterations at lower levels to maximize dynamic range. **Results:** A total of 402 pts had TBx and LBx (268 [67%] TF≥1% and 134 [33%] TF<1%). Without accounting for TF, the overall PPA for RAS and BRAFV600E was 77% and 83%, and the NPV was 65% and 99%. In LBx with TF≥1%, the PPA for RAS and BRAFV600E was 99% and 100%, and NPV was 98% and 100%. In contrast, for TF<1% samples, the PPA for RAS and BRAFV600E was 37% and 50%, and NPV was 57% and 97%. Of the 633 pts with LBx collected prior to Tx, 474 (75%) had TF≥1%. TF≥1% was independently associated with ECOG 1+ (odds ratio [OR]: 1.58, p=0.038), community oncology care site (OR: 2.58, p=0.002), elevated levels of AP (OR: 3.00, p<0.001) and LDH (OR: 3.97,p=0.002). TF≥1% was less likely to occur in right-side tumors (OR: 0.53, p=0.012) and metachronous disease (OR: 0.48, p<0.001). **Conclusions:** Approximately 70% of LBx samples from CRC pts have TF≥1%, with near-perfect concordance for RAS/BRAFV600E. For LBx with TF<1%, interpretation of negative results can be challenging due to low tumor shedding, especially for RAS mutations, where 43% may be false negatives and a subsequent TBx will provide more confidence in the negative results. Clinical factors such as right-sided tumors and metachronous disease may help anticipate low TF and guide decisions to pursue TBx. Research Sponsor: Foundation Medicine, Inc.

# Frequency and impact of a genome-wide homologous recombination deficiency signature (HRDsig+) on the genomic landscape of advanced colorectal cancer (CRC).

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**Background:** There have been significant advancements in novel therapies that target the DNA damage response pathway, and utilizing scar-based measures of homologous recombination deficiency (HRD) beyond genomic alterations (GA) in *BRCA1/2* may provide opportunities for additional biomarker-driven therapies for patients with advanced CRC. **Methods:** We identified 47,028 cases of advanced CRC for comprehensive genomic profiling (CGP) to examine all classes of GA and measure HRDsig+ status. HRDsig status was calculated using a broad set of genome-wide copy number features (PMID 37769224) and was correlated with microsatellite status, tumor mutation burden (TMB), genomic ancestry, trinucleotide mutational signatures, and PD-L1 IHC using the Dako 22C3 tumor proportional score (TPS) system. Categorical variables were compared using the Fisher exact test with the Benjamini-Hochberg adjustment to correct for false discovery rate. **Results:** A total of 796 (1.7%) advanced CRC tumors were HRDsig+ with a median age of 62 years for HRDsig+ and 63 years for HRDsig-. The HRDsig- cohort was more likely to be male compared to HRDsig+ (56.1% vs 50.4%; P=.004). Median GA per tumor was similar (range 5-6) in both groups. Genomic ancestry revealed significantly more African ancestry in the HRDsig- group (12.6% vs 8.8%; P=.002) and more European ancestry in the HRDsig+ group (75.8% vs 71.6%; P=.02). MSI-high status was more frequent in HRDsig- group as compared to the HRDsig+ group (5.5% vs 1.6%; P<.0001). Median TMB was higher in the HRDsig+ cohort (5.0 vs 3.6; P<.0001), as was frequency of TMB ≥ 10 mutations/Mb (16.2% vs 8.5%; P<.0001). Trinucleotide mutational signature distribution was similar in both cohorts with the exception that a tobacco signature was more frequent in the HRDsig+ group (1.5% vs 0.2%; P<.0001). Both PD-L1 low (1-49% TPS) expression (19.5% vs 13.1%; P=.005) and PD-L1 high (≥50% TPS) expression (4.4% vs 1.5%, P=.0021) were higher in HRDsig+ cases. GA in genes associated with HRD were *BRCA1* (13.6% vs 1.1%; P<.0001), *BRCA2* (8.4% vs 2.8%; P<.0001), *ATM* (8.0% vs 5.3%; P=.0003), and *RAD21* (6.7% vs 3.2%; P<.0001), which were more frequent in the HRDsig+ group. In the HRDsig- group, 83.7%/88.8% of *BRCA1/BRCA2* mutated CRC were not bi-allelic, likely non-driver GA. *ERBB2* GA were similar in both groups (5.3% vs 4.4%; P=0.31). *KRAS* GA were more frequent in the HRDsig+ group (49.1% vs 31.7%; P<.0001). **Conclusions:** This analysis is the largest assessment of HRD status in advanced CRC. HRDsig+ was 1.7% and is associated with co-occurring actionable GA compared to HRDsig- tumors and higher levels of PD-L1 TPS, which may provide opportunities to target these tumors with immunostimulatory therapies. Given the advances in novel DNA repair targeted therapies these results may provide insight for CRC clinical development for DNA repair combinations. Research Sponsor: None.

**239** **Poster Session**

# *TP53* Y220C mutations in colorectal carcinoma (CRC): A genomic landscape study.

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**Background:** Inactivating genomic alteration (GA) of the TP53 tumor suppressor gene leads to inactivation of p53 protein and is currently the most frequently associated GA across all cancers. Colorectal cancer is a common and lethal subtype of cancer, in the top 5 for incidence, prevalence and cancer-related deaths worldwide. TP53 is the second most frequent GA identified in clinically advanced CRC. It has been postulated that GA in TP53 not only affects the molecular biology of malignant cells but also impacts the tumor microenvironment in CRC. This GA is currently untargetable by approved drugs however, recently a novel approach designed to revert to wild type functioning of TP53 inactivated by Y220C base substitution has gained significant clinical interest. **Methods:** 67,301 cases of clinically advanced CRC underwent hybrid capture based comprehensive genomic profiling to assess all classes of GA. Cases were sequenced to a mean coverage depth of 650X with microsatellite instability (MSI) status and tumor mutation burden (TMB) determined from the sequencing data and PD-L1 expression measured by immunohistochemistry using the DAKO tumor proportional staining system (low positive set at 1-49% staining and high positive set at ≥50% staining). **Results:** 422 (0.6%) of the CRC featured the *TP53* Y220C GA (Y220C+). When compared with the CRC that were *TP53* Y220C negative (Y220C-), the Y220C+ cases were of similar age, gender and number of GA per tumor. The Y220C- cases has significantly higher frequencies of MSI status (6.1% vs 2.2%; p=.0007) and TMB levels greater >10 mutations/Mb (8.7% vs 4.7%; p=.004). The Y220C+ featured lower frequencies of GA in *PIK3CA* (12.9% vs 18.6%; p=.002), *BRAF* (8.8% vs 10.1%; NS) and *PTEN* (7.4% vs 8.3%, NS) and a higher frequency of *ERBB2* amplification (5.0% vs 2.7%; p=.004). GA in *APC*, *KRAS*, *NRAS*, *BRCA2*, *EGFR* and *MET* were similar in both Y220C+ and Y220C- CRC. PD-L1 low expression was similarly infrequent (12.1% to 16.7% range) in both groups. **Conclusions:** The potentially targetable *TP53* Y220C short variant mutation is uncommon in clinically advanced CRC but is associated with unique genomic and biomarker characteristics. Further research might elucidate subpopulations or CRC subtypes with a greater frequency of this mutation who might benefit from a novel targeted therapy. Research Sponsor: None.

| | CRC *TP53* Y220C+ (422 Cases) | CRC *TP53* Y220C- (66,879 Cases) | P Value |
|---|---|---|---|
| **Gender** | 41.2% F/ 58.8% M | 44.4% F/55.6% M | NS |
| **Median Age (range)** | 62 (23-89+) | 61 (8-89+) | NS |
| **GA/tumor** | 6.1 | 6.5 | NS |
| ***TP53* non-Y220C GA** | 16.1% | 75.7% | <.0001 |
| ***PIK3CA*** | 12.9% | 18.6% | .002 |
| ***ERBB2*** | 6.2% (5.0% amp) | 5.1% (2.7% amp) | .004 (amp) |
| **MSI High** | 2.2% | 6.1% | .0007 |
| **TMB > 10 mut/Mb** | 4.7% | 8.7% | .004 |
| **PD-L1 Low (1-49%TPS)/ PD-L1 High (>50% TPS)** | 16.7% (132 cases)/2.3% | 12.1% (20,759 cases)/1.4% | NS |

# Utilization of *SDC2* gene in conjunction with *SEPTIN9* gene methylation analysis for the diagnosis and efficacy assessment of colorectal cancer.

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**Background:** SDC2 gene encodes the transmembrane proteoglycan molecule Syndecan-2. Syndecan-2 protein affects the proliferation, migration and invasion of colorectal cancer cells by participating in the regulation of cell adhesion, tissue differentiation and angiogenesis. Plasma methylated SEPTIN9 gene has been shown to be a sensitive and specific biomarker for the detection of CRC, which is involved in apoptosis, pseudopod projection, tumor cell migration and invasion. The aim of this study is to validate the value of SDC2 and S9 gene methylation detection in the diagnosis and efficacy evaluation of CRC. **Methods:** A total of 102 patients were enrolled in the study. The methylation levels of fecal SDC2 gene and blood SEPTIN9 gene were detected by fluorescence PCR. The clinical diagnostic accuracy of SDC2 kit was evaluated by patients with benign gastrointestinal lesions, gastrointestinal tumors and healthy subjects. Combined detection of SDC2 and S9 to improve the detection rate of CRC. SDC2m levels were analyzed in 26 patients with partial remission or stable disease after palliative treatment and 32 patients with complete remission after radical surgery. **Results:** The detection rate of SDC2 in patients was 0.0% for UC and CD, 66.7% for AA and 60.0% for NA, 25.0% for HOP. For 8 healthy subjects with negative colonoscopies, the true negative rate was 100%. The detection rates of GC and EC were 50.0% and 0.0%, respectively. Data from 72 patients with CRC were evaluated, with a sensitivity of 89.1% (41/46,95%CI 0.798-0.985) for patients with untreated CRC. The sensitivity of SDC2 was 46.2% (12/26,95%CI 0.256-0.667) in patients with partial response or stable disease after palliative treatment. The difference was statistically significant ($P < 0.001$). The sensitivity of S9 for CRC detection was 88.2% (15/17, 95%CI 0.712-1.053). By two tests, 94.1%(16/17, 95%CI 0.816-1.066) of CRC cases could be detected.The sensitivity was 89.1% (41/46 95%CI 0.798-0.985) in untreated CRC and 46.2% (12/26 95%CI 0.256-0.667) in patients with PR and SD after chemotherapy/radiotherapy/targeted/immunotherapy. The difference rate of SDC2m between two groups was statistically significant ($P < 0.001$). Among 46 untreated patients with CRC, 32 patients with positive SDC2m underwent radical surgery, of which 30 CR patients turned SDC2m test negative after surgery. We found reduced SDC2m in stool from patients with clinical benefit (CR+PR+SD). **Conclusions:** Detection of SDC2m in untreated CRC patients has high sensitivity and has the potential to become a non-invasive diagnostic tool for CRC. Combination of S9 and fecal SDC2 could improve the detection rate of SDC2 in non-dominant population. The methylation level of SDC2 may be helpful for postoperative follow-up of CRC after radical surgery, prediction of recurrence, and monitoring the efficacy of a series of palliative treatments for CRC. Research Sponsor: Shanghai Shenkang Hospital Development Group Medical Enterprise Integration Innovation Synergy Project (SHDC2022CRT009); Preclinical Study of New Traditional Chinese Medicine by Shanghai Science and Technology Commission (21S21901500).

# Assessment of pathologic response in the colorectal cancer cohort of the IMHOTEP phase II trial of neoadjuvant pembrolizumab in dMMR/MSI tumors.

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**Background:** Neoadjuvant treatment with immune check point inhibitors has shown promising results in localized deficient mismatch repair/microsatellite instability (dMMR/MSI) colorectal cancer (CRC), notably in terms of pathologic response. However, pathologic response has rarely been precisely assessed on both primary tumor and its corresponding lymph nodes (LN) as an end-point. We performed this analysis on the CRC samples selected from the IMHOTEP trial, a multicenter, single-arm study evaluating peri-operative Pembrolizumab in 4 different cohorts of patients with localized resectable dMMR/MSI tumors. **Methods:** We analyzed pathologic response on the 56 surgical specimens available from the CRC cohort. Pathologic response was assessed on primary tumor according to the percentage of Residual Viable Tumor (RVT) score, encompassing 4 groups : 0% = pathologic Complete Response (pCR) ; ≤10% = Major Pathologic Response (MPR) ; >10% < 50% = partial Pathologic Response (pPR) ; > 50% = lack of response (noPR). Pathologic response was assessed on lymph node (LN) according to 4 groups : negative LN without sign of sterilization (yN0 reg-), negative LN with sterilization (yN0 reg+) ; metastatic LN without tumor regression (yN+ reg-) ; metastatic LN with tumor regression (yN+ reg+).The type of regression, including colloid, fibrotic and necrotic features was reported, as well as the presence of tertiary lymphoid structures (TLS). pCR and MPR corresponded to pathologic responder patients and pPR and noPR to pathologic non responder patients. **Results:** In thisCRC cohort pCR, MPR, pPR and noPR were observed in n= 33 (58.9%), 7 (12.5%), 5 (8.9%) and 11 (19.6 %) cases and yN0 reg-, yN0 reg+, yN+ reg-, yN+ reg+ in 41 (73.2%) , 7 (12.5%), 7 (12.5%), and 1 (1.8 % ) cases. There was a significant association between pathologic responders (pCR + MPR) and negative LN status encompassing sterilized LN (p=0.04). Pathologic responders were more frequently associated with classical (Lieberkuhnian) adenocarcinoma than with other CRC special subtypes (mucinous, signet ring-cell, poorly cohesive subtypes)(p<0.01). Colloid response was the most frequent pattern of regression, observed in 42 (75%) cases, and predominant (>80% of pathologic response areas) in 14 (25%) cases. TLS were seen in 46 (82%) cases, without any significant association with the RVT score. However TLS were observed in all CRC cases with MPR. **Conclusions:** These preliminary results show pembrolizumab efficacy as a neoadjuvant monotherapy in terms of pathologic response on both primary CRC and corresponding LN, highlighting the need for a new and more accurate pathologic score. It also illustrates the potential impact of the histological subtypes. Complementary data will be available after updated pathologic reviewing of additional specimens from CRC patients and will also be correlated with outcome. Clinical trial information: NCT04795661. Research Sponsor: MSD.

# A comprehensive study of clinical and molecular features in patients with metastatic colorectal cancer with and without liver mets.

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**Background:** Colorectal cancer (CRC) most frequently metastasizes to the liver, and patients (pts) with liver metastasis (mets) at the time of metastatic diagnosis tend to have poorer outcomes compared to pts with non-liver mets. However, the reason for this discrepancy is not well-defined. We thus aimed to identify any distinct molecular or clinical characteristics that could contribute to the poor outcomes associated with liver mets of CRC. **Methods:** Pts diagnosed with metastatic CRC from 2014-2022 were identified using the institutional molecular database of CRC. Demographic, clinical, and molecular data were collected by reviewing electronic medical records. Time to next treatment was defined as the time between 1st and 2nd line treatment. We utilized Fisher's exact tests for categorical variables and both Kaplan-Meier analysis and multivariate Cox proportional hazards models to analyze survival data, with a p-value threshold of $<0.05$ for statistical significance. **Results:** We identified a total of 217 pts, including pts with liver mets (N=146) and non-liver mets (N=71) at time of diagnosis. Pts with non-liver mets were more likely to have rectal primary than those pts with liver mets (40% vs. 15%, p=0.0001), indicating a distinct pattern of recurrence. Common driver mutations such as KRAS, BRAF, and TP53 were observed at equal frequencies in the liver and non-liver cohorts. In pts without liver mets, age had a significant effect on OS ($p < 0.05$, HR 1.1). Pts without liver mets with colon primary had worse OS compared to rectum primary (HR= 0.48, 95% CI: 0.24–0.96, p = 0.038). For pts without liver mets, KRAS was not a predictor of outcomes, while BRAF was associated with poorer OS ($p<0.001$, HR 7.09, 95%CI: 2.31-21.8). Contrarily, for pts with liver mets, KRAS mutation was significantly associated with worse survival (mOS 40.4 vs 61.7 mos, HR 2.04, 95% CI: 1.26-3.30, p=0.004). BRAF mutations were also associated with worse OS for this cohort (p=0.039 and HR 2.86, 95%CI: 1.06-7.77). Median time to the next treatment was significantly shorter in pts with liver mets (13.6 mos) than in pts with non-liver mets (22.6 mos) (HR 1.82, 95% CI 1.28-2.59; $p<0.001$). OS was longer for pts with non-liver mets compared to those with liver mets (mOS 52.9 mos vs 46.7 mos) though it did not reach statistical significance, likely due to the small sample size. **Conclusions:** Liver metastasis of CRC is associated with systemic therapy resistance with a shorter time on treatment, indicating treatment resistance associated with liver mets is not limited to immunotherapy. Given similar molecular characteristics in pts with and without liver mets, treatment resistance may be related to the inherent tumor microenvironment of the liver. In this study, pts with distinct molecular features had poorer outcomes, highlighting the importance of developing targeted therapeutics in this population. Research Sponsor: None.

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# Impact of baseline alterations in HER2-pathway oncogenic drivers on response to anti-HER2 therapy in patients (pts) with HER2-amplified advanced colorectal cancer (aCRC).

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**Background:** HER2-amplified aCRC is associated with poor prognosis despite new anti-HER2 regimens. Insight on biomarkers that predict response to these therapies is limited. This study aims to evaluate whether baseline alterations in oncogenic drivers along the HER2 pathway predict response to anti-HER2 therapy in HER2-amplified aCRC. **Methods:** In this international multi-center study involving Mayo Clinic, Duke Cancer Institute and National Cancer Center Hospital East (Japan), we collected baseline genomic alterations from tissue or blood samples using next-generation sequencing (NGS) prior to anti-HER2 therapy initiation. We defined PIK3CA, RAS, RAF, MAP2K, and MET as oncogenic drivers of resistance within the HER2 pathway. Overall survival (OS) was evaluated with the Kaplan-Meier method and a log-rank test was used to compare median OS (mOS) between pts with and without alterations in the above genes. A chi-square test (significance level 0.05) was performed to compare overall response rate (ORR) and disease control rate (DCR). **Results:** We identified 39 pts with HER2-amplified aCRC and baseline NGS. Median age was 57 (range 30-77), 54% were male. Baseline alterations in HER2 pathway oncogenic drivers were identified in 17 pts (43.5%). 18 (46%) received monoclonal antibodies (mAbs) plus tyrosine kinase inhibitors, 12 (30%) received dual mAbs and 7 (18%) received trastuzumab deruxtecan. The mOS was 41.9 months (mos) (95%CI 23.7-60.1) for other alterations vs 24.2 mos (95%CI 14.1-34.3) in pts with HER2 pathway alterations ($p = 0.1$). ORR was 47% vs 41% ($p = 0.7$) and DCR 82% vs 73% ($p = 0.4$), for pts with HER2 pathway alterations and without HER2 pathway alterations, respectively. 4 pts receiving trastuzumab/tucatinib had complete response, none of which had HER2 pathway alterations. **Conclusions:** Pts with baseline alterations in oncogenic drivers of the HER2 pathway had clinically worse outcomes compared to pts without such alterations despite receiving HER2-targeted therapy, though results lacked statistical significance. Larger studies are needed to confirm the lack of benefit from anti-HER2 therapy in this subgroup. Research Sponsor: None.

# Prognostic implications and therapeutic response in GNAS-mutated colorectal adenocarcinoma: A retrospective cohort analysis.

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**Background:** GNAS, a key regulator of the G-protein signaling pathway, is frequently mutated in colorectal adenocarcinomas and other malignancies. Despite its role in tumorigenesis, GNAS mutations remain understudied, with no established targeted therapies. Exploring the impact of GNAS alterations on therapeutic response and outcomes in colorectal cancer (CRC) is crucial for developing effective treatment strategies. **Methods:** We reviewed 531 CRC cases from our institutional Molecular Tumor Board database, identifying 8 patients with GNAS mutations. The prevalence ofGNASmutations in our cohort was 1.5%, which aligns with reported rates of 2-5% in various CRC cohorts. Comprehensive clinicopathologic data, including tumor histology, staging, treatment outcomes, and response rates, were analyzed. A detailed evaluation was conducted on stage IV patients to assess their response, progression-free survival (PFS), and overall survival (OS) in comparison to historical stage IV CRC benchmarks. **Results:** The cohort included 8 patients with a median age of 61 years (range: 33-73 years), with balanced gender representation (50% male, 50% female). The racial distribution was 50% Black or African American, 25% White, and 12.5% Asian or Pacific Islander. The majority of patients (62.5%) were diagnosed at stage IV, with primary tumor sites predominantly in the sigmoid colon and cecum. Among the 5 stage IV patients, while no patients achieved a complete or partial response, 3 exhibited stable disease, resulting in a 60% disease control rate (including stable disease). The median progression-free survival (mPFS) was 9 months, reflecting the duration of disease control before progression, while the median overall survival (mOS) was 15 months. These outcomes are notably lower than historical benchmarks for stage IV CRC, where response rates typically range from 40-50% with mOS of 20-24 months, highlighting the potentially adverse prognostic impact of GNAS mutations. **Conclusions:** Patients with GNAS-mutated colorectal adenocarcinoma appear to represent a unique subset with poorer therapeutic outcomes, even when achieving stable disease. The observed disease control in the absence of objective responses underscores the complexity of treating GNAS-mutated tumors and highlights the urgent need for targeted therapies tailored to this molecular subtype. Further research into the biological mechanisms underlying GNAS mutations and their impact on CRC treatment resistance is warranted to improve patient outcomes. Research Sponsor: None.

# Association between ctDNA levels, timing of blood collection, and overall survival in metastatic colorectal cancer.

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**Background:** While circulating tumor DNA (ctDNA) detection is influenced by various factors, the impact of blood collection timing remains understudied. This study investigated the relationship between ctDNA detection, and blood collection timing and prognosis in patients with metastatic colorectal cancer (mCRC) from the GOZILA study. **Methods:** GOZILA is a nationwide plasma genomic profiling study using Guardant360 CDx for advanced solid tumors. This study evaluated the association of ctDNA levels with blood collection timing and prognosis in patients with mCRC. ctDNA levels were determined by the maximum variant allele frequency. Blood collection timing was categorized into five groups: naive (before initiation of 1st line treatment), after treatment initiation (days 1-14 of treatment), during treatment (between "after treatment initiation" and "before progression disease [PD]"), before PD (from 28 days before PD to PD), and after PD (after disease progression but before the next line of treatment). Overall survival (OS) was compared across treatment lines using ctDNA level cutoff of 10. **Results:** We analyzed 2,398 blood samples from 1,994 patients with mCRC enrolled in the GOZILA between February 2018 and July 2023. ctDNA levels were significantly lower during treatment compared to the naive (median, 13.2 vs. 5.7, $p < 0.001$). No significant differences were observed between naive and other blood collection timing. Compared to blood collection before 1st line treatment, ctDNA levels significantly decreased before 2nd line (median, 13.5 vs. 8.3, $p = 0.02$) and 3rd line treatments (median, 13.5 vs. 7.6, $p = 0.01$). Patients with ctDNA levels<10 demonstrated significantly better OS across all treatment lines, including 1st line (median, NA vs. 23.9 months, HR 0.49, 95% CI, 0.34-0.72), 2nd line (median, 22.1 vs. 10.3 months, HR 0.42, 95% CI, 0.33-0.55), and later line treatments (FTD/TPI: median, 12.7 vs. 5.7 months, HR 0.51, 95% CI, 0.26-0.99; FTD/TPI+Bev: median, 9.3 vs. 6.5 months, HR 0.55, 95% CI, 0.37-0.82; Regorafenib: median, 13.5 vs. 7.4 months, HR 0.45, 95% CI, 0.26-0.80). **Conclusions:** In mCRC, blood collection during treatment is suboptimal due to decreased ctDNA levels. Patients with ctDNA levels <10 showed significantly better OS across all treatment lines. Optimal timing of blood collection can enhance ctDNA levels, enabling more accurate gene mutation assessment and stratifying outcomes. Research Sponsor: None.

# Effects of targeting CTMP on immunogenicity of MSI-L colon adenocarcinoma cells by metabolism and cell cycle modulation.

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**Background:** Regimen of metastatic colon cancer (mCOAD) is moving towards combination of immune checkpoint inhibitors (ICBs) and targeted therapy, however ICBs are rarely effective in MSI-L metastatic colon cancers, limiting options for backline treatments, new targeted therapy is on demand to improve immunogenicity of MSI-L mCOAD. As targeting fatty acid metabolism or cell cycle has shown potential in improving the efficacy of ICBs, our team focused on CTMP, a thioesterase (ACOT) family member, mainly modulates fatty acid β oxidation, meanwhile it binds to phosphorylation sites of AKT thus intervenes cell cycle, currently its role on immunogenicity of COAD is not clear. **Methods:** Correlation between CTMP and OS in MSI-L colon cancer patients was assessed by KM plotter in GEO database.11 Clinical samples of MSI-L colon cancer, who received a third-line treatment regimen of anti-PD-1 and fruquintinib, were collected then applied in immunofluorescence and western blot analyses to evaluate CTMP and MHC-I expression. In vitro, Lentiviral transfection of targeted genes were stably established,then applied in RNA-seq and metabolic assays. Proteomic analysis and co-IP were applied to explore CTMP-modulating pathway. In vitro, IFNγ was intratumorally injected. **Results:** KM curve showed CTMP was negatively correlated with OS in MSI-L COAD in GEO database. In clinical samples, lower CTMP expression and higher MHC-I expression were observed in partial response (PR) than progressive disease (PD) group. In vitro, CTMP-OE reduced MHC-I, increased AKT phosphorylation and immune checkpoints (PD-L1, IDO-1), promoted clone formation. Seahorse assay showed increased capacity of glycolysis and oxidative phosphorylation in CTMP-OE than vector. RNA-seq showed CTMP-OE reduced MHC-I augmentation under IFN-γ treatment, meanwhile glucose metabolism genes, ACOT family ,immune checkpoints were also up-regulated, indicating ACOT family related metabolism promotion and immunogenicity suppression of COAD. Proteomic analysis and co-IP assay indicated CTMP interacts REV7, a key mitotic regulatory protein. Enzyme activity assay showed REV7 negatively regulates CDK1. Furthermore, REV7-OE alone is capable to induce MHC I augmentation and G2 arrest, accordingly CTMP-KD enhanced REV7's negative modulation on CDK1,contributing to G2 arrest, that also augmented MHC I . In homologous mice tumor-bearing model, tumor shrinkage was observed in CTMP-KD/ REV7-OE group. CTMP-KD/ REV7-OE increases MHC I expression,CD8+ T cell infiltration under IFN γ intra-tumoral injection. **Conclusions:** In summary, our studies suggested that CTMP regulates the metabolism of colorectal cancer and intervenes MHC-I molecules expression, and targeting CTMP is potential in suppressing tumor cell cycle and promoting immunogenicity through its modulation on REV7. Research Sponsor: Shanghai Shenkang Hospital Development Group Medical Enterprises Interation Innovation Collaborative Special Project; SHD2022CRT009; Shanghai Municipal Science and Technology Commission of the preclinical research of new Chinese medicine; 21S901500.

# Landscape of *MET* genomic alterations in colorectal cancer (CRC).

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**Background:** MET signaling is implicated in the tumorigenesis of many solid tumors, including CRC. MET-targeted therapies are approved in non-small cell lung cancer with novel agents in clinical trials for tumors with *MET* exon 14 splice site mutations, *MET* amplifications, and recently MET expression. There is emerging therapeutic interest in targeting MET in CRC. **Methods:** Tumor samples from 50,500 cases of clinically advanced CRC were analyzed by hybrid capture-based comprehensive genomic profiling (CGP) that evaluated all classes of genomic alterations (GA). MSI-high status, tumor mutational burden (TMB), genomic ancestry, tri-nucleotide mutational signatures, and homologous recombination deficiency signature (HRDsig) were assessed for patients with activating *MET* GA. PD-L1 expression was determined by IHC (Dako 22C3 with TPS scoring system). Results were compared using the Fisher exact test with the Benjamini-Hochberg adjustment. **Results:** A total of 502 (1.0%) cases of activating GA of *MET* (METmut) were identified, including 418 cases (0.83%) with *MET* amplification (METamp), 10 (<0.1%) rearrangements, and 78 (0.1%) short variant (SV) mutations with 19 (<0.1%) cases of *MET* exon 14 splice site mutations. When compared with *MET* wild-type (METwt) CRC, the METmut CRC were of similar age (median 62 years), and numerically more likely to be male (61.4% vs 56.0%; P=.05). There were no differences in genomic ancestry (72% European ancestry for both groups). The METmut CRC featured more cases with a POLE signature (3.6% vs 0.4%; P=<0.0001) and more GA per tumor (7 vs 5; P<0.0001). MSI-high status was similar in both groups, with 4.2% in METmut and 6.0% in METwt (P=0.28). TMB levels were numerically higher in the METmut cases (TMB ≥ 10 mutations/Mb at 12.2% vs 9.0%; P=0.058). Low level PD-L1 expression (1-49% TPS) was similar in both groups (13.9% in METmut vs 13.3% in METwt), and while not statistically significant, high level PD-L1 expression (>50% TPS) was higher in the METmut group (4.3% vs 1.6%; P=0.056). The frequencies of HRDsig+ CRC were rare and similar in both groups with 1.5% in METmut and 1.7% in METwt. METmut CRC had lower frequencies of GA in *KRAS* (34.7% vs 48.8%; P<0.0001), *NRAS* (1.8% vs 4.2%; P=0.011), *APC* (72.5% vs 78.2%; P=0.009) and *PIK3CA* (12.2% vs 18.9%; P=0.0004), and higher frequencies of GA in *CDK6* (10.2% vs 0.6%; P<0.0001), *EGFR* (5.2% vs 0.2%; P=0.0002), *ERBB2* (9.6% vs 5.2%; P=0.0004), *HGF* (8.0% vs 0.8%; P<0.0001), *POLE* (4.2% vs 0.5%; P=0.005) and *TP53* (84.1% vs 75.9%; P<0.0001). In METmut CRC, ERBB2 alterations were 4.1% amplifications, 2.9% sequence mutations, 0.9% with more than one mutation. *BRAF* V600E was seen in 5.9% in METmut CRC and 8.4% of METwt. **Conclusions:** CGP reveals significant differences in the genomic landscapes of METmut CRC and METwt CRC, which may guide selection of potential rational drug combinations with novel MET-targeted therapies. Research Sponsor: None.

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# Co-formulated favezelimab plus pembrolizumab versus standard-of-care in previously treated, PD-L1-positive metastatic colorectal cancer: The phase 3, randomized KEYFORM-007 study.

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**The full, final text of this abstract will be available at meetings.asco.org on the day of presentation and in the online supplement to the February 1, 2025, issue of the *Journal of Clinical Oncology*.**

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# BTN1A1 expression in colon cancer: Implications for immunotherapy and patient stratification.

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**Background:** Butyrophilin 1A1 (BTN1A1) is a promising immune checkpoint protein offering a novel option for immunotherapy. We discovered that BTN1A1 and PD-L1 expression are mutually exclusive through the JAK/STAT pathway in various tumors. Recently, Nelmastobart, an antibody targeting BTN1A1, successfully completed a Phase 1 clinical trial at three sites in the United States and two in the Republic of Korea, spanning 98 weeks. This abstract focuses on the results from the Phase 1 trial in patients with colorectal cancer (CRC), emphasizing biomarker diagnostics and therapeutics derived from translational research. **Methods:** We conducted a first-in-human, multicenter, open-label Phase I clinical trial to evaluate the safety, tolerability, pharmacokinetics, and preliminary efficacy of hSTC810 monotherapy in CRC patients. Extensive analyses of BTN1A1 expression were performed using patient specimens through immunohistochemistry and multi-immunofluorescence techniques to identify critical biomarkers and accurately assess nelmastobart's anticancer activity. We then analyzed progression-free survival (PFS), overall survival (OS), objective response rate (ORR), and safety alongside translational research findings using patient tissue samples. **Results:** A total of 47 patients were enrolled in the study. The median PFS across all cohorts was 1.9 months, while the median Kaplan-Meier estimate for OS was 12.8 months. Among the 16 CRC patients enrolled, tissue samples were obtained and analyzed using IHC in 12 patients and OPAL staining in 7 patients. The objective response rate (ORR) for CRC was 2.3%. The median progression-free survival (PFS) and overall survival (OS) were 3.6 months and 11 months, respectively. Notably, PFS, OS, and ORR strongly correlated with BTN1A1 expression levels in CRC patients. High levels of BTN1A1 expression in colorectal tumor tissues correlated with clinical outcomes, identifying BTN1A1 as a potential biomarker for predicting patient responses to targeted therapies. The mutual exclusivity of BTN1A1 and PD-L1 expression suggests that BTN1A1 could serve as a diagnostic marker to identify patients who may be unresponsive to PD-1/PD-L1 blockade therapy and might benefit from BTN1A1 blockade therapy. **Conclusions:** This Phase 1 clinical trial demonstrated that antibody-mediated BTN1A1 blockade effectively suppresses tumor growth in patients with BTN1A1-positive CRC. These findings suggest that BTN1A1-targeted therapies could provide a viable treatment option for patients with colon cancers that are refractory or resistant to anti-PD-1/PD-L1 therapies. Clinical trial information: NCT05231746.
Research Sponsor: None.

# VEGF-A as a prognostic indicator in advanced gastric and colorectal cancers and correlation with immune suppression involving myeloid-derived suppressor cells (MDSC).

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**Background:** VEGF-A is important molecule should be controlled in cancer treatment since hypoxic and immunosuppressive conditions of tumor microenvironment are induced by VEGF-A. Myeloid-derived suppressor cells (MDSC), activated by chronic inflammation and VEGF-A, play critical roles in immunosuppression in patients with cancer. **Methods:** Serum levels of VEGF-A were measured by ELISA and circulating levels of MDSC were measured by Flow Cytometry (CD11b+, CD14-, CD33+). Patients with gastric and colorectal cancers were divided into two groups, high VEGF group (serum VEGF-A>330pg/ml) and lo group (≤330pg/ml). **Results:** Prognosis of stages III and IV gastric cancer with high VEGF-A was significantly worse than in low VEGF-A group, while difference was not significant in stages I and II. In colorectal cancer, the results were exactly same as found in gastric cancer. The levels of MDSC were significantly correlated with the levels of VEGF-A in patients with gastric and colorectal cancers. The levels of VEGF-A were also correlated with inflammatory markers including CRP and NLR (neutrophil to lymphocyte ratio), and inversely did to stimulation index of lymphocyte proliferation (SI: parameter of cellular immune reaction) and nutritional parameters such as albumin or rapid turnover protein. **Conclusions:** The observation in the present study suggest that VEGF-A is closely associated with inflammation-associated disease progression especially in advanced stages, and serve as useful marker of immunosuppression involving MDSC and prognosis in patients with gastric and colorectal cancers. Research Sponsor: None.

# Analysis of tumor microenvironmental features of primary and synchronous liver metastases from patients with colorectal cancer using a deep learning algorithm.

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**Background:** The development of colorectal liver metastases (CRLM) is associated with poor prognosis, and recent data suggest that metastasis to the liver is associated with resistance to immunotherapy. We characterized the microenvironment of primary colorectal carcinomas (CRCs) relative to their synchronous CRLM using a validated segmentation algorithm that quantifies 15 distinct morphologic tumor features. **Methods:** Adult CRC patients with synchronous CRLM (N=57) at Mayo Clinic were identified from the electronic health record using Epic Slicer Dicer and from an internal database. Tumor H&E sections were digitized and reviewed for quality control (RKP). QuantCRC (Aiforia, Inc) was applied to digitized images to extract 15 GI pathologist pre-defined morphological features. Tumor features were compared between primaries and CRLM using the Kruskal–Wallis test. The project was approved by the Mayo Clinic Institutional Review Board. **Results:** The study included 57 patients (median age 59 years [IQR 50, 73], 51% female) with CRC primaries and synchronous CRLM. Among primaries, 20 (35%) were right-sided and 37 (65%) were left-sided. QuantCRC identified 6 of 15 morphological features that differed significantly between primaries and their CRLM, including reduced stroma, more high-grade and necrosis, and a higher tumor: stromal ratio (TSR) in CRLM (Table). The increase in TIL density in CRLM vs the primary tumor was of borderline significance (p =0.053). Among patients with left-sided primary tumors, their CRLM had significantly higher TSR, percent high-grade, percent necrosis, and TIL density compared to the primary (all p values $\leq$ 0.02). Among right sided primaries, CRLM had a significantly reduced percent mature stroma (p=0.034) whereas percent necrosis was increased (p =0.01). **Conclusions:** Using deep learning, we identified tumor morphological features that differed significantly between primary CRC and their synchronous CRLM. This included higher TSR in CRLMs compared to primaries that is associated with epithelial-mesenchymal transition and has been shown to contribute to treatment resistance. Analysis of tumor morphological features with patient prognosis is ongoing. Research Sponsor: None.

**Deep learning-derived morphological features of primary CRC and CRLM.**

| Morphological feature (Median) | Primary tumor | Liver metastasis | P-value |
|---|---|---|---|
| **Tumor-Stroma Ratio** | 0.78 | 1.50 | **0.008** |
| **% High-grade** | 23.08 | 31.72 | **0.005** |
| **TIL Count** | 44.36 | 60.20 | 0.053 |
| **% Necrosis** | 6.35 | 19.32 | **<0.001** |
| **%Signet Ring Cells** | 0.25 | 0.13 | 0.088 |
| **TB/PDC** | 0.89 | 1.13 | 0.484 |
| **% Stroma** | 54.89 | 40.98 | **<0.001** |
| **% Immature Stroma** | 46.76 | 34.13 | **<0.001** |
| **% Mature Stroma** | 6.27 | 5.08 | **0.030** |

# Effect of homologous recombination deficiency on survival in patients with metastatic colorectal cancer with KRAS mutations.

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**Background:** Metastatic colorectal cancer (mCRC) patients exhibit intertumoral genetic heterogeneity, making it difficult to apply precision medicine. In the molecular genetic profile of mCRC, there are relatively few other gene alterations that significantly impact treatment decisions beyond KRAS and MSI status. We aim to investigate the prognostic impact of Homologous recombination deficiency (HRD)-related genetic alterations in mCRC patients, focusing on how mutation status influences on patient outcomes. **Methods:** The analysis included 364 mCRC patients who underwent comprehensive genomic profiling at Korea University Anam Hospital from Sep. 2016 to Dec. 2020. HRD was defined by the presence of pathogenic variants in genes including ATM, ATR, BARD1, BRCA1/2, BRIP1, CDK12, CHEK2, FANCA, MRE11A, PALB2, RAD50/51. Overall survival (OS) and progression-free survival (PFS) were evaluated using the Kaplan-Meier method. Multivariate Cox proportional hazards regression analyses were performed to assess the independent impact of HRD status. **Results:** Of the total cohort, 181 of patients (49.78%) harbored KRAS mutations. HRD gene mutations were identified in 38 of patients (10.4%) of patients, predominantly in ATM, BRCA 1/2, ARID1A genes. KRAS-HRD co-mutations were present in 18 of patients (4.7%). There is no significant difference in HRD status between the KRAS mutant and wild-type groups (p=0.73). While first-line treatment PFS was independent of KRAS mutational status (wt 10.8 vs. mut 9.6 months, p=0.31), it was significantly associated with HRD gene mutation status (16.2 vs. 10.6 months, p=0.001). OS was similar with KRAS mutations (34.7 vs. 32.9 months, p=0.72) and HRD mutations (34.7 vs. 39.1 months, p=0.73). However, patients with both KRAS and HRD mutations had significantly worse OS compared to those with either KRAS and HRD mutation (21.3 vs. 36.7 vs. 65.3 months, p=0.034, <0.001). A multivariate analysis was performed to evaluate factors affecting OS in mCRC patients. Among the variables, metastasectomy, which 44% of patients underwent, had a significantly low HR (=0.28) and would contributed to more favorable outcomes when evaluating OS. Neither KRAS nor HRD mutation alone was a significant prognostic factor, but their co-mutation was significantly associated with poor survival (HR=2.39, p=0.048). **Conclusions:** Poor OS in patients with KRAS-HRD co-mutations, despite metastasectomy, highlights the need for additional treatment strategies beyond surgery. Future studies with larger sample sizes and analyses stratified by metastasectomy status are necessary to strengthen these findings. Research Sponsor: None.

| Variables | HR (95% CI) | p-value |
|---|---|---|
| **Metastasis site** | | |
| **Liver** | 2.16 (1.60 – 2.92) | < 0.001 |
| **Peritoneum** | 1.52 (1.01 – 2.11) | 0.013 |
| **Bone** | 1.69 (0.96 – 2.97) | 0.067 |
| **Metastasectomy** | 0.28 (0.47 – 0.74) | < 0.001 |
| **KRAS** | 1.03 (0.77 – 1.37) | 0.826 |
| **HRD** | 0.75 (0.40 – 1.41) | 0.367 |
| **Co-mutation** | 2.39 (1.01 – 5.66) | 0.048 |

253 Poster Session

# Real-world analysis of the impact of timing of biomarker testing on first-line treatment choice in patients with metastatic colorectal cancer in British Columbia.

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**Background:** The treatment landscape for metastatic colorectal cancer (mCRC) has evolved significantly, shifting from chemotherapy alone to more personalized approaches that include immune checkpoint inhibitors and targeted therapies. Treatment decisions in mCRC are driven by molecular characteristics, and current guidelines recommend a minimum standard of biomarker testing, such as mismatch repair (MMR) status, and RAS and BRAF mutations, at the time of metastatic disease diagnosis to guide treatment selection. This study aimed to assess the real-world implementation of biomarker testing in a contemporary cohort of mCRC patients in British Columbia (BC) and its impact on treatment decisions by medical oncologists. **Methods:** We conducted a retrospective chart review of patients who received first-line systemic therapy for mCRC in BC between January 2022 and December 2023. Patient demographics, tumour characteristics, and treatment details were obtained from the BC Cancer provincial pharmacy database. Medical oncology documentation and pathology reports were reviewed to determine the availability of biomarkers, including MMR status, and RAS and BRAF mutations (OncoPanel), at the time of initial medical oncology consultation, and to assess how these results influenced treatment plan. **Results:** To date, 71 eligible patients have been identified, with data collection ongoing. Of these, 82% had proficient MMR status, 17% had deficient MMR status, and 1% had incomplete data. RAS mutations were present in 55% of cases, while BRAF mutations were found in 21% (Table). At the time of the initial medical oncology consultation, MMR status was unavailable in 14 patients (20%), two of whom were later found to have deficient MMR status. OncoPanel results were unavailable for 63 patients (88%) at the time of the initial consultation, and in 16 of these cases (25%), the systemic therapy plan was modified once the results were received. **Conclusions:** Even in contemporary practice with publicly funded biomarker testing, the results for RAS and BRAF mutations were not available at the time of initial medical oncology consultation for most patients. Biomarker testing is an essential tool for guiding therapeutic decisions in mCRC. This subsequently led to treatment plan changes in one-quarter of cases after results were received. These findings highlight the importance of timely biomarker testing to ensure optimal and efficient treatment decision-making for all mCRC patients. Research Sponsor: None.

**Metastatic colorectal cancer biomarker testing.**

| | Proficient/WT | Deficient/Mutated | Incomplete |
|---|---|---|---|
| **MMR** | 58 | 12 | 1 |
| **RAS** | 29 | 39 | 3 |
| **BRAF** | 53 | 15 | 3 |

# The prognostic impact of *SLFN11* gene expression in metastatic colorectal cancer (mCRC).

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**Background:** SLFN11 was recently identified as a dominant determinant of response to DNA-damaging anticancer agents. Several studies showed SLFN11 expression led to better sensitivity to topoisomerase inhibitors and PARP inhibitors. However, clinical evidence in mCRC has been lacking. **Methods:** Two independent cohorts (CALGB/SWOG 80405 trial and a real-world patient [Caris CODEai] cohort) were included in this study. Patients were stratified into *SLFN11*-high and -low groups based on the median *SLFN11* gene expression levels. Survival outcomes were compared between *SLFN11*-high and *SLFN11*-low in irinotecan-treated and oxaliplatin-treated patients. Median time was estimated using Kaplan-Meier method and Cox models were used for covariate adjustment. Endpoints of interest were progression-free survival (PFS) and overall survival (OS). **Results:** In the CALGB/SWOG 80405 trial cohort, 433 patients were included (110 treated with FOLFIRI-based chemotherapy and 323 treated with FOLFOX-based chemotherapy). In patients treated with FOLFIRI-based chemotherapy, *SLFN11*-low patients exhibited numerically better PFS (median PFS, 13.4 vs 11.2 months, log-rank $p$ = 0.03, adjusted hazard ratio [HR] = 0.86, 95% confidence interval [CI] 0.55-1.39, $p$ = 0.50) and OS (median OS, 39.6 vs 30.3 months, log-rank $p$ = 0.02, adjusted HR = 0.73, 95% CI 0.46-1.16, $p$ = 0.19) compared to *SLFN11*-high patients. In patients treated with FOLFOX-based chemotherapy, *SLFN11*-low patients exhibited numerically better PFS (median PFS 11.2 vs 9.8 months, log-rank $p$ = 0.03, adjusted HR = 0.83, 95% CI 0.65-1.05, $p$ = 0.12) and OS (31.1 vs 26.1 months, log-rank $p$ = 0.07, adjusted HR = 0.84, 95% CI 0.65-1.08, $p$ = 0.16) compared to *SLFN11*-high patients. No significant interaction was observed between treatment and *SLFN11* expression in terms of PFS ($p$ = 0.69) and OS ($p$ = 0.44). In the Caris CODEai cohort, a better OS in *SLFN11*-low patients than in *SLFN11*-high patients was observed in irinotecan-treated patients ($N$ = 2906) (median OS, 25.5 vs 21.7 months, HR = 0.90, 95% CI 0.82-1.00, $p$ = 0.048). However, no significant difference in OS between *SLFN11*-low and SLFN11-high was observed in oxaliplatin-treated patients ($N$ = 4428) (median OS, 39.5 vs 35.8 months, HR = 0.98, 95% CI 0.90-1.06, $p$ = 0.56). **Conclusions:** Low *SLFN11* expression showed a potential for better prognosis in mCRC patients receiving irinotecan- and oxaliplatin-containing chemotherapies. These findings are not consistently aligned with preclinical data and clinical evidence in other cancer types previously reported. Further investigation is warranted to better understand the biological role and clinical implication of SLFN11 in mCRC. Research Sponsor: National Cancer Institute; P30CA014089, U10CA180821, U10CA180882, U10CA180820, UG1CA233277, and U10CA180888; Pfizer and Genentech; Gloria Borges WunderGlo Foundation; Dhont Family Foundation; Victoria and Philip Wilson Research Fund; San Pedro Peninsula Cancer Guild; Daniel Butler Research Fund.

# Impact of *RAS* and *BRAF* heterogeneity on the efficacy of EGFR blockade in patients with metastatic colorectal cancer.

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**Background:** Epidermal growth factor receptor (EGFR) blockade can effectively shrink tumors in metastatic colorectal cancer (CRC) patients without *RAS* nor *BRAF* mutations. However, some patients without RAS or BRAF mutations in their primary tumors may develop these mutations in metastatic tumors (heterogeneity). Circulating tumor DNA (ctDNA) can reflect this heterogeneity in metastatic tumors. Here, we evaluated the efficacy of EGFR blockade in patients who had no *RAS* or *BRAF* mutations in their primary tumors but did have them in ctDNA. **Methods:** We prospectively enrolled 100 patients with confirmed metastatic CRC without *RAS* or *BRAF* mutations in their primary tumors. Patients were treated with first-line systemic chemotherapy including EGFR blockade. We obtained ctDNA from each patient before they started chemotherapy. *RAS, BRAF (V600E)*, and *PIK3CA* mutations were detected using digital PCR. **Results:** One of the 100 cases were excluded due to protocol violation. In the ctDNA obtained before starting chemotherapy, *RAS, BRAF*, and *PIK3CA* mutations were detected in 12, 4, and 6 patients, respectively. No patients had both *RAS* and *BRAF* mutations in their ctDNA; however, one patient had both *BRAF* and *PIK3CA* mutations and one had both *RAS* and *PIK3CA* mutations. Eighty-nine patients had measurable tumor lesions. Among these, 3 experienced a complete response (CR), 71 had a partial response (PR), 13 had stable disease (SD), and 2 had progressive disease (PD). The response rates for patients with *RAS* or *BRAF* mutations and for patients with neither mutation were 81% and 83%, respectively (P=0.73). Median progression-free survival (PFS) for patients with *RAS* or *BRAF* mutations and those without mutations were 251 days and 216.5 days, respectively (P=0.82). The presence or absence of PIK3CA mutations did not affect the response rate or PFS. **Conclusions:** Previously, we reported that the incidence of *RAS* and *BRAF* heterogeneity were 10% (1) and 4% (2). In the present study, *RAS* and *BRAF* heterogeneity were 12% and 4%, respectively. The heterogeneity of *RAS* and *BRAF* mutations had no effect on the response rate and PFS of EGFR blockade as first line chemotherapy. The effect of heterogeneity on the overall survival is of great interest; however, about half of the patients are still alive. 1. Yamada T, et al. Cancer Science 2016. 2. Ueda K, et al. Eur J Surg Oncol 2022. Clinical trial information: UMIN000031177. Research Sponsor: None.

# Prognostic value of circulating tumor DNA for progression-free survival in patients with locally advanced dMMR/MSI-H colorectal cancer managed without surgery.

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**Background:** Immunotherapy has shown the ability to induce a high rate of pathologic and clinical complete response in patients with deficient mismatch repair (dMMR)/microsatellite instability-high (MSI-H) colorectal cancer (CRC). The role of circulating tumor DNA (ctDNA) as a prognostic biomarker for survival in patients who receive non-operative management (NOM) is unknown. Among patients enrolled in a phase II trial (NCT04082572) that evaluated the efficacy of pembrolizumab in patients with locally advanced dMMR/MSI-H solid tumors, we evaluated the impact of ctDNA on progression-free survival (PFS) among 12 patients with locally advanced dMMR/MSI-H CRC who received NOM. **Methods:** ctDNA testing was performed in the CLIA-certified laboratory at MD Anderson using a 70-gene liquid biopsy panel (LBP-70), which used digital sequencing of cell-free DNA. Formalin-fixed paraffin-embedded tumor samples and germline peripheral blood mononuclear cells were profiled using next-generation sequencing. Mutations in ctDNA were considered tumor-specific if they were confirmed with matched tumor tissue profiling. The lower limit of detection of the LBP-70 assay was a mutational variant allele frequency of <0.3%. "ctDNA clearance" and "ctDNA(-)" status were defined as the absence of any detectable tumor-specific mutations in ctDNA. Two-year progression-free survival (PFS) was estimated using Kaplan-Meier techniques and compared with log-rank tests. **Results:** The median follow-up between the first cycle of pembrolizumab and last radiographic assessment was approximately three years (36.7 mo; range (range 1.2 – 51.0 mo). Forty-nine LBP-70 assays were performed among the 12 patients over the course of treatment with pembrolizumab. The median number of cycles was 16 (range 1 – 16 cycles). Six patients were ctDNA(-) prior to pembrolizumab, all of whom remained ctDNA(-) after. None of these six patients experienced disease progression (PD) at their last follow-up. The other six patients were ctDNA(+) prior to pembrolizumab (median baseline VAF 0.4%, range 0.3%–4.0%). Of these six patients, three experienced ctDNA clearance while receiving pembrolizumab. None of the three patients who cleared ctDNA experienced PD as of their last follow-up. Of the three patients who did not clear ctDNA, two experienced PD in their primary tumor (at two and six months after their first cycle, respectively), while the third patient remains without PD at their last follow-up. The two-year PFS rate was 100% among the nine patients who were ctDNA(-) after pembrolizumab compared to 33% among the three patients who were ctDNA(+) after pembrolizumab ($p = 0.03$). **Conclusions:** ctDNA may be a valuable tool for assessing treatment response and improving the ability to tailor organ-sparing, curative strategies to patients with locally advanced dMMR/MSI-H CRC. Research Sponsor: None.

# EORTC1527/JCOG1609INT/ESSO02: Diffusion-weighted magnetic resonance imaging (DW-MRI) assessment of initially unresectable liver metastasis to improve surgical planning (DREAM)—Primary analysis.

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**Background:** Disappearing liver metastases (DLMs) diagnosed on post-chemotherapy (Cx) computed tomography (CT) is a favorable prognostic factor in patients (pts) with colorectal liver metastases (CRLM). However, the optimal treatment of DLMs - whether they should be resected or left behind - is controversial. This is a prospective, multi-centred, international study examining the added value of MRI (DWI, T1/T2 and contrast-enhanced) to that of CT alone in accurate assessment of the viability of DLMs. **Methods:** Pts with initially unresectable CRLM downstaged to a planned liver resection after Cx were enrolled, based on the obligatory decision of a multidisciplinary team. Pts were imaged by both CT and MRI at baseline and presurgical timepoints. DLMs were defined as disappeared lesions diagnosed by CT alone, while confirmed DLMs (cDLMs) were defined as lesions that disappeared on both CT and MRI. cDLMs were either resected or followed-up for 2 years if not resected. All imaging scans were collected centrally for quality assurance. The primary endpoint was the negative predictive value (NPV) of MRI and CT in confirming the status of cDLMs using either pathological complete response or the absence of a local recurrence at the site of cDLMs during the 2 year follow up. The study was aimed at excluding a NPV ≤0.85 with a 1-sided alfa of 5% and power of 90% under the alternative that the NPV ≥0.95. The planned sample size was 92 evaluable (resected or left behind) cDLMs, assuming a within-patient correlation between cDLMs of 0.2 and an average number of 2 cDLMs per pt. **Results:** 233 pts were registered at 22 participating centres, and 112 were enrolled for analysis. A median of 8 cycles of Cx was delivered, and pts had a median of 7 CRLMs at baseline. A total of 203 cDLMs were identified while the number of DLMs diagnosed by CT was 296. Of these, 152 cDLMs and 227 DLMs, respectively, were evaluable according to the imaging protocol. Intraoperative ultrasound was performed for 195 cDLMs and, of these, 59 (30.2%) were still detected. The rate of R0/R1 resection was 95.5%. The NPV of evaluable cDLMs either resected or left behind was 62.5 % (95/152, 95% CI:50.8-74.2), which was lower than the prespecified threshold. The NPV of DLMs was 52.9% (119/227, 95% CI:42.7, 63.0). The NPV of resected cDLMs, and those left behind were 56.8% (50/88, 95% CI: 44.2, 69.5) and 70.3% (45/64, 95% CI: 48.6 -92.0), respectively. For DLMs, the NPV of resected DLMs and those left behind were 45.6% (72/158, 95% CI: 35.4-55.7) and 69.6% (48/69, 95% CI: 47.7-91.5), respectively. **Conclusions:** For pts with initially unresectable CRLM, cDLMs diagnosed by both CT and MRI do not correspond to tumor viability, even after highly effective chemotherapy. Ongoing survival analysis (to be presented at the annual meeting) may impact the treatment strategy of pts with DLMs. Clinical trial information: NCT02781935. Research Sponsor: Krebsforschung Schweiz; Ligue nationale contre le cancer; Anticancer Fund; Kom Op Tegen Kanker.

# Genomic alterations in circulating tumor DNA (ctDNA) and response to telisotuzumab adizutecan (ABBV-400) treatment in patients (pts) with colorectal cancer (CRC).

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**Background:** Telisotuzumab adizutecan is an antibody-drug conjugate comprising the c-Met–targeting antibody telisotuzumab conjugated to a topoisomerase 1 inhibitor payload, adizutecan. A phase 1 trial (NCT05029882) in advanced solid tumors reported higher response rates in pts with CRC and high c-Met expression. We present an analysis of responses based on genomic alterations in ctDNA. **Methods:** Pts with CRC that is refractory to standard treatment were enrolled. Telisotuzumab adizutecan was given IV Q3W. ctDNA was isolated from baseline (BL) and cycle 3, day 1 plasma samples and analyzed with the GuardantINFINITY assay. Circulating tumor fraction (cTF) was estimated on the basis of variant allele frequency of somatic mutations in a 74-gene panel and methylation signals across targeted regions of the GuardantINFINITY methylation panel. Molecular response (MR) was defined as 50% decrease in cTF from BL. Radiographic response (RR) was assessed by RECIST v1.1. **Results:** Overall, 113 pts had both BL ctDNA and RR data. Confirmed ORR was 18% (20/113). The most prevalent gene alterations included *TP53* (75%), *APC* (68%), and *KRAS* (66%) mut; *PTPRT* (56%), *ASXL1* (53%), and *FLT1* (52%) amp; *SMAD4* (44%) and *TP53* (43%) deletions; and *MET* (4%), *BRAF*, *FGFR3*, and *NRG1* (all 3%) fusions. The Table shows RRs in pts with specific CRC biomarkers, and MR using the 2 panels with corresponding clinical outcomes. RRs were seen in pts with positive plasma samples at BL for TMB, *RAS*, *BRAF*, and *HER2*. MRs were detected in 64% and 65% of pts using the 74-gene and methylation panel, respectively. ORR was higher in pts with MR (35%) than without MR (8%) using the 74-gene, and 35% vs 0% using the methylation panel, respectively. Median PFS (mPFS) was longer in pts with MR. **Conclusions:** Telisotuzumab adizutecan has promising efficacy. RRs were seen in CRC pts with heterogeneous genomic profiles, including pts positive for actionable biomarkers in the metastatic setting. More pts with MR experienced RRs and benefited from treatment. Clinical trial information: NCT05029882. Research Sponsor: AbbVie Inc; n/a.

| | Pts with ctDNA and radiographic response data (N=113) | | | | |
|---|---|---|---|---|---|
| **Radiographic cPR, n/N (%)** | 20/113 (18) | | | | |
| **Genomic alteration** | **TMB[a] high (≥20 mut/Mb)** | ***KRAS* mut** | ***KRAS* G12C mut** | ***BRAF* mut[b]** | ***HER2* amp** |
| **cPR, n/N (%)** | 11/48 (23) | 11/73 (15) | 1/5 (20) | 4/14 (29) | 2/9 (22) |
| **Pos** | 9/51 (18) | 9/40 (23) | 19/108 (18) | 16/99 (16) | 18/104 (17) |
| **Neg** | | | | | |
| **MR, n/N (%)** | **74-gene panel** 42/66 (64) | | **Methylation panel** 48/74 (65) | | |
| **ORR, n/N (%)** | **MR pos** 15/42 (36) | **MR neg** 2/24 (8) | **MR pos** 17/48 (35) | **MR neg** 0/26 (0) | |
| **mPFS, mo (95% CI)** | 6.1 (5.3, 6.9) | 3.5 (2.6, 4.3) | 5.9 (5.3, 6.8) | 3.5 (2.7, 4.1) | |
| **Events (n/N)** | 28/42 | 21/24 | 31/48 | 23/26 | |
| **MR in pts with SD, n/N (%)** | **74-gene panel** 27/45 (60) | | **Methylation panel** 31/53 (58) | | |
| | **MR pos** | **MR neg** | **MR pos** | **MR neg** | |
| **mPFS, mo (95% CI)** | 5.3 (4.5, 5.9) | 3.9 (2.8, 4.3) | 5.3 (4.5, 5.9) | 4 (2.8, 4.4) | |
| **Events (n/N)** | 21/27 | 16/18 | 23/31 | 19/22 | |

[a]14 pts not evaluable for TMB. Threshold determined by Guardant.
[b]1 pt had V600E mutation.
mo, months; neg, negative; pos, positive.

# Bragg peak proton irradiation and ATM inhibition for rectal cancer.

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**Background:** Rectal cancer patients with a complete response (CR) after chemotherapy and radiotherapy may opt for non-operative management. Currently only 20-40% of patients reach a CR. Escalating the dose of radiation may increase the risk of toxicity. ATM inhibitors (ATMi) are potent radiosensitizers in colorectal cancer (CRC). Proton irradiation exhibits an entrance portion of the beam (ENT) where dose is deposited sparsely and a Bragg Peak (BP) with virtually no 'exit dose' thereafter. ATMi renders cells uniquely hypersensitive to proton BP, and BP+ATMi may increase the response of Mismatch Repair proficient (pMMR) rectal cancer to immunotherapy. **Methods:** Tumor efficacy and immunogenicity was assessed using clonogenic survival, γH2AX foci, micronuclei, and cytoplasmic cGas assays after indicated IR modalities +/-ATMi. Normal tissue toxicity was assessed via murine models of non-tumor bearing BALB/c after whole-abdominal irradiation (WAI) after indicated IR modalities +/- ATMi via acute weight loss, FITC/Dextran assay, and Radiation Injury Score (RIS, Scale 0-18) in jejunal and rectal samples at 4 days post-IR. **Results:** ATMi decreased clonogenic survival across all types of radiation, but the effect was most pronounced with BP protons. BP+ATMi also significantly increased unresolved γH2AX foci, micronuclei, and cytoplasmic cGas compared to ENT+ATMi. Normal tissue toxicity studies demonstrated that BP+ATMi did not significantly increase acute weight loss, RIS in jejunal or rectal tissue; but did increase FITC/Dextran intestinal permeability. **Conclusions:** Inhibition of ATM preferentially radiosensitized BP protons compared to x-rays or ENT protons in CRC but not in normal tissue based on weight loss and histopathological evaluation. Research Sponsor: None.

**Differential tumor efficacy and normal tissue toxicity after x-ray, entrance and Bragg Peak Protons +/- ATM inhibitor.**

| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| **DMSO** | 1 | 1 | 1 | 2.9 | 1.5 | 8764 | 96.4 | 10 | 0.3 | 0.2 |
| **ATMi** | 0.873 | 0.842 | 0.723 | 8.2 | 3.0 | 16300 | 98.5 | 5.8 | 0.7 | 0.8 |
| **x-ray** | 0.615 | 0.751 | 0.109 | | | | 82.4 | 6.2 | 4.8 | 1.0 |
| **x-ray+ATMi** | 0.123 | 0.344 | 0.011 | | | | 80.0 | 14.4 | 5.2 | 1.5 |
| **ENT protons** | 0.293 | 0.515 | 0.080 | 10.2 | 5.8 | 10429 | 78.5 | 6 | 2.9 | 1.5 |
| **ENT+ATMi** | 0.091 | 0.066 | 0.013 | 15.1 | 7.2 | 12274 | 73.0 | 23.8 | 5.2 | 3.8 |
| **BP protons** | 0.213 | 0.340 | 0.032 | 12.0 | 12.6 | 7192 | 75.6 | 24.6 | 4.1 | 3.7 |
| **BP+ATMi** | 0.074 | 0.038 | 0.004 | 22.7 | 16.2 | 43677 | 73.6 | 310.4 | 5.3 | 4.3 |

Clonogenic survival of pMMR colorectal cancer cell lines after IR (4 Gy) +/- 10nM AZD1390 (ATM inhibitor) ofA) HT29, B) CT26, and C) SW480 cell lines. D) γH2AX foci at 24 hours post-IR in HT29. E) Micronuclei, and F) cytoplasmic cGas integrated density per cell via immunofluorescence in SW480 72 hours post-IR. WAI (10Gy/1 fraction) was delivered to non-tumor bearing BALB/c mice which were followed for G) acute weight loss (% from baseline), and H) FITC/Dextran permeability assay. Tissue samples were graded for Radiation Injury Score in I) jejunal, and J) rectal samples at 4 days post-IR (n=5 per cohort, 3 samples per mouse). Figures are means of triplicate data.

# A comparison of real-world outcomes by mismatch repair status in patients with stage II/III rectal cancer in the Netherlands.

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**Background:** A subset of rectal tumours (~3–5%) have mismatch repair deficiency (dMMR); the remaining are classified as MMR proficient (pMMR). A recent trial for stage II/III dMMR rectal cancer (RC) showed that 6 months of neoadjuvant treatment with dostarlimab, a programmed cell death protein 1 inhibitor, induced a 100% clinical complete response rate and allowed for organ preservation. Using real-world data from the Netherlands, a dMMR patient cohort was compared to a matched pMMR cohort to assess if clinical outcomes differed, even after matching for baseline characteristics. **Methods:** Utilizing data from the Netherlands Cancer Registry, this retrospective analysis assessed patients with stage II/III RC treated from 2015–2022 with known MMR status. The dMMR cohort was matched 1:2 to a cohort of pMMR patients on age, year of diagnosis, tumour clinical (cT) stage, and node clinical (cN) stage. Event-free survival (EFS; defined as first occurrence of locoregional failure, distant metastasis, a new primary colorectal tumour, or death from any cause) and overall survival (OS) were compared using Kaplan-Meier methodology and a univariable Cox model. **Results:** Among the 7939 patients included, 184 (2.3%) had dMMR tumours. After matching and a loss of 8 dMMR patients due to ineligibility, 176 dMMR patients were compared to 363 pMMR patients. Matching successfully removed differences in age, cT, and cN stage; however, dMMR patients continued to have more BRAF mutants (p=0.014), more poorly differentiated tumours ($p<0.001$) and more variation in histology (p=0.01). dMMR patients had a statistically significant lower risk of experiencing an event (hazard ratio [HR] 0.54 [95% confidence interval (CI) 0.38–0.77], $p<0.001$), with a 3-year EFS of 79.6% (95% CI 73.7–86.1) compared to pMMR patients (64.7% [95% CI 59.8–70.1]). For OS, there was no significant difference between the dMMR and pMMR cohorts (HR 0.73 [95% CI 0.47–1.16], p=0.181); however, within the small group of patients with an event, OS at 1 year for dMMR patients was 68.4% (95% CI 55.1–84.9) compared to 93.5% (95% CI 89.5–97.7) for pMMR patients (p=0.001). **Conclusions:** dMMR RC is rare, and tumour characteristics differ from those of patients with pMMR tumours (1). After controlling for clinical stage and age, this study suggests dMMR patients have improved EFS compared to pMMR patients. There is no significant difference in OS between groups but dMMR patients with an event have worse OS than pMMR patients, highlighting an additional need to prevent recurrence of disease within dMMR patients.1. Lunenberg, R et al. Poster presented at ESMO Congress (Presentation 569P), 13-17 Sept 2024, Barcelona, Spain. Research Sponsor: GSK; 219473.

261 Poster Session

# Leveraging large-scale PDO-based assays to optimize antibody-drug conjugate efficacy in CRC.

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**Background:** Clinical trials are one of the major barriers in contemporary drug development. Functional assays based on patient-derived organoids (PDO) are a promising new tool for derisking clinical development of new therapies, but their use has been limited due to small cohort sizes and the absence of systematic validation studies. Using the largest cohort of matched PDOs, longitudinal clinical data, transcriptomic and WES data in CRC, we explore the use of large PDOs collections to characterize ADC efficacy and affinity through systematic mapping of the relationship between target antigen affinity and payload efficacy. **Methods:** We have assembled a collection of 85 PDOs from colorectal cancer (CRC), generated from primary tumour samples and metastatic biopsies. Patients ranged from 32 to 89 years old and had experienced an average of 1.54 lines of treatment prior to the PDO-generating tissue sampling. Using gene expression levels of well known target antigens such as CEACAM5, ERBB2, MSLN and TROP2 as a proxy for protein concentration, we explore the relationship between the target antigen affinity and the efficacy of a subset of common payloads including topoisomerase and microtubule inhibitors. This allows us to identify the relationship between two of the three main vectors of ADC efficacy in CRC. **Results:** We show that topoismerase inhibitor and microtubule inhibitor efficacy relates to the RNA expression level of multiple target antigens commonly used in active clinical trials in CRC. These data match the target antigen and payload couples currently under development or approved in the clinic. We are developing a tool to optimize clinical trial design. This approach will allow us to match patients more accurately with the appropriate payload, ensuring a higher likelihood of response. **Conclusions:** We report that large-scale PDO-based assays can be a useful tool to anticipate clinical readouts. This work suggests that well-characterized PDO collections could also help redefine patient populations, increasing the likelihood of identifying those more likely to respond to a given ADC, and thus improving the success rates of clinical studies for new treatments in CRC. Research Sponsor: Fondation Jean-Jacques et Felicia Lopez-Loreta pour l'Excellence Académique; Banque Publique d'Investissement.

# Highly accurate detection of early-stage colorectal cancer using tumor and immune extracellular vesicles biomarkers.

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**Background:** Colorectal cancer (CRC) is one of the most common cancers worldwide, and early detection is critical for successful treatment and improved survival rates. However, precancerous and early-stage CRC present significant diagnostic challenges due to the small size of lesions and the very low expression of tumor-specific biomarkers in the bloodstream. Current genomics-based diagnostic methods struggle to detect these early-stage and precancerous lesions, making it imperative to develop more sensitive and specific approaches. Circulating extracellular vesicles (EVs) are emerging as a promising solution for early-stage CRC detection. Since EVs are produced by tumor cells as well as tumor microenvironment and host immune response cells, EVs offer the means to detect and monitor small early-stage tumors through both direct detection of tumor-associated biomarkers as well as tumor specific host response and tumor microenvironment biomarkers. **Methods:** To identify novel early-stage CRC specific biomarkers, we performed proteomics analysis on EVs purified from patient plasma using size exclusion chromatography and a proprietary buffer system that enhances EV and corona protein recovery. TrueDiscovery Data-independent acquisition (DIA) mass spectrometry (MS) analysis was conducted on 24 pre-cancer/stage 0 and 25 stage 1 CRC patients and 75 normal patient plasma samples. An in-house developed machine learning pipeline was used to identify differentially expressed proteins and model candidate multiplexes to identify those with extremely high diagnostic accuracy ($> 0.99$). **Results:** An average of ~2,500 proteins were identified per sample and included in bioinformatics analysis. Comparative analysis between control patient EVs and either pre-cancerous/Stage 0, or Stage 1 colon cancer EVs using an in-house Machine Learning pipeline identified 336 and 493 differentially expressed proteins for each group (FDR adjusted p-value $< 0.001$). Of these, the best 17 pre/stage 0 proteins and 53 stage 1 proteins were trained and tested using Support Vector Machine to assess all potential 3-plexes in order to identify those with mean diagnostic accuracy $> 99\%$. This yielded 5 3-plexes in precancer/stage 0 and 34 3-plexes in stage 1 with near perfect diagnostic accuracy. These plexes are currently being assessed in single analyte and multiplex immunoassays with the goal of translating candidate 3-plexes to the clinical. **Conclusions:** Key biomarkers from CRC patient EVs indicate the potential to detect and diagnose early-stage and low tumor burden cancers using our methods. Interestingly, the most accurate 3-plexes consist of proteins from immune, inflammatory and metabolic processes, suggesting that for the detection of early stage cancer it may be critical to include both tumor and host specific biomarkers. Research Sponsor: None.

# Correlation of mid-chemoradiation ctDNA results and clinical complete response to total neoadjuvant therapy (TNT) for locally advanced rectal adenocarcinoma.

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**Background:** The total neoadjuvant therapy (TNT) paradigm for locally advanced rectal cancer (LARC) is evolving with intensification and de-escalation neoadjuvant therapies. Chemoradiation (CRT) followed by consolidation chemotherapy offers the highest rate of organ preservation. Circulating tumor DNA (ctDNA) response during TNT may serve as a biomarker that allows for therapy personalization to better select candidates for rectal organ preservation (Watch-and-Wait, WW). **Methods:** We enrolled patients with mismatch repair proficient LARC eligible for TNT on a prospective protocol (NCT05108428). Patients underwent CRT to 50.4 Gy with a mid-treatment evaluation (27 Gy) on a MRI-linac followed by 8 cycles of consolidation FOLFOX. A sequential MRI-guided adaptive radiotherapy boost to 59.4 Gy was delivered if rectal tumor volume reduced > 30% at 27 Gy. Standard restaging with diagnostic MRI and endoscopy occurred after TNT to evaluate organ preservation candidacy. ctDNA was obtained at diagnosis, mid-CRT (27 Gy), completion of TNT, and regular intervals in surveillance. ctDNA analysis was performed using a clinically validated, personalized, tumor-informed assay (Signatera, Natera, Inc.). Descriptive statistics for clinical response for mid-treatment assessment are reported. Additional clinical outcomes will be reported as follow up matures. **Results:** A total of 20 patients were enrolled (70% male, 70% cN+) with average age of 58 (range: 30-86) and 40% with threatened or involved radial margin. Average follow up is 12 months (range: 4-24 months) from date of completion of TNT. Average initial tumor volume was 26cc (range: 8.6cc- 66.8cc) with an average tumor reduction of 68.5% at 27 Gy. Volumetric change based on mid treatment MRI >30% did not predict WW eligibility (2 patients with <30% volumetric change maintains clinical complete responses). All patients demonstrated positive ctDNA at diagnosis (average 7.08 MTM/mL, range: 0.08 MTM/mL -77.96 MTM/mL). 25% of patients had elevated CEA at diagnosis. Conversion to negative ctDNA at mid-CRT occurred in 53% of patients. 57% of patients with a positive ctDNA at this timepoint underwent formal oncologic mesorectal excision with adenocarcinoma confirmed in the specimens. 25% of patients with negative ctDNA at this timepoint underwent a local excision, with no histologic invasive cancer in both specimens. We observed a 100% rectal organ preservation rate when early conversion to negative ctDNA at the 27 Gy of CRT timepoint. **Conclusions:** ctDNA clearance during CRT may be a good early predictor for patients that may be a candidate for a WW protocol. This biomarker may also guide consolidation therapy escalation (FOLFIRINOX) or de-escalation (chemotherapy omission) for select patients. Longer follow up is needed to correlate with clinical outcomes and future studies with more patients is needed. Clinical trial information: NCT05108428. Research Sponsor: Natera, Viewray, Moffitt.

# KRAS status in lung oligometastases treated with SBRT.

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**Background:** Stereotactic Body Radiotherapy (SBRT) to lung metastases (LM) from colorectal cancer (CRC) has been applied to improve disease control in in the oligometastatic (OM) and oligoprogressive (OP) setting. The aim of this study is to assess the impact of KRAS mutational (KRASmut) status in the response to SBRT in LM. **Methods:** Data from a consecutive cohort of OM/OP CRC patients treated at our Institution with SBRT for LM from May 2019 to March 2024 were retrospectively reviewed. Patients (age, KRASmut in primary tumor, OM/OP disease, chemotherapy lines prior to SBRT) and treatment-related variables (dose regimen and Biologically Effective Dose: BED, toxicity) were collected. Local Control (LC) was defined as time from SBRT to local failure or last radiologic follow-up. **Results:** Sixty-two patients (median age: 74 years, range 49-85), accounting for 103 LM were included. LM were treated in the OM and OP setting in 65(63%) and 38(37%) cases, respectively. SBRT was delivered before chemotherapy (n=49,48%) or following first (n=31,30%) or further chemotherapy lines (n=23,22%). KRASmut was present ab initio in 49% of cases (n=50). Dose regimens included 48-60 Gy in 3-8 fractions, resulting in a median BED of 115.5 (range 76.8-151.2) $Gy_{10}$ : a BED>100$Gy_{10}$ was delivered in 92% (n=95) of treatment courses. Median follow-up was 16 (range 2-45) months. Median LC was not reached; 1- and 3 years LC rates were 85% and 67% respectively. At multivariate analysis (MVA), only a BED≤100$Gy_{10}$ (p=0.029, HR 2.99, IC95% 1.1- 9.5) and prior second or further chemotherapy line (p=0.031, HR 3.2, IC95% 1.1- 8.1) were independently correlated with impaired LC. No grade >2 toxicity was observed. Focusing on patients receiving no chemotherapy or first-line treatment before SBRT, 1- and 3 years LC rates were 92% and 77% respectively (median: NR). At MVA, while BED≤100$Gy_{10}$ (p=0.036, HR 6.96, IC95% 1.9-25.7) was associated with poorer LC, lack of KRASmut (p=0.037, HR 0.26, IC95% 0.07- 0.9) was independently correlated with improved LC. **Conclusions:** SBRT in LM from OM/OP CRC results in high LC rates, particularly with intensive regimens in non-heavily pretreated patients. No impact of KRASmut was observed. Due to possible occurrence of secondary mutations in the pretreated population, a subset analysis in the chemotherapy-naive/first-line subset suggests an impact of KRASmut in radioresistance that should be further explored in a dose-escalation perspective. Re-biopsy or liquid biopsy may be useful to identify de-novo mutations. Research Sponsor: None.

**265** **Poster Session**

# A retrospective study of colorectal cancer CT imaging features involving microsatellite instability.

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**Background:** This study aimed to identify correlations between CT imaging characteristics accessible to the radiologist with microsatellite instability (MSI) status among colorectal cancer. Noninvasive identification of MSI status can decrease time to treatment if pathology can be bypassed. **Methods:** 109 colorectal cancer patients were identified retrospectively from 2011 to 2018 with an average age of 61 years (58 male, 51 female) and had MSI pathological assessment. These subjects all had CT abdomen and pelvis obtained at the time of initial diagnosis. Imaging features of both the primary and metastatic lesions were assessed. Primary lesion features included: location, size, stage, attenuation compared to the liver, growth pattern, tumor margin, primary mass area, and the presence/absence of mesenteric infiltration. Metastatic lesion features included: size, attenuation relative to liver, and enhancement pattern. Patient age, sex, and tumor staging were obtained from patient medical records. Statistical analysis was performed using various methods, including chi-square, Mann-Whitney and Kruskal-Wallis tests. **Results:** 5 tumor characteristics displayed a statistically significant relationship with MSI status. MSI-H lesions were more likely to be clinical Stage 2, and MSI-L lesions were more likely to be Clinical Stage 3 ($p=0.012$). Tumors with distant metastasis were more likely MSI-L versus regional metastasis were more likely MSI-H ($p<0.001$). Primary mass area was larger in the MSI-H than MSI-L group ($p<0.001$). Tumors in the left colon are more likely to be MSI-L versus CRC tumors in the right colon are more likely to be MSI-H ($p<0.001$). MSI-H tumors had a lower primary tumor Hounsfield Unit standard deviation (SD) than MSI-L tumors ($p=0.002$). The remaining features were not statistically significant. For primary lesion, these include mesenteric infiltration ($p=0.189$), tumor margin ($p=1.000$), enhancement pattern ($p=0.498$), growth pattern ($p=0.127$), tumor CT density ($p=0.162$), and liver density ($p=0.105$). Enhancement of metastasis also had no difference ($p=0.376$). **Conclusions:** Multiple CT imaging characteristics were predictive of MSI classification in CRC tumors, holding potential clinical relevance that can help guide individualized therapy. While these results were statistically significant, more research must be done to validate these findings and to derive useful nomograms. Research Sponsor: None.

# Development of a methylation-based, tissue-agnostic test for the detection of molecular residual disease by circulating tumor DNA.

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**Background:** Clinical validation studies support tumor-informed molecular residual disease (MRD) as a prognostic biomarker for disease recurrence across multiple solid tumor types. However, these tests are not always feasible due to the occasional lack of tumor tissue. Here, we discuss the design of a test for tissue-agnostic MRD detection and its application to a cohort of patients with colorectal cancer (CRC). **Methods:** A targeted panel composed of differentially methylated regions was developed. A machine-learning model was trained on differential methylation patterns in order to classify plasma samples as MRD-positive or MRD-negative. Performance of the independently trained classifier was assessed in a cohort of 247 patients enrolled in the Bespoke CRC trial (NCT04264702). These patients had MRD results available using a tumor-informed circulating tumor DNA (ctDNA) assay (Signatera), of whom 163 were persistently MRD-negative without clinical progression and 84 had MRD-positive results. Tissue-agnostic MRD results were compared to the tumor-informed results by calculating the percent positive agreement (PPA). Additionally, the differentially methylated allele fraction (DMAF) from the tissue-agnostic test was compared with variant allele frequencies (VAFs) from the tumor-informed test. **Results:** In the Bespoke CRC clinical cohort (72% non-Hispanic White, 54% male, mean age 61.4±12.3 years), 71 (28%) patients had stage II CRC, and 147 (60%) had stage III CRC. Overall, PPA was 86% (95% CI: 70-100%) and specificity was 97% (95% CI: 93-100%). When categorizing based on tumor-informed MRD VAF levels, PPA was 97% for VAF >0.2%, 100% for VAF 0.1-0.2%, 89% for VAF 0.04-0.1%, and 68% for VAF <0.04%. The DMAFs strongly correlated with the VAFs from the tumor-informed test and the correlation was independent of disease stage, histology, age, and sex. **Conclusions:** This is the first study of its kind demonstrating high concordance between a tissue-agnostic MRD test and a clinically validated tumor-informed ctDNA assay. These findings demonstrate that in cases where tissue is not available or of inadequate quality, a methylation-based tissue-agnostic assay may serve as a potential alternative for MRD detection. Research Sponsor: None.

# Prognostic factors for thermo-ablation of liver metastases in colon cancer: A Maltese perspective.

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**Background:** Malta is a small Mediterranean archipelago with an annual colon cancer incidence of 300 patients per year and a cumulative 5-year overall survival of 58%. Despite major improvements in therapeutic strategies, colon cancer remains the third leading cause of cancer deaths locally with an annual mortality rate of circa 110 patients per year. Percutaneous thermal ablation of colorectal liver metastases involves exposure of metastatic deposits to temperature extremes using radiofrequency, microwave or cryotherapy. The role of this case series was to examine the prognostic value of clinico-pathological variables to provide insight into patient selection for ablative therapy. **Methods:** This retrospective analysis of 66 adult patients with colorectal liver oligometastatic disease who underwent thermal ablation between January 2014 and January 2022, reviews demographic and pathological criteria including tumour grade, presence of vascular invasion, mismatch repair and k-RAS & BRAF mutational status, CEA & Ca19.9 assay at time of treatment. Other parameters include metachronous versus synchronous disease, hepatic burden and concurrent chemotherapy. All these co-variables were analysed using the Cox Regression Hazard model to assess their impact on progression free- (PFS) and overall survival (OS). **Results:** For this patient cohort, the mean PFS and OS were 13.1 and 33.8 months. Our data indicates that most individuals experience progression within the first 10 months, with a gradual decline in PFS for the remaining individuals. A Ca19.9 level within normal limits (p=0.004, HR 0.157-0.700 95% CI) was related to an improved PFS but not OS. The presence of synchronous metastases (p=0.004, HR 0.168-0.715 95% CI) and ablation of one lesion (p=0.013 HR 0.191 (0.052-0.702 95 CI) were favourable prognostic markers for improved OS. k-RAS wt status was a prognostic adverse biomarker for PFS (p=0.002, HR 1.529, 6.686 95% CI). There was a paucity of data for BRAF and MMR status so these were excluded from the final analysis. No statistical survival benefit was noted for patients treated with chemotherapy, and other variables including gender, age, primary tumour sidedness, LVSI or CEA value. **Conclusions:** There are two surprising trends here- the paucity of survival benefit for the use of chemotherapy in conjunction with ablation and the worse outcome for k-ras wt tumours. We postulate that the absence of BRAF sub-stratification at the time of patient accrual, impacted statistical hazard. This case series suggests that the prognosis and survival of colorectal liver oligometastatic disease is multifactorial and complex, highlighting the need for further genomic and molecular insights. Research Sponsor: None.

**268** **Poster Session**

# Landscape of novel gene fusions in gastrointestinal tumors.

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**Background:** Gene fusions are recognized as critical drivers of cancer and targets for precision therapies. The 2022 ASCO provisional clinical opinion on tumor genomic testing includes recommendations for fusion testing and recognizes the strengths of RNA-based approaches. Targeted DNA/RNA sequencing panels can be used to detect well characterized, clinically actionable fusions, but their ability to detect rare novel fusions (i.e., those with uncharacterized fusion partners) can be limited by assay design. Whole-exome, whole-transcriptome sequencing (WES/WTS) may improve the identification of actionable gene fusions by detecting both known and novel fusions. This work describes clinically actionable novel fusions detected in gastrointestinal (GI) cancers. **Methods:** The study included samples from GI cancers that were analyzed using the OncoExTra test between May 2018 and May 2024. The assay uses tumor-normal WES/WTS to identify somatic alterations, including gene fusions with known and unknown partners and breakpoints.Clinical data were queried foractionable fusions with novel fusion partners that were unique in the entire laboratory database and have not, to our knowledge, been previously reported in the existing scientific literature or public databases. Actionable fusions were defined as those associated with FDA-approved targeted therapies or those that made patients eligible for a clinical trial. **Results:** Tumor samples from 2,776 patients with GI cancers were evaluated with a total of 75 actionable fusions identified in samples from 74 patients. Of these, 25.3% (n=19) were novel, and 74.7% (n=56) were previously described or recurrent fusions. Novel fusions were detected in small bowel (2 of 81), colorectal (4 of 1,621), gastric/gastroesophageal junction (GEJ) (2 of 259), pancreatic (7 of 466) and biliary (4 of 131) cancers. Thirteen (63.2%) of the novel fusions involved genes associated with FDA-approved therapies, including FGFR2 (n=6), BRAF (n=3), MET (n=2), FGFR1 (n=1) and NTRK2 (n=1). Of the fusions associated with FDA-approved therapies, 20% (8/40) of recurrent fusions and 38.5% (5/13) of novel fusions were detected only by RNA sequencing. No fusions were identified in appendiceal or esophageal cancers. **Conclusions:** Gene fusions with novel fusion partners represented one fourth of all actionable fusions detected in GI tumors. Use of a comprehensive WES/WTS testing approach that detects fusions agnostic of the fusion partner may affect clinical care in GI cancers by increasing the number of patients eligible for targeted therapies.
Research Sponsor: Exact Sciences.

# Impact of sirtuin genes expression on first-line treatment outcomes in patients (pts) with metastatic colorectal cancer (mCRC) enrolled in CALGB/SWOG 80405 (Alliance).

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**Background:** Sirtuins (SIRT1-7) are class III histone deacetylases with different roles in CRC as tumor suppressive and oncogenic, mainly influencing proliferation, migration, invasion, DNA damage and chemoresistance. In the present study, we explored the association between tumor gene expression of sirtuins and first-line treatment outcomes in mCRC pts. **Methods:** We evaluated the impact of the tumor gene expression of sirtuin genes (*SIRT1*-7) on treatment outcomes (overall survival: OS; progression free survival: PFS) in 433 pts enrolled in the randomized, phase III CALGB/SWOG 80405 (NCT00265850) trial. RNA extracted from FFPE tumor samples was sequenced using the HiSeq 2500 (Illumina). Subgroup analyses were based on treatment (bevacizumab [bev] (N = 226) vs. cetuximab [cet] (N = 207)). For each gene, expression was divided into tertiles of high, medium and low. The associations between tumor gene expression and outcomes (PFS and OS) were assessed using log-rank test in univariate analysis. Multivariable Cox proportional hazards regression model was used to compute hazard ratios (HR) and 95% confidence intervals while adjusting for baseline characteristics. Treatment-gene interactions were analyzed using the likelihood ratio test. **Results:** Of the seven genes analyzed, *SIRT3*$^{high}$ was associated with both longer PFS (HR: 0.59 [95% CI: 0.46-0.74], $P < 0.001$) and OS (HR:0.64 [95% CI: 0.50-0.83], $P = 0.002$) overall. Treatment-gene interaction testing found significant interactions for *SIRT1* and *SIRT5* with PFS (*P* values: 0.021 and 0.016, respectively) and OS (*P* values: 0.038 and 0.014, respectively) with targeted therapies (bev vs. cet). Multivariate analysis showed *SIRT1*$^{high}$ associated with longer PFS (HR: 0.62 [95% CI: 0.42-0.89], $P = 0.014$) and OS (HR: 0.59 [95% CI: 0.40-0.89], $P = 0.036$) in cet-treated pts; in contrast with shorter observed PFS (HR: 1.38 [95% CI: 0.96-1.99], $P = 0.22$) and OS (HR: 1.31 [95% CI: 0.90-1.93], $P = 0.23$) in bev-treated pts. *SIRT5*$^{high}$ showed similar results and associated with longer PFS (HR: 0.64 [95% CI: 0.45-0.91], $P = 0.039$) and OS (HR: 0.57 [95% CI: 0.39-0.83], $P = 0.013$) in cet-treated pts, with no similar associations in bev-treated pts. **Conclusions:** Our results highlight a potential role for *SIRT3* as prognostic and, *SIRT1* and *SIRT5* as predictive biomarkers for first-line treatment in mCRC, with differential effects based on targeted agents. Significant interaction favored cet treatment in *SIRT1*$^{high}$ and *SIRT5*$^{high}$ tumors. However, previous reports suggest an opposite role of *SIRT5* that has been linked to chemotherapy and/or cet resistance in CRC via the inactivation of succinate dehydrogenase complex subunit A. Therefore, further studies are warranted to dissect the mechanism behind the interaction of *SIRT5* with cetuximab activity and establish the potential of targeting *SIRT1/5* in CRC. Research Sponsor: National Cancer Institute; P30CA014089; National Cancer Institute; U10CA180821; National Cancer Institute; U10CA180882; SWOG; U10CA180888; National Cancer Institute; UG1CA180830; Genetech.

# Prediction of immunotherapy response in deficient mismatch repair colorectal cancers using an immune-enhanced exome and transcriptome platform.

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**Background:** While immune checkpoint blockade (ICB) is used to treat metastatic deficient mismatch repair (dMMR) colorectal cancers (CRCs), such tumors are heterogeneous and resistance to ICB is frequent. To date, biomarkers that can consistently predict patient response to ICB are lacking. We determined whether biomarkers that integrate tumor and immune-related molecular mechanisms could predict ICB response. **Methods:** Consecutive patients with metastatic dMMR CRCs (N=32) who had been treated with anti-PD-1 therapy were profiled using a validated immune-enhanced exome and transcriptome platform (ImmunoID NeXT, Personalis) using tumor and paired normal tissue. This platform provides antigen-specificity of T or B cells, indicated by the TCR or BCR respectively, neoantigen prediction, tumor mutation burden (TMB), immune profile, HLA allele-specific loss of heterogeneity (LOH) and a Neo-antigen Presentation Score (NEOPS) that combines a neoantigen prediction tool with mechanisms of immune evasion. Tumor variables were analyzed in relationship to objective tumor response, and a Cox proportional hazards model was fit to predict progression-free survival (PFS) or overall survival (OS). **Results:** In CRCs, TCR and BCR repertoire diversity were each significantly increased in tumors from responders versus those with stable or progressive disease after anti-PD-1 treatment (p=0.015 and p=0.048, respectively). Neoantigen burden score, TMB, and HLA allele-specific LOH (reflects loss of antigen presentation machinery) were each unable to distinguish responders from nonresponders. An immune checkpoint gene expression signature was increased in nonresponders relative to responders (P= 0.005). Neo-antigen presentation score (NEOPS) was prognostic when adjusted for tumor grade only for OS (p=0.048). **Conclusions:** The T-cell and B-cell receptor (TCR/BCR) repertoire, as well as NEOPS, were each significantly predictive of favorable response to anti-PD-1 therapy in dMMR tumors. In addition, an immune checkpoint gene expression signature was predictive of resistance to ICB. Research Sponsor: None.

# Differential gut microbiota profiles among normal colorectal mucosa, adenomatous and hyperplastic polyps in healthy volunteers: ColoMAR-1 study.

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**Background:** Increasing evidence indicates that the interaction between intestinal microbiota and host cells plays an important role in colorectal cancer oncogenesis through immune, inflammatory and/or metabolic alterations. However, there are no clear evidence in the very early adenoma-carcinoma sequence at which microbiota species influences future steps to colorectal carcinoma. This study aims do find a point in the adenoma-carcinoma sequence ad witch microbiota species suggest the risk of a future tumor to develop less invasive forms of early diagnosis. **Methods:** In this cross-sectional study, healthy volunteers above 45 years old were selected to participate. Exclusion criteria includes previous colonoscopy performed in the last 5 years. Stool sample were collected before a colonoscopy with biopsy. Samples of normal mucosa were systematically obtained, and all polyps found were removed. In fecal samples, microbiota species were identified using GoodGut test: *Bacteroides fragilis*, *Subdoligranulm variabile*, *Ruminococcus*, *Roseburia*, *Coprococcus*, eubacteria, *Faecalibacterium prausnitzii* and *Gemella morbillorum*. Number of genomic copies/μL were analyzed using the Wilcoxon test. **Results:** From February to June 2024, 29 healthy volunteers between 45 and 69 years were recruited. 62% were female. Adenomatous polyps or hyperplastic polyps were found in 9 volunteers. No case of colorectal cancer was found. For species *S. variabile*, *Ruminococcus*, *Roseburia*, *Coprococcus* and *F. prausnitzii*, higher counts were found in stool samples from volunteers with adenomatous or hyperplastic polyps than in those from volunteer with a normal colonoscopy, both in the total count and after correction for bacterial load ($p<0.001$). For *B. fragilis*, bacterial count was higher in samples from volunteers with adenomatous polyps, but not in the samples from people with hyperplastic polyps compared with those samples from volunteers with a normal colonoscopy. *G. morbillorum* counts were not related to any type of polyps. None of this counts met the laboratory-designated threshold related to colorectal cancer. **Conclusions:** The presence of *B. fragilis* could be related to the differential risk of malignancy of adenomatous compared to hyperplastic polyps. Research Sponsor: None.

**Bacterial count for each sample and specie.**

| Bacteria specie* | Normal mucosa | Adenoma | Hyperplastic |
|---|---|---|---|
| **B. fragilis** | 22.13 | 22.3 | 20.42 |
| **S.variable** | 6.06 | 7.29 | 7.53 |
| **Ruminococcus, Roseburia, Coprococcus** | 6.08 | 7.29 | 7.47 |
| **F. prausnitzii** | 4.7 | 6.28 | 7.09 |
| **G.morbillorum** | 20.91 | 20.54 | 20.84 |

*Number of genomic copies/μL.

# Biological and therapeutic impact of tumor mutational burden in microsatellite instability/mismatch repair-deficient (MSI/dMMR) metastatic colorectal cancer (mCRC) treated with immune checkpoint inhibitors (ICI).

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**Background:** Microsatellite instability (MSI) is linked to hypermutability and response to ICI(s), especially in colorectal cancer. The predictive impact of tumor mutational burden (TMB) in ICI-treated MSI/dMMR metastatic colorectal cancer remains controversial, with some studies suggesting low TMB as a potential biomarker of resistance to ICI(s). We aimed to investigate the biological relevance of TMB in MSI/dMMR mCRC and its impact on survival outcomes in ICI(s)-treated patients (pts). **Methods:** MSI/dMMR mCRC pts treated with ICI(s) from the NIPICOL trial (NCT03350126) and the immunoMSI prospective monocentric cohort with available data from whole exome sequencing (WES) and RNA-sequencing (RNA-seq) were included. All cases were centrally assessed for MSI and dMMR status. TMB was calculated as described (Dupain et al., BMC, 2024). The search for a cutpoint able to optimize the difference in survival between 2 groups of patients with low and high-TMB was applied. The main endpoint was progression free survival (PFS) by iRECIST criteria. The cohort (NIPICOL, immunoMSI) was included as a covariate in the analysis of differential gene expression and pathway enrichment, with various tumor mutational burden (TMB) thresholds evaluated, specifically 10 and 30 mutations per megabase (mut/Mb). **Results:** A total of 99 pts were analyzed, with 4 (4%) excluded after central MSI/dMMR status review. Of these, 52 received anti-PD1 plus anti-CTLA4 therapy, while 43 received anti-PD1 alone. Median follow-up was 25 months. Median TMB was 51.8 mut/Mb (IQR: 35.9-74.9). TMB was not predictive of patient PFS, whatever the TMB threshold applied, including cutpoints already reported in the literature, the 2-year PFS rates being 80% versus 80% for TMB $\leq$ 10 and $>$ 10 mut/Mb, and 79% versus 82% for TMB $\leq$ 30 and $>$ 30 mut/Mb, respectively. Multivariate analysis including treatment type confirmed no association between TMB and PFS (HR 1.00, 95% CI [0.97–1.04]). TMB levels were not linked with specific signalling pathways related to immune checkpoints or antigen presentation as evaluated by RNA-seq with Gene Set Enrichment Analysis (GSEA) at a TMB threshold of 10 or 30 mut/Mb. However pathways involved in cell proliferation (e.g., "cell cycle", "DNA replication") were increased in TMB high (adjusted p-value $<$0.05). RNA-seq signature quantification and deconvolution analysis revealed similar cell populations (e.g., fibroblasts, B cells, T cells) between high and low TMB groups at a TMB threshold of 10 and 30 mut/Mb. **Conclusions:** TMB has no survival impact for patients with ICI(s)-treated dMMR/MSI mCRC and does not discriminate two biologically distinct subpopulations. Other biomarkers should be developed for personalized medicine amongst these patients. Research Sponsor: GERCOR; Ligue nationale contre le cancer; BMS; MSD AVENIR.

**273** **Poster Session**

# Serial monitoring of circulating tumor DNA as a marker of response to immune checkpoint blockade in MSI-H gastrointestinal tumors.

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**Background:** Microsatellite instability- high (MSI-H) tumors have high rates of response to immune checkpoint blockade (ICB) with prolonged durability of response. However, predicting who will have a response is difficult and there are limitations to radiographic evaluation of response. As circulating tumor DNA (ctDNA) offers a noninvasive approach to monitoring response and therapeutic efficacy, we sought to correlate ctDNA change with response to ICB in MSI-H gastrointestinal (GI) tumors. **Methods:** We conducted a retrospective study of MSI-H GI patients receiving ICB who underwent serial ctDNA sequencing. Tissue-informed ctDNA monitoring utilized comprehensive genomic profiling results of pre-treatment tumor tissue from a clinical trial assay based on FoundationOne CDx, a variant selection algorithm to exclude non-tumor variants, and multiplex PCR of up to 16 patient-specific variants to detect and quantify ctDNA. ctDNA changes were correlated with radiographic response. **Results:** A total of 20 patients with MSI-H tumors were evaluated including 17 colon, 2 small bowel, and 1 pancreatic. For 12 patients, it was the first exposure to ICB whereas 8 had previously progressed and were being rechallenged with ICB. Response was seen in 9 patients with a best overall response (BOR) of complete response (CR) in 4 and partial response (PR) in 5 while 5 showed stable disease (SD) and progression (PD) in 6. All patients had baseline ctDNA samples and at least 1 serial ctDNA after ICB initiation (median number of samples was 3). The mean time from baseline to first on-treatment sample was 30 days (range 13-64 days). 4 patients had un-detectable ctDNA levels at baseline and remained undetectable on treatment. Of the 16 that had detectable ctDNA at baseline, 11 had a decrease in ctDNA levels at 1st on-treatment sample and 5 had an increase. The mean percent change of ctDNA at 1st on-treatment for those who had a response (CR + PR) was a 76% decrease while the mean percent change for those with SD was a 7% increase and for those with PD a 72% increase (p=0.05). Of the 6 patients with BOR of PD, 5 had an increase in ctDNA and 1 had undetectable levels at baseline. Of the 5 patients with BOR of SD, 2 had a decrease, 1 had an increase, and 2 were undetectable at baseline. Duration of SD did not correlate with initial ctDNA response. **Conclusions:** Early ctDNA change correlated with response in MSI-H GI tumors suggesting that serial ctDNA monitoring may provide early indication of response even prior to radiographic evaluation. Research Sponsor: None.

# Investigating the clinical and molecular characteristics of class II and III BRAF mutations and their response to anti-EGFR therapy in MSS CRC: A comprehensive analysis.

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**Background:** *BRAF* mutations (mts) are a heterogeneous group of molecular alterations seen in colorectal cancer (CRC). Class I *BRAF* mts (V600) are known to be associated with aggressive biology to CRC, but the knowledge of clinical characteristics of class II and III *BRAF* mts is limited. In this large cohort analysis, we evaluated the clinical and molecular features of class II and III *BRAF* mts and performed comparative analyses for their impact on survival outcomes. **Methods:** A total of 24402 MSS CRC samples were profiled by NGS (592-gene, NextSeq; WES, WTS NovaSeq) (Caris Life Sciences, Phoenix, AZ). *BRAF* mts were detected by NGS and classified using published literature (1). MAPK pathway activity score (MPAS) was calculated using RNA expression data. Real-world overall survival (OS) was obtained from insurance claims and calculated from tissue collection to last contact, while post-treatment survival from first treatment to last contact. KM estimates were calculated for defined patient cohorts. Significance was determined as $P<0.05$. **Results:** A total of 1270, 134, and 327 patients with class I, II, and III *BRAF* mts were identified. *BRAF* mts overall and class I *BRAF* mts were significantly less common among younger pts (age<50) compared to patients ≥ 50 (4.7% vs. 7.4 %; $P< 0.001$ and 3.2% vs. 5.7%, respectively, $P<0.001$). Class I *BRAF* mts were enriched with consensus molecular subtype 1 (CMS1) (class I, II, and III: 46% vs. 14% vs. 18% respectively) while class II and III mts had more CMS2 subtype compared to class I (2%, 30% and 29%, $p<0.05$). Higher MPAS scores were noted among patients with class I *BRAF* mts than those with class II and III mts (1.73 versus 0.38 vs 0.79). Class I *BRAF* mts and KRAS mts were nearly mutually exclusive (0.5%), while KRAS mts were relatively common among class II and III mts (12.8% and 27.4%, respectively). Median OS for patients with wild-type *BRAF*, class I, II, and III *BRAF* mts were significantly different (29.8, 17.3, 21.8, and 24.6 months, respectively, $p<0.01$). Similar OS was seen among patients who did not receive anti-EGFR therapy (28.3, 15.2, 20.7- and 21.9 months $P<0.001$). Patients with class III mts had significantly better OS compared to patients with class I mts (HR=1.28 CI: 1.09-1.48 p=0.002), and while no significant difference was noted for patients with class II vs class I mts, numerically more favorable outcomes were noted (mOS 21.8 vs 17.3 months P=0.133). Among treated with anti-EGFR, patients with class II and III *BRAF* mts had numerically better post-anti-EGFR survival compared to class I mts; however, this was not statistically significant (21.2 vs 14.2 vs. 10.4 respectively P=0.3). **Conclusions:** Class II and III *BRAF* mts represent a distinct biological subgroup of MSS CRC with distinct prognoses. Class II and III *BRAF* mts are associated with improved survival outcomes compared to patients with class I mts. 1. Sahin et al. JCO OP 2021. Research Sponsor: None.

# Identifying the optimal post-surgical timing of molecular residual disease (MRD) detection in colorectal cancer (CRC) using an ultra-sensitive assay: Interim results from the VICTORI study.

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**Background:** While detection of MRD using ctDNA is prognostic for recurrence in CRC, some patients still recur prior to MRD detection. VICTORI is prospectively investigating NeXT Personal, an ultra-sensitive NGS-based MRD assay, to profile patients with resected CRC. **Methods:** Patients with CRC treated with curative intent (all stages) are tested for MRD using NeXT Personal, a bespoke assay with up to ~1,800 tumor-informed single nucleotide variants (SNVs) identified from whole-genome sequencing. Plasma is collected prior to surgery, every 2 weeks post-surgery up to week 8 (MRD landmark window), and every 3 months for up to 3 years (surveillance). We present preliminary results on 397 samples from the first 62 patients. **Results:** A total of 62 patients (N=36 rectal [58%], N=26 colon [42%]; N=46 stage I-III [74%], N=16 stage IV [26%]) were included in our analysis. Baseline pre-surgical sensitivity (treatment naive) was 93.5% [N=29/31]. Pre-surgical positivity rate in patients who had received neo-adjuvant therapy and had residual cancer at the time of surgery was 67% [N=16/24]. 60 patients were evaluable for clinical outcomes. At a median follow-up of 355 days, 15 patients (25%) had a recurrence. Of these, all patients were ctDNA-positive prior to recurrence (100%, 14/14; 1 pt excluded due to lack of samples prior to recurrence). ctDNA detection preceded clinical relapse by a median of 194 days [range: 5-397]); ctDNA for 78.6% (N=11/14) were first detected in the MRD landmark window. All landmark-positive recurrences occurred within one year of surgery. The 14 recurrent cancers with samples were first detected at a median ctDNA concentration of 28.7 parts per million (PPM) (range 2.4-111,120), with 64.3% (9/14) of those detections in the ultra-low range of <100 PPM. MRD detection at week 4 and week 8 had the greatest reduction in RFS (HR 12.86 [2.74-60.28], p=0.0012 week 4; HR 16.14 [3.52-74.09], p=0.0004 week 8), with weeks 4, 6 and 8 having similar higher prevalence of ctDNA detection (36.0% [18/50], 35.3% [18/51], 38.5% [20/52] respectively). Week 2 detection rate was 17.0% (8/46). cfDNA concentration was highest at week 2 (4.63ng/ml vs. 2.22 at baseline, p=0.00028) and 4 (3.68 vs. 2.22, p=0.0081), returning to baseline levels at week 6 (2.28 vs. 2.22, p=0.67) and 8 (2.48 vs. 2.22, p=0.64). **Conclusions:** In this interim report after a median ~1 year follow-up, NeXT Personal detected MRD for all patients prior to disease recurrence. Most initial MRD detection was in the ultra-sensitive range <100ppm and detection at 4-8 weeks after surgery was highly prognostic for recurrence. Research Sponsor: Personalis, Inc.

# Clinical and molecular characterization of *FAP* and *SPP1* in colorectal cancer (CRC), CALGB (Alliance)/SWOG 80405 and real-world data.

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**Background:** *FAP* and *SPP1* contribute to immune modulation in the CRC tumor microenvironment (TME). FAP is expressed by cancer associated fibroblasts, aiding in tissue remodeling and tumor invasion. SPP1 is an integrin-binding protein expressed by tumor associated macrophages that promotes tumor growth, adhesion, and metastasis. We present a clinical and molecular characterization of *FAP* and *SPP1* in CRC. **Methods:** We analyzed 24,257 CRC samples tested at Caris Life Sciences (Phoenix, AZ) with WTS (Illumina NovaSeq), NextGen DNA sequencing (NextSeq), and PD-L1 expression (SP142, positive ≥ 2+, 5%). RNA deconvolution analysis estimated cell infiltration in the TME. Data from the phase 3 CALGB/SWOG 80405 trial (NCT00265850) on 433 metastatic CRC patients treated with bevacizumab (Bev, n = 226) or cetuximab (Cet, n = 207) in combination with first-line chemotherapy were also evaluated. RNA isolated from FFPE tumor samples were sequenced with HiSeq 2500 (Illumina). Overall survival (OS) and progression-free survival (PFS) were compared using Cox regression in categorical gene expression tertiles (high (T3), medium (T2), and low (T1)) for *FAP* and *SPP1*. **Results:** *FAP*-T3 and *SPP1*-T3 had increased PD-L1 positivity (q<0.05), while *SPP1*-T3 also demonstrated increased MSI-H (8.4% vs 5.3%) and TMB-H(>10 mt/mb, 15.0% vs 10.1%) status relative *SPP1*-T1 (all q<0.001). T3 of both genes of correlated with increased M1/M2 macrophages, NK cells and T cell inflamed score (q<0.001); dendritic cells and neutrophils were increased in *SPP1*-T3 and *FAP*-T1 tumors (q<0.05). *FAP*-T3 and *SPP1*-T3 had increased pathway activation of epithelial-mesenchymal transition (EMT), inflammatory response, TNF-a signaling, angiogenesis and *KRAS* signaling (all $q < .005$). In Caris cohorts, *FAP*-T3 demonstrated worse OS in Cet/panitumumab treated CRC (T3: 23.6 vs T1: 21.0 months [mo], $P = .005$; HR 0.85, 95% CI [0.77-0.95]), but *SPP1* did not correlate with OS. In 80405, *FAP*-T3 showed shorter PFS (T3: 9.5 vs T2: 11.5 vs T1: 12.6 mo, T3 vs T1 (reference) adjusted HR 1.27 [1.07-1.51]) and OS (25.2 vs 29.4 vs 35.5 mo, adjusted HR 1.31 [1.10-1.57]). Similarly, *SPP1*-T3 demonstrated worse PFS (9.0 vs 12.7 vs 14.0 mo, adjusted HR 1.29 [1.12-1.48]) and OS (20.9 vs 34.0 vs 36.3 mo, HR 1.24 [1.07-1.44], $P < 0.001$). Treatment interaction tests noted *FAP*-T1 CRC benefited from Cet over Bev with respect to PFS ($P = 0.003$) and OS ($P = 0.044$), while SPP1-T1 tumors demonstrated a PFS benefit with Cet ($P = 0.009$). **Conclusions:** Our results indicate that increased *FAP* and *SPP1* expression is associated with immune cell infiltration, EMT, and inflammatory signaling. Additionally, their expression may be prognostic and predictive of targeted therapy. These data support the evaluation of *FAP* and *SPP1* as predictive markers and therapeutic targets in CRC. Research Sponsor: P30CA014089, UG1CA180830 [HJL]; UG1CA180830 and U10CA180888 (SWOG).

# Acquired gene alterations potentially related to resistance to anti-HER2 therapy in HER2-positive metastatic colorectal cancer (mCRC).

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**Background:** While HER2-targeted therapies benefit patients (pts) with advanced HER2-positive mCRC, the development of resistance remains inevitable. This study aims to explore the landscape and frequency of genetic alterations putatively associated with acquired resistance to anti-HER2 therapy in HER2-positive mCRC. **Methods:** In this retrospective-multicenter international study involving Mayo Clinic, Duke Cancer Center and National Cancer Center Hospital East (Japan), we descriptively compared comprehensive genomic analyses from tissue or blood samples of pts with HER2-positive CRC pre-treatment and post-treatment with HER2-targeting agents. **Results:** A total of 36 pts were included in this study. 29 (80%) had HER2-positive mCRC as confirmed by immunohistochemistry (IHC) and fluorescent in situ hybridization (ISH), the remaining pts had HER2 amplification detected on blood or tissue . 34/36 (94%) pts were administered their first anti-HER2 agent after progressing on ≥2 prior lines of treatment. The most common anti-HER2 regimen administered was trastuzumab/pertuzumab (50%, 18/36), followed by trastuzumab-tyrosine kinase inhibitors (TKIs) (tucatinib or neratinib) (30%, 11/36) and trastuzumab deruxtecan (14%, 5/36). Putative genomic mechanisms of acquired resistance were detected in 72% (26/36), while the remaining pts lacked new genomic alterations that might explain resistance to anti-HER2 therapy, suggesting alternative escape mechanisms. Acquired alterations in downstream or alternate bypass pathways were identified in 72% (26/36) of pts; most commonly in the RAS/RAF pathway and in EGFR (34.6%, 9/26 each), followed by 27% (7/26) of pts with *MET* point mutations (mts) and amplification; activating muts in PI3K/AKT and in 27% (7/26), most commonly CDK12 (60%, 4/7). HER2 receptor alterations were detected in 3 pts (8.3%), all of whom received trastuzumab-tucatinib and included *HER2 L755S, HER2 V777L, HER2 D769Y, HER2 G727A* and *HER2 S310F*, which are known to cause resistance to , however, information about tucatinib resistance is not yet available. Other altered pathways included AR, RB, NOTCH and HRD. Two pts lost HER2 expression on blood and tissue NGS post anti-HER2 therapy. **Conclusions:** The emergence of resistance might be in part related to acquired alterations that constitutively activate signaling pathways parallel or downstream of HER2, bypassing HER2 inhibition. Additional histologic studies, deep mutational scanning, larger populations and correlative analysis are needed to validate these findings. Research Sponsor: None.

# Comprehensive landscape of FGFR variations and prognosis revelance in colorectal cancer from circulating tumor DNA and tissue gene analyses in 2083 patients.

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**Background:** FGFR genomic alterations have been identified, and several FGFR inhibitors are approved for various malignancies, including bladder and cholangiocarcinoma. However, limited studies have shown FGFR alterations and their prognostic impact in colorectal cancer (CRC). **Methods:** This retrospective analysis involved tissue and blood samples from a single-institute cohort (The Sixth Affiliated Hospital of Sun Yat-sen University) of stage I-IV CRC patients from 2014 to 2024. FGFR genomic alterations were analyzed using tissue NGS and plasma ctDNA. Disease-free survival (DFS) for stages I-III and progression-free survival (PFS) for stage IV were compared between FGFR-altered and negative CRC patients, matched in a 1:2 ratio. Validation of genomic alterations and their prognostic impact was conducted using cBioPortal and TCGA data. **Results:** FGFR alterations were found in 7.89% of 608 tissue samples and 10.95% of 2083 ctDNA samples. The most frequent alteration was FGFR1 amplification in all cohorts, along with FGFR2 p.R165W and FGFR3 p.A634T mutations in ctDNA. FGFR alterations correlated with younger age, left-sided colon tumors, adenocarcinomas, poor differentiation, advanced TNM staging, and multi-organ metastasis. Prognostic analysis showed a higher proportion of FGFR alterations in stage IV patients (66.23% vs. 59.14%), lower rates of no evidence of disease (NED) (6.62% vs. 17.68%), and significantly shorter PFS (median 17 vs. 20 months, HR 2.184, p=0.007). In stages I-III, FGFR-altered patients also had a lower NED rate (79.22% vs. 82.19%) and significantly shorter DFS (median 13 vs. 29 months, HR 2.833, p=0.002). Among frequent alterations, FGFR1 amplification was significantly associated with poor prognosis (PFS 17 vs. 20 months, HR 2.221, p=0.014; DFS 13 vs. 29 months, HR 2.964, p=0.002), while others showed no significant correlation. In stage IV patients receiving bevacizumab, FGFR1 alteration was linked to significantly shorter PFS (p=0.0021). KEGG analysis indicated FGFR1 was associated with angiogenesis pathways. FGFR alteration profiles and prognostic outcomes were consistent across tissue samples, ctDNA analyses, and databases. **Conclusions:** Our study outlines the genomic landscape and prognostic significance of FGFR in CRC, highlighting FGFR1 amplification as the dominant alteration. FGFR alterations were associated with poorer outcomes, as reflected in lower NED rates and shorter DFS and PFS. FGFR1 amplification was notably linked to reduced DFS and PFS, and it correlated with shorter PFS in patients receiving bevacizumab, indicating a potential link to resistance against bevacizumab. Research Sponsor: None.

| | ctDNA n=2083 | Tissue gene n=608 | cBioPortal Database n=6998 |
|---|---|---|---|
| **FGFR Total** | 228/10.95% | 48/7.89% | 528/8.0% |
| **FGFR1** | 76/3.65% | 29/4.77% | 264/4.0% |
| **FGFR2** | 78/3.74% | 10/1.64% | 125/1.9% |
| **FGFR3** | 104/4.99% | 10/1.64% | 139/2.1% |

# Clinical-pathologic characteristics in young patients with metastatic colorectal cancer: Experience from a specialized cancer institute in Peru.

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**Background:** Colorectal cancer (CRC) represents the third cause of malignant neoplasm in the world and the second in terms of mortality. CRC is a major cause of cancer-related morbidity and mortality in Peru. Diagnosis at younger ages represents a challenge. In this study, we evaluated prognostic factors in patients who were metastatic at diagnosis or progressed to metastatic disease during follow-up. **Methods:** A descriptive cross-sectional study, between 2013 and 2022, 66 patients under 40 years old and stage IV at diagnosis were analyzed. Parameters evaluated were age, gender, comorbidity, performance status and stage of the disease at the diagnosis, localization, surgery undergone, systemic treatment regimen, response to first-line treatment, incidence of K-RAS, sites and number of metastases, expression of tumor predictors (CEA levels), and survival times. A multivariate analysis was conducted with factors that were considered statistically significant in the univariate analysis. **Results:** Median age was 35 years [29.25-37] and the female/male ratio was 34/32. From 66 patients at stage IV disease, 30 patients (45.5%) were right-side, 36 (54.5%) were left-side colon cancer. At the end of follow-up, 51 patients had died, and 15 survived. Overall survival (OS) was significantly better in patients older than 35 years old ($p<0,05$), those who underwent primary tumor surgery ($p<0,001$), negative perineural invasion ($p<0.0045$). The treatment most common was CAPOX with a median OS of 17 months ($p<0.013$). The multivariate analysis shows that patients who did not undergo primary surgery had worse disease-specific OS (hazard ratio, CI 2.21.95% [1.15-4.25], $p< 0.018$) and there was no statistical difference between chemotherapy regimens and treatment line. **Conclusions:** In young patients with metastatic CRC, Primary tumor surgery and negative perineural invasion are significant factors influencing overall survival. Patients older than 35 years and those who underwent surgery showed better survival. These findings reflect the importance of early colorectal cancer screening and improve the risk assessment. Research Sponsor: None.

# Challenges in patient selection for surgical options in V600E BRAF–mutated metastatic colorectal cancer.

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**Background:** $^{V600E}$ BRAF mutated metastatic colorectal cancer (mCRC), which represents almost 10% of mCRC, is associated with a decreased disease-free interval and increased risk of recurrence compared to other mCRC molecular subtypes; however, data show wide outcome heterogeneity within this subgroup. Surgery is known to be associated with improved outcomes in patients with mCRC, particularly in those with liver-only disease. While many patients with $^{V600E}$BRAF have very aggressive disease and never have the chance to undergo surgery, it is not clear how to navigate the clinical heterogeneity and choose patients that might benefit the most from loco-regional treatment. **Methods:** We retrospectively evaluated$^{V600E}$ BRAF mutated mCRC patients from two independent high-volume institutions (MDACC and IOV IRCCS) who underwent metastectomy from Oct 2009 to Oct 2023. Patients with MSI-H disease and those who had R2 surgery were excluded. Clinicopathological features, such as sex, age, histology, radicality and site of surgery, were collected from institutional datasets. Relapse free survival (RFS) was defined as the time from surgery to first relapse or progression. Overall survival (OS) was defined as the time from surgery to the date of death or last follow up. Survival was estimated with the Kaplan Meier method, and the Cox proportional Hazard model was adopted for survival comparison. **Results:** We collected data from 71$^{V600E}$ BRAF mutated mCRC patients (N=37 MDACC, N=34 IOV IRCCS); 49% were female, 61% had right-sided primary tumor and 62% had synchronous metastatic disease. Median age at metastatic disease diagnosis was 62 years old (IQR: 61-28). For 52% of patients, surgery was performed on liver-only disease; 35% involved the peritoneum. There were no emergency interventions. Median RFS was 7.9 months (95% CI: 5.4 – 10.5) and median OS was 39 months (95% CI: 28.8 – 51.7). No clinicopathological features, including sex, sidedness, histology, differentiation, presentation (synchronous, oligometastatic), or site of intervention, were associated with RFS. However, radicality (R1) was associated with worse RFS (HR=3.5 [95% CI: 1.8 – 6.5], p=0.0003) and OS (HR=2.7 [95%CI: 1.3 – 5.6], p=0.0122). **Conclusions:** In this retrospective evaluation of patients with $^{V600E}$ BRAF mutated mCRC who underwent metastectomy, only surgical radicality, observed in the post-operative setting, was associated with recurrence and survival outcomes. Therefore, while surgery should be evaluated whenever radical resection is feasible, the identification of patients with $^{V600E}$ BRAF mutated mCRC who might benefit the most from metastasectomy during pre-operative prognosis evaluation remains challenging. Research Sponsor: None.

# Role of *RAS/BRAF* and *PIK3CA* mutations in tissue and plasma for prognostic assessment in metastatic colorectal cancer (mCRC).

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**Background:** Mutations in *RAS/BRAF* (*RAS/BRAFm*) and *PIK3CA* (*PIK3CAm*) genes have been described as mechanisms of resistance to anti-EGFR agents in mCRC. However, the utility of liquid biopsy in their baseline detection remains unknown. We aim to analyse oncological outcomes according to *RAS/BRAF* and *PIK3CA* mutational status in tissue & plasma in mCRC. **Methods:** We conducted aretrospective study in patients (pts) with mCRC from 2017 to 2024. *RAS/BRAF* and *PIK3CA* mutational status assessment was performed at baseline using Idylla (Biocartis) and Cobas PIK3CA kits, respectively. Cox models were used to analyse overall survival (OS) in *RAS/BRAFm* and *PIK3CAm* vs wild-type (WT) population. **Results:** We analysed 404 paired samples (tissue & plasma) from 202p with mCRC. *RAS/BRAFm* was described in 45% (91p) in tissue and in 43.6% (88p) in plasma. *PIK3CA* was tested in 84p, with *PIK3CAm* detection in 19% (16p) in tissue and 11.9% (10p) in plasma. Liver involvement was more frequent in *RAS/BRAFm* vs WT (80.6% vs 63.9%; p=0.02), and in *PIK3CAm* vs WT population (94.4% vs 63.6%; p=0.01). Concordance study (tissue & plasma) for *RAS/BRAF* and *PIK3CA* are presented (Table). Subjects with concordant results had worse OS in *RAS/BRAFm* (18.1m vs 51.1m, HR=2.60; p<0.001) regardless of liver involvement, and showed the same trend in *PIK3CAm* (22.6m vs 28.1m, p=0.81). Patients with discordant results and plasma *RAS/BRAFm* or plasma *PIK3CAm* showed worse OS (23.7 m vs 51.1m; p = 0.006 and 16.4 m vs 42.5 m; p=0.18, respectively) and a lower disease control rate with anti-EGFR therapy (p = 0.04 and p=0.42, respectively). *PIK3CAm* were evenly distributed in *RAS/BRAF* WT vs mutated tumors (50% vs 51.5%; p=0.92) and had no independent prognostic impact in OS after stratifying by *RAS/BRAF* mutational status; *RAS/BRAF* WT (p=0,93) and *RAS/BRAFm* (p=0,91). **Conclusions:** Baseline determination of *RAS/BRAFm* and *PIK3CAm* by liquid biopsy in mCRC provide valuable prognostic and predictive information, extending its utility beyond tissue-based analysis. Research Sponsor: None.

**Correlation study of *RAS/BRAF* and *PIK3CA* mutational status in tissue and plasma in mCRC.**

| ***RAS/BRAF* tissue** | | | | ***PIK3CA* tissue** | | | |
|---|---|---|---|---|---|---|---|
| ***RAS/BRAF* Plasma** | **Mutated n (%)** | **WT n (%)** | **Total n (%)** | ***PIK3CA* Plasma** | **Mutated n (%)** | **WT n (%)** | **Total n (%)** |
| **Mutated n (%)** | 72 (80) | 15 (13.4) | 87 (43.1) | Mutated n (%) | 7 (43.8) | 2 (2.9) | 9 (10.7) |
| **WT n (%)** | 18 (20) | 97 (86.6) | 115 (56.9) | WT n (%) | 9 (56.2) | 66 (97.1) | 75 (89.3) |
| **Total n (%)** | 90 (100) | 112 (100) | 202 (100) | Total n (%) | 16 (100) | 68 (100) | 84 (100) |
| | Positive agreement: 72/90: 80% Negative agreement: 97/112: 86.6% Overall agreement: 169/202: 83.6% KI*: 0.67 | | | | Positive agreement: 7/16: 43.8% Negative agreement: 66/68: 97.1% Overall agreement: 73/84: 86.9% KI: 0.49 | | |

KI: Kappa Index.

# Investigating survival outcomes in patients with early- vs late-onset colorectal cancer with liver metastases.

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**Background:** Approximately 50% of patients with colorectal cancer develop liver metastases (CRLM). The incidence of early onset colorectal cancer (EOCRC), defined as <50 years of age at time of diagnosis, is increasing. In this study, we aim to explore the prognosis of early vs late onset colorectal cancer patients with CRLM who receive curative-intent treatment. **Methods:** A retrospective study was conducted to assess clinical outcomes in patients with CRLM presenting with EOCRC compared to those with late onset colorectal cancer (LOCRC). All patients underwent surgical resection and/or ablation between 2009 and 2024 at a tertiary care center. Groups were defined by patient age at the time of diagnosis of the primary tumor $< 50$ years versus $\geq 50$ years. Demographics, tumor characteristics, and treatment modalities were compared between groups using Chi-square, independent samples t-test, and median test. Survival outcomes were assessed using Kaplan-Meier. **Results:** 342 patients met inclusion criteria. There were 85 (24.9%) EOCRC patients, with a median age at diagnosis of 44 years versus 63 years in the LOCRC group. EOCRC patients had fewer major medical comorbidities. EOCRC patients had a smaller proportion of right-sided (15.9% vs 29.6%, P=0.049) tumors. Rates of synchronous metastases (74.1% in EOCRC vs 72.4%, P=0.781), median size (28cm in EOCRC vs 25cm, P=0.421) median number of metastases (2 in EOCRC vs 2, P=0.385), and initial CEA levels (339.18 + 1276.48 ng/ml vs 328.87 + 2520.59 ng/ml, P=0.482) were similar between groups. However, EOCRC patients had higher rates of node-positive disease (75.6% vs 58.8%, P=0.010) and Fong Clinical Risk Scores 3-5 (57.0% vs 39.7%, P=0.009). Both groups underwent similar operative management with no differences in morbidity, but more EOCRC patients received neoadjuvant therapy (82.4% vs 66.1%, P=0.017). There were no significant differences in *RAS/RAF* (37.6% in EOCRC vs 34.2%, P=0.669) or *TP53* (25.9% vs 34.6%, P=0.98) mutational status or microsatellite instability (stable in 74.1% in EOCRC vs 69.3%, P=0.523). Both intra- (46.5% in EOCRC vs 50.6%, P=0.535) and extra-hepatic (47.7% in EOCRC vs 38.1%, P=0.129) recurrences occurred at similar rates, with no significant differences in post-recurrence treatments received. Patients with CRLM and EOCRC had significantly worse overall survival, with a median time of 80 months versus 116 months (P=0.042). **Conclusions:** Despite similar clinical, pathologic, and genetic features, fewer medical comorbidities, and receiving similar medical and surgical treatments, EOCRC patients had worse median overall survival than LOCRC patients. High risk Fong Clinical Risk Scores were more common in the EOCRC group, who also received neoadjuvant therapy at higher rates. Additional research is needed to understand the differences in survival. Research Sponsor: None.

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# Sex differences in the colorectal tumor immune microenvironment.

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**Background:** Males have increased risk and mortality rates of colorectal cancer (CRC) compared to females, but the reasons for these disparities are not well understood. Immune responses differ by sex, and a robust T cell response is an important prognostic indicator in CRC. We hypothesized that females exhibit a stronger CRC-associated T cell response, contributing to reduced mortality in females compared to males. **Methods:** The study cohort included 2,751 incident CRC patients (1,337 female; 1,414 male) from a population-based case-control study. T cell receptor (TCR) abundance and clonality were quantified using immunoSEQ, and a gastrointestinal pathologist scored tumor-infiltrating lymphocyte counts per high-powered field (TILs/hpf). We used logistic regression to estimate odds ratios (ORs) and 95% confidence intervals (CIs) for the associations between T cell metrics and sex, adjusting for age, anatomic site, and microsatellite instability (MSI) status. Cox proportional hazards regression was used to estimate hazard ratios (HRs) and 95% confidence intervals for the association between sex and 5-year all-cause and CRC-specific survival, adjusting for known prognostic indicators. We also conducted a formal test for interaction between sex and T cell features in relation to survival outcomes. **Results:** There were no associations between sex and TCR abundance (OR=1.05, 95% CI=0.98-1.13, p=0.14), TCR clonality (OR=0.99, 95% CI=0.91-1.06, p=0.72), or TILs/hpf (OR=1.05, 95% CI=0.88-1.25, p=0.60). We observed no differences in overall or disease-specific five-year survival between males and females. Further, the associations between T cell responses and survival did not significantly differ by sex. **Conclusions:** In a well-powered analysis, no robust sex differences were observed in the CRC immune microenvironment or in the relationship between T cell infiltration and CRC outcomes. The high burden of CRC and the emergence of immunotherapy as an important therapeutic approach suggest a need for further research investigating immunological sex differences in the setting of treatment response. Research Sponsor: James and Ruth Levitan Cancer Research Endowment from the Case Comprehensive Cancer Center; National Cancer Institute at National Institutes of Health; R01CA197350, R01CA081488, P30CA014089, U19CA148107, & P01CA196569; Gift from Daniel and Maryann Fong.

**Summary of associations between the colorectal tumor T cell response and sex.**

| T cell Features | Odds Ratio (Female vs. Male) | Lower 95% CI | Upper 95% CI | P Value |
|---|---|---|---|---|
| **TCR Abundance** | 1.05 | 0.98 | 1.13 | 0.14 |
| **TCR Clonality** | 0.99 | 0.91 | 1.06 | 0.72 |
| **TILs/hpf* <2** | 1.05 | 0.88 | 1.25 | 0.60 |

*A total of 99 patients (45 female; 54 male) with missing TILs/hpf data were excluded from the regression analysis.

# Circulating tumor DNA for predicting complete response to total neoadjuvant therapy in locally advanced rectal cancer: ENSEMBLE-2.

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**Background:** Total neoadjuvant therapy (TNT) has dramatically shifted the paradigm in the treatment of locally advanced rectal cancer (LARC), prolonging survival with high rates of pathologic complete response (pCR) and introducing non-operative management (NOM). However, no predictive biomarkers of TNT efficacy, the regrowth of NOM, or prognosis have been developed. Circulating tumor DNA (ctDNA) is a minimally invasive biomarker used to detect molecular residual disease (MRD) and predict recurrence in colon cancer after curative resection. The effectiveness of ctDNA MRD status as a predictive biomarker for TNT was evaluated in the Phase II TNT study, ENSEMBLE-2 (jRCTs071210143), conducted in Japan. **Methods:** Patients with LARC undergoing TNT were enrolled in ENSEMBLE-2. Protocol treatment was defined as total mesorectal excision (TME) following long course chemoradiotherapy (LCCRT: 50.4Gy, capecitabine) plus four cycles of CAPOX. NOM was allowed if a clinical complete response (cCR) was achieved in the evaluation after TNT. ctDNA MRD was measured by Signatera (Natera, Inc.) at the following time points: baseline, after LCCRT, after TNT, post operative 4w, 12w, 24, 36 and 48w in the GALAXY trial (UMIN000039205). **Results:** A total of 28 patients were enrolled in the study. Treatment was discontinued at the patient's request in one case. After completing TNT, TME and NOM were performed in 21 (77.8%) and 6 (21.4%) patients, respectively. ctDNA positivity rates were 96.4. % (27/28) at baseline, 14.8% (4/27) after LCCRT, and 34.6 % (9/26) after TNT. Post-LCCRT ctDNA status was not significantly associated with cCR + near CR (nCR) vs. incomplete clinical response (iCR), pCR vs. non-pCR (p=0.065 and p=0.539, respectively). In contrast, ctDNA status after TNT was significantly associated with cCR + nCR vs. iCR (p=0.028), as well as pCR vs. non-pCR (p=0.038). **Conclusions:** Our study indicates that post TNT ctDNA status may be a predictive biomarker for TNT response in LARC patients. ctDNA negativity upon completion of neoadjuvant treatment may indicate a favorable response. Clinical trial information: jRCTs071210143. Research Sponsor: None.

**Correlation with ctDNA, clinical response, treatment after TNT and pathological response.**

| | After LCCRT | p value | After TNT | p value |
|---|---|---|---|---|
| **Clinical Response** | cCR + nCR 18 / 1 | 0.065 | cCR + nCR 15 / 4 | 0.028 |
| **cCR + nCR vs. iCR (ctDNA -/+)** | iCR 5 / 3 | | iCR 2 / 5 | |
| **Treatment after TNT** | TME 17 / 3 | 0.438 | TME 11 / 9 | 0.063 |
| **NOM vs. TME (ctDNA -/+)** | NOM 6 / 0 | | NOM 6 / 0 | |
| **Pathological response after TME** | pCR 5 / 0 | 0.539 | pCR 5 / 0 | 0.038 |
| **pCR vs. non pCR (ctDNA -/+)** | non pCR 12 / 3 | | non pCR 6 / 9 | |

Fisher's exact test was used to statistically examine the correlation between ctDNA MRD status and the factors at each timing.

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# Evaluation of CSF1R as a potential prognostic biomarker in colorectal cancer (Alliance).

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**Background:** Immune checkpoint inhibitors (ICIs) have shown limited efficacy in colorectal cancer (CRC) outside of mismatch repair (MMR) deficient/microsatellite instability-high (MSI-H) tumors. Most CRC cases are MMR proficient /microsatellite stable and demonstrate resistance to ICIs, attributed to the tumor microenvironment. Colony Stimulating Factor 1 Receptor (CSF1R), a regulator of tumor-associated macrophages, has emerged as a potential target due to its association with poor prognosis in CRC. **Methods:** Data from 433 metastatic CRC (mCRC) patients in the CALGB/SWOG 80405 trial were analyzed. CSF1R RNA was extracted from formalin-fixed, paraffin-embedded tumor samples and sequenced. Patients were divided into tertiles of CSF1R expression (high, medium, low). Progression-free survival (PFS) and overall survival (OS) were compared using Kaplan-Meier curves, log-rank tests, and Cox proportional hazards models. Subgroup analyses were conducted based on liver metastases, treatment type (Bevacizumab, Cetuximab, FOLFOX, FOLFIRI), and MSI status. **Results:** Low CSF1R expression (CSF1R-L) was associated with significantly better survival outcomes. Median PFS was 12.9 months for CSF1R-L, compared to 11.2 and 9.2 months for CSF1R-M and CSF1R-H ($p = 0.003$). Median OS was 35.8 months for CSF1R-L versus 32.5 and 24.4 months for CSF1R-M and CSF1R-H ($p = 0.00086$). Stratification by liver metastases showed that CSF1R-L remained a strong prognostic factor only in patients with liver metastases. There were no significant survival differences based on CSF1R expression in bevacizumab-treated patients. However, in cetuximab-treated patients, CSF1R-L tumors had significantly longer PFS (12.9 vs 8.8 months, $p = 0.037$) and OS (35.8 vs 21.4 months, $p = 0.002$) than CSF1R-H tumors. Similarly, in FOLFOX-treated patients, CSF1R-L tumors showed longer PFS and OS ($p = 0.018$ and $p = 0.013$, respectively), and in FOLFIRI-treated patients, OS was significantly longer for CSF1R-L tumors (42.2 vs 24.7 months, $p = 0.022$) than CSF1R-H tumors. **Conclusions:** Our data suggests that CSF1R expression may serve as a prognostic biomarker in CRC, with lower expression potentially linked to improved survival, particularly in patients with liver metastases. These findings warrant further investigation of CSF1R as a possible prognostic and predictive biomarker, as well as a potential therapeutic target in CRC. Clinical trial information: NCT00265850. Research Sponsor: None.

# Safety and biomarker assessment of ST316, a novel peptide antagonist of ß-catenin, in patients with advanced solid tumors.

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**Background:** Constitutive activation of ß-catenin, resulting in its nuclear accumulation and deregulated transcriptional activity, is a key event in colorectal cancer (CRC). Additionally, pathway mutations correlate with immune exclusion in CRC and across different tumor types. Disruption of the interaction of ß-catenin and its co-activator BCL9 is sufficient to suppress oncogenic ß-catenin transcriptional activity, without impacting its homeostatic functions in normal tissue. ST316 is a first-in-class, cell-penetrating peptide antagonist of the ß-catenin and BCL9 interaction. Preclinical evaluation revealed significant bioavailability and potent activity in CRC models with no impact on ß-catenin homeostatic functions such as intestinal stem cell survival or bone morphology. **Methods:** A phase 1-2 (P1-2) escalation-expansion study has enrolled patients (pts) with advanced solid tumors likely to harbor abnormalities in the Wnt/ß-catenin pathway, to assess the safety, pharmacokinetics (PK), biomarker and preliminary activity of ST316, and to recommend a P2 dose (RP2D). Paired biopsies were collected when feasible and assessed for drug penetration and pharmacodynamic biomarkers. Peripheral blood (PB) collected pre- and post-treatment was assessed by flow cytometry for polymorphonuclear myeloid-derived suppressor cells (PMN-MDSCs). **Results:** As of Sep.24, 2024; 23 pts with CRC (N=14), NSCLC (N=4), PDAC (N=2), or Breast and Ovarian cancer (N=1, each), were treated in 6 cohorts (0.5-12mg/kg) with IV ST316, once weekly. Treatment related AEs were observed in 17/23 pts and were mostly low grade, except for 2 G3 AEs that resolved: Fatigue (1) and ALT/AST elevation (1). No dose-limiting toxicities (DLTs) were observed. Single agent ST316 showed disease stabilization in 4 pts. PK analysis showed dose-proportional increases in Cmax and AUC for doses up to 8 mg/kg. IHC analysis showed substantial tumor penetration of ST316 in all analyzed samples and evidence of a treatment-induced redistribution of β-catenin in a subset of pts. PB analysis indicated high baseline levels of the ß-catenin-driven immunosuppressive PMN-MDSCs , followed by a significant decrease after ST316 exposure ($p<0.005$). The RP2D selected for combination cohorts in CRC is 8mg/kg; additional PD assessments are ongoing. **Conclusions:** Monotherapy ST316 was well-tolerated with dose-proportional PK and substantial tumor uptake. Pharmacodynamic analyses demonstrate mechanistic evidence of antagonism of ß-catenin signaling, including redistribution of ß-catenin subcellular localization and significant depletion of the ß-catenin-driven immunosuppressive PMN-MDSC population. Based on these data, ST316 advanced into P2 in CRC pts in combination with SoC in 2$^{nd}$ and 3$^{rd}$ line. Clinical trial information: NCT05848739. Research Sponsor: None.

**287** **Poster Session**

# A multimodal deep learning approach for the prognostic stratification of patients with colorectal cancer (CRC).

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**Background:** Traditional prognostic systems like the AJCC TNM staging for colorectal cancer (CRC) often fall short in predicting long-term patient outcomes. These systems often rely on limited pathologic features, leading to generalized approaches to treatment despite diverse tumor heterogeneity, such as the complex interaction between the tumor-host microenvironment. There is therefore a pressing need for more accurate, scalable tools to enhance decision-making and predict outcomes for patients with CRC. **Methods:** Tumor sections from 191 CRC patients were comprehensively stained for CD8 to examine CD8-positive tumor-infiltrating lymphocytes (TILs). To account for tumor heterogeneity, multiple areas of the tumor were evaluated to include intertumoral and peritumoral areas. Slides were digitized and quantitatively analyzed using a cloud-based platform (SpatialX Diagnostics, Inc. USA). Additionally, clinicopathologic features including patient demographics, 8th edition AJCC staging, lymphovascular invasion (LVI), perineural invasion (PNI), mismatch repair (MMR) protein status, tumor location, presence/development of distant metastases, and patient follow-up were collected. Merging CD8 AI findings with clinicopathologic features, a complete data set was available for 175 patients. Cases were randomly assigned into 70% training and 30% validation. In the training set, unsupervised artificial intelligence (AI) models integrated clinicopathologic data and CD8-positive TIL counts using vision transformer models. The final model combined spatially resolved pathologic features with clinical variables to statistically predict overall survival (OS) and distant metastasis risk (DMR). **Results:** In the validation cohort, the model stratified patients into two groups based on a deep learning-derived risk score, with statistically significant differences between low-risk and high-risk groups for OS ($p<0.02$) and DMR prediction ($p<0.0001$), For OS prediction, the model selected tumor site, PNI, MMR status, epithelial and stromal CD8-positive TILs, and 5 deep vision features as the top 10 clinically significant features. For DMR prediction, the age, T- and N-stage, PNI, and 6 deep vision-derived features as key predictors. **Conclusions:** As captured by vision transformers, the distribution and spatial arrangement of CD8-positive TILs are critical factors in patient stratification for OS and DMR. AI-driven models like the one herein offer the potential for more personalized management strategies, advancing CRC management and improving patient outcomes beyond traditional prognostic staging systems and parameters. Research Sponsor: None.

# Deep learning–powered analysis of tumor-infiltrating lymphocytes (TILs) in colorectal cancer.

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**Background:** TILs are a promising biomarker in various cancers, including colorectal cancer (CRC). Advances in deep learning enable objective and efficient TIL assessment. This study applies a novel deep learning model to quantify TIL density in a large cohort of CRC tumors, examining its relationship with clinicopathologic factors and survival. **Methods:** We applied a deep learning model (Deep-ICP) to H&E-stained images from CRC patients treated at Dana-Farber Cancer Center to quantify TIL density (immune cells/mm$^2$). Kruskal-Wallis and Wilcoxon tests were used to compare categories. Overall survival (OS) was calculated from the date of diagnosis to the date of death or last follow-up. Cox proportional hazards models analyzed the association between TIL density tertiles (lower, middle, upper) and OS, adjusting for age, sex, stage, biopsy site, tumor mutational burden (TMB, mutations/megabase), and mismatch repair (MMR) status. **Results:** Of 1,477 tumor specimens tested, 1,039 (70%) were primary and 438 were metastatic. The most common sites of metastasis were liver (n=205), lung (n=86), and soft tissue (n=35). The median age at diagnosis was 56 (range 19–91) years and 706 were female (48%). Most tumors (1,214/1,300; 93%) were MMR proficient (MMRp); 86 were MMR deficient (MMRd). Median TIL density was 664 (interquartile range; 469–934) cells/mm$^2$ in primary tumors, 478 (358–685) cells/mm$^2$ in liver metastases, 415 (251–645) cells/mm2 in soft tissue ($p<0.001$). MMRd tumors had higher median TIL density than MMRp (881 vs. 620 cells/mm$^2$; $p<0.001$). After adjusting for covariates, patients with highest TIL tertile had significantly longer OS than patients with lowest TIL tertile (median OS: 75.4 [95%CI 67.4-87] months; 100.5 [83.9-116] months and 99.1 [92.4-157] months for lower, middle and upper TIL tertiles respectively; Table). In the MMRp subset, higher TIL density was still predictive of improved OS (p adj=0.010). **Conclusions:** This study demonstrates the feasibility of deep learning powered TIL quantification in CRC, potentially informing differences in tumor microenvironment and prognostication. Future prospective studies are needed to validate the results and explore their therapeutic implications in personalized CRC management. Research Sponsor: None.

| Variable | Hazard Ratio (95% CI) | P-value |
|---|---|---|
| **TIL upper tertile (vs. lower)** | 0.70 (0.50-0.99) | 0.04 |
| **TIL middle tertile (vs. lower)** | 1.04 (0.77-1.39) | 0.80 |
| **Age (continuous)** | 1.02 (1.00-1.03) | 0.008 |
| **Male (vs. female)** | 1.13 (0.86-1.47) | 0.38 |
| **Stage II (vs. stage I)** | 4.8 (0.59-40) | 0.14 |
| **Stage III (vs. stage I)** | 9.4 (1.20-73) | 0.03 |
| **Stage IVA (vs. stage I)** | 23 (2.94-178) | 0.003 |
| **Stage IVB (vs. stage I)** | 46 (5.6-369) | 0.0003 |
| **Liver specimen (vs. primary)** | 1.39 (1.00-1.92) | 0.050 |
| **Lung specimen (vs. primary)** | 0.78 (0.47-1.29) | 0.33 |
| **Lymph node specimen (vs. primary)** | 0.77 (0.27-2.17) | 0.62 |
| **Other metastasis (vs. primary)** | 1.23 (0.78-1.93) | 0.37 |
| **TMB≥10 (vs. <10)** | 1.17 (0.82-1.68) | 0.38 |
| **MMRd (vs. MMRp)** | 0.65 (0.29-1.46) | 0.30 |

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# The prevalence and prognostic significance of *KRAS G12C* mutation in metastatic colorectal cancer.

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**Background:** Kristen rat sarcoma virus (*KRAS*) mutations are reported in 40-45% of colorectal cancer (CRC), and they are associated with poor prognosis. *KRAS G12C* mutation is reported in 3-4% of CRC patients and may be associated with worse prognosis compared to *RAS non-G12C* mutations. Here, we explored the prevalence and prognostic significance of *KRAS G12C* mutation compared to *RAS non-G12C* and *BRAF V600E* in a cohort of CRC patients from a tertiary cancer center in the United States (US). **Methods:** The cBioCancer Genomics Portal was used to obtain genomic data. The prevalence of *KRAS G12C* mutation, *RAS non-G12C* (*KRAS* G12D, *KRAS* G13D, *KRAS* G12V, *KRAS* A146T, *KRAS* G12A, *KRAS* G12S, *KRAS* G13C, *KRAS* V14I, *KRAS* Q61H, *KRAS* G12R, *KRAS* Q61R, *KRAS* K117N, *NRAS Q61K*, *NRAS Q61R*, *NRAS Q61L*, *NRAS Q61H*, *NRAS G13D*, *NRAS G13C NRAS G13R NRAS G12D*, *NRAS G12C*, *NRAS G12V, NRAS G12V* and *NRAS G60V*) mutations and *BRAF V600E* mutations was calculated. The median overall survival (mOS) was explored in CRC (MSK, Cancer Cell 2018 cohort). Patients in this cohort were not treated with *KRAS G12C* targeted therapy. Survival analysis was performed using the Kaplan-Meier method with log-rank test comparisons. **Results:** In this cohort of patients with stage IV CRC (N=1134), the prevalence of *KRAS G12C* mutation in patients with stage IV CRC was 3.1%. In patients with *RAS non-G12C* mutations, *BRAF V600E* mutation, and *RAS/BRAF V600E* wild-type (WT), the prevalence was 41.7%, 7.5%, and 47.7% respectively. The mOS in *KRAS G12C* mutation, *RAS non-G12C* mutations, *BRAF V600E* mutation, and *RAS/BRAF V600E* WT was 65 months (m), 55m, 19m and 105m respectively. Compared to *RAS non-G12C*, the mOS of CRC patients with *KRAS G12C* was (65m vs 55m, p=0.54) . Compared to patients with *BRAF V600E* mutation, the mOS was better in patients with *KRAS G12C* (65m vs. 19m, $p < 0.0001$) and in patients with RAS non-G12C (55m vs. 19m, $p < 0.0001$). **Conclusions:** The prevalence of *KRAS G12C* mutation in this cohort was relatively similar to other reports at 3.1%. Consistent with several other reports, there was no statistically significant difference in the mOS between patients with *KRAS G12C* mutation and patients with *RAS non-G12C* mutations. Patients with BRAF V600E mutations had worse mOS compared to patients with *KRAS G12C mutation* and *RAS non-G12C* mutations. The reported high mOS in this cohort is possibly due to selection bias (patients able to afford and receive treatment at this tertiary center). Research Sponsor: None.

# Pan-cancer microsatellite instability testing using a multiplex assay including long mononucleotide repeats.

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**Background:** Microsatellite instability (MSI)/mismatch repair deficiency (dMMR) testing is a critical component of molecular testing for gastrointestinal cancers, among others. These analyses are important for the identification of patients with lynch syndrome and eligibility for treatment with immunotherapy. The current standard of care testing can identify most cases, however improved identification of MSI-H status is needed for non-colorectal cancers and for cancers with alterations outside MLH1/PMS2. A recent analysis indicated the potential for improved MSI-H detection with a multiplex assay including long mononucleotide repeats (LMR). **Methods:** Patients were selected based on MSI status, MMR gene variants, or the presence of a high tumor mutation burden and consented for participation as part of an IRB-approved protocol (UW15068). Formalin-fixed tissue slides were annotated by a pathologist for tumor and normal tissue. Slides were scraped and DNA isolated. The LMR MSI Analysis System (Promega, Madison, WI) and the OncoMate MSI Dx Analysis System (Promega) were performed on cancer DNA and normal tissue DNA when available per the manufacturer's protocols. Instability of microsatellites and the MSI score were reported and correlated with clinical MMR immunohistochemistry (IHC) and commercial next-generation sequencing (NGS). **Results:** A total of 96 cancer samples were tested across 94 patients with over 20 different cancer types, including lung, colon, rectal, gynecologic, and esophagogastric cancers. Tumors from 20 patients were known to have microsatellite instability based on clinical NGS testing. 27 cases were found to be MSI-H based on the OncoMate assay. All of these cases were also MSI-H per the LMR assay and an additional 3 cases were found to be MSI-H using only the LMR assay. Of these 3 cases all were microsatellite stable by NGS. This number includes 1 colorectal and 2 esophagogastric cancer cases. One of the cases was found to have loss of MSH6 by IHC with an MSH6 mutation (MSH6 L1061fs). One of the cases was MMR intact by IHC, though did have a MSH6 exon 7 splice acceptor alteration that failed NGS calling quality control. The last case did not have MMR IHC performed, no detected MMR gene alterations, and a tumor mutation burden of 6.3 mutations per megabase. **Conclusions:** In this limited dataset, the utility of various methods for determination of MSI, dMMR, and TMB including PCR, IHC, and NGS were demonstrated using a patient cohort spanning solid tumor types. Promising results were observed with the LMR MSI Analysis System. These findings need to be further validated in a larger dataset and correlated with the response to immunotherapeutics. Research Sponsor: Promega.

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# 5-hydroxymethylcytosine as a liquid biopsy biomarker in colorectal cancer.

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**Background:** 5-hydroxymethylcytosine (5hmC) is an epigenetic modification which regulates gene expression and is associated with active transcription. The optimization of 5hmC sequencing in cell-free DNA (cfDNA) could therefore enable assessment of gene activity through a liquid biopsy. We aimed to investigate the 5hmC landscape of colorectal cancer (CRC) in plasma-derived cfDNA to evaluate the potential of 5hmC modifications as a liquid biopsy-based biomarker of CRC. **Methods:** Genome-wide 5hmC modifications were analyzed with a low-input whole-genome 5hmC sequencing method based on selective chemical labeling in cfDNA from 72 CRC patients and 70 healthy samples. Differential 5hmC analysis between CRC and healthy samples was conducted using DESeq2 (padj $< 0.05$ and log2FC $<= -1$ and $>=1$). Low-pass whole genome sequencing (LP-WGS) was performed on all cfDNA samples. Tumor fraction was estimated using ichorCNA to classify samples into high ct-fraction ($>=0.1$) and low ct-fraction ($<0.1$) groups. Nucleosome profiling was performed on LP-WGS to reveal chromatin accessibility patterns. Elastic Net Regression (ENR) was applied to identify changes in 5hmC levels for classifying samples as cancerous or healthy. **Results:** Differential cfDNA 5hmC levels of 139 genes distinguished high ct-fraction CRC (n=16) and healthy samples (n=70). Although the remaining CRC samples (n=56) had similar ct-fractions as the healthy samples, as estimated by ichorCNA, we found a differentially hydroxymethylated gene signature could discriminate between these low ct-fraction CRC and healthy samples. Nucleosome profiling analysis with LP-WGS in cfDNA was used to identify active transcriptional drivers of the genes with increased 5hmC levels. The analysis revealed increased chromatin accessibility near the binding sites for the transcription factor caudal-related homeobox transcription factor 2 (*CDX2*), a master transcriptional regulator of intestinal cell fate and cancer, in CRC patients relative to healthy samples. Corroborating this, *CDX2* and its downstream targets had increased 5hmC levels in cfDNA in CRC relative to healthy samples, consistent with colorectal-derived tumor DNA in CRC plasma samples. **Conclusions:** Genes with differential 5hmC levels in cfDNA can discriminate CRC and healthy samples. Our 5hmC signature identified CRC samples that did not harbor copy number gains or losses; which is typically used to estimate ct-fraction and routinely included in targeted ctDNA sequencing assays. Integration of 5hmC and LP-WGS can be used to identify gene activation in disease. In the context of CRC, CDX2 displayed increased chromosome accessibility in the promoters of its target genes which in turn displayed increased 5hmC levels. Further studies are aimed at optimizing and validating 5hmC-based biomarkers throughout patient treatment. Research Sponsor: St. Louis Community Foundation; ClearNote Healthy.

# The role and molecular mechanism of GDF5 in regulating cell migration and angiogenesis via the Hippo-YAP signaling pathway in colorectal cancer.

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**Background:** Distant metastasis is a leading cause of poor prognosis and mortality in colorectal cancer (CRC) patients, with angiogenesis playing a crucial role in tumor progression. The TGF-β superfamily has been shown to significantly influence tumor angiogenesis, yet the biological function and molecular mechanism of Growth Differentiation Factor 5 (GDF5), a member of this superfamily, remain unexplored in CRC. Given that the efficacy of current anti-angiogenic therapies is limited to a subset of patients, identifying new therapeutic targets involved in tumor angiogenesis is of great clinical importance. GDF5, with its potential role in angiogenesis regulation, presents a promising candidate to bridge this therapeutic gap. **Methods:** In this study, we analyzed GDF5 expression in CRC tissue samples using both the TCGA dataset and samples from our center, revealing significantly lower GDF5 expression in tumors compared to adjacent normal tissues. Moreover, GDF5 downregulation correlated with lymph node metastasis and disease recurrence. Functional assays in vitro demonstrated that GDF5 overexpression reduced tumor cell migration and invasion, while its knockdown produced the opposite effect. Co-culture assays showed that GDF5 knockdown significantly enhanced endothelial cell migration and angiogenesis, whereas overexpression diminished these effects. Additionally, F-actin cytoskeletal staining indicated that GDF5 overexpression led to cytoskeletal remodeling in endothelial cells. Mechanistically, we found that GDF5 overexpression in tumor cells downregulated the epithelial-mesenchymal transition (EMT) marker Slug and upregulated E-cadherin, while downregulating N-cadherin. In endothelial cells, GDF5 inhibited the Hippo-YAP signaling pathway, increasing YAP phosphorylation and reducing nuclear YAP levels. Conversely, GDF5 knockdown suppressed YAP phosphorylation and promoted its nuclear translocation. **Results:** Our findings demonstrate that GDF5 functions as a tumor suppressor in CRC by regulating cell migration and angiogenesis. This study reveals that GDF5 exerts its anti-cancer effects by modulating both EMT in tumor cells and cytoskeletal changes in endothelial cells, leading to the inhibition of the Hippo-YAP signaling pathway and reduced angiogenic potential. These results identify GDF5 as a promising new target for anti-angiogenic therapy in CRC, offering potential avenues for improving treatment strategies in patients resistant to current therapies. **Conclusions:** GDF5 acts as a tumor suppressor in colorectal cancer by inhibiting cell migration and angiogenesis through modulation of the Hippo-YAP signaling pathway, making it a promising therapeutic target for anti-angiogenic treatment in patients resistant to current therapies. Research Sponsor: Shandong Province Natural Science Fundation; ZR2020MH216.

# Retrospective study of fluoropyrimidine chemotherapy dosing and toxicity in patients with screen-detected deleterious *DPYD* gene variants.

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**Background:** Fluoropyrimidine chemotherapy agents are commonly used to treat solid tumors. Fluoropyrimidines are degraded by the enzyme dihydropyridine dehydrogenase (DPD), encoded by the *DPYD* gene. Deleterious variants in *DPYD* leads to functional DPD deficiency and increased toxicity risk. **Methods:** We conducted a retrospective study of patients with screen-detected deleterious *DPYD* variants who were treated with fluoropyrimidine chemotherapy between 01/01/15-04/15/23. Outcomes of interest were initial fluoropyrimidine dosing, dose adjustments (cycles 2-6), and early toxicities. **Results:** We identified 12 patients with heterozygous *DPYD* variants, including *2A (n=5), c.2846A>T (3), c.1129-5923C>G (2) and *13 (2). Initial chemotherapy regimens included FOLFOX (7), mFOLFIRINOX (3) and CAPEOX (1). Of 11 patients receiving FOLFOX/mFOLFIRINOX, nine began treatment with a 50-60% reduction of the 5-FU infusion dose, one received a 25% reduction, and one received a standard dose (due to treatment initiation prior to return of the genotype result). Seven patients tolerated the initial 5-FU infusion dose without further dose reductions, including three who tolerated limited dose-escalation. Four patients required further dose reduction from the cycle 1 dose, due to toxicity. The patient who received a standard 5-FU infusion dose in cycle 1 experienced grade 3 toxicity, but tolerated subsequent treatment after 50% dose reduction. The patient treated with CAPEOX tolerated 4 planned cycles with a 50% capecitabine dose reduction. **Conclusions:** Most patients who screened positive for deleterious *DPYD* gene variants tolerated fluoropyrimidine chemotherapy with a 50% dose reduction. However, 3 of 12 patients experienced toxicity requiring treatment discontinuation or dose reduction to less than 50% of the standard 5-FU dose. Research Sponsor: None.

# UGT1A1 mutation rates in a minority-rich comprehensive cancer center: A retrospective pharmacogenomic study.

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**Background:** Irinotecan is a widely used and effective treatment for various gastrointestinal malignancies. Its toxic metabolite, SN-38, is inactivated by UDP glucuronosyltransferase-1A1 (UGT1A1). However, variability in individual clearance of irinotecan and SN-38 can cause unpredictable toxicities. Mutations in the *UGT1A1* gene reduces enzyme activity, resulting in dose-limiting neutropenia and diarrhea, hospitalizations, treatment delays, and dose reductions. The best characterized germline variant is a polymorphism in the UGT1A1*28 allele, but data in underrepresented groups remains particularly lacking. **Methods:** We conducted a retrospective cohort study of patients who were treated at our minority-rich academic medical center. A total of 63 patients with gastrointestinal malignancies were tested for the UGT1A1*28 polymorphism. Data for irinotecan treatment course, related toxicities, and dose reductions was available for 41 patients. Analysis was performed using Fisher's exact test, one-way ANOVA, and Welch's two-sample test. **Results:** 73% (n = 46) of patients were identified with the UGT1A1*28 polymorphism, including 63% (n = 29) with heterozygous genotype and 36.9% (n = 17) with homozygous genotype. There were significantly more patients of UGT1A1*28 polymorphisms carriers in the African American and Hispanic population as compared to the white population (p = 0.001, 0.01 respectively). The polymorphism frequency was 0.66 in the African American group [95% CI (0.49 – 0.81)], 0.48 in the Hispanic group [95% CI (0.34 – 0.62)], and 0.06 in the white group [95% CI (0.001 – 0.30)]. The frequency was significantly lower in the white population when compared to all other races (p-value = 0.0002), to African American (p- value $<$ 0.001), and to Hispanic (p-value = 0.003) populations. There were higher rates of overall toxicities of any grade in the heterozygotes (76%) and homozygotes (64%) when compared to the wild-type group (56%). 55% (n = 6) of the homozygous and 48% (n = 10) of the heterozygous patients developed G1-G4 neutropenia, as compared to the 22% (n = 2) of the wild-type patients. **Conclusions:** Ethnic differences are known to exist in UGT1A1*28 polymorphism, with the prevalence being significantly higher in African and Caucasian populations. Historical data based on large-scale genotyping report allele frequencies of 0.42 in African/African American, 0.37 in Hispanic American, and 0.29 in Europeans. Our local population has a higher rate overall in both the African American and Hispanic populations with significantly higher rates than the white population. Patients with the UGT1A1*28 polymorphism experienced more toxicities and neutropenia of any grade. *UGT1A1* genotyping is not routinely performed in patients initiating irinotecan treatment, however it may prove to be beneficial in certain demographics with higher rates of UGT1A1*28 polymorphisms. Research Sponsor: None.

# Real-world treatment patterns, healthcare resource utilization (HCRU), and costs in newly diagnosed patients with rectal cancer (RC) in the United States (US).

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**Background:** RC constitutes 30% of colorectal cancer cases, the fourth most common cancer in the US, and is associated with a substantial economic burden. This retrospective claims analysis examined patient (pt) characteristics, all-cause and RC-related HCRU, and costs in pts who received surgery, radiotherapy (RT), or systemic therapy (ST) for RC. **Methods:** Medicare Fee-for-Service and MORE$^2$ Registry claims data were used to select pts with RC (RC cohort) and matched, non-RC controls (non-RC cohort). Pts with RC were required to have a diagnosis of RC in the primary position of ≥1 inpatient claim or in any position of ≥2 outpatient claims between Jan 1, 2016, and Dec 1, 2021, and ≥1 claim for an RC surgery, RT, or ST after diagnosis. Pts (aged ≥18 y) also needed to have continuous enrollment for 12 months preceding and minimum 30 days following the treatment index date (earliest RC treatment claim). Non-RC controls were matched to pts with RC on birth year, sex, payer type, and Deyo-Charlson Comorbidity Index (DCI). All-cause HCRU and costs (per pt per month [PPPM]) were compared for the RC and non-RC cohorts. **Results:** A total of 20,953 pts were included in each cohort (median age, 70 y; 52.5% male; median DCI, 1.0 in both cohorts). Median follow-up time was 24 months. In the RC cohort, 88.5% of pts underwent ≥1 surgery after RC diagnosis and 51.3% had ≥4 surgeries during follow-up. Colonoscopy and biopsies (59.4%), total mesorectal excision (47.7%), and proctectomy (45.2%) were the most frequent surgeries/procedures. RT was used as neoadjuvant and adjuvant therapies in 21% and 15.7% of pts, respectively; 5-fluorouracil was the most common neoadjuvant (12.7%) and adjuvant (20.9%) ST. Among nonsurgical pts (11.5%), 72.8% were treated with RT and 49.9% with 5-fluorouracil; a small proportion received immunotherapy (pembrolizumab, 1.2%; nivolumab, 0.7%). All-cause HCRU was significantly greater ($P<0.001$) in the RC cohort vs the non-RC cohort (proportion of pts with ≥1 claim for physician office/clinic visit, 81.3% vs 73.6%; emergency room visits, 68.7% vs 45.3%; hospitalization, 71.1% vs 25.9%). The mean all-cause costs PPPM were $6043 for the RC cohort vs $1538 for the non-RC cohort ($P<0.001$); mean RC-related costs PPPM were $3119. Outpatient and inpatient services were the main cost drivers in the RC cohort, accounting for 57.1% and 36.6% of all-cause expenditure, respectively, and 58.1% and 40.5% of RC-related expenditure. **Conclusions:** Most pts newly diagnosed with RC received standard of care surgical treatment, often with neoadjuvant or adjuvant RT or ST. All-cause HCRU was significantly greater for pts with RC vs those without RC; all-cause costs were almost 4 times higher, with about half of these costs related to RC and driven by outpatient services and hospitalizations. These results show a high economic burden and a need for additional treatments for pts with RC. Research Sponsor: None.

# Effect of changes in body composition and sociodemographic factors on colon cancer survival.

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**Background:** Muscle mass, muscle quality, and adipose tissue volumes have been shown to affect colon cancer survival. Further research is required to understand how change in body composition during treatment impacts colon cancer outcomes, and how covariables such mental health and socioeconomic status affect this relationship. The aim of this project was to examine how changes in body composition during chemotherapy for colon cancer impact the duration of disease-free survival (DFS) and how sociodemographic covariables (age, sex, social isolation, depression, anxiety, income, community size) influence this relationship. **Methods:** Computed tomography (CT) scans from the time of diagnosis and the end of chemotherapy were obtained from individuals treated for stage III colon cancer with oxaliplatin at BC Cancer between 2012 and 2016. Whole body tissue volumes were estimated based on CT images at the level of the third lumbar vertebra, analysed using *DAFS Express* (Voronoi Health Analytics Inc., Vancouver BC). Muscle quantity was measured as skeletal muscle index (SMI), and muscle quality was measured as skeletal muscle density (SMD) and skeletal muscle gauge (SMG). Quantity of visceral adipose tissue (VAT), subcutaneous adipose tissue (SAT), intermuscular adipose tissue (IMAT), and total adipose tissue (TAT) were also measured. Social isolation, anxiety, and depression symptoms were measured using the validated Psychosocial Screen for Cancer–Revised. Neighbourhood income and community population were estimated using postal codes. Cox proportional hazard models were calculated for the effect of body composition variables on DFS, and interactions with sociodemographic variables. **Results:** Significant reductions in SMI, SMD, SMG, VAT, IMAT and TAT were observed between diagnosis and post-chemotherapy (n=282). Improved survival was associated with higher SMD and lower IMAT at diagnosis and post-chemotherapy, lower VAT at diagnosis, and an increase in VAT or TAT during chemotherapy. Body composition had a more significant effect on survival in individuals who were older, female, or lived in smaller communities. **Conclusions:** Significant reductions in SMI, SMD, SMG, VAT, IMAT and TAT were observed between diagnosis and post-chemotherapy (n=282). Improved survival was associated with higher SMD and lower IMAT at diagnosis and post-chemotherapy, lower VAT at diagnosis, and an increase in VAT or TAT during chemotherapy. Body composition had a more significant effect on survival in individuals who were older, female, or lived in smaller communities. Research Sponsor: None.

# Impact of COVID-19 pandemic on critically ill patients with colon cancer: A population-level study.

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**Background:** The COVID-19 pandemic strained healthcare systems and disrupted pre-pandemic levels and trends in the short-term mortality of colon cancer hospitalizations. Colon cancer is the fourth most diagnosed cancer in the US and Texas. The impact on short term mortality in Texas in the years since the COVID-19 pandemic have not been explored. **Methods:** We used publicly available, deidentified, and state-wide data to conduct a population-base cohort study of hospitalizations aged ≥ 18 years admitted to intensive care units at acute care hospitals in Texas during 2016-2022 with a diagnosis of colon cancer. Hospitalizations transferred to another acute care hospital were excluded from the study. Colon cancer was identified using ICD-10-CM codes C18x and C20 selected from Clinical Classification Software Refined category NEO015: Gastrointestinal cancers - colorectal. Short-term mortality was defined as in-hospital mortality or discharge to a hospice. The response variables are pre-pandemic time trend, level change, change in time trend, and final time trend. The primary analysis method was interrupted time series analysis (ITSA) based on multilevel-multivariable logistic regression with results reported as average marginal effects per quarter (AME) and 95% confidence intervals (95% CI) in changes to the absolute probability of mortality. The matched pairs t-test was applied to ITSA and counterfactual predicted probabilities to further elucidate the impact of the COVID-19 pandemic. Sensitivity analyses was performed for hospitalizations aged ≥ 65 years and for hospitalizations without a diagnosis of COVID-19. **Results:** A total of 49,007 hospitalizations were included in the study of which 21,995 (44.9%) occurred after the beginning of the pandemic. Unadjusted short-term mortality was lower in the pre-pandemic period compared to the post-pandemic period (18.1% *vs* 20.1%). On adjusted analysis, the post-pandemic period remained associated with higher mortality rates. The pre-pandemic short-term mortality time trend was negative (AME -0.19%/quarter, 95% CI [-0.28% to -0.10%], p = 0.0001). The change in time trend was positive (AME 0.39%/quarter, 95% CI [0.23% to 0.55%], $p < 0.0001$) and the change in level was also positive (AME 1.25%, 95% CI [0.0.04% to 2.47%], p = 0.0436). The final time trend was significant and positive (AME 0.20%/quarter, 95% CI [0.07% to 0.33%], p = 0.0023). The point estimate and 95% CI for the difference between ITSA predicted and counterfactual predicted probabilities of mortality was 3.45% (3.41% to 3.48%). Similar results were observed on sensitivity analyses. **Conclusions:** Since the onset of the pandemic there was a 2% increase in the absolute probability of short-term mortality and a change from negative to positive time trend in short-term mortality. The average impact of the COVID-19 pandemic on the absolute probability of short-term mortality is estimated at over 3%. Research Sponsor: None.

# Development and validation of a deep learning–based pathomics signature for prognosis and chemotherapy benefits in colorectal cancer: A retrospective multicenter cohort study.

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**Background:** The current TNM staging system fails to provide adequate information for prognosis and adjuvant chemotherapy benefits in colorectal cancer (CRC). Pathomics, an emerging field, shows promise in improving prognosis estimation and decision-making. In this study, we developed and validated a pathomics signature ($PS_{CRC}$) that directly analyzes hematoxylin and eosin–stained slides using deep learning to predict outcomes. **Methods:** A total of 883 whole slide images from two cohorts, Harbin Medical University Cancer Hospital and The Cancer Genome Atlas (TCGA), were retrospectively analyzed. An interpretable, multi-instance deep learning model was proposed to establish the $PS_{CRC}$. Shapley additive explanations were employed to interpret the model's decisions, and gradient-weighted class activation mapping was applied to visualise the pathological phenotypes of the $PS_{CRC}$. The transcriptomics data from TCGA cohort was used to explore the potential pathogenesis underlying the $PS_{CRC}$. **Results:** The $PS_{CRC}$ was identified as an independent prognostic factor associated with both overall survival and disease-free survival. Incorporating the $PS_{CRC}$ into the TNM stage model resulted in a significant improvement in prognosis estimation, as evidenced by a notable increase in net reclassification improvement and integrated discrimination improvement. Moreover, among stage II and III CRC patients with low levels of $PS_{CRC}$, satisfactory benefits from chemotherapy were observed. Notably, the main underlying features of $PS_{CRC}$ include tumor cell infiltration, adipocyte accumulation, fibrous tissue deposition, and stromal infiltration. Transcriptome analysis further support the relevance of $PS_{CRC}$ to tumor progression and immune suppression. **Conclusions:** Our finds highlight the significant potential of histopathology images-based deep learning in predicting prognosis and assess therapeutic response of CRC. The $PS_{CRC}$ could serve as an effective tool in clinical decision for CRC management, providing insights into the underlying pathogenic mechanisms. However, prospective studies are still necessary for further validation. Research Sponsor: None.

**299** **Poster Session**

# Man versus machine: Comparative analysis of ChatGPT's colon cancer guidance and expert opinion.

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**Background:** AI Chatbots, such as ChatGPT, are increasingly being utilized across various aspects of society, including healthcare. Despite this growing usage, their role within the field of oncology remains underdeveloped. The objective of this study is to evaluate ChatGPT's ability to accurately respond to patient inquiries regarding colon cancer by comparing its responses to assessments from expert clinical oncologists. **Methods:** Ten comprehensive questions were compiled by reviewing commonly asked questions from reputable sources, including the American Society of Colon & Rectal Surgeons, Mount Sinai, the National Cancer Institute, Mayo Clinic, and the American Cancer Society. The questions were categorized into two categories based on their content: General Oncology Characteristics (covering symptoms, screening, and prevention) and for Diagnosis & Treatment. These questions were then entered into ChatGPT, with prompts designed to simulate patient inquiries. The AI-generated responses were subsequently evaluated by oncology experts using a 5-point Likert scale to assess their accuracy and relevance, with scores reflecting the experts' level of agreement with the answers. **Results:** On a five-point Likert scale, with 1 representing "strongly disagree" and 5 representing "strongly agree," the mean score was 4.72. ANOVA analysis was performed and there was no statistically significant difference in the mean score across all raters ($p = 0.221$). However, ratings between the two categories were statistically significant ($p=0.034$). **Conclusions:** This study demonstrates that ChatGPT can provide accurate and relevant responses to patient inquiries about colon cancer, as assessed by medical oncology experts. With an average rating of 4.72 on a 5-point Likert scale, ChatGPT's responses closely align with expert opinion, particularly for general characteristics category which included symptoms, prevention, and screening. However, responses related to diagnosis and treatment showed a statistically significant difference between expert opinions and AI, indicating that the experts agreed less with this component of AI's responses. These findings highlight the potential of AI chatbots in supplementing patient education in oncology, though further research is necessary to explore its limitations and expand its clinical utilities. Research Sponsor: None.

**300** **Poster Session**

# Risk factors for mortality and financial burdens in patients with colon cancer.

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**Background:** Colon cancer remains one of the most common cancers of the digestive system and continues to rank high among the leading causes of cancer-related deaths, despite the increased prevalence of screening. In this study, we aimed to investigate risk factors for mortality and health care costs in hospitalized patients diagnosed with colon cancer. **Methods:** Patients admitted to the hospital in 2019 and 2020 were identified using the Nationwide Inpatient Sample database and categorized into two groups based on the presence of colon cancer (with relevant ICD-10 codes). To account for confounding factors, multivariate regression analysis was applied to calculate mortality. Statistical analyses were conducted to assess mortality rates, length of stay, and hospital charges. **Results:** A total of 169,805 patients were admitted with a primary diagnosis of colon cancer, with a mean age of 67.6 years. The majority were White (69.9%), and 50.2% were male. Age over 60 (OR: 1.67, $p < 0.001$) and sepsis (OR: 16.01, $p < 0.001$) were associated with increased in-hospital mortality among colon cancer patients, while higher household income (76th to 100th percentile, OR: 0.71, $p = 0.011$) and treatment at an urban teaching facility (OR: 0.54, $p < 0.001$) were linked to a lower risk of in-hospital mortality. Female gender (OR: 0.88, $p = 0.080$) and having inflammatory bowel disease (OR: 0.67, $p = 0.055$) did not significantly affect in-hospital mortality, nor did Black race (OR: 1.1, $p = 0.348$) or Hispanic race (OR: 1.15, $p = 0.287$) compared to White patients. Sepsis (+11.8 days, $p<0.001$; +$192,699, $p< 0.001$), age over 60 (+0.99 days, $p<0.001$; +$6,134, $p<0.001$), and care at urban teaching hospitals (+0.54 days, $p< 0.001$; +$32,230, $p< 0.001$) were associated with longer hospital stays and higher total charges. Conversely, female gender (-0.21 days, p=0.001; -$5,529, $p<0.001$) and higher income (-0.69 days, $p<0.001$; -$1,565, p=0.466) were linked to shorter hospital stays and lower total charges. **Conclusions:** Early recognition of sepsis in colon cancer patients could potentially reduce in-hospital mortality rates, as well as significantly decrease costs and length of stay. The higher mortality rates observed in patients with sepsis may be attributed to the immunocompromised state associated with cancer. Implementing targeted interventions to address the higher risks in older patients and those with sepsis could improve outcomes and reduce the overall burden of colon cancer-related hospitalizations. Research Sponsor: None.

**Impact of patient demographics, socioeconomic factors, and clinical conditions on mortality, length of hospital stay, and total hospital charges.**

| Data | Mortality | Length of Hospital stay | Total Charge |
|---|---|---|---|
| **Age>60** | 1.67 ($p< 0.001$) | 0.99 days, $p<0.001$ | $6428, $p< 0.001$ |
| **Race (Hispanic compared to White race)** | 1.15 (p=0.287) | 0.38 days, p= 0.004 | $14,127, $p<0.001$ |
| **Race (Black compared to White race)** | 1.1 (p=0.348) | 1.4 days, $p< 0.001$ | $5370, p=0.002 |
| **Female gender** | 0.88 (p=0.08) | -0.21 days, p= 0.001 | -$5529, $p< 0.001$ |
| **High house hold income** | 0.71 (p= 0.011) | -0.69 days, $p<0.001$ | -$1565, p= 0.466 |
| **Treatment at urban teaching facility** | 0.54 ($p<0.001$ | 0.54 days, $p<0.001$ | $32230, $p< 0.001$ |
| **Sepsis** | 16.01 ($p<0.001$) | 11.8 days, $p<0.001$ | $192699, $p< 0.001$ |
| **IBD** | 1.05 (p=0.888) | 0.67 days, p=0.055 | $5348, p=0.293 |

**301** **Poster Session**

# Exploring the link between liver cirrhosis and liver metastasis in colorectal cancer.

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**Background:** Colorectal cancer (CRC) is the third most common cancer and the second most common cause of cancer mortality in the world. Although it is now associated with improved outcomes due to advancements in treatment modalities, CRC continues to be the most common source of liver metastasis. Previous studies have shown cirrhotic livers to be at reduced risk of metastatic disease. Our study aims to investigate the association between colorectal cancer and liver metastasis in the presence of concomitant liver cirrhosis. **Methods:** The National Inpatient Sample (NIS) was queried to identify all hospitalizations with colorectal cancer utilizing ICD-10 codes C18.x, C19.x, and C20.x from 2016 to 2020 and were further classified based on the presence or absence of cirrhosis. Demographic and clinical data were analyzed using chi-squared tests, independent sample t-tests, and binary logistic regression (adjusted for age, gender, and Charlson comorbidity index or CCI). The primary outcome studied was liver metastasis. Secondary outcomes were hepatic encephalopathy, portal hypertension, and spontaneous bacterial peritonitis. Statistical significance is indicated by a p-value less than 0.05. **Results:** A total of 255,599 hospitalizations with colorectal cancer were identified. Out of these, 5,370 (2.1%) had liver cirrhosis and 250,229 (97.9%) did not. Hospitalizations with CRC and concomitant cirrhosis had higher comorbidities as per the Charlson Comorbidity Index (10.5 vs 8.1, p-value$<$0.001) and higher mortality rates (8.5% vs 4.2%, p-value$<$0.001). Higher rates of liver metastasis (26.5 vs 23.4, p $<$0.001) and portal hypertension (26.1% vs 0.3%, p$<$0.001) were observed in the cirrhosis subgroup. This subgroup also has a significantly higher risk of SBP (2.8% vs 0.2%, p $<$0.001). **Conclusions:** This is a large retrospective study analyzing the impact of cirrhosis on liver metastasis in patients with colorectal cancer. Patients with concomitant cirrhosis consistently have increased liver metastasis and worse outcomes. The results of this study imply that diligent CRC screening is of utmost importance in patients with cirrhosis. Patients with CRC and coexisting cirrhosis would benefit from routine surveillance such as carcinoembryonic antigen (CEA) screening or CT scans to allow early detection of liver metastases which could potentially improve outcomes. Research Sponsor: None.

| | Colorectal Cancer Patients | | |
|---|---|---|---|
| | No Cirrhosis (N=250,229) | Cirrhosis (N=5,370) | p-value |
| **Age (in years)** | 65.79 ± 13.8 | 65.04 ± 10.9 | $<$.001 |
| **Sex (Female)** | 47.3% | 36.8% | $<$.001 |
| **Charlson Comorbidity Index** | 8.1 ± 3.6 | 10.5 ± 3.7 | $<$.001 |
| **Mortality** | 4.2% | 8.5% | $<$.001 |
| **Length of Stay** | 6.6 ± 7.2 | 7.4 ± 7.2 | $<$.001 |
| **Liver Metastasis** | 23.4% | 26.5% | $<$.001 |
| **Hepatic Encephalopathy** | 0.0% | 0.2% | $<$.001 |
| **Portal hypertension** | 0.3% | 26.1% | $<$.001 |
| **Spontaneous Bacterial Peritonitis (SBP)** | 0.2% | 2.8% | $<$.001 |

# Circulating tumor DNA as a marker of recurrence risk in stage III colorectal cancer: The $\alpha$-CORRECT study.

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**Background:** Patients with stage III CRC undergo surgery and adjuvant chemotherapy. Evaluation of MRD using a highly sensitive and specific, tumor-informed ctDNA assay may help predict which patients will have a recurrence. **Methods:** Patients with stage III CRC were enrolled after surgery in an observational study across 19 Pennsylvanian hospitals between 2016-2020. Tumor tissue and serial blood samples (every 3 months for years 1-3, biannually for years 4-5) were collected. Following whole exome sequencing (WES) of the tumor and selection of 50 – 200 somatic variants, a personalized MRD assay was used to measure ctDNA status of each sample. Three timepoints were used to analyze the association of ctDNA with recurrence: the single sample post-surgical (PS) and post-definitive therapy (PDT) timepoints, and the surveillance period, which included the PDT and subsequent samples. Primary endpoint was the association of ctDNA status during surveillance with recurrence-free survival (RFS). **Results:** A total of 124 patients with 1029 ctDNA results were analyzed. Almost all patients (96.0%) received adjuvant chemotherapy, primarily with an oxaliplatin-based regimen. On average, patients had 10 (SD = 4.5) blood draws and the median follow-up was 4.8 years. WES identified 172 (range 50 – 200) somatic variants per patient on average. Patients who were ctDNA positive during surveillance had a significantly higher likelihood of recurrence than patients who remained ctDNA negative (HR 49.6, 95% CI: 16.6 – 148.3; ctDNA status as time-dependent variable). At all timepoints, ctDNA detection was strongly associated with recurrence (Table). In multivariable analyses, ctDNA was strongly prognostic for recurrence but CEA was not. The median lead time between positive ctDNA result and a clinical diagnosis of recurrence was 10.4 months. **Conclusions:** In this cohort of patients with stage III CRC with long-term follow-up, a tumor-informed ctDNA assay using up to 200 variants was strongly prognostic for recurrence at all timepoints. Research Sponsor: National Cancer Institute; U01CA152753; Exact Sciences Corporation.

**Hazard ratios (HR), sensitivity, and specificity, with 95% confidence intervals (95% CI), for the association of ctDNA status with recurrence-free survival (RFS) at the surveillance, postsurgical (PS), and post-definitive therapy (PDT) timepoints (three-year RFS for the PS and PDT timepoints are also shown).**

| Timepoint | Statistic | Result |
|---|---|---|
| **Surveillance** | HR | 49.6** |
| **(N = 111)** | (95% CI) | (16.6 – 148.3) |
| | Sensitivity | 20/22 = 90.9% |
| | (95% CI) | (72.2 – 97.5%) |
| | Specificity | 82/87 = 94.3% |
| | (95% CI) | (87.2 – 97.5%) |
| **PS** | HR | 9.6** |
| **(N = 88)** | (95% CI) | (3.2 – 29.5) |
| | Sensitivity | 14/18 = 77.8% |
| | (95% CI) | (54.8 – 91.0%) |
| | Specificity | 53/66 = 80.3% |
| | (95% CI) | (69.2 – 88.1%) |
| | RFS at 3 yrs ctDNA(+), ctDNA(-) | 54.5%, 96.1% |
| **PDT** | HR | 16.7** |
| **(N = 97)** | (95% CI) | (6.9 – 40.3) |
| | Sensitivity | 10/21 = 47.6% |
| | (95% CI) | (28.3 – 67.6%) |
| | Specificity | 75/76 = 98.7% |
| | (95% CI) | (92.9 – 99.8%) |
| | RFS at 3 yrs ctDNA(+), ctDNA(-) | 18.2%, 90.0% |

*P< 0.001, **P<0.0001.

# Systematic literature review (SLR) of prognostic factors (PFs) in locally advanced rectal cancer (LARC) and mismatch repair deficient (dMMR)/microsatellite instability-high (MSI-H) rectal cancer (RC).

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**Background:** Innovative immunotherapies arebeing developed to improve clinical outcomes in patients (pts) withdMMR/MSI-H LARC who respond poorly to the current standard of care. A comprehensive overview of PFs in dMMR/MSI-H LARC may provide insight into the natural history of this disease to inform clinical study design and pt risk stratification. **Methods:** Systematic searches of bibliographic databases (EMBASE, MEDLINE, Cochrane; 2013–2023) and congress websites (2021–2023) were conducted to identify PFs for tumor response and survival outcomes in adult pts with LARC (SLR1) and dMMR/MSI-H RC (SLR2). Given the expected high volume of evidence, SLR1 was limited to large observational studies (≥1000 pts) and clinical trials (≥100 pts); SLR2 had no such limits. **Results:** 95 of 2307 publications met the inclusion criteria for SLR1. Of the PFs reported in ≥20 studies, older age, more lymph node involvement or fewer nodes assessed, higher T-stage or larger tumor size, higher histologic grade, and more comorbidities were most frequently significantly associated with poorer patient prognosis (Table). 3 of 289 publications met the inclusion criteria for SLR2. Neoadjuvant chemoradiation therapy (CRT) compared to chemotherapy (CT) was associated with statistically significant worse disease-free survival (DFS) and distant metastasis-free survival (DMFS), while T- or N-stage, age, and sex did not significantly impact these outcomes. Low vs middle tumor localization was also associated with worse DMFS. Stage III and IV disease were associated with worse RC-specific survival, while surgical procedure (segmental vs extended) showed no difference in overall survival (OS). The addition of neoadjuvant CRT or adjuvant CRT/CT to surgery alone had no impact on OS but was associated with worse DFS. **Conclusions:** Evidence regarding PFs in dMMR/MSI-H RC is scarce and may limit its interpretability, highlighting the need for further research that may be informed by the PFs identified from SLR1. The results from these SLRs may be helpful for optimizing patient management and treatment decision making, and to further contextualize RC clinical trials. Research Sponsor: GSK; 221776.

**PFs reported in ≥20 studies with ≥75% of studies reporting significant results in pts with LARC.**

| PF | Studies reporting PF (n) | Studies reporting significant* impact of PF on prognosis (%) | Poorer prognosis group | Top 2 most frequently reported outcomes |
|---|---|---|---|---|
| **Age** | 59 | 81.4 | Older | OS, DFS |
| **Lymph node involvement/ assessment** | 55 | 78.2 | Yes or higher N stage | OS, DFS |
| **T-stage or tumor size** | 53 | 79.2 | Higher or larger | OS, DFS |
| **Histologic grade** | 37 | 75.7 | Higher | OS, pathological complete response (pCR) |
| **Comorbidities** | 23 | 87.0 | More | OS, pCR |

*p-value $<0.05$ was considered significant as reported in the identified studies.

# Impact of synchronous vs metachronous primary tumor resection on prognosis of BRAF-V600E mutant metastatic colorectal cancer (mCRC).

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**Background:** The goal of this study is to evaluate if resection of the primary tumor in synchronous versus metachronous metastatic disease predicts survival of BRAF-V600E mCRC in the era prior to combination BRAF/EGFR inhibitors. **Methods:** This is a retrospective study of patients treated at the Ottawa Hospital Cancer Centre with BRAF-V600E mutation mCRC who did not receive combination BRAF/anti-EGFR therapy, diagnosed with colorectal cancer prior to March 31, 2021. Demographic, clinical, pathologic and cancer characteristics were abstracted from electronic medical records. Outcomes of interest included median duration of therapy and overall survival (OS). **Results:** 71 patients were included with a median age of 70 years:, 37 (52%) were females, 49 (69%)had right-sided tumors and 19 (27%) were mismatch repair deficient (dMMR). The primary tumor was unresected in 22 (30%) and resected in 49 (70%), completely resected in 44 (62%), and partially resected in 5 (7%) patients. Overall, median OS was 12.9 months 95% CI (8.4, 19) with 17 months 95% CI (11, —) mOS in patients with resected primary and 6.3 95% CI (4.8, 15) months in patients with unresected primary, HR 2.48 95% CI (1.42, 4.32), p=0.001. In univariate analysis: signet ring cell (p=0.007), good performance status (p<0.001), distant lymph nodes (p<0.001), peritoneal metastasis (p=0.008), metasetectomy (p=0.002), Charlston Comorbidity Index (p=0.003) were significant prognostic factors. Synchronous versus metachronous primary tumour was not a significant prognostic factor. **Conclusions:** Real world data confirms that patients with BRAF-V600E mutant mCRC have poor clinical outcomes. Our analysis suggests that resection of the primary tumor did not seem to impact on survival of BRAF-V600E mutant mCRC patients. This suggests that prognosis is dictated by the systemic disease and not complications of the primary tumour itself. Research Sponsor: None.

**Of the patients with unresected primary tumour (n=21), 6 were synchronous.**

| Metastatic status | Primary Tumor Resected | Primary Tumor unresected | Total |
|---|---|---|---|
| **Synchronous** | 23 | 22 | 45 |
| **Metachronous** | 26 | 0 | 26 |
| **Total** | 49 | 22 | 71 |

# Impact of inpatient palliative care on end-of-life care among patients with early-onset colorectal cancer.

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**Background:** Palliative care has proven benefits in elderly patients with advanced cancer; however, the objective benefits of palliative care in younger patients with cancer remain under-studied. We sought to examine the impact of inpatient palliative care consultation on end-of-life (EOL) care among hospitalized patients with early-onset colorectal cancer (EO-CRC). **Methods:** This study analyzed all admissions in the National Inpatient Sample (NIS) between 2016 and 2020 for patients younger than 50 years old with a diagnosis of CRC and death during the admission. Patients with CRC were identified utilizing ICD-10 codes C18.x, C19.x, and C20.x. Patient cohorts were made based on if they received inpatient palliative care consultation (ICD-10 code Z515). We assessed these encounters for patient demographics, Charlson comorbidity index (CCI), utilization of mechanical ventilation, vasopressors or blood transfusion, and presence of a Do Not Resuscitate (DNR) order. **Results:** Among 4,060 patients with EOCRC, 2,425 (59.7%) received inpatient palliative care consultation. There was no significant difference in age or length of stay between groups. The cohort of patients who received palliative care consultation had more female patients (44.3% vs 39.8%, p=0.004) and had higher comorbidity burden measured by the CCI (8.6 vs 8.3, p<.001). Patients receiving inpatient palliative care consultation at EOL were significantly more likely to have a DNR order (83.3% vs 44.6%, p<.001) in place and less likely to receive aggressive interventions, including mechanical ventilation (20.4% vs 43.4%, p<.001), blood transfusion (14.8% vs 22.6%, p<.001), vasopressor use (6.2% vs 8.6%, p=.004), and surgical intervention (3.0% vs 7.3%, p<.001). There was no difference in chemotherapy use between cohorts (2.8% vs 2.9%, p=0.8). Admissions including palliative care consultation were also associated with significantly lower average hospital charges ($99,387 vs $131,993, p<.001). **Conclusions:** Inpatient palliative care consultation at EOL among patients with EOCRC was associated with lesser use of aggressive interventions and higher rates of DNR code status. We also noted significantly lower costs of hospitalization among patients receiving inpatient palliative care consultation at EOL. These results underscore the importance of integration of inpatient palliative care consultation among patients with EOCRC at EOL. Research Sponsor: None.

| | No Palliative N=1635 | Palliative N=2425 | p-value |
|---|---|---|---|
| **Age (in years)** | 41.9 ± 6.1 | 42.1 ± 6.5 | .28 |
| **CCI** | 8.3 ± 2.6 | 8.6 ± 2.3 | **<.001** |
| **Total Charges** | 131993 ± 206404 | 99387 ± 151864 | **<.001** |
| **DNR** | 730 (44.6%) | 2020 (83.3%) | **<.001** |
| **Blood transfusion** | 370 (22.6%) | 360 (14.8%) | **<.001** |
| **Mechanical ventilation** | 710 (43.4%) | 495 (20.4%) | **<.001** |
| **Vasopressor** | 140 (8.6%) | 150 (6.2%) | **.004** |
| **Chemotherapy** | 45 (2.8%) | 70 (2.9%) | .800 |
| **Surgery** | 125 (7.3%) | 75 (3.0%) | **<.001** |

**TPS306** **Trials in Progress Poster Session**

# Trial in progress for a colorectal cancer detection blood test.

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**Background:** Blood-based biomarkers for cancer screening have garnered much anticipation in recent years. Current colorectal cancer screening methodologies such as colonoscopy and stool-based tests are plagued with low compliance rates. While scientific advancement calls for clinical trial recruitment from diverse backgrounds to ensure health equity, traditional clinical trial design has led to the underrepresentation of such populations, resulting in a gap in our understanding of disease conditions, preventative measures, and appropriate treatment across populations. Here, we describe an ongoing, decentralized, prospective sample collection clinical study, PROCEED-CRC, to develop a blood-based CRC screening test in an average-risk population. **Methods:** The primary endpoint of the study is to achieve the successful enrollment of up to 1500 evaluable patients. Key eligibility criteria include participants aged ≥40 years, planning to undergo an asymptomatic screening colonoscopy, willing to provide a blood sample within 120 days prior to a colonoscopy procedure, and an informed consent prior to participation in the study. Exclusion criteria include individuals who have had a prior malignancy, have undergone diagnostic colonoscopy in the past 9 years, or any other recent CRC screening tests, precancerous findings on recent colonoscopy, or are at high risk for CRC based on germline carrier status. Primary analysis would include summarizing the number and percentage of participants classified to have CRC, advanced adenoma, or non-advanced neoplasia. In order to reach a diverse study cohort that is representative of this average-risk US population aged over 45 years, we implemented several non-traditional recruitment outreach efforts, such as engaging a pharmacy provider to whom individuals may have previously consented to receive messages/notifications. Additional outreach efforts include digital advertisements and referral letters from healthcare professionals and community advocacy groups. Through these efforts, interested participants will have access to a link, a web address, or QR code to be screened for the study. After informed consent and enrollment, participants may schedule a blood draw via a mobile phlebotomy center based on their availability. Participants are considered evaluable if they meet the inclusion criteria, provide informed consent, provide an adequate blood draw, complete the screening colonoscopy within 120 days of the blood draw, and have medical records and pathology reports provided. The data from this study is expected to be completed in May 2025. Research Sponsor: None.

**TPS307** **Trials in Progress Poster Session**

# Colorectal cancer legal and administrative burden support (COLLABS): A pilot clinical trial.

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**Background:** Patients with colorectal cancer often have health-related social needs (e.g. food insecurity, housing instability, employment loss). Failure to adequately address these is associated with financial hardship, worse quality-of-life, psychosocial distress, lower rates of treatment receipt, and worse survival. Cancer clinical care teams often lack the resources, knowledge, and processes to resolve these issues, many of which are actually legal in nature. Engagement with legal teams—medical-legal partnerships—can screen for, prevent, and resolve health-related social needs, as well as address related legal issues such as insurance denials, employment rights and protections, disability benefits, debt management, and estate planning. Cancer Legal Care (CLC) is a nonprofit providing free legal services to persons affected by cancer in MN. Past CLC clients have indicated a preference for proactive connection to CLC to prevent crises and overcome barriers to access such as concerns regarding perceived costs of legal services. **Methods:** This is a single-arm, pilot study to evaluate the feasibility, acceptability, and preliminary efficacy of proactive and free legal care support for patients with colorectal cancer at the University of Minnesota. The enrollment goal is 20 patients and the trial duration is 6 months. Inclusion criteria include English-speaking adults with advanced stage colorectal cancer and estimated life expectancy of >6 months. There is no requirement for baseline health-related social need(s) or time from diagnosis. After participants complete baseline surveys, we will connect them to a CLC attorney for an initial "legal care checkup"-— the timing and method of connection (in-person, video visit, or phone call) is based on patient preference. Patients can bring up self-identified or active issues themselves, while CLC will also screen for needs. CLC will provide any required legal services and engage with the cancer care team as needed. CLC will follow-up with participants at 3 and 6 months, though services may continue after the study period off trial. Participants will complete surveys at baseline, 3 months, and 6 months. Surveys will assess patient-reported outcomes on financial toxicity, quality-of-life, comfort levels with accessing cancer care, status of health-related social and legal needs, perceived stress, life mastery and engagement, relationship with CLC contact, and acceptability of the intervention. The primary outcomes are feasibility (rates of initial engagement, continued engagement, and intervention completion; feasible if >50% on each metric) and acceptability. We will conduct exit qualitative interviews. We will collect sociodemographics, rates of treatment adherence/ completion, legal services provided, outcome of legal issues, and monetary savings or recovery for patients. Enrollment opened August 2024 and we have enrolled 15/20 participants. Clinical trial information: NCT06475664. Research Sponsor: Minnesota Colorectal Cancer Research Foundation.

**TPS308** **Trials in Progress Poster Session**

# Phase 2 randomized study evaluating safety, efficacy, and optimal dose of ABBV-400 in combination with fluorouracil, folinic acid, and bevacizumab in previously treated patients with metastatic colorectal cancer.

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**Background:** Colorectal cancer (CRC) is the third most common cancer. In the metastatic (m) setting, the 5-year relative overall survival is approximately 15%. Conventional treatment comprises fluorouracil (5-FU)–based chemotherapy. Recently, targeted therapies have been studied for specific molecular subtypes. c-Met overexpression frequently occurs in a variety of tumors, including CRC. ABBV-400 is a c-Met–directed antibody-drug conjugate composed of the monoclonal antibody telisotuzumab conjugated to a potent topoisomerase 1 inhibitor payload. Preliminary data from the first-in-human study of ABBV-400 in patients with advanced solid tumors indicate encouraging efficacy of ABBV-400 monotherapy in patients with third-line or later mCRC. This phase 2 randomized study evaluates the safety, efficacy, and optimal dose of ABBV-400 in combination with 5-FU, folinic acid (FA), and bevacizumab (bev) in patients with mCRC with progression after first line (1L) treatment. **Methods:** Global, open-label, phase 2 randomized controlled study (NCT06107413). Eligible patients (≥18 years) have confirmed unresectable mCRC and measurable disease per RECIST v1.1, are microsatellite stable or mismatch repair proficient, *BRAF* V600E wild type, and have progression after 1L combination chemotherapy ± an anti-vascular endothelial growth factor or anti-epidermal growth factor receptor antibody. Primary objectives are (a) optimize ABBV-400 dose in combination with 5-FU, FA, and bev; (b) evaluate the efficacy of the combination, using objective response and progression-free survival as dual primary endpoints; (c) evaluate the safety and tolerability of the combination. Approximately 206 patients planned for enrollment in 2 stages: safety lead-in dose escalation (stage 1; n=30) and dose optimization (stage 2; n=176). In stage 1, patients receive escalating doses of ABBV-400 either every 2 weeks (Q2W; 0.8–2.4 mg/kg) or every 4 weeks (Q4W; 1.6–3.0 mg/kg) in combination with Q2W 5-FU (2400 mg/m$^2$ infusion), FA (200 mg/m$^2$), and bev (5 mg/kg) in 28-day cycles. Dose escalation of ABBV-400 uses a Bayesian optimal interval design, with target toxicity rate of 30%. Dose-limiting toxicities (DLT) are assessed during cycle 1, with ≥6 DLT evaluable patients required to declare a dose safe for the dose-optimization stage. In stage 2, patients are randomized to up to 4 ABBV-400 dose cohorts (2 with Q2W and 2 with Q4W ABBV-400 schedule; all in combination with Q2W 5-FU, FA, and bev) and a comparator cohort (irinotecan [180 mg/m$^2$] + 5-FU [400 mg/m$^2$ bolus and 2400 mg/m$^2$ infusion] + FA [200 mg/m$^2$] + bev [5 mg/kg]; all Q2W). Patients are treated until progression, unacceptable toxicity, or other discontinuation criteria are met. Enrollment was initiated in November 2023, with 3 patients enrolled as of January 4, 2024. Clinical trial information: NCT06107413. Research Sponsor: None.

**TPS309** **Trials in Progress Poster Session**

# Alliance A022101/NRG-GI009: A pragmatic randomized phase III trial evaluating total ablative therapy for patients with limited metastatic colorectal cancer—Evaluating radiation, ablation, and surgery (ERASur).

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**Background:** Patients with oligometastatic colorectal cancer (CRC) often experience long-term progression-free survival and overall survival (OS) following aggressive local treatment of isolated metastases, especially in the liver. However, the effectiveness of local ablative therapies, such as microwave ablation (MWA) and stereotactic body radiation therapy (SBRT), in patients with inoperable limited metastatic CRC or those with additional metastases outside the liver or lungs, remains uncertain. Despite the historical use of local therapy for oligometastatic CRC, often influenced by provider bias rather than strong evidence, questions persist about the potential benefits of applying this approach to patients with more extensive disease. This trial aims to address this clinical challenge by evaluating the safety and efficacy of adding total ablative therapy (TAT) of all disease sites to standard systemic treatment in patients with limited metastatic CRC. **Methods:** A022101/NRG-GI009 is a randomized phase III study within the National Clinical Trials Network, set to enroll 364 patients with newly diagnosed metastatic CRC (BRAF wild-type, microsatellite stable). Patients with liver-only metastatic disease are excluded. Pre-registration during first-line systemic therapy induction is encouraged. Eligible lesions must be treatable with surgical resection, MWA, and/or SBRT. Patients without significant radiographic progression after 12-39 weeks of first-line systemic therapy, with 4 or fewer disease sites post-induction chemotherapy, will be randomized 1:1. Randomization will be stratified by the number of metastatic organ sites (1-2 vs. 3-4), timing of metastatic disease diagnosis (de novo vs. secondary), and the presence of metastatic disease outside the liver/lungs in at least one site. Patients in Arm 1 will receive TAT, targeting all metastatic sites with a combination of surgical resection, MWA, and/or SBRT, followed by standard systemic therapy. Patients in Arm 2 will continue with standard systemic therapy alone. The choice of systemic therapy is at the discretion of the medical oncologist and may include continued therapy, maintenance chemotherapy, or observation. The primary endpoint is OS, with secondary endpoints including event-free survival, treatment-related toxicities, and local recurrence. Exploratory biomarker analyses will also be conducted. The study requires 346 evaluable patients across the two arms to demonstrate an OS improvement with a hazard ratio of 0.7, providing 80% power with a one-sided alpha of 5%. The trial employs a group sequential design with two interim analyses (at 25% and 50% of events) to assess futility. The trial began in January 2023, and patient recruitment is ongoing. Clinical trial information: NCT05673148. Research Sponsor: National Cancer Institute.

**TPS310** **Trials in Progress Poster Session**

# NRG-GI008: Colon adjuvant chemotherapy based on evaluation of residual disease (CIRCULATE-North America).

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**Background:** Currently, there are no biomarkers validated prospectively in randomized studies for resected colon cancer (CC) to determine need for adjuvant chemotherapy (AC). However, circulating tumor DNA (ctDNA) represents a highly specific and sensitive approach (especially with serial monitoring) for identifying minimal/molecular residual disease (MRD) post-surgery in CC patients (pts), and may outperform traditional clinical and pathological features in prognosticating risk for recurrence. CC pts who do not have detectable ctDNA (ctDNA-) are at a much lower risk of recurrence and may be spared the toxicities associated with AC. Furthermore, for CC pts with detectable ctDNA (ctDNA+) who are at a very high risk of recurrence, the optimal AC regimen has not been established. We hypothesize that for pts whose CC has been resected, ctDNA status may be used to risk-stratify for making decisions about AC. **Methods:** In this prospective phase II/III trial, up to 1,912 pts with resected stage IIB, IIC, and III CC will be enrolled. Based on the post-operative ctDNA status using personalized and tumor-informed assay (Signatera, bespoke assay), those who are ctDNA- (Cohort A) will be randomized to immediate AC with fluoropyrimidine (FP)+oxaliplatin (Ox) for 3-6 mos per established guidelines *v* serial ctDNA monitoring. Patients who are ctDNA+ post-operatively, or with serial monitoring (Cohort B), will be randomized to FP+Ox *v* more intensive AC with addition of irinotecan (I) for 6 mos. One cycle of chemotherapy is allowed while awaiting ctDNA testing results for cohort assignment. The primary endpoints for Cohort A are time to ctDNA+ status (phase II) and disease-free survival (DFS) (phase III) in the immediate *v* delayed AC arms. The primary endpoint for Cohort B is DFS in the FP+Ox *v* FP+Ox+I arms for both phase II and phase III portions of the trial. Secondary endpoints include prevalence of detectable ctDNA post-operatively, time-to-event outcomes (overall survival and time to recurrence) by ctDNA status, and the assessment of compliance to adjuvant therapy. Biospecimens including archival tumor tissue, as well as post-operative plus serial matched/normal blood samples, will be collected for exploratory correlative research. Active enrollment across the NCTN started in June 2022 with CCTG sites joining in August 2023. Current accrual (as of 9-20-2024): 500/1,912. Clinical trial information: NCT05174169. Research Sponsor: National Cancer Institute; U10CA180868; National Cancer Institute; U10CA180822; National Cancer Institute; UG1CA189867; Natera, Inc.

**TPS311** **Trials in Progress Poster Session**

# Colorectal cancer metastatic dMMR immunotherapy (COMMIT) study: A randomized phase III study of atezolizumab (atezo) monotherapy versus mFOLFOX6/bevacizumab/atezo in the first-line treatment of patients with deficient DNA mismatch repair (dMMR) or microsatellite instability-high (MSI-H) metastatic colorectal cancer 9NRG-GI004/SWOG-S1610).

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**Background:** Approximately 45% of dMMR/MSI-H metastatic colorectal cancer (mCRC) in the immunotherapy arm progressed at 12 mos (KEYNOTE 177). We hypothesize that dMMR/MSI-H mCRC patients (pts) may be more effectively treated with the combination of PD-1/PD-L1 (PD-1) pathway blockade and mFOLFOX6/bevacizumab (bev) rather than with anti-PD-L1 therapy (atezo) alone. Preclinical work demonstrated synergistic effects between anti-PD-1/anti-VEGF as well as between oxaliplatin/anti-PD-1 in murine CRC models, and phase II data showed activity of anti-PD-1/anti-VEGF in chemotherapy refractory colon cancer. Within the Atezo-TRIBE 8-pt dMMR CRC subgroup treated with FOLFOXIRI+bev+atezo, median PFS was not reached, with the first progression event at ~16 mos. Additionally, in other solid tumor malignancies, anti-PD-1 plus anti-VEGFr (i.e., HCC and RCC) as well as anti-PD-1 plus chemotherapy (i.e., gastroesophageal and lung cancers) combinations are standard first-line treatments. **Methods:** This two-arm prospective phase III open-label trial randomizes (1:1) mCRC dMMR/MSI-H to atezo monotherapy *v* mFOLFOX6/bev+atezo combination. Key inclusion criteria have been simplified on recent amendments to better mirror clinical practice for pts receiving mFOLFOX6/bev+atezo: One cycle of FOLFOX or CAPOX, with or without bev (or biosimilar) prior to enrollment allowed, dMMR tumor determined by local CLIA-certified IHC assay (MLH1/MSH2/MSH6/PMS2) or MSI-H by local CLIA-certified PCR or NGS panel; pts with total bilirubin ≤4.0 x ULN; duration of therapy for up to two years for both arms; imaging frequency on post-treatment follow-up has been reduced; as has measurable disease per RECIST. Primary endpoint is PFS. Assuming the atezo monotherapy control arm has a 48% PFS at 24 mos as assessed by site investigator, we have 80% power to detect a hazard ratio of 0.6 (equivalent to 64.4% PFS at 24 mos) with alpha 0.025 one-sided. Stratification factors include BRAFV600E status, metastatic site, and prior adjuvant CRC therapy. Secondary endpoints include overall survival, objective response rate, safety profile, disease control rate, and duration of response. Archived tumor tissue and blood samples will be collected for correlative studies. Harmonization of translational analyses is planned between GI004 (COMMIT) and A021502 (ATOMIC). Sample size has been modified with the accrual goal of 120 pts randomized between the two immunotherapy arms needed for study completion. Enrollment actively continues at U.S. sites. Current accrual (as of 9-20-2024): 100/120. Clinical trial information: NCT02997228. Research Sponsor: National Cancer Institute; U10 CA180868; National Cancer Institute; U10 CA180822; National Cancer Institute; U10 CA180888; National Cancer Institute; UG1 CA189867; National Cancer Institute; U24 CA196067; Genentech, Inc.

**TPS312** **Trials in Progress Poster Session**

# Total neoadjuvant therapy followed by total mesorectal excision and selective lateral lymph node dissection for locoregionally advanced low rectal cancer: A phase III randomized controlled trial (JCOG2207).

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**Background:** For locally advanced low rectal cancer (RC), total neoadjuvant therapy (TNT) followed by total mesorectal excision (TME) confers a survival advantage compared with neoadjuvant chemoradiotherapy (nCRT) in the West, even though the local control rate was similar between TNT and nCRT. The Japan Clinical Oncology Group (JCOG) Colorectal Cancer Study Group has been developing therapies for RC based on recurrence risk according to presence/absence of clinical lateral lymph node metastasis (cLLNM). Our previous study in Japan showed that the standard of care (SOC) for RC is TME with lateral lymph node dissection (LLND) plus adjuvant chemotherapy (CTx). However, the prognosis for cStage III low RC remains poor regardless of cLLNM status. Therefore, new treatment strategies are needed for this population. In cLLNM-positive cStage III low RC patients, previous research suggests that adding nCRT to LLND would reduce local recurrence and improve survival. TNT and LLND may further improve outcomes. In cLLNM-negative cStage III low RC, several reports have suggested that nCRT as an alternative addition to LLND has comparable efficacy in terms of local control. Thus, in these populations, if TNT can replace LLND, adverse events due to LLND can be avoided while improving prognosis. We hypothesize that TNT + TME + selective LLND would be the most promising strategy for cLLNM-positive and -negative cStage III low RC. **Methods:** Eligibility criteria include low rectal adenocarcinoma or adenosquamous carcinoma, cT2-cT4 tumor depth, clinically positive lymph node metastasis, no distant metastasis, no history of pelvic irradiation or rectal surgery, age 18-75 years, and sufficient organ function. Eligible patients are randomized (1:1) with adjustment factors of institution, sex, and cLLNM status. The experimental treatment arm receives TNT, short-course radiotherapy (25 Gy/5 fractions to whole pelvis), and consolidation chemotherapy consisting of 6 courses of CAPOX (oxaliplatin 130 mg/m2 on day 1 and capecitabine 2000 mg/m2/day on days 1-14). After completing TNT, TME + selective LLND are performed for cLLNM-positive cases or TME alone for cLLNM-negative cases. The SOC arm receives TME + LLND followed by adjuvant chemotherapy with 8 cycles of CAPOX (same as above). The primary endpoint is overall survival (OS) in the intention-to-treat population. To detect an increase in 5-year OS from 76% to 84%, corresponding to a target hazard ratio of 0.64, 408 patients (108 events) would achieve 75% power at a one-sided $\alpha$ of 0.05, an expected accrual period of 4 years, and a follow-up period of 5 years. The total sample size was set at 420 patients to account for patients lost to follow-up. Accrual in JCOG2207 began in October 2023. Clinical trial information: s031230415. Research Sponsor: National Cancer Hospital Japan.

**TPS313** **Trials in Progress Poster Session**

# Combining low-dose regorafenib with pembrolizumab for patients with MSI-H colorectal cancer: REGPEM-CRC-01.

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**Background:** Currently, pembrolizumab is one of the front-line therapies for patients with MSI-H CRC. However, approximately 40% of patients who received pembrolizumab experienced disease progression early in the course of disease (KEYNOTE 177). Therefore, there is still an unmet need to enhance the efficacy of checkpoint inhibitors in MSI-H CRC. MSI-H CRC has a higher level of expression of VEGF in blood compared to patients compared to its MSS counterpart (Hansen et al. Colorectal Dis. 2011). Consistently, exploratory analysis of CALBG-80405 and PARADIGM trial showed that patients with MSI-H CRC were more likely to benefit from anti-VEGF therapy than anti-EGFR therapy regardless of the side of the tumor. NSABP C-08 also suggested that anti-VEGF therapy may have biological activity even in as adjuvant therapy for patients with MSI-H colon cancer. Regorafenib is a potent VEGF inhibitor, with preclinical evidence showing its immune modulatory effect in the tumor microenvironment. In this trial, we hypothesize that adding low-dose regorafenib to pembrolizumab may induce synergistic activity beyond their independent clinical efficacy and create deep and durable responses for patients with MSI-H CRC. **Methods:** In the lead arm of this prospective randomized study, 22 patients will be enrolled through Hoosier Cancer Research Network (HCRN-GI23-643). Patients with treatment naïve MSI-H CRC will be enrolled in this front-line trial. One cycle of pembrolizumab and up to 3 cycles of chemotherapy prior to determination of MMR-D/MSI-H is allowed. Patients will receive regorafenib 60 mg daily in combination with pembrolizumab 200mg IV in cycle 1, followed by regorafenib 90mg in subsequent cycles to improve treatment tolerance. The primary outcome that will be measured is ORR, defined as the percentage of partial or complete response to the treatment within 12 months. ORR will be measured using RECIST 1.1. criteria. A formal one-sided hypothesis test will be conducted for futility, assuming that we will reject the null hypothesis of a target ORR only if we have strong evidence. In this study, we assume a null hypothesis that ORR is 0.60, which would reflect significant clinical improvement over the current standard of ORR = 0.43 from KEYNOTE 177. The alternative hypothesis is that ORR is less than 0.60. For the lead-in phase of the study, the emphasis is on controlling Type I error to be small, approximately 0.05. The test statistic will be the number of ORRs in the 22 patients, which we assume to follow a binomial distribution. Clinical trial information: NCT06006923. Research Sponsor: None.

**TPS314** **Trials in Progress Poster Session**

# A phase II, randomized study of JMT101 in combination with irinotecan and SG001 versus regorafenib in patients with metastatic colorectal adenocarcinoma (mCRC).

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**Background:** Epidermal growth factor receptor (EGFR) inhibitorcombined withchemotherapy based on5-FU with oxaliplatin and irinotecanis the first-line treatment of wild-type RAS mCRC.JMT101 is a humanized monoclonal IgG1 antibody targeting EGFR. Previous trials showed that JMT101 monotherapy or in combination with chemotherapy had demonstrated a promising anti-tumor activity in advanced solid tumors with good safety profiles. EGFR inhibitor might have a synergetic effect with immunocheckpoint inhibitor (ICI) by activating innate and adaptive immune response. SG001 is a fully humanized and high-affinity immunoglobulin G4 monoclonal antibody that targets programmed death-1 (PD-1), which also had encouraging efficacy in advanced solid tumors. Therefore, EGFR antibody in combination with chemotherapy plus ICI is worthy exploration in patients with mCRC. **Methods:** This is a phase II, multicenter, open-label, two-part study. Part I is a safety run-in, followed by part II randomized treatment. Eligible patients are histologically confirmed mCRC without MSI-H/dMMR, wild-type RAS and BRAF, who progressed on at least 2 prior systemic therapy with chemotherapy based on 5-FU with oxaliplatin and irinotecan, with or without cetuximab and bevacizumab (both parts); no prior treatment with regorafenib, fruquintinib, or TAS-102 (part II). Safety run-in data will be analyzed after 3-6 patients have received JMT101 6 mg/kg plus irinotecan 180 mg/m$^2$ plus SG001 240 mg Q2W. Upon dose confirmation, 90 patients with mCRC will be randomized into three arms (1:1:1) to receive JMT101 plus irinotecan plus SG001, or JMT101 plus irinotecan, or regorafenib monotherapy. Primary endpoints include incidence of dose limiting toxicities (DLT) and adverse events (part I) and objective response rate (ORR) as per RECIST v1.1 by investigator (part II). Secondary endpoints include JMT101 and SG001 pharmacokinetic profile, antidrug antibodies (ADA) (both parts), disease control rate (DCR), duration of response (DOR), progression free survival (PFS), and overall survival (OS), as well as safety (part II). Prespecified goal for safety run-in part was met; randomized treatment part accrual began in February 2024. Clinical trial information: NCT06089330. Research Sponsor: Shanghai JMT-Bio Technology Co., Ltd.

**TPS315** **Trials in Progress Poster Session**

# Chemoradiotherapy (CRT) followed by sequential tislelizumab (TIS) + CAPOX and TIS monotherapy for organ preservation in locally advanced low rectal cancer.

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**Background:** CRT followed by total mesorectal excision (TME) is the standard care for patients with locally advanced low rectal cancer (≤ 5cm from the anal margin). However, TME may lead to permanent colostomy and severely impact quality of life. Organ preservation is, therefore, a critical unmet need for these patients. The proportion of patients achieving clinical complete response (cCR) after CRT is relatively low, with the literature reporting around 20%. Total neoadjuvant therapy (TNT) has been investigated to increase cCR rate. Several clinical trials demonstrated that PD-1 inhibitor combined with TNT could result in significate improvement in pathological complete response rate (pCR rate, 39.8% vs 15.3%, $p < 0.001$ in UNION study) or complete response rate (pCR+cCR rate, 44.8% vs 26.9%, $p = 0.031$ in a randomized phase 2 study) compared with TNT alone in rectal cancer patients. The RELIEVE-01 study (NCT06390982) was designed to evaluate the efficacy in organ preservation with CRT followed by sequential TIS + CAPOX and TIS monotherapy in patients with low rectal cancer. **Methods:** This multicenter, single-arm, phase 2 study will enroll 46 pts with histopathologically confirmed rectal adenocarcinoma (≤ 5cm from the anal margin), pMMR/MSI-L/MSS, cT1-3N1M0/T2-3N0M0, and ECOG PS ≤1. Pts will initially receive 6 weeks of CRT (50.4 Gy/28F, capecitabine 825 mg/m$^2$, bid, d1-5 each week), followed by 4 cycles of TIS + CAPOX (TIS, 200 mg, IV, d1; capecitabine 1000 mg/m$^2$, bid, d1-14; oxaliplatin 130 mg/m$^2$, d1) in a 21-day cycle. Then, clinical response will be assessed to determine the subsequent treatment. Patients with cCR will receive TIS + CAPOX (4 cycles) and TIS (up to 9 cycles), and then be managed using the W&W strategy. Patients with non-cCR will undergo TME. Patients with near-cCR will be given local excision, and those achieving pCR will receive the same treatment as cCR patients, and for those with non-pCR after local excision, TME will be performed. The primary endpoint is rate of CR (cCR plus pCR post local resection). Secondary endpoints include organ-preserving rate, event-free survival rate and OS rate at 1, 2 and 3 years, respectively. Recruitment is ongoing. Clinical trial information: NCT06390982. Research Sponsor: None.

**TPS316** **Trials in Progress Poster Session**

# Cetuximab and vemurafenib plus liposomal irinotecan (II), leucovorin and fluorouracil for BRAF$^{V600E}$-mutated advanced colorectal cancer (IMPROVEMENT2): A multicenter, open-label, randomized controlled trial.

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**Background:** Current first-line treatment regimen for advanced colorectal cancer with BRAF$^{V600E}$ gene mutation is chemotherapy combined with anti-angiogenic drug, but the efficacy is somewhat limited. Theoretically, targeting BRAF$^{V600E}$ mutation is an effective treatment strategy and has achieved success in lung cancer and melanoma. However, the objective response rate (ORR) is below 5% in colorectal cancer, suggesting that BRAF$^{V600E}$ mutation in colorectal cancer may have different feedback regulation mechanisms compared to other cancer types. Combining chemotherapy (such as 5-fluorouracil plus irinotecan) with BRAF/EGFR targeted therapy may potentially achieve good efficacy. Subsequently, we conducted the IMPROVEMENT (NCT03727763) study, which showed an objective response rate (ORR) of 85% and a disease control rate of 100% with this regimen, significantly higher than those with chemotherapy combined with anti-angiogenic drug. The recent BREAKWATER study showed similar trends. To further validate the efficacy and safety of this regimen, we designed a randomized controlled trial. **Methods:** IMPROVEMENT2 (NCT06603376) is a multicenter, open-label, randomized controlled trial in patients with histologically or cytologically confirmed advanced colorectal cancer. Adult patients who are previously untreated in the metastatic setting, have a confirmed BRAF$^{V600E}$ mutation, measurable disease per Response Evaluation Criteria In Solid Tumors (RECIST) version 1.1, and Eastern Cooperative Oncology Group performance status of 0-1 are eligible. Patients who received previous treatment in the adjuvant setting but experienced metastasis or recurrence within 12 months are permitted. A total of 75 eligible patients will be randomly assigned (2:1) to receive liposomal irinotecan(II) 60 mg/m$^2$, leucovorin 400 mg/m$^2$, fluorouracil 2800 mg/m$^2$, and cetuximab 500 mg/m$^2$ on day 1 of each 14-day cycle plus oral vemurafenib 720 mg twice daily, or liposomal irinotecan(II) 60 mg/m$^2$, leucovorin 400 mg/m$^2$, fluorouracil 2800 mg/m$^2$, and bevacizumab 5 mg/kg on day 1 of each 14-day cycle. Treatment will be continued until disease progression or unacceptable toxicity. The primary endpoint is ORR per RECIST 1.1 by blinded independent central review (BICR). Key secondary endpoints are depth of response, progression-free survival per RECIST 1.1 by BICR, overall survival, and safety. The study is actively enrolling patients. Clinical trial information: NCT06603376. Research Sponsor: None.

**TPS317** **Trials in Progress Poster Session**

# A phase 2, multicenter, open-label study of AlloStim in-situ cancer vaccine immunotherapy as 3L therapy in MSS/pMMR metastatic colorectal cancer.

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**Background:** Neoantigen therapeutic cancer vaccines have had limited clinical benefit in metastatic tumor settings. Tumor immunoediting selects tumor clones over time that are non-immunogenic. In addition, the metastatic tumor microenvironment (TME) becomes profoundly immunosuppressive. Microsatellite stable (MSS) and mismatch repair proficient (pMMR) metastatic colorectal cancer (mCRC) tumors do not accumulate mutations, resulting in low mutational burden (TMB) and a lack of neoantigen targets. Consequently, therapeutic vaccine approaches for mCRC have had difficulty in demonstrating clinical effectiveness. Novel approaches that boost the immune response in low TMB tumors may benefit mCRC patients. AlloStim is an allogeneic, non-genetically manipulated, living Th1-like cell immunotherapy. The Th1-like cells are ex-vivo differentiated and expanded from healthy blood donors and activated prior to infusion. These cells are designed to be rejected upon intradermal (ID) injection. Repeated ID injections elicit high titers of allo-specific Th1 memory cells in circulation. In allo-primed patients, intravenous (IV) infusion of AlloStim causes a rejection response that releases cytokines, such as IL-12, which non-specifically activate circulating memory T cells and NK cells. Activated host memory T-cells and NK cells extravasate and traffic to metastatic tumor sites. This creates the conditions for "in-situ vaccination". Activated NK cells cause disruption of the cell membrane and release of chaperoned neoantigens into the tumor microenvironment, a process known as immunological cell death (ICD). The release of neoantigens into the microenvironment in the context of inflammation mediated by infiltrating Th1 memory cells creates conditions for *in-situ* vaccination. *In situ* vaccination represents an alternative immunotherapy approach in which a therapeutic cancer vaccine is generated *in vivo* without the need to previously identify and isolate tumor neoantigens. **Methods:** To investigate this novel in-situ vaccine mechanism, a phase 2, multicenter, single-arm clinical trial in 3L MSS/pMMR mCRC subjects that have been previously treated with oxaliplatin-containing and irinotecan-containing chemotherapy regimens, anti-EGFR (RAS wt and left-sided) and anti-VEGF was initiated. Patients were administered weekly AlloStim in three 28-day cycles consisting of 4 ID injections and a combination ID and IV infusion on the final week in each cycle to create self-amplifying waves of allo-specific and tumor-specific host immune cells in circulation which traffic to tumors upon IV infusion. The primary end-point is overall survival. 21 evaluable patients have been enrolled and are currently being followed for maturity of the survival data. Clinical trial information: NCT04444622. Research Sponsor: Mirror Biologics, Inc.

**TPS318** **Trials in Progress Poster Session**

# A phase 2, multicenter, open-label study of AlloStim combined with anti-PD-L1 immunotherapy as 4L therapy in patients with MSS/pMMR metastatic colorectal cancer.

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**Background:** Microsatellite stable (MSS)/mismatch repair proficient (pMMR) metastatic colorectal cancer (mCRC) are immunologically 'cold' tumors that do not respond to immunotherapy. Cold tumors represent a formidable hurdle for checkpoint blockade immunotherapy approaches. Therefore, there is a need for novel strategies that could reprogram mCRC cold tumors to become inflamed hot tumors in a manner that might potentially convert low tumor mutational burden (TMB) tumors into checkpoint blockade responsive tumors. AlloStim is an experimental allogeneic, non-genetically manipulated, Th1-like activated living cell immunotherapy derived from healthy unrelated blood donors. The mechanism of action is designed to convert immunologically cold tumors into inflamed hot tumors. A case study of a single 3L MSS/pMMR mCRC patient primed with AlloStim followed by combination anti-PD-1/anti-CTLA4 checkpoint blockade resulted in a rare objective tumor response (Hirschfeld, et al, Translational Medicine Communications (2024) 9:15). The AlloStim priming results in an abundance of infiltrating interferon-gamma producing Th1 cells. Interferon-gamma has been shown to increase expression of PD-L1 in the TME. Therefore, it is hypothesized that the combination of AlloStim with an anti-PD-L1 checkpoint inhibitor might optimally evoke anti-tumor immune responses in this immunotherapy-refractory MSS/pMMR mCRC population. **Methods:** The study is a phase 2, multicenter, single-arm clinical trial of MSS/pMMR mCRC subjects in 4L that have been previously treated with an oxaliplatin-containing and irinotecan-containing chemotherapy regimen, anti-EGFR (RAS wt and left-sided), anti-VEGF and have progressed on 3L TAS-102 +/- bevacizumab or regorafenib or fruquinitinib. The study is evaluating weekly AlloStim priming from day 0 to day 49 followed by combination of AlloStim IV infusion (q2w) followed a week later by anti-PD-L1 (avelumab 800mg/q2w) from days 63 to day 98. Objective tumor response and disease control rate are the primary end-points. Overall survival is the secondary end-point. Clinically stable patients at day 112 can continue in an expansion phase with another round of combination AlloStim and avelumab from day 119 to day 154. CT scans are conducted at baseline, day 56 after AlloStim priming, day 112 after combination of alternating q2w infusions of AlloStim and avelumab and day 168 in all patients. CT scans will be analyzed by independent central readers using RECIST 1.1. Up to 50 patients will be enrolled with an interim analysis performed to assess efficacy after 20 patients become evaluable. The study is open and has not yet enrolled any patients at time of submission. Clinical trial information: NCT06557278. Research Sponsor: Mirror Biologics, Inc.

**TPS319** **Trials in Progress Poster Session**

# A phase 2 study of intravenous pembrolizumab and intratumorally injected autologous dendritic cells (DCs) in refractory colorectal cancer (CRC).

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**Background:** Immune checkpoint inhibitors (ICIs) have shown limited benefit in CRC outside the microsatellite instability high (MSI-H) population. Deficient cytotoxic T lymphocyte (CTL) trafficking to the tumor microenvironment (TME) and excess pro-tumorigenic cytokines and immunosuppressive cells may contribute to primary ICI resistance in microsatellite stable (MSS) CRC. Selective enrichment of T cells with specific chemokine receptors may enhance antitumor immunity. Studies highlight the role of IL-12 in improving progression-free survival (PFS) and overall survival (OS) in patients on DC therapies. Our group developed ST-aDC1s, optimized for high production of high levels of IL-12 and CTL-attracting chemokines when injected intratumorally. In preclinical models, ST-aDC1s combined with anti-PD-1/PD-L1 therapy showed therapeutic synergy, suggesting ST-aDC1s as a promising alternative to traditional DC vaccines for inducing CTL entry into tumors. **Methods:** We designed an open-label, non-randomized, phase 2 study of intratumoral autologous ST-αDC1 combined with pembrolizumab in recurrent/metastatic unresectable MSS CRC. Eligible patients must have prior treatment with, or contraindication to fluoropyrimidine, irinotecan, oxaliplatin, anti-VEGF, and anti-EGFR therapy (if RAS wild type), be anti-PD-1/PD-L1 therapy naïve, have ECOG PS ≤1, and have at least two target lesions, one amendable to biopsy and injection. Leukapheresis will collect monocytes for ST-αDC1 generation. Patients will receive intratumoral ST-αDC1s (6 million) on days 1 and 8 with biopsies, and an optional dose on day 50 before a third pembrolizumab dose. Pembrolizumab 200 mg IV will start on day 8, given every 3 weeks for 4 doses during the study period. Pembrolizumab will continue as conventional care afterwards. CT scans will occur after 2 and 4 doses of pembrolizumab. A safety lead-in cohort will include six patients, with up to 17 patients enrolled. The primary endpoint is objective response rate (ORR) at 12 weeks per RECIST 1.1. Secondary endpoints are safety, PFS, OS, and ORR per iRECIST. Correlative analysis will assess baseline levels and post-treatment changes in infiltrating T cells, CD4/CD8 ratios, FoxP3+ Tregs, and Teff-attracting and Treg-favoring chemokines from biopsy specimens, and immune cell subsets and cytokine levels from peripheral blood. With an expected ORR under 5% for standard treatment, 17 patients are required to show at least a 25% ORR with 81.2% power and a one-sided type I error rate of 0.05. A Simon two-stage Optimal design will enroll 9 patients in stage 1, and if ≥1 response occurs, 8 more patients will be added in stage 2. If ≥3 responses are observed, the treatment will be deemed promising for further study. Clinical trial information: NCT05518032. Research Sponsor: Roswell Park Comprehensive Cancer Center internal funding.

**TPS320** **Trials in Progress Poster Session**

# BXQ-350, a novel sphingolipid metabolism modulator, in combination with mFOLFOX7 and bevacizumab in newly diagnosed metastatic colorectal cancer (mCRC) patients: A phase 1b/2 randomized study.

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**Background:** Ceramides and sphingosine-1-phosphate (S1P) are key bioactive signaling molecules. Ceramides are proapoptotic and mitigate chemoresistance. Conversely, S1P promotes cancer cell proliferation, activates multiple oncogenic pathways, and stimulates immunosuppressor cell populations promoting a pro-tumoral microenvironment. Several studies in colorectal cancer patients have shown high levels of ceramides are associated with improved survival, while high S1P levels are associated with a poor prognosis. Hence, modulation of sphingolipid metabolism could be a promising therapeutic approach. BXQ-350 is a nanovesicle of Saposin C, an allosteric activator of sphingolipid metabolism, that lowers systemic S1P and increases C18 ceramide. BXQ-350 was investigated in a Phase 1 dose-escalation safety study in cancer patients with advanced solid malignancies (NCT02859857). BXQ-350 was safe and well-tolerated (no DLT, no MTD). Also, 13 patients (~17.8% of evaluable patients) had a clinical benefit up to cycle 6 and beyond (PR, SD). Among patients with PFS > 6 months, there were 4 recurrent CRC patients: 1 patient had a PFS of ~12 months, 2 of ~18 months, and 1 is still on study after 7 years suggesting potential long-term benefit. Furthermore, there were signs that BXQ-350 may alleviate symptoms of CIPN as 4 out of 10 patients with chronic CIPN at time of enrollment experienced an improvement of their symptoms. **Methods:** BXQ-350 is being investigated in a Phase 1b/2 study in combination with mFOLFOX7 and Bevacizumab in newly diagnosed mCRC patients (NCT05322590) to assess the efficacy and safety of BXQ-350. Design of the Phase 1b (open label study): 1) A safety dose escalation part to establish the RP2D: patients will initially receive 1.8 mg/kg BXQ-350 in combination with mFOLFOX7 and Bevacizumab. If safe (no MTD), dose of BXQ-350 will be increased to 2.4 mg/kg and 9 additional patients will be entered at this dose level. If safe, then this dose will be the RP2D and 21 additional patients will be enrolled, completing a 30 patient expansion cohort 2) Efficacy will then be evaluated for all patients entered at the RP2D. Primary objectives of the Phase 1b are to assess safety, identify RP2D, and assess preliminary efficacy of BXQ-350 in this combination. A secondary objective is to determine if BXQ-350 decreases CIPN. Enrollment in the expansion cohort of 30 patients is nearly complete, with 26 patients enrolled as of September 2024. Clinical trial information: NCT05322590. Research Sponsor: Bexion Pharmaceuticals.

**TPS321** **Trials in Progress Poster Session**

# Phase 1/2 study of the HER3-directed antibody-drug conjugate patritumab deruxtecan (MK-1022) as monotherapy in patients with gastrointestinal cancers.

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**Background:** A significant medical need exists for more effective and tolerable therapeutic options for patients with advanced gastrointestinal cancers (colorectal cancer [CRC], biliary tract cancer [BTC], and hepatocellular carcinoma [HCC]). Human epidermal growth factor receptor (HER3) expression is high in CRC and BTC, and low-to-intermediate in HCC. Patritumab deruxtecan (HER3-DXd; U3-1402/MK-1022) is a HER3-directed antibody-drug conjugate (ADC) comprising a fully human anti-HER3 immunoglobulin G1 monoclonal antibody (patritumab), which includes a plasma-stable, selectively cleavable linker, and a potent topoisomerase I inhibitor payload (DXd; released payload) that leverages the clinically validated deruxtecan (a derivative of exatecan) technology. DXd is cell membrane permeable and enables a bystander antitumor effect after DXd ADC internalization and linker cleavage, resulting in elimination of both target and neighboring tumor cells. Patritumab deruxtecan has shown antitumor activity in phase 1 and 2 studies of EGFR-mutated advanced non–small cell lung cancer and advanced breast cancer and in preclinical studies of CRC. This non-randomized, open-label, phase 1/2 study (NCT06596694) will examine the safety and efficacy of patritumab deruxtecan in select gastrointestinal cancers. **Methods:** Approximately 130 adults (aged ≥18 years) will be allocated to 1 of 3 cohorts based on their disease. Patients in cohort 1 (CRC; n = 40) must have previously treated unresectable or metastatic colorectal adenocarcinoma. Patients in cohort 2 (BTC; n = 40) must have previously treated locally advanced or metastatic BTC. Patients in cohort 3 (HCC; n ≤50) must have previously treated advanced HCC. Prior topoisomerase I inhibitor therapy is not allowed. Patients in cohorts 1 and 2 will receive patritumab deruxtecan intravenously (IV) as monotherapy. Cohort 3 includes a dose-escalation phase to determine the recommended phase 2 dose of patritumab deruxtecan monotherapy and an efficacy phase. In the efficacy phase, patients will receive the recommended phase 2 dose of patritumab deruxtecan. The primary objectives are to evaluate safety and tolerability of patritumab deruxtecan and to evaluate the confirmed objective response rate per RECIST v1.1 assessed by blinded independent central review (BICR) for each cohort. Secondary objectives are to evaluate duration of response and progression-free survival per RECIST v1.1 by BICR and overall survival, and to characterize the pharmacokinetics of patritumab deruxtecan. Clinical trial information: NCT06596694. Research Sponsor: Funding for this study was provided by Daiichi Sankyo Company, Limited and Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA.

**TPS322** **Trials in Progress Poster Session**

# Phase 1/2 study of FOG-001, a first-in-class direct β-catenin:TCF inhibitor, in patients with colorectal cancer, hepatocellular carcinoma, and other locally advanced or metastatic solid tumors.

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**Background:** Wnt/B-catenin pathway activation, which is frequently present in most colorectal cancers and a variety of other solid tumors, is associated with poor prognosis and resistance to immunotherapy. FOG-001 is a Helicon peptide that competitively inhibits the interaction between β-catenin and the T-cell factor (TCF) family of transcription factors. Experiments in patient-derived-xenograft (PDX) models of colorectal cancer (CRC) and hepatocellular carcinoma (HCC) have demonstrated that FOG-001 can inhibit tumor growth and promote tumor regression when administered as monotherapy or in combination with immune checkpoint inhibitors or standard of care therapies, including bevacizumab and 5-FU. **Methods:** This first-in-human, Phase 1/2, multicenter, open-label, dose escalation (Part 1) and dose-expansion (Part 2) study evaluates the safety/tolerability, pharmacokinetics (PK), pharmacodynamics (PD), and anti-tumor effects of FOG-001 as monotherapy and in combination with other anti-cancer therapies. Eligible patients must have received at least one prior systemic anti-cancer therapy and either progressed on, not responded to, or be unfit for available therapies. In Part 1, FOG-001 is administered intravenously, either weekly or every other week, at escalating dose levels evaluated sequentially in a standard 3+3 design as monotherapy in patients with: (1a) microsatellite stable (MSS) CRC or any solid tumor with a documented Wnt/b-catenin pathway activating mutation (WPAM) and (1c) HCC with WPAM. Part 1b will evaluate PD effects in cohorts of approximately six patients with MSS CRC. Upon selection of the preliminary recommended Phase 2 dose from Part 1a for weekly dosing, combination dose escalation (1d) evaluates the safety/tolerability, PK and anti-tumor effects of FOG-001 in CRC patients combined with FOLFOX+bevacizumab, anti-PD-1/PD-L1 (also includes other solid tumors with known WPAM), or trifluridine/tipiracil+bevacizumab. Part 2 dose expansion evaluates FOG-001 monotherapy in MSS CRC, HCC, and other solid tumors with documented WPAM. Combination dose expansion evaluates the combinations initially evaluated in Part 1d. Primary endpoints are safety/tolerability and preliminary anti-tumor effects (objective response rate). Secondary endpoints are PK, PD, and other anti-tumor responses (e.g., best objective response, duration of response, and progression free survival). Enrollment in Parts 1 and 2 is ongoing in the USA. Clinical trial information: NCT05919264. Research Sponsor: Fog Pharmaceuticals.

**TPS324** **Trials in Progress Poster Session**

# Phase II single-arm, single-center clinical trial of all-trans-retinoic acid, bevacizumab, and atezolizumab in refractory microsatellite stable colorectal cancer.

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**Background:** A critical clinical challenge in microsatellite stable (MSS) metastatic colorectal (mCRC) patients is to identify strategies to overcome lack of response to immune checkpoint inhibitors. An immunosuppressive tumor microenvironment comprising myeloid-derived suppressor cells (MDSC), endothelial cells, regulatory T cells, tumor associated macrophages, and cancer associated fibroblasts promotes immune evasion and resistance to these agents. MDSCs are bone marrow derived myeloid cells that suppress T-cell function and promote tumor growth. Among all cancers, mCRC patients have one of the highest frequency of MDSC (CD11b+, CD33+, CD14+ HLA-DRneg) in blood (1). It was demonstrated that altering the MDSC% in the tumor microenvironment by ATRA and anti-VEGF therapy (bevacizumab) can enhance the effects of immune checkpoint inhibitors (2-4). The hypothesis of the current clinical trial is that the combination of ATRA, bevacizumab and atezolizumab will lead to a decrease in MDSC population in tumor microenvironment leading to a clinically meaningful improvement in response rates among refractory MSS mCRC patients. **Methods:** This is a single-arm, open-label, phase 2 clinical trial combining ATRA, bevacizumab, and atezolizumab in refractory MSS mCRC patients. It will enroll a total of 21 patients over 24 months at UT Southwestern Medical Center that are MSS by PCR or NGS testing or are proficient in immunohistochemical expression of all four mismatch repair enzymes (MLH1, MSH2, MSH6, PMS2). ATRA will be administered orally at 45 mg/m2/day in 2 divided doses on days 1-7 and repeated every 14 days; atezolizumab will be given intravenously on day 1 at 840 mg dose every 14 days, and bevacizumab will be administered intravenously on day 1 at 10 mg/kg dose every 14 days. The first six patients enrolled on this study will contribute to the safety lead-in phase of this study. The primary outcome is to assess the overall response rate by RECIST v1.1. Secondary outcomes include assessment of disease control rate and frequency of adverse events using CTCAE v5.0. Exploratory outcomes include assessment of PFS and OS and collecting blood and tissue samples at defined timelines to study the changes in the MDSC population among responders and non-responders (NCT05999812). 1. Kobayashi, M. et. al., Clin Cancer Res, 2019. 2. Mirza, N., et al., Cancer Res, 2006. 3. Tobin, R.P., et al., Int Immunopharmacol, 2018. 4. Tobin, R.P., et. al., Clin Can Res, 2023. Clinical trial information: NCT05999812. Research Sponsor: Eugene Frenkel Award; BMS Foundation.

**TPS325** **Trials in Progress Poster Session**

# Randomized study to assess colonic microbiome changes in response to energy drink consumption (ROSANNA Trial).

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**Background:** Colorectal cancer (CRC), the second leading cause of cancer deaths in the US, is increasingly diagnosed in individuals under 50. This rise in early-onset CRC (eoCRC) led the U.S. Preventative Services Task Force to recommend a reduction in the age to begin colorectal cancer screening in average-risk persons, from 50 to 45. The etiology of eoCRC is multifaceted, with one postulated theory pointing to alterations in the young adult colonic microbiome. Hydrogen sulfide ($H_2S$) producing bacteria, like *Bilophila wadsworthia*, *Fusobacterium nucleatum* and *Atopobium parvulum*, are typically minor gut microbiota components but are overrepresented in CRC cases, linked to inflammation, and may promote a pro-carcinogenic environment. These bacteria preferentially use taurine, an essential amino acid, as a primary energy source. Energy drinks represent one of the largest dietary sources (6-16x normal daily intake) of taurine in contemporary diets. Our hypothesis is that high taurine levels in energy drinks could exacerbate CRC risk by promoting preferential growth and metabolic activities of already present $H_2S$-producing bacteria, contributing to the rise of eoCRC. **Methods:** This randomized open-label trial is actively enrolling up to 60 young adults (18-40 years) without personal or family history of CRC, other major GI disease/distress, and baseline infrequent energy drink consumption. Group 1 (Arm A; n=30), will consist of subjects consuming at least one of two protocol-specified energy drinks daily for 4 weeks. Group 2 (Arm B; n=30), will consist of usual dietary intake. Enrollments are randomized 1:1 and stratified, by gender, age (18-25 vs. 26-40), and baseline energy drink consumption (rare/none vs. 1-2 per week). Every subject will complete: 1) a daily dietary and fitness log, 2) a weekly wellness log, 3) collection of stool/saliva/skin/urine at 3 sequential time points (baseline, after 2 weeks, after 4 weeks), and 4) blood collection at the same time points. The primary endpoint is the change in $H_2S$-metabolizing bacterial communities between baseline and 1 month of energy drink exposure within Arm A and also compared to those in Arm B using microbiome analysis (diversity, taxonomy, sulfur-enriched gene pathways). Several secondary and exploratory assessments include changes related to subject demographic and dietary differences, concordance of microbiome changes with dietary logs and in other biospecimens beyond stool, $H_2S$-production measurements, and analyses of subject metabolic and immune function. Trial opened in February 2024 with active enrollment ongoing. Updated data on trial status and any amendment modifications will be presented at the meeting. Clinical trial information: NCT06137248. Research Sponsor: McJunkin Family Charitable Trust; National Cancer Institute.

**TPS326** **Trials in Progress Poster Session**

# A phase 3 study of first-line sotorasib, panitumumab, and FOLFIRI versus FOLFIRI with or without bevacizumab-awwb for patients with *KRAS* G12C–mutated metastatic colorectal cancer (CodeBreaK 301).

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**Background:** CodeBreaK 300 (NCT05198934) evaluated sotorasib, a selective inhibitor that irreversibly binds to the Kirsten rat sarcoma (*KRAS*) G12C mutant protein, in combination with panitumumab, an anti–epidermal growth factor receptor inhibitor, and showed that 960 mg sotorasib + panitumumab provided a statistically and clinically significant improvement in progression-free survival compared with trifluridine/tipiracil or regorafenib in patients with chemorefractory metastatic colorectal cancer (mCRC) (1). In CodeBreaK 101 (NCT04185883) subprotocol H expansion cohort 2F, the addition of folinic acid, fluorouracil, and irinotecan (FOLFIRI) to this combination in the first-line setting demonstrated a manageable safety profile and promising response rate of 78% in patients with *KRAS* G12C–mutated mCRC (2). The current study aims to evaluate whether the combination of sotorasib, panitumumab, and FOLFIRI is superior to FOLFIRI with or without bevacizumab-awwb, a VEGF inhibitor, in the first-line setting for mCRC. **Methods:** CodeBreaK 301 (NCT06252649) is a phase 3, multicenter, randomized, open-label, active-controlled study of sotorasib, panitumumab, and FOLFIRI versus FOLFIRI with or without bevacizumab-awwb as first-line therapy for patients with *KRAS* G12C–mutated mCRC. Key eligibility criteria include mCRC with *KRAS* G12C mutation confirmed by central molecular testing, and no prior therapy for metastatic disease. Approximately 450 patients will be randomized 1:1 to receive either sotorasib, panitumumab, and FOLFIRI, or FOLFIRI with or without bevacizumab-awwb. The primary endpoint is progression-free survival assessed by a blinded independent central review of disease response per Response Evaluation Criteria in Solid Tumors Version 1.1 (RECIST v1.1). The key secondary endpoint is overall survival, and additional secondary endpoints include objective response rate and depth of response. Global enrollment is ongoing. 1. Fakih, NEJM 2023. 2. Siena, ESMO 2024. Clinical trial information: NCT06252649. Research Sponsor: Amgen Inc.